Advanced Microeconomic Theory[1]

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[1]This book draft is for my teaching and convenience of my students in class. Please not distribute it.

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Preface

This book is based on my lecture notes of advanced microeconomic theory, which I used and revised during my teaching at many universities, including Texas A&M University, Shanghai University of Finance and Economics, Hong Kong University of Science and Technology, Tsinghua University, and Renmin University of China in the past thirty years. A Chinese version was originally published in 2016.

Economics is a discipline that is seemingly simple, but hard to learn and master well. The main reason why economic issues are challenging to understand and solve is that, besides the common behavior pattern that economic agents at any level (as nation, firm, household, or person) pursues their own interests under normal circumstances, there is information asymmetry in the vast majority of cases given that economic agents are usually privately informed about relevant economic characteristics. For example, a person may say something, but whether or not he or she is telling the truth is unclear; and even if someone is staring directly at you, seeming to be listening attentively, one cannot be certain if he or she is really listening. Indeed, how best to deal with these two most significant objective realities, and what kinds of economic systems, institutions, incentive mechanisms, or policies should be adopted have become core issues and topics in all branches of economics.

At the same time, economics frequently involves normative statements, i.e., subjective value judgments. Different individuals have dissimilar values and varied opinions. For instance, some people emphasize the efficiency of resource allocations, while others highlight the equality of resource allocations. Individuals also frequently hold divergent views on economic reforms and policies. Consequently, it is easy to cause major controversy, making it challenging to understand and master economics.

In addition, the usage of economics is likely to generate strong externality, either positive or negative. Unlike in the case of medical treatment, those who suffer, or even die, from poor medical skills are only a few individuals, while poor economic application may influence all aspects of the economy. As a consequence, a rigorous and systematic training of economics, and especially the main content of microeconomics discussed in this textbook, is not only important for theoretical innovations, but also critical for practical applications. Once faults are made and inappropriate economic policies or rules are implemented, not only will individuals be harmed, but also economic development at the national level. As such, in addition to learning economics well, when making policy suggestions, we should not make decisions with too much concern about your personal interests. Indeed, we should not only be courageous in taking responsibility for being thorough, but also possess a strong sense of social conscience and responsibility.

Modern economics constitutes a dynamically developing and markedly inclusive open discipline. Indeed, it has far surpassed (benchmark) neoclassical economics, and economic practice can provide rich realities for the innovative development of economic theory. From the author's perspective, as long as rigorous inherent logical analysis (not necessarily a mathematical model) is used and rational assumptions (including bounded rationality assumptions, which implies that they will attempt to avoid unfavorable choices) are adopted, such research fits in the category of economics. Under the framework of modern economics, microeconomics primarily concerns the theory of how individuals make decisions, thereby laying the microfoundation of macroeconomics, as well as almost all other branches of economics. It is a theory about how markets operate, as well as how specific market mechanisms should be modified in the case of market failure. It also focuses on the study of how limited resources are allocated among different uses to better meet the various needs of humanity. Therefore, it can assist one to understand the decisive role of market in resource allocation, and the critical role of government in making the market orderly and efficient, providing public goods and services, and guaranteeing social fairness.

Knowing what an economic theory is, it is also necessary to be cognizant of the scope of its applications. If this is not the case, once it is broadly used to guide the formulation of economic policies, it will produce highly negative social impacts. With limited training of theoretical logic and empirical quantifications, and the lack of understanding in the premise of theory, blind applications will indeed lead to negation of the role of economics and the assertion that the basic assumptions of mainstream economics are too strong, and too much focus is given to mathematics and rigor. Consequently, a false conclusion will be reached that economic theories cannot explain and solve practical problems. In extreme cases, one will disavow the basic role of economics in economic development and market-oriented reforms.

In fact, in most cases, people who hold this view do not know the preconditions of a theory, and thus they are unaware of the scope of application of the theory. Once proven to be incorrect, they blame the theory for being fallacious and of no value. In fact, just like the theories of any discipline, any rigorous economic theory provides preconditions, i.e., it is not valid in all situations. Therefore, unless a theory itself possesses one or more logical inconsistencies, there is nothing right or wrong about the theory, but only whether or not the theory is suitable for a particular economic environment. If there is no rigorism, how can one always reach the result of inherent logic? This is as pointed out by Dani Rodrik [2]. These accusations usually come from laymen or certain unorthodox marginalists. Indeed, individuals who hold such opinions are often those who have a limited understanding of economic theory and its methodology. Overall, whether the goal is to conduct original research or applied research, it is necessary to learn economics well, thoroughly understand its analytical framework and methods, and adapt the rigor of logical analysis.

As a high-level course in microeconomics, the purpose of advanced microeconomics teaching is to elucidate the inherent logic existing behind common principles and concepts, and to cultivate students' ability and way of thinking to analyze theoretical issues in a rigorous manner. It also aims to teach students how to characterize the nature of complex economic behaviors and economic phenomena (i.e., through modeling) in order to carry out rigorous economic analyses. This book systematically expounds on the content of modern microeconomics from basic theory, benchmark theory, analytical framework, research methods, to the latest frontier topics. Therefore, all or part of the chapters and sections can be selected for use by doctoral, postgraduate, and senior undergraduate students majoring in economics, finance, statistics, management, applied mathematics, and related disciplines according to the course requirements. This book can also serve as an useful reference book for teachers and scholars engaged in economics teaching and research.

The characteristics of this book

A full coverage of the frontiers of microeconomic theory: The seminal graduate-level texts of microeconomic theory were written during the 1980s-1990s — essentially, Kreps (1990), Varian (1992), and Mas-Colell et al. (1995). Since then, microeconomic theory has developed rapidly, which is especially so for decision theory, matching theory, auction theory, game theory, and mechanism design theory. This textbook therefore systematically embraces these new developments in microeconomic theory, which are important materials for both students and researchers, but cannot be covered by the standard microeconomics textbooks. In particular, matching theory, auction theory, dynamic mechanism design, and cooperative game theory, which consist of 4 chapters in this text, are relevant materials not introduced by the current textbooks, such as Jehle and Reny (2011), and Muñoz-Garcia (2017). Additionally, each chapter of this text includes a Reference section

[2]DANI RODRIK. "Economists vs. Economics", http://www.project-syndicate.org /commentary/economists-versus-economics-by-dani-rodrik-2015-09.

that provides the related and up-to-date literature — books and papers. Therefore, this text should be especially attractive for PhD students in programs with a strong focus on economic theory.

The content is rich and suitable for multiple purposes: This book is rich in content and comprises a full range of topics, including the most typical themes in modern microeconomic theory — consumer theory, producer theory, choice under uncertainty, noncooperative game theory, market theory, general equilibrium theory, social choice theory, and theories of public goods and externality — up to those at the frontier, such as matching theory, auction theory, and mechanism design theory. For each topic, this textbook not only introduces the standard models with theoretical results and intuitive (numerical) examples, but also provides the formal proofs of key lemmas, propositions, and theorems, as well as a discussion of more modern developments in the subfield. Therefore, by selecting different chapters and/or sections, it can serve as the text for a graduate or an upper-level undergraduate course in microeconomics, a sequential course of advanced microeconomics for doctoral students, a course in advanced topics in microeconomic theory, or even a course in mathematical economics. Additionally, it can be used as an important and user-friendly reference book for scientific researchers in the fields of economics, finance, political science, and other social sciences.

A special emphasis on the organic balance between economic thoughts and scientific rigor throughout the text: While the mainstream of the economics profession prior to World War II focused mainly on economic thoughts and lacked scientific rigor, the most attention is currently paid to techniques and rigor, and the profound economic thoughts that exist behind economics as well as and the intuition behind assumptions and results are largely neglected. It is the case that many learners get lost in mathematical models, and may not know the underlying assumptions and profound insights of these economic theories. Why can we not achieve the dialectical unity of rigorous academics "with deep thoughts and for deep thoughts"? In fact, rigorous and original research can achieve this unity, numerous highly technical theories and methods also contain profound economic thoughts and insights, and models can reflect them (e.g., general equilibrium theory, mechanism design theory). As such, when studying economics, one should know well not only the academic contents of economics, but also its systems, in order to master the underlying profound thoughts and wisdom. Although it may not be crucial for it to become a norm for the public to understand these thoughts in a mature modern market system, it remains extremely important in institutional transitional countries where the direction of economic and social transformation is not clear. Combining both, this textbook attempts to advocate pursuing academics both "with deep thoughts and for deep thoughts".

A nontechnical and comprehensive discussion on the nature of mod-

ern economics as preliminary knowledge: Modern economics is regarded as a social science that uses and develops fancy and sophisticated scientific methods to address the behavior and interactions of economic agents and how economies work, leading to both rigorous theories and wide applications in practice. Meanwhile, criticism and controversy about the methodology that modern economics adopts always exist. To this end, Chapter 1 presents an elaborate, systematic and comprehensive exposition of the nature, ideas, and methods of modern economics, the basic scope of economics, as well as an overall framework of understanding for further discussions, clarifying possible misconceptions and misunderstandings, as well as responding to the usual criticisms. Economics is a discipline of knowledge that can be applied to the real world. A good mastery of economics, a deep understanding and a comprehensive application of its profound ideas and methods may lead to wiser thinking, broader vision and more comprehensive ability to deal with different things. Success often relies on five comprehensive elements: grand vision, ambitious goal, great passion, hardworking spirit, and unremitting perseverance. These five factors also play a vital role for any achievements, despite the varied level of talent, including learning well and learning readily. Therefore, this part is especially useful and informative for students majored or interested in economics.

Recognition of ancient Chinese thoughts on the market as well as their connections to modern economics: This textbook connects with the profound philosophy and wisdom of ancient China, in an effort to achieve more comprehensive ideological and academic thinking, realize ideological and organic academic integration, and consequently attain pragmatic wisdom. To the best of our knowledge, this textbook is the first time to explicitly point out that many key thoughts and ideas of modern economics had been proposed much earlier than the emergence of economics *per se*. For instance, Confucius affirmed that the pursuit of personal material interests based on social ethics is justifiable, and *Sima Qian*'s thinking contains great wisdom regarding the philosophy of state governance, the importance of economic freedom, and the priority of several basic institutional arrangements. Therefore, what Adam Smith discussed had already been addressed by ancient Chinese philosophers much earlier. However, one should be noted that those ancient Chinese insights and thoughts were just summaries of experience; they did not constitute rigorous scientific themes or disciplines, provide boundary conditions or scope for conclusions, or represent logically inherent analyses.

A collection of the biographies of 44 representative microeconomists: This text has comprehensively compiled the biographies of 44 economists who made pioneering contributions to the development of modern microeconomics, in order to enhance understanding of the origin, development, and inheritance of the various economic theories (including the economic theory expressed in advanced mathematics) discussed in this textbook. This was also done to increase readers' interest in learning economic theory, to expand the comprehensiveness of knowledge, and to balance scientific rigor and profound thoughts well.

Structure of this textbook

Microeconomics focuses on the study of economic phenomena and economic issues from the analysis of individual economic behavior, by developing various benchmark theories and then relatively realistic theories under given or designed institutional arrangements, especially under the market system. This textbook arranges thematic chapters according to this logic. Prior to this, however, the textbook presents an introduction to the requisite knowledge and methods to study advanced microeconomic theories.

The textbook is divided into seven parts. Part I comprises the general introduction and preparatory knowledge of the textbook. Parts II-IV mainly introduce the benchmark models and theories in ideal economic environments in which a spontaneous market works well, as well as analytical frameworks, methods, and tools. Parts V-VII address market failure in terms of efficiency. They primarily examine how to modify the market in the presence of economic externalities, public goods, and especially private information, in order to achieve efficient allocation of resources. The contents of the textbook are as follows:

There are two chapters in Part I. Chapter 1 primarily introduces the nature and methods of economics, the scope of the textbook, as well as the preparatory knowledge and methods of mathematics. It commences with an overview of the nature, scope, thoughts, analytical framework, and research methods of economics, especially microeconomic theory, as well as the similarities between economic thoughts and Chinese wisdom. These contents aim to increase readers' inclusiveness to all subdisciplines of economics, which is not only essential for the development of various branches of economics, but also within disciplines, such as benchmark theories and relatively realistic theories. Chapter 2 provides almost all of the commonly used mathematical analytical tools and methods in economics, and particularly in this textbook. It can be used as the basic text or primary reference textbook for courses of mathematical economics for studying advanced macro/micro economics. It can also serve as a manual reference for basic mathematic tools that are needed to study economics.

Part II comprises three chapters, which largely discuss individual decision-making, including consumer theory, producer theory, and individual choice under uncertainty. The individual decision-making theory constitutes the micro foundation for the establishment of numerous theoretical models in economics, and it occupies a central position in the way that economists think about problems. Many choices are made in risk and uncertain situations. Since people frequently need to avoid some uncertainties, for example, by purchasing insurance, choice under risk and uncertainty is a critical aspect of economics.

Part III consists of four chapters. It mainly discusses game theory and market theory, including basic game theory, repeated game and reputation mechanism, cooperative game, and market theory of various market structures. Game theory has become an extremely important subdiscipline in mainstream economics, a core field in microeconomics, and the most foundational analytical tool for analyzing various interactive decision problems in economics. For instance, monopolistic competition, and especially the discussion of oligopolistic markets, necessitates the use of a great amount of knowledge and results of game theory, and thus it is discussed together as applications.

Part IV comprises five chapters. It mainly discusses the benchmark market theory in the frictionless situation of perfect competition: general equilibrium theory and social welfare, including the positive theory of competitive equilibrium, the normative theory of competitive equilibrium, economic core, fair allocation, social choice theory, and general equilibrium theory under certainty. General equilibrium theory is one of the most important theories in the history of economic theory development in the past 100 years. It provides a core reference and benchmark for better studying and solving practical problems. This part describes the nature of competitive equilibrium, and explores how to achieve the fair allocation of resources, in order to further demonstrate the universality, optimality, and rationality of the market economy system.

The above four parts mainly describe benchmark models and theories under ideal economic environments in which the market results in efficient allocations, as well as the analytical framework, methods, and tools. However, in many cases, the market is not omnipotent and frequently a natural market fails in term of efficiency, especially from the micro- and information-perspective. Indeed, it can face numerous problems, leading to "market failure". Therefore, it is very important to elucidate where the market fails and precisely what corrective actions should be taken by governments, regulators, and/or designers. As a consequence, the next three parts of this textbook focus on how to deal with problems caused by market failures, which is closer to reality.

Part V consists of two chapters. It discusses theories of externalities and public goods, including the typical market failures of externalities and public goods provision. In this part, it will be shown that, in general, these "non-market goods" or "harmful goods" will result in Pareto inefficient allocations, and produce market failure. Due to externalities and the provision of public goods, a primary market is generally not an ideal mechanism for allocating resources.

Part VI has five chapters. The most important three keywords in economics are information, incentive, and efficiency. Therefore, this part is devoted to discussing incentive, information and economic mechanism design, including principal-agent theory under hidden information and moral hazard, mechanism design under complete information and incomplete information, and dynamic mechanism design. Mechanism design does not attempt to change human nature, which is essentially immutable. As such, mechanism design theory investigates whether and how to design a set of mechanisms (i.e., rules of games or systems) to achieve desired goals under the conditions of individuals pursuing their self-interest, free choice, voluntary exchange, incomplete information and decentralized decision-making, and to compare and assess the advantages and disadvantages of a particular mechanism. Whether or not information is symmetric and incentive is compatible constitute the root causes of different performances of alternative competing mechanisms.

Part VII comprises two chapters, which discusses market design of auction theory and matching theory that are the two frontier subfields of microeconomic theory. Market design, as a relatively new field, can be regarded as the specific expansion and extension of the general mechanism design theory in Part VI, and has a wide range of applications in the real world.

Teaching tips

As mentioned above, the content of this textbook may be taught at many different levels, and is intended for courses of microeconomics for graduate and advanced undergraduate students, topics of advanced microeconomic theory, or courses in mathematical economics. The rich and fully expounded-upon content in this textbook also provides a broad range of choices and space for the teaching of microeconomics at different levels, and for the aspects that teachers consider to be the most important. Moreover, teachers can flexibly select relevant chapters and sections to teach according to their teaching needs. The second chapter, about which concerns the knowledge of mathematics, can be also be used as a teaching material or an important reference for mathematical economics.

The following are some suggestions for instructors who choose to use this textbook. (1) For microeconomics sequence I for graduate students and senior undergraduates, consider the following chapters for the teaching of one semester: Chapters 3-5 on individual decision-making and Chapters 6-8 on game theory. (2) For microeconomics II for graduate students, consider the following chapters for the teaching of one semester: Chapter 9 on market theory, Chapters 10-13 on general equilibrium theory, Chapters 14-15 on externalities and public goods, and Chapters 16-17 on principal-agent theory. (3) For advanced topics in microeconomic theory for graduate students, depending on different foci, the instructor can choose Chapters 18-20 on mechanism design theory or Chapters 21-22 on market design. Of course, the selection of chapters needs to be based on the preference of instructors, research interests, and constraints on course time. In addition, in teaching advanced microeconomics I, II, or advanced topics in microeconomic theory, it is crucial for students to know the context of Chapter 1 on the nature, category, analytical framework, thoughts and methodologies of economics, and thus students should first either self-study Chapter 1 or be taught it by instructors. In addition, for undergraduates to obtain a general understanding of the scope, ideas, analytical framework, and research methods of economics, they are strongly recommended to read Chapter 1 and the introduction sections of the other chapters.

Doing exercises is the most reliable and effective way to master the content of teaching materials. The exercises at the end of each chapter were written by myself, adapted from classical textbooks and the Doctoral Qualification Examination Question Bank of the Economics Department of world-class universities, or are examples or basic conclusions of original academic papers, indicating their sources whenever possible. I would also like to express my sincere gratitude to the anonymous authors of many exercises in this textbook.

Research tips

This textbook can also be used as a reference for research. Whether it is to carry out original theoretical, applied or policy-oriented research, it is necessary to learn economics well, master its basic analytical framework and research methods, and establish a solid theoretical and methodological foundation. From the content of textbook, it is possible to learn the most advanced microeconomic theory and engage in relevant research. In general, the study of economics can be broadly divided into two categories. The first category is the study of basic, original, and common theories and tools, most of which have no borders and can generally be applied to any region. The various theories introduced in this textbook basically fall into this category. The second is the attempt to solve pragmatic or practical issues, i.e., applying the basic principles, analytical frameworks, research methods, and analytical tools of economics to study and solve real-world problems of a nation or region. Indeed, these two are dialectically unified, and it is crucial not to negate the latter by the former, or vice versa. This is the same situation as in basic research in the natural sciences and technological innovation in industry: they are complementary, equally important, and indispensable.

In addition, it should be pointed out that economics, and especially microeconomic theory, mainly provides two categories of theories, both of which possess stringent prerequisites. The first category is to provide various benchmark theories. Most of the theories introduced in Parts II-IV of this textbook belong in this category. The second category is to deliver various relatively more realistic theories. Parts V-VII are more closely associated with reality, and are primarily theories that are proposed to solve practical problems. The first category of theories is primarily based on the economic environments of mature market economies or nations, and provides basic theory in ideal situations. Although it serves an important role in guiding improvement or reform orientation, it deviates from reality, especially concerning economic environments that are still in the process of transition. As a consequence, it is necessary to revise the benchmark theory and consider situations that have frictions and are closer to reality, in order to develop a relatively more realistic economic theory that solves specific practical problems, draws accurate conclusions, and makes predictions with intrinsic logic. For these reasons, these two categories of theories exist in a progressive relationship in the development of disciplines. The basic analytical framework, thoughts, and research methods of economics described in this textbook are not only common to the study of these two types of theories, but can also be employed to better apply economic theory to investigate real-world economic problems. They can even establish a solid theoretical and methodological foundation for the development of economic theories that are suitable for studying transitional economies. This requires the special attention of researchers when using this textbook.

Acknowledgements

Economics is a mature discipline and, therefore, it is little wonder that this book may remind the readers of the other textbooks which have already been written and published. As such, I did not intend to be fully original. Instead, I tried to benefit from books already in existence and adapted some interesting examples and worthy pieces of models presented there. I have included a useful, but hardly exhaustive reference guide at the end of each chapter. In addition, this textbook incorporates, in almost every chapter, some results of my papers published in the past 30 years. Here, first of all, I would like to express my deep gratitude to many collaborators for inspiring my research.

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ticipated in the practice of the reform and development of China's economic education, including the introduction of a large-scale, organized-system of overseas high-level talents, and the reform of the curriculum system with advanced international standards. By now, the School of Economics at SUFE is internationally well-known. According to the Worldwide Economics Departments Research Ranking by Tilburg University, SUFE ranks constantly the first among all the universities in mainland China, top three in Asia, and 50s worldwide in last eight years. During that period, almost every year, I have given doctoral students at SUFE course, entitled Advanced Microeconomics II. Without this opportunity, this textbook would not have been possible.

In short, this textbook is a crystallization of many people's wisdom in a sense. Of course, I am fully responsible for the deficiencies and errors of this textbook.

Guoqiang Tian
In my Xingkong Study in Texas
December, 2021

Part I

Preliminary Knowledge and Methods

In order for readers to grasp the content in this textbook more effectively, learn economics, and rigorously understand its profound economic thoughts, theoretical models and proofs, this part introduces the preliminary knowledge and methods of economics and mathematics.

Chapter 1 briefly discusses the nature, essence, category, thoughts, and methods of the discipline of economics. We will fully discuss the ideas and methods of economics, especially those that are involved in this textbook. While the mainstream of the economics profession prior to World War II focused largely on qualitative analysis and economic concepts, to a great extent lacking scientific rigor and quantitative analysis, it currently seems that primary attention is given to techniques and rigor, and the profound economic thoughts behind economics are largely neglected. When studying economics, one should know not only the academic contexts of economics well, but also its systems, in order to master its deep thoughts and wisdom. Combining both, this textbook attempts to advocate pursuing academics “with deep thoughts and for deep thoughts”.

Chapter 2 introduces the basic knowledge and results of mathematics that are indispensable for studying economics, in general, and advanced microeconomic theory, in particular. They are used for rigorous analysis of economic problems and for derivation and proofs of theoretical results. In other words, they provide requisite mathematical knowledge and tools for formalization, axiomatization, and scientification of microeconomic theory.

Chapter 1

Nature of Modern Economics

In this chapter, we discuss the nature, role, and methodology of economics, categories of theories, and the scope, preliminary knowledge, concepts, and methods involved in the textbook. We will present the basic terminologies, core assumptions, standard analytical frameworks, methodologies and techniques used in economics, and discuss its research object of the market system, its connection with ancient Chinese economic thought, as well as some key points.

Methodologies and techniques for studying economics include the following: providing benchmarks, establishing reference systems, setting up studying platforms, developing analytical tools, conducting positive and normative analyses, mastering the basic requirements of learning economic theory, understanding the role of economic theory, clarifying necessary and sufficient conditions for a statement, elucidating the role of mathematics and statistics in economics, and becoming familiar with the conversion between economic and mathematical languages.

1.1 Economics and Modern Economics

1.1.1 What is economics about?

To learn economics well, one must first know its definition and understand its nature, connotation, scope, and concerns.

Economics is a social science that studies how people interact with value in the face of resource scarcity and/or information asymmetries. Specifically, it studies economic behavior and phenomena, and how rational individuals (agents, households, firms, nations, organizations, and government agencies) make trade-off choices with limited resources.

In fact, economics could only come into being due to the fundamental inconsistency and conflict that exists between resource scarcity and individuals' unlimited desires (or wants). The core idea is that individuals, who are under the basic constraint of limited resources (i.e., limited information, capital, time, capacity, freedom etc.) and driven by unlimited desires, must make trade-off choices in resource allocation to make optimal utilization of limited resources to maximize satisfaction of their needs.

As a discipline of social science, economics investigates the problem of choices based on logical analysis and scientific tools, and establishes itself via systematic exploration of the specific topic of choice. Such exploration not only involves the building of theory, but also provides analytical tools for the testing of economic data.

1.1.2 Four Basic Questions in Economics

For any economic system, regardless of whether it is a planned economy, wherein the government plays a decisive role, a free economy, wherein the market plays a decisive role, or a semi-market and semi-planned mixed economy, wherein the state-owned economy plays a leading role, all face the following four basic questions regarding the allocation of resources:

(1) What should be produced, and in what quantity?

(2) How should the product be produced?

(3) For whom should it be produced, and how should it be distributed?

(4) Who makes the decisions?

Although these questions must be answered in all economic systems, different economic institutional arrangements provide varied answers. Whether an institutional arrangement can effectively resolve these problems depends on whether it can effectively deal with the issues induced by information and incentives, and result in efficient or equitable allocations of resources.

Two basic economic institutional arrangements have been used in the real world:

(1) The institutional arrangement of planned economies: All of the four questions are answered by the government, who determines most economic activities and monopolizes decision-making processes and all sectors. The government makes all decisions on market access, product catalogue, infrastructure investment allocation, individual job assignment, product pricing and wages, and it bears all of the risk.

(2) The institutional arrangement of market economies: Most economic activities are organized through the free market. The decisions about what to produce, how to produce and for whom to produce are mainly made by decentralized firms and consumers, and the risk is borne by individuals.

While almost every real-world economic system exists somewhere in between these two extremes, the key factor is which extreme is in the dominant position. Due to the pursuit of personal interests and the presence of private or incomplete information, the fundamental flaw of the planned economic system is that it cannot effectively resolve problems induced by information and incentives, which in turn results in inefficient allocation of resources. On the other hand, the free-market economic system provides a viable solution in these respects in most situations. This is the fundamental reason why countries that once adopted a planned economic system inevitably failed, and why China carried out market-oriented reforms and strove to have the market play the decisive role in resource allocation.

1.1.3 What is Modern Economics?

Modern economics, which has developed rapidly since the 1940s and was constructed on the basic recognition of individuals pursuing their self-interest, systematically studies individuals' economic behavior and economic phenomena by intensively using mathematical tools and adopting scientific methods for rigorous thinking. Specifically, it makes historical and empirical observations of the real world, utilizes the observations towards the formation of theory through rigorous logical analysis, and then again tests the theory in the continuing real world. As a consequence, it is a branch of science equipped with a scientific analytical framework and research methodology. This systematic inquiry not only involves the form of theory, but also provides analytical tools for testing economic data. For brevity, modern economics is simply termed the economics. In the following we will use the both terms interchangeably.

Scientific economic analysis, especially aimed at studying and solving major practical problems affecting the overall situation, is inseparable from “**three dimensions and six natures** ”, among which the “three dimensions” are “**theoretical logic, practical knowledge, and historical perspective**” and the “six natures” are **scientific, rigorous, realistic, pertinent, forward-looking, and thought-provoking** ”.

Social economic issues generally cannot be studied by only using real society and performing experiments. Studying and especially solving major social economic problems need not only theoretical logic analysis and empirical tests, but also the confirmation of historical experiences. ***Only using theory and practice is insufficient, and may cause shortsightedness***, because the short-term optimum does not necessary equate to the long-term optimum. As a consequence, historical comparisons from a broad perspective are also requisite for gaining experience and drawing lessons, since many basic laws, norms, human behavior patterns and values are often exogenously shaped by historical and repeated lessons learned, and are thus difficult to be drawn from logical analysis or reality. Of course, with the change of time, if merely relying on experience, men's cognition will be deficient and stick in the mud, it will be difficult to have innovation, and would be hard for economic and social development. As such, it requires both a theoretical logical analysis of deductive or inferred results under given assumptions, and a justification of the plausibility of the assumptions through historical experience and practical knowledge of the results or inferences derived from a logical or quantitative analysis. Therefore, only through the three dimensions of "theoretical logic, practical knowledge, and historical perspective" can we guarantee that a solution or reform measure satisfies the "six natures". Therefore, the "three dimensions and six natures" are indispensable as scientific and wise decision-making.

Indeed, all knowledge is presented as history, all science is exhibited as logics, and all judgment is understood in the sense of statistics. This is in line with the great wisdom of the "three Yi" principle in the Book of Changes (I Ching or Yi Jing), an ancient Chinese divination text and one of the earliest Chinese classics. The "three Yi" refers to Bian Yi (changing), Jian Yi (simple) and Bu Yi (unchanging). "Bian Yi" indicates that nothing in the universe is eternal, so there is the need for adjustment to changing circumstances and the courage to innovate; that is, "different prescriptions for different diseases", rather than sticking to established rules. How to achieve this? The answer is to "simplify". As many things are beyond the existing knowledge and understanding, complex problems need to be simplified to seize the essence. Once we understand the underlying laws and principles, the problems are relatively easy to solve, such as in logical inference and empirical analysis, and in theory combining with practice. "Bu Yi" refers to the essence, the unchanged intrinsic rules, which should be followed as they are laws of development honored by time and history, and are not liable to change. This is why Joseph Schumpeter (see Section 2.12.2 for his biography) asserted that the difference between an economic scientist and a general economist lies in whether adopting the following three elements when conducting economic analysis: the first element is theory for logical analysis; the second is history for historical analysis; and the third is statistics for empirical analysis with data.[1]

For theoretical innovations and practical applications, it is of critical importance to correctly understand and master general knowledge of economics and the content of this textbook. It is useful for studying and analyzing economic problems, interpreting economic phenomena and in-

[1]In his inaugural speech titled "Science and Ideology" when assuming the position of President of the American Economic Association in 1949, Schumpeter pointed out that science is knowledge processed by special skills. Economic analysis, i.e., scientific economics, involves skills of history, statistics, and economic theory. Refer to Schumpeter, Joseph A. (1984). "Science and Ideology" in Daniel M. Hausman, eds., *The Philosophy of Economics*, Cambridge: Cambridge University Press, 260-275.

dividuals' economic behavior, setting up goals, and identifying the direction of improvements. More importantly, with the support of comparative analysis from the historical perspective and quantitative analysis based on data, we can draw reliable conclusions of inherent logic and make relatively accurate predictions through rigorous inference and analysis.

Economics — a key part of social sciences, is referred to as the "crown" of social sciences due to its extremely general analytical framework, research methods, and analytical tools. Its basic ideas, analytical framework, and research methodologies are potent for studying economic problems and phenomena that occur in different countries, regions, customs and cultures, and can be applied to almost all social sciences. It can even be beneficial if one strives to be a good leader with strong leadership ability, management and work ethic. Indeed, it is lightheartedly referred to as "economic imperialism" or an "omnipotent discipline" due to a bunch of influential works by Gary S. Becker (1930-2014, see Section 13.7.2 for his biography), who applied economic analysis to the entire spectrum of human behavior, including areas previously considered more or less the exclusive domain of sociology, psychology, criminology, demography and education.

1.1.4 Economics vs. Natural Science

There are three major differences between social science, especially modern economics, and natural science:

(1) Natural science does not consider subjective phenomena, such as thought, intention, and will, but basically measures and calculates, while economics considers both objective phenomena and subjective factors, especially human behavior. The Austrian economist Ludwig von Mises captured the difference with the following statement: "You throw a rock in water, it sinks; throw a stick in water, it floats; but throw a man into water, and he must **decide** to sink or swim." Economics studies human choices and needs to impose certain behavioral assumptions, especially in a theoretical analysis where behavioral assumptions are exogenously given; whereas, natural science, in general, does not involve the intention and behavior of human beings (of course, such distinction is not absolute; for example, biology and medicine sometimes involve human behavior. However, these involvements are not from the perspective of rationality, while economics considers human behavior primarily from the perspective of individual motivations). Once individuals are involved, information is highly incomplete and private (asymmetric) and easy to obscure because their behavioral choices are unpredictable or they reveal information strategically, making it highly challenging to deal with.

(2) In the discussion and study of economic problems, positive analysis of description and normative analysis of value judgment are both needed. As people possess dissimilar values and self-interests, controversies

frequently emerge, while natural science generally makes descriptive positive analysis only and the conclusions can be verified through experiments or practice.

(3) Society cannot be simply experimented upon or subjected to tests to form conclusions in economics because policies have broad impacts and large externalities. However, this does not constitute a problem for almost all branches of natural science.

These three differences may make the study of economics more complex and harder. In order to study and solve practical economic problems, one must start from the reality, combine theory with practice, establish the overall and systematic thinking, and adhere to the comprehensive governance concept with general equilibrium analysis as the core, rather than simple controlled experiment (although it is the first step of scientific research). It is essential to adopt the research methodology of “three dimensions and six natures” mentioned above.

1.1.5 Modern Economics vs. Other Social Sciences

Economics is an important component of social science. Social science studies human interests and social phenomena, so that the demarcation among its various branches is relatively vague, without a unified criterion. Traditional classification of social science disciplines is based on their fields of study. Roughly speaking, history is concerned with people and events in the past, anthropology with people of different races or regions, law with regulation, institutions and laws, political science with the world and the state, sociology with the activities and phenomena of social groups (collectives), and economics with individual economic activities.

Economics differs from other disciplines in methodology. Economics emphasizes microscopic (individual) analysis as its starting point. Modern economics takes a more comprehensive approach to scientific research, including the methodology of "three dimensions and six natures" mentioned earlier, while other social sciences only focus on certain parts of the "three dimensions and six natures". Meanwhile, modern economics provides a methodological cornerstone for other disciplines, such as finance, management, sociology, and political science. Economics not only emphasizes a historical perspective and a sense of mission for the country and the people, but also attaches importance to understanding the logic behind human behavior and economic phenomena. It possesses the rational and rigorous thinking like natural science, such that it is the social science closest to natural science. Therefore, modern economics is more neutral and universal than other social sciences with strong ideological and value judgment.

It is the methodological individualism and the diversity of research fields and methods that makes modern economics the key part of social science. Economics is referred to as the “crown” of social sciences or "economic imperialism" due to the wide application of its methodology in other social sciences. The basic ideas, analytical framework, and research methods of modern economics are potent for studying economic problems and phenomena that occur in different countries, regions, customs and cultures, and can be applied to almost all social sciences. It can even be beneficial if one strives to be a good leader with strong leadership ability, management and work ethic. Indeed, it is lightheartedly referred to as an "omnipotent discipline" due to a bunch of influential works by Nobel Laureate Gary S. Becker (1930-2014, see Section 13.7.2 for his biography), who applied economic analysis to the entire spectrum of human behavior, including areas previously considered more or less the exclusive domain of sociology, psychology, criminology, demography and education.

1.2 Two Categories of Economic Theory

Modern economic theory is an axiomatic way to study economic issues. Similar to mathematics, it relies on logic deductions from presupposed assumptions. It further consists of assumptions/conditions, analytical frameworks and models, and conclusions (interpretations and/or predictions). Since these conclusions are strictly derived from the assumptions and analytical frameworks and models used, it constitutes an analytical method with inherent logic. This analysis method is highly advantageous for clearly elucidating the problem and can avoid unnecessary complexities and disputes. Economics aims to explain and evaluate observed economic phenomena and make predictions based on economic theory.

1.2.1 Benchmark Theory and Relatively Realistic Theory

Economic theory can be divided into two categories by function. One is benchmark economic theory, which provides various benchmarks or reference systems[2], which is relatively remote from reality and deals with ideal situations. Parts II-IV of the textbook provide such benchmark theories. The second category is a relatively realistic theory that aims to solve practical issues, so that assumptions are closer to reality, which are usually modifications to the benchmark theory. Parts V-VII provide such relatively applied theories. As such, both types of theories are essential, and can be used to draw logical conclusions and make predictions. In addition, a progressive and complementary relationship of development and extension exists between these two categories. The second category of realistic theories is developed by revising the first category of benchmark theories as the

[2]We will come back to discuss the role of the benchmark or reference system in more detail.

reference frame, thus making the theoretical system of economics complete and proximal to the real world.

The benchmark theories are largely built on the economic environment of mature market economies and ideal situations. Their great significance should not be underestimated, misunderstood, or denied. They have demonstrated their critical importance in at least two aspects.

Firstly, although theoretical results of this category do not exist and cannot be realized in practice, they do play a critical role in providing guidance, orientation, and benchmarks. When we tackle a problem, it is necessary to first determine what to do and whether it should be done, and then proceed to the question of how to do it. Benchmark theories answer the question of what to do, or provide the direction and goals of improvement towards the ideal situation. Although it is the case that sometimes only a relatively better result can be achieved, the optimal outcome can be approached through the process of continually comparing the outcomes with the benchmark or reference system. This is why it is correctly claimed that it is only learning from the best and comparing with the best, can we perpetually improve. Therefore, the benchmark theory provides necessary criteria for judging what is better and whether it constitutes the right direction, without which what we are doing may not be moving us towards our goals at all.

Secondly, it establishes the necessary foundations for developing the other category of realistic theories. Any theory, conclusion, or statement can only be considered relatively; otherwise, there will be no basis for analysis or evaluation. It is for this reason that benchmark theories are required. This is true for both physics, which is a natural science, and economics, which is a social science. For instance, a world with friction is relative to a world with no friction, information asymmetry is relative to information symmetry, monopoly is relative to competition, technological progress and institutional changes are relative to technological and institutional lock-in, etc. Consequently, we must first develop the benchmark theory under rather ideal situations. This is analogous to basic laws and principles in physics, which only hold under an ideal situation without friction, and do not exist in reality, but nevertheless remain fundamentally important because they provide requisite benchmarks for solving physics problems in reality. Similarly, to study real economic behaviors and phenomena, which include "friction", it is necessary to first be clear about the ideal situation without "friction", and then use it as a benchmark and reference system. Indeed, the rapid development of economics would be impossible without benchmark economic theories.

As an important part of economics, neoclassical economics assumes the regularity conditions of complete information, zero transaction costs, and convexities of consumer preference and production sets, and thus falls into the first category of benchmark economic theory. Neoclassical economics considers ideal situations; although there is no artificially designed social goal, it contends that, as long as individuals are self-interested, the market of free competition will naturally lead to the efficient allocation of resources. This is regarded as a rigorous statement of the "invisible hand" proposed by Adam Smith (1723-1790, see 1.17.1 for his biography). It is thus set as the reference system for us to determine the direction and goals for reforms in order to improve the economic, political, and social environment, establish the competitive market system, and let the market play a decisive role in the allocation of resources. The boundary conditions for the market to work well also inform us about when the market will fail, at which time the government will need to step in and play a guiding role.

One may assert that an ideal reference system is far removed from the real economy, and thus deny the role of neoclassical economics and reject the instructive role of economics in economic reform. This, however, constitutes a serious misunderstanding, as it fails to acknowledge that the great gap between reality and the benchmark/reference system only shows the necessity for a nation, such as China, to implement market-oriented reforms and to continuously improve the efficiency of resource allocation. This kind of opinion, which refutes the role of benchmark theory, is similar to the denial of physics by a junior high school student who has just learned several formulas of Newton's three laws and criticizes them for postulating conditions that are totally dissimilar from those in the real world. In these cases, however, the role of benchmark theory is not understood correctly. Indeed, without the benchmark theory in physics concerning free fall and uniform motion, how could we know the magnitude of frictional force so that we could construct a house that is stable ? Furthermore, how could we determine how much frictional force should be overcome to solve problems regarding the taking-off and landing of airplanes or the launching of satellites? It is clear that, without benchmark theory, applied physics cannot be developed. The study of economics follows the same logic, and thus we need directions, structures, goals, and certain fundamentals, which is especially the case for implementing reforms in transitional economies.

To facilitate reforms for transitional economies, goals must be established, and thus benchmarks and reference systems are required for the reform to orientate itself. For the social economic development of a country, it is necessary to transcend rational thinking,theoretical discussion and theoretical innovation, and determine the direction and goals of the reform in the first place. Moreover, it must be acknowledged that the fundamental institutions that determine the rule of collective decision-making, legal systems, strategies and policies play decisive roles in this process. If basic institutions of the rule of law, legal systems, politics, economy, society, and culture that concern a nation's path of development and long-term stability are not determined, economic theories in the present state-of-the-art may accomplish nothing, and may even have deleterious effects. In the discipline of economics, there is not a universal economic theory that is always applicable for all development stages, but rather there is an optimal one that is best suited for certain development stage under certain institutional environments.

For market-oriented reforms, it is natural and necessary to set neoclassical economic theory - especially economic theory of the first category, such as general equilibrium theory, that demonstrates the market as the optimal economic system - as a benchmark and the competitive market as a reference system for the orientation of these reforms. In this way, the results of the reforms will be continuously improved towards reaching the best possible outcome.

According to the economic environment defined by these benchmarks, reforms of deregulation and delegation for competition neutrality must be carried out, including ownership neutrality, liberalization, privatization, and marketization or reforms against government monopoly of resources and control of market access. In particular, the general equilibrium theory defines the applicable range of market mechanisms and identifies the environments under which the market may fail. With such knowledge, it is possible to identify the areas in which governments could establish rules and institutions to correct market failures.

Therefore, the study of economic problems and reforms, and especially determination of the direction of reforms, must commence from the benchmark of economics. Reforms that run counter to common sense in economics will end up in failure. The benchmark and reference system present the premises on which the market will lead to a more efficient allocation and result in a more prosperous market economy, thereby revealing the direction of the reforms. The fourth part of this textbook stresses the Arrow-Debreu general equilibrium theory (Kenneth Joseph Arrow, 1921-2017, see Section 10.8.2 for his biography; Gerald Debreu, 1921-2004, see Section 11.9.2 for his biography) and the rational expectations theory of macroeconomics (referred to as neoclassical macroeconomics), both of which are standard theories of neoclassical economics and rigorously demonstrate that markets of free competition lead to efficient allocation of resources.

Alternative benchmarks and reference systems under different value judgements and goals could lead to markedly divergent outcomes. For example, when students regard a “pass” grade as their benchmark, the result is frequently a failure because a test comprised of questions is a random variable to the students. Just as Confucius and Sun Zi asserted, those who aim at the superior get the medium, those who aim at the medium get the inferior, and those who aim at the inferior lose entirely, which illustrates the critical importance of the choice of benchmark. Meanwhile, given that many benchmark theories are established under ideal conditions, they cannot be simply adopted to solve real problems in practice. In other words, a well-trained economist will not mechanically apply economic theories in the first category. However, concerning the economic reforms implemented in some transitional economies, we may come across some economists who do not analyze the dynamics in the transition, consider only developed countries but not developing countries, and neglect the objective law and constraints in special development stages. As such, both the target and the path selection and schedule arrangement for accomplishing the target are of crucial importance in practice. In addition, although some short-term economic and social problems might be resolved through promoting growth and development per se, market-oriented reforms should not be delayed; otherwise, an emerging economy is likely to be trapped in the transitional status.

It should be pointed out that a benchmark economic theory is established under an exogenously given economic environment that constitutes a relatively ideal situation. In so doing, we are able to address the key issues and draw some benchmark conclusions; otherwise, no question can be scientifically discussed if we do not control for many factors. Therefore, in neoclassical theories and numerous other economic theories, an economic environment that comprises basic institutions, individual preferences and production technologies must be exogenously-given. Elaborating further, if achieving the market system under the ideal situation is our goal, it is logical to set it as a given institutional arrangement in order to thoroughly elucidate its desired properties. However, this does not mean that economics only investigates situations in which the institution is given. In fact, many theories in economics are specialized in addressing how the economic environment changes, such as the study of institutional evolution, economic transition, endogenous preferences and technological progress. As such, one should not misinterpret economics as a sort of discipline that provides only benchmark economic theories established under ideal situations.

Theories in the second category constitute relatively more realistic economic theories that aim haw to solve practical problems, and are constructed on presupposed assumptions that more closely approach reality and are modifications of the benchmark theories. According to their functions, they can be further divided into two types: the first kind provides an analytical framework, method, or tools for solving practical problems, such as game theory, mechanism design theory, principal-agent theory, auction theory, matching theory; and the second kind offers specific policy suggestions, such as the Keynesian theory, rational expectations theory, and growth theory in macroeconomics.

1.2.2 The Domain and Scientific Rigor of Economics

It can be seen from the definition of the above two categories of economic theories, modern economics is a highly inclusive and open discipline in dynamic development, which far exceeds the scope of neoclassical economics. Through weakening the assumptions of benchmark theories and standardized axiomatic formulation of descriptive theories, economics continuously develops the second category of economic theories, which grants itself great insight, explanatory power, and predictability. In the author's opinion, *as long as a study involves rigorous logical analysis (not necessarily using mathematical models) and adopts rationality assumption (bounded rationality assumption included), it falls into the category of (modern) economics*.

Modern economics originated from classical economics, which was developed based on integration of Adam Smith's work by Thomas Robert Malthus (1766-1834, see Section 4.6.1 for his biography) and David Ricardo (1772-1823, see Section Section 1.17.2 for his biography), including not only benchmark theories, such as neoclassical marginal analysis economics established by Alfred Marshall (1842-1924, see Section 3.11.1 for his biography) and Arrow-Debreu's general equilibrium theory, but also many more realistic economic theories. For instance, the new institutional economics by Douglass C. North (1920-2015, see Section 5.5.1 for his biography) and mechanism design theory by Leonid Hurwicz (1917-2008, see Section 16.10.2 for his biography) have both developed neoclassical theory in a revolutionary manner. Specifically, while neoclassical theory takes institutions as given, North and Hurwicz endogenized institutions, viewing them as changeable, shapeable and designable, and thus formulated various institutional arrangements by complying with human nature for different environments. Indeed, they have both become crucial components of economics. Furthermore, the development of new political economics has, to a large extent, borrowed the analytical methods and tools of the second category of economic theory.

It is important to note that, because theories of the second category strive to develop analytical frameworks, methods, and tools for solving practical problems and offer specific policy recommendations largely based on a mature modern market system, their application must be approached cautiously. In fact, every rigorous economic theory in modern economics possesses a self-consistent logic and consequently must have boundaries and scopes within which they are applicable. This is true for not only original theory but also analytical tools, and for not only the first category of theory that offers benchmarks or reference systems but also the second category of theory that aims to solve practical economic problems. As a consequence, much mathematics is frequently needed, which incurs a common criticism of economics' overemphasis on model details and increasingly heavy involvement of mathematics and statistics, making economic questions and conclusions per se even more opaque and challenging to understand.

The major reason that modern economics uses a substantial amount of mathematics and statistics is that economists must scientifically (both qualitatively and quantitatively) identify the applicable scope of an economic theory, which is especially the case when proposing economic policies based on the economic theory. Once a theory is adopted for making policy, great negative externalities may come into being if the boundary conditions of the theory are not known. In particular, it is the economists who make policy proposals, rather than the policy-makers and the public, that must have a good knowledge of the details or premises of a rigorous theoretical analysis. To this end, mathematics is needed to thoroughly identify the boundary conditions and applicable scopes of economic theories; simultaneously, economists equipped with the knowledge of mathematics can lay a strong foundation for continuous theoretical innovations. Moreover, the application of a theory or the formulation of a policy will usually require the use of statistics and econometrics for quantitative analyses or empirical tests.

In most cases, as real society cannot be simply used for experiments, a larger historical perspective is useful for viable vertical and horizontal comparisons. In addition, many superfluous disputes can be avoided in the exploration and discussion of certain questions. Hurwicz, for example, believed that the biggest shortcoming of traditional economic theories is the imprecise explanation of concepts, while the greatest significance of the axiomatic method lies in its formalization of the theory, providing a commensurable research paradigm and analytical framework for both discussion and criticism.

As a result, as the basic theoretical foundation for market economies, economics relies heavily on the introduction of research methodology and analytical framework of natural sciences to study social economic behaviors and phenomena. Indeed, using mathematical models as basic analytical tools, it stresses the inherent logic from assumptions and derivations to conclusions, along with the use of statistics, econometrics and computer simulations for data-driven empirical research, laboratory experimental research, as well as field research. As such, comparing to other humanities and social sciences, modern economics exhibits strong characteristics of positivism and pragmatism, and is more likely to have the flavor of natural sciences.

1.2.3 The Roles of Economic Theory

Economic theory has at least three roles.

The first role is to provide various benchmarks and reference systems to establish desired goals in order to create directions to be pursued for improvement. Through reforms, transitions and innovations guided by theory, the economy in the real world is driven increasingly proximal to the ideal state.

The second role is to be used to learn and understand the real economic world, and to explain economic phenomena and economic behaviors in order to solve real problems. Indeed, this is the major content of economics.

The third role is to be used to make logically inherent inferences and predictions. Practice is the sole criterion for testing truth, but not the sole criterion for predicting it. In many cases, problems may still arise if only historical examination and existing data are employed in making economic predictions, and thus theoretical analysis with inherent logic is imperative. Through logical analysis of economic theory, it is possible to make logically inherent inferences and (short, medium, and long-term) predictions on possible outcomes under given economic environments (behavioral assumptions and economic institutions). In so doing, we can solve real economic problems in a better way. Of course, whether they conform to reality requires the verification of the plausibility or falsification of the assumptions through historical experience and practical knowledge of the results or inferences derived from a logical or quantitative analysis. As long as the pre-assumptions in a theoretical model are roughly met, scientific conclusions can be obtained and essentially correct predictions and inferences can be made accordingly, so that we may know the outcomes. For instance, the theoretical inference that a planned economy is unfeasible proposed in 1920s by Friedrich Hayek (1899-1992, see Section 2.12.1 for his biography) possesses this kind of insight. A good theory can deduce logically inherent results without requiring social experiments, which can somehow overcome the shortcoming that economics cannot carry out experiments on real society to a great extent. For example, we are not allowed to issue currency just for examining the relationship between inflation and unemployment. Similar to the case of astronomers and biologists, most of the time economists can only rely on existing data and phenomena to test and revise theories. What is necessary is to check whether the assumptions made on economic environments (e.g., human behavior) are reasonable (experimental economics, which has become popular in recent years, is mainly engaged in fundamental theoretical research on testing individual behavioral assumptions).

Of course, as indicated above, it would not be helpful to exaggerate the role of economic theory, and expect it to solve key and fundamental problems. What is fundamental, key, and decisive is the basic constitution and institution that determine a nation's fundamental path of development. If the underlying system governing the direction of the country and long-term prosperity in politics, economy, society, and culture is not yet built, the application of economic theory may even lead to deleterious results.

1.2.4 Microeconomic Theory

A notable feature of microeconomic theory is that it sets up theoretical models for economic activities of self-interested individuals, especially in market economies, conducts rigorous analysis and examines how the market works on such a basis.

Microeconomics deals with the core issue of pricing. It focuses on such questions as: which factors affect pricing? Do enterprises have market (pricing) power? How can an enterprise get the power of pricing ? How can an enterprise set the optimal price? To elucidate the answers to such a large issue, it is necessary to study the demand, supply, characteristics and functions of the market, and pricing in all kinds of markets and economic environments. As a result, microeconomics is also called the price theory.

Microeconomics constitutes the core of economics and the theoretical foundation of all branches of economics. It enables us to employ simplified assumptions for in-depth analyses of various aspects of the complex world in order to get some useful insights. It also assists us to extract the most useful information from unrelated entities and consider various issues using the method of economics to develop explanations and predictions that conform to reality. It is in this sense that all other branches of economics, such as macroeconomics, finance, applied economics, etc., call for support from microeconomic theory.

1.3 Economics and Market System

A main purpose of economics is to investigate the objective laws of market and individuals' (e.g., consumers and firms) behavior in the market. Specifically, it examines how self-interested individuals coordinate their economic activities and make optimal choices in the market, how the market allocates social resources, how economic stability and sustainable growth are achieved, etc. Therefore, for the purpose of studying economics comprehensively, one should possess a general understanding of the functions and advantages of modern market mechanisms.

1.3.1 Market and Market Mechanism

Here, we briefly introduce the operation and basic functions of the market and how the market coordinates individuals' economic activities without requiring excessive participation or intervention of the government.

Market: The market constitutes a modality of trade in which buyers and sellers conduct voluntary exchanges. It refers not only to the location where buyers and sellers conduct exchanges, but also to all forms of trading activity, such as auction and bargaining mechanisms.

When studying microeconomics, it is crucial to keep in mind that any transaction in the market has both buyers and sellers. In other words, for a buyer of any good, there is a corresponding seller. The final outcome of the market process is determined by the rivalry of relative forces of sellers and buyers in the market. Three forms of competition exist in such a rivalry: consumer-producer competition, consumer-consumer competition, and producer-producer competition. Throughout this textbook, readers will find that the bargaining position of consumers and producers in the market is circumscribed by these three sources of competition in economic transactions. Competition in any form is like a disciplinary mechanism that guides the market process and has varied impacts on different markets.

Market mechanism: The market mechanism or price mechanism is an economic institution in which individuals make decentralized decisions guided by price. It is worth noting, however, that this is usually a narrow definition of market mechanism. In fact, a mature market mechanism or market system constitutes the set of all systems and mechanisms closely related with the market (including the system of market laws and regulations). As a form of economic organization featuring decentralized decision-making, as well as voluntary cooperation and voluntary exchange of products and services, it is one of the greatest inventions in human history and by far the most successful means for human beings to solve their economic problems. Indeed, the establishment of the market mechanism is not a conscious, purposeful human design, but rather a product of the natural process of evolution. In the opinion of Hayek, market order is a spontaneous order of the economy which has evolved through long-term choices and processes of trial and error. The emergence, development, and further extension of economics are mainly based on the study of the market system. In aggregate, the operation of the market appears wonderous and beyond comprehension. It is genuinely awe-inspiring that, in the market system, decisions on resource allocation are independently made by producers and consumers who pursue their own interests under the guidance of market price without the imposition of any command or order. The market system unknowingly solves the previously mentioned four basic questions which must be faced by all economic systems: what to produce, how to produce, for whom to produce, and who makes the decisions.

Under the market system, firms and individuals make the decisions on voluntary exchange and cooperation. Consumers seek maximal satisfaction of their demands, while firms pursue profits. In order to maximize profits, firms must have meticulous plans for the most efficient utilization of resources. In other words, for resources with similar usage or quality, firms will choose the ones with the lowest possible cost. Although “making the best use of everything” may differ in meaning from the point of view of firms and that of the economy, price makes them related which, as a result, harmonizes the interests of firms and those of the entire society, and leads to the efficient allocation of resources. *The price level reflects the supply and demand of resources in the economy and the degree of scarcity of resources.* For example, in the case of an inadequate timber supply and ample steel supply in the economy, timber will be expensive while steel will be inexpensive; consequently, to reduce expenses and make more profits, firms will strive to use more steel and less timber. In doing so, firms do not take the interests of society into consideration, but the outcome is precisely in accordance with social interests, and it is the role of resource price that achieves this. Resource price coordinates the interests of firms and those of the overall society, and solves the problem of how to produce. The price system also guides firms to make production decisions in the interest of society. Indeed, it is the consumer who has the final say about what to produce. Firms only need consider how to produce products that have a higher price. Yet, in the market system, the price level exactly reflects social needs. For instance, poor harvests and the corresponding rising grain price will encourage farmers to produce more grain. As such, profit-pursuing producers "come to the rescue" under this guiding force, and the problem of what to produce is solved. Moreover, the market system also addresses the problem of how to distribute products among consumers. If a consumer really needs a shirt, the consumer will offer a higher price for it than will others. Profit-pursuing producers will certainly aim to sell the shirt to the consumer who offers the highest price. In this way, the problem of for whom to produce is solved. Furthermore, all of these decisions are made by producers and consumers in a decentralized manner, and thus the problem of who makes the decision is also resolved.

As such, the market mechanism easily coordinates seemingly incompatible individual interests and public interests. As early as over 200 years ago, Adam Smith, the Father of modern Economics, identified the harmony and wonder of the market mechanism in his masterpiece, *The Wealth of Nations* (Adam Smith, 1776). He regarded the competitive market mechanism as an "invisible hand". Under the guidance of this invisible hand, individuals solely pursuing their own interests move towards a common goal, and thus achieve maximization of social welfare:

> ... every individual necessarily labours to render the annual revenue of the society as great as he can. He generally, indeed, neither intends to promote the public interest, nor knows how much he is promoting it ... he intends only his own gain, and he is in this, as in many other cases, led by an invisible hand to promote an end which was no part of his intention. Nor is it always the worse for the society that it was no part of it. By pursuing his own interest, he frequently promotes that of the society more effectually than when he really intends to promote it.

Smith meticulously examined how the market system combines the self-interestedness of individuals with social interests and the division and cooperation of labor. The core of Smith's paradigm is that, if the division of labor and exchange of goods are totally voluntary, then exchange will only occur when we realize that the result of the exchange is mutually beneficial to both parties of exchange. Indeed, as long as there are benefits, individuals driven by self-interest will cooperate voluntarily. In other words, external pressure is not a requisite condition for cooperation. Even if language barriers exist, as long as mutual benefits can be obtained, exchange can take place. In most cases, the market mechanism works so harmoniously that individuals are not even cognizant of its existence. With the metaphor of "the invisible hand", Smith highlighted the importance of voluntary cooperation and voluntary exchange in economic activities. It is worth noting, however, that the claim that the welfare of society can be achieved by the market system is not yet recognized universally, nor was it during Smith's lifetime. The Arrow-Debreu general equilibrium theory, which will be discussed in this textbook, contains a formal statement of Smith's "invisible hand", and rigorously demonstrates how the market of free competition can lead to the maximization of social welfare, and proves the optimality of the market in allocating resources.

1.3.2 Three Functions of Price

As discussed above, the normal operation of the market system is realized via the price mechanism. As analyzed by Milton Friedman (1912-2006, see Section 4.6.2 for his biography), the Nobel Laureate in Economics, price performs three functions in organizing rapidly changing economic activities involving hundreds of millions of individuals:

(1) Transmitting information: price transmits production and consumption information in the most efficient manner;

(2) Providing incentive: price provides incentives for individuals to carry out consumption and production in an optimal way;

(3) Determining income distribution: endowment of resources, price, and the efficiency of economic activities determine the income distribution.

In fact, as early as the Han dynasty of China, Sima Qian (a Chinese historian of the Han dynasty who is considered the father of Chinese historiography) observed and summarized the law of commodity price fluctuation, stating that, for all commodities, "when an article has become extremely expensive, it will surely fall in price, and when it has become extremely cheap, then the price will begin to rise". Therefore, in the effort to become rich, individuals will make good use of this law to "look for a profitable time to sell".

Function 1 of Price: Transmitting Information

Price guides the decision-making of participants and transmits information about changes in supply and demand. When the demand for a certain commodity increases, sellers will notice the increase of sales and thus place more orders with wholesalers. These wholesalers will then place more orders with manufacturers, causing the price to rise, and the manufacturers will then invest more factors of production to produce this commodity. In this way, the message of increasing demand for this commodity is received by all related parties.

The price system also transmits information in a highly efficient manner, and it only transmits information to those who need it. Moreover, the price system not only transmits information, but also produces a certain incentive to ensure the smooth transmission of information so that information will not remain in the hands of those who do not need it. Those who transmit information are internally motivated to look for those who are in need of that information, while those who need information are internally driven to acquire information. For example, ready-to-wear apparel manufacturers are continually striving to obtain the best kind of cloth and looking for new suppliers. Meanwhile, cotton cloth manufacturers are also always reaching out to clients to attract them with the high quality and inexpensive price of their products by various means of marketing and publicity. Those who are not involved in such activities will surely be indifferent to the prices and supply and demand of cotton cloth. The mechanism design theory, as discussed in Chapter 18 of this textbook, will demonstrate that the competitive market mechanism is the most efficient mechanism in the utilization of information because it requires the least amount of information and thus the lowest transaction cost. In the 1970s, Hurwicz and his collaborators had already proved that, for the neoclassical economic environment of pure exchange, no other economic mechanism can achieve as efficient resource allocation using less information than does the competitive market mechanism.

Function 2 of Price: Providing Incentive

Price can also provide incentives, so that individuals will react to changes of supply and demand. When the demand of a commodity decreases, an economic society should provide certain incentives so that manufacturers of the commodity will increase production. One of the advantages of the market price system is that prices not only transmit information, but also provide incentives for individuals to respond to the information voluntarily out of self-interest, so that consumers are driven to consume in an optimal way while producers are driven to conduct production in the most efficient manner. The incentive function of price is closely related to the third function of price: determining income distribution. As long as the increased gain brought by increased production (i.e., marginal revenue) exceeds the increased cost (i.e., marginal cost), producers will continue to increase production until the two are equal, and thus maximum profits are realized.

Function 3 of Price: Determining Income Distribution

In a market economy, an individual's income depends on the resource endowment that the individual's owns (e.g., assets, labor) and the outcomes of economic activities in which the individual is engaged. Concerning income distribution, it is always desired to separate price's function of income distribution from its other functions, in the aim of attaining a more equal income distribution without affecting the other two functions of transmitting information and providing incentive. The three functions, however, are closely related and complementary. Indeed, once price no longer influences income, its functions of transmitting information and providing incentive will disappear. If one's income does not depend on the price of labor or commodities that one offers to others, why would one bother to acquire the information of price and market demand and supply and respond to such information? If one receives the same income irrespective of how much work one performs, then who would strive to do an excellent job? If no benefits are given for innovation and inventions, who would be willing to invest effort in this endeavor? If price has no impact on income distribution, it will also lose its other two critical functions.

1.3.3 The Superiority of the Market System

The modern market system is a sophisticated and delicate economic institution that has emerged, gradually taken shape, and been constantly improved in the long-term evolution of human society. The fundamental and decisive role of the market mechanism in resource allocation is the key to the market economy's capacity for optimal resource allocation. Optimality here has the same meaning as the Pareto optimality (efficiency) proposed by Vilfredo Pareto (1848–1923, see his biography in Section 11.9.1) which will be discussed in Chapter 11 in more detail. It means that, under given resource constraints, no other feasible allocation of rescouses exists that can make some participants better off without harming the welfare of others. Even though Pareto optimality fails to consider the issue of social fairness and justice in terms of equality, it provides a basic criterion of whether or not a resource is wasted concerning social benefit for an economic system, and evaluates social economic effects regarding feasibility. According to this criterion, if an allocation is not efficient, space exists for such allocation to be improved.

Two fundamental theorems of welfare economics, which we will discuss in Part IV on general equilibrium theory, provide a rigorous formal expression of Adam Smith's assessment. As the formal statement of his 'invisible hand', the theorems prove that the free competitive market can maximize social welfare and achieve market optimality in terms of resource allocation. The First Fundamental Theorem of Welfare Economics demonstrates that when individuals pursue their own self-interest, and if economic agents have unlimited or locally non-satiable desire, the competitive market system can achieve Pareto-efficient allocation for economic environments with private divisible goods, complete information, and no externality. The Second Fundamental Theorem of Welfare Economics, on the other hand, proves that for neo-classical economic environments, any Pareto-efficient allocation can be achieved by reallocating initial endowments and competitive market equilibrium without the need to introduce other economic systems to replace the market mechanism. Precise statements and rigorous proofs of these theorems will be given in Chapter 11.

The economic core equivalence theorem in Chapter 12, from another perspective, proves that the market system can benefit social stability, and is optimal and unique in terms of resource allocation, being the result of natural selection with objective inherent logic regarding economic activities. The competitive market mechanism can not only lead to efficient allocation of resources, but also solve the problem of social stability and orderliness well. Moreover, it is the product of free and full competition. The basic connotation of the economic core is that, when the allocation of social resources possesses the core property, there will not be any coalition (i.e., a group of agents) that is dissatisfied with the allocation, and wants to improve their welfare by controlling and utilizing their own resources. In this sense, no powers or groups exist that pose a threat to society, and thus society will be relatively stable.

Under the fundamental fact of individuals' pursuit of self-interest, the economic core equivalence theorem reveals that: the equilibrium allocation by competitive market mechanism has the core property; on the contrary, under some regularity conditions, such as monotonicity, continuity, and convexity (diminishing marginal rate of substitution) of preference, as long as individuals are given enough economic freedom (i.e., the freedom to cooperate and exchange voluntarily) and perfect competition, the outcome will be identical to that of competitive market equilibrium without establishing any institutional arrangements in advance, so they are equivalent. Therefore, the market mechanism is not an invention but an intrinsic economic law and spontaneous order as objective as a natural law. The policy implication is that the competitive market should be allowed to solve the problem of optimal resource allocation when it can do so; other mechanisms should be designed to compensate for the failure of the market mechanism only when the competitive market is impotent.

Furthermore, the competitive market mechanism is not only optimal and unique concerning social stability maintenance and resource allocation, but also most efficient in the transmission of information. Some scholars regard that only the Austrian School has developed the market process theory, which is absent in mainstream modern economics, emphasizing that market players can only accept, rather than affect, existing market conditions. This misunderstanding is caused by the lack of a comprehensive understanding of the evolution of modern economics. The mechanism design theory pioneered by Hurwicz includes two sub-branches, one of which is the realization theory that concerns the mechanism design of decentralized information and transmission efficiency (including the learning, discovery, and transmission of dispersed knowledge).

In the 1970s, Hurwicz and his collaborators characterized the adjustment process of the market's transmission of information (including knowledge), and proved that, for neoclassical pure exchange economies, there is no other economic mechanism which can lead to such efficient allocation of resources using less information than the competitive market mechanism. In 1982, Jordan further proved that, in pure exchange economies, the market mechanism is the only mechanism that achieves efficient allocation using the least amount of information. Tian (2006) also demonstrated that this conclusion is true not only in pure exchange economies, but also for economies with production, and that the market mechanism is unique. Consequently, an important inference follows: irrespective of whether in a command planned economy, state-owned economy or mixed economy, the amount of information that is needed to realize the efficient allocation of resources is more than that in a competitive market mechanism, and thus those economic systems are not informationally efficient. This conclusion provides a key theoretical explanation for why China needs market-oriented economic reform and privatization of its state-owned economy. The uniqueness result of information efficiency will be explored further in Chapter 18.

Even though the market mechanism alone cannot perfectly solve the problem of social fairness manifested in the large income and wealth gap between the rich and the poor, as long as the government strives to provide a level playing field with equality of opportunity and equal value of resources for all individuals and allows the market to play its role instead of controlling it, equity and efficient resource allocation can be achieved by the market, as the fairness theorem in Chapter 12 indicates. The above statements and proofs about the optimality, uniqueness, and fairness of the free, competitive modern market system in resource allocation and its contribution to social stability are all core aspects of the general equilibrium theory.

Joseph Schumpeter also discussed the optimality of the market mechanism from the perspective of how interactions (dynamic game) between competition and monopoly lead to innovation-driven growth. His innovation theory informed us that valuable competition is not merely price competition, but more importantly, competition in new commodities, new technologies, new markets, new supply sources, and new combinations of ideas, knowledge, and resources. As a consequence, the root of the long-term vitality of the market economy is innovation and creativity, which stems from entrepreneurship and entrepreneurs' constant, creative destabilization of the market equilibrium, which he refers to as 'creative destruction'. Profit-pursuing entrepreneurs and private economies play the principal role in cultivating and protecting the institutional soil for innovation.

In fact, competition and monopoly, like supply and demand, can form an astonishing unity of opposites through the power of the market, thus revealing the true beauty and power of the market system. Indeed, market competition and enterprise innovation are inseparable. If there is no competition, there will also be no motivation for innovation; this is what occurs in state-owned enterprises in state monopolies. It is the case, of course, that competition results in profit decline. Overall, the fiercer the competition becomes, the more rapidly corporate profits decrease. This provides profit-seeking private enterprises with strong incentives to innovate with new knowledge and develop new products, so as to gain high profits by setting the price of new products above the competitive equilibrium price. However, other firms in the same industry will soon develop similar products in order to share the profits. Such market competition will lower profits and thus urge enterprises to innovate further. Innovating enterprises may gain a monopoly position, which implies monopoly profits, which will attract more enterprises to participate in the competition. As a consequence, a repeated cycle of "competition $\rightarrow$ innovation $\rightarrow$ monopoly profit $\rightarrow$ competition" is established, in which market competition tends to achieve an equilibrium, but innovation disrupts it. The market continually goes through such cycles to inspire enterprises to pursue innovation. Through this dynamic process, the market maintains its vitality, and greater economic development and social welfare are obtained. Therefore, in order to encourage innovation, the government should strictly enforce laws regarding intellectual property rights protection.

While people may realize the importance of entrepreneurs and entrepreneurship, not all fully recognize the fundamental importance of the institutional basis for the emergence of entrepreneurs and entrepreneurship, and that innovation and development need institutional support. This is because entrepreneurs and entrepreneurship do not appear randomly, but rather entrepreneurship is derivative and superficial. Therefore, it must be built on a basic meta-institution, which necessitates a conducive institutional environment as a prerequisite. As such, Baumol (1990) extended Schumpeter's innovation theory, and argued that *innovation and entrepreneurship depend on institutional choice, and are therefore endogenous variables*. If the rules of the game that affect the choice of entrepreneurial behaviors are abnormal, institutional innovation is lagging behind, or even disruptive, innovation and entrepreneurship cannot achieve their full potential. Indeed, so far, three industrial revolutions have occurred in the world: the industrial revolution in Britain; the second industrial revolution led by the United States and Germany; and the third industrial revolution, which takes artificial intelligence manufacturing as the core. The occurrence and development of each industrial revolution are closely related to institutional innovation, which provides critical support for the smooth development of industrial revolutions.

For instance, in order to encourage competition and form positive externalities of technological innovation, anti-monopoly legislation should be enacted and enforced. In addition, protection of intellectual property rights should not last forever, but rather should be limited to a certain number of years so that they will not establish a perpetual oligopoly or monopoly. Therefore, technological innovations operate on the basis of institutional innovations. These two kinds of innovations behave like an action-reaction pair. Specifically, a good institution can reduce the transaction costs of innovation, create conditions for cooperation, provide incentives for innovation, and facilitate internalization of the benefits of innovation. One goal of constructing a technological innovation system is to promote effective interaction and cooperation among innovative elements.

Innovation is the process of breaking rules and conventions, which inevitably carries high risks. In high-tech innovation, the risks are particularly high, and successful venture capital investments are relatively few. However, a successful venture brings huge returns, which attract increased investment. Nevertheless, it is impossible for state-owned enterprises to take such high risks due to the fact that they inherently lack the incentive mechanisms that would enable the assumption of such risks. In contrast, it is private enterprises that typically dare to take the most risk out of a strong incentive to pursue their self-interest, and consequently they are the most creative and innovative entities. Therefore, entrepreneurial innovation (not fundamental scientific research) largely takes place within the context of private business. In fact, Sima Qian stated that competition and survival of the fittest are natural tendencies. He wrote the following in The Records of the Grand Historian (Shiji): “There is no fixed road to wealth, and money has no permanent master. It finds its way to the man of ability like the spokes of a wheel converging upon the hub, and from the hands of the worthless it falls like shattered tiles.”

With the emergence of Internet finance innovation, the deviation of the real economy from its ideal state will be constantly mitigated. Instead, it will be pushed towards its ideal state as described by Smith, Hayek, Arrow–Debreu, Coase, and others, as they all demonstrated the optimality of the market—whether it be the competitive market’s role as the invisible hand described by Smith, or the Arrow–Debreu General Equilibrium Theory in a perfectly competitive market, or Coase (1910-2013, see Section 14.6.2 for his biography)'s theory of zero transaction costs in a perfectly competitive market, or Schumpeter's theory that market competition favors innovation. The basic conclusion is that a perfectly competitive market leads to efficient resource allocation and social welfare maximization. However, the theory of a perfectly competitive market has served only as a reference point or an ultimate goal: The greater the market competition and information symmetry, the better. Of course, the perfectly competitive market merely provides a reference system or an ultimate goal, which means that the more competitive the market is, the better it is; and the more symmetric the information is, the better it is. Such a perfectly competitive market system, however, does not exist in reality because communication costs, transaction costs, and financing costs cannot be zero (although they may approach it).

With the Internet as a medium for finance and big data method, transaction costs are becoming increasingly diminutive. Due to the disruptive innovation and development of financial technology and big data method, to some extent, the perfectly competitive market, as considered by the first category of economic theory, is not just an ideal state but also tends to increasingly approach reality. In particular, financial technology and big data method will greatly reduce the cost of information communication in reality in order to make market economic activities closer to the ideal state of perfect competition and thus more efficient.

1.4 Government, Market, and Society

The theoretical conclusion concerning market optimality relies on an implicit assumption about the critical importance of fundamental institutions. In other words, there should be a mature governance structure to regulate the government, the market, and the society.

1.4.1 Three Elements of State Governance and Development

State governance involves three dimensions: **government, market, and society**. The market mechanism may give an incorrect impression that, in a market economy, one can do whatever one desires in the pursuit of self-interest; this, however, is not the case. In the world, there is no completely laissez-faire market economy that is totally independent of the government. A well-functioning market requires appropriate and effective integration of government, market and society, which constitutes a three-dimensional structure of state governance. A completely laissez-faire market without governance and regulation is also not omnipotent. As we shall discuss in

Parts V-VII of this textbook, the market frequently fails in certain circumstances, such as monopoly, unfair income distribution, polarization of rich and poor, externality, unemployment, inadequate supply of public goods and information asymmetry, thus resulting in inefficient allocation of resources and various social problems.

The three basic institutional arrangements of government, market and society are the **three elements of state governance and benign development**: 1) inclusive economic institution; 2) state capacity to plan and implement policies and laws; and 3) an inclusive and transparent civil society with democracy, the rule of law, fairness and justice. Either in short-term response or in long-term governance, the three are all indispensable and essentially necessary and sufficient conditions for a nation's sustainable and benign economic development, social harmony and stability, and long-term peace and stability. Indeed, the practice at all times and all over the world has repeatedly shown that all economic and social achievements or progress were due to the improvement of some aspects of these three elements, and the problems were inevitably caused by the lack of some of them. As such, these three elements are the most fundamental comprehensive governance elements to identify whether a state's governance system and capacity are good or not and whether it can cope with crisis in the short term and maintain stability in the long term. A modern state governance system must be a system that properly deals with the relationship between government and market and between government and society, so that each functions in its respective position and interacts effectively with one another.

Benign development and governance exhibit an inherent dialectical relationship, and must be accurately understood. Economic development primarily focuses on the improvement of a nation's hard power; whereas, governance stresses the construction of soft power. Of course, governance should be all-dimensional from different aspects, including governance systems of the government and the market, social equity and justice, culture, values, etc. How the relationship between the government and the market and between the government and the society is handled frequently determines the effect of state governance and development. If they cannot be well balanced, a series of major problems and crises may occur, including poor development, an excessive gap between the rich and the poor, unequal opportunities, etc., preventing an inclusive market economy and a tolerant and harmonious society from coming into existence. In this way, in the logic of governance, there is a "good" kind and a "bad" kind of governance that will lead to a good or bad market economy and good or bad social norms, respectively. Therefore, governance should not be taken as being equivalent to rules, controls or regulations, or regarded as the opposition of development, making it difficult to attend to both governance and development simultaneously. To achieve and maintain an efficient market and harmonious society, a limited government should be built that is capable, accountable, effective and caring, leading towards a desired governance that features the principle of the rule of law.

1.4.2 Good Market Economies vs. Bad Market Economies

A market economy can be classified into "**good market economy**" and "**bad market economy**". Whether it is good or bad depends on the system of state governance and whether the governance boundaries among the government, the market, and the society are clearly and appropriately delineated. In a good market system, the government enables the market to fully play its role, and in case of market failures, the government can take certain remediative actions. This does not mean that the government should directly intervene in economic activities, but rather the government is expected to enact appropriate the design of rules or institutions to correct market failures in order to achieve incentive-compatible outcomes (i.e., individuals can achieve the best outcomes just by acting according to their true preferences) so that individual interests and social interests are consistent.

Good institutions are often deemed evolutionary and thus cannot be designed, which is quite debatable from both theory and practice. Even Hayek argued that "spontaneous evolution is a necessary if not a sufficient condition of progress". As shown in his discussion about the constitutional design in **Law, Legislation, and Liberty**, he lacked sufficient confidence in spontaneous evolution towards good institutions, but hoped for certain top-level institutional design. In Hayek's view, "societies form but states are made". In fact, evolution is not equal to progress. The direction of evolution may be randomly divergent, but not necessarily unidirectionally progressive. As a prominent member of the modern Austrian School following Mises and Hayek, Israel Kirzner acknowledged that free market is a spontaneous process of discovering and using decentralized knowledge to achieve benign outcomes, especially the entrepreneurial discovery process where "each entrepreneur seeks to outdo his rivals in offering goods to consumer", which leads to the emergence of dynamic equilibrium of the market. He did not regard evolution as a necessary condition for good institutions, and held that there does not exist a systematic discovery process like the market for institutional evolution. As such, it is difficult to ensure that all knowledge is fully utilized, and thus a good institution emerges.

In fact, even the free market cannot achieve benign outcomes and evolve towards a modern market economy merely through the discovery and use of decentralized knowledge. This section focuses on the conclusion that government's proper role in the market through institutions is indispensable. One thing to be noted is that although most of the nations in the world have a market economy, few of them have established a good market economy system. The market process "occurs within the framework of rules and institutions conducive to the protection of its certain operational characteristics", and the formation of these rules and institutional frameworks is inseparable from intentional mechanism design under rationality, rather than an unconditional natural evolutionary process. Therefore, government has an important role to play in this regard, that is, "to modify and develop abstract rules by forcing people to obey these rules" to ensure good functioning of the market. One of the most successful examples of institutional design is the enaction of the basic constitution of the U.S. at its founding, which made the U.S. become the most powerful nation in the world within approximately 100 years.

Therefore, a good, inclusive, and efficient modern market economy is based on rules and institutions that support and maintain the benign operation of the market. It should protect the private interests of individuals to the greatest extent through institutions or laws, and simultaneously limit and counterbalance the government and its public powers as much as possible. In this sense, it is a contractual and rule-of-law economy constrained by an agreement regarding commodity exchange, rule of market operation, and reputation, thus leaving little room for rent-seeking through power. Under the constraints of individuals' pursuit of self-interests, of resources and of information asymmetries in an economic society, to build a strong state with prosperous people, individuals should first be endowed with private rights, the core of which includes the basic rights to survival, freedom of choice for one to pursue happiness, and private property rights. Through their participation in full competition, voluntary cooperation, and voluntary exchange of the market mechanism and their pursuit of self-interest, efficient allocation of resources and maximization of social welfare can be attained. Therefore, the modern market economy is established upon the basis of the rule of law, which works in two ways: first, it is of fundamental importance to restrict arbitrary government intervention in market economic activities; and second, it further supports and promotes the market in certain ways, including the definition and protection of property rights, enforcement of contracts and laws, maintenance of fair market competition, etc., in order to allow the market to play the decisive role in resource allocation and give full range to the three basic functions of price, i.e., transmitting information, providing incentives, and determining income distribution. In addition, a good market calls for good social norms, for one's pursuit of self-interest should exist on the premise of respect for others' pursuit of self-interest, and self-interest and fair competition function in parallel to one another with no conflict. The spirit of compromise and respect for others' standards of values constitute the foundation for the normal process of exchange.

On the other hand, in a bad market economy, in which there is a lack of adequate ruling and governance capacity in economic and social transitions, and the government is unable to provide sufficient public goods and services to compensate for market failures. The government's excessive economic activities result in public powers that are not effectively counterbalanced, property rights of state-owned enterprises that are not clearly defined, and the government that is involved in numerous rent-seeking and corrupt behaviors so that equity and justice are greatly diminished. This breeds the so-called the "**State Capture**", which refers to the phenomenon that, by providing personal interests for government officials, economic agents interfere in decisions on laws, rules and regulations, and thus, without going through fair competition, they convert their personal interests to the basis of rules of the game of the whole market economy. This then leads to policies that produce high monopoly profits for specific individuals at the expense of enormous social costs and the decrease of government credibility. As a result, an inefficient balance in public choice continues over a long period of time. The behavior of striving for social and governmental resources by means of unfair rent-seeking instead of fair competition will not only produce market failures, but more importantly gradually result in bad social norms in the long term. These poor social norms produce a distortion of resource allocation and social values, the decline in moral values, an absence of good faith, the popularity of "false, big, and empty" words and deeds, the frivolity of society and increased factors of instability, which finally result in enormous explicit and implicit transaction costs. Some sociologists refer to such social state as "**social corruption**", meaning that the social cells of the social organism are dead and experiencing functional failures.

Therefore, in the three-dimensional framework of government, market and society, the government, as an institutional arrangement with great positive and negative externalities, plays a vital role. Indeed, it can make the market efficient, become the impetus for economic development, assist to construct a harmonious society, and realize sustainable development. On the other hand, it may also make the market inefficient, lead to various social conflicts, offer tremendous resistance against the benign development of the society and economy, and exert deleterious social impacts. Although almost all countries in the world have adopted a market economy, a majority have not achieved sound and rapid development. Among numerous reasons for this, the most fundamental one is the lack of reasonable and clearly delineated governance boundaries between the government, the market and the society, so that there is over-playing, under-playing, and mis-playing of the government role. Only when the government appropriately tightens its omnipresent "visible hand", and the functions and governance scope of the government are appropriately and reasonably defined, can it be expected to define a proper and scientific governance boundary between the government, the market, and the society.

1.4.3 The Boundaries of Government-Market-Society

How can governance boundaries be best defined among the government, the market, and the society? The answer is to allow the market to do whatever it can do well, while the government does not participate in economic activities directly (however, it is necessary for the government to maintain market order and guarantee strict implementation of contracts and rules). Regarding what the market cannot achieve, or in circumstances in which it is not appropriate for the market to be involved, such as in cases affecting national security, the government can directly participate in economic activities. In other words, when considering the construction of an inclusive and transparent civil society and benign development of an economy, or when transforming government functions and innovating modes of management, the boundaries of the government, the market, and the society should be carefully considered. For instance, the government should exit from competitive sectors. Only in the case of market failure should the government solve problems by itself or be in cooperation with the market. However, the basic dictum is that the government should not directly intervene in economic activities, but instead enact conducive rules and institutions to correct market failures. Because individuals' pursuit of self-interest and private information are at the core of many economic activities, direct intervention in economic activities (e.g., large numbers of state-owned enterprises, and arbitrary restriction of market access and interference with commodity prices) would not frequently generate desired outcomes. In this respect, mechanism design theory can play a key role in making the market more efficient and solving the problem of market failures. In Hurwicz's opinion, "law-making by the U.S. Congress or other legislative bodies equals to designing new mechanisms".

Under a modern market economy, the basic and sole functions of government can be distilled as "maintenance" and "service", i.e., making fundamental rules to ensure national security, social stability and economic order, as well as providing public goods and services. Just as Hayek asserted, the government has two basic functions: firstly, the government must be responsible for law enforcement and defense against its enemies; and secondly, the government must provide services that the market is unable to provide or unable to fully provide. He also stated that "it is indeed most important that we keep clearly apart these altogether different tasks of government and do not confer upon it in its service functions the authority which we concede to it in the enforcement of the law and defence against enemies." [3] This requires that, in addition to undertaking necessary functions, the government should separate its powers regarding market and society. Abraham Lincoln provided the following clear and incisively-

[3] See Von Hayek F. *Law, Legislation, and Liberty* (Vol. 3). Chicago, IL: The University of Chicago Press, 1979, pp42.

defined functions of the government:

> The legitimate object of government, is to do for a community of people, whatever they need to have done, but can not do, at all, or can not, so well do, for themselves—in their separate, and individual capacities. In all that the people can individually do as well for themselves, government ought not to interfere.[4]

Meanwhile, a good-inclusive-efficient modern market economy and state governance mode require an independent autonomous civil society with a strong ability to coordinate interests as an auxiliary informal institutional arrangement. Otherwise, the explicit and implicit transaction costs of economic activities would be prohibitive, and it would be very challenging to establish the most basic norm of social trust.

In summary, a reasonable and clearly defined governance boundary among the government, the market, and the society is a prerequisite for establishing a good-inclusive-efficient market economic system and achieving benign development that is characterized by efficiency, equity, and harmony. Of course, the transition to an effective modern market system often constitutes a long and arduous process. Due to various constraints, governance boundaries of the government, the market, and the society cannot be clearly defined in a single attempt, but rather a series of transitional institutional arrangements are frequently requisite. However, with the deepening of transitions, some transitional institutional arrangements may decline in efficiency, and may even degenerate into invalid institutional arrangements or negative ones. If governance boundaries of the government, the market, and the society cannot be timely and appropriately clarified while some temporary, transitional institutional arrangements (e.g., government-led economic development) function as permanent and ultimate institutional arrangements, it is then impossible to achieve an efficient market and an inclusive, transparent society. With the development of modern economics, its analytical framework and research methods play an essential role in the study of how to reasonably and clearly define governance boundaries among the government, the market and the society, and how to carry out comprehensive governance.

1.5 Comprehensive Governance by the Three Arrangements

As discussed above, in order to establish a well-functioning and efficient modern market system, it is necessary to coordinate and integrate the relationship of the three basic institutional arrangements, i.e., the government, the market and the society, in order to regulate and guide individuals'

[4] Abraham Lincoln's Quotes on Government (1854).

economic behavior and conduct comprehensive governance. The government, the market, and the society correspond precisely to the three basic elements of governance, incentive, and social norms in an economy, respectively. Mandatory public governance and formal institutional arrangements, such as market mechanism as an incentive scheme, as two elements of comprehensive governance which overlap, through long-term interactions, may assist to guide and form normative informal institutional arrangements, enhance the predictability and certainty of social and economic activities, and markedly decrease transaction costs. The informal institutional arrangement mentioned here is social norms and culture. The same as with enterprises, the first-class enterprise performs branding, in which its corporate culture plays the key role, the second-class enterprise develops technologies, and the third-class enterprise produces. Similarly, it is the position of the government that plays the crucial role in the establishment of good or bad social norms and market economy.

Expressed in another way, the three basic institutional arrangements for comprehensive governance are employed to "enlighten with reason, guide with benefits, and persuade with emotions", all of which are realized and implemented mainly by the government, the market, and the society, respectively, on the level of state governance. To "enlighten with reason" means to work through the stimulus of legal principles and reason; to "guide with benefits" means to link up economic activities with revenue through the stimulus of rewards and penalties, thus becoming an incentive mechanism; and to "persuade with emotions" means to work through the stimulus of emotions and shared beliefs, as sometimes relationships, friendships and emotions, and especially shared beliefs and concepts, can assist to solve large problems, which as a kind of social culture will greatly decrease transaction costs.

1.5.1 Governance on Rules

Governance on rules, as the basic institutional arrangement and management rule, is about enforcement. The basic criterion for whether such rules and regulations shall be formulated is whether or not they facilitate a clear definition (according to the possibility of information transparency and symmetry), and whether the costs of information acquisition, supervision, and enforcement are high. If a regulation is too costly concerning supervision, it will not be feasible for enforcement. Protection of property rights, contract implementation, and appropriate supervision all call for relevant rules, which thus require a third party to oversee enforcement of those rules. The third party is then a government agency. In order to maintain market order, the role of the government is inevitable. As the government is also an economic agent, it functions as both a judge and a player, and thus it exerts an enormous influence. This requires procedures and rules to constrain the behavior of the government. Regulations on other economic agents and the market should be the opposite of this. Due to information asymmetry, regulations in this regard should not be too detailed, so as not to interfere too much with the freedom of choice of economic agents.

Institutional arrangements that "enlighten with reason" and deter people with the threat of sanctions are similar to the thoughts of Legalism in ancient China. However, a major problem of Legalism was that it only aimed to govern economic agents, but applied no restrictions on the government. In this way, it is rule by law, but not rule of law. Moreover, this kind of institutional arrangement only considered cruelty in individuals' struggle for power and profit, but neglected the influence of individuals' consciences and emotions upon behavior. The employment of the heavy hand of Legalism without considering other institutional arrangements will frequently lead to coercive powers, and hinders the formation of a modern market economy.

1.5.2 Incentive Mechanism

Incentive mechanisms, such as market mechanisms, are inducing by their nature, have the widest applications, and are a main concern of this textbook. Due to information asymmetry and the high cost of information acquisition, as well as the fact that human nature, especially the nature of self-interest, can hardly be altered, specific operation rules need to mobilize individuals' enthusiasm through incentive mechanisms, such as market mechanisms, to realize incentive compatibility in order to make individuals, who pursue their own interests, strive to achieve the aims of the mechanism designer, as well. As shown in Chapter 7 on repeated game theory, reputation and integrity, we will show that reputation and integrity under the market incentive mechanism are one kind of punishment incentive mechanism. Integrity is essential in engaging in business activities; this does not, however, mean that business owners behave with integrity out of their own will, but rather that they have no choice because, otherwise, they will be forced out of the market. It is also the case that integrity can save economic costs and lower transaction costs.

Institutional arrangements that "guide with benefits" are similar to the precepts of Taoism in ancient China. It asserts that individuals are all self-interested and have a limited ideological realm, and these two assertions should be universally recognized. Although it is not difficult to stop "talking" about interests, in reality, it is challenging not to "value" interests. Taoism contends that "the law of nature is beneficial, but not harmful", and emphasizes compliance with natural tendency and governance by non-interference. However, it neglects two necessary conditions for governing by non-interference, i.e., the establishment of basic institutions and the proper role of government.

1.5.3 Social Norms

A key factor in understanding institutions and other long-term relationships is the role of shared expectations of behavioral norms and cultural beliefs, as well as the role of sanctions in ensuring compliance with the "rules". Behavioral norms and cultural beliefs mentioned here are the so-called social norms. They are informal institutional arrangements that are implemented by neither forces nor incentives. Solving problems with mandatory laws and inducing incentives in the long term will become a kind of social norms, values, beliefs, and culture that requires neither enforcement nor incentive, such as corporate culture, folk customs, religious faith, ideology, sentiments and concepts. This is an efficient way to minimize transition costs. Especially when the force of collective consciousness is well developed, problems will be much easier to resolve, and working efficiency will be greatly augmented. Otherwise, even if one problem is solved through mandatory commands, inducing incentive mechanisms or personal relationships, new problems will continue to arise in the same way, which will produce large implementation costs. Chapter 7 will discuss the important role of social norms in stimulating voluntary cooperation among individuals.

Even so, social norms that espouse morality and "persuade with emotions" remain largely constrained by the reality of individuals' ideological limitations. Relying on improvement of humanity, social norms lack the power of constraint and possess a limited scope of governance. This kind of institutional arrangement is similar to the philosophy of Confucianism in ancient China. The concept of the "rule of virtue" in Confucianism overemphasizes ethical relations among people, and deliberately overlooks economic relations. Overall, it is successful in managing families or small communities, but biased in the governance of a nation. Benevolence and morality can be dominant, or at least considerably important, in a family or a small group, but may not be very effective in the lives of general people. Indeed, benevolence is highly personal, and its effect diminishes with the enlargement of the realm. Those who rely on others' benevolence to acquire the necessities of life cannot be satisfied, in most cases. Although all people may need assistance from others, they cannot depend on benevolence alone to meet their needs. Consequently, the experience of managing a family cannot be simply extended to all economic and social activities; otherwise, manifold problems and even disasters may ensue. Especially under the environment of a modern market economy and people's limited ideological development, reliance solely on internal ethical norms, and an absence of external laws and incentive mechanisms, will cause the market economy to move towards a perilous state.

Therefore, all three of these mentioned institutional arrangements have two sides: positive and negative. They also possess varied functions, ranges of application, and limitations. Moreover, if emphasis is placed too heavily on only one of them, serious negative consequences will occur; it is necessary for the three of them to play their own specific roles and complement each other. This condition can be elucidated with the example of friendship. If friends are made solely on the basis of interests, the friendship will disintegrate when the interests are exhausted; if friends are made based on influence, the friendships will end when the influence fails; if friends are established on the basis of power, the friendships will fail when the power dissipates. Of course, only when one makes friends with a sincere heart can the friendship last forever. Making friends with a sincere heart is the best way, but it can be very difficult, and even many couples may not be able to sustain this.

1.5.4 The Hierarchical Structure of the Three Arrangements

Even though this textbook primarily discusses the problem of economic incentives, governance on rules or institution remains the most essential and fundamental among the three kinds of arrangements since it establishes the most basic institutional environment, has strong positive and negative externalities, and determines whether or not the role of government is appropriate, thereby determining the effect of incentive mechanism design and the formation of good or bad social norms. In addition, for the formulation of institutional arrangements of both regulatory governance and incentive mechanisms, the principle should not, and essentially cannot, change the self-interested nature of human beings. Instead, it should make use of individuals' immutable self-interestedness to guide them to perform actions that are beneficial to society. In other words, the design of an institution should conform to the self-interested nature of individuals, rather than attempting to alter it. Moreover, individuals' self-interestedness cannot be simply deemed to be either good or evil, but instead what kinds of institutions that are employed and towards what directions they are guided must be taken carefully into account. Different institutional arrangements will result in individuals' different responses to incentives and various tradeoff choices, thereby leading to markedly dissimilar consequences. As Deng Xiaoping contended, "Good institutions can make bad people unable to run amok arbitrarily, and bad institutions can make good people unable to do good enough, or even go to the opposite side." [5] Expressed in a colloquial way, bad institutions can turn good people into bad people, but good institutions can turn even bad people to do good things.

Therefore, the utilization of "reason, interest, and emotion," should be synthesized, and vary with individuals, matter, place, and time to analyze

[5] *Selected Works of Deng Xiaoping (Second Version).* Beijing: People's Publishing House. Volume 2, 333.

and solve problems case by case. The criterion for deciding which aspect to use is determined by the importance of regulations, the degree of information symmetry, and the cost of supervision and law enforcement. All three institutional arrangements possess their own boundary conditions. To "enlighten with reason" should depend on the availability of information symmetry and difficulty of legal supervision. Indeed, the law will be meaningless if its cost of supervision and execution is too high.

To sum up this and the previous sections, a well-functioning market needs the government, the market, and the society to be optimally situated so that the three-dimensional structure of state governance can be effectively interconnected and integrated. Defining the boundaries between the government and the market and between the government and the society involves two levels. The first level is defining boundaries. It is first necessary to know the appropriated boundaries among them. The prerequisite for an efficient market and a normal society is to build a limited government that is capable, accountable, effective and caring, and thus the reasonable position of government is vital. The principle here is that the market should be allowed to do whatever it can do, while the government should do what the market cannot do or cannot do well. Consequently, the function of government can be generalized as maintenance and service.

The second level is the identification of priorities. What is the top priority? The answer is fundamental institutions. After knowing the boundaries of the three, it is then necessary to sort them out. Who can best sort them out? The answer is the government. Then, what is the best entity to regulate the position of the government? It must be the *rule of law that mainly restricts the government, but not rule by law that mainly restricts the people*. Regulatory governance (or institutions) is the most important and foundational arrangement, which establishes the most basic institutional environment, possesses great positive or negative externalities, assesses whether the position of the government is appropriate, and thus determines the effect of the incentive mechanism design and the formation of good or bad social norms. However, is the government willing to limit its power? In general, absolutely not. Therefore, power requires further partitioning, and separation of the responsibilities and power of administrative departments, law-making departments, and judicial departments.

For all of these reasons, the underlying governance system is of determinant importance. Only when the governance boundaries between the government and the market and between the government and the society are reasonably defined through comprehensive governance by the three dimensions of institution, the rule of law and civil society, can government power be regulated, restricted, and supervised, problems of economic efficiency, social equity and justice be resolved, the phenomena of corruption and bribery be eradicated, and healthy relations among the government, the market, the society, enterprises, and individuals be established. Indeed, in this way, relations of benign interactions are established between all of them. When benign interactions are realized, the government can then augment the efficiency of the market through continuous enaction and enforcement of laws, in order to truly promote the long-term peace and stability of a nation.

1.6 Ancient Chinese Thoughts on the Market

Many people think that concepts such as the market economy and market-determined commodity prices as well as basic economic principles such as the importance of economic freedom, private property rights, and restricted government intervention originate in the West. However, this is not true. As early as the prehistoric period, China advocated a simple free market economy and believed that prices should be determined by the market. China originated a great number of insights regarding the market economy and incentive-compatible, dialectical strategies of state governance. Almost all the fundamental ideas, core assumptions, and basic conclusions of economics have been discussed by ancient Chinese philosophers, such as self-interest; economic freedom; governance by the invisible hand; the social division of labor; and the intrinsic relationships between national prosperity and individual wealth, development and stability, and government and the market. Some examples are discussed below.

Over 3,000 years ago, Jiang Shang (also known as Jiang Ziya and Jiang Taigong), an ancient Chinese strategist and adviser during the Zhou dynasty, argued that “averting risks and pursuing interests” is the innate nature of human beings:

> In general, people hate death and take pleasure in life. They love virtue and incline to profit.

He proposed the people-centered notion of the dialectical unity between national prosperity and stability and individual wealth, as well as the fundamental law and strategy of state governance:

> All under Heaven is not one man’s domain. All under Heaven means just that, all under Heaven. Anyone who shares profit with all the people under Heaven will gain the world.

This implies that government should respect the common interests, risks, welfare, and livelihoods of the populace as its own, so as to obtain an incentive-compatible outcome in which the populace and government have the same interests and losses.

Jiang Shang also offered an incisive view on the relationship between the wealth of the state and that of the people:

> The true king enriches the people. The hegemon enriches the gentry. The state which barely survives enriches its grand ministers. The state which perishes enriches its coffers and fills up its treasuries.

His advice was taken by King Wen of the Zhou dynasty, who opened his granary to help the poor and reduced taxes to enrich the people. The Western Zhou subsequently grew in power.

Over 2,600 years ago, Guan Zhong (a Legalist chancellor and reformer of the State of Qi) offered many profound economic insights. The core of his economic thought was “the theory of self-interest.” In his book Guan Zi: Jinzang (On Maintaining Restraint), he explained the nature of social economic activities with individuals pursuing self-interest vividly and profoundly:

> All men pursue interests and avert harm. When doing business, merchants hasten on the way day and night and make light of traveling from afar because, for them, interests are on their way. When fishermen go fishing in the sea, though the sea is hundreds of meters deep, they sail against the current for hundreds of miles day and night because, for them, interests are in water. So as long as there is interest, people would climb the mountain regardless of its height and go to sea regardless of its depth. Therefore, if those who know well how to govern the origin of interest, people will naturally admire the state and settle. The governor does not need to push them to go or lead them to come. Without being bothered or disturbed, people will get rich in a natural course. It is like a bird incubating eggs, the process of which is invisible and silent, but the result is noticeable when it is done.

This was a very vivid demonstration of Smith’s invisible hand, made more than 2,000 years earlier. In Guan Zi, Guan Zhong presented the law of demand:

> The devaluation comes from the excess, while the value from the scarcity.

He also drew the basic conclusion that individuals’ wealth leads to national stability, security, prosperity, and power:

> Only at times of plenty will people observe the etiquette. Only when they are well-clad and fed will they have a sense of honor and shame.

He further pointed out:

> State governance must start with enriching the people. When the people become well-off, the state will be easy to govern. If they are in poverty, the state will be hard to govern. ⋯ Usually, an orderly state is abundant in prosperity, while a disorderly one is deep in poverty. So, a king versed in ruling a state must give priority to making people wealthy over governance itself.

Comprehensive governance is another essential point in Guan Zhong’s thoughts on state management. For example, speaking of vassal kings, Guan Zhong suggested:

> Restrain them with interest, associate with them with trust, admonish them with military power [so that vassal kings] won’t dare to defy the king and will accept his interest, trust his benevolence, and fear his military force.

There are obvious links between the “interest, trust, and military power” mentioned here and the three institutional arrangements discussed above.

More than 2,500 years ago, Sun Tzu (a military general, strategist, and philosopher in ancient China) wrote a book entitled The Art of War on military strategies and tactics. Its principles and ideas are very similar to market behavior and business decision making. The ideas in the first chapter, “Detail Assessment and Planning,” coincide broadly with the basic analytical framework of modern economics and can be fully adopted in the context of business tasks. They can be applied to accomplishing ambitious goals, making correct decisions, and succeeding in governing a state or managing an enterprise. Sun Tzu also posited the basic theory of information economics—that it is possible to achieve the optimal outcome (“the best is first best”) only under complete information and that we can at most obtain a suboptimal outcome (“the best is second best”) under asymmetric information:

> If you know your enemies and know yourself, you will not be imperiled in a hundred battles; if you do not know your enemies but do know yourself, you will win one and lose one; if you do not know your enemies nor yourself, you will be imperiled in every single battle.

Even more remarkably, Lao Tzu (a famous philosopher living in the same period and the founder of Taoism) presented the supreme law of comprehensive governance:

> “Rule a kingdom by the Normal. Fight a battle by (abnormal) tactics of surprise. Win the world by doing nothing.” (Chapter 57, Tao Te Ching)

This is the essential way of governing a state or administering an organization. It can be summarized as follows: The government should be righteous in deed, flexible in practice, and minimal in intervention, and thus realize governance via non-intervention.

Lao Tzu considered Tao to be the invisible inner law of nature and Te (meaning "inherent character," "integrity," "virtue") the concrete embodiment of Tao. He taught that the governance of both state and people should follow the Way of Heaven (referring to the objective laws governing nature or the manifestations of the heavenly will), the Virtue of Earth, and the Principle of Non-intervention:

> Man models himself after the Earth; The Earth models itself after Heaven; The Heaven models itself after Tao; Tao models itself after nature. (Chapter 25, *Tao Te Ching*)

He also pointed out:

> The difficult (problems) of the world must be dealt with while they are yet easy; the great (problems) of the world must be dealt with while they are yet small. (Chapter 63, *Tao Te Ching*)

Thus, whatever we do, success lies in the details. All these statements demonstrate that Lao Tzu's non-intervention does not mean "doing nothing" in its literal sense as is commonly believed. The non-intervention discussed by Lao Tzu is a relative concept that requires non-intervention in major aspects but action and care in specific details. In other words, we should never lose sight of the general goal and must begin by tackling the small practical problems at hand.

Over 2,300 years ago, Shang Yang (an important Chinese philosopher and statesman) of the State of Qin used the example of a hare to illustrate the utmost importance of establishing private property rights and show how clear, well-defined property rights can "determine ownership and set disputes" and help establish market order. He came to this conclusion 2,300 years before Coase:

> When a hare is running, a hundred men chase after it; this is not because the hare can be divided into one hundred shares, but its ownership is not yet determined. On the other hand, hares are sold on the market, but even a thief dare not take one because ownership has been determined. Thus, if ownership is not definite, sages like Yao, Shun, Yu, and Tang would also chase after it; when ownership is definite, even a greedy thief will not dare to take it. *Shang Jun Shu* (*The Book of Lord Shang*)

Thus, the hare is chased because people are motivated to strive for its ownership, and even sages would do the same. Contrariwise, the ownership of a captured hare in the market is determined, so others cannot take it.

About 2,100 years ago, Sima Qian (a Chinese historian of the Han dynasty who is considered the father of Chinese historiography) made a remarkable statement in his work *Records of the Grand Historian: Biographies of the Money-makers* :

> Jostling and joyous, the whole world comes after profit; racing and rioting, after profit the whole world goes,

This was in the same vein as Guan Zhong' s thoughts on self-interest. He also discussed achieving social welfare through the social division of labor based on self-interest, which is similar to Smith' s views. Sima Qian investigated the development of social and economic life and realized the importance of the social division of labor. He wrote:

> All of them are commodities coveted by the people, who according to their various customs use them for their bedding, clothing, food, and drink, fashioning from them the goods needed to supply the living and bury the dead. ... Society obviously must have farmers before it can eat; foresters, fishermen, miners, etc., before it can make use of natural resources; craftsmen before it can have manufactured goods; and merchants before they can be distributed.

Moreover, he believed that the whole social economy composed of agriculture, forestry, industry, and commerce should develop in a natural course without the constraint of administrative orders:

> What need is there for government directives, mobilizations of labour, or periodic assemblies? Each man has only to be left to utilize his own abilities and exert his strength to obtain what he wishes. Thus, when a commodity is very cheap, it invites a rise in price; when it is very expensive, it invites a reduction. When each person works away at his own occupation and delights in his own business then, like water flowing downward, goods will naturally flow forth ceaselessly day and night without having been summoned, and the people will produce commodities without having been asked. Does this not tally with reason? Is it not a natural result?

In addition, Sima Qian's thinking contains great wisdom regarding the philosophy of state governance, the importance of economic freedom, and the priority of several basic institutional arrangements. Sima Qian provided an insightful conclusion in the *Biographies of the Money-makers*:

> The highest type of ruler accepts the nature of the people, the next best leads the people to what is beneficial, the next gives them moral instruction, the next forces them to be orderly, and the very worst kind enters into competition with them.

Confucius affirmed that the pursuit of personal material interests based on social ethics is justifiable:

> When a country is well governed, poverty and a mean condition are things to be ashamed of. When a country is ill governed, riches and honor are things to be ashamed of. (*The Analects: Tai Bo*)

Thus, Confucius encouraged people to pursue only decent material wealth. The Analects also record Confucius complimenting his disciple Zigong (Duanmu Ci), who was a merchant:

> The Master said, 'There is Hui! He has nearly attained to perfect virtue. He is often in want. Ci does not acquiesce in the appointments of Heaven, and his goods are increased by him. Yet his judgments are often correct.' (*The Analects: Xian Jin*)

Here, Confucius compared Yan Hui, his favorite disciple, with Zigong. The former was almost perfect in morality, but frequently lived in poverty, which did not seem to be the right way of living; whereas, the latter, who did not follow the arrangement of destiny and went into business, turned out to be excellent at predicting the market.

These ancient economic precepts are profound. What Adam Smith discussed had already been addressed by ancient Chinese philosophers much earlier. However, those ancient Chinese insights were just summaries of experience; they did not constitute rigorous scientific themes or disciplines, provide boundary conditions or scope for conclusions, or represent logically inherent analyses. As a result, because ancient Chinese economic thinking was not subject to a scientification process driven by knowledge accumulation and theoretical innovation over generations, it was never systematized into a body of knowledge that was intelligible to, or digestible by, the West.

In the remaining parts of this chapter, we will present a rough discussion on core assumptions, key points, analytical frameworks, and research methodology of economics to assist you to understand the rigorous analysis of the content covered by this textbook.

1.7 A Cornerstone Assumption in Economics

Every social science discipline imposes assumptions on individual behavior as the logical starting point of its theoretical system. As discussed above, the essential distinction of social science and natural science is that the former studies individuals' behavior and needs to make certain assumptions about individuals' behavior, while the latter investigates natural environments and objects. Economics is an important social science discipline since it not only studies and elucidates economic phenomena and enables positive analyses, but also closely examines individual behavior in order to make accurate predictions and value judgments.

1.7.1 Self-love, Selfishness, and Self-interest

When discussing individuals' behavior, three terms are often mentioned: self-love, selfishness, and self-interest, which are related to, and also distinct from, each other. Self-love means one's esteem and affection for oneself, which can be positive, as it encourages individuals to live a righteous and productive life, and can also be negative, since it may lead to a tendency towards narcissism, and even self-harm or self-deceit. Self-love can also generate self-interestedness and selfishness.

Selfishness refers to caring only for one's own welfare or advantage at the expense of or in disregard of others. It can lead to manifold harms. For example, being selfish makes one greedy; being greedy makes one overly ambitious; being overly ambitious makes one vain and arrogant; and being vain makes individuals lose themselves, while being arrogant may make individuals ruthless and offensive.

Self-interest means benefiting oneself without at the expense of others while it may or may not benefit others. It may even be altruistic out of justice and sympathy for a certain purpose (such as life goals or a sense of mission). Therefore, self-interest makes one judicious and rational, while selfishness produces greed. In other words, it may be altruism for the sake of self-interest. To pursue and obtain one's interest, individuals have to choose altruistic behavior. This constitutes the self-interest assumption adopted in the study of economics. When self-love and self-interest complement one another, individuals will possess clear self-knowledge; whereas, when self-love is related with selfishness, individuals will experience moral decline. Consequently, when individuals are dominated by self-love, it does not necessarily mean that they will disregard others because self-love may also be combined with self-interest.

1.7.2 Practical Rationality of Self-interested Behavior

In this textbook, especially in the proof that competitive market mechanisms lead to optimal resource allocation, a key assumption is that individual behavior is driven by self-interest in normal situations. In fact, this is the most basic assumption in economics and forms its cornerstone. Self-interest is not only a basic assumption but also the most significant reality.

It can apply to any country, nation, group, family, or politician; this is what we mean when talking about the interests of a country, a nation, a group, or a family. For example, when dealing with the relation between two nations, as a citizen, one must protect the interests of one's own nation and speak and act from the standpoint of that nation, and may be subject to penalties if one divulges state secrets. When dealing with the relation between enterprises, as an employee, one must protect the interests of one's organization, and if the person leaks firm secrets to competitors, one may face negative consequences. The self-interest assumption is frequently questioned with the following: why are there families if individuals are rationally self-interested and pursue personal interests? In fact, concerning the family, individuals act in the interest of their own families. In other words, under normal circumstances, individuals care about their own families more than they do about the families of others. The discussion on the relationships between individuals follows the same reasoning. In practice, misunderstandings of this assumption are common, one of which is that this applies to only individuals in every case.

It is necessary to assume that individuals are self-interested because this assumption conforms to reality. More importantly, the risk is minimal: If the self-interest assumption were not true, it would not lead to serious consequences. Contrariwise, if we adopted an altruistic behavior assumption and it were proven wrong, the consequences would be much more serious than those of a false self-interest assumption. As pointed out at the beginning of this chapter, the analytical framework of economics does not exclude a discussion of institutional arrangements and individual trade-offs on the premise of the altruistic behavior assumption. In fact, the rules of the game adopted under the self-interest assumption also apply to altruistic individuals in most cases, and the institutional arrangements, rules, and trade-offs under the altruistic behavior assumption are much simpler. However, a false altruistic behavior assumption could be disastrous. It is also very important to make correct judgments on individuals' behavior in daily life. Just imagine your costs if you mistook a selfish, cunning person for a simple, selfless, honest person and you will understand the serious consequences of such a wrong assumption.

Acknowledging that individuals are self-interested shows a realistic and responsible attitude towards solving social and economic problems. This is why we need certain government legislation to prevent opportunists from taking advantage of loopholes in institutions under the altruism assumption. As already discussed, if altruism is used as the premise to solve social and economic problems, the consequences may be disastrous. As an additional example, in the organization of production, if we deny the crucial importance of individuals' self-interest and only motivate individuals by emphasizing their contribution to the nation and the group, the result would be that everyone would aim to take advantage of the institutions to benefit from others' contributions, thereby leading to the free-rider problem. In this case, how could any nation become prosperous?

1.7.3 Boundaries of Self-interest and Altruism

It should be specially noted that, although the self-interest assumption holds true in most cases, it exhibits certain boundaries in application. For instance, under abnormal circumstances, such as natural or man-made disasters, wars, earthquakes and other major crises, individuals will often demonstrate their altruistic and selfless character, make sacrifices to fight for the nation, and go to great lengths to assist others. This constitutes another form of rationality, i.e., altruism, with which individuals are willing to sacrifice their lives (even certain animals possess this instinct). For example, when a nation is invaded by another nation, many citizens are willing to sacrifice their lives to protect their country. Under a normal and peaceful environment, however, when engaged in economic activities, individuals normally pursue their own interests. These illustrations demonstrate that self-interest and altruism are not oppositional to each other, but rather constitute natural responses to different situations and environments.

Therefore, we can view self-interest and altruism as relative terms to circumstances or purposes (such as pursuing dreams or life goals). An individual's purpose can be self-interested (self-regarding) or altruistic, and self-interest does not exclude the motivation of altruism. Both are reactions to different situations and environments, and are by no means contradictory. In fact, such duality can also be witnessed in animals. For example, when wild goats are chased to the edge of a cliff, the old goats sacrifice themselves by making the first jump, so that the goatlings could jump on them and have a greater chance of survival.

Adam Smith not only wrote the foundational work of *The Wealth of Nations*, but also *The Theory of Moral Sentiments*, which lays the foundation for his views about self-interest and the common good, and points out that we humans have a strong altruistic tendency to be other-regarding in our actions. These two works complement each other in the overall philosophy of Adam Smith. The basis of self-interest is purpose. It is an individual's own purpose that motivates her actions, which can be self-interested or altruistic. It is the case that, under the reality of an individual's self-love and self-interest, morality should be a kind of balance, an equilibrium outcome, and a convention realized through the social division of labor and cooperation. Under the guidance of appropriate institutions, individuals voluntarily divide the work by choosing varied specializations and cooperate in order to establish a harmonious, civilized, stable, and orderly society. It is against human nature to regard self-interest as immoral and runs counter to reality; it is not selfish. In fact, the organic combination of moral ethics, altruism and self-interest can actually enhance sympathy for others and a sense of justice, and promote social civilization and individuals' decency. The biggest advantage of the modern market is its utilization of the power of self-interest to counteract the weakness of benevolence so that hard-workers can be rewarded. As a consequence, we should not neglect the role of benevolence and morals in the formation of the modern market system. The progress of society cannot be sustained by those who always want to hurt others. Therefore, *The Theory of Moral Sentiments* has laid an ethical foundation for the rationality of market economy. However, many market economies today have overlooked the ethical basis of the market system.

Overall, self-interested individuals can be benevolent, altruistic, and moral. However, "self-interest" should not be "at others' cost". There are limitations and boundaries for self-interest and altruism, while selfishness that benefits oneself at the expense of others is the origin of maleficence and greed. Rationally self-interested behavior will conform to social norms as a requisite constraint. As a society, we collectively agree to educate individuals to pursue their personal interests without violating public order and to protect the public interest based on individual rationality. However, we also disagree with economic idealism that grounds policies in ignorance of the driving force of personal interest in a wayward attempt to protect the public interest. It is the case that self-interested behavior under appropriate legal constraints must be distinguished from selfishness that violates laws and/or harms the interests of others. The former should be protected whereas the latter opposed.

It should also be mentioned that, even in the case of self-interested behavior, there are differences in extent. Under ideal situations, the less self-interested are individuals, the better are the results produced. However, it is also impossible to eliminate self-interest completely. We can prudently state that self-interest is the logical starting point for economics. If all human beings are unselfish and always considerate to others, then economics involving human behavior would be useless, and industrial engineering or input-output analysis may be sufficient. It is from the starting point that self-interest is an objective reality and individuals tend to pursue their own interests concerning economic activities that China has carried out the reform and opening-up, and made the transition from a planned economy to a market economy.

1.8 Key Points in Economics

When investigating economic issues, economists commonly use the following basic assumptions, constraints, axioms, and principles:

(1) **Scarcity of resources**;

(2) **Information asymmetry and decentralization**: individuals prefer decentralized decision-making;

(3) **Economic freedom: voluntary cooperation and voluntary exchange**;

(4) **Decision-making under constraints**;

(5) **Incentive compatibility**: economic institutions or mechanisms should be used to solve the problem of conflicting interests among individuals or economic units, i.e., provide individuals with the incentive to do what the institution or mechanism designer wants them to do;

(6) **Well-defined property rights**;

(7) **Equality of opportunity**;

(8) **Efficient allocation of resources**.

The above assumptions and constraints are crucial because relaxing any one of them may result in different outcomes. The consideration and application of these assumptions, constraints, and principles are also useful for people in their daily lives. Although they may appear to be simple, thoroughly understanding and skillfully utilizing them in reality is not easy. In the following, we briefly discuss these key points, conditions, axioms, and principles, respectively.

1.8.1 Scarcity of Resources

Economics stems from the fact that resources are limited in the world (at least the Earth's mass is finite). As long as an individual is self-interested, and the person's material desire is unlimited (i.e., the more one possesses, the better), it is impossible to realize distribution according to wants. Therefore, the problem of precisely how to employ limited resources to optimally satisfy wants must be addressed, and economics is needed to achieve this.

1.8.2 Information Asymmetries and Decentralization

In addition to the major objective reality of the self-interested nature of individuals, another fundamental objective reality is that, in most cases, information is private or asymmetric among economic agents, so that the effect of institutional arrangements adopted may be inadvertently neutralized. This is a fundamental reason why some core economic problems are difficult to solve. For example, although a person's words may be righteous, it may be challenging to discern if one actually means what is said; listeners may appear to concentrate on what you are saying, but you do not know if the message is actually being well-received by them. The fundamental reason for such phenomena is the presence of private information. Private information, together with the self-interested nature of individuals, frequently leads to conflicts of interest among economic agents. If there is no appropriate governance system to equitably reconcile these differences, dishonesty, cheating, and selfishness in fighting for limited resources may become rampant. This is also the reason why the social sciences, and especially economics, are more complex and challenging to study and master than the natural sciences. Also due to information asymmetry, centralized decision-making is often inefficient; whereas, decentralized decision-making, such as the use of market mechanisms, is required to solve economic problems.

Only when complete information is acquired can the outcome be the first best. However, private information is frequently hard to obtain, and thus an incentive mechanism is needed to obtain truthful information. Since information acquisition also incurs costs, only the second-best outcome can be obtained in most situations. This is a basic conclusion in principal-agent theory, optimal contract theory and optimal mechanism design theory, which will be discussed in Part VI of this textbook. Without appropriate institutional arrangements, incentive distortion will exist, in which inducing information revelation inevitably incurs costs. As a consequence, it is particularly important to achieve information symmetry for the first-best outcome, without which many misunderstandings and inefficiencies may arise. By communicating openly with others, you enable others to understand you (signaling) and also get to know them (screening), which makes information more symmetric, clears up misunderstandings and enables consensus, which is the requisite condition of obtaining desired outcomes.

Excessive intervention of governments in economic activities and the over-playing of governments' role will lead to inefficiency, which is essentially resultant from information asymmetry. Many problems exist regarding governments' information acquisition and discrimination. If decision-makers are able to possess all relevant information, centralized decision-making featuring direct control would not be problematic, i.e., all that would be required is optimal decision-making. However, it is impossible for decision-makers to have all related information. This is why we prefer decentralized decision-making. This is also the reason why economists call for the design of various incentive mechanisms, and that a decentralized decision-making method featuring indirect control should be used to stimulate individuals to do as decision-makers desire or to achieve the goals that decision-makers want to be attained. We will focus on the issue of information and incentive in Parts VI and VII.

It is worth mentioning that centralized decision-making also offers certain advantages, especially concerning decision-making involving rapid major changes. For instance, centralized decision-making is more efficient when a nation, group, or enterprise is establishing a vision, orientation and/or strategies, or making major decisions. Such major changes, however, might bring about enormous successes or disasters. For example, the decision of adopting the reform and opening-up policy has led to the rapid development of the Chinese economy. In contrast, the decision of the Cultural Revolution almost drove the Chinese economy to collapse. One viable solution to this problem is to take public opinion into deep consideration when making decisions and select great leaders.

1.8.3 Economic Freedom and Voluntary Exchange

Due to economic agents' pursuit of their own interests and information asymmetry, institutional arrangements of the mandatory "stick" style are often not effective. Consequently, it is necessary to provide individuals with more freedom of economic choice, which is the most important right among the three private rights (right to survival, freedom of choice for one to pursue happiness, and private property rights). To achieve this, we should mobilize economic agents with economic freedom based on voluntary cooperation and exchange through inductive incentive mechanisms, such as the market. Therefore, the freedom of economic choice ("deregulation") plays a vital role in market mechanisms, with decentralized decision-making ("decentralization") being a prerequisite for the normal operation of market mechanisms and also a core precondition to ensure efficient allocation of resources under competitive market mechanisms.

In fact, the Economic Core Equivalence Theorem, which will be discussed in Chapter 12, reveals that once full economic freedom is given and free competition, voluntary cooperation and exchange are allowed, even without the establishment of any institutional arrangement in advance, the outcome of resource allocation driven by the self-interested behavior of individuals will be theoretically consistent with equilibrium allocation of a perfectly competitive market. The essence of the Economic Core Equivalence Theorem can be summarized as follows: under the rationality assumption, as long as economic freedom and competition are given, even if institutional arrangements are not considered, the economic core obtained will constitute a competitive market equilibrium.

China's reform and opening-up over the past 40 years have confirmed this assertion in practice. When analyzing the reasons for China's remarkable economic achievements, the critical factor is the provision to individuals of more freedom of economic choice. Indeed, reform practices from rural to urban areas indicate that wherever there are looser policies and a greater degree of economic freedom provided for producers and consumers, higher levels of economic efficiency prevail. China's so-called miraculous economic growth stems from the government's delegation of powers to the market; whereas, its imperfect market today is the result of excessive government intervention and inadequate or inappropriate government regulation and institutional arrangements.

1.8.4 Acting under Constraints

Acing under constraints is one of the most fundamental principles in economics, and is well expressed in the saying that "one must bow under the eaves". In fact, everything has its own objective constraints. Individuals, then, make trade-off choices under these existing constraints. Moreover, individuals' choices are determined by both objective constraints and subjective preferences. Constraints include material constraints, information constraints and incentive constraints, all of which can make it difficult for economic agents to achieve their goals. In economics, one embodiment of the basic idea of constraints is the budget set (or opportunity set) of consumer theory, as will be discussed in Chapter 3, which states that an individual's budget is constrained by the prices of the commodities and the individual's income. For an enterprise, constraints include available technologies and prices of inputs, under which the goal of maximum profit requires firms to determine the quantity of production, technologies to be adopted, the quantity of each input, pricing for products, responses to competitors' decisions, etc. The development of a person, or even a nation, must contend with various constraints, including political, social, cultural, environmental, and resource constraints. Furthermore, if constraint conditions are not clearly identified and understood, it is difficult to perform tasks and achieve goals.

When introducing a reform measure or an institutional arrangement, it is essential to consider feasibility and meet the objective constraints. In addition, the implementation risk is expected to be reduced to a minimum so that social, political, and economic turmoil will not result. Therefore, feasibility is a requisite condition to judge whether a reform measure or institutional arrangement is conducive to economic development and the smooth transformation of economic systems. In a nation's economic transition, to make a feasible institutional arrangement, it must conform to the institutional environment of the specific stage of the country's development.

Participation constraint is critical when considering optimal contract design (cf. Chapters 16 and 17), which means that an economic agent can benefit, or at least will not experience harm, from economic activities; otherwise, the agent will not participate in, or may even oppose, the rules or policies to be implemented. Individuals who pursue the maximization of self-interest will not automatically accept an institutional arrangement, but instead will make a choice between acceptance and refusal. Only under an institutional arrangement in which the individual's benefit is not less than the individual's reserve level (or the individual does not accept the arrangement) will the individual be willing to work, produce, trade, distribute, and consume. Moreover, if a reform measure or an institutional arrangement does not meet the participation constraint, individuals may give up. Of course, if everyone is reluctant to accept the reform measure or institutional arrangement, it cannot be successfully implemented. Mandatory reform may also arouse opposition and cause social instability, so that development will not be possible. Consequently, participation constraint is closely related to social stability, and is an essential factor of social stability in development.

1.8.5 Incentives and Incentive Compatibility

The incentive is one of the core concepts in economics. Each individual has the own self-interest; to obtain interests from some activity, one must also pay the corresponding cost. Through a comparison between benefits and costs, individuals may be willing (have an incentive) to get something done or do it well, or be reluctant or unwilling to get it done or do it well, and thus will have a rational incentive response to the rules of the game. This, however, frequently leads to incentive-incompatible conflicts of interest among individuals or between individuals and society, and produces chaos. The concept of incentive compatibility has been discussed by Adam Smith in *The Theory of Moral Sentiments*, in which he stated that *"...in the great chess-board of human society, every single piece has a principle of motion of its own, altogether different from that which the legislature might choose to impress upon it. If those two principles coincide and act in the same direction, the game of human society will go on easily and harmoniously, and is very likely to be happy and successful. If they are opposite or different, the game will go on miserably, and the society must be at all times in the highest degree of disorder."* [6] The reason for this is that, under given institutional arrangements or rules of the game, individuals will make optimal choices according to their own interests, but such choices will not automatically satisfy the interests or goals of others and society. In addition, information asymmetry makes it challenging to implement social optimum by command. A good institutional arrangement or rule is able to guide self-interested individuals to act subjectively for themselves, but objectively for others, making individuals' social and economic behavior beneficial to the nation and the individuals, as well as to themselves. This is core content of economics.

Everything that an individual does involves interests and costs (i.e., benefits and costs), making incentive a ubiquitous issue that must be dealt with in daily work and life. As long as the benefits and costs are not equal, there will be different incentive reactions. To maximize profits, an enterprise has the incentive to use resources in the most efficient way and provide incentives to guide employees to expend the greatest effort. Outside of the enterprise, changes of profits provide an incentive for resource holders to modify their ways of using the resource; whereas, inside of the enterprise, incentive influences the way that the resource is used and the ef-

[6] Adam Smith: *The Theory of Moral Sentiments*.

fort that employees invest in the work. To make management efficient, the role of incentive in organizations must be clear, as well as how to construct incentives to guide subordinates to exert the greatest effort in the work.

Since the interests of individuals, society and economic organizations cannot be identical, how can self-interest, mutual benefits, and social interests be organically combined? This requires incentive compatibility, with which the reform measures and institutional arrangements adopted can drive individuals' incentive for production and work. As a consequence, to implement a goal of one's own or that of society, appropriate rules of the game must be defined, under which when individuals pursue their self-interest, the goal can also be achieved. In other words, it unifies the self-interest of individuals and mutual benefits among individuals so that when pursuing one's self-interest, each individual can assist in attaining the goal intended by society or other individuals. We will focus on the issue of how to achieve incentive compatibility in Part V of the textbook.

1.8.6 Property Rights as an Incentive Scheme

Property rights are an important component of a market economy. In the previous section about ancient Chinese thoughts on the market, we mentioned that over 2,300 years ago, Shang Yang of the State of Qin utilized the example of the hare to expound on the crucial importance of establishing private property rights, and how a clear definition of property rights can "determine ownership and settle disputes", and assist to establish the market order.

Property rights determine how a resource or economic good is owned and used. A clear definition of property rights will enable a clear definition of the attribution of profits, thus providing incentives for property owners to consume and produce in the most efficient way, to provide quality products and good services, to build reputation and credibility, and to maintain their own commodities, housing, and equipment. If property rights are not unequivocally defined, however, the enterprises' incentive will be diminished, giving rise to incentive distortion and moral risk. For example, unclear property rights in state-owned enterprises will lead to inefficiency, induce wide-spread corruption, such as rent-seeking and interest transfer, squeeze the private economy, impede innovation, and lead to unfair competition. In the market mechanism, incentives are given to individuals mainly in forms of property possession and profit acquisition. The Coase Theorem, which will be discussed in Chapter 14, is a benchmark theorem in property rights theory. It claims that when there is neither transaction cost nor income effect, as long as property rights are clearly defined, an efficient allocation of resources can be achieved through voluntary coordination and cooperation.

1.8.7 Equality of Opportunity and Equity in Outcome

“**Equality in outcome**” is a goal that an ideal society aims to achieve. However, for human society with self-interested behavior, this “outcome equality” often produces low efficiency. Then, in what sense can equality or equity be consistent with economic efficiency? The answer is that if one uses “**equality of opportunity**” as the value judgment standard, equity and efficiency can be consistent. “Equality of opportunity” means that no barrier should exist to hinder individuals from pursuing their goals, and there should be a level playing field for every individual. The Fairness Theorem, to be discussed in Chapter 12, informs us that as long as individuals' initial endowments are of equal value, through the operation of a competitive market, allocation of resources that is not only efficient, but also equitable, can be attained even if individuals pursue self-interest. A concept similar to “equality of opportunity” is “individual equality” (also known as “all men are created equal”), which means that, although individuals are born with different values, genders, physical conditions, cultural backgrounds, capacities, ways of life, etc., “individual equality” requires deep respect for such individual differences.

As individuals hold heterogeneous preferences, the seemingly equal distribution of, for example, milk and bread, may not satisfy everyone, and then the difference between equality and equity must be emphasized. Although both promote fairness, equality achieves this through treating everyone the same irrespective of need, while equity achieves this through treating people differently depending on need. Therefore, beside equal allocation that defines equality as absolute equalitarianism, the concept of **equity** in other senses should also be used in the discussion of economic issues. For instance, **equitable allocation**, which will be discussed in Chapter 12, considers both subjective needs and objective factors, such as income, which means that everyone is satisfied with their own allotment.

1.8.8 Efficient Allocation of Resources

Whether resources are efficiently allocated is a basic criterion to evaluate the effectiveness of an economic system. In economics, efficient allocation of resources usually refers to Pareto efficiency/optimality, which means that no other feasible allocation exists, such that at least one individual is better off without making any one worse off. As such, it requires not only efficient consumption and production, but also production of products that can best meet the needs of consumers.

It is worth mentioning that, concerning economic efficiency, we should distinguish between three types of efficiency: a firm's production efficiency; industrial production efficiency; and allocation efficiency of an economy. By stating that a firm's production is efficient, we mean that, with a given input, the output is a maximum, or with a given output, its inputs are a minimum. Industry is the aggregation of all firms' production for a particular commodity, the efficiency of which can be similarly defined. However, the efficiency of a firm does not imply the efficiency of an industry. The reason for this is that if the production materials of firms with outdated technologies are given to those firms with advanced technologies, there will be more outputs for the whole industry. At the same time, even if production of the entire industry is efficient, the allocation of resources may not be (Pareto) efficient.

The concept of Pareto efficiency in allocating resources is applicable to any economic institution. Indeed, it provides a basic criterion of value judgment for an economic institution from the perspective of social welfare, and assesses the economic performance from the perspective of feasibility. It can also be applied to a planned economy, a market economy, or a mixed economy. The First Fundamental Theorem of Welfare Economics, which will be introduced in Chapter 11, proves that when individuals pursue their own self-interest, a perfectly competitive market will lead to the efficient allocation of resources.

1.9 Understanding Economics Properly

An accurate and thorough understanding of economics can assist individuals to correctly use basic principles and analytical methods of economics to study various economic issues under different economic environments, behavioral assumptions, and institutional arrangements. The different schools and theories of economics per se show inclusiveness, specificity, universality, and generality of the analytical framework and methodologies of economics. Under different economic environments, various assumptions and specific models are required. Only in this way will the developed theory be able to explain different economic phenomena and individuals' behavior, and more importantly, make logically inherent analyses, draw conclusions of inherent logic, and make scientific predictions and reasoning under various economic environments that are close to the theoretical assumptions. However, due to the complexity of economic environments, for the sake of semantic and logic clarity, economics employs various rigorous mathematical tools to construct economic models in order to develop various economic theories. Rigorous mathematical tools are often difficult to master, which results in frequent misunderstandings and criticisms of economics. In addition to the common misunderstandings of benchmark theories discussed previously, misunderstandings exist about economics concerning the following several aspects.

1.9.1 On the Scientification of Economics

One of the primary misunderstandings is that economics is not a science and includes numerous conflicting theories. A common criticism is that too many different economic theories exist that come from different economic schools in economics, making it difficult to discern which is correct and which is not. Those who hold such an opinion, however, fail to fully comprehend that all economic theories do not diverge from the two basic categories of theory discussed previously. Indeed, it is precisely due to the complexity of objective reality, and dissimilar economic, political, social and cultural environments in different countries and regions, different thoughts and preferences of people, and various economic goals that people pursue, that different theoretical economic models and economic institutional arrangements need to be developed.

It may be easy to understand that different economic theories or models should be developed for different economic, social, and political environments. However, it is challenging for many to comprehend why different economic theories are developed under the same economic environment. Consequently, some individuals, while disavowing economics and its scientific characteristics, sharply criticize economists by stating that "100 economists will have 101 opinions". Actually, it is they who do not realize that it is analogous to the fact that we need different maps for different purposes, such as traffic, travel, military, etc., even though there is only one Earth. It is the case that, in the same given economic environment, we may need to develop distinct economic theories and different economic institutional arrangements for various goals.

The fact that different opinions of economists will arise for the same problem merely demonstrates the precision and thoroughness of economics because, when the premises and the environment change, the conclusions should vary accordingly; this is especially true in the case of the second category of economic theory that aims to solve practical problems. Furthermore, as different individuals will also have dissimilar subjective judgements of values, there are few universally correct conclusions that satisfy everyone and apply in all situations. Just as medicines will vary according to the disease, when considering economic problems, case-by-case analyses must be conducted according to the specific time, place, people, and occasion involved. The major difference in the analogy to medicine is that, while an incorrect prescription for a condition may result in serious health problems or the death of an individual, an incorrect choice of economic policy, with large externalities, will influence an entire group of individuals or even a nation.

Although different economic theories and models exist, both the benchmark economic theory, which provides a benchmark or reference system, and the second category of economic theory, that aims to solve practical problems, they definitely do not constitute different “economics”.

The basic analytical framework and methodologies of (modern) economics, just like those of mathematics, physics, chemistry, engineering, etc., are not bounded by regions or nations. The foundational principles, methodologies, and analytical framework can be used to investigate a variety of economic issues under all economic environments and institutions, and to study economic behavior and phenomena in specific areas and time periods. The analytical framework and methodologies that will be introduced later can be used to conduct comparative analyses on almost every economic phenomenon and issue. In fact, this is precisely where the power and wonder of the analytical framework of economics lies: its essence and core require that the economic, political, and social environment conditions at a specific time and place must be considered and clearly defined when carrying out research. Economics can be used to investigate economic issues and phenomena under human behavior as manifested in different nations, regions, and cultures. Its basic analytical framework and methodologies can also be applied to elucidate other social phenomena and human decision-making processes. Indeed, it has been proven that, due to the universality and generality of the analytical framework and methodologies of economics, in the past few decades, various analytical methods and theories have been successfully extended to other disciplines, including political science, sociology, and the humanities.

1.9.2 On the Mathematical Feature of Economics

In addition to the criticisms that the assumptions of economic theory do not conform with reality and economics is not a science, another common criticism is that economics pays too much attention to details and involves an increasing amount of mathematics, statistics and models, and this makes questions more incomprehensible. However, the reason that economics uses so much mathematics and statistics is to achieve rigor and quantifiability of empirical studies. Although decision-makers in government and the general public do not need to understand details or premises of rigorous theoretical analysis, economists who make policy suggestions must thoroughly understand these. As economic theory will generate great externalities once adopted, blind application without considering premises may produce serious problems and perhaps catastrophic consequences. This is why mathematics should be employed to rigorously define the boundary conditions of a theory. In addition, the application of a theory or enactment of a policy often requires the tools of statistics and econometrics to carry out quantitative analyses or empirical tests. Moreover, because the real economic society is so complicated, the use of mathematical models in economic theories can assist to depict the real economic world for people to understand problems to be solved in reality. We will discuss the role of mathematics in economics later in this chapter.

1.9.3 Misunderstandings on Economic Theory

Each theory or model in economics that aims to solve practical problems adopts axiomatic analysis, which comprises a set of presupposed assumptions on economic environments, behavior patterns and institutions, and conclusions based on these assumptions. Considering the complexity of economic environments and the diversity of individual preferences in the real world, the more general are the presupposed assumptions of a theory, the more powerful is the theory. If the presupposed assumptions of a theory are too restrictive, the theory will lose generality, and thus offer less utility in reality. As economics is employed to serve the society and the government, the theory must be of some breadth and depth. Consequently, a necessary requirement for a good theory is its generality, in which the more general it is, the larger explanatory power it will have and the more useful it will be. The general equilibrium theory, which studies competitive markets, constitutes such a theory. It proves that the existence of competitive equilibrium leads to efficient resource allocation under the condition of very general individual preferences and production technologies.

Even so, theories of social sciences, and especially those of economics, like all theorems in mathematics, are subject to boundary conditions. As discussed previously, because of the great externalities of economic theories, when discussing or applying an economic theory, it is necessary to focus on its presupposed assumptions and application scope. This is because the conclusions of any economic theory are not absolute, but rather only hold when the assumptions are satisfied. Whether or not this point is recognized when discussing economic issues is a basic criterion of whether an economist is well-trained. As economic issues are closely correlated with daily life, even ordinary people can give their opinions about economic problems, such as inflation, business climate, balance or imbalance of supply and demand, unemployment, the stock market, and the housing market. For this reason, many people do not regard economics as a science. Indeed, economics would not be a science if it did not take into account constraints or base itself on accurate data and rigorous theoretical logical analysis. A well-trained economist always discusses issues based on economic theories and is fully cognizant of the boundary conditions of the relationship among economic variables and the inherent logic of the corresponding conclusions. It is crucial to fully understand the boundary conditions of economic theories; otherwise, one will not be able to distinguish between the theory and the reality, but instead tend towards one of two extremes: either simply applying the theory to reality, irrespective of constraints in reality; or completely denying the value of economic theory.

The first extreme viewpoint overestimates the role of theory and misuses it. For example, some people disregard the realistic objective constraints that face a nation. They blindly or mechanically apply the two categories of economic theories to solve the nation's problem, and indiscriminately copy models to study the problem, assuming that the inclusion of mathematical models will produce valid theories. Conclusions and suggestions obtained by copying economic theories or models, whether they be benchmark economic theory, or the second category of economic theory that closely reflects reality, if blindly applied without taking into consideration various constraints and boundary conditions created by the nation's actual situation and economic institutional environment will frequently lead to serious problems. In fact, irrespective of how general the theory and behavioral assumption are, they always have application boundaries and limitations, and thus must not be used indiscriminately. Indeed, especially for those theories developed based on an ideal state that exists far from reality and primarily for the purposes of establishing a reference system, benchmark and goal, they should not be applied directly or incorrect conclusions will result. In addition, without a sense of social responsibility or strong training in economics, a person may overestimate the role of theory and blindly apply economic theory to the real economy without taking the premises into account, which could bring severe consequences, negatively influence social and economic development, and lead to great negative externalities. For instance, the conclusion of the First Fundamental Theorem of Welfare Economics that a competitive market leads to the efficient allocation of resources is subject to a series of preconditions, and its misapplication will produce severe policy errors and damage to the real economy.

The other extreme viewpoint completely denies the role of theory. People holding such a point of view underestimate or deny the pragmatic utility of economics, including its behavioral assumptions, analytical framework, basic principles, and research methodology. In fact, just as in the case of the benchmark theory of economics, no discipline in the world exists whose assumptions and principles coincide with reality perfectly (like the concepts of free fall without air resistance and fluid motion without friction in physics). However, this is not a reason to deny the scientific features and usefulness of a discipline, including economics. We learn economics to acquire not only its basic principles and utility, but also its approaches towards thinking, asking, and solving questions. As discussed previously, the value of benchmark theories lies not in the direct explanation of reality, but in the provision of a study platform and reference system for developing new theories to explain the real world. With these methods, economists can become enlightened on how to solve economic problems in reality. Furthermore, as discussed in the previous section, a theory applicable to one nation or region may not be applicable to another due to different environments. Instead of applying the theory mechanically and indiscriminately, it is necessary to modify the original theory to develop new theories according to given economic environments and individual behavior patterns.

There is also another extreme viewpoint, in which certain people contend that market failures may occur under any circumstance, and that the market has externalities. In claiming this, they deny the practical significance of economic theory, believing that economics is highly hypothetical, and that these assumptions are superfluous because the market does not have boundaries.

It is also sometimes stated that a certain theory or conclusion has been overturned. As not all of the conditions of a certain theory are in accordance with reality, the theory is deemed to be incorrect and thus supplanted. In general, this statement is not correct. Assumptions, even those in the second category of theories that aim to solve practical problems, cannot fully coincide with reality or cover every possible case. Indeed, a theory may be applicable to the economic environment of one location but inapplicable to that of another. As long as there is no inherent logic error present, however, it cannot be concluded that the theory is fallacious and needs to be abandoned. It may only be stated that it is not applicable to a certain place or a particular time.

Another common mistake is attempting to draw a general theoretical conclusion based solely on certain specific examples. This constitutes a methodological error. Of course, we do not deny the unique role of history, culture, and paradigms of each country in the establishment of its own discourse of economics.

1.9.4 On Experiments in Economics

Another criticism about economics is that it is not an experimental science at all, and thus the scientific feature of economics is negated. Such viewpoint constitutes a misunderstanding. First of all, as with the rapid development of experimental economics in recent years, economics is using increasing numbers of laboratory experiments, field experiments, and computer simulation experiments to investigate various economic problems. Experimental economics also tests individuals' behavior and rationality of behavioral assumptions through these experimental methods, and consequently constitutes an important tool to determine whether or not an economic theory fits objective reality. Theorists have also obtained crucial information from experiments in order to promote the advancement of theories (important discussions of numerous economists on how to understand experiments in economics can be found on the website of Al Roth). Furthermore, experiments in economics are already transitioning more from the laboratory toward field experiments (see relevant discussion by John List).

Indeed, from an empirical perspective, in real economic activities, experiments in economics provide an indispensable advantage to verify policies and systems, especially concerning the need for institutional transitions. After continual exploration by early scholars, and the systematic synthesis of methodologies and tools of experiments in economics by Vernon Smith, winner of the 2002 Nobel Memorial Prize in Economic Sciences, experimental economics as an important empirical tool has received increasing attention in market mechanism design. When external environments change rapidly and new technologies abound, reform becomes inevitable; however, the strategic risks and social costs of various policy suggestions must be carefully considered. Therefore, it is a difficult and key objective in institutional reform to identify a way to comprehensively examine problems in advance that might arise out of new proposals regarding institutions. For instance, in the earlier days of the reform and opening-up, China took various measures, including the "special economic zone policy'", "pioneering pilot scheme", "typical example as the lead", etc. Although economic experiments are consistent with such measures in the guiding aim of lowering the risk and cost of the reform, major differences in methodology remain between economic experiments and pilot experiments for the accumulation of experience.

In comparison with the pilot method, firstly, the economic experiment is liable to focus on a single research question, so that each experiment examines the effects of only one policy and the characteristics of only one mechanism. Secondly, the economic experiment employs relatively normal techniques and tools. Indeed, there are multiple influential factors in real economic activities, while the methodology of economic experiments is able to control factors that are irrelevant to the research question in order to focus more closely on the effect of one factor on certain economic phenomena. In comparison with the pilot method, it is also less costly to carry out economic experiments. It is also worth noting that the relation between genetics and economic behavior is meaningful and promising. Once the relation between the two is accurately determined, the foundation will be established for economics to become a discipline of science, just as a natural science.

It must also be admitted that some economic theories, such as general equilibrium theory that can be used for comprehensive analysis cannot easily be tested by social experiments since policy mistakes may result in enormous risks to economy and society. This is the greatest difference from natural sciences. As in natural sciences, natural phenomena and objects can be studied through experiments, and theories can be tested and further developed in the laboratory. Astronomy might be the only exception to this, but it involves no individual behavior. In all cases, once individual behavior is involved, the situation will at least become relatively more complicated. Moreover, extreme precision can be attained in the application of theories in natural sciences. For instance, in the construction of buildings or bridges, and the manufacturing of missiles or nuclear weapons, accuracy of any degree is attainable since all of the parameters are controllable, and the interrelationship between variables is amenable to experimentation. In economics, however, numerous factors affecting economic phenomena are uncontrollable.

Economists are also frequently criticized for inaccurate economic forecasts. This can be explained from two perspectives. From the subjective perspective, it is due to whether or not specific economists have had systematic and rigorous training in economics. If not, they may be unable to discern the main causes of problems or make correct logical analyses and inferences when they discuss and attempt to solve economic problems, and thus will prescribe solutions to economic problems that will be unsuccessful. From the objective perspective, some economic factors that influence economic results may suddenly and uncontrollably change, thus causing predictions to be inaccurate, even though they may be made by well-trained economists with excellent economic insight. Moreover, an economic issue involves not only human behavior, which increases complexity, but numerous other uncontrollable factors, as well. Although an economist might be knowledgeable and talented, the economist predictions may still deviate because certain uncontrollable factors that influence economic results may change. For instance, even for a good economist with sound judgment, the economist economic forecasts might become inaccurate once sudden changes occur in the economic, political, or social environment. It is also the case that some people assert that no matter how economics develops and what the reason may be, inaccurate prediction is the norm, and accurate prediction is attributable to chance. In principle, this is true since economic fluctuations are random variables. However, since the probability for an event to occur and the capability of economists vary, a well-trained economist can better judge the probability of an event and be more likely to be accurate in the prediction. This constitutes the subjective factor of inaccurate prediction as mentioned above.

How can the problem that economic theory cannot experiment on society under many circumstances be overcome? The answer is logically inherent analysis, based on which inherent conclusions and inferences can be drawn, and comparisons and empirical data tests can be carried out through both horizontal and vertical perspectives of history. In this way, when conducting economic analysis or providing policy suggestions, according to the three dimensions of economic analysis discussed previously, there should first be theoretical analysis of inherent logic to define the applicable boundary conditions and scope. Meanwhile, tools of statistics, econometrics and/or experimental economics should be used to perform empirical quantitative analyses or tests, and the invaluable perspective of history should be taken into account to conduct vertical and horizontal comparative analysis. Therefore, when carrying out economic analysis or identifying policy suggestions, there must be theoretical analysis of inherent logic, historical comparative analysis, and empirical quantitative analysis based on data and statistics; and all three are indispensable, which makes the analysis have six natures: scientific, rigorous, realistic, pertinent, forward-looking, and thought-provoking. This three-in-one study methodology, to a large extent, overcomes the challenge that many economic theories cannot or cannot easily experiment on society.

The method of logically inherent analysis of economics aims to fully understand and characterize relevant circumstances (economic environment, situation, and status quo) for a problem to be solved to ascertain what the problem is and its causes, apply appropriate economic theories accordingly, draw conclusions, and make accurate forecasts and inferences or give the direction of reforms. As long as the status quo accords with the causes (economic environments and behavioral assumptions) presupposed in the economic model, logically inherent conclusions can be reached according to economic theories, and thus solutions can be obtained or certain institutional arrangements can be suggested for different circumstances (varying with time, place, individual, and case). The method of logically inherent analysis can assist to make academic inferences on possible results under circumstances in which real economic and social environments, behavior patterns of economic agents, and economic institutional arrangements are given, and thus provide useful guidance for solving real economic problems. In other words, once the problem and its causes are clearly identified, and the appropriate economic theory is applied, if such a theory exists, then the optimal solution can be implemented, and logically inherent conclusions can be drawn in order to make accurate predictions and inferences or provide appropriate reform measures. Otherwise, severely harmful consequences might arise.

It is true that the result of an economic theory may not be tested by experiments on society under many circumstances, and data do not provide the sole basis for analysis. Although practice is the sole criterion for testing truth, but not for predicting it. It is logically inherent analysis and historic comparison that should be relied upon, and thus theory becomes indispensable. Like a doctor prescribing medications for a patient or a mechanic repairing an automobile, the most challenging and crucial task is the accurate diagnosis of the disease or the cause of the failure. The criterion for a good physician lies in whether one can accurately find the true cause of the problem. Once the cause is identified and a remedy identified, it is relatively easy to administer appropriate treatment. For economic problems, however, the prescription is economic theories. Once we truly understand the characteristics of an economic environment, thoroughly investigate the situation, and reasonably characterize individuals' behavior, the likelihood of success should rise dramatically.

1.10 Basic Analytical Framework of Modern Economic-s

For performing most tasks in life, basic laws exist. The way that economics studies and solves problems is similar to how people deal with personal, household, economic, political, and social affairs. In order to do something well and establish and maintain good relationships with others, the first task is to understand national conditions and customs, i.e., to know the real environment, behavior, and personality of the persons with whom you interact. On such a basis, one determines the optimal way of dealing with them and performing tasks, and make an incentive response after weighing the advantages and disadvantages to obtain the best outcome. Finally, one must make a value judgement on the choice, and evaluate the rules of the particular game being played. The basic analytical framework and research methods of economics follow this mode precisely to study economic phenomena, human behavior, and how people assess trade-offs and make decisions. Of course, a major difference between these two is the rigorous reasoning of economics, which uses formal models to identify the logical relationship between presupposed assumptions and conclusions. Such analytical framework offers great generality and consistency.

A standard academic work needs first to delineate all of the problems to be studied and/or resolved, or the economic phenomena to be elucidated. In other words, economists should first identify research objectives and their significance, provide readers with information regarding an overview and progress of the issues under investigation through a literature review, and illustrate the work's innovation concerning technical analyses and/or theoretical conclusions. Subsequent to this, they should discuss how to address the issues raised and draw conclusions.

Although economic issues under study may be quite dissimilar, the basic analytical framework used is essentially identical, which is an axiomatic approach to investigate economic problems. The analytical framework for a standard economic theory in economics consists of the following five steps: (1) specifying economic environments; (2) making behavioral assumptions; (3) establishing institutional arrangements; (4) determining equilibrium outcomes; and (5) making evaluations. All economics papers written with clarity and logical consistency comprise these five parts, especially the first four parts, irrespective of the conclusion and whether or not the author realized it. In this way, writing an economics paper constitutes innovative writing with a logical structure and analysis in such steps. Once these components are understood, the basic writing pattern of academic economics papers is known, and it will be substantially easier to learn economics. These five steps are also quite beneficial for understanding economic theory and its proofs, and finding research topics and conducting research.

Prior to discussing the five components in detail, it is first necessary to define the term "institution". An institution is usually defined as a set of rules related to social, political, and economic activities that dominate and restrict the behavior of various social classes (Schultz, 1968; Ruttan, 1980; North, 1990). When people consider an issue, it is normal to consider certain factors as exogenously-given variables or parameters, and others as endogenous or dependent variables. These endogenous variables depend on the exogenous variables, and thus are functions of those exogenous variables. In line with the classification method of Davis-North (1971, pp 6-7) and the issue to be studied, any institution can be divided into two categories: institutional environment and institutional arrangement. An institutional environment is the set of a series of basic economic, political, social, and legal rules that form the basis for establishing production, exchange, and distribution rules. Among these rules, the basic rules and policies that govern economic activities, and property and contract rights, constitute the economic institutional environment. An institutional arrangement is the set of rules that dominate potential cooperation and competition existing among economic participants. It can be interpreted as the generally known the rules of the game, with different rules of the game leading to dissimilar incentive reactions of individuals. In the long term, institutional environment and institutional arrangement will affect each other and evolve. Yet, in most cases, as Davis-North points out, people usually regard economic institutional environments as an exogenously-given variable, and consider economic institutional arrangements (e.g., the market system) as exogenous or endogenous, depending on the issue under study or discussion.

1.10.1 Specifying Economic Environment

The primary component of the analytical framework of modern economics is specifying the economic environment wherein the issue or object to be studied is situated. An economic environment is usually composed of economic individuals and their characteristics, such as their preferences, initial endowments, production possibility set, the institutional environment of the economic society, and the information structure.

The specification of an economic environment can be divided into two levels: an objective and realistic description of economic environments; and a concise and acute characterization of the essential features. The former is a science and the latter is an art. The specification of an economic environment requires combining the two levels. In most economic studies, economic environments are assumed to be exogenously given but not derived from theoretical models; otherwise, discussions would be impossible because some economic factors or variables need to be given as parameters.

Description of economic environment: The first step in formulating any modern economic theory is to approximately depict the economic environment of the issue or object to be studied. The more clear and accurate the description, the greater the likelihood of obtaining correct theoretical conclusions. A reasonable, useful economic theory should properly depict the specific economic environment. Though economists use the same basic analytical framework and research methods, different countries and regions have different economic environments, which usually lead to different conclusions.

Characterization of economic environment: When describing the economic environment, it is important to depict it clearly and accurately; it must also be depicted concisely and acutely, in order to capture the essence of the problem. Real economic environments involve many aspects and are very complicated. For example, an individual, as a main object of economic studies, can be characterized by height, weight, gender, age, wealth, intelligence, hobbies, behavior, and other factors, while different firms may use different production technologies and different combinations of factors of production to produce different commodities. If all these aspects are depicted, we have probably described the economic situation or environment accurately and truthfully. However, a simple listing of all the elements in the environment cannot capture the key points or essence of the problem, and people may be confused by a large quantity of complicated details.

Therefore, to avoid trivial matters and focus on the most critical issues, we need to characterize economic environments specifically according to the issue to be studied. For example, when discussing consumer behavior in Chapter 3, it is simply assumed that consumers are described thoroughly by noting their consumption set, preference relation (or utility function), initial endowment, and information structure (if uncertainty is considered) regardless of gender, age, race, or beauty. When discussing producer theory in Chapter 4, the economic characteristics of a firm are depicted by noting its production possibility set or production function. The economic characteristics of all economic agents constitute the economic environment. Similarly, when studying a regional economy, we need to describe the regional economic environment and its economic characteristics; when studying economic issues of transitional economies, we also need to characterize the basic features of the economic environment under irregular economic institutions. Thus, though we still use the basic analytical framework and methodologies of modern economics in the study of transitional economies under immature economic institutions, we cannot simply copy and apply the conclusions derived for regular economic environments since the environments of transitional economies are quite different.

A common criticism of economics is that it is useless because it relies on a few simple assumptions to summarize complex situations. In fact, this is also the basic research methodology of controlled experiments. In studying the relationship between two physical variables, both theoretical research and experimental testing will fix the other variables influencing the object of study. In many cases, clarifying every aspect (even unrelated aspects) is unnecessary and may even lead to a loss of focus. Moreover, different economic institutional environments require different economic theories. Different economic theories may be required even for the same economic environment because the objectives may differ. This is just like drawing maps for different purposes: People need a tourist map for travelling, a traffic map for driving, and a military map for war. Each map describes some of the characteristics of a region, but it does not present the whole picture. Why do people need a tourist map, traffic map, and military map? Because different maps are for different purposes.

Therefore, economics is both a science and the art of abstracting and characterizing real economic environments. Economics uses the concise and profound characterization of economic environments to describe the causes of problems, conduct analyses of inherent logic, and thereby obtain logical conclusions and inferences. A good economist takes a sophisticated approach to accurately grasping the essential characteristics of the economic situation. Only when we have clarified the causes for a current situation can we solve its problems using the proper remedy (i.e., economic theory). Of course, to achieve this, one needs basic training in economic theory.

Determining how to describe the economic environment frequently leads to divergent theoretical results, and can even produce new schools of thought. Because economic environments are markedly complex, in many cases, economics can not only carry out descriptive analysis, like in natural science, but must also refine and characterize the behavior of the economic environment in an abstract manner to identify the most important characteristics. This, however, often makes economists possess a degree of subjective judgment. Different subjective judgments will lead to dissimilar specifications of economic environments, which will in turn lead to various economic theories, economic schools, and/or theoretical results. For example, there are many schools in macroeconomics: the Keynesian school, the post-Keynesian school, the rational expectations school (or neoclassical macroeconomics), the monetarism school, the supply school, and the new institutional school. In fact, the antagonism between these schools is not as great as commonly believed by non-economists and the media. Indeed, the schools have numerous factors in common: the same basic analytical framework, the same research method (using the economic model and market equilibrium to analyze the market), and the same object (studying the interaction and change law of macroeconomic variables under the arrangement of the market system). It is also the case that they all believe in the market system. Moreover, it is agreed upon that, in terms of the long-term or general trend of economic operation, there will be an optimal market equilibrium. The differences among these theories are primarily resultant from the differences in describing the economic environment, especially whether the impact on or interference to the economic system comes from the demand side or the supply side, whether information about economic fluctuation is sufficient, and whether the differences are caused by different assumptions on the time effect of the interference, such as lag or instant.

1.10.2 Imposing Behavioral Assumptions

The second basic component of the analytical framework of economics is making assumptions about economic individuals' behavior. Such assumptions are critical and fundamental. Whether an economic theory is convincing and feasible and whether an institutional arrangement or economic policy is conductive to sustainable and rapid economic development depend mainly on whether the assumed individual behavior truly reflects the behavior of most individuals and whether the institutional arrangements are incentive compatible with individuals' behavioral patterns (i.e., whether individuals' reactions to the incentives are beneficial to others and to society).

In general, in a given real environment and rules of the game, individuals will make trade-offs according to their behavioral pattern. Thus, when deciding on the rules of the game, policies, regulations, or institutional arrangements, we need to consider the behavioral patterns of the players and make correct judgments accordingly. As when dealing with people in daily life, we need to know whether they are selfless or honest, for example. Different rules should be instituted depending on the players. The rules required for honest and upright people who tend to tell the truth may be comparatively simple. One does not need to invest much energy in designing the rules of the game for someone of the utmost integrity; the rules may be simple and even unimportant. On the contrary, dealing with a cunning and dishonest person will be quite different; the rules will be much more complicated and require much more attention. Thus, making the right judgments about individual behavior is a very important step in studying and solving realistic problems involving how individuals react to incentives and make trade-off choices. In studying economic issues (e.g., economic choice, the interaction among economic variables, and their patterns of change), defining individuals' behavioral patterns is also very important.

As mentioned above, under normal circumstances, a logical and realistic assumption about individuals' behavior used by economists is the self-interest assumption, or the stronger **rationality** assumption, , which implies that they will attempt to avoid unfavorable choices, i.e., economic agents pursue the maximization of benefits. Bounded rationality means to make the best choice according to the knowledge and information possessed by an agent, which belongs to the category of rationality assumption. In consumer theory, which will be discussed later, we assume that consumers pursue utility/satisfaction maximization; in producer theory, we assume that producers pursue profit maximization; and in game theory, various equilibrium solution concepts have been introduced to describe the behavior of economic agents, which are given based on different behavioral assumptions. Overall, any economic agent, in contacting with others, implicitly assumes others' behavior.

The assumption of (bounded) rationality is largely reasonable. From a practical point of view, as mentioned previously, there are three basic kinds of institutional arrangements: mandatory regulation (for situations with small operation costs and relatively easy information symmetry); incentive mechanism (for cases of information asymmetry); and social norms (composed of ideology, beliefs, morals, customs, etc., that are norms of self-discipline). If individuals are all selfless and highly developed ideologically, there will be no need for rigid "stick-style" regulations that "enlighten with reason" or the flexible market system that "guides with interest".

1.10.3 Determing Institutional Arrangements

The third basic component of the analytical framework of economics is determining institutional arrangements, which are usually referred to as the "rules of the game." Different institutions (rules) should be adopted for different situations, different environments, and different individuals with different behavioral patterns. When the situation or environment changes, the rules of the game will often change accordingly. Determining the rules is of utmost importance. Different rules will lead to different reactions to incentives, trade-off decisions, and results. This also applies to the study of economics. When an economic environment is given, agents need to decide the economic institutional arrangements. Modern economics—especially mechanism design theory, information economics, contract theory, and auction theory, which have developed over the last five decades—studies and describes various economic institutional arrangements for different economic environments and behavioral assumptions. Depending on the issue under consideration, an economic institutional arrangement can be exogenously given or endogenously determined.

As discussed previously, there are three basic kinds of institutional arrangements that guide individuals' behavior: mandatory regulatory governance or government intervention; institutional norm of incentive mechanism; and didactic social norm. The three means play different roles and possess respective applicable ranges and limitations. Didactic social norm relies on the improvement of humanity and lacks a constraining force; mandatory regulatory governance or government intervention incurs high information costs, and over-intervention will harm individual freedom;

compared with the other two means, the incentive mechanism is the most effective. This is why economists focus closely on institutions.

Therefore, for the enaction of an institutional arrangement, irrespective of whether regulatory governance or incentive mechanism is present, the purpose is not to change the self-interested nature of individuals, but rather to make use of such immutable self-interestedness in order to guide individuals to engage in action that will objectively benefit society. In other words, mechanism design should follow the nature of individuals, but not attempt to change such nature. There is no good or evil in the discussion of individuals' self-interestedness, and the key lies in how to guide it with institutions. Different institutional arrangements will induce dissimilar incentive reactions and trade-off choices, and thus could produce markedly divergent results.

Any theory of economics involves economic institutional arrangements. Standard economics mainly focuses on the market system and studies how individuals make trade-off decisions in a market system (e.g., consumer theory, producer theory, and general equilibrium theory) and under what economic environments market equilibrium exists. It also makes value judgments on the results of resource allocation under different market structures (the criterion is based on whether resource allocation is efficient and equitable). In these investigations, the market system is normally assumed to be exogenously-given. In this way, it is possible to simplify the problem in order to focus on the study of individuals' economic behavior and how individuals make trade-off choices.

Of course, as the exogeneity assumption of institutional arrangements is not entirely reasonable in many cases, different economic institutional arrangements should be established, depending on particular economic environments and individuals' behavioral patterns. As will be discussed in Parts V-VII of the textbook, there will be market failures (i.e., inefficient allocation of resources and non-existence of market equilibrium) in many situations, and thus we will need to find an alternative or better economic mechanism. In that case, it is necessary to treat institutional arrangements as endogenously-determined by the economic environment and individual behavior. As a consequence, economists should provide a range of alternative economic institutional arrangements for various purposes.

When studying the economic behavior and choice issues of a specific economic organization, economic institutional arrangements should especially be endogenously-determined. New institutional economics, transition economics, theory of the firm, and in particular the economic mechanism design theory, information economics, optimal contract theory, auction theory, and matching theory that have developed in the last few decades, provide various economic institutional arrangements for a broad range, from the state to individuals, according to different economic environments and behavioral assumptions. Part VI presents detailed discussions on the issue of incentive design in economic institutional arrangements.

1.10.4 Choosing Equilibrium

The fourth basic component of the analytical framework of economics is to make trade-off choices and determine the "optimal" outcome. Given a certain economic environment, institutional arrangement (rules of the game), and specific constraints that must be adhered to, individuals will respond to incentives based on their own behavior, and assess and choose an outcome from available and feasible outcomes. Such an outcome is termed an **equilibrium**. This concept is not hard to understand. It simply means that individuals need to choose one among various feasible choices, the one chosen is called an equilibrium. Those who are self-interested will choose the optimal one for themselves; whereas, those who are altruistic may choose an outcome that is favorable to others. For this reason, equilibrium, which refers to a state without deviation incentives for all economic agents, is a static concept.

The equilibrium defined above may be the most general definition in economics. It embraces equilibria in textbooks that are reached by independent decisions under the drive of self-interested motivation and manifold technology or budget constraints. For instance, under the market system, for the producer, a profit-maximizing production plan under the constraint of production technology is called an equilibrium production plan; for the consumer, a utility-maximizing consumption bundle established under the budget constraint is called an equilibrium consumption bundle. When producers, consumers, and their interactions reach a state at which there is no incentive for deviations, a competitive market equilibrium for each commodity is obtained.

It should be noted that equilibrium is also a relative concept. The equilibrium outcome depends on economic environments, participants' behavior patterns (whether regarding rationality assumption, bounded rationality assumption, or other behavioral assumptions), and the rules of the game by which individuals react to incentives. Indeed, it constitutes the "optimal" choice relative to these factors. Moreover, due to bounded rationality, it may not be the optimal choice in objective reality, but is nevertheless the "optimal" one chosen by individuals according to their preferences and the information that they possess at the time.

1.10.5 Making Evaluations

The fifth basic component of the analytical framework of economics is making evaluations and value judgments on equilibrium outcomes and institutional arrangements. After making their choices, individuals usually hope to evaluate the equilibrium that then arises and compare it with the ideal outcome (such as efficient allocation, equitable allocation, incentive compatibility, informational efficiency), in order to make further assessments and value judgments on economic institutional arrangements. They also strive to determine whether or not the adopted economic institutional arrangement has led to a certain "optimal" outcome, and test whether the theoretical result is consistent with the empirical evidence, whether it can provide accurate predictions, and whether it is of pragmatic significance. Finally, they assess the economic institution and rules adopted to discern if there is room for improvement. Overall, in order to achieve better results, after completing the task, we should evaluate the effects, whether it is worth continuing, and whether there is a possibility for improvement. Therefore, it is necessary to make evaluations and value judgments on equilibrium outcomes under institutional arrangements and trade-off choices in order to identify the institutions that are best suited to the development of a nation.

One of the most important criteria used in evaluating an economic mechanism or institutional arrangement is whether the institutional arrangement is in accordance with the principle of efficiency. Certainly, as economic environments and individuals' behavioral patterns, as well as science and technology, continue to change, the precise Pareto optimality may never be fully realized. Just like Newton's three laws of motion, free fall, and fluid flow without friction in physics, Pareto optimality is an ideal state and provides the direction for improvement regarding economic efficiency. As long as the improvement of economic efficiency is desired, individuals will continually strive to approach this goal as closely as possible. With the ideal standard of Pareto optimality, we have a benchmark against which to compare, measure, and evaluate various economic institutional arrangements in the real world. Furthermore, it enables us to determine how far they are from this ideal goal for improvement of economic efficiency in order to continue to approach Pareto optimality.

Nonetheless, Pareto optimality is not the sole criterion for social value, equality or equity is also used. The market system achieves efficient allocation of resources, but it also faces numerous problems, such as social injustice resultant from an enormous wealth gap. There are a variety of definitions of equality, equity, and fairness. Equitable allocation, which will be introduced in Chapter 12, takes both objective equality and subjective factors into consideration, and more importantly, it can achieve equitable and efficient outcomes simultaneously. This is the basic conclusion of the Fairness Theorem, which will be discussed in Chapter 12. Another important criterion for evaluating an economic institutional arrangement is incentive compatibility.

In summary, the five components discussed above constitute the analytical framework underlying almost all standard economic theories and models, regardless of how much mathematics is used, or whether the institutional arrangement is exogenously-given or endogenously-determined. In the study of economic issues, one should first define the economic environment, and then examine how the self-interested behavior of individuals affects each other under exogenously-given or endogenously-determined mechanisms. Economists usually take "equilibrium", "efficiency", "information", "incentive compatibility" and "equity" as key aspects to observe the effects of different mechanisms on individual behavior and economic organizations, explain how individual behavior achieves equilibrium, and evaluate and compare the equilibrium. Using such a basic analytical framework in the analysis of economic issues is not only compatible in methodology, but may also lead to surprising (but logically consistent) conclusions.

1.11 Basic Research Methodologies in Economics

We have discussed the five components of the basic analytical framework of economics: (1) specifying economic environment; (2) making behavioral assumptions; (3) establishing institutional arrangements; (4) determining equilibrium; and (5) making evaluations. In general, any economic theory consists of these five aspects. Discussion of the five components naturally leads to the question of how to combine them appropriately, gradually deepen the study of various economic phenomena, and develop new economic theories. This is what we will discuss in this section: the basic research methodology and key points, which include setting up benchmarks, establishing reference systems, building studying platforms, developing analytical tools, constructing rigorous models, and conducting positive and normative analyses.

The research methodology of economics aims to firstly provide basic studying platforms for all levels and aspects, and then establish benchmarks and reference systems in order to present the criteria to evaluate equilibrium outcomes and institutional arrangements. Building a studying platform, setting benchmarks, and establishing reference systems are of great importance to the construction and development of any discipline, and economics is no exception to this.

1.11.1 Setting up a Benchmark

Evaluations or judgements can only be relative, and not absolute, and thus there should be a benchmark, which also applies to the discussion of economic issues. In economics, *benchmark refers to a relatively ideal state or simple economic environment*. As discussed in the first section of this chapter, to investigate a realistic economic issue and develop a new theory, we usually need to first consider it under a relatively ideal economic environment to develop a simple result or theory. Subsequently, we discuss the result in a non-ideal economic environment, which is closer to reality, develop a more general theory, and compare it with that developed under the benchmark situation.

In this sense, benchmarks are relative to non-ideal economic environments and new theories to be developed that are closer to reality. For instance, a complete information environment is the benchmark for the study of incomplete information. When investigating economic issues under private information, the situation of complete information must first be understood (even though it is highly unrealistic). Only when we are clear about the situation of complete information can we adeptly study economic issues taking place under the circumstance of private information. This is the case with theoretical research in economics. We start from the ideal state or simple scenarios prior to considering more realistic or general scenarios. In addition, we learn from others' research results before innovating the existing theories. In fact, new theories are always developed on the basis of prior research findings and results. An example of this is that Newton's mechanics makes Einstein's theory of relativity possible, while the theory of relativity makes it possible for Chen-Ning Yang and Tsung-Dao Lee to put forward the non-conservation of parity.

1.11.2 Establishing a Reference System

A reference system refers to economic theories or systems generated in an ideal situation, such as the general equilibrium theory, i.e., a perfectly competitive market will lead to efficient resource allocation. Establishing a reference system is of key importance to the construction and development of a new discipline, including economics. Although economic theories set as the reference system may include many assumptions that do not accord with reality, at least they can assist in the following: (1) simplifying the issue and capturing its characteristics directly; (2) establishing a comparative measurement criterion that is conductive to the assessment of the gap between the existing system and the ideal system, understanding the reality, and establishing the direction of improvement; and (3) studying how to reform or carry out further theoretical innovation, which can be used as a reference system for additional analysis.

Although an economic theory as a reference system might possess numerous unrealistic assumptions, they remain highly useful and can serve as a reference for further analysis. The importance of the reference system does not lie in whether or not it accurately describes the real world, but rather in establishing the measurement for a better understanding of reality. This is similar to our practice of setting role models in life. In addition, like a mirror, it assists to reveal the gap between theoretical models or realistic economic institutions and the ideal state. It is critically important in the sense that it identifies the direction of efforts and adjustments, and the extent of those adjustments. If a person has no goal and is unaware of the gap and approximate direction that should be followed, how can one make improvements or reach any goal?

General equilibrium theory is such a reference system. As we know, a perfectly competitive market will lead to efficient allocations of resources. Even if there is no such market in reality, if we make efforts in that direction, efficiency will be enhanced. This is why we have institutional arrangements, such as anti-trust laws, to protect market competition. By virtue of the reference system with perfect competition and complete information as the benchmark, we can study what outcomes can be obtained from economic institutional arrangements that are closer to reality (e.g., some monopolistic or transitional economic institutional arrangements) where assumptions in the general equilibrium theory are not valid (incomplete information, imperfect competition, externalities), and compare them with the results obtained from general equilibrium theory in the ideal state. In this way, we will know whether an economic institutional arrangement (be it theoretical or realistic) is efficient in allocating resources and using information, and how far away the economic institutional arrangement adopted in reality is from the ideal situation. Based on this, we can make appropriate policy suggestions or reform measures. In this sense, general equilibrium theory also serves as a reference theory that provides a criterion for evaluating institutional arrangements and the corresponding economic policies or pointing out the direction of reforms in practice.

Therefore, we should hold lofty ideals, which let us know our goals and the best direction to go. Even if we cannot get there eventually, by only learning from the best and comparing our performance with the best, we can do better and better, and continue to at least approach the ideal.

1.11.3 Building Studying Platforms

A studying platform in economics consists of certain basic economic theories, models or methods, which provide the basis for deeper analysis. The methodology of economics is very similar to that of physics, i.e., simplifying the issue first to capture the core essence of the issue. In cases in which many factors produce an economic phenomenon, the impact of every factor must be elucidated. This can be achieved by investigating the effect of one factor at a time, while holding all other factors constant. The theoretical foundation of economics is microeconomics, and the most fundamental theory in microeconomics is individual choice theory; individual choice theory further comprises consumer theory and producer theory, which are the basic studying platform and cornerstone of economics. This is why almost all economics textbooks start from the discussion of consumer theory and firm theory. They provide the fundamental theories that explain how individuals make choices as consumers and firms, and establish the studying platform for further study of individual choice.

In general, the equilibrium choice of an individual depends not only on one's own choice, but also on others' choices. In order to investigate individual choice, it is necessary to determine what are the most important factors in an individual's decision making in the absence of influence by other agents. Consumer theory and producer theory are developed through this approach. It is assumed that economic agents are in the institutional arrangement of a perfectly competitive market. Therefore, every agent will take price as given, and individual choice will not be influenced by others' choices. The optimal choice is determined by subjective factors (e.g., the pursuit of utility or profit maximization) and objective factors (e.g., budget line or production constraints).

This methodology that involves simplification and idealization of the question establishes a basic platform for deepening research. It is like the approach in physics: in order to study a problem, we examine the essence first, start with the simplest situation that excludes frictions from consideration, and then gradually deepen the research and consider more general and complicated cases. In microeconomics, theories of market structures, such as monopolies, oligopolies and monopolistic competition, are generated from more general cases, in which producers can influence each other. To study the choice issue under the more general situation in which economic agents can influence each other's decision-making, economists have developed a very powerful analytical tool: game theory.

General equilibrium theory is a more sophisticated studying platform based on consumer theory and producer theory. Consumer theory and producer theory provide a fundamental platform for investigating individual choice problems; whereas, general equilibrium theory provides a core platform for analyzing how to reach market equilibrium through interactions of individuals and all markets. Mechanism design theory provides an even higher-level platform for studying, designing, and comparing various institutional arrangements and economic mechanisms. It can be used to study and prove the optimality and uniqueness of perfectly competitive market mechanisms in resource allocation and information requirement, and more importantly, in the case of market failure, it offers methods of how to design alternative mechanisms. Under some regularity conditions, the institutional arrangement of a perfectly competitive market will not only lead to efficient allocation of resources, but also prove the most efficient concerning information requirements (mechanism operation cost, transaction cost), as it requires the least amount of information. Under other circumstances of market failure, it is necessary to design a variety of alternative mechanisms for different economic environments. Furthermore, the studying platform also creates conditions for providing reference systems for evaluating various kinds of economic institutional arrangements. In other words, it provides a criterion for measuring the gap between reality and the ideal state.

1.11.4 Developing Analytical Tools

For the research of economic phenomena and economic behavior, we also need various analytical tools besides the analytical framework, benchmark, reference system, and studying platform. economics necessitates not only qualitative analysis, but also quantitative analysis, to find the boundary conditions for each theory to be true, so that the theory will not be misapplied. Consequently, a series of powerful analytical tools should be provided, which are usually given as mathematical models or diagrams. The power of these tools lies in their ability to support us to deeply analyze intricate economic phenomena and economic behavior through a simple and clear diagram or mathematical structure. Examples of this include the demand-supply curve, game theory, principal-agent theory for investigating information asymmetry, Paul A. Samuelson's (1915-2009, see Section 3.11.2 for his biography) overlapping generations model, dynamic optimality theory, etc. Of course, exceptions exist which are not expressed with analytical tools. For instance, Coase's theory is established and demonstrated through words and basic logical deduction only.

1.11.5 Constructing Rigorous Analytical Models

Logical and rigorous theoretical analysis is needed when we explain economic phenomena or economic behavior, and make conclusions or economic inferences. As mentioned above, each theory holds true under certain conditions. We need to establish rigorous analytical models that clearly identify the conditions under which a theory holds true. A lack of related mathematical knowledge will make it difficult to achieve an accurate understanding of the connotations of a definition, or to have discussions on related issues, as well as defining boundary conditions or constraints. Therefore, it is not surprising that mathematics and mathematical statistics are used as basic analytical tools; indeed, they are also among the most important research methods in economics.

1.11.6 Making Positive and Normative Analysis

According to the research method, economic analysis can be divided into two categories: positive or descriptive analysis; and normative or value judgement analysis. Another major difference between economics and natural science is that the latter essentially only performs positive analysis, while the former involves both positive and normative analysis.

Positive analysis explains how an economy operates. It only gives facts and provides explanations (thus verifiable), but does not make value judgements about economic phenomena or offer means of revision. For example, an important task of economics is to describe, compare, and analyze such phenomena as production, consumption, unemployment and price, and to predict possible outcomes of different policies. Consumer theory, producer theory, and game theory are typical examples of the positive approach.

Normative analysis, on the other hand, makes value judgments on economic phenomena. It not only explains how an economy operates, but also attempts to identify the means of revision. As a consequence, it always involves the value judgments and opinions of the economists, and is thus not verifiable through facts. For instance, certain economists may place more emphasis on economic efficiency, while others may focus on equality or social justice. If we are cognizant of the differences between the two methods, many disputes can be avoided when discussing economic issues. Economic mechanism design theory is a typical example of the normative approach.

Positive analysis is the foundation for normative analysis, while normative analysis is the extension of positive analysis. In this sense, the foremost task of economics is to make positive analysis, and then normative analysis follows. General equilibrium theory includes both the former (e.g., the existence, stability, and uniqueness of competitive equilibrium) and the latter (the First and Second Fundamental Theorems of Welfare Economics).

1.12 Practical Role of the Analytical Framework and Methodologies

The most basic analytical framework and research methods of economics that have been discussed are of practical utility. Even though these analytical frameworks and methodologies may appear simple, it is not easy to comprehend and use them in our lives, studies, and research. However, once the idea is mastered, lifelong benefits will ensue. For example, one will be able to think scientifically. It will also be possible to investigate esoteric pure economic theories, as well as come up with viable solutions to solve practical issues in one's life and work.

First of all, from the perspective of studying economics, if the analytical framework and methodologies are mastered, abstract models and intricate mathematics will not be confusing. This is because, irrespective of the profundity of the mathematics used, how many formulae are employed, and how complicated are the economic models that are used in an economic theory, it still uses the above analytical framework and methodologies. If the basic analytical framework and methodologies are grasped and employed as the main paradigm, you will be able to maintain your focus and understand its general idea. Therefore, incomprehensible technical details can be temporarily put aside, and the framework and conclusion understood first; only then can we fully comprehend the details. In other words, it is necessary to first grasp the primary point and general idea, know its objectives, line of thinking as well as conclusions, and then consider the details. Moreover, once the basic analytical framework and methodologies are mastered, one possesses the correct idea of economics and cannot be misled. This, of course, can have a markedly positive effect on your study of economics. The reason why some people criticize economics and its methodologies is that their judgements are mostly not based on methods of scientific analysis, and may even rely only on subjective assumptions. If the basic analytical framework and methodologies are not understood, it is possible that their judgments will make you confused, lose direction when studying economics, and/or resist or even disregard the study of economics.

Second, regarding economic research, once the basic analytical framework and methodologies are understood, it will be easier to carry out research. It is the case that many people who strive to do economic research, even though they understand economics quite well and have read a large number of related papers, they still find it difficult to conduct research. Either they do not know how to do research, or their research findings are not significant or widely recognized. In fact, research would become relatively easier, as long as the basic analytical framework and methodologies are understood, and one possesses basic mathematic knowledge and the ability to carry out logical analysis. Conducting research is, to a certain extent, step-by-step writing with inherent logic according to the five components of the basic analytical framework. The basic analytical framework and methodologies can greatly assist you to improve your research and innovative abilities.

For example, if you are interested in a specific economic issue or phenomenon, or you want to put forward a new theory with stronger explanatory power to guide the resolution of a pragmatic economic issue, it is necessary to reasonably and precisely describe the economic environment and economic agent's behavior, employ existing analytical tools or develop new ones to build as simple a model as possible, and then perform the deduction and proof. On the other hand, if you only intend to extend and improve the original theory, it is necessary to determine whether the original assumptions about the economic environment, behavior, and models fit the reality and whether or not the assumptions can be relaxed to derive novel or more general conclusions or even pathbreaking theories. In this case, it might be easier to carry out the work of extensions and improvements. Of course, it is also possible to modify the specification of economic environments or other components in order to reach a different, or perhaps more important, conclusion. This is how the school of macroeconomics and numerous theories under information asymmetry were developed. If an economic theory is to be criticized, one should criticize which parts of its analytical framework are unreasonable, illogical, or unrealistic, and in what way, rather than censuring economics and its methodologies in aggregate.

Finally, understanding economics, its methodologies, and analytical framework may assist us to have a greatly improved thinking process, and deal with everyday concerns and other individuals in a much better manner. Indeed, it can make you more thoughtful, insightful, and capable in your work. It is frequently stated that economics is esoteric and metaphysical since it involves so much difficult mathematics that seems remote from practice. Some people wonder “What will it be used for?” In fact, the basic approach of dealing with others and events in our daily lives is similar to the basic framework in economic analysis. For example, when one is in a new place preparing to perform a task or cooperate with others, the first thing one needs to do is to become familiar with the local environment and situation, which is similar to “specifying the economic environment” in the framework. It is then necessary to know the local culture and customs, the behavioral patterns and personalities of your counterparts, etc., which is similar to “making behavioral assumptions”. Subsequently, by taking all of the information together, one can decide on one’s rules for dealing with people, which is similar to “establishing economic institutional arrangements”. The next step is to choose the optimal scheme by making trading-offs among feasible options, which is similar to the “determining equilibrium”. The final step is to summarize and reflect on your decisions, actions, and your ways of dealing with people, circumstances, and events to assess whether they constitute the most effective approaches, whether they achieve the best outcomes, whether they are fair and reasonable, whether individuals’ enthusiasm is engendered, whether people have reactions to the incentive, and whether the intended goal of incentive compatibility is achieved, which is similar to “making evaluations”. In addition, when the environment and conditions are changed, or the subject with which you are working is altered, the rules should be changed accordingly. If you can act in accordance with the five aspects, and adjust the rules with changes in conditions, better results will certainly be produced. Not only may this be one of the best ways to deal with daily life and work, certain conclusions of economic theories can also assist you to better think about and solve problems.

1.13 Requirements for Learning Economics

There are three basic requirements for learning and mastering economic theory:

1. The first requirement is to master the basic concepts and definitions, which is a reflection of logical thinking and a clear mind. This is the prerequisite not only for the discussion and analysis of questions and logically inherent analysis, but also for a strong command of economics. Otherwise, different definitions of terms may produce substantial confusion and lead to unnecessary controversies.

2. One should be able to clearly state all theorems or propositions, and also be definitive about the basic conclusions and their conditions. Otherwise, even a small misunderstanding about an economic theory in application to the analysis of issues may lead to serious problems. Since any theory or institution has its proper scope of application, if we go beyond that, problems are liable to arise, bringing about tremendous negative externalities. In many cases, social or economic problems occur solely because economists misapply some theories without a solid understanding of their applicable conditions. In this sense, a good economist is similar to a qualified physician who needs to master the properties and efficacy of different kinds of medicines when prescribing for his or her patients.

3. One must also grasp how the basic theorems or propositions are proven (ideas and processes). A good economist, like a good physician, should know what the problem is and why it is present, understand its pathology, as well as be able to determine appropriate medicines. Then, one can gain a deeper understanding and a better command of the theories that one has learned.

If these requirements are met, it would be easy to refresh your memory, even if the proofs of some conclusions are forgotten. Economics relies primarily upon the analysis of inherent logic, which also constitutes the power of economic theory. This is why it is so crucial to accurately grasp the theories and their scopes of application.

1.14 Distinguishing Sufficient and Necessary Conditions

When discussing economic issues, it is also important to distinguish between sufficient conditions and necessary conditions, which can assist people to think clearly and avoid superfluous debates. A necessary condition is a condition that is indispensable in order for an assessment to be true. A sufficient condition, on the other hand, is a condition that guarantees that the assessment is true. For instance, some people often negate the market economy based on examples of severely underdeveloped countries, which adopt a market economy but remain poor. From this, they contend that a nation should not embark on the path of a market economy. These people, however, do not realize the difference between sufficient and necessary conditions: the adoption of a market economy is a necessary condition, rather than a sufficient condition, for a nation to become developed and prosperous. In other words, if a country aims to be prosperous, it must adopt a market economy. Indeed, it is the case that, without exception, all wealthy nations in the world are market economies. One must also admit, however, that the market mechanism does not necessarily lead to national prosperity. To be sufficient, other supplementary systems are needed, such as the rule of law, strong state capacity, a suitable political system, etc.

Indeed, as discussed previously, a distinction exists between a good market economy and a bad market economy. The reason for this is that, although (based on observations of reality) the market mechanism is critical for national prosperity, many other factors also influence the prosperity of a nation, such as the degree of government intervention, the political system, the legal framework, religions, cultures and social structures, all of which contribute to the labelling of the market mechanism as either good or bad.

1.15 The Role of Mathematics and Statistics in Economics

Mathematics and statistics are of extreme significance for people to possess a thorough knowledge of nature and to successfully manage their daily affairs. As the well-known statistician, C. Radhakrishna Rao, pointed out, mathematics is a type of logic employed to deduce results on a given premise; whereas, statistics is a rational method acquired through experience and a kind of logic used to verify the premise with a given result. Rao believes:

> All knowledge is, in final analysis, history. All sciences are, in the abstract sense, mathematics. All judgements are, in their rationale, statistics.[7]

This assertion profoundly and comprehensively depicts the significance of mathematics and statistics, and their respective connotations.

Mathematics and statistics are also essential in economics. Almost every branch in economics uses mathematics, statistics, and econometrics to a greater or lesser extent. The reasons for this include the following: economics is increasingly becoming a science; mathematical analytical tools

[7] C. R. Rao, *Statistics and Truth: Putting Chance to Work*, World Scientific, Singapore, 2nd edition, 1997.

are being increasingly used; and social systems are becoming increasingly complex and influential. Therefore, when investigating economic problems, it is necessary to have a rigorous theoretical model for inherently logical analysis, and determine the range for a theoretical conclusion to be true; then, empirical tests for the given results must be performed by means of econometric analysis. As a consequence, it is not surprising that mathematics and mathematical statistics are used as basic analytical tools, and have become the most important analytical tools in economics. Those who study and conduct research on economics must possess a thorough knowledge of both mathematics and mathematical statistics.

Modern economics mainly adopts the mathematical language to make assumptions about economic environments and individual behavior patterns, uses mathematical expressions to illustrate logical relations between economic variables and economic rules, constructs mathematical models to study economic issues, and finally follows the logic of mathematical language to deduce conclusions. Without related mathematical knowledge, it is difficult to grasp the essence of concepts and discuss related issues, and certainly conduct research and determine necessary boundaries or constraints when reaching conclusions. Consequently, it is necessary to master sufficient mathematical knowledge if one intends to learn economics well, engage in research on economics, and become a good economist.

People with little knowledge of mathematics are unlikely to be able to master the basic theories and analytical tools of economics or understand advanced economic textbooks or papers. In fact, they may use certain excuses, such as it is more important to produce economic thoughts, or mathematics is not generally needed to approach practical economic issues. Of course, no one could refute the importance of economic thoughts since they constitute the output of research. Without the tools of mathematics, however, how could the boundary conditions and the applicable scope of economic thoughts or conclusions be identified? Without knowledge of conditions and scope, how could we protect against the misapplication of economic thoughts or conclusions? Moreover, what are the odds for us to develop such profound economic thoughts without using mathematical models, as Adam Smith and Ronald H. Coase did? Even so, economists have never ceased investigating what conditions are required for their conclusions to hold. Without strict arguments, the thoughts or conclusions could not be widely acknowledged. As mentioned above, the economic thoughts presented by philosophers in ancient China, such as Jiang Shang, Lao Tzu, Sun Tzu, Guan Zhong and Sima Qian, are extremely profound, and some of the ideas of Adam Smith had already been put forward by these philosophers much earlier. However, their thoughts have never been known to the outside world because they were just observations or conclusions of experience that did not form a scientific system or involve logically inherent analyses using scientific methods.

There is another misunderstanding that research of economic issues with mathematics is remote from reality. In fact, however, most mathematical knowledge is developed on the basis of practical demands. People who possess basic knowledge of physics, physical science history, or history of mathematical thought will know that both primary and advanced mathematics originated from the demands of scientific development and reality. As such, why cannot mathematics be used to investigate practical economic issues? The foundation of mathematics and economics is absolutely essential for one to be a good economist. If one understands mathematics well and masters fundamental analytical frameworks and research methodologies of economics, economics can be learned more easily and study efficiency will be markedly improved.

The primary functions of mathematics in the theoretical analysis of economics are as follows: (1) it makes the language more precise and the statement of assumptions clearer, which can reduce superfluous disputes resulting from ambiguous definitions. (2) It makes the analytical logic more rigorous and makes it possible to definitively state the boundary, application scope and conditions for a conclusion to hold true, and accurately identify the direction of theoretical innovation beyond the limitations of the original theory. Otherwise, the abuse of a theory may occur. For example, when discussing the issue of property rights, some people may quote the Coase theorem, assuming that as long as the transaction cost is zero, there will be efficient allocation of resources. Surprisingly, many people still exist (including Coase himself when providing his assessments) who do not know that this conclusion is normally false if the utility (payoff) function is not quasi-linear. (3) Mathematics can assist to obtain results that cannot be easily attained through intuition. For example, from intuition, according to the laws of supply and demand, competitive markets will achieve market equilibrium through the adjustment of market prices according to the "invisible hand" as long as the supply and the demand are not equal. Yet, this conclusion does not always hold. For example, Scarf (1960) gave counterexamples of market instability. (4) It helps to improve and extend existing economic theory. Examples of this are manifold in the study of economic theory. For instance, economic mechanism design theory constitutes an improvement and extension of general equilibrium theory.

Qualitative theoretical analysis and quantitative empirical analysis are both requisite for studying economic problems. Statistics and econometrics play a key role in these analyses. Statistics focuses more on data collection, description, sorting and providing statistical methods; econometrics identifies economic structures through economic theory, and focuses more on testing economic theory, evaluating economic policy, making economic forecasts, and identifying causal relationships between economic variables; big data analysis can process data faster, better and more efficiently, and use these data to discover new opportunities, provide new products and services, and has great cost advantages. In order to better estimate economic models and reach more accurate predictions, theoretical econometricians and statisticians have continually developed increasingly powerful econometric tools and big data analytic tools.

It is, however, worth noting that economics is not mathematics. Mathematics in economics is used as a tool to elucidate economic problems. Economists employ mathematics to express their opinions and theories more rigorously, and to assess interdependent relationships among economic variables. With the metrication of economics and the precision of various assumptions, economics has become a veritable social science with a rigorous system.

However, a person with a strong knowledge of mathematics does not necessarily become a good economist. This also requires a deep understanding of the analytical frameworks and research methodologies of economics, and acute intuitions of actual economic environments and economic issues. The study of economics not only calls for the understanding of terms, concepts and results from the perspective of mathematics (including geometry), but more importantly, one must strive vigorously to grasp economic meanings and underlying economic thoughts. This is, of course, also the case when those are given by mathematical language and/or geometric figures. As a consequence, confusion must be avoided regarding mathematical formulas or symbols in the study of economics. For all of these reasons, we often state that, in order to become a good economist, one needs to pursue academics with a good mastery of both scientific rigor and underlying profound thoughts.

1.16 Conversion between Economic and Mathematical Languages

The product of economic research is economic inferences and conclusions. A standard economics paper usually consists of three parts: (1) it raises questions, states the significance, and identifies the research objective; (2) it establishes economic models and rigorously expresses and proves the inferences; and (3) it uses non-technical language to explain the conclusions and provides policy suggestions. In other words, an economic conclusion is usually obtained through the following three stages: non-mathematical language stage $\rightarrow$ mathematical language stage $\rightarrow$ non-mathematical language stage. The first stage proposes economic ideas, concepts or conjectures, which may stem from economic intuition or historical and foreign experience. As they have not yet been proven by theories, they can be regarded as primary products of general production. The first stage is critical because it is the origin of theoretical research and creation.

The second stage verifies whether or not the proposed economic ideas or conjectures hold true. The verification requires economists to give formal and rigorous proofs through axiomatic analysis by introducing economic models and analytical tools, and if possible, to test them with empirical data. The conclusions and inferences obtained are usually expressed in mathematical language or technical terms, which may not be understandable to non-experts. Therefore, they may not be adopted by the public, government administrations, or policy-makers. For this reason, these conclusions expressed by technical language can be regarded as intermediate products of general production.

Economic studies should serve the real economic world. Therefore, the third stage aims to express the conclusions and inferences in non-expert language rather than expert language, making them more decipherable to the general public. Policy implications and profound meanings of conclusions and insightful inferences conveyed through non-expert language constitute the final products of economics. It is notable that, in both the first and the third stages, economic ideas and conclusions are presented in common, non-technical and non-mathematical language; whereas, the third phase is a kind of enhancement of the first phase. In fact, the three-stage form of common language - technical language - common language is a normal research method that is widely adopted by numerous disciplines.

1.17 Biographies

1.17.1 Adam Smith

Adam Smith (1723-1790) is well-known as the father of economics. Adam Smith finished his study of Latin, Greek, mathematics, and ethics at the University of Glasgow in Britain. Subsequently, he worked at the University of Glasgow as a Professor in Logics and Moral Philosophy, and held the honorary position of Lord Rector. *The Wealth of Nations*, published in 1776, is Smith's most influential work and also a great contribution to the establishment of economics as an independent discipline. This book is widely considered to be the most influential work among all publications in the field of economics. His main academic thinking was strongly influenced by Bernard de Mandeville (1670-1731), Francis Hutcheson (1694-1764), David Hume (1711-1776), J. Vanderlint (year of birth unknown, died in 1740), George Berkeley (1685-1753), and others.

Smith suggested that the economic development of human society was the outcome of the spontaneous actions of tens of millions of individuals whose behavior followed the power of instinct and was driven by their self-interested nature. Smith regarded this power as the "invisible hand", which is his idea of allowing the rule of the market to play the decisive role in the organization of economic society. *The Wealth of Nations* denied the attention to land in physiocracy, but valued labor as the most important aspect and believed that the division of labor could increase the efficiency of production.

Thomas Robert Malthus and David Ricardo focused on summarizing Smith's theories into a theory known as the classical economics in the twentieth century (where economics thus originated). Malthus further extended Smith's theories to the problem of surplus of population. Ricardo, on the other hand, put forward the iron law of wages, suggesting that the surplus of population may lead to the consequence that workers' livelihood cannot be guaranteed. Smith assumed that the increase in wages would accompany the increase in production, which seems more correct from today's perspective. Theories involved in his book not only set up the division of labor theory, but also pioneered in certain areas, including monetary theory, theory of value, theory of distribution, capital accumulation theory, theory of taxation, etc. In addition, Marx's labor theory of value, which was built upon the basis of Ricardo's political economy, also received indirect influence from Smith's theory.

Before the foundational work, *The Wealth of Nations*, Smith wrote *The Theory of Moral Sentiments* (first published in 1759). In this book, he mainly argued that people should have sympathy and a sense of justice, which may occur particularly under unusual circumstances (for example, when people are suffering or when a nation is invaded). He strove diligently to prove how self-interested individuals controlled their emotions and behavior, and especially their selfish sentiment and behavior, so that incentive compatibility between social interest and self-interest could be achieved. Indeed, the system of economic theory built by Smith in *The Wealth of Nations* is based on his arguments in *The Theory of Moral Sentiments*.

Smith worked on the two works simultaneously, and revised them repeatedly until his death. They became two organic and complementary components in his academic thinking system. *The Theory of Moral Sentiments* states the problem of morality, while *The Wealth of Nations* states the problem of economic development. Smith regarded *The Wealth of Nations* as a continuation of his thinking in *The Theory of Moral Sentiments*. The two books, however, differ in mood, scope of discussion, structural arrangement, and emphasis. For example, the control for self-interested behavior in *The Theory of Moral Sentiments* relied on sympathy and a sense of justice; whereas, in *The Wealth of Nations*, it relied on the competitive mechanism. Nonetheless, the discussions about the motivation of self-interested behavior were essentially identical. It can be suggested that individuals' sympathy, sense of justice, and pursuit of self-interest just constitute different reactions to different circumstances (unusual or usual). In *The Theory of Moral Sentiments*, Smith regarded "sympathy" as the core of judgement, but when it is viewed as the motivation of an individual's behavior, a completely different outcome will result.

1.17.2 David Ricardo

David Ricardo (1772-1823), a representative of Classical Political Economy, together with Thomas Robert Malthus (1766-1834), integrated Smith's theory into classical economics. Ricardo was born to a Jewish family, and his father was a stock broker. He attended a business college in the Netherlands at 12-years-old and worked on a stock exchange with his father at 14. He was engaged in stock exchange work independently in 1793, and by the time he was 25, he had amassed a wealth of two million pounds. After that, he began to study mathematics and physics. When he first read Adam Smith's *The Wealth of Nations* in 1799, he began to study economic issues. At the age of 37, he published his first paper in economics and did very well in this field. During the 14 years of his short academic life, he left numerous works, papers, notes, letters, and speeches. Among those, the most famous one is the *Principles of Political Economy and Taxation* published in 1817. It has been claimed that Ricardo was somewhat conceited. He stated that his point of view was different from Smith and Malthus, who enjoyed great prestige then and, in Britain, there would be less than 25 people who could fully comprehend his book. In 1819, Ricardo was elected as a member of parliament.

Starting with Bentham's version of utilitarianism, Ricardo established a theoretical system based on the labor theory of value and centered on the theory of distribution. He insisted in the principle which stated that the value of a commodity was determined by the labor cost contained in it. He also criticized Smith's theory of value by contending that labor that could decide value constituted socially necessary labor. In addition, labor that could decide the value of a commodity was not only direct living labor, but also labor in factors of production. Ricardo suggested that all value was produced by labor and was distributed among three classes: wages that were decided by the value of the essential means of subsistence of workers; profit was the surplus when wages were deducted; and land rent was the surplus when wages and profit were deducted.

Based on the labor theory of value, Ricardo established the theory of comparative advantage. In *On the Principles of Political Economy and Taxation*, he clearly stated that "The value of a commodity, or the quantity of any other commodity for which it will exchange, depends on the relative quantity of labour which is necessary for its production". He further asserted that "the exchangeable value of these commodities, or the rule which determines how much of one shall be given in exchange for another, depends almost exclusively on the comparative quantity of labour expended on each". The profit of each party in international trade is also fully correlated with the exchangeable value of all commodities in the international market, i.e., the relative price level.

Ricardo regarded the free flow of factors of production, such as capital and labor, among regions and industries within a country as the fundamental reason of equalized rate of profit. The flow of factors between nations, however, would inevitably be interrupted by force or even totally stopped due to various reasons. Ricardo concluded that it was the immobility of factors between countries that decided "the same rule which regulates the relative value of commodities in one country, does not regulate the relative value of the commodities exchanged between two or more countries". Since there are numerous reasons for different relative values of one commodity in different countries, there is room for profit for all of the participating countries in international trade. Its main premise, however, is that each country is cognizant of its advantages compared with others, i.e., they are certain about their own comparative advantages.

1.18 Exercises

Exercise 1.1 (Economics and the three dimensions of scientific economic analysis) Answer the following questions:

1. What is the definition of economics?
2. What are the two great objective realities facing the study of economic problems?
3. What is modern economics?
4. What are the "three dimensions and six natures"?
5. Why does scientific economic analysis need the "three dimensions and six natures"?

Exercise 1.2 (Differences between economics and natural science) Answer the following questions:

1. What are the main differences between economics and natural science?
2. Why do these differences make economic research more complex and difficult?

Exercise 1.3 (Two basic categories of economic theory) Answer the following questions:

1. Which two categories of economic theories can be classified according to their functions?
2. Describe the connotation and function of each category, as well as the relationship between these two categories.

3. How should we correctly regard and deal with the interaction between these two kinds of economic theory?

Exercise 1.4 (The fundamental functions of economic theory) Answer the following questions:

1. What are the three main functions of economic theory?

2. Why is there not a best kind of economic theory that is always right and fits every development stage, but instead a kind that fits certain institutional environments the best?

3. What are two common misunderstandings of economic theories?

Exercise 1.5 (Market and market mechanism) Answer the following questions:

1. From the perspectives of information and incentive, what are the advantages of the market economic system compared with the planned economic system?

2. Under the condition of the market economy, what are the three basic functions of price?

3. What is the superiority of the market system?

4. What are the three development stages that an economy will experience? How can efficiency-driven and further innovation-driven development be realized? What is the basic economic institution behind this?

Exercise 1.6 (The dialectical relationship between competition and monopoly) Answer the following questions:

1. Why do people want to introduce the competitive mechanism in the view of social resource allocation, while enterprises desire monopolies? Please also state the dialectical relationship between competition and monopoly.

2. What does the *Innovation Theory* of Schumpeter tell us? Please state the importance of innovation-driven development.

3. Why do innovation and entrepreneurship depend on institutional choice, and therefore constitute endogenous variables?

Exercise 1.7 (The boundaries among the government, the market, and the society) Answer the following questions:

1. Why is it necessary to reasonably define the boundaries between the government and the market and between the government and the society?

2. How should the boundaries between the government and the market and between the government and the society be generally defined?

3. Why does a well-governed nation need to reasonably define not only the boundaries between the government and the market, but also those between the government and the society?

Exercise 1.8 (The three basic institutional arrangements for state governance and benign de Answer the following questions:

1. What are the three elements of state governance and benign development?

2. What are the three basic institutional arrangements for state governance?

3. State the range of application and limitation of each of the three basic institutional arrangements. Which one is the most basic and important?

Exercise 1.9 (The logic of development and governance) Answer the following questions:

1. How shall we correctly understand the logic of development and governance and the dialectical relationship between the two?

2. Explain the achievements and limitations of economic reform in China according to this framework.

Exercise 1.10 (Ancient Chinese Economic Thought) Answer the following questions:

1. Provide five examples that indicate the thought of the market economy in ancient China.

2. Why could not those numerous deep economic thoughts in ancient China form a scientific economic theory?

Exercise 1.11 (The cornerstone assumptions of economics) Answer the following questions:

1. State the relationship and distinction between self-love, selfishness, and self-interest.

2. Why does economics use the self-interest assumption as the most basic, important, and central assumption?

3. How shall we regard self-interestedness and altruism?

Exercise 1.12 (Key points in economics) Answer the following questions:

1. What are the key points of economics? Please state each of them generally.

2. State the meanings, advantages, and limitations of centralized and decentralized decision-making.

3. Why are economic freedom and competition crucial to economic development?

4. Why is acting under constraints one of the most fundamental principles in economics?

5. What is the relationship between incentive and information?

6. Why are clearly defined property rights crucial to the efficient allocation of resources?

7. Discuss the differences between equity in outcome and equality of opportunity. Which one does not conflict with efficiency?

Exercise 1.13 (Proper understanding of economics) Answer the following questions:

1. How shall we regard the scientific nature of economics?

2. How shall we regard the mathematical nature of economics?

3. How shall we regard economic theory correctly?

4. How shall we regard the critics that contend that economics cannot be tested through experimentation?

Exercise 1.14 (Basic analytical framework of (modern) economics) Answer the following questions:

1. What are the components that constitute the basic analytical framework of a standard modern economic theory?

2. Why do different economic environments need different economic theories?

3. Why are different economic theories needed even for the same economic reality or environment under many circumstances?

4. Why should evaluation be included in the analytical framework?

5. Taking Coase theorem as an example, expound on its economic analytical framework.

6. What are practical usages of the basic analytical framework and research methodologies of economics?

Exercise 1.15 (Benchmark and reference system) Answer the following questions:

1. What are the definitions of benchmark and reference system?

2. Why are setting-up benchmarks and establishing reference systems the premise of discussing economic problems?

3. What are typical examples of reference systems?

Exercise 1.16 (Methodologies) Answer the following questions:

1. When considering the problem of economic reform, why is it important to distinguish necessary conditions from sufficient conditions?

2. Why are both positive and normative analysis needed when discussing economic problems?

3. How shall we regard the role of mathematics and statistics in economics?

4. How shall we complete the conversion between economic and mathematical languages?

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Chapter 2

Preliminary Knowledge and Methods of Mathematics

2.10 Basic Probability

Risk and uncertainty, as well as some of their basic operations, are broadly used in economics. This section briefly introduces knowledge involved in the textbook.

2.10.1 Probability and Conditional Probability

Compared with other fields of mathematics, the development of probability theory occurred relatively late. However, probability theory has developed rapidly and become a very important field in mathematics since its axiomatization.

When dealing with probabilities, we must clearly define the probability space. Classical probability (i.e., explaining probability as the same possibility) is often associated with permutation and combination. Since statistics requires obtaining data from sampling, the randomness in probability theory is revealed.

Let the probability of random variable X_a taking on value X_{as} be π_s, $s \in S$, where S can be either discrete or continuous. When $S = \{1, \cdots, n\}$, where n can be finite or infinite, this situation is about a discrete random variable. If S is an interval of the real space, then it is called a continuous random variable.

If there is a correlation between two random variables, then the value of a random variable provides information for the value of the other random variable, which gives the concept of conditional probability.

When the two random variables X_a and X_b taking on X_{as} and X_{bs}, the joint probability distribution $\pi_{ss'}$ can be expressed as

$$\begin{aligned}\pi_{ss'} &\equiv P(X_{bs}) \times P(X_{as'}|X_{bs}) \\ &\equiv P(X_{as}) \times P(X_{bs'}|X_{as}),\end{aligned}$$

where $P(X_{as}) = \sum_{t' \in S} \pi_{st'}$ and $P(X_{bs}) = \sum_{t \in S} \pi_{ts}$. Therefore, given X_{as}, the conditional probability of $X_{bs'}$ is

$$\begin{aligned}P(X_{bs'}|X_{as}) &\equiv \frac{\pi_{ss'}}{\sum_{t' \in S} \pi_{st'}} \\ &\equiv \frac{P(X_{bs}) \times P(X_{as'}|X_{bs})}{P(X_{as})} \\ &\equiv \frac{P(X_{bs}) \times P(X_{as'}|X_{bs})}{\sum_{t' \in S} P(X_{bt'}) \times P(X_{as'}|X_{bs})}.\end{aligned}$$

This formula is also called the Bayes rule, which is widely used in economics, especially in studying dynamic games of incomplete information and mechanism design.

2.10.2 Mathematical Expectation and Variance

The **(mathematical) expectation** of random variable X_a is the weighted average of all possible values, and is defined and denoted by

$$E(X_a) \equiv \bar{X}_a = \sum_{s \in S} \pi_s X_{as},$$

which in a continuous case is defined by integral instead of summation, and it will be discussed in the next subsection.

The operation rule of expected utility is that if X_a and X_b are two random variables, then we have

$$E(aX_a + bX_b) = a\bar{X}_a + b\bar{X}_b.$$

The **variance** of a random variable X_a measuring the degree of variation of its value is defined as

$$\mathrm{Var}(X_a) \equiv \sigma^2_{X_a} = \sum_{s \in S} \pi_s (X_{as} - \bar{X}_a)^2.$$

Therefore, the larger is the variance, the greater is the variation degree.

There may be some correlations between the two random variables, X_a and X_b. Suppose that the value space of X_a is $\{X_{as}\}_{s \in S}$, and that of X_b is $\{X_{bs'}\}_{s' \in S'}$. Then, their **covariance** measures the correlations between their values.

Let $\pi_{ss'}$ be the probability of $X_a = X_{as}$ and $X_b = X_{bs'}$. **Covariance**, denoted by $\mathrm{Cov}(X_a, X_b)$, is defined as:

$$\mathrm{Cov}(X_a, X_b) = \sum_{s \in S, s' \in S'} \pi_{ss'}(X_{as} - \bar{X}_a)(X_{bs'} - \bar{X}_b)$$

or

$$\mathrm{Cov}(X_a, X_b) = E(X_a - \bar{X}_a)(X_b - \bar{X}_b) = E(X_a X_b) - E(X_a)E(X_b).$$

If two random variables X_a and X_b are independent, then $\pi_{ss'} = \pi_s \pi_{s'}$, and thus we have $\mathrm{Cov}(X_a, X_b) = 0$.

The following operation is for deriving the variance of linear combinations:

$$\mathrm{Var}\left(\sum_{a \in A} \alpha_a X_a\right) = \sum_{a \in A, b \in A} \alpha_a \alpha_b \mathrm{Cov}(X_a, X_b).$$

2.10.3 Continuous Distributions

When a random variable X takes values over $[a, b]$, its **probability distribution function** F on support $[a, b]$ is defined by

$$F(x) = Prob[X \leqq x],$$

which is the probability that X takes values not exceeding x. By definition, the function F is nondecreasing and satisfies $F(a) = 0$ and $F(b) = 1$. Here, a and b can be any real number, and thus it is possible that $a = -\infty$ and $b = \infty$.

The derivative of F is called the **probability density function** and is denoted by $f \equiv F'$. We assume that f is continuous, and $f(x) > 0$ for all $x \in (a, b)$.

The expectation of X is then defined by

$$E(X) = \int_a^b x f(x) dx.$$

If $u : [a, b] \to \mathcal{R}$ is an arbitrary function, the expectation of $u(X)$ is defined by

$$E[u(X)] = \int_a^b u(x) f(x) dx,$$

which can also be written as

$$E[u(X)] = \int_a^b u(x) dF(x).$$

The **conditional expectation** of X given that $X < x$ is

$$E[X | X < x] = \frac{1}{F(x)} \int_a^x t f(t) dt,$$

and thus

$$F(x)E[X|X < x] = \int_a^x tf(t)dt = xF(x) - \int_a^x F(t)dt,$$

in which the second equality is obtained by integrating by parts.

2.10.4 Common Probability Distributions

Next, we review some common distributions, as well as their expectations and variances.

Binomial Distribution

Assume that there are many balls in a box with two colors. The proportion of red balls is p, and that of black balls is $1 - p$. The value of the random variable X is 1 if a red ball is drawn; otherwise, it is 0. If it is taken only once, the probability distribution of the random variable is $p(X = 1) = p$ and $p(X = 0) = 1 - p$.

The expectation and variance are:

$$E(X) = p; \quad \mathrm{Var}(X) = p(1 - p).$$

If we draw n times (the ball is put back into the box each time), the random variable is defined as the number of times that a red ball is drawn.

The probability distribution of random variables is

$$p(X = k) = \frac{n!}{k!(n-k)!} p^k (1 - p)^{n-k}.$$

Its expectation and variance are:

$$E(X) = np; \quad \mathrm{Var}(X) = np(1 - p).$$

Poisson Distribution

If the probability of a random variable X is

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!},$$

then X follows a **Poisson distribution** with parameter λ, and its expectation and variance are:

$$E(X) = \lambda; \quad \mathrm{Var}(X) = \lambda.$$

Uniform Distribution

If the probability density function of a random variable X is

$$f(x) = \frac{1}{b-a},\ x \in [a,b],$$

then X follows a **uniform distribution** over $[a,b]$. Its expectation and variance are:

$$E(X) = \frac{b+a}{2}; \quad \mathrm{Var}(X) = \frac{(b-a)^2}{12}.$$

Normal Distribution

If the probability density function of a random variable X is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}},\ x \in (-\infty, \infty),$$

then X follows a **normal distribution** with parameters (μ, σ^2).

The expectation and variance are:

$$E(X) = \mu; \quad \mathrm{Var}(X) = \sigma^2.$$

Exponential Distribution

If the probability density function of a random variable X is

$$f(x) = \lambda e^{-\lambda x}, x \in [0, \infty),$$

then X follows an **exponential distribution** with parameter λ, and its expectation and variance are

$$E(X) = \frac{1}{\lambda}; \quad \mathrm{Var}(X) = \frac{1}{\lambda^2}.$$

2.11 Stochastic Dominance and Affiliation

2.11.1 Order Stochastic Dominance

First-Order Stochastic Dominance

Definition 2.11.1 (First-Order Stochastic Dominance) Given two distribution functions F and G with support $[a,b]$, we say that F **first-order stochastically dominates** G if for all $x \in [a,b]$, $F(x) \leqq G(x)$.

First-order stochastic dominance means that for any outcome x, the probability of obtaining at least x under $F(\cdot)$ is at least as high as that under $G(\cdot)$. For example, considering two assets, first-order stochastic dominance means that when two assets are greater than a certain constant return, the probability of one asset's return is higher than that of the other. This is analogous to the monotonicity concept under certainty.

There is another test criterion for F to first-order stochastically dominate G. The following theorem shows that these two criterions are equivalent.

Theorem 2.11.1 *$F(\cdot)$ first-order stochastically dominates $G(\cdot)$ if and only if for any nondecreasing function $u : [a, b] \to \mathcal{R}$, we have*

$$\int_a^b u(z)dF(z) \geqq \int_a^b u(z)dG(z).$$

PROOF. Define $H(z) = F(z) - G(z)$. We need to prove that $H(z) \leqq 0$ if and only if $\int_a^b u(z)dH(z) \geqq 0$ for any increasing and differentiable function $u(\cdot)$.

Sufficiency: We prove this by way of contradiction. Suppose that there is a $\hat{z}$, such that $H(\hat{z}) > 0$. We choose a weakly increasing and differentiable function $u(z)$ as

$$u(z) = \begin{cases} 0, & z \leqq \hat{z}, \\ 1, & z > \hat{z}, \end{cases}$$

then immediately $\int_a^b u(z)dH(z) = -H(\hat{z}) < 0$, which is a contradiction.

Necessity: Since a monotonic function is differentiable almost everywhere, in finding integration, we may just assume that u is differentiable, and we have

$$\int_a^b u(z)dH(z) = [u(z)H(z)]_a^b - \int_a^b u'(z)H(z)dz = 0 - \int_a^b u'(z)H(z)dz \geqq 0,$$

in which the first equality is obtained by integration by parts, the second equality is based on

$$F(a) = G(a) = 0, \ F(b) = G(b) = 1,$$

while the inequality is based on the assumptions that $u(\cdot)$ is weakly increasing ($u'(\cdot) \geqq 0$) and $H(z) \leqq 0$. $\square$

For any two probability distributions F and G, as long as an agent's utility is (weakly) increasing in outcomes, the agent prefers the one that first-order stochastically dominates the other one.

Second-Order Stochastic Dominance

Definition 2.11.2 (Second-Order Stochastic Dominance) Given two distribution functions F and G defined on $[a, b]$, which have the same expectation, we say that $F(\cdot)$ **second-order stochastically dominates** $G(\cdot)$ if

$$\int_a^z F(r)dr \leqq \int_a^z G(r)dr$$

for all z.

It is clear that first-order stochastic dominance implies second-order stochastic dominance. In addition, second-order stochastic dominance implies not only monotonicity, but also lower risk. To show this, we introduce the notion of "mean-preserving spreads".

Suppose that X is a random variable with distribution function F. Let Z be a random variable whose distribution conditional on $X = x$, $H(\cdot|X = x)$, is, such that for all x, $E[Z|X = x] = 0$. Suppose that $Y = X + Z$ is the random variable obtained from first drawing X from F and then for each realization $X = x$, drawing a Z from the conditional distribution $H(\cdot|X = x)$ and adding it to X. Let G be the distribution of Y so defined. We will then say that G is a **mean-preserving spread** of F.

While random variables X and Y have the same mean, i.e., $E[X] = E[Y]$, variable Y is "more spread-out" than X since it is obtained by adding a "noise" variable Z to X. Now, suppose that $u : [a, b] \to \mathcal{R}$ is a concave function. Using Jensen's inequality, we have

$$\begin{aligned} E_Y[u(Y)] &= E_X[E_Z[u(X + Z)]|X = x] \\ &\leqq E_X[u(E_Z[X + Z|X = x])] \\ &= E_X[u(X)]. \end{aligned}$$

As such, similar to Theorem 2.11.1, we have the following conclusion for second-order stochastic dominance.

Theorem 2.11.2 *If distributions $F(\cdot)$ and $G(\cdot)$ defined on $[a, b]$ have the same mean, then the following statements are equivalent.*

(1) *$F(\cdot)$ second-order stochastically dominates $G(\cdot)$;*

(2) *for any nondecreasing concave function $u : \mathcal{R} \to \mathcal{R}$, we have $\int_a^b u(z)dF(z) \geqq \int_a^b u(z)dG(z)$;*

(3) *$G(\cdot)$ is a mean-preserving spread of $F(\cdot)$.*

PROOF. **(3)⇒(2)**: It is obtained by using

$$\begin{aligned}\int_a^b u(z)dF(z) &= \int_a^b u\left(\int_a^b (x+z)dH_z(x)\right)dF(z)\\ &\geqq \int_a^b \left(\int_a^b u(x+z)dH_z(x)\right)dF(z)\\ &= \int_a^b u(z)dG(z),\end{aligned}$$

in which the inequality follows from the concavity of $u(\cdot)$.

(1)⇒(2): For expositional convenience, we set $b = 1$. We have

$$\begin{aligned}&\int_a^b u(z)dF(z) - \int_a^b u(z)dG(z)\\ =&-u'(1)\int_a^b (F(z)-G(z))dz + \int_a^b \left(\int_a^z (F(x)-G(x))dx\right)u''(z)dz\\ =&\int_a^b (\int_a^z (F(x)-G(x))dx)u''(z)dz\\ \geqq& 0,\end{aligned}$$

in which the inequality follows from the definition of second-order stochastic dominance, i.e.,

$$\int_a^z F(r)dr \leqq \int_a^z G(r)dr,$$

and also $u''(\cdot) \leqq 0$ for any z. We thus have

$$\int_a^b u(z)dF(z) - \int_a^b u(z)dG(z) \geqq 0.$$

(1)⇒(3): We just show the case with discrete distributions.

Define

$$\begin{aligned}S(z) &= G(z) - F(z),\\ T(x) &= \int_a^x S(z)dz.\end{aligned}$$

By the definition of second-order stochastic dominance, we have $T(x) \geqq 0$ and $T(1) \geqq 0$, which imply that there exists some $\hat{z}$, such that $S(z) \geqq 0$ for $z \leqq \hat{z}$ and $S(z) \leqq 0$ for $z \geqq \hat{z}$.

Since the random variable follows a discrete distribution, $S(z)$ must be a step function. Let $I_1 = (a_1, a_2)$ be the first interval over which $S(z)$ is positive, and $I_2 = (a_3, a_4)$ be the first interval over which $S(z)$ is negative. If no such $I_1 = (a_1, a_2)$ exists, then $S(z) \equiv 0$, and thus statement (3) is immediate. If $I_1 = (a_1, a_2)$ does exist, then $I_2 = (a_3, a_4)$ must exist, as well.

Therefore, $S(z) \equiv \gamma_1 > 0$ for $z \in I_1$, and $S(z) \equiv -\gamma_2 < 0$ for $z \in I_2$. By $T(x) \geqq 0$, we must have $a_2 < a_3$. If $\gamma_1(a_2 - a_1) \geqq \gamma_2(a_4 - a_3)$, then there exist $a_1 < \hat{a}_2 \leqq a_2$ and $\hat{a}_4 = a_4$, such that $\gamma_1(\hat{a}_2 - a_1) = \gamma_2(\hat{a}_4 - a_3)$. If $\gamma_1(a_2 - a_1) < \gamma_2(a_4 - a_3)$, then there exists $a_3 < \hat{a}_4 \leqq a_4$, such that $\gamma_1(\hat{a}_2 - a_1) = \gamma_2(\hat{a}_4 - a_3)$.

Letting

$$S_1(z) = \begin{cases} \gamma_1, & \text{if } a_1 < z < \hat{a_2}, \\ -\gamma_2, & \text{if } a_3 < z < \hat{a_4}, \\ 0, & \text{otherwise.} \end{cases}$$

If $F_1 = F + S_1$, then F_1 is a mean-preserving spread of F. Letting $S^1 = G - F_1$, we can similarly construct $S_2(z)$ and F_2. Since $S(z)$ is a step function, then there exists an n, such that $F_0 = F, F_n = G$, and F_{i+1} is a mean-preserving spread of F_i. Furthermore, a finite summation of mean-preserving spreads is still a mean-preserving spread. □

Although a continuous function can be arbitrarily approximated by step functions, the formal proof is complicated. Rothschild and Stiglitz (1971) provided a complete proof for the case with continuous distributions.

2.11.2 Hazard Rate Dominance

Let F be a distribution function with support $[a, b]$. The hazard rate of F is the function $\lambda : [a, b) \to \mathcal{R}_+$ defined by

$$\lambda(x) \equiv \frac{f(x)}{1 - F(x)}.$$

If we interpret F as the probability that some event will occur prior to time x, then the hazard rate at x represents the instantaneous probability that the event will happen at x, given that it has not occurred until time x. S-ince the event may be the failure of some component, e.g., a lightbulb, it is sometimes also called the “failure rate”.

Solving for F, we have

$$F(x) = 1 - \exp\left(-\int_a^x \lambda(t)dt\right). \tag{2.11.1}$$

This shows that any arbitrary function $\lambda : [a, b) \to \mathcal{R}_+$, such that for all $x < b$,

$$\int_a^x \lambda(t)dt < \infty, \quad \lim_{x \to b} \int_a^x \lambda(t)dt = \infty,$$

is the hazard rate of *some* distribution that is given by (2.11.1).

Definition 2.11.3 (Hazard Rate Dominance) For any two distributions F and G with hazard rates λ_F and λ_G, respectively, we say that F **dominates** G **in terms of the hazard rate** if $\lambda_F(x) \leqq \lambda_G(x)$ for all x. This order is also referred to in shortened from as **hazard rate dominance**.

If F dominates G in terms of the hazard rate, then

$$F(x) = 1 - \exp\left(-\int_a^x \lambda_F(t)dt\right) \leqq 1 - \exp\left(-\int_a^x \lambda_G(t)dt\right) = G(x),$$

and thus F first-order stochastically dominates G. Therefore, hazard rate dominance implies first-order stochastic dominance.

2.11.3 Reverse Hazard Rate Dominance

A closely related concept to the hazard rate is the reverse hazard rate $\sigma : (a, b] \to \mathcal{R}_+$ given by

$$\sigma(x) \equiv \frac{f(x)}{F(x)},$$

and is sometimes referred to as the **inverse of the Mills' ratio**. Similarly, solving for F gives

$$F(x) = \exp\left(-\int_x^b \sigma(t)dt\right). \tag{2.11.2}$$

This shows that any arbitrary function $\sigma : (a, b] \to \mathcal{R}_+$, such that for all $x > a$,

$$\int_x^b \sigma(t)dt < \infty \text{ and } \lim_{x\to 0}\int_x^b \sigma(t)dt = \infty.$$

is the "reverse hazard rate" of some distribution that is given by (2.11.2).

Definition 2.11.4 (Reverse Hazard Rate Dominance) For two distributions F and G with reverse hazard rates σ_F and σ_G, we say that F **dominates** G**in terms of the reverse hazard rate** if $\sigma_F(x) \geqq \sigma_G(x)$ for all x. This order is also referred to in shortened form as **reverse hazard rate dominance**.

If F dominates G in terms of the reverse hazard rate, then

$$F(x) = \exp\left(-\int_x^b \sigma_F(t)dt\right) \leqq \exp\left(-\int_x^b \sigma_G(t)dt\right) = G(x),$$

and thus, again, F first-order stochastically dominates G. Therefore, reverse hazard rate dominance also implies first-order stochastic dominance.

2.11.4 Likelihood Ratio Dominance

Definition 2.11.5 (Likelihood Ratio Dominance) We say that the distribution function F dominates G in terms of the **likelihood ratio** if for all $x < y$,

$$\frac{f(x)}{g(x)} \leqq \frac{f(y)}{g(y)}, \tag{2.11.3}$$

which means that $\frac{f}{g}$ is a nondecreasing function. As such, we refer to this order as **likelihood ratio dominance**.

Rewriting (2.11.3) gives

$$\frac{f(y)}{f(x)} \leqq \frac{g(y)}{g(x)},$$

and then for all x, we have

$$\int_x^b \frac{f(y)}{f(x)} dy \leqq \int_x^b \frac{g(y)}{g(x)} dy,$$

which, in turn, implies that

$$\frac{1-F(x)}{f(y)} \leqq \frac{1-G(x)}{g(y)}.$$

Therefore, likelihood ratio dominance implies hazard rate dominance.

Similarly, rewriting (2.11.3) gives

$$\frac{f(x)}{f(y)} \leqq \frac{g(x)}{g(y)},$$

and then for all x, we have

$$\int_a^y \frac{f(x)}{f(y)} dx \leqq \int_a^y \frac{g(x)}{g(y)} dx,$$

which implies that

$$\frac{F(y)}{f(y)} \leqq \frac{G(y)}{g(y)}.$$

Therefore, likelihood ratio dominance implies reverse hazard rate dominance.

Summarizing the above discussions, one can see that likelihood ratio dominance is the strongest, which implies both hazard rate dominance and reverse hazard rate dominance, both of which, in turn, imply first-order s-tochastic dominance that again implies second-order stochastic dominance.

2.11.5 Order Statistics

Let $X_1, X_2, \cdots, X_n$ be n random variables independently and randomly drawn from a distribution F with density f. Let $Y_1^{(n)}, Y_2^{(n)}, \cdots, Y_n^{(n)}$ be a rearrangement of these, and thus

$$Y_1^{(n)} \geqq Y_2^{(n)} \geqq \cdots \geqq Y_n^{(n)},$$

where $Y_k^{(n)}$, $k = 1, 2, \cdots, n$ are referred to as **order statistics**.

Let $F_k^{(n)}$ denote the distribution of $Y_k^{(n)}$, with corresponding probability density function $f_k^{(n)}$. If there is no confusion, we simply denote them as Y_k, F_k and f_k. In auction theory, we will typically be interested in properties of the highest and second highest order statistics, i.e., Y_1 and Y_2, respectively.

Highest Order Statistic

The distribution of the **highest order statistic** Y_1 can be obtained as follows. The event that $Y_1 \leqq y$ is equivalent to the event: $X_k \leqq y$ for all k. Since X_k is independently drawn from the same distribution F, we have that

$$F_1(y) = F(y)^n.$$

The density function is

$$f_1(y) = nF(y)^{n-1}f(y).$$

Note that if F stochastically dominates G, and F_1 and G_1 are distributions of the highest order statistics of n draws from F and G, respectively, then F_1 stochastically dominates G_1.

Second-Highest Order Statistics

The distribution of the **second-highest order statistic** Y_2 can also be easily derived. The event that $Y_2 \leqq y$ is the union of the following disjoint events: (1) all X_k's are less than or equal to y; and (2) $n-1$ of the X_k's are less than or equal to y, and one is greater than y. There are n different ways in which (2) can occur. Therefore, we have

$$\begin{aligned} F_2(y) &= F(y)^n + nF(y)^{n-1}(1 - F(y)) \\ &= nF(y)^{n-1} - (n-1)F(y)^n. \end{aligned}$$

The probability density function is then

$$f_2(y) = n(n-1)(1 - F(y))F(y)^{n-2}f(y).$$

Again, one can verify that if F stochastically dominates G and also F_2 and G_2 are distributions of the second-highest order statistics of n draws from F and G, respectively, then F_2 stochastically dominates G_2.

2.11.6 Affiliation

Affiliation is a basic assumption employed to study auctions with interdependent values in which random variables are non-negatively correlated.

Definition 2.11.6 Suppose that random variables $X_1, X_2, \cdots, X_n$ are distributed on some product of intervals $D \subseteq \mathcal{R}^n$ according to the joint density function f. $X = (X_1, X_2, \cdots, X_n)$ are said to be **affiliated** if for all $\mathbf{x}', \mathbf{x}'' \in \mathcal{X}$,

$$f(\mathbf{x}' \vee \mathbf{x}'')f(\mathbf{x}' \wedge \mathbf{x}'') \geqq f(\mathbf{x}')f(\mathbf{x}), \tag{2.11.4}$$

in which

$$\mathbf{x}' \vee \mathbf{x} = (\max(x_1', x_1), \cdots, \max(x_n', x_n))$$

denotes the component-wise **maximum** of $\mathbf{x}'$ and $\mathbf{x}''$, and

$$\mathbf{x}' \wedge \mathbf{x}'' = (\min(x_1', x_1''), \cdots, \min(x_n', x_n''))$$

denotes the component-wise **minimum** of $\mathbf{x}'$ and $\mathbf{x}''$. If (2.11.4) is satisfied, then we also say that f is **affiliated**.

Suppose that the density function $f : D \to \mathcal{R}_+$ is strictly positive in the interior of D and twice continuously differentiable. One can verify that f is affiliated if and only if, for all $i \neq j$,

$$\frac{\partial^2}{\partial x_i \partial x_j} \ln f \geqq 0,$$

which means that the off-diagonal elements of the Hessian of $\ln f$ are non-negative.

Proposition 2.11.1 *Let $X_1, X_2, \cdots, X_n$ be random variables, and $Y_1, Y_2, \cdots, Y_{n-1}$ be the largest, second largest, ..., smallest order statistics from among $X_2, X_3, \cdots, X_n$. If $X_1, X_2, \cdots, X_n$ are symmetrically distributed and affiliated, then we have*

(1) *variables in any subset of $X_1, X_2, \cdots, X_n$ are also affiliated;*

(2) *$X_1, Y_1, Y_2, \cdots, Y_{n-1}$ are affiliated.*

Monotone Likelihood Ratio Property

Suppose that the two random variables X and Y have a joint density $f : [a, b]^2 \to \mathcal{R}$. If X and Y are affiliated, then for all $x' \geqq x$ and $y' \geqq y$, we have

$$f(x', y)f(x, y') \leqq f(x, y)f(x', y') \Leftrightarrow \frac{f(x, y')}{f(x, y)} \leqq \frac{f(x', y')}{f(x', y)} \tag{2.11.5}$$

and

$$\frac{f(y'|x)}{f(y|x)} \leqq \frac{f(y'|x')}{f(y|x')},$$

so the **likelihood ratio**

$$\frac{f(\cdot|x')}{f(\cdot|x)}$$

is increasing, and this is referred to as the **monotone likelihood ratio property.**

Using the same arguments as for order stochastic dominance in the previous subsection, it can be deduced that for all $x' \geqq x$, $F_Y(\cdot|x')$ dominates $F_Y(\cdot|x)$ in terms of the likelihood ratio, and the other dominance relationships then follow as usual. We have the following conclusions.

Proposition 2.11.2 *If X and Y are affiliated, the following properties hold:*

(1) *For all* $x' \geqq x$, $F(\cdot|x')$ *dominates* $F(\cdot|x)$ *in terms of* **hazard rate**, *i.e.*,

$$\lambda(y|x') \equiv \frac{f(y|x')}{1 - F(y|x')} \leqq \frac{f(y|x)}{1 - F(y|x)} \equiv \lambda(y|x).$$

Or equivalently, for all y, $\lambda(y|\cdot)$ *is nonincreasing.*

(2) *For all* $x' \geqq x$, $F(\cdot|x')$ *dominates* $F(\cdot|x)$ *in terms of the* **reverse hazard rate**, *i.e.*,

$$\sigma(y|x') \equiv \frac{f(y|x')}{F(y|x')} \leqq \frac{f(y|x)}{F(y|x)} \equiv \sigma(y|x),$$

or equivalently, for all y, $\sigma(y|\cdot)$ *is nondecreasing.*

(3) *For all* $x' \geqq x$, $F(\cdot|x')$ **first-order stochastically dominates** $F(\cdot|x)$, *i.e.*,

$$F(y|x') \leqq F(y|x),$$

or equivalently, for all y, $F(y|\cdot)$ *is nonincreasing.*

(4) *For all* $x' \geqq x$, $F(\cdot|x')$ **second-order stochastically dominates** $F(\cdot|x)$, *i.e., for all* y

$$\int_a^y F(r|x')dr \leqq \int_a^y F(r|x)dr$$

or equivalently, for all y, $\int_a^y F(y|\cdot)$ *is nonincreasing.*

All of these results extend in a straightforward manner to the case in which the number of conditional variables is more than one. Suppose that $Y, X_1, X_2, \cdots, X_n$ are affiliated and let $F_Y(\cdot|\mathbf{x})$ denote the distribution of Y conditional on $X = \mathbf{x}$. We can then also obtain the above dominance relations.

2.12 Biographies

2.12.1 Friedrich August Hayek

Friedrich August Hayek (1899-1992), one of the greatest thinkers of the 20th century and a representative of the Austrian school, won the 1974 Nobel Memorial Prize in Economic Sciences for his contributions to the theory of money and economic cycles, as well as his penetrating analysis of the interdependence of economic, political, and institutional phenomena. The Nobel Prize Committee believed that Hayek's in-depth analyses of economic fluctuations and business cycles made him one of the few economists who had warned about a possible great economic depression prior to 1929. In fact, both academically and practically, the 20th century was characterized by competition between the market economy system and the planned economy system, and disputes concerning their respective advantages and disadvantages. Hayek's penetrating analysis of different economic systems pointed out that the planned economy is not feasible from the perspectives of information efficiency, incentive compatibility, and resource allocation efficiency. Indeed, pragmatic results proved Hayek's insightful judgment. Finally, the planned economy system experienced its demise, which made him one of the most influential economists of the 20th century.

Hayek was born into an intellectual family in Vienna and received a doctorate from the University of Vienna (1921-1923). When attending the University of Vienna, he attended classes taught by Ludwig von Mises (1881-1973). Mises' thorough critique of socialism published in 1922 eventually pulled Hayek out of the Fabian socialist ideological trend. The best way to understand Hayek's great contribution to economics and classical liberalism is to analyze it from the perspective of Mises' paradigm of social collaboration. Hayek taught at the London School of Economics and Political Science (1931-1950), the University of Chicago (1950-1962), and Freiburg University (1962-1968). At the University of Chicago, Hayek was a professor of social and ethical science in the "Committee on Social Thought" and did not obtain a teaching post in the Department of Economics. Milton Friedman, a friend of his in the economics department, was also critical of Hayek's books on economics. When arriving at the University of Chicago, Hayek first conducted political studies and did not engage in economics research, and held a negative attitude towards some research methods that were being used at the Department of Economics. Even so, Hayek interacted frequently with some of the members of the Chicago School of Economics, and his political views were consistent with many of the Chicago School. Hayek made a remarkable contribution to the University of Chicago. He strongly supported Aaron Director, a Chicago School economist and the founder of law and economics, to carry out the "Law and Society" project at the University of Chicago Law School. Meanwhile, Professor Director persuaded the University of Chicago Press to publish Hayek's *The Road to Serfdom* , which later became popular globally. Hayek also collaborated with Friedman and others on the establishment of the International Forum of Liberal Economists.

Hayek involved two profound debates in his life. One was the "socialist controverse" in the 1920s and 1930s, which was the debate between Mises-Hayek and Lange-Lerner on the theoretical feasibility of efficient allocations under socialism. He criticized the drawbacks of the planned economy from the perspective of information and incentives. He held that the planned economy was theoretically impracticable, and emphasized the importance of a spontaneous social order based on freedom, competition, and rules. This internally logical judgment was verified prior to his death. The second was the theoretical debate with Keynes in the 1930s. He pointedly criticized Keynes's theoretical claims and academic viewpoints put forward in *A Treatise on Money* , and argued that Keynes's economic proposition of achieving full employment by lowering the interest rate and increasing the money supply was fundamentally incorrect. In 1947, Hayek advocated for the establishment of the Pilgrimage Mountain Society, an important liberal academic organization. He advocated thorough economic freedom and opposed any form of state intervention, calling for "non-nationalization" of currency issuance.

Hayek's profound philosophy of revealing the importance of institutions will undoubtedly continue to influence and guide the world, especially the next step of reform in China.

2.12.2 Joseph Alois Schumpeter

Joseph Alois Schumpeter (1883-1950), an Austrian American political economist (but not a member of the Austrian School), was profoundly influential, and is hailed as the originator of the theory of innovation. Known as one of the greatest economists in history, his name is closely associated with most of the concepts and knowledge about market economies and innovation. Indeed, he proposed the four most representative and well-known economic terms, i.e., innovation, entrepreneurship, corporate strategy, and creative destruction. In addition, he held that creative destruction constitutes a double-edged sword which can engender economic growth, but can also impair some values that people have traditionally cherished. He expressed this by stating "What poverty brings is a tragic life, while it is difficult for prosperity to maintain peace of mind."

In 1883, Schumpeter was born into the family of a weaving factory owner in Triesch, Habsburg Moravia (now part of Czech, and thus Schumpeter is sometimes considered to be a Czech-American), Austria-Hungary. He enrolled in an elite middle school in Vienna. He studied law and sociology at the University of Vienna from 1901 to 1906, and received his doctoral degree in law in 1906. In 1908, with his instructor's recommendation, he became an associate professor at the University of Czernowitz just at the beginning of his journey as an economist. Czernowitz is a remote city, but a suitable place for learning with its tranquility outside of modern industrial civilization. Here, Schumpeter wrote his first masterpiece, *The Theory of Economic Development* published in 1912, which touches on "innovation" and its role in economic development, and had a great impact in the economics community. According to statistics, the concept of "creative destruction" proposed by Schumpeter was cited ubiquitously, second only to the "invisible hand" of Adam Smith. *The Theory of Economic Development* has become one of the classical economic works of the 20th century. Later, Schumpeter emigrated to the United States, and taught at Harvard University until the end of his life.

Schumpeter argued in his famous book *History of Economic Analysis* that an economic scientist differs from an average economist by simultaneously adopting the following three elements in the process of economic analysis. The first one is economic theory with inherently logical analyses. The second is historical perspectives. The third is statistics with data and quantitative analyses. Schumpeter's theory of innovation is also frequently mentioned, and his name appears in most discussions about innovation. Indeed, as the founder of the innovation theory and the research of business history, Schumpeter' s influence is also currently being "rediscovered".

Innovation refers to an economic process that recombines and integrates old production factors into new production methods in order to reduce costs and increase efficiency. In Schumpeter's economic model, those who can successfully innovate can survive the dilemmas of diminishing returns; whereas, those who fail to recombine production factors will be the first to be eliminated by the market. The creative destruction of capitalism means that when the economy cycles to the bottom, it is the time when some entrepreneurs have to exit the market and others must innovate to survive. As long as excess competitors are excluded or some successful "innovations" occurred, the economy's production efficiency will increase. When an industry becomes profitable again, however, it will attract new investments. Then, the process of diminishing returns begins anew and returns to the previous state. Therefore, every depression implies the possibility of another technological innovation, or it can be stated that the result of technological innovation is another expected depression. In Schumpeter's view, the creativity and destructiveness of capitalism are homologous. However, Schumpeter did not believe that the superiority of capitalism is due to its own impetus which can promote its own development continually. Instead, he contended that the capitalist economy will eventually collapse because it cannot withstand the energy of its rapid expansion. The business cycle, also known as the economic cycle, is Schumpeter' s most quoted economic term. Schumpeter's concept of creative destruction has had a great influence on the development of modern economics. The combination of the dynamic market mechanism and R&D economics provides economists with a crucial perspective of endogenous technological change. Schumpeter's technological innovation has become a core element of the theory of endogenous growth in macroeconomics.

In *Capitalism, Socialism and Democracy*, Schumpeter gave the following modern definition of democracy: "The democratic method is that institutional arrangement for arriving at political decisions in which individuals acquire the power to decide by means of a competitive struggle for the people's vote". He held that democracy constitutes a process in which political elites compete for power, and that the people choose political leaders. The essence of democracy lies in a competitive election process. Political elites occupy political power and implement their rule, but their legitimacy comes from the choice of the people. Schumpeter also held the view that, in the political market with democracy, politicians propose political programs and policies according to voters' preferences, and compete freely in elections to attract voters. Schumpeter's definition of democracy symbolizes the great transformation of democratic theory from the classical direct democracy to the modern representative democracy.

The ideas and theories of Hayek and Schumpeter play a crucial role in theoretically guiding and clarifying the way that the two transition stages occur.

Hayek's economic thought on the fundamental importance of the market and institutions deeply affected economic development in the 20th century; likewise, Schumpeter's economic thought on the critical importance of innovation will undoubtedly continue to exert remarkable influence, as we have already witnessed during the first 20 years of the 21st century.

2.13 Exercises

Exercise 2.1 Consider an economy with two sectors: the industrial sector and the monetary sector, characterized by the following equations:

$$Y = C + I + G,$$

$$C = a + b(1 - t)Y,$$

$$I = d - ei,$$

$$G = G_0,$$

where Y, C, I and i (i is the interest rate) are endogenous variables, G_0 is an exogenous variable, and a, b, d, e and t are all structure parameters.

In the newly introduced monetary market, we have:
the equilibrium conditions: $M_d = M_s$,
the money demand: $M_d = kY - li$,
and the money supply: $M_s = M_0$,
where M_0 is the exogenous variable of money stock, and k and l are parameters. Given this economy, please solve the following problems: (using Cramer's rule)

1. Equilibrium income Y^*;

2. Money supply multiplier;

3. Government expenditure multiplier.

Exercise 2.2 Q represents the set of rational numbers, and as a metric space, its distance is defined by $d(p, q) = |p - q|$, where $p \in E = \{p \in Q : 2 < p < 40\} \subseteq Qp \in E = \{p \in Q : 2 < p < 40\} \subseteq Q$.

1. Prove that E is bounded in Q.

2. Prove that E is not compact.

3. Is E open in Q? If yes, why?

Exercise 2.3 Given a metric space X, consider a series of open sets $\{E_n\}_{n \in N}$ in X.

1. Prove that $\bigcup_{n \in N} E_n$ is an open set.

2. Prove that it may not be true that the intersection of a series of open sets is open (please provide an example of this).

Exercise 2.4 Prove the following theorems:

1. The difference of an open set and a closed set is also open, while the difference of a closed set and an open set is also closed.

2. Each closed set is the intersection of a countable number of open sets; each open set is the union of a countable number of closed sets.

Exercise 2.5 Let $S \subseteq \mathcal{R}^L$. Prove that the following propositions are equivalent:

1. S is compact.

2. S is bounded and closed.

3. Every sequence in S has a convergent subsequence with the limit point in S.

4. Every infinite subset of S has a cluster point in S.

5. Each closed subset of the set S with finite intersection property (i.e., the intersection over any finite subcollection is nonempty) is nonempty.

Exercise 2.6 Prove the following propositions:

1. Every closed subset of a compact set is compact.

2. If $f : X \to Y$ is continuous and K is compact in X, then $f(K)$ is compact in Y.

3. S_i is compact, $i \in I$, if and only if $\prod_{i \in I} S_i$ is compact.

4. S_i is compact, $i = 1, 2, \cdots, m$, if and only if $\sum_i^m S_i$ is compact.

Exercise 2.7 (Shapley-Folkman Theorem) Prove the theorem: Let $S_i, (i = 1, \cdots, n)$ be n non-empty subsets of R^m and $S = \sum_{i=1}^{n} S_i$. Then, each $x \in Co(S)$ has a representation $x = \sum_{i=1}^{n} x_i$, such that $x_i \in Co(S_i)$ for all i, and $x_i \in S_i$ for at least $(n - m)$ indices i.

Exercise 2.8 Prove the following theorems:

1. If f is a differentiable function defined on $\mathcal{R}^1$, then f is concave if and only if the first-order condition $f'(x)$ is non-increasing.
2. If f is a twice differentiable function defined on $\mathcal{R}^1$, then f is concave if and only if the second-order condition $f''(x)$ is non-positive.
3. If f is a differentiable function defined on $\mathcal{R}^1$, then f is concave if and only if $f(y) \leqq f(x) + f'(x)(y - x)$ for any $x, y \in \mathcal{R}^1$.

Exercise 2.9 Suppose that $f(\mathbf{x}) = \dfrac{1}{2}\mathbf{x}^T A\mathbf{x} + b^T\mathbf{x} + c$, where $\mathbf{x} \in \mathcal{R}^n$, $\mathbf{x}^T$ is the transpose of vector $\mathbf{x}$, A is an $n \times n$ symmetric matrix , **b** is an n-dimensional vector, and c is a constant.

1. Prove that if A is a positive semi-definite matrix, then $f(\mathbf{x})$ is a convex function.
2. Prove that if A is a positive definite matrix, then $f(\mathbf{x})$ is a strictly convex function.

Exercise 2.10 Determine whether Kuhn-Tucker conditions are applicable for the following optimization problems and solve them.

$$\begin{aligned} &\max\ x_1 \\ \text{s.t.}\quad & x_1^3 - x_2 \leqq 0, \\ & x_2 \leqq 0. \end{aligned}$$

Exercise 2.11 Solve the following optimization problems using Kuhn-Tucker conditions.

$$\begin{aligned} &\max\ xyz \\ \text{s.t.}\quad & x^2 + y^2 + z^2 \leqq 6, \\ & x \geqslant 0, y \geqslant 0, z \geqslant 0. \end{aligned}$$

Exercise 2.12 The maximization problem is as follows:

$$\begin{aligned} &\max f(\boldsymbol{x}) \\ \text{s.t.}\qquad & g^1(\boldsymbol{x}) = 0, \cdots, g^m(\boldsymbol{x}) = 0, \end{aligned}$$

where $f : \mathcal{R}^n \longrightarrow \mathcal{R}$ and $g^j : \mathcal{R}^n \longrightarrow \mathcal{R}$ are increasing functions with respect to $\boldsymbol{x}$, and $m < n$. Prove that: If f is quasi-concave and all g^j are quasi-convex functions, then any local optimum is the global optimal solution.

Exercise 2.13 Let $u : \mathcal{R}^n \longrightarrow \mathcal{R}$ be a function, $\boldsymbol{p}, \boldsymbol{x} \in \mathcal{R}^n$, and $y \in \mathcal{R}$. Consider the following optimization problem:

$$\max_{\boldsymbol{x}} u(\boldsymbol{x})$$
$$\text{s.t.} \quad \boldsymbol{px} = y.$$

Suppose that there is an optimum solution $x^*(\boldsymbol{p}, y) > 0$, such that $v(\boldsymbol{x}, y) = u(\boldsymbol{x}^*(\boldsymbol{p}, y))$.

1. Prove that $v(\boldsymbol{p}, y)$ is homogeneous of degree zero.

2. Prove that $v(\boldsymbol{p}, y)$ is a quasi-concave function.

Exercise 2.14 Suppose that a Cobb-Douglas utility function $u : \mathcal{R}^2 \longrightarrow \mathcal{R}$ is defined as:

$$u(x_1, x_2) = x_1^\alpha x_2^\beta, \ \alpha, \ \beta > 0.$$

Prove:

1. If $\alpha + \beta \leqq 1$, then u is a concave function.

2. If $\alpha + \beta > 1$, then u is a quasi-concave function, but not concave.

3. For any $\alpha > 0$ and $\beta > 0$, $h(x_1, x_2) = \ln(u(x_1, x_2))$ is a concave function.

Exercise 2.15 Suppose that $\overline{X}$ is a nonempty, closed, and convex set in $\mathcal{R}^n$, $x_0 \notin \overline{X}$. Prove that the following propositions are true.

1. There is a point $a \in \overline{X}$, such that $d(x_0, a) < d(x_0, x)$ for all $x \in \overline{X}$, and $d(x_0, a) > 0$.

2. There is a point $p \in \mathcal{R}^n$, $p \neq 0$, $||p|| \equiv (\sum_{i=1}^n p_i^2)^{1/2} < \infty$ and $\alpha \in \mathcal{R}$, such that

   $$p \cdot x \geqq \alpha, \text{ for all } x \in \overline{X} \text{ and } p \cdot x_0 < \alpha.$$

   Specifically, $\overline{X}$ and x_0 is separated by a hyperplane $H = \{x : p \cdot x = \alpha, x \in \mathcal{R}^n\}$.

Exercise 2.16 Consider following functions:

(1) $3x^5y + 2x^2y^4 - 3x^3y^3$.

(2) $3x^5y + 2x^2y^4 - 3x^3y^4$.

(3) $x^{3/4}y^{1/4} + 6x + 4$.

(4) $\dfrac{x^2 - y^2}{x^2 + y^2} + 3$.

(5) $x^{1/2}y^{-1/2} + 3xy^{-1} + 7$.

(6) $x^{3/4}y^{1/4} + 6x$.

1. Find homogeneous functions among them and determine their orders of degree.

2. Test whether the above functions satisfy the Euler's Theorem.

Exercise 2.17 There is a simple application of upper (lower) hemi-continuity of correspondences. Suppose that $f : X \times Y \longrightarrow \mathcal{R}$,

$$G(x) = \{y \in \Gamma(x) : f(x,y) = \max_{y \in \Gamma(x)} f(x,y)\}.$$

1. Suppose that $X = \mathcal{R}$, $\Gamma(x) = Y = [-1,1]$. For all $x \in X$, $f(x,y) = xy^2$. Draw the graph of $G(x)$ and prove that $G(x)$ is upper hemi-continuous at $x = 0$, but not lower hemi-continuous.

2. Suppose that $x = \mathcal{R}$ and $\Gamma(x) = Y = [0,4]$ for all $x \in X$. Define that

$$f(x,y) = \max\{2 - (y-1)^2, x + 1 - (y-2)^2\}.$$

   Draw the graph of $G(x)$ and prove that: $G(x)$ is upper hemi-continuous but not lower hemi-continuous, and specify at which points it is not lower hemi-continuous.

3. Suppose that $X = \mathcal{R}_+$, $\Gamma(x) = Y = \{y \in \mathcal{R} : -x \leqq y \leqq x\}$. For all $x \in X$, define that $f(x,y) = \cos(y)$, then draw the graph of $G(x)$ and prove that: $G(x)$ is upper hemi-continuous but not lower hemi-continuous, and specify at which points it is not lower hemi-continuous.

Exercise 2.18 Let $S = \{x \in \mathcal{R}^2 :\| x \|= 4\}$ be the boundary of a circle with a radius of 2. The mapping $\psi : \mathcal{R}^2 \to S$ is defined as:

$$\psi(x) = \arg\min_{x' \in S} d(x,x'),$$

specifically, $\psi(x)$ contains the closest point in S to x. Discuss the upper and lower hemi-continuity of $\psi(x)$.

Exercise 2.19 Consider a correspondence $\Gamma : D \subseteq \mathcal{R}^l \longrightarrow \mathcal{R}^k$, of which the graph is defined as

$$G(\Gamma) = \{(x,y) \in D \times \mathcal{R}^k : y \in \Gamma(x)\}.$$

If $G(\Gamma)$ is a closed set, then we say Γ has a closed graph; if $G(\Gamma)$ is a bounded and closed set, we call Γ compact-valued. Suppose that Γ is compact-valued. Prove:

1. If Γ is upper hemi-continuous, then it has a closed graph.

2. If Γ is locally bounded and its graph is closed, then Γ is upper hemi-continuous. (Hint: The definition of locally bounded correspondence Γ: $G(\Gamma) = \{(x, y) \in D \times \mathcal{R}^k : y \in \Gamma(x)\}$ is **locally bounded**, if for each $x \in D$, there is an $\epsilon > 0$ and a bounded set $Y(x) \subseteq \mathcal{R}^k$, such that for all $x' \in N_\epsilon(x) \bigcap D, \Gamma(x') \subseteq Y(x)$.)

Exercise 2.20 Suppose that $X \subseteq \mathcal{R}_+$ is a nonempty compact set. Prove that:

1. If $f : X \longrightarrow X$ is a continuous increasing function, then f has a fixed point.

2. Specially, suppose that $X = [0, 1]$. If $f : X \longrightarrow X$ is an increasing function (not necessarily continuous), does f have another fixed point?

Exercise 2.21 Suppose that X is a complete metric space, and T is the mapping from X to X. Denote

$$a_n = \sup_{x \neq x'} \frac{d(T^n x, T^n x')}{d(x, x')}, n = 1, 2, \cdots .$$

Prove that: If $\sum_{n=1}^{\infty} a_n < \infty$, then the mapping T has a unique fixed point.

Exercise 2.22 Consider $n \in N$ and an nth order square matrix $A = (a_{ij})_{n \times n}$. For any $x \in \mathcal{R}^n$, we have

$$Ax = (\sum_{j=1}^{n} a_{1j} x_j, \sum_{j=1}^{n} a_{2j} x_j, \cdots, \sum_{j=1}^{n} a_{nj} x_j)^T.$$

Suppose that f is a differentiable mapping from $\mathcal{R}$ to $\mathcal{R}$, such that

$$s = \sup\{|f'(t)| : t \in \mathcal{R}\} < \infty.$$

Define a mapping F from $\mathcal{R}^n$ to $\mathcal{R}^n$, i.e.,

$$F(x) = (f(x_1), \cdots, f(x_n))^T.$$

For a given n-dimensional vector w, we can solve the following system of nonlinear equations:

$$z = AF(z) + w. \qquad (*)$$

1. Prove that: if $\max\{\sum_{j=1}^{n} |a_{ij}| : i = 1, \cdots, n\} < \frac{1}{s}$, then there is a unique $z \in \mathcal{R}^n$ satisfying the above system of equations $(*)$.

2. Prove that: if $\sum_{i=1}^{n}\sum_{j=1}^{n}|a_{ij}| < \frac{1}{s^2}$, then there is a unique $z \in \mathcal{R}^n$ satisfying the above system of equations. $(*)$.

Exercise 2.23 Suppose that h is a mapping from $\mathcal{R}_+$ to $\mathcal{R}_+$, and $H : \mathcal{R}_+ \times \mathcal{R} \longrightarrow \mathcal{R}$ is a bounded function, such that there is a $K \in (0, 1)$,

$$|H(x, y) - H(x, z)| < K|y - z|, \text{for any} x \geqq 0,\ y,\ z \in \mathcal{R}.$$

Prove that there is a unique bounded function, $f : \mathcal{R}_+ \longrightarrow \mathcal{R}$, such that

$$f(x) = H(x, f(h(x))), \text{for any } x \geqq 0.$$

Exercise 2.24 Find the extremum curve of the following functional:

1. $V(y) = \int_0^1 (t^2 + y'^2)dt, y(0) = 0, y(1) = 2;$
2. $V(y) = \int_0^1 (y + yy' + y' + 0.5y'^2)dt, y(0) = 2, y(1) = 5;$
3. $V(y) = \int_0^T (1 + y'^2)^{0.5} dt, y(0) = A, y(T) = Z.$

Exercise 2.25 Solve the following optimal control problem:

$$\begin{array}{c} \max \int_0^3 (x-2)^2(x'(t)-1)^2 dt \\ \text{s.t.}\quad x(0) = 0, x(3) = 2. \end{array}$$

Exercise 2.26 Consider the following optimal control problem. Write the Hamilton equation and solve the optimal function.

$$\begin{array}{c} \max \int_0^1 (x + u)dt \\ \text{s.t.}\quad x'(t) = 1 - u^2, x(0) = 1. \end{array}$$

Exercise 2.27 Consider the following optimization problem:

$$v(q) = \max_{x \in \mathcal{R}^+} \ln(2x + q) - 6x + 2q,$$

where $q \in (0, 2)$.

1. Solve $v(q)$ and its derivative $v'(q)$.
2. Verify that the Envelope Theorem holds.

Exercise 2.28 Find the general solution to the extremum curve of the following functional:

$$V(y, z) = \int_a^b (y'^2 + z'^2 + 3y'z')dt.$$

Exercise 2.29 In the problem of functional $\int_0^T F(t, y, z, y', z')dt$, suppose that $y(0) = A$, $z(0) = B$, $y_T = C$, $z_T = D$, T are free, and A, B, C, and D are constants.

1. How many transversal conditions are required for the problem? Why?
2. Write these transversal conditions.

Exercise 2.30 The integrand function of the target functional is $F(t, y, y') = 4y^2 + 4yy' + y'^2$.

1. Write the Euler equation.
2. Is the above Euler equation sufficient for maximization or minimization problems? Why?

Exercise 2.31 Solve the paths of $y(t)$ and $z(t)$ of extremum curves of $V(y, z) = \int_0^T (y'^2 + z'^2)dt$ subject to $y - z' = 0$.

Exercise 2.32 Solve the optimal paths of control variables, state variables, and costate variables as follows:

1. $\max \int_0^T -(t^2 + 2u^2)dt$ subject to $y' = u, y(0) = 2, and\ y(T) = 3$, and T is free.
2. $\max \int_0^T -(u^2 + y^2 + 3uy)dt$ subject to $y' = u, y(0) = y_0$, and $y(t)$ is free.
3. $\max \int_0^4 2ydt$ subject to $y' = y + u, y(0) = 3, y(4) \geqq 200$.

Exercise 2.33 Find the optimal consumption path of the following exhaustible resource problem:

$$\begin{aligned} & \max \int_0^T \ln qe^{-\delta t}dt \\ \text{s.t.} \quad & s' = -q, s(0) = s_0, s(t) \geqq 0. \end{aligned}$$

Exercise 2.34 Using the revised transversal conditions expressed by the present value Hamilton function, solve the problems

1. with the end curve $y_T = \phi(t)$.
2. with truncated vertical end line.
3. with truncated horizontal end line.

Exercise 2.35 In a maximization problem, there are two known state variables, (y_1, y_2), two control variables (u_1, u_2), an inequality constraint, and an inequality integral constraint. The initial state is fixed, but the final state is free at fixed T.

1. State the maximization problem.

2. Define the Hamilton's equation and the Lagrange function.

3. Suppose that there is an interior solution, and then write the conditions of the maximum principle.

Exercise 2.36 Consider the problem of "eating cake" as follows. The agent has $A_0 > 0$ units of cake for consumption in period 0 and can save the cake to the next period without costs, and its utility function is $\sum_{t=0}^{\infty} \beta^t \ln c_t$.

1. Write the Bellman equations of the problem.

2. Define the state variable and control variable.

3. Find the value function.

Exercise 2.37 Consider the following problem of "tree cutting" : the growth of a tree can be represented by the function h, i.e., $k_{t+1} = h(k_t)$, where k_t is the scale of the tree at time t.

There is no cost for cutting trees, and the timber price is $p = 1$. Interest rate r remains unchanged, $\beta = 1/(1 + r)$.

1. Assuming that trees cannot be replanted, we write the maximization problem of present value as $v(k) = \max\{k, \beta v[h(k)]\}$. Under what conditions about h is there a simple rule that can be used to describe when to cut trees?

2. Suppose that another tree can be planted where the original tree is cut down, and the replanting cost $c \geqq 0$ remains unchanged for a long period of time. Under what conditions about h and c is there a simple rule that can be used to characterize when to cut trees?

Exercise 2.38 Solve the following dynamic programming problems by three methods: value function iteration, guessing value function, and guessing policy function, respectively:

$$\begin{aligned} & \max_{\{c_t.k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \ln c_t \\ \text{s.t.} \qquad & c_t + k_{t+1} = Ak_t^{\alpha}, \end{aligned}$$

where k_0 is given.

Exercise 2.39 Solve the following differential equations:

1. $y' = t^2 y$.

2. $y'' - 4y' + 5y = 0$.

3. $y'' - 2y' - 3y = 9t^2$.

Exercise 2.40 Consider the following two-dimensional autonomous differential equations:

$$\frac{dx}{dt} = x(4 - x - y),$$
$$\frac{dy}{dt} = y(6 - y - 3x).$$

1. Solve the equilibrium of the power system.

2. Verify the stability of each equilibrium.

Exercise 2.41 Solve the following difference equations:

1. $y(t+1) - 2y(t) = 4^t$.

2. $y(t+2) + 3y(t+1) + 2y(t) = 0$.

3. $y(t+2) - y(t+1) - 6y(t) = t + 2$.

Exercise 2.42 Suppose that X_1, X_2, $\cdots$, and X_n are n independent and identically distributed random variables. The distribution function is F, and the probability density function is f. Let $Y_1^{(n)}, Y_2^{(n)}, \cdots, Y_n^{(n)}$ be the corresponding order statistics satisfying $Y_1^{(n)} \geqq Y_2^{(n)} \geqq \cdots \geqq Y_n^{(n)}$.

1. Find the distribution function and probability density function of $Y_n^{(n)}$.

2. Find $E(Y_n^{(n)})$ and $\mathrm{Var}(Y_n^{(n)})$.

3. Find $\mathrm{Cov}(Y_1^{(n)}, Y_n^{(n)})$.

Exercise 2.43 Let X and Y be two random variables in the range $[a, b]$. Prove:

1. If X first-order stochastically dominates Y, then X necessarily second-order stochastically dominates Y.

2. If X second-order stochastically dominates Y, then $EX \geqq EY$.

3. If X second-order stochastically dominates Y and $EX = EY$, then $Eu(X) \geqq Eu(Y)$ for all concave and twice differentiable functions (whether increasing or not).

4. If X second-order stochastically dominates Y and $EX = EY$, then $Var(X) \leqq Var(Y)$.

Exercise 2.44 Suppose that X is a non-negative random variable, and the distribution function and density function are F and f, respectively. The risk rate of random variable X is defined as

$$\lambda_X : \mathcal{R}_+ \longrightarrow \mathcal{R}_+, \quad \lambda_X(t) = \frac{f(t)}{1 - F(t)}.$$

If $\lambda_X(\cdot) \leqq \lambda_Y(\cdot)$, we say that the random variable X stochastically dominates random variable Y in terms of risk rate. Suppose that G and g are, respectively, the distribution function and density function of random variable Y. If $f(\cdot)/g(\cdot)$ is a non-decreasing function, then we say that X stochastically dominates Y in terms of likelihood ratio. Prove the following statements:

1. $\lambda_X(\cdot) \leqq \lambda_Y(\cdot)$ if and only if $1 - G(t)/[1 - F(t)]$ is a non-increasing function.

2. If X dominates Y in terms of likelihood ratio, then X must stochastically dominate Y in terms of risk rate.

Exercise 2.45 Prove that: if $X_1, X_2, \cdots, X_N$ are correlated, and $\gamma(\cdot)$ is an increasing function, then for $x_1' > x_1$, we have

$$E[\gamma(Y_1)|X_1 = x_1'] \geqq E[\gamma(Y_1)|X_1 = x_1].$$

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Part II

Information, Incentives, and Mechanism Design

The previous chapters focus on the role of the competitive market system in resource allocation, discuss how it works, and identify its advantages and limitations. In reality, people are always exploring whether there is a better economic system, which is especially true when the market fails. We thus need a general analytical framework to study the design of economic mechanisms, and compare the advantages and disadvantages of all economic mechanisms under certain criteria.

Information economics, principal-agent theory (also called contract theory), mechanism design theory, and the theory of market design (mainly consisting of auction theory and matching theory) have become very important and active research fields in economics in the last five decades. They have a wide range of applications in various disciplines such as finance, management, corporate law, and political science. Because of this, more than 20 economists who are founding contributors of mechanism design and associated fields of game theory have so far won the Nobel Prize in Economics, including Friedrich A. Hayek, Kenneth Arrow, George J. Stigler, Gerand Debreu, Ronald Coase, Herbert Simon, John Nash, Reinhard Selten, William Vickrey, James Mirrlees, George Akerlof, Joseph Stiglitz, Michael Spence, Robert Auman, Leo Hurwicz, Eric Maskin, Roger Myerson, Peter Diamond, Oliver Williamson, Alvin E. Roth, Lloyd S. Shapley, Jean Tirole, Oliver Hart, Bengt Holmstrom, and Paul Milgrom.

The core issues in economic theory and practice are information and incentive. As mentioned, pursuing individual interest (their own utilities or profits) is not only the most basic assumption of economics, but is also the most relevant reality, besides informational incompleteness and asymmetry. As such, if a self-interested agent has said something, others cannot tell if this is the truth. Indeed, even if someone's eyes are focused on us while we are speaking, we do not know if the person is actually listening to us. Accordingly, real-world economic problems can be extremely complicated and challenging to cope with, and economics, which focuses on investigating economic phenomena and economic human behavior, has become a highly challenging discipline. Thereby, how can we solve the problem of incentive distortions caused by individual self-interestedness and informational asymmetry? In other words, when information is asymmetric, how can we induce self-interested individuals to report their economic characteristics and actions truthfully through proper mechanism design so that the goal of the society or the designer is achieved? If this goal can be achieved, are individual interest and social interest incentive-compatible?

In dealing with real economic activities, there are two basic economic institutional arrangements: Government and market, or prosaically, “want me to do” and “I want to do”. The former is a commanding, passive, enforcing institutional arrangement, but because of information asymmetry, it is easy to cause incentive incompatibility, which is often not implementable. The latter is an inductive, active and self reinforcing incentive compatible institutional arrangement, which is the main content covered by mechanism design theory. Incentive or incentive-compatibility has become one of the most basic and core concepts in modern economics. To many economists, economics is to a large extent a matter of incentives: Incentives to work hard, to produce good quality products, to study, to invest, to save, etc.

Until the 1970s, traditional economic analysis took economic mechanisms as given, and was mostly concerned with understanding the theory of value in large economies. A central question asked in general equilibrium theory is whether a certain mechanism (especially the competitive mechanism) generates Pareto efficient allocations, and if so, for what categories of economic environments. In a perfectly competitive market, the competition pressure solves the problem of incentives for consumers and producers. The major task of understanding how prices are formed in competitive markets can proceed without being concerned about incentives.

The question was then reversed: Instead of regarding mechanisms as given and seeking the class of environments for which they work, one seeks mechanisms which will implement some desired objectives (especially those that result in Pareto efficient and individually rational allocations) for a given class of environments without removing participants' incentives, and have a low cost of operation and other desired properties. In a sense, the theorists returned to the basics.

The reverse question was stimulated by two major lines in the history of economics. Within the capitalist/private-ownership economics literature, a stimulus arose from studies focusing on the failures of the competitive market to function as a mechanism for implementing efficient allocations in various kinds of non-neoclassical economic environments, such as the presence of externalities, public goods, indivisible goods, incomplete information, imperfect competition, and increasing returns to scale. At the beginning of the 1970s, works by Akerlof (1970), Hurwicz (1972), Spence (1974), and Rothschild and Stiglitz (1976) showed in various ways that asymmetric information was posing a much greater challenge, and could not be satisfactorily imbedded in a proper generalization of the Arrow-Debreu general equilibrium theory.

A second stimulus arose from the socialist/state-ownership economics literature, as evidenced in the "socialist controversy", i.e., the debate between Mises-Hayek and Lange-Lerner in the 1920s and 1930s. This controversy was provoked by von Mises's skepticism regarding even a theoretical feasibility of rational allocation under socialism. This debate clarified numerous problems and reached many consensuses. However, because both sides of the debate lacked precise definitions of key economic terms, with unclear and/or inconsistent definitions, plus the lack of scientific quantitative analysis tools, they were unable to provide formal and clear conclusions with rigorous mathematical logic. It is Hurwicz who was the first to distill the most essential issues in this debate, one of which is information in economic systems (especially, informational requirement to run an economic system), and the other is incentive. He studied these two issues by establishing corresponding rigorous analytical frameworks.

The structures of incentives and information are thus two basic features of any economic system. The study of these two features is attributed to these two major lines, culminating in the theory of mechanism design. The theoretical framework of economic mechanism design that was originated by Hurwicz is very general. All economic mechanisms and systems (including those that are known and unknown, private-ownership, state-ownership, and mixed-ownership systems) can be studied with the framework.

At the micro-level, the development of the theory of incentives has constituted a major advance in economics in the last 50 years. Prior to this, by treating the firm as a black box, the theory remained silent on how the owners of firms succeed in aligning the objectives of its various members, such as workers, supervisors, and managers, with profit maximization. When economists began to examine the firm more carefully, either in agricultural or managerial economics, incentives became the central focus of their analysis. Indeed, delegation of a task from the principal to an agent who has different objectives is problematic when information about the agent is asymmetric or incomplete. This problem is the essence of incentive questions. Thus, *conflicting objectives and decentralized information are the two basic constituents of incentive theory*.

We will discover that, in general, these informational problems prevent a principal or society from achieving the **first-best** that is obtained in a world of complete information (and so it is unnecessary to consider incentive-compatibility). Thus, the best is only the **second-best** but not the first-best.[1] Additional costs that must be incurred for inducing private information of agents could be viewed as one category of transaction costs. Although they do not exhaust all possible transaction costs, economists have been relatively successful in modeling and analyzing these types of costs, and providing a good understanding of the limits set by these on the allocation of resources. This line of research also provides a whole set of insights on how to begin to take into account agents' responses to incentives

[1]In incentive theory, the first-best refers to the best that one could do when information is complete so that information rent is zero, and the **second-best** is the best that one can do if agents have to reveal their private information at a cost. More generally, the second-best concerns what occurs when one or more optimality conditions cannot be satisfied. The notions of first-best and second-best were first defined by Canadian economist Richard Lipsey and Australian economist Kelvin Lancaster, who showed in a 1956 paper that if one optimality condition in an economic model cannot be satisfied, it is possible that the next-best solution involves changing other variables to be further away from the ones that are usually assumed to be optimal, see: Lipsey and Lancaster (1956), "The General Theory of Second Best". *Review of Economic Studies* 24(1): 11-32.

provided by institutions.

The three terms, i.e., mechanism, institution and contract are largely synonymous.[2] They all mean "rules of the game," which describe what actions the parties can undertake, and what outcomes these actions would obtain. In most cases, the rules of the game are given by designers, such as in chess and basketball. The rules are designed to achieve better outcomes. However, these synonyms also have certain differences. The mechanism design theory may be able to answer "big" questions at a social level, such as "market economy vs. planned economy" and "socialism vs. capitalism." A designer may not participate in economic activities and pay more attention to the efficiency of the whole social allocation. The principal-agent theory/contract theory, on the other hand, is to answer the micro problems at the management level. A designer (also known the principal, especially for contract design) participates in economic activities and concerning specific voluntary contracting practices and mechanisms that maximize the principal's own benefits. Because the contract is voluntary, they need to meet both the incentive compatibility conditions and the participation conditions.

Therefore, mechanism design is normative economics, in contrast to game theory which is positive economics. Game theory is important because it predicts how a given game will be played by agents. The theory of mechanism design, however, goes one step further: Given the physical environment and the constraints faced by a designer, what goal can be realized or implemented? What mechanisms are optimal among those that are feasible? In designing mechanisms, one must take into account incentive constraints (e.g., consumers may not report truthfully how many pairs of shoes they need or how productive they are).

This part considers the design of economic mechanism, in which one or more parties have private characteristics or hidden actions. The party who designs the mechanism is called the designer or principal, while the other parties are called agents, individuals, or participants. For the most part, we will focus on the situation in which the designer has no private information and the agents do. This framework is called **screening**, because the principal will, in general, attempt to screen different types of agents by inducing them to choose different outcomes. The opposite situation, in which the designer has private information and the agents do not, is called **signaling**, since the designer could signal her type with the design of her contract or mechanism.

We will discuss mechanism design theory in five chapters. Chapters 16 and 17 discuss contract theory/principal-agent theory with mainly only one agent. Thereby, we only consider the situation in which the princi-

[2]However, a significant difference is that a contract must be voluntary, while a mechanism may not require this constraint. The term "contract" is usually used at the micro-level.

pal delegates an action to a single agent with exogenous or endogenous private information and subject to two kinds of constraints: **Participation constraint** and **incentive-compatibility constraint**. This private information of the agent can be of two classes: Either the agent has some private information about his cost or valuation unknown to the principal, which is the case of adverse selection or hidden information; or the agent can take an action that is unobserved by the principal, which is the case of moral hazard or hidden action. The design of the principal's optimal contract can be simply regarded as a constrained optimization problem. This simple focus will turn out to be sufficient to highlight the various trade-offs between **allocative efficiency and the extraction of information rents** for the case of adverse information, or the trade-off between **incentives and risk** (consequently, the trade-offs between **efficiency and insurance**) for moral hazard arising under incomplete information. The mere existence of informational constraints may generally prevent the principal from achieving allocative efficiency. We will characterize the allocative distortions that the principal finds desirable to implement in order to mitigate the impact of informational constraints. The many discussions of these two chapters refer to the monographs of Laffont and Martimort (2002) and Bolton and Dewatripont (2005). The main focus is on whether first-best is implementable.

Chapter 18 will consider situations with multiple agents and strategic interactions between them under complete information. Then, organizational contexts require a solution concept of equilibrium, which describes the strategic behavior of agents. The solution concepts of equilibrium that we consider are primarily dominant equilibrium and Nash equilibrium. The primary focus is on whether efficiency/Pareto efficiency of outcomes is implementable.

Chapter 19 will consider situations with multiple agents and strategic interactions between them under incomplete information, in which asymmetric information may not only affect the relationship between the principal and each of her agents, but it may also harm the relationships between agents. As such, agents do not know each other's characteristics, and it is necessary to consider a Bayesian incentive-compatible mechanism.

Chapter 20 will briefly discuss dynamic mechanism design. We will discuss the long-term incentive contract in a dynamic principal-agent setting, mainly with one agent and adverse selection. We will first consider the case in which the principal (designer) can commit to a contract forever, and then consider what takes place when she cannot commit to not modifying the contract as new information arrives. After that, we will discuss the design of efficient dynamic mechanisms in multi-agent environments with strategic interactions.

Chapter 16

The Principal-Agent Model: Hidden Information

16.1 Introduction

The principal-agent theory is also called the optimal contract theory. This chapter mainly discusses the optimal contract design with only one agent. When a principal assigns a task to an agent who has private information, the agent's utility, technology, cost, and ability are all likely to be his private information. By misreporting the private information, the agent may obtain additional benefits, and then the incentive issue appears. We call such a principal-agent situation the "**hidden information**" or "**adverse selection**" problem.

Some Examples

The principal-agent problem exists almost everywhere. Here are some specific examples.

(1) An environmental protection agency wants to reduce air pollution, but does not know the cost of pollution abatement of a firm.

(2) A township government in China entrusts collective land to farmers, while the farmers possess farming skills and know how to observe local weather conditions.

(3) Shareholders delegate daily decisions of a company to a professional manager who knows more about markets and production technologies.

(4) Professors teach students, but do not know the students' learning ability.

(5) A client delegates her defense to an attorney who has professional knowledge of various legal provisions and the difficulty of the case.

(6) Venture capital companies lend funds to a founder of a high-tech company who will be the only one that masters a certain new technology.

(7) The central government delegates administrative power to a local government that knows more about local situations.

(8) The National Development and Reform Commission (NDRC) of China entrusts petroleum resources to the China National Petroleum Corporation (CNPC), but NDRC does not participate in the specific utilization of these petroleum resources.

(9) An insurance company provides automobile insurance to an agent whose driving skill is his private information.

(10) A regulatory agency contracts with a public utility company without having complete information about its technology.

These examples show that the difference in information between a principal and an agent has a significant influence on contract design. In order to make optimal use of resources and to allow the agent to have incentives to reveal private information, the agent needs to be given some information rent. Thereby, the principal needs to make a trade-off between allocative efficiency and rent extraction when designing an optimal contract. An implicit assumption is that the contractual relationship is conducted within a certain legal framework. As the contract can be enforced by the courts, the agent is bound by the terms of the contract.

This chapter aims to characterize the optimal rent extraction-efficiency trade-off faced by the principal when designing her contract offer to the agent under incentive-feasible constraints: **Incentive-compatibility** and **participation constraints**. If an incentive constraint binds at the optimal solution, then adverse selection limits the efficiency of the transaction. The main lesson of this chapter is that **the second-best contract calls for a distortion in the volume of trade away from the first-best**, and the principal must transfer some information rent to the more efficient type.

16.2 Basic Model

In this section, we introduce the simplest canonical model to capture the essence of adverse selection. This problem was first studied by Mirrlees in 1971 (see Section 17.11.2 for Mirrilees' biography). We assume that there are two types of participants: One is called the principal, who has some sort of monopoly power in the transaction; and the other is called the agent, who has private information in the transaction. Different from the market equilibrium discussed previously, here we focus on how information distribution affects the efficiency of market transactions.

16.2.1 Economic Environment

For ease of understanding, we discuss the issue of adverse selection in a monopolistic market. This problem was first proposed by Mussa and Rosen (1978), and a more deeper discussion was conducted by Maskin and Riley (1984).

Consider an economy in which a monopolist (seller) offers a product to a consumer (buyer). For the case of a continuum of consumers, the proportion of a certain type of consumers can be considered as the probability that there is one such consumer. According to the law of large numbers, these two approaches are equivalent.

Since the consumer's valuation of the product is private information that cannot be observed by the monopolist, she typically screens buyers of different types, i.e., induces them to select different consumption bundles. This situation is known as monopolistic screening, or second-degree price discrimination. The purpose of the monopolist is to obtain maximum surplus from the consumer, and this depends on how she distinguishes the consumer's type from the consumer's behavior. If we put this example into the principal-agent framework, the monopolist corresponds to the principal, and the consumer corresponds to the agent. Determination of the role of participants in the principal-agent framework is usually based on the fact that the principal provides a menu of contracts, and the agent chooses one from these contracts.

Assume that the consumer's utility function is

$$u(q, T, \theta) = \theta v(q) - T,$$

where q is the quantity of good purchased by the consumer; T is the consumer's transfer; and θ is the type of consumer's valuation on the good with $\theta \in \{\theta_H, \theta_L\}$ and $\theta_H > \theta_L$. Let $\triangle\,\theta = \theta_H - \theta_L$ denote the valuation gap of the two types. Assume that the probability that θ_L occurs is β, which means that the probability of the utility level $u(q, T, \theta_L) = \theta_L v(q) - T$ is β, and the probability of the utility level $u(q, T, \theta_H) = \theta_H v(q) - T$ is $1 - \beta$. The private information is exogenously given. The utility function defined above is quasi-linear, which usually satisfies the following assumptions:

$$v(0) = 0,\ v'(q) > 0,\ v''(q) < 0.$$

The monopolist's objective function is

$$\pi = T - cq,$$

where c is the production cost per unit.

16.2.2 Outcome Space and Contracting Variables

The contracting variables are the quantity q produced by the principal and the transfer T received by the agent. Let $\mathcal{A}$ be the set of feasible outcomes:

$$\mathcal{A} = \{(q, T) : q \in \mathcal{R}_+, T \in \mathcal{R}\}. \tag{16.2.1}$$

These variables are both observable and verifiable by a third party, such as a court of law.

16.2.3 Information Structure and Timing

The information structure can be divided into three categories according to the timing when the principal and/or the agent knows the type. It is called the ex-ante stage when both participants do not know the agent's type. The state at which the agent knows his type while the principal does not know is called the interim stage. When both participants know the agent's type, this is called the ex-post stage.

In the principal-agent theory, contracts are often offered at the interim stage. Unless explicitly stated, two participants will proceed in the following order:

- $t = 0$: The consumer is informed about his own type θ;
- $t = 1$: The monopolist provides a contract;
- $t = 2$: The consumer accepts or rejects the contract;
- $t = 3$: The contract is executed.

16.2.4 Optimal Contract under Complete Information

To analyze the influence of adverse selection on decisions, we usually use the complete information case as a benchmark in which the first-best outcome can be implemented.

If the monopolist knows the consumer's type, she will provide a separate contract for each type, i.e., the contract (T_i, q_i) corresponds to θ_i, $i \in \{H, L\}$. The goal of the monopolist is to extract the consumer's surplus as much as she can, given that the consumer accepts the contract. The consumer's reservation utility level is $\bar{u} = 0$ when rejecting the contract. Thus, the monopolist's goal is to solve the following problem:

$$\max_{T_i, q_i} T_i - cq_i,$$

subject to participation constraint:

$$\theta_i v(q_i) - T_i \geqq 0.$$

Since the principal prefers a T_i that is as high as possible, the constraint is binding at the optimum:

$$T_i = \theta_i v(q_i).$$

By substituting the constraint into the objective function, the problem becomes

$$\max_{T_i, q_i} \theta_i v(q_i) - cq_i,$$

which is *exactly the total surplus-maximization problem*, and thus the principal chooses a socially efficient output level (in fact, it is a Pareto efficient level since the budget balance is also satisfied since the sum of the principal's transfer T and the agent's transfer $-T$ equals zero.). The first-best outputs under this complete information are then obtained from the first-order condition:

$$\theta_i v'({q_i}^*) = c,$$

i.e., the marginal social value of a unit of consumption is equal to the marginal social cost, hence the outcome is indeed socially efficient.[1] At the same time, the monopolist obtains the maximum surplus

$$T_i{}^* = \theta_i v({q_i}^*) - cq_i^*,$$

and the consumer's surplus is 0.

This benchmark result without contraction cost (typically constituted by information cost) offers the simplest possible demonstration of the Coase Theorem, which says that private bargaining among parties in the absence of transaction costs results in Pareto efficient outcomes. However, as discussed in next subsection, private information cost constitutes a transition cost, hence it could prevent the parties from achieving efficiency.

Implementation of the First-Best Contract

The monopolist can design the following contract to reach the maximum profit level. First, it necessitates that the consumer obtains a utility level no lower than the reservation utility $\bar{u}$, i.e., $\theta_i v(q_i) - T_i \geqq \bar{u} = 0$. Second, since the principal can distinguish different types under complete information, the monopolist can make a take-it-or-leave-it offer (T_i^*, q_i^*) to the consumer. It will be accepted by the type-θ_i consumer because the resulting utility is no less than his reservation utility.

Graphical Representation of Optimal Contract

The optimal contract under complete information is the first-best, which is given by points A^* and B^* shown in Figure 16.2. Figure 16.1 depict-

[1] This is true just for this simple model. We may not obtain a socially efficient outcome for a general setting.

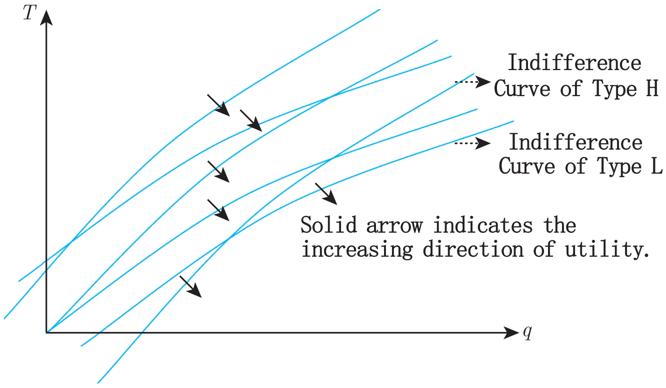

Figure 16.1: Indifference curves of two different types of consumers

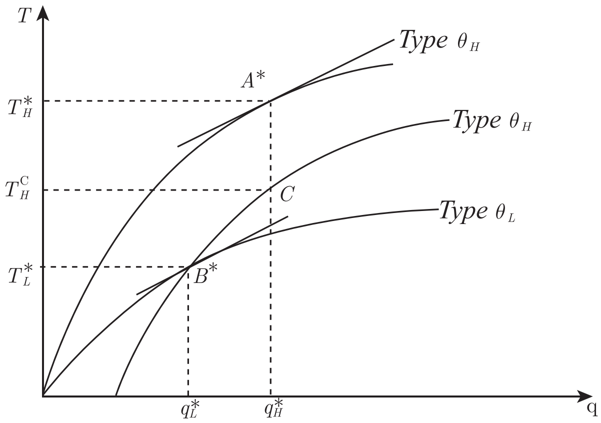

Figure 16.2: The first-best contract

s the indifference curves of two types of consumers. Compared to θ_L, the indifference curve of θ_H has a larger slope; moreover, the two types of indifference curves intersect only once. This feature is called the **single crossing property** or the **Spence-Mirrlees property**.

In the optimal contract, each type of consumer only obtains his reservation utility, and at the same time, the slopes of the optimal points equal the marginal cost c as shown in Figure 16.2.

16.2.5 Incentive-feasible Contracts

We now analyze the optimal contract in the presence of private information. We will see that, under incomplete information, the optimal contract will lead to incentive incompatibility issues, and then the optimal contract is, in general, just a second-best contract.

When the monopolist cannot observe the consumer's type, the above contract (T_i^*, q_i^*), $i \in \{H, L\}$ is not implementable. From Figure 16.2, one can see that, under incomplete information, the low-type will choose point B^*, but the high-type (the θ_H type) will also choose point B^* for a higher utility, instead of choosing point A^* that is optimal for the high-type under complete information, which means that the high-type has an incentive to misreport himself as the low-type. This is because, for the high-type, the utility of choosing (T_H^*, q_H^*) is 0, but the resulting utility level of choosing (T_L^*, q_L^*) is:

$$u({T_L}^*, {q_L}^*, \theta_H) = \theta_H v({q_L}^*) - {T_L}^* > \theta_L v(q_L) - T_L = 0.$$

Thus, under asymmetric information, (T_i^*, q_i^*), $i \in \{H, L\}$, cannot be implemented by the "take-it-or-leave-it" contract. The high-type θ_H will mimic the low-type θ_L.

Then, an appropriate contract must be designed so that each type of consumer has the incentive to choose the contract proposed for his type, i.e., the incentive-compatibility constraint must be met.

Definition 16.2.1 A menu of contracts $A_i = (q_i, T_i)$, $i \in \{H, L\}$ is **incentive-compatible** if it satisfies:

$$u(q_i, T_i, \theta_i) \geqq u(q_j, T_j, \theta_i), \forall i, j \in \{H, L\}.$$

An incentive-compatible contract means that the contract chosen by each type of consumer is exactly the one specified by the principal for his true type. If so, it is said to be **implementable**. Under asymmetric information, when a monopolist designs a contract, the contract must satisfy the following two sets of constraints:

$$U_H = \theta_H v(q_H) - T_H \geqq \theta_H v(q_L) - T_L, \tag{16.2.2}$$
$$U_L = \theta_L v(q_L) - T_L \geqq \theta_L v(q_H) - T_H, \tag{16.2.3}$$
$$U_H = \theta_H v(q_H) - T_H \geqq 0, \tag{16.2.4}$$
$$U_L = \theta_L v(q_L) - T_L \geqq 0. \tag{16.2.5}$$

We call constraints (16.2.2) and (16.2.3) the **incentive-compatibility constraints**, and constraints (16.2.4) and (16.2.5) the **participation constraints**, of types θ_H and θ_L, respectively.

Definition 16.2.2 A menu of contracts $A_i = (q_i, T_i)$, $i \in \{H, L\}$ is **incentive-feasible** if it satisfies both incentive-compatibility and participation constraints (16.2.2) through (16.2.5).

Solving for the asymmetric information optimal contract actually constitutes solving the following problem:

$$\max_{(q_i,T_i)} \beta(T_L - cq_L) + (1-\beta)(T_H - cq_H),$$

subject to the incentive-compatibility constraints (16.2.2) and (16.2.3), and the participation constraints (16.2.4) and (16.2.5).

Prior to formally solving the problem, we first want to know that the set of incentive-feasible contracts is not empty. There are two special contracts that are incentive-feasible.

(1) **Bunching or pooling contracts**: These occur when $T_L = T_H = T, q_L = q_H = q$, and both types accept the contract. In this case, the incentive-compatibility constraints are automatically established. Thus, it is only the participation constraints that matter. If the monopolist specifies the same T for both types, $\theta_L v(q) - T \geqq 0$ is the only constraint.

(2) **Shutdown of the low type (or the least efficient type)**: The non-zero contract (T_H, q_H) is provided for type θ_H, the null contract $(T_L, q_L) = (0, 0)$ is for type θ_L, and it also satisfies incentive-compatibility constraint since $\theta_L v(q_H) - T_H \leqq 0$. With such a contract, the low-type will not be served.

Like the pooling contract, the null contract $(0, 0)$ reduces the number of constraints since the incentive and participation constraints take the same form. The cost of such a contract may be an excessive screening of types, leading to non-optimal contracting. Therefore, these two contracts may not optimal, although they are incentive-feasible.

16.2.6 Monotonicity Condition

Incentive-compatibility constraints that reduce the set of feasible outcomes since output levels must satisfy a **monotonicity condition**: $q_H \geqq q_L$, which is not required under complete information. This condition is satisfied if the Spence-Mirrlees single-crossing condition

$$\frac{\partial^2 u}{\partial\theta\partial q} > 0$$

holds, which means that two different types of indifference curves intersect at most once. Its economic implication is that, as a consumer's utility level is increasing in θ_i, the consumption of the agent is also increasing. Concerning the basic model discussed in this section, this condition is automatically satisfied.

Indeed, adding (16.2.2) and (16.2.3), we immediately have:

$$\Delta\theta(v(q_H) - v(q_L)) \geqq 0.$$

Since $v'(q) > 0$, when $q_H \neq q_L$, we must have $q_H > q_L$, which means that the consumption of the low-type will not be greater than the consumption of the high-type. In fact, this condition is not only necessary, but also sufficient, for implementability.

In order to derive sufficiency, assume $q_H \geqq q_L$. We need to show that there exist transfers T_H and T_L, such that the incentive constraints hold. This can be achieved by choosing transfers that satisfy

$$\theta_L(v(q_H) - v(q_L)) \leqq T_H - T_L \leqq \theta_H(v(q_H) - v(q_L)). \qquad (16.2.6)$$

16.2.7 Information Rents

To understand the characteristics of the optimal contract under asymmetric information, it is useful to introduce the concept of **information rent**, which is the difference between the utility level under asymmetric information and the utility level under complete information.

Under complete information, since the monopolist has full negotiating power, she can fully extract consumer surplus so that the consumer only gets reservation utility levels U_H^* and U_L^* which are zero:

$$U_H{}^* = \theta_H v(q_H{}^*) - T_H{}^* = 0,$$

$$U_L{}^* = \theta_L v(q_L{}^*) - T_L{}^* = 0.$$

Then, the utility levels

$$U_H = \theta_H v(q_H) - T_H \ (\geqq 0)$$

and

$$U_L = \theta_L v(q_L) - T_L \ (\geqq 0)$$

can be regarded as the information rent of the high-type θ_H and the low-type θ_L, respectively.

Under asymmetric information, how much benefit (utility) would a type-θ_H agent obtain by mimicking a type-θ_L agent? The high-type would get:

$$\theta_H v(q_L{}^*) - T_L{}^* = \Delta\theta v(q_L{}^*) + U_L.$$

Even if the expected utility obtained by the low-type is equal to the reservation utility, i.e., $U_L = \theta_L v(q_L^*) - T_L^* = 0$, in order to have an incentive-compatible contract $(T_H, q_H; T_L, q_L)$, the high-type should obtain the information rent that is greater than or equal to the benefit from mimicking the low type, i.e., it should satisfy

$$U_H \geqq \triangle\theta v(q_L^*).$$

Thus, as long as the principal insists on a positive output for the low type, $q_L^* > 0$, the principal must give up a positive rent to the high-type to have an incentive-compatible contract $(T_H, q_H; T_L, q_L)$. This information rent is generated by the informational advantage of the high-type agent over the principal. The problem for the monopolist is thus to choose an optimal way to transfer a certain amount of information rent to certain types in order to obtain the maximum profit.

16.2.8 Optimal Contract under Asymmetric Information

According to the timing of the contractual setting, the monopolist must offer a menu of contracts prior to knowing the type of the consumer. Thus, the monopolist wants to maximize the expected payoffs under incentive-compatible contract $(T_H, q_H; T_L, q_L)$, and then the monopolist's problem can be written as:

$$\begin{aligned} & \max_{(T_H, q_H; T_L, q_L)} (1-\beta)(T_H - cq_H) + \beta(T_L - cq_L) \\ \text{s. t.} \quad & \theta_H v(q_H) - T_H \geqq \theta_H v(q_L) - T_L, \\ & \theta_L v(q_L) - T_L \geqq \theta_L v(q_H) - T_H, \\ & \theta_H v(q_H) - T_H \geqq 0, \\ & \theta_L v(q_L) - T_L \geqq 0. \end{aligned}$$

Using the notation of information rents, $U_H = \theta_H v(q_H) - T_H$ and $U_L = \theta_L v(q_L) - T_L$, we can replace the transfers T_H and T_L with information rents, for which the monopolist's problem is to solve for $(U_H, q_H; U_L, q_L)$. The focus on information rents enables us to assess the distributive impact of asymmetric information, and the focus on outputs allows us to analyze its impact on allocative efficiency and the overall gains from trade. Thus, an allocation corresponds to a level of output and a distribution of the gains from trade between the monopolist (principal) and the consumer (agent).

With the change of variables, the principal's objective function can be rewritten as:

$$\begin{aligned} & \max (1-\beta)(\theta_H v(q_H) - cq_H) \\ & \qquad + \beta(\theta_L v(q_L) - cq_L) - ((1-\beta)U_H + \beta U_L) && (16.2.7) \\ \text{s.t.} \quad & U_H \geqq U_L + \Delta\theta v(q_L), && (16.2.8) \\ & U_L \geqq U_H - \Delta\theta v(q_H), && (16.2.9) \\ & U_H \geqq 0, && (16.2.10) \\ & U_L \geqq 0. && (16.2.11) \end{aligned}$$

Expression (16.2.7) can be divided into two parts:

$$\underbrace{(1-\beta)(\theta_H v(q_H) - cq_H) + \beta(\theta_L v(q_L) - cq_L)}_{\text{total expected surplus}} - \underbrace{((1-\beta)U_H + \beta U_L)}_{\text{information rent of high-type}}. \tag{16.2.12}$$

The first term denotes total expected surplus which measures allocative efficiency, and the second term denotes total expected information rent. This expression implies that the principal is ready to accept some distortions away from efficiency in order to decrease the agent's information rent. We denote the solution to the problem (16.2.7) with a superscript SB, meaning second-best.

16.2.9 The Rent Extraction-Efficiency Trade-Off

For the above optimization problem (16.2.7), a technical difficulty is determining how to deal with these constraints. One way is to employ the Lagrangian method to incorporate the constraints to form a Lagrange function through introducing Lagrangian multipliers, and then to solve the optimization problem with inequality constraints according to the Kuhn-Tucker theorem. Another way that is more convenient to solve for the optimal contract is to analyze these inequality constraints by identifying which are the equality (binding) constraints, and which are the strict inequality constraints so that they do not need to be considered. Let us first consider contracts without shutdown, i.e., $q_L > 0$. This is true when the so-called Inada condition $v'(0) = +\infty$ is satisfied plus the condition $\lim_{q\to 0} v'(q)q = 0$.

At least one of the high-type's incentive-compatibility constraint (16.2.8) and participation constraint (16.2.10) must be binding at the optimum; otherwise, the monopoly profit can be increased by slightly reducing the information rent U_H of the high-type. Similarly, at least one of the low type's incentive-compatibility constraint (16.2.9) and participation constraint (16.2.11) must be binding at the optimum.

From constraint (16.2.8), we know that $U_H = U_L + \triangle\theta v(q_L) > 0$, resulting in a strict inequality for the high-type's participation condition (16.2.8). Thus, the incentive-compatibility constraint for the high-type (16.2.8) must be binding (i.e., an equality constraint).

By the monotonicity condition $q_H > q_L$ and the binding constraint (16.2.8), we obtain that the low type's incentive-compatibility constraint (16.2.9) must be a strict inequality constraint, and thus the low type's participation constraint (16.2.11) must be binding.

Therefore, in the problem (16.2.7), the binding constraints facing the monopolist are: The participation constraint for the low-type (16.2.11), and the incentive-compatibility constraint for the high-type θ_H. We then have:

$$U_H = \Delta\theta v(q_L), \tag{16.2.13}$$

$$U_L = 0. \tag{16.2.14}$$

Substituting these two equality constraints into (16.2.7) gives us:

$$\max_{q_H, q_L} (1-\beta)(\theta_H v(q_H) - cq_H) + \beta(\theta_L v(q_L) - cq_L) - (1-\beta)(\Delta\theta v(q_L)), \tag{16.2.15}$$

which leads to the following first-order conditions:

$$\theta_H v'(q_H^{SB}) = c, \tag{16.2.16}$$

$$\theta_L v'(q_L^{SB}) = \frac{c}{1 - (\frac{1-\beta}{\beta}\frac{\Delta\theta}{\theta_L})} > c. \tag{16.2.17}$$

Obviously, an interior solution satisfies: $q_H^{SB} = q_H^*$ and $q_L^{SB} < q_L^*$.

Summarizing the above discussion, when a monopolist provides a positive amount of products to both types, there is no allocation distortion for the high-type θ_H in the separating equilibrium, but the principal must give up some information rent to the high-type. For the low-type θ_L, his consumption is lower than his first-best consumption so that the principal can pay less information rent to the high-type, and thus there is an allocation distortion with no information rent to the low type. Formally, we have the following proposition:

Proposition 16.2.1 *Under asymmetric information, the optimal contracts entail:*

(1) *No output distortion for the high-type with respect to the first-best, $q_H^{SB} = q_H^*$; a downward output distortion for the low-type, $q_L^{SB} < q_L^*$ with:*

$$\left[\theta_L - \left(\frac{1-\beta}{\beta}\right)\Delta\theta\right] v'(q_L^{SB}) = c. \tag{16.2.18}$$

(2) *Only the high-type gets some positive information rent:*

$$U_H^{SB} = \Delta\theta v(q_L^{SB}). \tag{16.2.19}$$

(3) *The second-best transfers are given by:*

$$T_H^{SB} = \theta_H v(q_H^*) - \Delta\theta v(q_L^{SB}), \tag{16.2.20}$$

$$T_L^{SB} = \theta_L v(q_L^{SB}). \tag{16.2.21}$$

The characteristics of the optimal contracts are correlated with each other. In order to induce the high-type to choose the level of consumption designed for him, the principal should give him some information rent that is determined by the consumption q_L^{SB} of the low-type and the valuation gap of the two types, $\theta_H - \theta_L$. The reason for the principal to reduce the consumption of the low-type is to minimize the information rent. In addition, the distortion of the low-type depends on the valuation gap between the two types.

When $\theta_H - \theta_L \to 0$, the high-type's information rent goes to zero, and the low type's consumption will go to the efficient level. Moreover, when $\theta_H - \theta_L \to \infty$, the high-type's information rent goes to infinity. In this case, the monopolist may adopt an exclusive contract to shut down the low-type to avoid paying a high information rent. Thus, under asymmetric information, the monopolist faces a basic **trade-off** when choosing contracts: *The* *extraction of high-type's information rent and the efficiency concern of low-type's consumption distortion.*

The points A^{SB} and B^{SB} in Figure 16.3 specify the second-best contract for θ_H and θ_L, respectively. There is no allocation distortion in A^{SB} at which the consumer θ_H obtains an information rent of $T_H^* - T_H^{SB}$; and in B^{SB}, the consumption of θ_L is lower than his first-best consumption level under complete information.

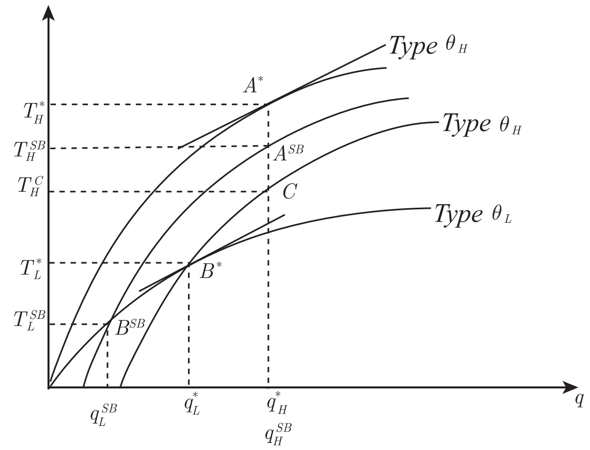

Figure 16.3: Second-best contract

The above proposition gives one of the most important conclusions of the adverse selection problem: *The optimal contract under incomplete information, in general, is only a second-best, but not a first-best, contract*, and *there is no distortion at the top*. In other words, under asymmetric information, the consumption of the high-type is not distorted with respect to the first-best contact under complete information, while the consumption of the low-type may be distorted downward. This rule exists in almost all asymmetric information environments. For example, a capital owner (bank) faces a large number of borrowers, but does not know their operating efficiency, and thus the owner can only set loan thresholds or differentiate interest rates based on their "declarations."

In order to prevent the high-type from mimicking the low-type, the loan threshold or interest rate level of the low-type must be distorted. This is why small corporations often do not have as sufficient credit support as large corporations. They frequently face stricter approval and collateral conditions, or need to pay higher interest rates. This phenomenon is called credit rationing (see the classic paper by Stiglitz and Weiss (1981) for more details).

Remark 16.2.1 As early as 2,600 years ago, Sun Tzu, an ancient Chinese general, strategist, and philosopher, already had insight into the basic idea of the principal-agent theory. The above basic conclusions are implied by *The Art of War*, which established the earliest known theory of war. It is often referred to as the "Bible of Military Science", and is the earliest known work on military strategies and soft war in the world. Sun Tzu considered the critical importance of information and its symmetry, and provided the basic conclusion of information economics: In the complete information situation, the best is first-best; when information is asymmetric, the best is second-best. The saying goes that "if you know the enemy and yourself, you can fight 100 battles with no danger of defeat; if you only know yourself, but not your opponent, you may win or may lose; and if you know neither yourself nor your enemy, you will always endanger yourself."

16.2.10 Shutdown Policy

When $\beta < \frac{\triangle\theta}{\theta_L}$, there is no positive solution in (16.2.17) so that $q_L^{SB} = 0$. In other words, the monopolist will choose to shutdown the low-type. In addition, if the difference of the two types is sufficiently large and there is a positive low-type consumption, it is necessary to provide a high information rent to the high-type. The monopolist may also choose to shut down the low-type market, and the constraint condition becomes $T = \theta_H v(q)$. The monopolist's optimal choice is a solution to the following problem:

$$\max_{q} \theta_H v(q) - cq.$$

Then, we have the exclusive contract (q^c, T^c) that satisfies

$$\theta_H v'(q^c) = c, \quad T^c = \theta_H v(q^c).$$

Then, the consumption of θ_H is efficient with zero information rent, while the low-type does not have any consumption.

We thus conclude that the contracting outcome under asymmetric information is, in general, only the second-best. Of course, there are some ways to improve allocative efficiency, such as signaling, or restricting to the ex-ante situation in which the agent does not have any information advantage.

16.3 Applications

This section discuss two classical models in applying the basic model above. Introducing adverse selection in each of these contexts has proven to be a significant improvement of standard microeconomic analysis. The discussions on these models are drawn from Laffont and Martimort (2002).

16.3.1 Regulation

In some market conditions, such as natural monopoly, the occurrence of redundant construction, and the emergence of industry monopolies, government regulations may improve market efficiency, but it is also necessary to pay attention to the problem of information asymmetry. Unlike traditional regulation theory, since the 1980s, the new regulation theory has discussed the government's optimal regulation policies under asymmetric information. We use the principal-agent theory with adverse selection to discuss the issue of optimal regulation. Baron and Myerson (1982) were among the earliest contributors to incentive regulation theory.

In the Baron and Myerson (1982) regulation model, the principal is a regulator (e.g., the government) who maximizes a weighted average of the consumer's surplus $S(q)-t$ and of a regulated monopoly's profit $U = t-\theta q$, where θ is marginal cost, and therefore the smaller is θ, the more efficient is the firm. There are two types for the firm, θ_H, θ_L, where the probability of the θ_L type is ν, and also $\triangle\theta = \theta_H - \theta_L$. With a weight $\alpha < 1$ for the firm profit and a weight 1 for a consumer's net revenue, the principal's objective function is now written as

$$V = (S(q) - t) + \alpha U = S(q) - \theta q - (1 - \alpha)U.$$

Because $\alpha < 1$, it is socially costly to give up a rent to the firm. The monopoly firm owns private information about the cost. To encourage the monopolist to truthfully report the cost, the principal needs to offer a second-best incentive contract. Let $(t_H, q_H; t_L, q_L)$ be the incentive contract and $U_i = t_i - \theta_i q_i$ be the utility level of the θ_i type.

The government's problem is

$$\max\{\nu[(S(q_L) - \theta_L q_L - (1 - \alpha)U_L] + (1 - \nu)[S(q_H) - \theta_H q_H - (1 - \alpha)U_H]\}$$

subject to

$$U_L \geqq U_H + \Delta\theta q_H;$$

$$U_H \geqq U_L - \Delta\theta q_L;$$

$$U_L \geqq 0;$$

$$U_H \geqq 0.$$

The analysis is the mirror image of that of the standard model discussed in the previous section. Now, the efficient type is the one with the lower marginal cost θ_L for producing the good. Therefore, the incentive-compatibility constraint for the efficient type and the participation constraint for the inefficient type are the only binding constraints. Using the previous method, we can get $U_H = 0$ and $U_L = \triangle\theta q_H$.

Thus, there is no output distortion (with respect to the first-best outcome), $q_L^{SB} = q_L^*$, for the efficient type, but a downward distortion of the inefficient type's output, i.e., $q_H^{SB} < q_H^*$, which is determined by:

$$S'(q_H^{SB}) = \theta_H + \frac{\nu}{1-\nu}(1-\alpha)\Delta\theta. \tag{16.3.22}$$

Note that a higher value of α reduces the output distortion, because the government is less concerned about the distribution of rents within society as α increases. If $\alpha = 1$, the firm's rent is no longer costly, and the government behaves as a pure efficiency maximizer implementing the first-best outcome.

Laffont and Tirole (1986, 1993) discussed a related, but different, incentive regulation issue. The project cost of the firm depends on the unobservable types of the firm and observable effort in reducing the cost. At this point, in order to encourage the firm to reduce the cost of project, the regulator needs to design an optimal incentive contract.

Consider a public project. The value of the society is S, and the cost of the project completed by the regulated firm is $C = \theta - e$, where e is the effort invested by the firm in reducing costs, assuming that the cost function for effort is $\psi(e)$, which satisfies $\psi' > 0$ and $\psi'' > 0$. θ is the type of firm. Suppose that there are two types θ_H and θ_L, where the probability of the θ_L type is ν. Let T be the transfer to the firm, and then the utility of the firm is $U = T - \psi(e)$. λ is the shadow cost of public funds, which means that if one unit is paid to the firm, the social cost is $1 + \lambda$. The regulator's goal is to maximize the social welfare

$$S - (1+\lambda)(T + \theta - e) + T - \psi(e) = S - (1+\lambda)(\theta - e + \psi(e)) - \lambda U.$$

The regulator designs an incentive-compatible contract that stipulates the cost of the firm and the compensation to the firm, $(T_H, C_H; T_L, C_L)$, which satisfies $U_i = T_i - \psi(\theta_i - C_i) \geqq T_j - \psi(\theta_i - C_j)$. Using the previous principle's trade-off between rent extraction and efficiency, we obtain:

$$\begin{aligned}
&\psi'(\theta_L - C_L) = 1, \text{ or } e_L = e^*,\\
&\psi'(\theta_H - C_H) = 1 - \frac{\lambda}{1+\lambda}\frac{\nu}{1-\nu}\Phi'(\theta_H - C_H) < 1,
\end{aligned}$$

where $\Phi \equiv \psi(e) - \psi(e - \triangle\theta)$. In other words, for the efficient type, the effort to reduce cost under the optimal contract is socially optimal; whereas, for the inefficient type, there exists a distortion in effort provision.

With regard to the application of the principal-agent model in regulation theory, interested readers may refer to the classic book of Laffont and Tirole (1993) for additional details.

16.3.2 Financial Contracts

Asymmetric information significantly affects financial markets. For instance, the difficulty of loaning small- and medium-sized firms is a worldwide problem. In the case of asymmetric information, this situation is even worse. Freixas and Laffont (1990) discussed the issue of optimal credit decision when information is asymmetric. The principal is a bank, and the agent is an investor that has private information about project profitability.

The bank also has a borrowing cost in the credit process, such as paying interest to depositors. Assume that the interest cost per unit is R. Assume that the bank provides a loan of size k to an investor who has one of two types of productivity: $\Theta = \{\theta_H, \theta_L\}$. The probabilities of taking θ_L and θ_H are $1-\nu$ and ν, respectively. The cost of the bank is Rk. Let $T(k)$ be the credit return to the bank within the agreed time. Therefore, the lender's utility function is $V = T(k) - Rk$, where we do not consider the actual factors of some financial markets, such as credit risk and default, but only focus on the issue of credit incentives. The bank gives the investor an incentive-compatible credit contract $(T_L, k_L; T_H, k_H)$.

Let $U_i = \theta_i f(k_i) - T_i$ denote the utility of the investor θ_i under the incentive contract. If $(T_L, k_L; T_H, k_H)$ is an incentive-feasible contract, then it needs to satisfy:

$$U_i = \theta_i f(k_i) - T_i \geqq \theta_i f(k_j) - T_j = U_j + (\theta_i - \theta_j) f(k_j)$$

and

$$U_i \geqq 0.$$

Using the trade-off between rent extraction and efficiency in adverse selection, we easily get: For the high-type, the optimal borrowing scale is $k_H^{SB} = k_H^*$, such that $\theta_H f'(k_H^*) = R$, and the for low type, there is a downward distortion in the size of optimal capital lending:

$$k_L^{SB} < k_L^*$$

with

$$\theta_L f'(_L^{SB}) = \frac{R}{1 - \dfrac{v}{1-v}\dfrac{\Delta\theta}{\theta_L}} > R = \theta_L f'(_L^*).$$

In this way, the loan given to a low-productivity firm is even lower when information is asymmetric. Because of economic scale, small- and medium-sized firms have lower productivity compared with large-scale firms. Coupled with large repayment risks, loans are more difficult. The principal-agent model can then explain the difficulty of small- and medium-sized firms in borrowing.

16.4 The Revelation Principle

In the above analysis, the consumer (agent) is required to report his own "type." A natural question is whether a higher utility level of the principal could be achieved with a more complex contract allowing the agent possibly to choose among more options. The answer is negative. Indeed, ask the agent to report more general information does not result in a better outcome, but only increases the complexity of the contract.

The revelation principle introduced below ensures that there is no loss of generality in restricting the principal to offer a simple menu of contracts that has at most as many options as the cardinality of the type space. This simple menu of contracts is actually an example of **direct revelation mechanism**.

Definition 16.4.1 A **direct revelation mechanism** is a mapping $g(\cdot)$ from the type space Θ to the outcome space$\mathcal{A}$.

In the principal-agent model, a direct revelation mechanism can then be written as $g(\theta) = (q(\theta), T(\theta)), \forall \theta \in \Theta$. If the agent reports that his type is $\tilde{\theta} \in \Theta$, then the quantity of goods purchased by the consumer is $q(\tilde{\theta})$ and the transfer is $T(\tilde{\theta})$. In a direct revelation mechanism, an important concept is "truth-telling."

Definition 16.4.2 A direct revelation mechanism $g(\cdot)$ is **truthful** if the agent reports his type truthfully, i.e., the following incentive-compatibility constraints are satisfied:

$$\theta_H v(q(\theta_H)) - T(\theta_H) \geqq \theta_H v(q(\theta_L)) - T(\theta_L), \tag{16.4.23}$$
$$\theta_L v(q(\theta_L)) - T(\theta_L) \geqq \theta_L v(q(\theta_H)) - T(\theta_H). \tag{16.4.24}$$

Letting $q_H = q(\theta_H), q_L = q(\theta_L); T_H = T(\theta_H), T_L = T(\theta_L)$, we get back to the original form.

When communication between the principal and the agent is more complicated (the principal is doing more than just asking the agent to report his type), we can obtain a more general mechanism. Let M be the message space of the agent under the more general mechanism.

Definition 16.4.3 A **general mechanism** $\langle M, \tilde{g} \rangle$ is composed of the mapping $\tilde{g}(\cdot)$ from the message space M to $\mathcal{A}$:

$$\tilde{g}(m) = (\tilde{q}(m), \tilde{T}(m)), \forall m \in M. \tag{16.4.25}$$

Under such a mechanism, the agent with type θ will report the optimal message $m^*(\theta)$ that is determined by:

$$\theta v(\tilde{q}(m^*(\theta))) - \tilde{T}(m^*(\theta)) \geqq \theta v(\tilde{q}(\tilde{m})) - \tilde{T}(\tilde{m}), \quad \forall \tilde{m} \in M. \tag{16.4.26}$$

Then, the mechanism $\langle M, \tilde{g}(\cdot) \rangle$ determines the allocation rule $a(\theta) = (\tilde{q}(m^*(\theta)), \tilde{T}(m^*(\theta)))$ that maps Θ to $\mathcal{A}$.

A pertinent question is whether we can have a more complex mechanism to make the principal more profitable. The answer is negative. Consider a general mechanism $\langle M, \tilde{g}\rangle$. By choosing an optimal decision, the outcome can be compounded into a direct revelation mechanism through a composite function. This is illustrated by the "**revelation principle**" of the single agent scenario below. This result significantly reduces the complexity of finding an optimal mechanism. We only need to restrict our attention to direct revelation mechanisms.

Proposition 16.4.1 (Revelation Principle) *Any allocation rule $a(\theta)$ obtained by a general mechanism $\langle M, \tilde{g}\rangle$ can also be implemented with a truthful direct revelation mechanism.*

PROOF. The indirect mechanism $\langle M, \tilde{g}\rangle$ induces an allocation rule $a(\theta) = (\tilde{q}(m^*(\theta)), \tilde{T}(m^*(\theta)))$ from Θ into $\mathcal{A}$. By composition of $\tilde{g}(\cdot)$ and $m^*(\cdot)$, we can construct a direct revelation mechanism $g(\cdot)$ mapping Θ into $\mathcal{A}$, i.e., $g = \tilde{g} \circ m^*$, or more precisely $g(\theta) = (q(\theta), T(\theta)) \equiv (\tilde{q}(m^*(\theta)), \tilde{T}(m^*(\theta))), \forall\theta \in \Theta$.

We now verify that the direct revelation mechanism $g(\cdot)$ is truthful. Indeed, since (16.4.26) is true for all $\tilde{m}$, it holds in particular for $\tilde{m} = m^*(\theta'), \forall\theta' \in \Theta$. Then, we have

$$\theta v(\tilde{q}(m^*(\theta))) - \tilde{T}(m^*(\theta)) \geqq \theta v(\tilde{q}(m^*(\theta')) - \tilde{T}(m^*(\theta')),\ \forall(\theta, \theta') \in \Theta^2. \tag{16.4.27}$$

Finally, by the definition of $g(\cdot)$, we get

$$\theta v(q(\theta)) - T(\theta) \geqq \theta v(q(\theta')) - T(\theta'), \quad \forall(\theta, \theta') \in \Theta^2. \tag{16.4.28}$$

Therefore, the direct revelation mechanism $g(\cdot)$ is truthful. □

The revelation principle can be illustrated by Figure 16.4.

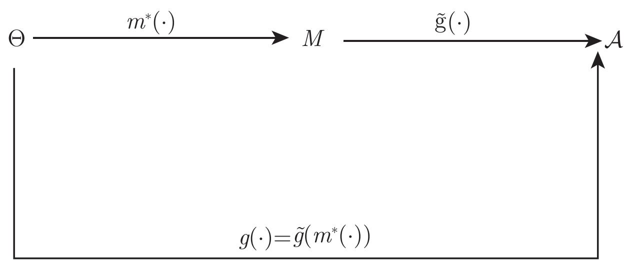

Figure 16.4: Revelation principle

Thus, the revelation principle significantly simplifies contract design, which allows us to restrict the search to truthful direct revelation mechanisms.

16.5 Extensions of the Basic Model

In this section, we extend the basic model to the one with a more general type space. We first consider the economic environment with a finite number of discrete types, and then discuss the case of continuum types.

16.5.1 Finite Types

We again discuss the monopolist's choice. The consumer's objective function is the same as previously:

$$u(q, T, \theta_i) = \theta_i v(q) - T.$$

However, here we assume that there are at least three types:

$$\theta_n > \theta_{n-1} > \cdots > \theta_1, \quad n \geqq 3.$$

Assume that the probability of taking θ_i is β_i. Consider that the monopolist offers different contracts (q_i, T_i) for different types, and the monopolist's optimal choice is determined by the following problem:

$$\max_{\{(q_i,T_i)\}} \sum_{i=1}^{n}(T_i - cq_i)\beta_i \tag{16.5.29}$$

$$\text{s.t.} \quad \theta_i v(q_i) - T_i \geqq \theta_i v(q_j) - T_j, \quad \forall i, j, \tag{16.5.30}$$

$$\theta_i v(q_i) - T_i \geqq 0, \quad \forall i, \tag{16.5.31}$$

where (16.5.30) and (16.5.31) are the incentive-compatibility constraint and the participation constraint, respectively, for type θ_i. Below, we focus on the separating equilibrium.

Adding the incentive-compatibility conditions for types θ_i and θ_j leads to:

$$(\theta_i - \theta_j)(v(q_i) - v(q_j)) \geqq 0.$$

Under the Spence-Mirrlees single-crossing condition, as the consumer's utility level is increasing in θ_i, the consumption of the agent is also increasing. Thus, when $\theta_i > \theta_j$, we must have $q_i \geqq q_j$.

Definition 16.5.1 A menu of contracts satisfies the **local downward incentive constraints** if

$$\theta_i v(q_i) - T_i \geqq \theta_i v(q_{i-1}) - T_{i-1}, \forall \theta_i > \theta_1. \tag{16.5.32}$$

It satisfies the **downward incentive constraints** if

$$\theta_i v(q_i) - T_i \geqq \theta_i v(q_j) - T_j, \forall \theta_i > \theta_j.$$

Let $q_0 = T_0 = 0$ denote a contract that does not provide goods. Then, θ_1's participation constraint can be written in the form of local downward incentive constraint:

$$\theta_1 v(q_1) - T_1 \geqq \theta_1 v(q_0) - T_0 \equiv 0.$$

When the incentive-compatibility constraints and the participation constraint of θ_1 hold, we have

$$\theta_i v(q_i) - T_i \geqq \theta_i v(q_1) - T_1 \geqq \theta_1 v(q_1) - T_1 \geqq 0,$$

which means that the participation constraints of other types $\theta_i > \theta_1$ are also satisfied.

θ_1's participation constraint must be binding. If not, a small amount added for all T_i will still make all of the participation constraints and incentive-compatibility constraints satisfied, but this will increase the monopolist's profit.

In the following, we show that, as long as local downward incentive constraints hold, global incentive constraints must also be satisfied. We first confirm that if local downward incentive constraints and the monotonicity condition hold, then downward incentive constraints are satisfied, i.e., $\theta_i v(q_i) - T_i \geqq \theta_i v(q_j) - T_j, \forall \theta_i > \theta_j$.

Indeed, if the local downward incentive constraints hold for $\theta_i, \cdots, \theta_j > \theta_1$, then for all $\forall \theta_k \in \{\theta_i, \cdots, \theta_{j+1}\}$, we have

$$\theta_k v(q_k) - T_k \geqq \theta_k v(q_{k-1}) - T_{k-1}.$$

Since $\theta_k \leqq \theta_i$, from monotonicity condition $q_k \leqq q_i$ and the above inequality, we have

$$\theta_i v(q_k) - T_k \geqq \theta_i v(q_{k-1}) - T_{k-1}$$

for all $k \in \{i, i-1, \cdots, j+1\}$.

Letting $k = i, i-1, \cdots, j+1$, and making summation, we have:

$$\theta_i v(q_i) - T_i \geqq \theta_i v(q_j) - T_j, \ \forall \theta_i > \theta_j.$$

Therefore, downward incentive constraints are satisfied.

We further verify that for all $\theta_i > \theta_1$, the local downward incentive constraints must be binding:

$$\theta_i v(q_i) - T_i = \theta_i v(q_{i-1}) - T_{i-1};$$

otherwise, for all $j \geqq i$, we can increase T_j slightly, and all incentive-compatibility constraints and participation constraints still hold, but will increase the monopolist's profit.

Finally, we verify that if all local downward incentive constraints are binding and the monotonicity condition is satisfied, then all upward incentive constraints also hold. This is because

$$\theta_k v(q_k) - T_k = \theta_k v(q_{k-1}) - T_{k-1},$$

or

$$\theta_k [v(q_k) - v(q_{k-1})] - T_k = -T_{k-1},$$

$q_{k-1} \leqq q_k$, $\theta_k \geqq \theta_i$ and $v(q_{k-1})$ is monotonic, then for all $k \in \{i+1, \cdots, j\}$, we have

$$\theta_i v(q_k) - T_k \leqq \theta_i v(q_{k-1}) - T_{k-1}.$$

Letting $k = i+1, \cdots, J$ in the above inequality and making the summation, we have

$$\theta_i v(q_j) - T_j \leqq \theta_i v(q_i) - T_i, \ \forall \theta_j > \theta_i.$$

Thus, provided local downward incentive constraints hold, global incentive constraints must also be satisfied.

Therefore, for finite types of the agent, the monopolist's optimal choice of contract is actually a solution to the following problem:

$$\max_{\{(q_i, T_i)\}} \sum_{i=1}^{n} (T_i - cq_i)\beta_i \tag{16.5.33}$$

$$\text{s.t.} \quad \theta_i v(q_i) - T_i = \theta_i v(q_{i-1}) - T_{i-1}, q_0 = T_0 = 0, \ \forall i, \tag{16.5.34}$$

$$q_i \geqq q_j, \ i > j. \tag{16.5.35}$$

The Lagrange function is

$$L = \sum_{i=1}^{n} [T_i - cq_i]\beta_i + \sum_{i=1}^{n} \lambda_i [\theta_i v(q_i) - \theta_i v(q_{i-1}) - T_i + T_{i-1}],$$

and we then have the following first-order conditions:

$$\begin{aligned} (\lambda_i \theta_i - \lambda_{i+1}\theta_{i+1}) v'(q_i) &= c\beta_i, \quad i < n; \\ \lambda_n \theta_n v'(q_n) &= c\beta_n; \\ \beta_i &= \lambda_i - \lambda_{i+1}, \quad i < n; \\ \beta_n &= \lambda_n. \end{aligned}$$

Therefore, we have the following conclusions:

Proposition 16.5.1 *Under asymmetric information, for the case of a finite number of types, the optimal contracts entail:*

(1) For the highest-type consumer, we have $\theta_n v'(q_n) = c$, and thus there is no consumption distortion.

(2) For other types $\theta_i < \theta_n$, there is consumption distortion

$$\theta_i v'(q_i) = c \frac{\lambda_i - \lambda_{i+1}}{\lambda_i - \lambda_{i+1}\frac{\theta_{i+1}}{\theta_i}} > c.$$

(3) For the lowest-type consumer, there is no information rent, while other types' information rents are positive, and as the type changes from the lowest to the highest, its information rent also increases. For $\theta_i > \theta_1$, the information rent is $\sum_{j=2}^{i}(\theta_j - \theta_{j-1})v(q_{j-1})$.

From two types to more than two types, we find that the basic trade-off between information rent extraction and allocative efficiency still holds.

16.5.2 Continuum of Types

We now consider the economic environment with a continuum of types. Most work in the principal-agent literature is done within this setting.

Consider the standard model with θ in the interval $\Theta = [\underline{\theta}, \overline{\theta}]$. Since the revelation principle is still valid with a continuum of types, and we can restrict the analysis to direct revelation mechanisms $\{(q(\theta), T(\theta))\}$.

Assume that the type of consumer (agent) follows the density function $f(\theta)$ (distribution function denoted by $F(\theta)$), and the range of θ is $[\underline{\theta}, \bar{\theta}]$. The principal chooses the contract that solves the following optimization problem:

$$\max_{\{(q(\theta),T(\theta))\}} \int_{\underline{\theta}}^{\bar{\theta}} (T(\theta) - cq(\theta))f(\theta)d\theta \tag{16.5.36}$$

$$\text{s.t.} \qquad \theta v(q(\theta)) - T(\theta) \geqq \theta v(q(\hat{\theta})) - T(\hat{\theta}),\ \forall(\theta, \hat{\theta}) \in \Theta^2, \tag{16.5.37}$$

$$\theta v(q(\theta)) - T(\theta) \geqq 0, \quad \forall \theta \in \Theta. \tag{16.5.38}$$

As the same as before, the participation constraints given by (16.5.38) is satisfied if the participation constraint of the lowest type is satisfied:

$$\underline{\theta} v(q(\underline{\theta})) - T(\underline{\theta}) \geqq 0.$$

For the incentive-compatibility constraints, from the inequality (16.5.37), we have:

$$(\theta - \hat{\theta})(v(q(\theta)) - v(q(\hat{\theta}))) \geqq 0.$$

Thus, incentive-compatibility alone requires that $q(\cdot)$ must be nondecreasing, which implies that $q(\cdot)$ is differentiable almost everywhere. Therefore, we will restrict the analysis to differentiable outcome functions. Then, the monotonicity condition implies $\frac{dq(\theta)}{d\theta} \geqq 0$.

The incentive-compatibility requirement is equivalent to the following problem:

$$\theta = \text{argmax}_{\hat{\theta}}[\theta v(q(\hat{\theta})) - T(\hat{\theta})].$$

The first-order necessary condition is:

$$\theta v'(q(\theta))\frac{dq(\theta)}{d\theta} - T'(\theta) = 0. \tag{16.5.39}$$

(16.5.39) is also called the **local incentive-compatibility condition**.

The second-order condition is:

$$\theta v''(q(\theta))\left(\frac{dq(\theta)}{d\theta}\right)^2 + \theta v'(q(\theta))\frac{d^2q(\theta)}{d\theta^2} - T''(\theta) \leqq 0. \tag{16.5.40}$$

Differentiating (16.5.39) with respect to θ, we have

$$\theta v''(q(\theta))\left(\frac{dq(\theta)}{d\theta}\right)^2 + v'(q(\theta))\ dfracdq(\theta)d\theta + \theta v'(q(\theta))\frac{d^2q(\theta)}{d\theta^2} - T''(\theta) = 0. \tag{16.5.41}$$

By comparing (16.5.40) with (16.5.41) and using $v'(q(\theta)) > 0$, the second-order condition turns out to be equivalent to the monotonicity condition $\frac{dq(\theta)}{d\theta} \geqq 0$.

We now show that the local incentive-compatibility condition implies that the global incentive-compatibility constraints must be satisfied:

$$\theta v(q(\theta)) - T(\theta) \geqq \theta v(q(\hat{\theta})) - T(\hat{\theta}), \quad \forall(\theta, \hat{\theta}) \in \Theta^2. \tag{16.5.42}$$

By the first-order condition (16.5.39) and integration by parts, we have:

$$\begin{aligned} T(\theta) - T(\hat{\theta}) &= \int_{\hat{\theta}}^{\theta} \tau v'(q(\tau))\frac{dq(\tau)}{d\tau}d\tau \\ &= \theta v(q(\theta)) - \hat{\theta}v(q(\hat{\theta})) - \int_{\hat{\theta}}^{\theta} v(q(\tau))d\tau, \end{aligned} \tag{16.5.43}$$

or

$$\theta v(q(\theta)) - T(\theta) = \hat{\theta}v(q(\hat{\theta})) - T(\hat{\theta}) + \int_{\hat{\theta}}^{\theta} v(q(\tau))d\tau - (\theta - \hat{\theta})v(q(\hat{\theta})). \tag{16.5.44}$$

Since $q(\cdot)$ is non-decreasing, $\int_{\hat{\theta}}^{\theta} v(q(\tau))d\tau - (\theta - \hat{\theta})v(q(\hat{\theta})) \geqq 0$. Thus, as long as the local incentive-compatibility constraint (16.5.39) holds, the global incentive-compatibility constraint (16.5.42) also holds.

Within the above settings, continuum incentive-compatibility constraints (16.5.42) simplify to a differential equation and a monotonicity constraint. As such, *the first-order condition (16.5.39) and the monotonicity condition* $\frac{dq(\theta)}{d\theta} \geqq 0$ *fully characterize the truthful revelation mechanism*.

The monopolist's problem can then be rewritten as follows:

$$\max_{\{(q(\theta),T(\theta))\}} \int_{\underline{\theta}}^{\bar{\theta}} (T(\theta) - cq(\theta))f(\theta)d\theta \tag{16.5.45}$$

$$\text{s.t.} \quad \theta v'(q(\theta))\frac{dq(\theta)}{d\theta} - T'(\theta) = 0, \quad \forall(\theta, \hat{\theta}) \in \Theta^2, \tag{16.5.46}$$

$$\frac{dq(\theta)}{d\theta} \geqq 0, \quad \forall \theta \in \Theta, \tag{16.5.47}$$

$$\underline{\theta}v(q(\underline{\theta})) - T(\underline{\theta}) \geqq 0, \quad \forall \theta \in \Theta. \tag{16.5.48}$$

To solve the above problem, the usual procedure is to ignore the monotonicity constraints at the moment. Define the information rent function for θ:

$$U(\theta) = \theta v(q(\theta)) - T(\theta) = \max_{\hat{\theta}} \theta v(q(\hat{\theta})) - T(\hat{\theta}).$$

Using the envelope theorem, we obtain

$$\frac{dU(\theta)}{d\theta} = v(q(\theta)).$$

By monotonicity of objective function, the participation constraint of the lowest type is binding, which implies that his information rent is 0, i.e., $U(\underline{\theta}) = 0$. Thus, we have

$$U(\theta) = \int_{\underline{\theta}}^{\theta} v(q(\tau))d\tau.$$

Since $T(\theta) = \theta v(q(\theta)) - U(\theta)$, we can write the monopolist's objective function as:

$$\Pi = \int_{\underline{\theta}}^{\bar{\theta}} \left[\theta v(q(\theta)) - \int_{\underline{\theta}}^{\theta} v(q(\theta))d\theta - cq(\theta)\right] f(\theta)d\theta.$$

By integrating by parts, we have

$$\begin{aligned}
\Pi &= \int_{\underline{\theta}}^{\bar{\theta}} \{[\theta v(q(\theta)) - cq(\theta)]f(\theta) - v(q(\theta))[1 - F(\theta)]\}f(\theta)d\theta \\
&= \int_{\underline{\theta}}^{\bar{\theta}} \left[\left(\theta - \frac{1 - F(\theta)}{f(\theta)}\right) v(q(\theta)) - cq(\theta))\right] f(\theta)d\theta \qquad (16.5.49)
\end{aligned}$$

The first-order condition for $q(\theta)$ is then given by

$$\left[\theta - \frac{1 - F(\theta)}{f(\theta)}\right] v'(q(\theta)) = c. \qquad (16.5.50)$$

Therefore, we obtain the basic conclusion:

For the highest type $\theta = \bar{\theta}$, there is no distortion in consumption, while for other types of $\theta < \bar{\theta}$, there is a downward distortion in consumption.

Let

$$\nu(\theta) = \theta - \frac{1 - F(\theta)}{f(\theta)},$$

which is known as the **virtual valuation function**. The first-order condition (19.6.14) can then be simply written as:

$$\nu(\theta)v'(q(\theta)) = c,$$

which means that for the principal to maximize her profit, she will use the virtual value $\nu(\theta)$ instead of the true type of the agent in the presence of asymmetric information. The virtual value of the agent is less than his true value because the latter includes the information rent of the agent and the principal cannot extract all of the information rent.

Finally, it is necessary to provide the condition that can guarantee the monotonicity of $q(\theta)$ for fulfilling a complete characterization of the optimal contract. Recall that $h(\theta) \equiv \frac{f(\theta)}{1-F(\theta)}$ in the above first-order condition is termed the hazard rate. A sufficient condition for the monotonicity constraint is $\frac{dh(\theta)}{d\theta} \geqq 0$.

To see this, differentiating the first-order equation with respect to θ, we get

$$\frac{dq(\theta)}{d\theta} = -\frac{\nu'(\theta)v'(q(\theta))}{v''(q(\theta))g(\theta)}.$$

Thus, as long as $\frac{dh(\theta)}{d\theta} \geqq 0$, then $\nu'(\theta) \geqq 0$, hence the monotonicity condition is satisfied. For many continuous distributions, such as uniform distribution, normal distribution, exponential distribution, and other frequently used distributions, monotonicity of the hazard rate is satisfied.

16.5.3 Bunching and Ironing

We now discuss how to handle the optimal choice problem if the monotonicity of $q(\theta)$ is not satisfied. The usual method is to “**iron out**” intervals that do not satisfy monotonicity.

Without considering monotonicity, the previous optimal contract $q^*(\theta)$ satisfies:

$$\left[\theta - \frac{1-F(\theta)}{f(\theta)}\right] v'(q^*(\theta)) = c.$$

Suppose that in the interval $[\hat{\theta}_1, \hat{\theta}_2]$, $q^*(\theta)$ does not satisfy monotonicity, as shown in Figure 16.5.

The appearance of this non-monotonic contract may be due to the fact that the hazard rate $h(\theta)$ is not monotonic. The monopolist's optimal contract $\bar{q}(\theta)$ is the solution to the following problem:

$$\max_{\{(q(\theta),T(\theta))\}} \int_{\underline{\theta}}^{\bar{\theta}} \left[\theta v(q(\theta)) - cq(\theta) - \frac{v(q(\theta))}{h(\theta)}\right] f(\theta) d\theta$$

$$\text{s.t.} \quad \frac{dq(\theta)}{d\theta} = \mu(\theta),$$

$$\mu(\theta) \geqq 0.$$

The Hamiltonian is:

$$H(\theta, q(\theta), \mu, \lambda) = \left[\theta v(q(\theta)) - cq(\theta) - \frac{v(q(\theta))}{h(\theta)}\right] f(\theta) + \lambda(\theta)\mu(\theta).$$

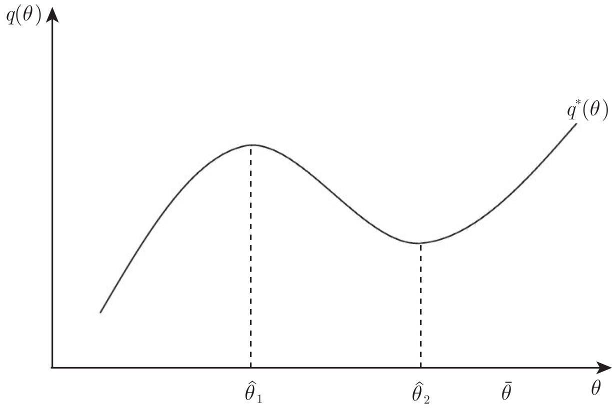

Figure 16.5: Contracts that violate the monotonicity condition

Using Pontryagin's maximum principle, the optimal solution $(\bar{q}(\theta), \bar{\mu}(\theta))$ satisfies the following conditions:

$$H(\theta, \bar{q}(\theta), \bar{\mu}(\theta), \lambda(\theta)) \geqq H(\theta, q(\theta), \mu(\theta), \lambda(\theta)); \tag{16.5.51}$$

$$\frac{d\lambda(\theta)}{d\theta} = -\left[\left(\theta - \frac{1}{h(\theta)}\right) v'(\bar{q}(\theta)) - c\right] f(\theta), \tag{16.5.52}$$

$$\lambda(\underline{\theta}) = \lambda(\bar{\theta}) = 0 \quad \text{(transversality conditions)} \tag{16.5.53}$$

for all continuous points.

From Kamien and Schwartz (1991), when $H(\theta, q(\theta), \mu, \lambda)$ is a concave function with respect to $q(\theta)$, the above necessary conditions are also sufficient conditions.

Applying the formula of integration by parts to (16.5.52) shows:

$$\lambda(\theta) = -\int_{\underline{\theta}}^{\theta} \left[\left(\theta - \frac{1}{h(\theta)}\right) v'(\bar{q}(\theta)) - c\right] f(\theta) d\theta.$$

The transversality conditions are:

$$0 = \lambda(\underline{\theta}) = \lambda(\bar{\theta}) = -\int_{\underline{\theta}}^{\bar{\theta}} \left[\left(\theta - \frac{1}{h(\theta)}\right) v'(\bar{q}(\theta)) - c\right] f(\theta) d\theta.$$

The first-order condition for optimization requires $\bar{\mu}(\theta) \geqq 0$ when maximizing $H(\theta, q(\theta), \mu, \lambda)$, which implies that $\lambda(\theta) \leqq 0$ or

$$\int_{\underline{\theta}}^{\theta} \left[\left(\theta - \frac{1}{h(\theta)}\right) v'(\bar{q}(\theta)) - c\right] f(\theta) d\theta \geqq 0.$$

As long as $\lambda(\theta) < 0$, we must have

$$\bar{\mu}(\theta) = \frac{d\bar{q}(\theta)}{d\theta} = 0.$$

We then have the complementary slackness condition

$$\bar{\mu}(\theta)\lambda(\theta) = 0,$$

which implies that

$$\frac{d\bar{q}(\theta)}{d\theta}\int_{\underline{\theta}}^{\theta}\left[\left(\theta - \frac{1}{h(\theta)}\right)v'(\bar{q}(\theta)) - c\right]f(\theta)d\theta = 0.$$

From the above equation, we know that, if $\bar{q}(\theta)$ is increasing in an interval, $\bar{q}(\theta)$ must be the same as $q^*(\theta)$. Indeed, if $\bar{\mu}(\theta) = \frac{d\bar{q}(\theta)}{d\theta} > 0$, then $\lambda(\theta) = 0$, and so $\frac{d\lambda(\theta)}{d\theta} = 0$. Then, we have $[(\theta - \frac{1}{h(\theta)})v'(\bar{q}(\theta)) - c] = 0$, which is the first-order condition for $q^*(\theta)$. Thus, we must have $\bar{q}(\theta) = q^*(\theta)$.

What we need to do is to determine under which interval $\bar{q}(\theta)$ is constant. As shown in Figure 16.6, for θ_1 on the left side of $\hat{\theta}_1$ and θ_2 on the right side of $\hat{\theta}_2$, we have $\lambda(\theta) = 0$ and $\bar{\mu}(\theta) = \frac{d\bar{q}(\theta)}{d\theta} = \frac{dq^*(\theta)}{d\theta} > 0$ for $\theta < \theta_1$ or $\theta > \theta_2$. However, for $\theta \in [\theta_1, \theta_2]$, we have $\lambda(\theta) < 0$ and $\bar{\mu}(\theta) = \frac{d\bar{q}(\theta)}{d\theta} = 0$, which means that $\bar{q}(\theta)$ is constant on $[\theta_1, \theta_2]$.

By the continuity of $\lambda(\theta)$, we know that $\lambda(\theta_1) = \lambda(\theta_2) = 0$, and thus

$$\int_{\theta_1}^{\theta_2}\left[\left(\theta - \frac{1}{h(\theta)}\right)v'(\bar{q}(\theta)) - c\right]f(\theta)d\theta = 0. \tag{16.5.54}$$

On the other hand, by the continuity of $\bar{q}(\theta)$, we have

$$q^*(\theta_1) = q^*(\theta_2). \tag{16.5.55}$$

From equations (16.5.54) and (16.5.55), we can determine two unknowns, θ_1 and θ_2, respectively. The new optimal contract $\bar{q}(\theta)$ is shown in Figure 16.6.

Therefore, the optimal contract actually "irons" the original $q^*(\theta)$ in the interval $[\theta_1, \theta_2]$. This interval is called the bunching interval since $\bar{q}(\theta)$ is constant on it. For further discussion, see Bolton and Dewatripont (2005).

In addition, another extension is a multi-dimensional adverse selection issue. One of the classic problems is the monopoly pricing of multi-products. For example, in practice, numerous merchants often bundle different products. The multi-dimensional case of adverse selection provides an analytical framework for understanding similar bundled sales mechanisms. A detailed discussion of this can be found in Bolton and Dewatripont (2005), as well as the survey by Rochet and Stole (2003).

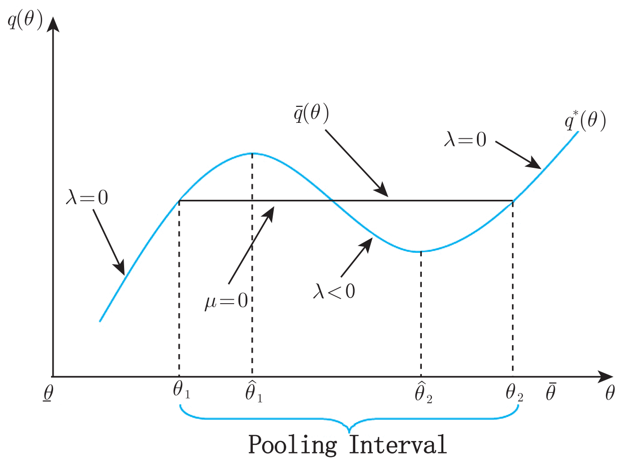

Figure 16.6: Optimal contract with bunching property

16.6 Ex-Ante Participation Constraints

The contracts that we have considered so far are offered at the interim stage, i.e., the agent already knows his type. However, sometimes the principal and the agent can contract ex-ante, i.e., before the agent is informed of his type. For instance, the contracts of the firm may be designed before the agent receives the exact information on his productivity, e.g., a newly graduated student does not know his own productivity, or whether he is competent for a particular job.

Moreover, in the above example of monopoly sales, if the monopoly upstream firm provides a new production tool for the downstream firm, the downstream firm does not know the true effectiveness of this new production tool prior to signing a sales contract, and their interactions will have different characteristics than previously. In this section, we characterize the optimal contract for this alternative timing structure under various assumptions about the risk attitude of the participants.

Unlike the previous model, prior to signing the contract, the buyer did not know the type of value that he had for the product, and the timing of interaction is:

- $t = 0$: The seller provides a contract;
- $t = 1$: The buyer accepts or rejects the contract;
- $t = 2$: The buyer is informed of his own type θ;
- $t = 3$: The contract is executed.

To simplify the discussion, we still assume that there are only two types, $\theta \in \{\theta_H, \theta_L\}$. The agent's attitude toward risk will affect the contract design. For this reason, we discuss the optimal incentive-compatible contract when the agent is risk-neutral and risk-averse, respectively.

16.6.1 Risk Neutrality

Suppose that the principal and the agent meet and contract ex-ante. If the agent is risk-neutral, his **ex-ante participation constraint** is now written as

$$(1-\beta)U_H + \beta U_L \geqq 0. \tag{16.6.56}$$

This ex-ante participation constraint replaces the two interim participation constraints.

Since the principal's objective function is decreasing in the agent's expected information rent, the principal wants to impose a zero expected rent to the agent so that (16.6.56) is binding. Moreover, the principal must structure the rents U_H and U_L to ensure that the two incentive constraints remain satisfied.

$$U_H \geqq U_L + \Delta\theta v(q_L), \tag{16.6.57}$$

$$U_L \geqq U_H - \Delta\theta v(q_H). \tag{16.6.58}$$

A typical optimal contract of such a rent distribution that is both incentive-compatible and satisfies the ex-ante participation constraint with an equality is

$$\begin{aligned} U_H^* &= \beta\Delta\theta v(q_H^*) > 0; \\ U_L^* &= -(1-\beta)\Delta\theta v(q_H^*) < 0. \end{aligned} \tag{16.6.59}$$

With such a rent distribution, the optimal contract implements the first-best outputs without cost from the principal's point of view, as long as the first-best is monotonic as required by the implementability condition. In the contract defined by (16.6.59), the agent is rewarded when he is efficient and punished when he turns out to be inefficient.

In fact, the principal has many more options in structuring the rents U_H and U_L in such a way that the incentive-compatibility constraints hold and the ex-ante participation constraint (16.6.56) is binding. As shown in Figure 16.7, the information rent distribution is not unique, and in fact, there are infinitely many possibilities.

In summary, we then have the following proposition.

Proposition 16.6.1 *When the agent is risk-neutral and contracting takes place ex-ante, the optimal incentive contract implements the first-best outcomes.*

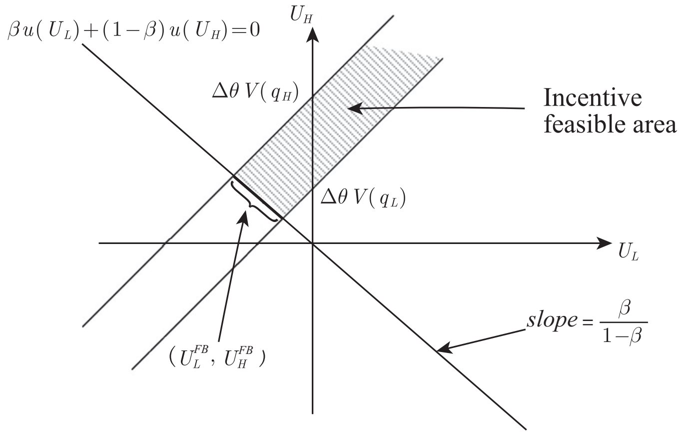

Figure 16.7: Information rent of a risk-neutral agent

Consider the following more common contracts with a lump-sum payment to the principal: $\{(T_H^*, q_H^*); (T_L^*, q_L^*)\}$, where q_H^* and q_L^* are optimal productions, $T_H^* = cq_H^* + T^*$ and $T_L^* = cq_L^* + T^*$, with T^* being a lump-sum payment to be defined below. By definition of q_H^*, we have

$$\begin{aligned} \theta_H v(q_H^*) - T_H^* &= \theta_H v(q_H^*) - cq_H^* - T^* \\ &> \theta_H v(q_L^*) - cq_L^* - T^* = \theta_H v(q_L^*) - T_L^*. \quad (16.6.60) \end{aligned}$$

By definition of q_L^*, we have:

$$\begin{aligned} \theta_L v(q_L^*) - T_L^* &= \theta_L v(q_L^*) - cq_L^* - T^* \\ &> \theta_L v(q_H^*) - cq_H^* - T^* = \theta_L v(q_H^*) - T_H^*. \quad (16.6.61) \end{aligned}$$

Thus, the contracts are incentive-compatible.

Note that the incentive-compatibility constraints are now strict inequalities. Moreover, the fixed-fee T^* can be used to make the agent's ex-ante participation constraint binding by choosing

$$T^* = \beta(\theta_L v(q_L^*) - cq_L^*) + (1-\beta)(\theta_H v(q_H^*) - cq_H^*).$$

This implementation of the first-best outcome amounts to having the principal selling the benefit of the relationship to the risk-neutral agent for a lump-sum payment T^*. The agent benefits from the full value of the good and trades off the value of any production against its cost just as if he was an efficiency maximizer. We will assume that the agent is a residual claimant for the profit. From a later discussion, it is known that even a principal who is risk-averse will obtain the same result. In practice, many contracts comprise such features.

(1) A building owner leases a building to a renter to do business, the renter pays a certain rent, and the remaining profits or losses are borne by the renter himself or herself.

(2) A bank lends funds to enterprises at a fixed loan interest rate, and companies bear business risks. Profits or losses are all their own.

(2) After reform and opening-up in China, contracts were implemented for contracting land and households, and production teams contracted the farmland to the farmers. The farmers needed to pay a certain amount of products to the government each year, and the rest was kept by the farmers. The subsequent production responsibility system is the same.

16.6.2 Risk-Aversion

We just showed that the implementation of the first-best is feasible with risk-neutrality. Then, what happens if the agent is risk-averse?

Now, consider a risk-averse agent with a Von Neumann-Morgenstern utility function $u(\cdot)$ defined on his monetary gains $v(q)-T$, such that $u' > 0$, $u'' < 0$ and $u(0) = 0$. Again, the contract between the principal and the agent is signed before the agent knows his type. The incentive-compatibility constraints are the same as previously, but the agent's ex-ante participation constraint is now written as:

$$\beta u(U_L) + (1-\beta)u(U_H) \geqq 0. \tag{16.6.62}$$

As usual, one can check that the incentive-compatibility constraint for the low-type is slack (not binding) at the optimum, the high-type's incentive-compatibility constraint (16.6.57) is binding, and the ex-ante participation constraint (16.6.62) is also binding, so that

$$\beta u(U_L) + (1-\beta)u(U_L + \Delta\theta v(q_L)) = 0. \tag{16.6.63}$$

and thus the principal's program now reduces to

$$\max_{\{(U_H,q_H);(U_L,q_L))\}} (1-\beta)(\theta_H v(q_H) - cq_H - U_H) + \beta(\theta_L v(q_L) - cq_L - U_L)$$

subject to (16.6.57), (16.6.58) and (16.6.62).

We then have the following proposition.

Proposition 16.6.2 *When the agent is risk-averse and contracting takes place ex-ante, the optimal menu of contracts entails:*

(1) No output distortion for the high-type, $q_H^{SB} = q_H^*$. A downward output distortion for the low type, $q_L^{SB} < q_L^*$, with

$$\theta_L v'(q_L^{SB}) = \frac{c}{1 - \frac{(1-\beta)[u'(U_L) - u'(U_H)]}{\beta u'(U_L) + (1-\beta)u'(U_H)} \frac{\Delta\theta}{\theta_L}}. \tag{16.6.64}$$

(2) The high-type's incentive-compatibility constraint (16.6.57) and the ex-ante participation constraint (16.6.62) are the only binding constraints. The high (resp. low) type receives a strictly positive (resp. negative) ex-post information rent, $U_H^{SB} > 0 > U_L^{SB}$.

The rent distribution is shown in Figure 16.8. Note that when the agent is risk-neutral, the second term in the right of (16.6.64) is zero, and thus we obtain the same conclusion as in Proposition 16.6.1: The optimal incentive contract implements the first-best outcome.

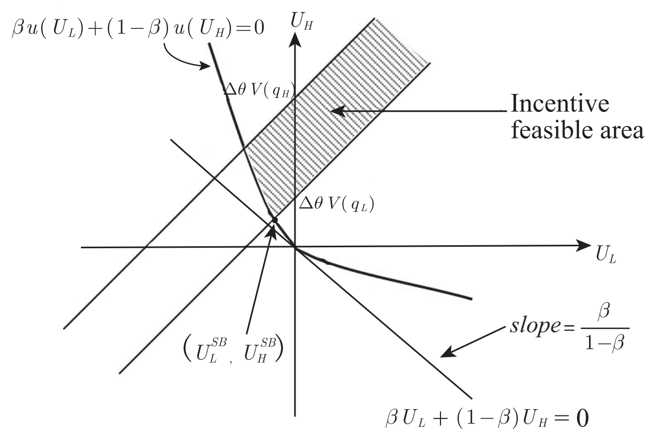

Figure 16.8: Information rent of the risk-averse agent

Thus, with risk-aversion, the principal can no longer costlessly structure the agent's information rents to ensure that the high-type's incentive-compatibility constraint is satisfied. Creating a difference between U_H and U_L to satisfy the incentive-compatibility constraints makes the risk-averse agent bear some risk. To guarantee the participation of the risk-averse agent, the principal must now pay a risk premium. Reducing this premium calls for a downward reduction in the low type's output, so that the risk borne by the agent is lower. As anticipated, the agent's risk-aversion leads the principal to weaken the incentives.

When the agent becomes infinitely risk-averse, the agent's ex-ante individual rationality constraint is the same as his ex-post individual rationality constraint for the worst state of the world, such that $U_L^{SB} = 0$ but not

$U_L^{SB} < 0$. As such, in the limit, the agent's outputs q_H^{SB} and q_L^{SB} and the utility levels U_H^{SB} and U_L^{SB} all converge toward the same solution as for the interim case discussed in the previous sections. Therefore, the previous model at the interim stage can also be interpreted as a model with an ex-ante infinitely risk-averse agent at the zero utility level.

16.6.3 Risk-Averse Principal

Now, consider a risk-averse principal with a von Neumann-Morgenstern utility function $v(\cdot)$ defined on her monetary gains from trade, $T - cq$, such that $v' > 0$, $v'' < 0$ and $v(0) = 0$. Again, the contract between the principal and the risk-neutral agent is signed before the agent knows his type.

In this context, the first-best contract obviously calls for the first-best outputs q_H^* and q_L^*. It also calls for the principal to be fully insured between both states of nature and for the agent's ex-ante participation constraint to be binding. This leads to the following two conditions that must be satisfied by the agent's rents U_H^* and U_L^*:

$$\theta_H v(q_H) - U_H^* - cq_H = \theta_L v(q_L) - U_L^* - cq_L \tag{16.6.65}$$

and

$$\beta U_L^* + (1-\beta)U_H^* = 0. \tag{16.6.66}$$

Solving this system of two equations with two unknowns (U_H^*, U_L^*) yields

$$U_H^* = \beta\{[\theta_H v(q_H^*) - cq_H^*] - [\theta_L v(q_L^*) - cq_L^*]\}, \tag{16.6.67}$$

$$U_L^* = -(1-\beta)\{[\theta_H v(q_H^*) - cq_H^*] - [\theta_L v(q_L^*) - cq_L^*]\}. \tag{16.6.68}$$

By the definition of q_H^*,

$$U_H^* - U_L^* = [\theta_H v(q_H^*) - cq_H^*] - [\theta_L v(q_L^*) - cq_L^*] > \Delta\theta v(q_L^*).$$

By the definition of q_L^*,

$$U_L^* - U_H^* = [\theta_L v(q_L^*) - cq_L^*] - [\theta_H v(q_H^*) - cq_H^*] > -\Delta\theta v(q_H^*).$$

Therefore, the profile of rents U_H^* and U_L^* is incentive-compatible, and the first-best allocation is easily implemented in this framework. We can thus generalize the proposition for the risk-neutral case as follows:

Proposition 16.6.3 *Suppose that the principal is risk-averse over the monetary gains $T - cq$, the agent is risk-neutral, and contracting takes place ex-ante. Then, the optimal incentive contract implements the first-best outcome.*

Remark 16.6.1 It is interesting to note that U_H^* and U_L^* obtained in (16.6.67) and (16.6.68) are also the levels of rent obtained in (16.6.60) and (16.6.61), respectively. Indeed, the lump-sum payment $T^* = \beta(\theta_L v(q_L^*) - cq_L^*) + (1-\beta)(\theta_H v(q_H^*) - cq_H^*)$, which *allows the principal to make the risk-neutral* *agent residual claimant for the hierarchy's profit, also provides full insurance to the principal*. By making the risk-neutral agent the residual claimant for the value of trade, *ex-ante* contracting allows the risk-averse principal to get full insurance and implement the first-best outcome despite the informational friction.

Of course, this result no longer holds if the agent's interim participation constraint must be satisfied. In this case, the incentive-compatibility constraint (16.6.58) is slack at the optimum, and then the principal's program reduces to:

$$\max(1-\beta)\nu(\theta_H v(q_H) - cq_H - U_H) + \beta\nu(\theta_L v(q_L) - cq_L - U_L), \quad (16.6.69)$$

subject to the high-type's binding incentive-compatibility constraint (16.6.57), as well as the low type's binding participation constraint:

$$U_L = 0. \quad (16.6.70)$$

By substituting U_H and U_L obtained from (16.6.57) and (16.6.70) into the principal's objective function and maximizing the objective function, we have $q_H^{SB} = q_H^*$, which is the same as the result in the risk-neutral buyer situation. The high-type buyer has no consumption distortion, while the low-type buyer has a downward output distortion, $q_L^{SB} < q_L^*$, which satisfies

$$\theta_L v'(q_L^{SB}) = \frac{c}{1 - \dfrac{(1-\beta)\nu'(V_H)\Delta\theta}{\beta\nu'(V_L)\theta_L}}, \quad (16.6.71)$$

where $V_H^{SB} = \theta_H v(q_H^{SB}) - cq_H^{SB} - \Delta\theta v(q_L^{SB})$ and $V_L^{SB} = \theta_L v(q_L^{SB}) - cq_L^{SB}$ are the returns of the principal in two states. By the definition of q_H^*, we have

$$\theta_H v(q_L^{SB}) - cq_L^{SB} < \theta_H v(q_H^{SB}) - cq_H^{SB} = \theta_H v(q_H^*) - cq_H^*.$$

Thus, it can be verified that $V_L^{SB} < V_H^{SB}$. In particular, it is easy to see that the distortion on the right side of equation (16.6.71) is always less than $\frac{c}{1-\frac{(1-\beta)\Delta\theta}{\beta\theta_L}}$ in the case of a risk-neutral principal. The economic implication of this result is obvious. By increasing q_L^{SB}, the gap between V_H^{SB} and V_L^{SB} becomes smaller, which provides the principal with a certain amount of insurance and increase her ex-ante earnings.

16.7 Extensions of the Canonical Model

As shown in the preceding sections, the canonical principal-agent model makes several simplifying assumptions. First, there is no externality among agents, i.e., the agent's behavior is assumed to have no impact on the welfare of others; and second, the agent is assumed to obtain a minimal type-independent level of utility if he rejects the offer made by the principal. Under these assumptions, the optimal contract exhibits no-distortion for the "best" type agent and downward distortions for all other types.

However, as Meng and Tian (2009, 2021) showed, challenges to this "no distortion at the top" convention may arise if we relax either of these two assumptions. To introduce these results, in this section we will discuss these two "challenges" in an integrated nonlinear pricing model.

16.7.1 Network Externalities

An externality is present whenever the well-being of a consumer or the production possibilities of a firm are directly affected by the actions of others in the economy.

A special case of externality is the so-called "**network externalities**" that might arise for any of the following reasons: First, because the usefulness of the commodity depends directly on the size of the network (e.g., telephones, emails, Facebook (now known Meta), Tik Tok, WeChat, etc.); second, because of the **bandwagon effect**, which means the desire to be "in style", i.e., to have a good because of the perception that almost everyone else has it; or third, indirectly through the availability of complementary goods and services (often known as the "hardware-software paradigm") or after-sale services (e.g., for automobiles). Although "network externalities" are often regarded as a positive impact on each other's consumption, it may exhibit negative properties in some cases. For example, an individual has the desire to own exclusive or unique goods, which is called the "**snob effect**". The quantity demanded of a "snob" good is higher, the fewer is the number of people who own it (e.g., garments specially designed by a well-known fashion designer, special designs of watches, automobiles, buildings, etc.).

Consider a principal-agent model, in which the principal is a monopolist of a network good with marginal production cost c and total output q who faces a continuum of consumers. The principal's payoff function is given by $V = t - cq$, where t is the transfer received from a consumer. Consumers' preference of the good is characterized by $\theta \in \Theta = \{\underline{\theta}, \overline{\theta}\}, \Pr(\theta = \underline{\theta}) = v, \Pr(\theta = \overline{\theta}) = 1 - v$, and then v can be regarded as the frequency of type $\underline{\theta}$ by the Law of Large Numbers.

A consumer of θ type is assumed to have a utility function of $U(\theta) = \theta V(q(\theta)) + \Psi(Q) - t(\theta)$, where $q(\theta)$ is the amount of the good that is consumed, $Q = v\underline{q} + (1 - v)\overline{q} = E[q(\theta)]$ is the total amount of consumption (network size), and $t(\theta)$ is the tariff charged by the principal. $\theta V(q(\theta))$ is the *intrinsic value* of consuming, while $\Psi(Q)$ is the *network value*. Note that we assume that the network effect is homogeneous among all of the consumers, i.e., the network value is independent of individual preference θ and individual consumption $q(\theta)$. It is assumed that $V'(q) > 0$ and $V''(q) < 0$.

Definition 16.7.1 The network is **decreasing, constant**, or **increasing** if $\Psi''(Q) < 0$, $\Psi''(Q) = 0$, or $\Psi''(Q) > 0$, respectively.

Remark 16.7.1 When the network capacity is large and the maintaining technology is sufficiently advanced, an increase in one consumer's consumption will increase the marginal utilities of others, and thus $\Psi''(Q) > 0$. However, when the network capacity and the maintaining technology are limited, consumers are rivals to one another in the sense that an increase in one consumer's consumption will decrease the marginal utilities of others, and thus $\Psi''(Q) < 0$. When network expansion benefits all of the consumers with a constant margin, the network value term is a linear function: $\Psi''(Q) = 0$.

The objective of the monopolist is to design a menu of incentive-compatible and self-selecting quantity-price pairs $\{q(\hat{\theta}), t(\hat{\theta})\}$ to maximize her own revenue, where $\hat{\theta} \in \Theta$ is the consumer's announcement. The network magnitude is $Q = v\underline{q} + (1-v)\overline{q}$. Under complete information, the monopolist's problem is:

$$
(\text{P2})\begin{cases} \max\limits_{\{(\underline{U},\underline{q});(\overline{U},\overline{q})\}} v\Big[\underline{\theta}V(\underline{q}) - c\underline{q}\Big] + (1-v)\Big[\overline{\theta}V(\overline{q}) - c\overline{q}\Big] + \Psi(Q) - \Big[v\underline{U} + (1-v)\overline{U}\Big] \\ s.t.\ IR(\underline{\theta}) : \underline{U} \geqslant 0, \\ \qquad IR(\overline{\theta}) : \overline{U} \geqslant 0. \end{cases}
$$

The first-best consumption is thus:

$$
\begin{cases} \underline{\theta}V'(\underline{q}^{FB}) + \Psi'\left(v\underline{q}^{FB} + (1-v)\overline{q}^{FB}\right) = c, \\ \overline{\theta}V'(\overline{q}^{FB}) + \Psi'\left(v\underline{q}^{FB} + (1-v)\overline{q}^{FB}\right) = c. \end{cases} \tag{16.7.72}
$$

Under asymmetric information, two incentive-compatibility constraints should be added to the above program. Then, we obtain:

$$
(\text{P3})\begin{cases} \max\limits_{\{(\underline{U},\underline{q});(\overline{U},\overline{q})\}} v\Big[\underline{\theta}V(\underline{q}) - c\underline{q}\Big] + (1-v)\Big[\overline{\theta}V(\overline{q}) - c\overline{q}\Big] + \Psi(Q) - \Big[v\underline{U} + (1-v)\overline{U}\Big] \\ s.t.\ IR(\underline{\theta}) : \underline{U} \geqq 0, \\ \qquad IR(\overline{\theta}) : \overline{U} \geqq 0, \\ \qquad IC(\underline{\theta}) : \underline{U} \geqq \overline{U} - \Delta\theta V(\overline{q}), \\ \qquad IC(\overline{\theta}) : \overline{U} \geqq \underline{U} + \Delta\theta V(\underline{q}). \end{cases} .
$$

The incentive-compatibility constraint $IC(\overline{\theta})$ of the high-demand type and the participation constraint of the low-demand type $IR(\underline{\theta})$ are binding. Then, the consumptions in the optimal contract (i.e., second-best) are characterized by the following first-order conditions:

$$\begin{cases} \left(\underline{\theta} - \dfrac{1-v}{v}\Delta\theta\right) V'(\underline{q}^{SB}) + \Psi'\left(v\underline{q}^{SB} + (1-v)\overline{q}^{SB}\right) = c, \\ \overline{\theta}V'(\overline{q}^{SB}) + \Psi'\left(v\underline{q}^{SB} + (1-v)\overline{q}^{SB}\right) = c. \end{cases} \tag{16.7.73}$$

We synthesize the complete and incomplete information maximization problems by considering them as solutions to the following maximization problem

$$\max_{\{\underline{q},\overline{q}\}} \Pi(\underline{q}, \overline{q}, \alpha) \tag{16.7.74}$$

where

$$\Pi(\underline{q}, \overline{q}, \alpha) = v\left[\alpha V(\underline{q}) - c\underline{q}\right] + (1-v)\left[\overline{\theta}V(\overline{q}) - c\overline{q}\right] + \Psi(Q).$$

Note that we have the first-best contract under complete information given in (16.7.72) when $\alpha = \underline{\theta}$, and the second-best contract under asymmetric information given in (16.7.73) when $\alpha = \underline{\theta} - \frac{1-v}{v}\Delta\theta$.

We then have the following proposition.

Proposition 16.7.1 *In the presence of network externalities and asymmetric information, the direction of distortion in consumption depends on the sign of* $\Psi''(Q)$*.*

1. *If the network is mildly increasing, i.e.,* $\Psi''(Q) > 0$ *but is not too large such that* Π_{qq} *is negative-definite for all* $\alpha \in [\underline{\theta} - \frac{1-v}{v}\Delta\theta, \underline{\theta}]$*, then the consumption of all types exhibits one-way distortion:* $\underline{q}^{SB} < \underline{q}^{FB}$ *and* $\overline{q}^{SB} < \overline{q}^{FB}$*.*

2. *If the network is decreasing, i.e.,* $\Psi''(Q) < 0$*, then the consumption exhibits two-way distortion:* $\underline{q}^{SB} < \underline{q}^{FB}$ *and* $\overline{q}^{SB} > \overline{q}^{FB}$*.*

3. *If the network is constant, i.e.,* $\Psi''(Q) = 0$*, then the rules "no distortion on the top" and "one-way distortion" in canonical settings are still available:* $\underline{q}^{SB} < \underline{q}^{FB}$*,* $\overline{q}^{SB} = \overline{q}^{FB}$*.*

For all of these cases, the network magnitude is downsized: $Q^{SB} < Q^{FB}$*.*

PROOF. The first-order condition to (16.7.74) is:

$$\Pi_q(q, \alpha) = 0, \tag{16.7.75}$$

this implies,

$$\begin{cases} \alpha V'(\underline{q}) + \Psi'\left(v\underline{q} + (1-v)\overline{q}\right) = c, \\ \overline{\theta}V'(\overline{q}) + \Psi'\left(v\underline{q} + (1-v)\overline{q}\right) = c. \end{cases} \tag{16.7.76}$$

Differentiating (16.7.75) with respect to parameter α, we obtain:

$$\Pi_{qq}\frac{\mathrm{d}q}{\mathrm{d}\alpha} + \Pi_{q\alpha} = 0, \tag{16.7.77}$$

i.e.,

$$\begin{pmatrix} \alpha v V''(\underline{q}) + v^2\Psi''(Q) & v(1-v)\Psi''(Q) \\ v(1-v)\Psi''(Q) & (1-v)\overline{\theta}V''(\overline{q}) + (1-v)^2\Psi''(Q) \end{pmatrix} \begin{pmatrix} \frac{\mathrm{d}\underline{q}}{\mathrm{d}\alpha} \\ \frac{\mathrm{d}\overline{q}}{\mathrm{d}\alpha} \end{pmatrix} + \begin{pmatrix} vV'(\underline{q}) \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Solving the above equations, we have

$$\begin{cases} \dfrac{\mathrm{d}\underline{q}}{\mathrm{d}\alpha} = \dfrac{-V'(\underline{q})\left[\overline{\theta}V''(\overline{q}) + (1-v)\Psi''(Q)\right]}{\alpha V''(\underline{q})\left[\overline{\theta}V''(\overline{q}) + (1-v)\Psi''(Q)\right] + v\overline{\theta}V''(\overline{q})\Psi''(Q)}, \\ \dfrac{\mathrm{d}\overline{q}}{\mathrm{d}\alpha} = \dfrac{vV'(\underline{q})\Psi''(Q)}{\alpha V''(\underline{q})\left[\overline{\theta}V''(\overline{q}) + (1-v)\Psi''(Q)\right] + v\overline{\theta}V''(\overline{q})\Psi''(Q)}, \\ \dfrac{\mathrm{d}Q}{\mathrm{d}\alpha} = v\dfrac{\mathrm{d}\underline{q}}{\mathrm{d}\alpha} + (1-v)\dfrac{\mathrm{d}\overline{q}}{\mathrm{d}\alpha} = \dfrac{-v\overline{\theta}V'(\underline{q})V''(\overline{q})}{\alpha V''(\underline{q})\Big[\overline{\theta}V''(\overline{q}) + (1-v)\Psi''(Q)\Big] + v\overline{\theta}V''(\overline{q})\Psi''(Q)}. \end{cases} \tag{16.7.78}$$

From the assumption that the Hessian matrix Π_{qq} is negative definite, it can be verified that the $2th$ diagonal element of Π_{qq} is negative, and thus

$$\overline{\theta}V''(\overline{q}) + (1-v)\Psi''(Q) < 0, \tag{16.7.79}$$

and the determinant of Π_{qq} is positive,

$$\det(\Pi_{qq}) = v(1-v)\Big\{\alpha V''(\underline{q})\left[\overline{\theta}V''(\overline{q}) + (1-v)\Psi''(Q)\right] + v\overline{\theta}V''(\overline{q})\Psi''(Q)\Big\} > 0. \tag{16.7.80}$$

The signs of derivatives in (16.7.78) can be determined, which are $\frac{\mathrm{d}\underline{q}}{\mathrm{d}\alpha} > 0$ and $\frac{\mathrm{d}Q}{\mathrm{d}\alpha} > 0$, which means $\underline{q}^{SB} < \underline{q}^{FB}$ and $Q^{SB} < Q^{FB}$. The sign of $\frac{\mathrm{d}\overline{q}}{\mathrm{d}\alpha}$, and thus the distortion direction of $\overline{q}$ depends on the sign of $\Psi''(Q)$: If $\Psi''(Q) > 0$, $\frac{\mathrm{d}\overline{q}}{\mathrm{d}\alpha} > 0$, then $\overline{q}^{SB} < \overline{q}^{FB}$; if $\Psi''(Q) < 0$, $\frac{\mathrm{d}\overline{q}}{\mathrm{d}\alpha} < 0$, then $\overline{q}^{SB} > \overline{q}^{FB}$; if $\Psi''(Q) = 0$, $\frac{\mathrm{d}\overline{q}}{\mathrm{d}\alpha} = 0$, then $\overline{q}^{SB} = \overline{q}^{FB}$. □

Remark 16.7.2 $\Psi''(\cdot) > 0$ implies that the marginal value from an increase in individual consumption is greater at a higher level of others' consumption: $\frac{\partial^2 U_i}{\partial q_i \partial q_j} > 0$ for $i \neq j$. Its interpretation is that the externalities in a bigger network are larger than in a small network, so that an agent is more eager to consume more when other agents consume more. This is consistent with the critical "strategic complementarity" assumption in Segal (1999, 2003) and Csorba (2008). This condition allows one to characterize optimal contracts by applying monotone comparative static tools, pioneered by Topkis (1978) and Milgrom and Shannon (1994).[2]

[2]By Topkis (1978) and Milgrom and Shannon (1994), a twice continuously differentiable

16.7.2 Countervailing Incentives

We now discuss another cause of the failure of “no distortion on the top” and “one-way distortion” rules: The countervailing incentive problem. We assume that consumers can bypass the network offered by the incumbent firm and enter a competitive market, including many homogenous firms. All of these firms are potential entrants of the market. Let ω denote the marginal production cost of the entrants. We assume that the goods or services offered by the entrants are incompatible with those of the incumbent monopolist,[3] and they have not yet formed their own consumer network. In the competitive outside market, each firm's unit charge equals its marginal cost ω, and thus the representative consumer's utility derived from consuming the entrants' goods is $\theta V(q) - \omega q$.

Let $G(\theta) = \max_q [\theta V(q) - \omega q]$, $\underline{G} = G(\underline{\theta})$, $\overline{G} = G(\overline{\theta})$, and $\Delta G = \overline{G} - \underline{G}$. Throughout this section, we assume that the network is congestible.

The entry threat gives the consumer non-zero type-dependent reservation utilities, and thus the problem of the incumbent network supplier can be represented as

$$
(\text{P5})\begin{cases}
\max\limits_{\{(\underline{U},\underline{q});(\overline{U},\overline{q})\}} v\left[\underline{\theta}V(\underline{q}) - c\underline{q}\right] + (1-v)\left[\overline{\theta}V(\overline{q}) - c\overline{q}\right] + \Psi(Q) - \left[v\underline{U} + (1-v)\overline{U}\right] \\
s.t.\ IR(\underline{\theta}) : \underline{U} \geqq \underline{G}, \\
\quad IR(\overline{\theta}) : \overline{U} \geqq \overline{G}, \\
\quad IC(\underline{\theta}) : \underline{U} \geqq \overline{U} - \Delta\theta V(\overline{q}), \\
\quad IC(\overline{\theta}) : \overline{U} \geqq \underline{U} + \Delta\theta V(\underline{q}).
\end{cases}
$$

Note that (P5) is the same as (P3), except for the non-zero type-dependent reservation utilities $\underline{G}$ and $\overline{G}$. The following proposition characterizes the optimal entry-deterring contract.

Proposition 16.7.2 *The optimal entry-deterring contract depends on the marginal cost of potential entrants. Specifically, there exist positive values $\omega_1 < \omega_2 < \omega_3 < \omega_4$ such that:*

1. *If $\omega > \omega_4$, then $\Delta G < \Delta\theta V(\underline{q}^{SB})$, and consequently the pricing contract is: $\underline{q} = \underline{q}^{SB}$, $\overline{q} = \overline{q}^{SB}$, $\underline{U} = \underline{G}$, and $\overline{U} = \underline{G} + \Delta\theta V(\underline{q}^{SB})$.*
2. *If $\omega_3 \leqslant \omega \leqslant \omega_4$, then $\Delta\theta V(\underline{q}^{SB}) \leqslant \Delta G \leqslant \Delta\theta V(\underline{q}^{FB})$, and consequently the optimal consumption level $\underline{q}$ and $\overline{q}$ are determined*

function $\Pi = \Pi(q_1, q_2, \cdots, q_n; \epsilon)$ defined on a *lattice* Q is supermodular if and only if for all $i \neq j$, $\frac{\partial^2 \Pi}{\partial q_i \partial q_j} > 0$; furthermore, if $\frac{\partial^2 \Pi}{\partial q_i \partial \epsilon} > 0, \forall i$, then function Π has strictly increasing differences in (q, ϵ). Let $q(\epsilon) = \max\limits_{q \in Q} \Pi(q, \epsilon)$. Then, for a supermodular function with increasing differences in (q, ϵ), $q_i(\epsilon)$ is a strictly increasing function of ϵ for all i.

[3] Otherwise, the entrants can share the present network with the incumbent monopolist.

by

$$\begin{cases} \underline{q} = V^{-1}\left(\dfrac{\Delta G}{\Delta\theta}\right), \\ \overline{\theta}V'(\overline{q}) + \Psi'\left(v\underline{q} + (1-v)\overline{q}\right) = c, \end{cases} \tag{16.7.81}$$

where $\underline{q} \in \left[\underline{q}^{SB}, \underline{q}^{FB}\right]$ *and* $\overline{q} \in [\overline{q}^{FB}, \overline{q}^{SB}]$. *The consumers' information rents are* $\underline{U} = \underline{G}$ *and* $\overline{U} = \overline{G}$.

3. *If* $\omega_2 < \omega < \omega_3$, *then* $\Delta\theta V(\underline{q}^{FB}) < \Delta G < \Delta\theta V(\overline{q}^{FB})$, *and consequently the pricing contract is* $\underline{q} = \underline{q}^{FB}$, $\overline{q} = \overline{q}^{FB}$, $\underline{U} = \underline{G}$, *and* $\overline{U} = \overline{G}$.

4. *If* $\omega_1 \leqslant \omega \leqslant \omega_2$, *then* $\Delta\theta V(\overline{q}^{FB}) \leqslant \Delta G \leqslant \Delta\theta V(\overline{q}^{CI})$, *and consequently the optimal consumption level* $\underline{q}$ *and* $\overline{q}$ *are given by*

$$\begin{cases} \overline{q} = V^{-1}\left(\dfrac{\Delta G}{\Delta\theta}\right), \\ \underline{\theta}V'(\underline{q}) + \Psi'\left(v\underline{q} + (1-v)\overline{q}\right) = c, \end{cases} \tag{16.7.82}$$

where $\underline{q} \in [\underline{q}^{CI}, \underline{q}^{FB}]$ *and* $\overline{q} \in [\overline{q}^{FB}, \overline{q}^{CI}]$, *"CI" denotes "countervailing incentives". The consumers' information rents are* $\underline{U} = \underline{G}$ *and* $\overline{U} = \overline{G}$.

5. *If* $0 < \omega < \omega_1$, *then* $\Delta G > \Delta\theta V(\overline{q}^{CI})$, *and consequently the optimal contract is* $\underline{q} = \underline{q}^{CI}$, $\overline{q} = \overline{q}^{CI}$, $\underline{U} = \overline{G} - \Delta\theta V(\overline{q}^{CI})$, $\overline{U} = \overline{G}$. $\underline{q}^{CI}$ *and* $\overline{q}^{CI}$ *are given by:*

$$\begin{cases} \underline{\theta}V'(\underline{q}^{CI}) + \Psi'\left(v\underline{q}^{CI} + (1-v)\overline{q}^{CI}\right) = c, \\ \left(\overline{\theta} + \dfrac{v}{1-v}\Delta\theta\right)V'(\overline{q}^{CI}) + \Psi'\left(v\underline{q}^{CI} + (1-v)\overline{q}^{CI}\right) = c. \end{cases} \tag{16.7.83}$$

PROOF. In (P5), we have as many regimes as combinations of binding constraints among $IR(\underline{\theta})$, $IR(\overline{\theta})$, $IC(\underline{\theta})$, and $IC(\overline{\theta})$. To reduce the number of possible cases, we first give the following lemmas.

Lemma 16.7.1 *A pooling contract with* $\underline{q} = \overline{q}$ *and* $\underline{t} = \overline{t}$ *can never be optimal.*

PROOF. Suppose that the optimal contract is pooling with $\underline{q} = \overline{q} = q$ and $\underline{t} = \overline{t} = t$. There are two cases to be considered.

(i) $\overline{\theta}V'(q) > c$. Then, increase $\overline{q}$ by ε and the transfer by $\overline{\theta}V'(q)\varepsilon$, and the $\overline{\theta}-$type can remain indifferent. Since at (q,t) the marginal rate of substitution between q and t is higher for $\overline{\theta}-$type, this new allocation is incentive-compatible. This raises the firm's revenue by $(1-v)[\overline{\theta}V'(q) - c]\varepsilon$.

(ii) $\overline{\theta}V'(q) \leqslant c$ and $\underline{\theta}V'(q) < c$. Then, decrease $\underline{q}$ by ε and adjust $\underline{t}$ so that $\underline{\theta}-$type can remain on the same indifference curve. Then, the firm's total charge will be increased by $[c - \underline{\theta}V'(q)]\varepsilon$.

Thus, in both cases, it contradicts the fact that (q, t) is an optimal contract. □

Lemma 16.7.2 *If the two types are offered two different contracts, the two incentive constraints cannot be simultaneously binding.*

PROOF. Suppose by way of contradiction that both ICs are binding. From $\underline{\theta}V(\underline{q})-\underline{t}+\Psi(Q) = \underline{\theta}V(\overline{q})-\overline{t}+\Psi(Q)$ and $\overline{\theta}V(\overline{q})-\overline{t}+\Psi(Q) = \overline{\theta}V(\underline{q})-\underline{t}+\Psi(Q)$, we have $\underline{q} = \overline{q}$ and $\underline{t} = \overline{t}$. However, this is impossible by Lemma 1. □

Lemma 16.7.3 *The IC and IR constraints of the same type cannot be simultaneously slack.*

PROOF. If $IR(\theta)$ and $IC(\theta)$ are both slack, increasing $t(\theta)$ by a tiny increment will not violate all of the constraints, but the firm's charge will be increased. □

Applying the above three lemmas, only five possible regimes need to be considered, which are summarized in the following table.

Table1 – The five possible regimes

Constraints	Regime 1	Regime 2	Regime 3	Regime 4	Regime 5
$IR(\underline{\theta})$	B	B	B	B	S
$IR(\overline{\theta})$	S	B	B	B	B
$IC(\underline{\theta})$	S	S	S	B	B
$IC(\overline{\theta})$	B	B	S	S	S

Here, "B" denotes "binding" , and "S" denotes "slack" .

The regimes are ordered from 1 to 5 when ΔG increases, and ΔG itself is determined by the entrants' marginal cost ω. To determine how ω affects the nonlinear pricing contract of the incumbent firm, we give the following two lemmas. Lemma 16.7.4 states the change of ΔG in different regimes, and Lemma 16.7.5 shows how ΔG is affected by ω.

Lemma 16.7.4 *The optimal pricing contracts and the utility difference ΔG in different regimes are:*

1. *In regime 1, the optimal solution to (P5) is $\underline{q} = \underline{q}^{SB}$, $\overline{q} = \overline{q}^{SB}$, $\underline{U} = \underline{G}$, and $\overline{U} = \underline{G} + \Delta\theta V(\underline{q}^{SB})$. The value of utility difference satisfies $\Delta G < \Delta\theta V(\underline{q}^{SB})$.*
2. *In regime 2, the optimal consumption level $\underline{q}$ and $\overline{q}$ are determined by*

$$\begin{cases} \underline{q} = V^{-1}\left(\dfrac{\Delta G}{\Delta\theta}\right), \\ \overline{\theta}V'(\overline{q}) + \Psi'\left(v\underline{q} + (1-v)\overline{q}\right) = c, \end{cases} \tag{16.7.84}$$

where $\underline{q} \in \left[\underline{q}^{SB}, \underline{q}^{FB}\right]$ and $\overline{q} \in [\overline{q}^{FB}, \overline{q}^{SB}]$. The consumers' information rents are $\underline{U} = \underline{G}$ and $\overline{U} = \overline{G}$. The utility difference satisfies $\Delta\theta V(\underline{q}^{SB}) \leqslant \Delta G \leqslant \Delta\theta V(\underline{q}^{FB})$.

3. *In regime 3, the optimal solution to (P5) is $\underline{q} = \underline{q}^{FB}$, $\overline{q} = \overline{q}^{FB}$, $\underline{U} = \underline{G}$, $\overline{U} = \overline{G}$, and $\Delta\theta V(\underline{q}^{FB}) < \Delta G < \theta V(\overline{q}^{FB})$.*

4. *In regime 4, the optimal consumption level $\underline{q}$ and $\overline{q}$ are determined by*

$$\begin{cases} \overline{q} = V^{-1}\left(\dfrac{\Delta G}{\Delta\theta}\right), \\ \underline{\theta}V'(\underline{q}) + \Psi'\left(v\underline{q} + (1-v)\overline{q}\right) = c, \end{cases} \tag{16.7.85}$$

where $\underline{q} \in [\underline{q}^{CI}, \underline{q}^{FB}]$ and $\overline{q} \in [\overline{q}^{FB}, \overline{q}^{CI}]$. The consumers' information rents are $\underline{U} = \underline{G}$ and $\overline{U} = \overline{G}$. The utility difference satisfies $\Delta\theta V(\overline{q}^{FB}) \leqslant \Delta G \leqslant \Delta\theta V(\overline{q}^{CI})$.

5. *In regime 5, the optimal contract is $\underline{q} = \underline{q}^{CI}$, $\overline{q} = \overline{q}^{CI}$, $\underline{U} = \overline{G} - \Delta\theta V(\overline{q}^{CI})$, and $\overline{U} = \overline{G}$. The utility difference $\Delta G > \theta V(\overline{q}^{CI})$. $\underline{q}^{CI}$ and $\overline{q}^{CI}$ are determined by:*

$$\begin{cases} \underline{\theta}V'(\underline{q}^{CI}) + \Psi'\left(v\underline{q}^{CI} + (1-v)\overline{q}^{CI}\right) = c, \\ \left(\overline{\theta} + \dfrac{v}{1-v}\Delta\theta\right)V'(\overline{q}^{CI}) + \Psi'\left(v\underline{q}^{CI} + (1-v)\overline{q}^{CI}\right) = c. \end{cases} \tag{16.7.86}$$

PROOF.

- In regime 1, the constraints $IR(\underline{\theta})$ and $IC(\overline{\theta})$ are binding. Solving (P5) in the same way as (P3), we obtain the second-best solution

$$\left\{\underline{q} = \underline{q}^{SB}, \overline{q} = \overline{q}^{SB}; \underline{U} = \underline{G}, \overline{U} = \underline{G} + \Delta\theta V(\underline{q}^{SB})\right\},$$

with $\Delta G < \Delta U = \Delta\theta V(\underline{q}^{SB})$.

- In regime 2, the constraints $IR(\underline{\theta})$, $IR(\overline{\theta})$ and $IC(\overline{\theta})$ are binding. The optimal contract set is thus given by

$$\left\{(\underline{q}, \overline{q}, \underline{U}, \overline{U}) : \Delta\theta V(\underline{q}) = \Delta G, \overline{\theta}V(\overline{q}) + \Psi'(Q) = c; \underline{U} = \underline{G}, \overline{U} = \overline{G}\right\}.$$

Substituting $IR(\underline{\theta})$ and $IR(\overline{\theta})$ into the objective function, the Lagrange function is constructed as

$$\begin{aligned} L(\underline{q}, \overline{q}) = {} & v\left[\underline{\theta}V(\underline{q}) - c\underline{q}\right] + (1-v)\left[\overline{\theta}V(\overline{q}) - c\overline{q}\right] \\ & + \Psi(Q) - \left[v\underline{G} + (1-v)\overline{G}\right] + \lambda\left[\Delta G - \Delta\theta V(\underline{q})\right], \end{aligned}$$

where $\lambda > 0$ is the Lagrange multiplier of the binding constraint $IC(\overline{\theta})$. Then, $\underline{q}$ and $\overline{q}$ are determined by:

$$\begin{cases} \left(\underline{\theta} - \dfrac{\lambda}{v}\Delta\theta\right) V'(\underline{q}) + \Psi'\left(v\underline{q} + (1-v)\overline{q}\right) = c, \\ \overline{\theta}V'(\overline{q}) + \Psi'\left(v\underline{q} + (1-v)\overline{q}\right) = c. \end{cases} \tag{16.7.87}$$

Because $\underline{\theta} - \frac{\lambda}{v}\Delta\theta < \underline{\theta}$, from formula (16.7.78) it is easy to verify that $\underline{q} < \underline{q}^{FB}$ and $\overline{q} > \overline{q}^{FB}$. Substituting $IR(\underline{\theta})$ and $IC(\overline{\theta})$ into the objective function of (P5) and letting $\delta > 0$ be the Lagrange multiplier associated with the binding constraint $IR(\overline{\theta})$, we obtain the following Lagrange function:

$$\begin{aligned} L(\underline{q}, \overline{q}) &= v\left[\underline{\theta}V(\underline{q}) - c\underline{q}\right] + (1-v)\left[\overline{\theta}V(\overline{q}) - c\overline{q}\right] \\ &\quad + \Psi(Q) - \left[v\underline{G} + (1-v)(\underline{G} + \Delta\theta V(\underline{q}))\right] + \delta\left[\Delta\theta V(\underline{q}) - \Delta G\right]. \end{aligned}$$

The optimal consumptions $\underline{q}$ and $\overline{q}$ are determined by:

$$\begin{cases} \left(\underline{\theta} - \dfrac{1-v-\delta}{v}\Delta\theta\right) V'(\underline{q}) + \Psi'\left(v\underline{q} + (1-v)\overline{q}\right) = c, \\ \overline{\theta}V'(\overline{q}) + \Psi'\left(v\underline{q} + (1-v)\overline{q}\right) = c. \end{cases} \tag{16.7.88}$$

Because $\underline{\theta} - \frac{1-v-\delta}{v}\Delta\theta > \underline{\theta} - \frac{1-v}{v}\Delta\theta$, from expression (16.7.78) we have $\underline{q} > \underline{q}^{SB}$ and $\overline{q} < \overline{q}^{SB}$. It then suffices to show that $\Delta\theta V(\underline{q}^{SB}) \leqslant \Delta G \leqslant \Delta\theta V(\underline{q}^{FB})$.

- In regime 3, $IR(\underline{\theta})$ and $IR(\overline{\theta})$ are binding. Then, the optimal contract is given by
$$\left\{\underline{q} = \underline{q}^{FB}, \overline{q} = \overline{q}^{FB}; \underline{U} = \underline{G}, \overline{U} = \overline{G}\right\}.$$
From the slack ICs, we can verify that ΔG satisfies $\Delta\theta V(\underline{q}^{FB}) < \Delta G < \Delta\theta V(\overline{q}^{FB})$.

- In regime 4, $IR(\underline{\theta})$ $IR(\overline{\theta})$ and $IC(\underline{\theta})$ are binding. The optimal contract is:
$$\left\{(\underline{q}, \overline{q}, \underline{U}, \overline{U}) : \Delta\theta V(\overline{q}) = \Delta G, \underline{\theta}V(\underline{q}) + \Psi'(Q) = c; \underline{U} = \underline{G}, \overline{U} = \overline{G}\right\}.$$
Substituting $IR(\underline{\theta})$ and $IR(\overline{\theta})$ into the objective function, and letting $\mu > 0$ be the multiplier associated with the binding constraint $IC(\underline{\theta})$, we have the following Lagrange function:
$$\begin{aligned} L(\underline{q}, \overline{q}) &= v\left[\underline{\theta}V(\underline{q}) - c\underline{q}\right] + (1-v)\left[\overline{\theta}V(\overline{q}) - c\overline{q}\right] \\ &\quad + \Psi(Q) - \left[v\underline{G} + (1-v)\overline{G}\right] + \mu\left[\Delta\theta V(\overline{q}) - \Delta G\right]. \end{aligned}$$

Thus, $\underline{q}$ and $\overline{q}$ are determined by

$$\begin{cases} \underline{\theta}V'(\underline{q}) + \Psi'\left(v\underline{q} + (1-v)\overline{q}\right) = c, \\ \left(\overline{\theta} + \frac{\mu}{1-v}\Delta\theta\right)V'(\overline{q}) + \Psi'\left(v\underline{q} + (1-v)\overline{q}\right) = c. \end{cases} \tag{16.7.89}$$

Substituting $IR(\overline{\theta})$ and $IC(\underline{\theta})$ into the objective function, and letting $\eta > 0$ be the multiplier associated with the binding constraint $IR(\underline{\theta})$, we have the Lagrange function:

$$\begin{aligned} L(\underline{q}, \overline{q}) = {} & v\left[\underline{\theta}V(\underline{q}) - c\underline{q}\right] + (1-v)\left[\overline{\theta}V(\overline{q}) - c\overline{q}\right] + \Psi(Q) \\ & - \left[v(\overline{G} - \Delta\theta V(\overline{q})) + (1-v)\overline{G}\right] + \eta\left[\Delta G - \Delta\theta V(\overline{q})\right]. \end{aligned}$$

Then, the $\underline{q}$ and $\overline{q}$ are determined by:

$$\begin{cases} \underline{\theta}V'(\underline{q}) + \Psi'\left(v\underline{q} + (1-v)\overline{q}\right) = c, \\ \left(\overline{\theta} + \frac{v-\eta}{1-v}\Delta\theta\right)V'(\overline{q}) + \Psi'\left(v\underline{q} + (1-v)\overline{q}\right) = c. \end{cases} \tag{16.7.90}$$

To compare the different consumption levels, we perform some comparative static analysis. To do so, let

$$\begin{cases} \underline{\theta}V'(\underline{q}) + \Psi'\left(v\underline{q} + (1-v)\overline{q}\right) = c, \\ \beta V'(\overline{q}) + \Psi'\left(v\underline{q} + (1-v)\overline{q}\right) = c. \end{cases} \tag{16.7.91}$$

If $\beta = \overline{\theta}$, it is the expression of $\underline{q}^{FB}$ and $\overline{q}^{FB}$; if $\beta = \overline{\theta} + \frac{v}{1-v}\Delta\theta$, it coincides with the countervailing incentives solution $\underline{q}^{CI}$ and $\overline{q}^{CI}$.

Differentiating these two equations with respect to β leads to:

$$\begin{cases} \left[\underline{\theta}V''(\underline{q}) + v\Psi''(Q)\right]\frac{\mathrm{d}\underline{q}}{\mathrm{d}\beta} + (1-v)\Psi''(Q)\frac{\mathrm{d}\overline{q}}{\mathrm{d}\beta} = 0, \\ v\Psi''(Q)\frac{\mathrm{d}\underline{q}}{\mathrm{d}\beta} + [\beta V''(\overline{q}) + (1-v)\Psi''(Q)]\frac{\mathrm{d}\overline{q}}{\mathrm{d}\beta} = -V'(\underline{q}). \end{cases} \tag{16.7.92}$$

Thus, when $\Psi''(Q) < 0$, we have

$$\begin{cases} \frac{\mathrm{d}\underline{q}}{\mathrm{d}\beta} = \frac{(1-v)V'(\underline{q})\Psi''(Q)}{\beta V''(\underline{q})[\underline{\theta}V''(\underline{q}) + v\Psi''(Q)] + (1-v)\underline{\theta}V''(\underline{q})\Psi''(Q)} < 0, \\ \frac{\mathrm{d}\overline{q}}{\mathrm{d}\beta} = \frac{-V'(\underline{q})[\underline{\theta}V''(\underline{q}) + v\Psi''(Q)]}{\beta V''(\underline{q})[\underline{\theta}V''(\underline{q}) + v\Psi''(Q)] + (1-v)\underline{\theta}V''(\underline{q})\Psi''(Q)} > 0. \end{cases} \tag{16.7.93}$$

Because $\overline{\theta} + \frac{\mu}{1-v}\Delta\theta > \overline{\theta}$ and $\overline{\theta} + \frac{v-\eta}{1-v}\Delta\theta < \overline{\theta} + \frac{v}{1-v}\Delta\theta$, from formula (16.7.93), it can be verified that $\overline{q} > \overline{q}^{FB}$, $\underline{q} < \underline{q}^{FB}$, $\overline{q} < \overline{q}^{CI}$, and $\underline{q} > \underline{q}^{CI}$. Thus, $\Delta G = \Delta\theta V(\overline{q}) \in [\Delta\theta V(\overline{q}^{FB}), \Delta\theta V(\overline{q}^{CI})]$.

- In regime 5, $IR(\overline{\theta})$ and $IC(\underline{\theta})$ are binding constraints. Substituting $\overline{U} = \overline{G}$ and $\underline{U} = \overline{G} - \Delta\theta V(\overline{q})$ into the objective function, we obtain the following first-order conditions:

$$\begin{cases} \underline{\theta}V'(\underline{q}) + \Psi'\left(v\underline{q} + (1-v)\overline{q}\right) = c, \\ \left(\overline{\theta} + \dfrac{v}{1-v}\Delta\theta\right)V'(\overline{q}) + \Psi'\left(v\underline{q} + (1-v)\overline{q}\right) = c. \end{cases} \tag{16.7.94}$$

It is the countervailing incentives consumption level. Note that $\overline{\theta} + \frac{v}{1-v}\Delta\theta > \overline{\theta}$, and thus, from (16.7.93) $\underline{q}^{FB} > \underline{q}^{CI}$, $\overline{q}^{FB} < \overline{q}^{CI}$. The difference of reservation utility satisfies $\Delta G > \Delta U = \Delta\theta V(\overline{q}^{CI})$.

□

Lemma 16.7.5 *Suppose that $V(0) = 0$, $V'(\cdot) > 0$, $V''(\cdot) < 0$, and $V(\cdot)$ satisfies the standard Inada conditions: $\lim_{q\to+\infty}V'(q) = 0$, and $\lim_{q\to 0}V'(q) = +\infty$. Then, the utility difference across different states $\Delta G = \overline{G} - \underline{G}$ is a decreasing function of the marginal cost ω, $\lim_{\omega\to 0}\Delta G = +\infty$, and $\lim_{\omega\to+\infty}\Delta G = 0$.*

Proof. The first-order condition of $G(\theta) = \max_q [\theta V(q) - \omega q]$ is given by $\theta V'(q^*) = \omega$. Therefore, the maximized utility derived from network bypassing is: $G(\theta) = \theta[V(q^*(\theta)) - q^*(\theta)V'(q^*(\theta))]$. Let $\Phi(q) = V(q) - qV'(q)$. Then, $\Delta G = G(\overline{\theta}) - G(\underline{\theta}) = \overline{\theta}\Phi(\overline{q}^*) - \underline{\theta}\Phi(\underline{q}^*)$, and its derivative with respect to marginal cost ω is:

$$\begin{aligned} \frac{\mathrm{d}\Delta G}{\mathrm{d}\omega} &= \overline{\theta}\Phi'(\overline{q}^*)\frac{\mathrm{d}\overline{q}^*}{\mathrm{d}\omega} - \underline{\theta}\Phi'(\underline{q}^*)\frac{\mathrm{d}\underline{q}^*}{\mathrm{d}\omega} \\ &= -\overline{\theta}\overline{q}^*V''(\overline{q}^*)\frac{\mathrm{d}\overline{q}^*}{\mathrm{d}\omega} + \underline{\theta}\underline{q}^*V''(\underline{q}^*)\frac{\mathrm{d}\underline{q}^*}{\mathrm{d}\omega} \\ &= -\overline{\theta}\overline{q}^*V''(\overline{q}^*)\frac{1}{\overline{\theta}V''(\overline{q}^*)} + \underline{\theta}\underline{q}^*V''(\underline{q}^*)\frac{1}{\underline{\theta}V''(\underline{q}^*)} \\ &= -\overline{q}^* + \underline{q}^* < 0. \end{aligned}$$

It is easy to verify that when the conditions $V(0) = 0$, $V'(\cdot) > 0$, $V''(\cdot) < 0$, $\lim_{q\to+\infty}V'(q) = 0$, and $\lim_{q\to 0}V'(q) = +\infty$ are satisfied, $\lim_{\omega\to 0}\Delta G = +\infty$ and $\lim_{\omega\to+\infty}\Delta G = 0$. Figure 6 depicts the relationship of ΔG and ω. □
Figure 16.9 illustrates the relationship between ΔG and ω.

From the above lemmas, one can see that if the potential entrants' competitiveness increases, the utility differences will increase from zero to infinity. Thus, there exist positive values $\omega_i, i = 1, 2, 3, 4$, such that $\omega_1 < \omega_2 < \omega_3 < \omega_4$ corresponding to $\Delta\theta V(\overline{q}^{CI})$, $\Delta\theta V(\overline{q}^{FB})$, $\Delta\theta V(\underline{q}^{FB})$ and $\Delta\theta V(\underline{q}^{SB})$, respectively, where $\overline{q}^{CI}$, $\overline{q}^{FB}$, $\underline{q}^{FB}$, and $\underline{q}^{SB}$ are given in expressions (16.7.83), (16.7.72), and (16.7.73), respectively.

Thus, by combining Lemmas 16.7.4 and 16.7.5, we have proven Proposition 16.7.2. □

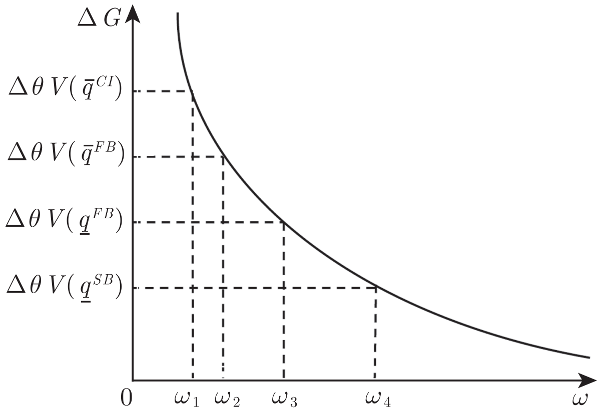

Figure 16.9: The impact of ω on ΔG

Remark 16.7.3 When $\omega > \omega_4$, the second-best contract is also entry-deterring. This means that when the outside competitors are not sufficiently efficient to give high-demand consumers enough utility exceeding their information rents acquired from the present network, the outside market is only attractive to low-demand consumers. As a consequence, the incumbent firm need not change its pricing contract when facing the entry threat of a firm with low competitiveness.

When $\omega_3 \leqslant \omega \leqslant \omega_4$, we have $\underline{q} \in [\underline{q}^{SB}, \underline{q}^{FB}]$ and $\overline{q} \in [\overline{q}^{FB}, \overline{q}^{SB}]$. This means that when the marginal cost ω decreases to the extent that the utility difference ΔG is sufficiently large to attract high demand consumers bypassing the present network, the monopolist must give up more information rent to him by increasing the consumption level of low-demand consumers. The consumption level of high demand consumers themselves should also be lowered accordingly due to network effects. In this case, the sharper competitiveness of outside competitors makes the allocations less distorted.

When $\omega_2 < \omega < \omega_3$, asymmetric information imposes no distortion on both types' allocation. As ω decreases and ΔG increases further, $\underline{q}$ will reach the first-best level. At this point, it is suboptimal for the monopolist to increase the high-type's information rent at the cost of distorting the consumption level of the low-demand consumers upward. In this case, the main task for the firm toward the high-type is to prevent them from bypassing the incumbent market instead of preventing them from misreporting. In other words, the participation constraints are more difficult to be satisfied than the incentive-compatibility constraints. Therefore, only the IRs are binding, and the first-best allocation is attained.

When $\omega_1 \leqslant \omega \leqslant \omega_2$, we have $\underline{q} \in [\underline{q}^{CI}, \underline{q}^{FB}]$, and $\overline{q} \in [\overline{q}^{FB}, \overline{q}^{CI}]$. The high difference of utilities induces the low-type to pretend to be a high-type, from which the countervailing incentives problem arises. The $IC(\underline{\theta})$, $IR(\underline{\theta})$, and $IR(\underline{\theta})$ in (P5) are binding. Again, the allocations of the two types will be distorted in opposite directions. However, it is different from the distortions in cases 1 and 2. Specifically, the monopolist then distorts $\underline{q}$ downward to curb the rent of high-demand consumers. In this case, however, information rent has to be given to low-demand consumers to elicit them to report their types truthfully. The information rent is a decreasing function of high-demand consumers' consumption $\overline{q}$. Consequently, $\overline{q}$ has to be distorted upward to reduce the information rent gained by low-demand consumers, while a certain amount of the low-demand type's consumption is "crowded out" of the network so that $\underline{q}$ is distorted downward.

When $0 < \omega < \omega_1$, the allocations remain at the countervailing incentives level: $\underline{q} = \underline{q}^{CI}$ and $\overline{q} = \overline{q}^{CI}$. The decrease in marginal cost ω demands further upward distortion on the consumption of high-demand consumers (the consumption of low-demand consumers will be distorted downward accordingly). Thus, the participation constraint of the low-type has to be slackened, which means that a certain amount of information rent should be given to consumers with a low willingness to pay. In this case, only $IC(\underline{\theta})$ and $IR(\overline{\theta})$ are binding constraints, and the low-type gets information rent $\overline{G} - \Delta\theta V\left(\overline{q}^{CI}\right)$. The incumbent firm keeps reducing the tariffs ($\underline{t}$ and $\overline{t}$ keep decreasing in this case), instead of distorting allocations to prevent high-demand consumers from bypassing and low demand consumers from misreporting.

Changes in consumption with marginal cost ω are summarized in the following Figure 16.10.

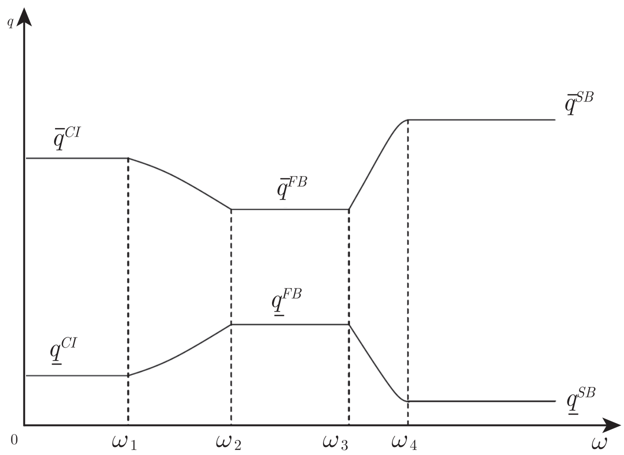

Figure 16.10: Impact of potential entrant marginal cost ω on second-best consumption

16.8 Adverse Selection in a Competitive Market

In this section, we mainly discuss the issue of adverse selection in a competitive market. Different from the above discussions, in the competitive market, there are numerous suppliers offering goods to customers. However, due to the asymmetry of information, the market is inefficient, and even competitive equilibrium may fail to exist.

George Akerlof (1940-, for his biography, see Section 15.5.2) first introduced information asymmetry in market analysis in 1970. In a used car market, a seller knows more about the car performance than a buyer. For each possible market price, used automobiles of good quality may be withdrawn from the market, and those of poor quality will enter the market. As the price drops, the quality of used automobiles in the market will become increasingly worse, which may eventually lead to market collapse. Here, we use the insurance market to discuss the impact of asymmetric information on market equilibrium.

In a competitive insurance market, each insured customer has his own private information, which is not known by insurance companies. For example, in health insurance, customers have information about their own health. In property insurance, there is private information about the possibility of accidents related to behavior. We introduce a simple continuous situation.

Assume that the probability of an accident $p \in [\underline{p}, \bar{p}] \subseteq [0,1]$ is private information with a density function $f(p)$, the accidental loss is L (which is identical for all types), and each customer's utility function $u(w)$ for wealth is the same, which is strictly concave.

Under complete information, for customers of type p, in the competitive market, the premium rate is also p. The premium of loss L countervailing is $I = pL$. Since the customers are risk-averse,

$$u(w - pL) > pu(w - L) + (1 - p)u(w).$$

Under incomplete information, there is only one type of contract in the market in the Akerlof (1970) model. Consider a simplified scenario, assuming that the contract is full loss insurance, the insurance company has no customer's private information, the premium rate is r, and the cost of insurance is $I = rL$. If market equilibrium exists, it has the following structure: There exists $\hat{p}$ such that

$$\begin{aligned} u(w - I(\hat{p})) &< pu(w - L) + (1 - p)u(w), \quad \text{if } p < \hat{p}, \\ u(w - I(\hat{p})) &> pu(w - L) + (1 - p)u(w), \quad \text{if } p > \hat{p}, \\ u(w - I(\hat{p})) &= \hat{p}u(w - L) + (1 - \hat{p})u(w), \quad \text{if } p = \hat{p}, \end{aligned}$$

where $I(\hat{p}) = \frac{\int_{\hat{p}}^{\bar{p}} pf(p)dp}{1 - F(\hat{p})} L$.

Let $\tilde{p} = \int_{\underline{p}}^{\bar{p}} pf(p)dp$. If $u(w - I(\tilde{p})) < \underline{p}u(w-L) + (1-\underline{p})u(w)$, it means that $p \in [\underline{p}, \hat{p})$ types cannot be covered in the insurance market, leading to market failure. In some extreme cases, only some customers with the highest probability of accidents will be covered by the market, resulting in the phenomenon of Gresham's Law: "**Bad money drives out good money**." The following example reveals the possibility of a collapse of the second-hand goods market, which is discussed in more detail in Akerlof (1970).

Example 16.8.1 Suppose that in a used car market, the seller knows the quality of the car, and it is characterized by θ, which follows a uniform distribution over $[0, \bar{\theta}]$. For a θ type vehicle, the seller's rating is θ, and the buyer's rating is $k\theta$, which satisfies $1 < k < 2$. Obviously, under complete information, all used cars (except $\theta = 0$) will be traded. If the transaction price is $p(\theta)$, after the transaction, the utility of the two parties is $p(\theta) - \theta$ and $k\theta - p(\theta)$, respectively, and the market equilibrium will be Pareto optimal. However, if the used car quality information is asymmetric, when the market price is p, the used car with $\theta \in [0, p]$ will still be on the market. At this time, the average quality is $\frac{p}{2}$. If consumers purchase, their expected utility is $\frac{kp}{2} - p < 0$, and the used car market does not exist at all.

16.8.1 Screening Mechanism of Competitive Market under Asymmetric Information

When there is more than one type of contract, different sellers may choose different contracts to cater to different consumers. In this case, market equilibrium will have new features, such as the absence of pooling equilibrium. Rothschild and Stiglitz (1976) first introduced a contract screening mechanism in the competitive market. Below, we briefly discuss the various types of contract equilibrium in the competitive market.

For the sake of simplicity, suppose that there are two types of consumers in the market. The probability of accidents is $p_L < p_H$. There are many insurance companies in the market, and they can provide consumers with different contracts. Assume that the distribution probability of the low-accident type is λ. The contract form provided by insurance company i is (I_i, D_i), where I_i is the premium and D_i is the amount of compensation. If the consumer of type i accepts the contract, his expected utility is:

$$p_i u(w - D_i - I_i) + (1 - p_i)u(w - I_i).$$

Assume that the insurance process is divided into two steps. Firstly, each insurance company provides insurance contracts; secondly, the customer chooses one of the most favorable insurance contracts. We also assume that an insurance company has the ability to fulfill its contract so that even if the insurance contract is loss-making, it will be implemented.

In addition, in the competitive environment, an insurance company's expected profit is zero, and imagine that insurance companies compete in a Bertrand-like manner.

In equilibrium, there may be two forms of pure strategies. The first form is known the **pooling equilibrium**, i.e., all insurance companies choose the same contract which attracts all types of customers. The second one is known the **separating equilibrium**, i.e., there are two different types of insurance contracts in the market, and different types of customers choose different contracts.

We first discuss the pooling equilibrium. Assume that $\alpha = (I, D)$ is the insurance contract. At this point, for both types, their expected benefits are:

$$U^i \equiv p_i u(w - D - I) + (1 - p_i)u(w - I).$$

The zero profit condition implies:

$$I = [p_L\lambda + p_H(1 - \lambda)](L - D).$$

In the pooling contract, the low-accident type of consumers subsidize the high-accident type. Then, there are positive profit opportunities in the insurance market. New insurance companies can design new contracts $\beta = (\hat{D}, \hat{I})$ to attract the low-accident type of consumers only if they meet:

$$\begin{aligned} p_H u(w - D - I) + (1 - p_H)u(w - I) &> p_H u(w - \hat{D} - \hat{I}) + (1 - p_H)u(w - \hat{I}), \\ p_L u(w - D - I) + (1 - p_L)u(w - I) &< p_L u(w - \hat{D} - \hat{I}) + (1 - p_L)u(w - \hat{I}), \\ \hat{I} &> p_L(L - \hat{D}). \end{aligned}$$

Let $W_1 = u(w - I)$ and $W_2 = u(w - D - I)$. Figure 16.11 shows such a new contract.

When $\hat{D} < D, \hat{I} < I$, the low-accident type of consumers is less sensitive to deductibles, but is more sensitive to insurance costs; however, for the high-accident type of consumers, the opposite is true. The insurance company can profit from the low-accident type of customers.

For the separating equilibrium, there are two insurance contracts in the market: One is for the low-accident type $\alpha^L : (D_L, I_L = p_L(L - D_L))$, and the other is for the high-accident type $\alpha^H : (D_H, I_H = p_H(L - D_H))$. Rothschild and Stiglitz (1976) found that separating equilibrium may not exist under certain situations.

We know that the high-accident type of consumers is very sensitive to deductibles, and thus it is assumed that the high-accident contract is $\alpha^H = (0, I_H = p_H L)$, while the low-accident contract $\alpha^L = (D_L, I_L = p_L, L - D_L))$ satisfies:

$$\begin{aligned} & U^L = p_L u(w - D_L - I_L) + (1 - p_L)u(w - I_L) \\ \text{s.t.} \quad & u(w - I_H) \geqq p_H u(w - D_L - I_L) + (1 - p_H)u(w - I_L). \end{aligned}$$

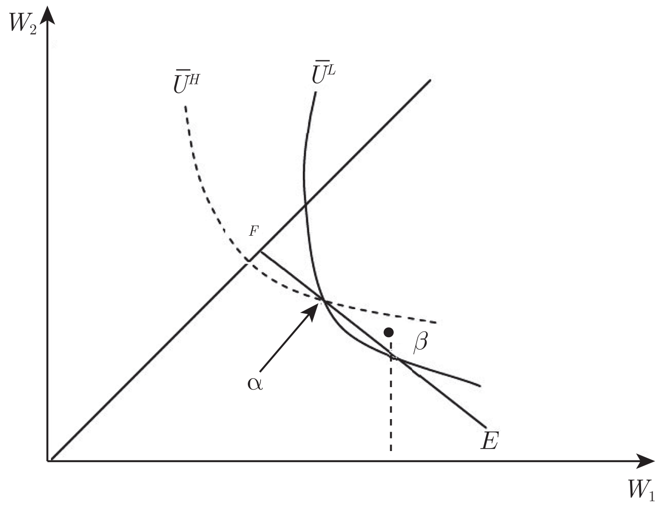

Figure 16.11: Pooling equilibrium does not exist

In the separating equilibrium, the low-accident contract is not related to the distribution of the two types of participants in the market, but is greatly affected by $p_H - p_L$, because it is necessary to avoid the high-accident type choosing the low-accident contract. This, however, makes the low-accident type of customer face high risks. If the probability of the low-accident type in the market is high, there will be a positive profit opportunity in the market, which will allow the new insurance company to provide a pooling contract that caters to both types of customers (the contract γ in Figure 16.12). Figure 16.12 depicts such a new contract.

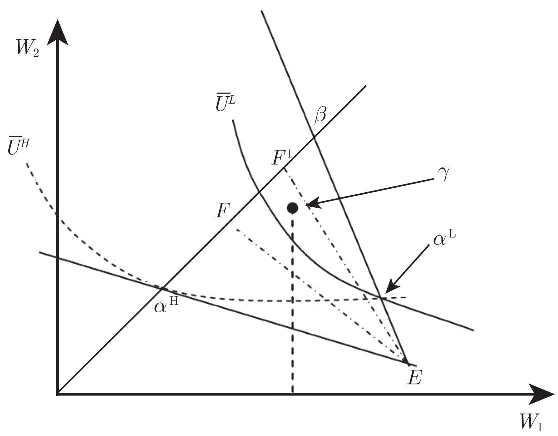

Figure 16.12: Separating equilibrium does not exist

To have an equilibrium, Wilson (1977) introduced "**anticipatory equilibrium**." Under this concept of equilibrium, if a new contract will result in a loss after the old contract is withdrawn, then the new contract will not be introduced. Wilson proved that under such constraint, a market equilibrium exists, and a pooling equilibrium may also exist. Riley (1979) introduced "**reactive equilibrium**." Under this concept of equilibrium, if a series of chain reactions will be triggered after the introduction of a new contract and the original new contract loses money, the new contract will not be introduced. Similarly, under such constraint, a market equilibrium also exists.

Hellwig (1987) and Engers and Fernandez (1987) provided the game theoretic foundations for anticipatory equilibrium and reactive equilibrium, respectively. For the new equilibrium's ability to interpret real-world problems, Kreps (1990) believed that this is closely related to the market regulatory environment. If the regulatory agency does not allow the initial contract (even if it loses money) to exit, the reactive equilibrium will be closer to reality. If the regulatory agency allows the original contract to be withdrawn from the market, then the anticipatory equilibrium will be closer to reality.

16.8.2 Signaling under Asymmetric Information

We discussed the signal mechanism of education in Chapter 6 on game theory. Different employees reveal their ability information through evidence of their education level. Below, we use an example (Tirole, 1988) to discuss how producers can reveal their quality information through some mechanism in the presence of asymmetric information.

Let s be the quality of the product, and the utility level of the consumer at price p be $\theta_s - p$. Assume $s \in \{0, 1\}$ and the production cost per unit of enterprise is c_s ($c_0 < c_1$). We primarily discuss the monopoly situation. There are two periods of repeated purchases of the product. Assume that the quality of the product cannot be changed, and consumers will know the quality of the product after using the product. Let δ be the time discount rate. The monopolist can display quality information by pricing, and p_1 shows high-quality price information. The profit of the high-quality type is:

$$\pi_1 = (p_1 - c_1) + \delta(\theta - c_1).$$

In order to avoid imitation of the low-quality type, it is required that:

$$p_1 \leqq c_0,$$

which implies:

$$\pi_1 \leqq \delta(\theta - c_1) - (c_1 - c_0).$$

If $\delta(\theta - c_1) > (c_1 - c_0)$, there is a high-quality revelation equilibrium. The monopolist asks $p_1 = c_0$ for overall profit by obtaining a monopoly profit in the second stage to make up for the first-stage profit loss. If $\delta(\theta - c_1) \leqq (c_1 - c_0)$, there is no separating equilibrium of high-quality goods. $\delta(\theta - c_1)$ is the benefit obtained by the high-quality type in revelation equilibrium, and $c_1 - c_0$ is the cost of revealing high quality. For this reason, a separating equilibrium can exist if and only if the high-quality type reveals that its benefits outweigh its costs.

In the separating equilibrium, we found that the monopolist established her reputation through initial price concessions. In practice, some firms choose low prices to attract customers when they open a business, and the underlying logic is related to this. However, there are many ways for a company to reveal signals. For example, a company can reveal its type through extravagantly large advertising expenditures.

Let A be a larger advertisement expenditure. Assume that it is a signal of quality. When $A = \theta - c_0$, we find that low-quality companies have no incentive to imitate because

$$\pi_0 \leqq (\theta - c_0) - (\theta - c_0) = 0,$$

while the profit of a high-quality type under advertising investment is:

$$\pi_1 = (\theta - c_1) + \delta(\theta - c_1) - (\theta - c_0) = \delta(\theta - c_1) - (c_1 - c_0).$$

Therefore, when $\delta(\theta - c_1) > (c_1 - c_0)$, the advertisement becomes a signal for displaying the quality information of the company.

In addition, the company can also display quality information through some warranties, after-sales services, etc. The mechanism behind this is that high-quality companies have higher returns than low-quality companies in terms of displaying signals, such as those achieved by repeated purchases in the future.

16.9 Further Extensions

The main theme of this chapter was to determine how the fundamental conflict between rent extraction and allocative efficiency could be solved in a principal-agent relationship with adverse selection. In the models discussed in this chapter, because there is only one binding incentive constraint and one binding participation constraint, this conflict is relatively easy to understand. Here, we mention several possible directions for extension. We can consider two-dimensional adverse selection models, models that involve random participation constraints, limited liability constraints, or models of supervision of the agent by the principal. For a detailed discussion of these topics and their applications, interested readers can refer to Laffont and Martimort (2002).

16.10 Biographies

16.10.1 Oskar Lange

Oskar Ryszard Lange (1904-1965) was a Polish economist, politician, and diplomat. He earned a Ph.D. in law in 1928, and taught statistics and economics at the University of Kraków, Poland. He also taught statistics and economics at the University of Michigan and the University of Chicago from 1938 to 1945. From 1945 to 1946, he was the Polish ambassador to the United States. From 1946 to 1947, he also served as Poland's delegate to the United Nations Security Council. After 1948, he was a professor, and taught statistics and economics at the University of Warsaw, Poland. When the Polish Workers' Party and the Socialist Party merged into the Polish United Workers Party, he was elected as a member of the Central Committee.

Lange's economic theory is of great significance for observing and studying socialist economic relations and exploring the theory of modern socialist economic operation. Lange wrote many books in his life, of which the most influential ones are "On the Economic Theory of Socialism" (1936-1937), "Price Flexibility and Employment" (1944), "Introduction to Econometrics" (1958), "The Political Economy of Socialism" (Volume 1) (1958), "Optimal Decisions: Principles of Programming" (1964), and "Problems of Political Economy of Socialism" (1965).

Lange is best known for advocating the use of market pricing tools in socialist systems. Lange proposed the decentralized model of the socialist economy, i.e., the so-called "Lange model", leading to the famous debate with Mises (1881-1973) and Hayek (1899-1992) in the mid-1930s. Using this model, he argued that prices in the socialist economy are not arbitrarily determined, but are as objective as the market prices in a free competition system. Although the means of production are nationalized in his model, the prices of consumer goods and labor are still priced through the market, while the price of the means of production is simulated by a planning agency, following the "trial and error method", similar to that in a competitive market. Lange thus created a precedent for the analysis of the operation of the market mechanism in socialist economies.

During World War II, Lange studied the issue of price and employment in capitalist economies. He started from the general equilibrium theory and analyzed the currency effect. He pointed out that price elasticity can result in the automatic maintenance or restoration of the supply and demand balance of production factors only under special conditions. However, he believed that the possibility of reaching this special condition in a capitalist economy since the beginning of World War I in 1914 is very small.

Lange also carried out a great deal of pioneering work in applying econometrics to planning the socialist national economy and applying the method of control theory to economic research. He regarded the scientificization of central planning as the theme of the theory of socialist economic operation, which formed modern scientific planning theory.

16.10.2 George Akerlof

George A. Akerlof (1940–) was born in New Haven, the United States. He received a Ph.D. degree at the Massachusetts Institute of Technology in 1966 and has been the Koshland Professor of Economics at the University of California, Berkeley since 1980. In 2001, he was awarded the Nobel Memorial Prize in Economic Sciences with two other American economists, Michael Spence and Joseph Stiglitz, for their "analyses of markets with asymmetric information".

Akerlof's paper, "The Market for 'Lemons': Quality Uncertainty and the Market Mechanism" published in 1970, was one of the most important research works in the information economics literature. In this paper, he introduced a well-known model in the study of information economics– "The Market for Lemons", which was primarily used to describe a situation in which a seller had more information on the quality of goods than a buyer, and poor quality goods would expel good quality goods, resulting in a reduction in the average quality of goods in the market. Akerlof's theory was widely used in different subfields of modern economics, such as health insurance, financial markets, and employment contracts.

Today, a "lemon", the colloquial expression for a defective used car, has become a famous metaphor in economists' theoretical vocabulary. He believed that the problem of information asymmetry may lead to the collapse of the entire market, or contraction of the market, so that the market is flooded with only inferior goods. Akerlof also pointed out that similar information asymmetries were particularly common in developing countries, and had a significant impact on them. He used India's credit market in the 1960s as an example to illustrate the issue of adverse selection. The interest rate set by lenders in small locations in India was twice the interest rate in large cities. A lender who borrowed in cities and then lent out in rural areas did not know the borrower's reputation, and was therefore vulnerable to heavy losses. A key insight in the "Lemon thesis" is that economic agents have strong incentives to offset the adverse effects of information problems on market efficiency. Akerlof thought that many market institutions could be seen as ways to solve asymmetric information problems.

Akerlof believed that economic theorists, in a sense, are similar to French chefs in regard to food, and it is necessary to develop styled models whose ingredients are constrained by unwritten rules. Just as traditional French cooking does not utilize raw fish or seaweed, neoclassical economic models do not make assumptions from psychology, anthropology, or sociology. He disagrees with any rules that limit the nature of the ingredients in economic models. Therefore, in addition to research on asymmetric information, Akerlof also developed economic theory that borrows from sociology and social anthropology. The most noteworthy contribution in this regard was his concern about the efficiency of the labor market.

Akerlof's wife is his colleague Janet L. Yellen, an economist at the University of California, Berkeley. They collaborated in numerous fields of research. Yellen commented that, "he has real flashes of insight into human problems, into what may explain social phenomena. It's wonderful to work with him; he's so original". Yellen served as chair of the Federal Reserve from 2014-2018 and officially replaced Ben Bernanke as the first female chair in the Fed's history, and is now the Secretary of the Treasury in the U.S.

16.11 Exercises

Exercise 16.1 (General Cost Function) Consider the principal-agent model with an unobserved cost function of the agent. Different cost functions $C(q, \theta)$ represent different types. Assume that $C_q > 0$, $C_\theta > 0$, $C_{qq} > 0$ and $C_{q\theta} > 0$, where q is the perfectly observed output of the agent. Assume that there are two types of agents, $\theta \in \{\theta_L, \theta_H\}$, with the probability ν and $1-\nu$, respectively, and $\Delta\theta \equiv \theta_H - \theta_L > 0$. Output is evaluated at $S(q)$ by the principal, where $S' > 0$, $S'' < 0$ and $S(0) = 0$. Let T denote the transfer from the principal to the agent. The payoff function of the agent is $T - C(q, \theta)$. The principal and the risk-neutral agent signed the contract at the interim stage.

1. Write out the principal's optimization problem satisfying incentive-feasible constraints.

2. What is the optimal transfer of T_H and T_L, and economic rent for each type?

3. Determine the first-order conditions for the problem of the principal's incentive-compatibility. What do they mean?

4. Let $S(q) = q$, $C(q, \theta) = \theta q^2/2$, $\theta_L = 1$, $\theta_H = 2$, $\nu = 2/3$. Solve for the optimal contract.

Exercise 16.2 (Risk-Averse Principal) Under the same setup as that in the previous question, now consider a risk-averse principal with a Von Neumann-Morgenstern utility function $v(\cdot)$ defined on her monetary gains from trade $S(q) - t$ such that $v' > 0$, $v'' < 0$ and $v(0) = 0$.

1. Write down the principal's optimization problem satisfying incentive-compatibility constraints.

2. What is the optimal transfer of T_H and T_L, and economic rent for each type?

3. Find the first-order conditions of the principal's problem. Compare the output distortions of the risk-averse principal and the risk-neutral principal. Which one is greater and why?

4. Suppose that $v(x) = \frac{1-e^{-rx}}{r}$. Determine the first-order condition for the problem. What happens when $r \to 0$? Does the solution converge to the distortion of the situation with a risk-neutral principal and an interim participation constraint? What occurs when r approaches infinity? Can the first-best output be implemented?

Exercise 16.3 (More than One Good) Consider a principal-agent model in which the agent is producing a whole vector of goods $q = (q^1, ..., q^n)$ for the principal. The agents' cost function becomes $C(q, \theta)$ with $C(\cdot)$ being strictly convex in q and $C_{q^i\theta} > 0$ holds for each output i. Assume that there are two types of agents, $\theta \in \{\theta_L, \theta_H\}$, with the probability ν and $1 - \nu$, respectively, and $\Delta\theta \equiv \theta_H - \theta_L > 0$. The value for the principal of consuming this whole bundle is $S(q)$ with $S(\cdot)$ being strictly concave in q. Let T denote the transfer from the principal to the agent. The payoff function of the agent is $T - C(q, \theta)$. The principal and the risk-neutral agent signed the contract at the interim stage.

1. Write out the principal's optimization problem satisfying incentive-feasible constraints.

2. Determine the first-order conditions for the problem of the principal's incentive-compatibility.

3. What is the optimal transfer of T_H and T_L, and economic rent for each type?

4. Without further specifying the value and cost functions, can we still have the second-best outputs define a vector of outputs with some components q_H^{iSB} less than q_H^{i*} for all i?

Exercise 16.4 Suppose that a firm hires two types of workers θ_H and θ_L to produce a product. The proportion of workers of type θ_L is λ. When a type θ worker pays T dollars to consume x units of goods, the utility is $u(x, T) = \theta v(x) - T$, where $v(x) = \frac{1-(1-x)^2}{2}$. The firm is the only producer of this product, and its production cost per unit is $C > 0$.

1. Consider a non-discriminatory monopolist. Derive the monopolist's pricing strategy. Prove that when θ_L or λ is sufficiently large, the monopolist will provide products to these two types of consumers.

2. Now, assume that the monopolist can distinguish between these two types of consumers, but she can only request a uniform price of p_i for each type of θ_i. Describe the optimal pricing of the monopolist.

3. Fine the optimal nonlinear pricing.

Exercise 16.5 (Shutdown Contract) Under the setting of Exercise 16.1, assume that the cost function is $C(q, \theta) = \theta q$. Solve for the optimal contract with the shutdown property.

1. Under what conditions will the optimal contract shut down the inefficient participant?

2. Assume that the reservation utility is U_0. Prove that the conditions for the shutdown contract are $\frac{v}{1-v}\Delta\theta\overline{q}^{SB} + U_0 \geqq S(\overline{q}^{SB}) - \overline{\theta}\overline{q}^{SB}$.

Exercise 16.6 (State-dependent fixed costs) Under the same settings as those in Exercise 16.1, assume that the cost function is given by $C(q, \theta) = \theta q + F(\theta)$, and the fixed cost satisfies $F(\theta_L) > F(\theta_H)$, where higher marginal costs are associated with lower fixed costs, and vice versa.

1. Write down the principal's optimization problem satisfying incentive-feasible constraints.

2. What is the optimal transfer of T_H and T_L, and economic rent for both types?

3. Find the first-order conditions of the principal's incentive-compatibility problem.

4. Discuss the different intervals of the solution, based on the different positions of $F(\theta_L) - F(\theta_H)$, $\Delta\theta q_H$, and $\Delta\theta q_L$.

Exercise 16.7 (Three types of agent) Under the setting of Exercise 16.1, assume that there are three possible types θ of agent, $\{\theta_1, \theta_2, \theta_3\}$, and $\theta_3 - \theta_2 = \theta_2 - \theta_1 = \Delta\theta$, with probability v_1, v_2, v_3, respectively. The direct revelation mechanisms are $\{(t_1, q_1), (t_2, q_2), (t_3, q_3)\}$, respectively. The other settings are the same as those in Exercise 16.1. The cost function is $C(q, \theta) = \theta q$.

1. Write down the economic rents, participation constraints, and incentive-compatibility constraints for the three types.

2. Simplify the participation constraints and the incentive-compatibility constraints.

3. If $v_2 > v_1 v_3$, prove that the monotonicity condition is binding and solve for the optimal solution.

4. If $v_2 \leqq v_1 v_3$, solve for the optimal contract.

Exercise 16.8 (Agent with any Number of Types) Under the setting of Exercise 16.1, assume that there are n types of agent, $\theta_n > \cdots > \theta_2 > \theta_1$. The probability of each type θ_i is β_i.

1. Give the Spence-Mirrlees single crossing condition.
2. Write down the local downward incentive-compatibility constraints.
3. Prove that the single-crossing condition implies monotonicity and the local incentive-compatibility constraints.
4. Prove that, at the optimal solution, all local downward incentive-compatibility constraints (LDICs) are binding.
5. Prove that all LDICs are binding, plus the monotonicity condition, guarantee that all local upward incentive-compatibility constraints (LUICs) are satisfied. Moreover, prove that global incentive-compatibility constraints hold.
6. Solve the simplified optimal contract problem.

Exercise 16.9 (Continuum Types of Agent) In the above question, the number of types is finite. It is now assumed that there is a continuum of types, i.e., θ no longer takes a finite number of values, but is defined on the interval $[\underline{\theta}, \overline{\theta}]$, and the density function is $f(\theta)$. Due to the revelation principle, we only consider the direct revelation mechanism $\{q(\theta), T(\theta)\}$.

1. Give the Spence-Mirrlees single-crossing condition.
2. Write down the local downward incentive-compatibility constraint.
3. Write down the monotonicity condition, the local incentive-compatibility constraints, and the Hamiltonian function of the optimal contract problem.
4. Solve the optimization problem and explain the results. Compare the optimal contract problem with a continuum of types to the case of two types in Exercise 16.1.

Exercise 16.10 Consider a screening model. The seller's cost function is $C(q) = q^{1/2}$. If a buyer consumes q units of goods, his utility function is $\theta \ln q$. The buyer's type θ is private information and follows a uniform distribution on $[0, 1]$.

1. Write down a monopolist's optimization problem.
2. Solve for the optimal contract $(q(\theta), t(\theta))$.

Exercise 16.11 A monopolist wants to sell a single item to a consumer. The latter's willingness-to-pay for this item is t_1 or $t_2(> t_1)$. The seller knows that the buyer is subject to the liquidity constraint, and the maximum transfer for the item is $t \in \{t_1, t_2\}$. The buyer's willingness-to-pay is private information, and the seller only knows the probability $\mu(t_i)$ of type t_i. This item is worth $c < t$ to the seller. Suppose that both the buyer and the seller are risk-neutral. Let v denote the probability that the transaction occurs, h denote the payment to the seller, and t_i be the value of the buyer. The buyer's revenue function is:

$$vt_i - h,$$

and the seller's revenue function is:

$$h - vc.$$

In addition, regardless of type, the buyer's reservation utility is 0.

1. Give the participation constraints, incentive-compatibility constraints, and liquidity constraints required by a direct mechanism.

2. Determine the direct mechanism for maximizing the expected return of the seller.

Exercise 16.12 (Bunching and Ironing) In the agency problem with a continuum of types, it is generally assumed that the hazard rate condition, i.e., the hazard rate $h(\theta) = \frac{f(\theta)}{1-F(\theta)}$, is increasing in θ.

1. Prove that if the hazard rate condition holds, the monotonicity condition is satisfied in the implementation of the optimal contract.

2. If the monotonicity condition does not hold, then in most cases it is necessary to re-solve the optimization problem. One can refer to the content of Section 16.5.3 of the textbook, and it is necessary to redefine the Hamiltonian function and the corresponding constraints.

3. Use the Pontryagin maximum principle to re-solve the optimal contract problem.

Exercise 16.13 Suppose that two persons consider trading some kind of asset at a price p. This asset can only be used as a store of wealth. Person 1 currently owns this asset. Everyone's evaluation of this asset is only known to himself, and each person only cares about the expected value of the asset after one year. Assume that only when both sides think that the transaction can make their situation better will they trade at the price of p. Prove that the probability of a transaction is zero.

Exercise 16.14 Consider the following process. First, nature determines the type of a worker. It is continuously distributed over the interval $[\underline{\theta}, \overline{\theta}]$. Once the worker knows his type, he can choose whether to participate in an exam without cost. The exam accurately reflects the worker's ability. After observing whether the worker took the test and the performance of the worker who took the test, the two companies start bidding on the service of the worker. Prove that in any subgame perfect Nash equilibrium of this model, all worker types take the exam, and the company does not provide any salary greater than $\underline{\theta}$ for any worker who does not take the exam.

Exercise 16.15 (Lending with Adverse Selection) Suppose that there is a continuum of risk-neutral borrowers with personal wealth and limited liability. A proportion ν of borrowers (called type 1) has risk-free projects with return h for an investment of 1. A proportion $1-\nu$ of borrowers (called type 2) has stochastically independent projects with return h with probability θ belonging to $(0, l)$ and return 0 with probability $1-\theta$, for an investment of 1. If he does not apply for a loan, the borrower has an outside opportunity that generates a utility of u. There is a single risk-neutral bank available for loans which has a financing cost of r. The bank offers contracts to maximize its expected profit. For simplicity, we assume that all projects are socially valuable, i.e.,

$$\theta h > r + u.$$

1. Explain why there is no loss of generality in considering the menu of contracts, (r_1, P_1) and (r_2, P_2), where P_i is the probability of obtaining a loan and r_i is the repayment to the bank when the investment succeeds, whenever the borrower announces that he is of type i.

2. Write down the maximization program of the bank which chooses the contract menu, (r_1, P_1) and (r_2, P_2), to maximize its expected profit under the borrower's participation and incentive-compatibility constraints (for simplicity, assume that if a borrower applies for a loan, he loses the outside opportunity).

3. Show that the optimal contract entails a non-random allocation of loans, i.e., P_i is either 0 or 1, for $i = 1, 2$. Characterize the optimal contract and discuss it.

Exercise 16.16 (Bribery Game) Consider a regulation agency that provides a service to citizens with a fixed period of delay. Under the normal operation of the regulation agency, citizens can obtain the utility of u_0 (depending on their valuation of time). Officials who devote extra effort can shorten the delay period. Officials can shorten the delay period of q at the cost of $\frac{(q-Q)^2}{2}$, where Q is a constant. Assume that there is ν (or $1-\nu$) of type 1 (type 2) citizens rating q as $\theta_L q$ ($\theta_H q$). Citizens are willing to bribe officials to shorten the delay period. Write down the optimal bribe contract that officials provide to citizens and find the optimal contract.

Exercise 16.17 A monopolist can produce a product at different quality levels. The cost required to produce a unit with a mass of s is s^2. A consumer purchases up to one unit of product. If he consumes a product with unit mass s, the resulting utility level is $u(s|\theta) = s\theta$. The monopolist can determine the price and quality of the product. The consumer observes the quality and price of the product, and decides what quality of product to purchase.

1. Solve for the optimal solution.

2. Suppose that seller cannot observe θ, and $prob(\theta = \theta_H) = 1-\beta, prob(\theta = \theta_L) = \beta$, where $\theta_H > \theta_L > 0$. Solve for the second-best outcome and information rent of the consumer.

3. Suppose that θ follows a uniform distribution on the interval $[0, 1]$. Find the second-best contract.

Exercise 16.18 (Labor Contract) Consider the following setting. A firm faces a worker. The worker's utility function is $U^A = u(c) - \theta l$, where c is consumption and l is labor supply. $\theta \in \{\theta_L, \theta_H\}$ is a parameter whose value is privately known by the worker. $\theta_L < \theta_H$, and $u(\cdot)$ is an increasing and concave function. The probability of the worker having a lower disutility ($\theta = \theta_L$) is ν. The agent's optimal choice must satisfy the budget constraint $c \leq T$, where T is the transfer received from the employer. The utility function of the employer is $U^P = f(l) - T$, where $f(l)$ is a production function with decreasing returns to scale.

1. Suppose that the employer knows θ. Give the solution for the employer to maximize her utility subject to the worker's participation constraint. This solution is called the first-best solution.

2. Suppose that the employer can observe and verify the labor supply, but cannot observe U and θ. Prove that the first-best solution cannot be achieved. The employer can now provide the contract menu, (T_L, l_L) and (T_H, l_H), where (T_H, l_H) is the contract selected by type θ_H and (T_L, l_L) is the contract selected by type T_L. Derive the optimal contract and compare the second-best solution with the first-best solution.

3. Suppose that the worker with a lower disutility of the effort has an outside opportunity that can bring him the utility of V. Compare the first-best solution with the second-best solution in this case. Note that

the solution will depend on V. Consider, respectively, $l_H^{SB}\Delta\theta \geqq qV$ (SB denote "second-best"), $l_H^{SB}\Delta\theta \leq V \leq l_H^{FB}\Delta\theta$ ($l_H^*\Delta\theta \leqq qV \leqq ql_L^*\Delta\theta$, denote "the first-best") and $V \geqq ql_L^*\Delta\theta$. What are the binding constraints in each situation? What type of distortion is needed?

Exercise 16.19 (Labor Contract with Adverse Selection) Consider a principal-agent relationship, in which the principal is the employer and the agent is the worker. For the worker of type $\theta \in \{\theta_L, \theta_H\}$, the work disutility for production y is $\psi(\theta y)$. In other words, the worker of type θ must work on l units ($l = \theta y$), and the resulting disutility is $\psi(l)$. If the worker receives a compensation of t from the employer, the net utility is $U = t - \psi(\theta y)$, and the employer's utility function is $V = y - t$.

1. Refer to the revelation principle, and write down the characteristics of the direct mechanism.

2. Suppose that ν (or $1-\nu$) is the probability that the worker is of type θ_L (or θ_H). Find the contract that maximizes the expected utility of the employer, while satisfying the workers' incentive-compatibility and participation constraints.

Exercise 16.20 (Information and Incentive) An agent (natural monopoly manufacturer) produces q units of output with the variable cost function θq ($\theta \in \{\theta_L, \theta_H\}$, $\Delta\theta = \theta_H - \theta_L$). The utility obtained by the principal from production is $S(q)$ ($S' > 0$ and $S'' < 0$), and the transfer to the agent is T. The utility function of the principal is $V = S(q) - T$, and the agent's utility function is $U = T - \theta q$. In addition, the agent's current utility is normalized to 0.

1. When the principal has complete information on θ, write down the principal's first-best contract.

2. Suppose that θ is private information of the agent and $\nu = Pr(\theta = \theta_L)$. Write down the optimal contract for the principal that satisfies the agent's participation constraints at the interim stage (suppose that the value of the project is sufficiently large, and thus the principal is always willing to have a positive output).

3. Suppose that, with information technology, the principal can obtain signals $\sigma \in \{\sigma_L, \sigma_H\}$ with

$$\mu = Pr(\sigma = \sigma_L | \theta = \theta_L) = Pr(\sigma = \sigma_H | \theta = \theta_H) \geqq \frac{1}{2}q.$$

   Derive the principal's updated beliefs about the agent's efficiency, i.e., $\hat{\mu}_1 = Pr(\theta = \theta_L | \sigma = \sigma_L)$ and $\hat{\mu}_2 = Pr(\theta = \theta_H | \sigma = \sigma_H)$. Solve for the optimal contract for each σ.

4. Prove that an increase in μ will lead to an increase in the expected utility of the principal.

Exercise 16.21 A government agency signs a procurement contract with a firm. The marginal cost of the firm producing q units is c, and thus its profit is $P - cq$. The cost of the firm is private information, which may be high cost or low cost ($0 < c_L < c_H$). The government's prior belief about the cost is $Prob(c = c_L) = \beta$, and it makes a take-it-or-leave-it offer to the firm whose reservation profit is zero.

1. Let $B(q)$ represent the revenue function of the government when it obtains q units. What is the government's second-best contract?

2. Compare the second-best contract with the first-best contract.

3. Suppose that c follows a uniform distribution over $[0, 1]$. Solve for the first-best contract and the second-best contract.

Exercise 16.22 (Lemon Market) Consider a used car market. There are many sellers in the market. Each seller has a used car ready for sale. It will be sold with the car's quality $\theta \in [0, 1]$ that follows a uniform distribution over $[0, 1]$. If a seller of type θ sells a car at price p, then his utility is $u_s(p, \theta)$; if he does not sell a car, the utility obtained is 0. If a buyer buys a used car, his utility function is $\theta - p$; if not, it is 0. The quality of the used car is the seller's private information, and the buyer does not know it beforehand. Assume that there are not enough used cars in the market for all possible buyers.

1. Prove that for competitive equilibrium under asymmetric information, $E(\theta|p) = p$.

2. Prove that if $u_s(p, \theta) = p - \frac{\theta}{2}$, then any $p \in (0, 1/2)$ is an equilibrium price.

3. If $u_s(p, \theta) = p - \sqrt{\theta}$, find the equilibrium price and indicate which kind of used cars will be traded in equilibrium.

4. Suppose that $u_s(p, \theta) = p - \theta^3$. Solve for the equilibrium price, and determine how many equilibria there are in this case.

5. Are the above outcomes Pareto efficient? If possible, propose a Pareto improvement scheme.

Exercise 16.23 (Insurance Market) Consider the following insurance market. There are two types of consumers: High risk and low risk. The initial wealth of each consumer is W, but it is possible to lose L due to fire. For high-risk and low-risk consumers, probabilities of fire occurrences are p_H

and p_L, respectively, where $p_H > p_L$. Both types are pursuing the largest expected utility, and the utility function is $u(W)$, which satisfies neoclassical properties (e.g., differentiability, concavity, monotonicity, etc.). There are two insurance companies, and both are risk-neutral. Any insurance contract consists of a premium of M and a compensation of R that the insurance company pays according to claims.

1. Suppose that each consumer purchases up to one insurance contract. Prove that the insurance contract specifies the wealth of the insured in the "no loss" state and the "in loss" state.

2. Suppose that each insurance company provides a contract at the same time, and each company can provide any finite number of insurance contracts. What is the subgame perfect equilibrium for this problem? Does equilibrium necessarily exist?

Exercise 16.24 (Pollution Regulation) Consider a manufacturer whose revenue is R. The manufacturer generates x units of pollution in production. The damage caused by the pollution is $D(x)$, with $D'(x) > 0$ and $D''(x) \geqq 0$. The cost function of the manufacturer is $C(x, \theta)$, satisfying $C_x < 0$ and $C_{xx} > 0$. θ is the parameter that the manufacturer privately observes, for $\theta \in \{\theta_L, \theta_H\}$, while $\nu = Pr(\theta = \theta_L)$ is public knowledge.

1. The first-best pollution $x^*(\theta)$ under complete information is given by:

$$D'(x) + C_x(x, \theta) = 0.$$

   Prove that if the regulator does not have to meet the manufacturer's participation constraint, she can implement $x^*(\theta)$ by giving the manufacturer a transfer equal to the damage of the pollution.

2. Suppose that the manufacturer can now refuse to accept regulation (in this case the manufacturer utility is 0), and the regulator's objective function is as follows:

$$W = -D(x) - (1 + \lambda)T + T - C(x, \theta),$$

   where T is the transfer from the regulator to the manufacturer, and $(1 + \lambda)$ is the opportunity cost of social fund spending. For any θ, the regulator must satisfy the manufacturer's participation constraint:

$$T - C(x, \theta) \geqq q_0.$$

   Write down the decision rule $\hat{x}(\theta),\ \theta \in \{\theta_L, \theta_H\}$ that maximizes W under complete information, and compare it with problem 1.

3. Suppose that $C_\theta < 0$ and $C_{x\theta} < 0$. Find the menu of contracts (T_L, x_L) and (T_H, x_H) that maximizes the expected W under participation and incentive-compatibility constraints.

4. Suppose that θ is distributed in the interval $[\theta_L, \theta_H]$ according to the distribution function $F(\theta)$ and the density function $f(\theta)$, satisfying:

$$\frac{d}{d\theta}\left(\frac{1-F(\theta)}{f(\theta)}\right) < 0,$$

and $C_{\theta\theta x} \leqq q_0$. Answer question 3 under these new conditions.

Exercise 16.25 (Life Insurance Demand and Adverse Selection) A group of consumers have an income of y_0 in period 1 and no income in period 2. Each consumer knows his probability of death, π_i. Insurance companies can neither observe the probability of the death of consumers nor observe the behavior of consumers. Insurers offer insurance at a price of p: If a consumer purchases x units of insurance, the insurance company will receive px of income for the period and pay x when the consumer dies. The consumer can choose to purchase the insurance amount of x and save the amount of s. Savings of the current period are added to the income of the consumer when he is alive in the next period. Assume that there are many consumers. The goal of consumers is to maximize the income of each period to increase the consumption of family members. The consumer's utility function is of Bernoulli form.

1. If a consumer's utility function is:

$$u(c_1, c_2a, c_2d) = \log c_1 + (1-\pi_i)\log c_2a + \pi_i \log c_2d.$$

Calculate the demand price elasticity of life insurance.

2. Suppose that the competition between insurance companies means that consumers' expectation payments are equal to insurance premiums. Prove that p is greater than the average of π_i.

Exercise 16.26 (Self-Managing Manufacturer) A manufacturer's production function is $y = \theta l^{1/2}$, where l is the number of workers (here considered as a continuous variable), and $\theta > 0$ is the parameter only known to the manufacturer. The fixed cost of production is A, and thus p represents the price of the manufacturer's product in the competitive market.

1. Suppose that the manufacturer is managed by the workers, and the objective function is:

$$U^{SM} = \frac{py - A}{l}.$$

Find the optimal number of workers for this worker self-managing manufacturer.

2. Let w represent the wage rate under a perfectly competitive labor market, i.e., w is the opportunity cost of labor in this economy. What is the optimal allocation of labor? What happens if w is too large? Why is the scale of the self-managed manufacturer generally not optimal?

3. Suppose that the government knows θ. Consider the situation in which w is sufficiently small to make the self-managed company relatively reasonable. Find the unit product tax τ that can restore the first-best labor allocation. Prove that the same goal can be achieved by imposing a fixed tax of T on the manufacturer (assuming that the size of the manufacturer is negligible compared to the entire economy).

4. Suppose that the government does not know θ, but only knows that θ may be θ_L or θ_H, $\Delta\theta = \theta_H - \theta_L > 0$. The government uses a mechanism $(l(\tilde{\theta}), t(\tilde{\theta}))$ specifying a labor input of $l(\tilde{\theta})$ and a transfer of $t(\tilde{\theta})$ for the type of manufacturer report $\tilde{\theta}$. The manufacturer's objective function is now:

$$U^{SM} = \frac{p\theta(l(\tilde{\theta}))^{1/2} + t(\tilde{\theta}) - A}{l(\tilde{\theta})}.$$

   Write a regulatory mechanism that leads to truthful-type reporting.

5. Suppose that $\nu = Pr(\theta = \theta_L)$. The government wants to maximize

$$U^G = p\theta l^{1/2} - wl.$$

   Prove that even if the transfer is costless to the government, the first-best outcome cannot be implemented. At this point, suppose that the opportunity cost of the manufacturer is 0. What is the optimal regulatory mechanism?

Exercise 16.27 (Insurance Contract) In a continuum economy, the economic individual's production function is $q = \theta l$, where θ is the productivity parameter, and the probability density function is $f(\theta)$,$\theta \in [\underline{\theta}, \overline{\theta}]$. The utility function of the economic individual is $u = u(c) - l$, which is a concave function.

1. If it is an autarky economic environment, solve for the distribution of output and consumption in the economy.

2. Suppose that the insurance contract is signed in advance, i.e., all economic individuals sign the insurance contract before they know their productivity parameter θ. If θ and l can be observed afterwards, solve for the optimal insurance contract. If only θ can be observed afterwards, solve for the optimal insurance contract.

3. In the previous question, if θ and l are not observable afterwards, and $f(\theta)/[1-F(\theta)]$ monotonically increases, what is the optimal insurance contract?

Exercise 16.28 Consider an educational investment signaling model. There are two possible types of employee: $\theta \in \{\theta_H, \theta_L\}$ with $\theta_H > \theta_L$. Given $i \in \{H, L\}$, the ex-ante probabilities of this employee's type θ_i is β_i. The reservation utility for all employees is $\overline{u} = 0$. Each type θ can produce θ for the firm. The firm is willing to hire an employee at the salary level of w if and only if that employee's expected productivity can at least offset wages. Type θ can receive e year of education at a cost of $c(e, \theta) = \frac{e}{\theta}$. The educational investment cost function $c(e, \theta)$ satisfies the single crossover property for (e, θ), i.e., if $e > e'$, then $c(e, \theta_L) - c(e', \theta_L) > c(e, \theta_H) - c(e', \theta_H)$. Given a salary level of w and an education level of e, type θ has a reward function of $u(w, e|\theta) = w - c(e, \theta)$. Consider the following sequential actions:

- The employee observes his own type that is his private information;
- The employee chooses an education investment level;
- The firm observes the level of the employee's education, but cannot observe his type;
- The employee proposes a salary offer to the firm;
- The firm either rejects the offer or accepts the offer, and employs the employee at the salary level.

It is assumed that education investment can promote a low-type employee to a high-type employee. Specifically, suppose that the probability of type θ_L investing in a e year education to become type θ_H is $p(e)$, which satisfies the following properties: $p(0) = 0$, $\lim_{e \to \infty} p(e) = 1$, $p' > 0$, $p'(0) = \infty$, $\lim_{e \to \infty} p'(e) = 0$ and $p'' < 0$. Once type θ_L is invested in education and converted to type θ_H, he can observe the type change prior to entering the labor market. First, it is assumed that there is no asymmetric information between the firm and its employees. Answer the following questions:

1. Write down the optimization problem that can find the optimal salary and the optimal education investment level.
2. Solve this problem. Particularly, prove that the educational investment level of type θ_H is 0, while the education investment level of the low-type employee is strictly positive.
3. Suppose that investment in education now becomes more efficient, i.e., the probability that type θ_L invests e in education and becomes type θ_H with the probability $q(e)$. It satisfies $q(e) > p(e)$ for any $e > 0$.

What is the optimal education investment level at this time? What is the salary of type θ_L at this time? Provide a corresponding intuitive explanation.

Exercise 16.29 (Educational Investment Signaling Model 1) Assume that there is asymmetric information between a firm and its employees, and we adopt a refined Bayesian pure strategic equilibrium solution concept. Consider a separating equilibrium that satisfies $e_H \neq e_L$. Answer the following questions:

1. In any segregated equilibrium, type θ_L employee selects the education level $e_L = e_{FB}$, where FB represents the first-best.

2. Describe the education investment level of type θ_H, which satisfies $e_H \neq e_{FB}$.

3. Explain whether separation equilibrium always exists.

4. Suppose that investment in education is now becoming more efficient. In other words, the probability that a worker of type θ_L invests in e year education and becomes type θ_H is $q(e)$. It satisfies $q(e) > p(e)$ for any $e > 0$. Does the employee of type θ_H change the level of education investment in equilibrium as described above? Explain your conclusions.

Exercise 16.30 (Educational Investment Signaling Model 2) Consider a pooling equilibrium that satisfies $e_H = e_L = e^*$. Answer the following questions.

1. Given educational investment e^* under a pooling equilibrium, what is the salary of the employee?

2. Describe the pooling equilibrium e^*.

3. Does the pooling equilibrium $e^* = 0$ always exist? If $e^* = e_{FB}$, does the pooling equilibrium always exist?

4. Suppose that investment in education now becomes more efficient. In other words, the probability that a worker of type θ_L invests in e year education and becomes type θ_H is $q(e)$, which satisfies $q(e) > p(e)$ for any $e > 0$. Describe the influence of the change on the pooling equilibrium e^*. Explain your conclusions.

5. Given the assumption of effective education investment, is there a pooling equilibrium satisfying the intuitive criterion?

Exercise 16.31 (Supervision Cost) There are two types of economic individuals in the economy, banks and firms, respectively, indexed by $i = 1, 2, 3, \cdots$. Each bank receives one unit of investment endowment in every period, which can be invested or lent to firms. If a bank makes an investment, it can receive t_i units of a certain benefit. A firm does not have any endowment in every period. However, every firm has an investment project, which can produce a random benefit of $w_i \in [0, \overline{w}]$ with one unit investment, where w_i is independently identically distributed. The density function is $f(w)$, and the distribution function is $F(w)$. f is continuous and differentiable, which is common knowledge. For every firm i, w_i can be observed without cost. The bank needs to pay γ units of effort cost to observe the investment benefit of a firm. The firm and bank sign a loan contract ex-ante, with the interest rate being x. The firm makes a signal to the bank $w^s \in [0, \overline{w}]$ after observing the return of w. If $w^s \in S \subseteq [0, \overline{w}]$, the bank engages in supervision; if $w^s \notin S$, there is no supervision. According to the contract, the bank lends one unit of investment to the firm, and the firm should return $R(w)$ units if $w^s \in S$, where $0 \leqq R(w) \leqq w$. If $w^s \notin S$, the firm should return $K(w)$ units. In order to guarantee that all borrowing demand is satisfied, assume that the ratio of the banks is $1/2 < \alpha < 1$.

1. Prove that $R(w) = w$ and $K(w) = x$ are the optimal payment plan.
2. For the optimal payment plan, rewrite the profit of the bank and the firm (a function of x), and prove that the profit of the firm is monotonously decreasing with x.
3. Write down the optimization problem of the firm, including the participation constraint of the bank.
4. Suppose that the profit function of the bank is concave. Prove that Nash equilibrium exists in the credit allocation of the economy.
5. Compared with Stiglitz and Weiss (1981), where all firms are homogenous ex-ante, why does credit allocation still exist?

Exercise 16.32 (Costly State Verification) Suppose that there exist two types of agent, with different cost function $C(q, \theta)$. Assume that $C_q > 0$, $C_\theta > 0$, $C_{qq} \geq 0$ and $C_{q\theta} > 0$, where q is the output observed perfectly by the principal. $\theta \in \{\theta_L, \theta_H\}$, with the probabilities of v and $1 - v$, and $\Delta\theta \equiv \theta_H - \theta_L > 0$. The output q is valuated via $S(q)$ by the principal, where $S' > 0$, $S'' < 0$, and $S(0) = 0$. Let t denote the transfer from principal to agent. The benefit function of the agent is $t - C(q, \theta)$, and the cost function is $C(q, \theta) = \theta q$.

The principal has an auditing technology. The principal can observe the true type of agent with the probability of p by paying a cost of $c(p)$. For the audit cost, assume that $c(0) = 0$, $c' > 0$, and $c'' > 0$. To insure the existence of an interior solution, suppose that the cost function satisfies

Inada conditions. The incentive mechanism includes four variables: Transfer $t(\tilde{\theta})$; output $q(\tilde{\theta})$; the probability of observing the true type by auditing $p(\tilde{\theta})$; and if $\tilde{\theta}$ that the agent reported is different from the true θ, the principal can exert the punishment $P(\theta, \tilde{\theta})$ on the agent. In equilibrium, the revelation principle indicates that the report is truthful.

1. Write down the participation and incentive-compatibility constraints.
2. Suppose that the punishment is exogenous, $P(\underline{\theta}, \overline{\theta}) \geqq l$, and $P(\overline{\theta}, \underline{\theta}) \geqq l$. Solve for the optimal contract.
3. Suppose that the punishment is endogenous, $P(\underline{\theta}, \overline{\theta}) \geqq \bar{t} - \underline{\theta}\overline{q}$ and $P(\overline{\theta}, \underline{\theta}) \geqq \underline{t} - \overline{\theta}\underline{q}$. Solve for the optimal contract.
4. Compare the difference of the above two results.

Exercise 16.33 (Financial Contract) Suppose that there are two types of participants, with the bank acting as a principal and the firm acting as an agent. The profit of the investment project operated by the firm is θ. Among them, a high profit of $\overline{\theta}$ is obtained with probability of v, and a low profit of $\underline{\theta}$ is obtained with probability of $1 - v$. The principal has a random auditing technique with a probability of $p(\theta)$. The cost of using this technique is $c(p)$.

1. Prove that if the type of profit reported by the agent is a high-type, the optimal probability of auditing is 0.
2. Write down the incentive-compatibility and participation constraints for this problem, as well as the principal's objective function.
3. Solve for the optimal contract.
4. Under the optimal contract, will there be credit rationing in the economy?

Exercise 16.34 (Termination Threat) Suppose that there are two types of participants, with the bank acting as a principal and the firm acting as an agent. The profit of the investment project operated by the firm is θ, where a high profit of $\overline{\theta}$ is obtained with probability of v and a low profit of $\underline{\theta}$ is obtained with probability of $1 - v$. The agent does not have any endowment at the beginning. If the agent needs to invest, he must invest I units. Suppose that $v\overline{\theta} + (1 - v)\underline{\theta} > I$ and $\underline{\theta} > I$. Assume that the relationship of the contract lasts for two periods, and the profit type θ in both periods is independent and the time discount is 1. After knowing the type of the agent's report in the first period, the principal can terminate the contract relationship with the agent at the end of the first period. The probability is $\overline{p}$ and $\underline{p}$, respectively, for these two types.

1. Write down the incentive-compatibility and participation constraints for this two-stage problem.

2. Suppose that the agent has limited liability, $\bar{t} \leqq \bar{\theta}$ and $\underline{t} \leqq \underline{\theta}$. Solve for the optimal contract.

3. Compare the difference between the contract and the one with the random auditing technique.

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Chapter 17

The Principal-Agent Model: Moral Hazard

17.1 Introduction

In the previous chapter, we stressed that the delegation of tasks creates an information gap between the principal and the agent when the latter has some private information relevant to determining the efficient level of trade. The principal thus needs to design appropriate incentive contracts to induce the agent to truthfully reveal his types. However, it is costly to do so. As such, the first-best outcome, in general, cannot be achieved, and only the second-best outcome is possible. The optimal contract obtained under asymmetric information is different from that under complete information. The basic trade-off in adverse selection is between the extraction of information rent and the efficiency of allocation.

Adverse selection is not the only problem that arises under asymmetric information. In many situations, the principal usually has no control over the agent's actions, and the agent's behavior is also unobservable, so the agent can always act in a way that is beneficial to himself and unfavorable to the principal. Such a type of asymmetric information is called the **hidden action** or **moral hazard**.

Here are some examples.

(1) The bank does not know the credit rating of the borrowing company (hidden information), nor does it know whether the company will misspend the money after obtaining the loan.

(2) The employer does not know the employee's ability to work, nor does she know if the employee is lazy, or is careful with machines and equipment.

(3) A driver may not be particularly careful after buying full car

insurance.

(4) A publicly-owned firm may not particularly concerned with job performance and innovation. The profit or loss belongs to the public, but the risk is its own and it would not take risks to innovate.

(5) Duplicity: saying yes, but meaning no. Saying one thing, while acting differently.

(6) One does not take as good care of public property as one treats one's own private property.

(7) A government official may not be accountable, and is reluctant to take risks. In general, more work implies more erroneous decisions, less work implies fewer erroneous decisions, and no work implies no mistakes. Why would they get work done?

(8) Absolute equalization, such as equal pay regardless of performance of work before the reform and opening-up in China made workers and peasants have no incentives to work diligently.

(9) A teacher does not know whether or not the student is listening. Even if the student is staring directly at the teacher, he may be thinking about other things.

(10) If there is no assessment or pressure on students, they may not work diligently.

The leading candidates for describing such moral hazard actions are effort variables. In this chapter, we present moral hazard with the agent's effort. The basic analysis of moral hazard issues is as follows. The agent's level of effort will affect the principal's revenue, but the level of effort is not observable. As the effort will bring a disutility (effort cost) to the agent, it may cause the agent not to make effort. At the same time, effort cannot be deduced from the outcomes of effort, since they are also uncertain due to external shocks. Effort may bring good outcomes (e.g., high output) and may also bring bad outcomes (e.g., low output). However, although the outcome is a random variable, the level of effort will affect the probabilities of the outcome level. For instance, the output of a crop field depends on the amount of time that the tenant has spent selecting the best crops or the quality of his harvesting. Similarly, the probability that a driver has a car accident depends on how safely he drives, which also affects the demand for insurance. In addition, a regulated firm may have to perform a costly and unobservable investment to reduce its cost of producing a socially valuable good.

Therefore, the principal needs to design an appropriate contract to give the agent an incentive to make an effort instead of being lazy. What we want to examine is what kind of contracts will make the agent work hard. In the case of complete information, it is easy to make the agent work hard through a strict reward and punishment system. In the case of incomplete information, incentives are needed and the principal needs to pay a certain amount of information rent to induce the agent to work diligently. For example, the salary, bonuses, and other benefits that a firm provides in order to make an employee work hard constitute information rents. However, it is obviously impossible for such an incentive contract to be based on unobservable efforts, but only on the performance of the agent.[1] The performance is affected by the level of effort of the agent, and it is also disturbed by some random factors.[2]

What the principal needs to do is to design a payment contract based on observable performance in order to induce the agent's optimal effort level to maximize his own gain. In this process, the principal needs to face a *trade-off between incentives and risk*, consequently, the *trade-off between efficiency and insurance*. On the one hand, a high-intensity incentive contract can induce the agent to work hard, and thus increase the benefits of the principal. This is called the **incentive effect**. On the other hand, because of noise factors in the performance indicators, the high-intensity incentive will amplify the uncertainty caused by noise. Thus, the uncertainty resultant from noise increases the risk that an agent needs to bear. In this way, the risk-neutral principal must provide insurance for a risk-averse agent by paying more risk premium, which is the **risk effect**.[3] What the principal needs to do is to make the incentive-risk or trade-off an efficiency-insurance to determine the optimal incentive intensity.

It should be emphasized that one of the differences between adverse selection and moral hazard is that for the principal, the uncertainty in the case of adverse selection is exogenous, but it is endogenous in the case of moral hazard. Therefore, the probability under moral hazard is different from the probability in a natural state. This is because the probability of moral hazard is partially determined by the level of effort of the agent. For instance, careful driving will reduce the probability of car accidents. This uncertainty is critical to understanding the contractual issues of moral hazard. If the correspondence between the agent's effort and performance is completely determined, there is no difficulty for the principal and the court to infer the agent's level of effort based on the observed outcomes. Even if the agent's effort cannot be directly observed, it may also be indirectly stip-

[1]For instance, the research achievements of researchers, the number of products produced by workers on the production line, etc.

[2]For instance, agricultural outputs depend not only on the efforts of farmers, but also on climate, hydrology, and numerous other uncontrollable human factors.

[3]If the relationship between a performance indicator and effort is weak and stochastic, it is undoubtedly to guide agents to engage in a gambling activity with high-intensity incentives on such performance indicators.

ulated by the contract because the outcome itself is both observable and verifiable.

We will discuss the properties of incentive schemes that induce a costly effort. Such schemes must thus satisfy an incentive-compatibility constraint and a participation constraint. Among such schemes, the principal prefers the one that implements the optimal level of effort at minimal cost. By minimizing the cost of the incentive scheme, it is possible to derive the second-best cost for the implementation of the level of effort. In general, the second-best cost is greater than the first-best cost at the observable level of effort. An allocative inefficiency emerges as a result of the conflict of interests between the principal and the agent, resulting in **trade-off between incentives and risk** (i.e., the **trade-off between efficiency and insurance)**. As such, the best outcome one can obtain is only second-best, but not first-best.

17.2 Basic Model

We discuss the interaction between principal and agent by establishing the simplest form of moral hazard. The form of interaction is a contract in which the principal does not know the actual action of the agent. In the interaction, the principal has a task delegated to the agent, and the agent needs to devote effort to, and bear the cost of the effort. After the task is completed, the agent's payoff comes from the principal's payment to the agent. There may be a potential conflict of interest between the principal and the agent (the principal wants the agent to work hard and the agent does not want to). However, the principal can use the design of the contract to ease their differences of interest and stimulate the agent to work diligently. Encouraging the agent to work hard is the core issue to be solved.

17.2.1 Setting

Consider a simplest situation in which the completion of a task has two possible outcomes of performance: $q = 1$ on success and $q = 0$ on failure. The returns to the principal are $\bar{S}$ and $\underline{S}$, respectively, with $\bar{S} > \underline{S}$. The agent has two possible actions e, which are working and shirking, corresponding to $e = 1$ or 0, respectively. In general, the outcomes of the task depend on other external factors in addition to the effort. Therefore, the agent's action does not determine the outcomes completely, but only affects the probabilities of the outcomes.

Let π_e denote the probability that the agent succeeds when the agent chooses action e. The harder the agent works, the higher is the probability of a successful task (i.e., $\pi_1 > \pi_0$). Let $\Delta\pi = \pi_1 - \pi_0$ denote the increase in the probability that the agent will choose to work harder to succeed.

Note that effort improves production in the sense of first-order stochastic dominance, i.e., the probability $\Pr(\tilde{q} \leqq q^*|e)$ is decreasing with e for any given production q^*, or equivalently, the probability $\Pr(\tilde{q} \geqq q^*|e)$ is increasing with e for any given production q^*. Indeed, we have $\Pr(\tilde{q} \leqq \underline{q}|e = 1) = 1 - \pi_1 < 1 - \pi_0 = \Pr(\tilde{q} \leqq \underline{q}|e = 0)$ and $\Pr(\tilde{q} \leqq \bar{q}|e = 1) = 1 = \Pr(\tilde{q} \leqq \bar{q}|e = 0)$.

In addition, the disutility of effort is denoted as $\psi(1) = \psi > 0$ and $\psi(0) = 0$. The principal cannot directly observe the agent's action. The compensation to the agent can only rely on the outcome q. For this reason, the compensation contract is in the form of $t(q)$. $\bar{t} = t(1)$ and $\underline{t} = t(0)$ are the compensations for success and failure, respectively.

Assume that the principal is risk-neutral, and her utility function is $V(y) = y$. The agent's utility function for compensation and effort is separable: $U(t, e) = u(t) - \psi(e)$, where $u' > 0$ and $u'' \leqq 0$. $h = u^{-1}$, which will be used to find the solution later. Obviously, h is a strictly increasing and convex function: $h' > 0, h'' \geqq 0$. Under such a utility function, $h(\psi)$ can be regarded as the first-best monetary cost C^{FB} of implementing the positive effort level.

If the agent's effort level is $e = 1$, the principal's expected utility function is:

$$\pi_1(\bar{S} - \bar{t}) + (1 - \pi_1)(\underline{S} - \underline{t}). \tag{17.2.1}$$

If the agent's effort level is $e = 0$, the principal's expected utility function is:

$$\pi_0(\bar{S} - \bar{t}) + (1 - \pi_0)(\underline{S} - \underline{t}). \tag{17.2.2}$$

If the agent chooses $e = 1$, his expected utility is:

$$\pi_1 u(\bar{t}) + (1 - \pi_1)u(\underline{t}) - \psi. \tag{17.2.3}$$

The question that we need to answer is that, in the presence of moral hazard, if the principal wants the agent to make the effort, can she always choose an appropriate contract so that the agent's choice is consistent with the principal's goal?

The timing of contracting under moral hazard is given as follows:

- $t = 0$: the principle offers $\{\underline{t}, \bar{t}\}$;
- $t = 1$: the agent accepts or rejects the contract;
- $t = 2$: the agent decides on effort input;
- $t = 3$: realize output q;
- $t = 4$: the contract is executed.

17.2.2 Benchmark Case

In order to study the optimal contract under incomplete information, we first consider the benchmark case of the optimal contract under complete information. In this case, the agent's actions can be put into the contract and can be executed by a third party, such as a legal system. The problem of the principal is to design an optimal contract subject to the participant constraint. In other words, the contract sets the action of the agent, and the expected utility obtained by the agent that is not less than the reservation utility. For simplicity, let the reservation utility be 0.

If the principle wants to induce effort $e = 1$, her optimization problem is:

$$\begin{aligned} &\max_{\{(\bar{t},\underline{t})\}} \pi_1(\bar{S} - \bar{t}) + (1 - \pi_1)(\underline{S} - \underline{t}) \\ \text{s.t.} \quad &\pi_1 u(\bar{t}) + (1 - \pi_1)u(\underline{t}) - \psi \geqq 0. \end{aligned}$$

Since the action can be determined by the contract, the optimal contract just needs to specify that if the agent does not choose $e = 1$, he will be given a serious punishment. Then, the agent will abide by the contract. The above inequality constraint is called the **participation constraint**. If the agent accepts the contract, his expected utility cannot be lower than the reservation utility. Obviously, in this optimization problem, this constraint must be binding. Let λ be the Lagrangian multiplier involved in the constraint. Then, we have the Lagrange function

$$L(\lambda, \bar{t}, \underline{t}) = \pi_1(\bar{S} - \bar{t}) + (1 - \pi_1)(\underline{S} - \underline{t}) + \lambda[\pi_1 u(\bar{t}) + (1 - \pi_1)u(\underline{t}) - \psi].$$

The first-order conditions for $\bar{t}$ and $\underline{t}$ are:

$$-\pi_1 + \lambda \pi_1 u'(\bar{t}^*) = 0, \tag{17.2.4}$$

$$-(1 - \pi_1) + \lambda(1 - \pi_1)u'(\underline{t}^*) = 0. \tag{17.2.5}$$

From equations (17.2.4) and (17.2.5), the pooling contract can be immediately derived:

$$\underline{t}^* = \bar{t}^*.$$

Thus, the agent can avoid all risks (full insurance) in the optimal contract. Since the participation constraint is binding, the first-best contract is $\underline{t}^* = \bar{t}^* = h(\psi)$. In other words, the transfer to the agent is equal to the monetary cost of effort. In this case, as long as the agent works hard, whether the outcome is good or bad, the principal should pay the same amount to the agent since the difference is caused by nature, and it is not the agent's fault. For the principal, the expected return is:

$$\pi_1 \bar{S} + (1 - \pi_1)\underline{S} - h(\psi).$$

If the principal wants the agent to choose $e = 0$, then the contract at this time is the solution to the following optimization problem:

$$\max_{\{(\bar{t},\underline{t})\}} \pi_0(\bar{S} - \bar{t}) + (1 - \pi_0)(\underline{S} - \underline{t})$$
$$\text{s.t.} \quad \pi_0 u(\bar{t}) + (1 - \pi_0)u(\underline{t}) \geqq 0.$$

Similar to the reasoning above, we obtain the contract $\underline{t}' = \bar{t}' = 0$, and the agent's expected return is

$$\pi_0\bar{S} + (1 - \pi_0)\underline{S}.$$

Thus, if inducing effort is optimal to the principal, it requires that $\pi_1\bar{S} + (1 - \pi_1)\underline{S} - h(\psi) \geqq \pi_0\bar{S} + (1 - \pi_0)\underline{S}$, or

$$\Delta\pi(\bar{S} - \underline{S}) \geqq h(\psi). \tag{17.2.6}$$

Equation (17.2.6) means that the gain of increasing effort from $e = 0$ to $e = 1$ is greater than or equal to the first-best cost of inducing effort.

Under condition (17.2.6), it is clear that the principal stipulates that the agent's action in the first-best contract is $e = 1$. Figure 17.1 depicts the level of effort under the first-best contract.

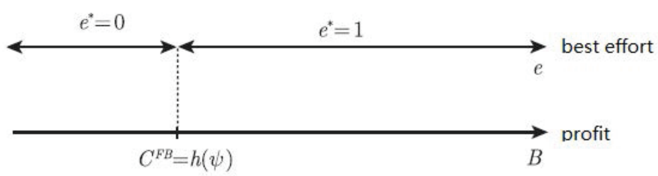

Figure 17.1: The level of effort under the first-best contract

17.2.3 Incentive-feasible Contracts

If the principal cannot observe the agent's action, the principal and the agent are unable to agree on the agent's action in the contract. In this situation, the contract can only be based on verifiable outcomes. Then, the agent's contractual form of avoiding all risks in incomplete information is not implementable. This is because, if $\underline{t}^* = \bar{t}^*$, the agent has no incentive to choose the action $e = 1$, otherwise it is worse off. As such, the principal should not provide the pooling contract to the agent.

An implementable contract then needs to introduce a new constraint:

$$\pi_1 u(\bar{t}) + (1 - \pi_1)u(\underline{t}) - \psi \geqq \pi_0 u(\bar{t}) + (1 - \pi_0)u(\underline{t}). \tag{17.2.7}$$

We call the expression (17.2.7) the **incentive-compatibility constraint**. Under this condition, even if the principal cannot specify the agent's actions, the agent will have the incentive to choose the principal's desired action. Note that the pooling contract $\bar{t} = \underline{t}$ under complete information does not satisfy the incentive-compatibility constraint (17.2.7).

In addition, the participation constraint must be met:

$$\pi_1 u(\bar{t}) + (1-\pi_1)u(\underline{t}) - \psi \geqq 0. \tag{17.2.8}$$

Note that the agent's participation constraint must be established before the output variations are realized.

The principal hopes that each level of effort taken by the agent corresponds to a contract that guarantees that the agent's moral hazard incentive-compatibility constraint and participation constraint are established.

Definition 17.2.1 (Incentive-feasible Contract) A contract is said to be **incentive-feasible** if it satisfies the incentive-compatibility and participation constraints (17.2.7) and (17.2.8).

17.2.4 Risk Neutrality and First-Best Implementation

If the agent is risk-neutral, we have (up to an affine transformation) $u(t) = t$ for all t and $h(u) = u$ for all u. The principal who wants to induce effort must thus choose the contract to solve the following optimization problem:

$$\max_{\{(\bar{t},\underline{t})\}} \pi_1(\bar{S} - \bar{t}) + (1-\pi_1)(\underline{S} - \underline{t}) \tag{17.2.9}$$

$$\text{s. t.} \quad \pi_1\bar{t} + (1-\pi_1)\underline{t} - \psi \geqq \pi_0\bar{t} + (1-\pi_0)\underline{t}; \tag{17.2.10}$$

$$\pi_1\bar{t} + (1-\pi_1)\underline{t} - \psi \geqq 0. \tag{17.2.11}$$

The incentive-compatibility constraint (17.2.10) can be written as:

$$\Delta\pi\bar{t} \geqq \Delta\pi\underline{t} + \psi. \tag{17.2.12}$$

We can see that in the above optimization problem, the participation constraint (17.2.11) must be binding. This is because one of (17.2.10) and (17.2.11) must be binding; otherwise, the principal can reduce $\bar{t}$ a little bit to increase her expected utility. If (17.2.11) is not binding, the principal can reduce $\bar{t}$ and $\underline{t}$ by the same small amount. Under the above two constraints, the principal's expected utility can be increased. Of course, (17.2.10) may not be binding. The shaded area in Figure 17.2 depicts the incentive-feasible transfers. The optimal contracts that are the first-best are the portion of the participation constraint line that also satisfies the incentive-compatibility constraint in Figure 17.2. This is because the objective function (17.2.9) can be written as:

$$\max_{\{(\bar{t},\underline{t})\}} \pi_1\bar{S} + (1-\pi_1)\underline{S} - [\pi_1\bar{t} + (1-\pi_1)\underline{t}],$$

or equivalently:

$$\min_{\{(\bar{t},\underline{t})\}} \pi_1 \bar{t} + (1-\pi_1)\underline{t}.$$

Among all optimal contracts, the one that satisfies both the incentive-compatibility constraint (17.2.10) and the participation constraint (17.3.17) is given by

$$\underline{t}^{SB} = -\frac{\pi_0}{\Delta\pi}\psi, \tag{17.2.13}$$

$$\bar{t}^{SB} = \frac{1-\pi_0}{\Delta\pi}\psi. \tag{17.2.14}$$

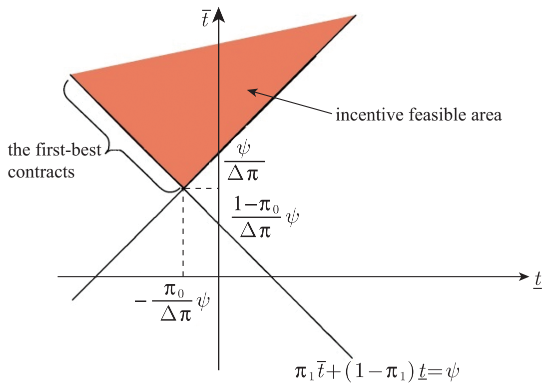

Figure 17.2: The optimal contract is first-est when the agent is risk-neutral.

The principal's expected payoff is $\pi_1\bar{S} + (1-\pi_1)\underline{S} - \psi$. This result is consistent with the complete information situation and the expected return of the agent is 0, so that the optimal contract is first-best. Even so, the reward for different outcomes is not the same. This is different from the pooling contract under complete information, although the expectation is equal to 0. We then have the following proposition:

Proposition 17.2.1 *Moral hazard is not an issue with a risk-neutral agent, despite the unobservability of effort. The first-best level of effort is still implementable.*

Remark 17.2.1 One may find the similarity of these results with those in the previous chapter. In both cases, when contracting takes place ex-ante, the incentive-compatibility constraint, under either adverse selection or moral hazard, does not conflict with the ex-ante participation constraint with a risk-neutral agent, and the first-best outcome is still implementable.

However, inefficiencies in effort provision due to moral hazard will arise when the agent is no longer risk-neutral or constrained by limited liability. There are then two alternative ways to model these transaction costs. One is to maintain risk-neutrality for positive income levels but to impose a limited liability constraint, which necessitates transfers not to be too negative. The other is to let the agent be strictly risk-averse. In the following, we analyze these two contractual environments and the different trade-offs that they imply.

17.3 Optimal Contract under Limited Liability and Risk-Aversion

17.3.1 Optimal Contract under Limited Liability

In the previous section, we found that the imposition of the incentive-compatibility constraint still results in the first-best outcome of the contract, provided that the agent is risk-neutral. The incentive-compatibility constraint requires the optimal contract must be **separating**, i.e., the agent's transfer must be different between the success and failure of the task: $\bar{t} - \underline{t} \geqq \frac{\psi}{\Delta\pi}$. This implies that the agent will be punished for failure.

In the optimal contract shown in Figure 17.2, since $\underline{t} \leqq -\frac{\pi_0}{\Delta\pi}\psi$, the punishment could be arbitrarily large. However, for various reasons (e.g., factors of ability to make a large transfer or make a transfer legal), the agent's loss that he can bear is limited. For example, in a limited liability company, the investor's loss limit in the case of failure is limited to the amount of investment, and he does not bear unlimited liability. In the following discussion under limited liability (or a limited degree of punishment for the agent), only the second-best outcome is achievable, even if the agent is risk-neutral.

Let l be the upper bound of punishment, so that $\underline{t}, \bar{t} \in [-l, \infty)$. These constraints, together with the incentive-compatibility and participation constraints, may prevent the principal from implementing the first-best level of effort even if the agent is risk-neutral. Indeed, when the principal wants to induce a high effort, her program can be written as

$$\max_{\{(\bar{t},\underline{t})\}} \pi_1(\bar{S} - \bar{t}) + (1 - \pi_1)(\underline{S} - \underline{t}) \tag{17.3.15}$$

$$\text{s. t.} \quad \pi_1\bar{t} + (1 - \pi_1)\underline{t} - \psi \geqq \pi_0\bar{t} + (1 - \pi_0)\underline{t}; \tag{17.3.16}$$

$$\pi_1\bar{t} + (1 - \pi_1)\underline{t} - \psi \geqq 0; \tag{17.3.17}$$

$$\bar{t} \geqq -l; \tag{17.3.18}$$

$$\underline{t} \geqq -l. \tag{17.3.19}$$

The second-best contract can be analyzed with the following figures. There are two cases to be considered.

Case 1. $l \geqq \frac{\pi_0}{\Delta\pi}\psi$. In this case, there are a portion of first-best transfers under the failure outcome that satisfy $-l \leqq \underline{t} \leqq -\frac{\pi_0}{\Delta\pi}\psi$. Then, the first-best outcome is still implementable, although the set of first-best outcomes

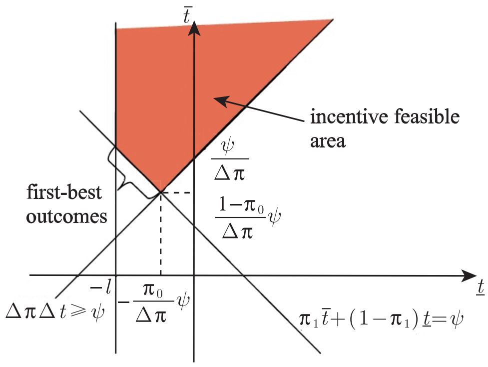

Figure 17.3: The first-best outcomes still prevail when $l \geqq \pi_0\psi/\Delta\pi$.

is shrinking. This situation can be illustrated by Figure 17.3: Because of the existence of the upper bound of penalty, the set of incentive-feasible contracts has been restricted. Since the agent's participation constraint is binding, it results in the first-best outcomes in the range of optimal contracts.

Case 2. $0 \leqq l < \frac{\pi_0}{\Delta\pi}\psi$. In this case, $-\frac{\pi_0}{\Delta\pi}\psi < -l$, and thus all first-best transfers obtained without limited liability under the failure outcome are less than $-l$, which is not feasible under limited liability. Then, at the optimal contract, the limited liability constraint must be binding, but the agent's participation constraint (17.2.11) is slack (inequality constraint), and thus we can only have the second-best contract. We graphically show this in Figure 17.4, in which the region of incentive-feasible contracts is restricted. In this new region, the principal's optimal decision is to minimize the expected information rent in the viable area, i.e.,

$$\min_{\{(\bar{t},\underline{t})\}} \pi_1\bar{t} + (1-\pi_1)\underline{t}$$

subject to the limited liability constraint and incentive-compatibility constraint.

As such, $\underline{t}^{SB} = -l$, at which the limited liability affects the transfer (penalty) for failure. Then, $\bar{t}^{SB} = -l + \frac{\psi}{\Delta\pi}$, and thus the expected utility of the principal is

$$\pi_1\bar{S} + (1-\pi_1)\underline{S} + l - \psi\frac{\pi_1}{\Delta\pi},$$

and the agent's expected utility is

$$EU^{SB} = \pi_1\bar{t}^{SB} + (1-\pi_1)\underline{t}^{SB} - \psi = -l + \frac{\pi_0}{\Delta\pi}\psi > 0.$$

We can further obtain that the principal induces the agent to choose $e = 1$ if and only if

$$\Delta\pi\Delta S \geqq \psi\frac{\pi_1}{\Delta\pi} = \psi + \pi_0/\Delta\pi\psi.$$

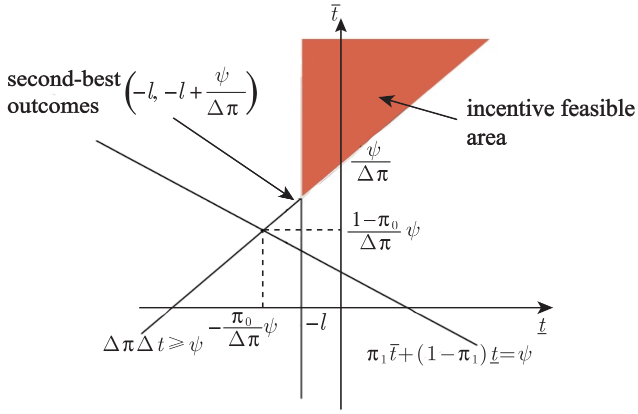

Figure 17.4: Only the second-best outcomes prevail when $0 \leqslant l < \pi_0\psi/\Delta\pi$.

Thus, only when the failure state occurs, the limited liability constraint may be binding. Because exerting effort requires that $\bar{t}$ and $\underline{t}$ be different, we must have $\bar{t} > -l$ when $\underline{t} \geqq -l$. When the upper bound of penalty is smaller than the payment under the original failure, the principal's punishment for the agent is limited. In this case, the agent can at most pay a penalty of l. When the task is successful, the agent receives a reward of $-l + \psi/\Delta\pi$. Therefore, the agent obtains a positive expected liability rent $EU^{SB} > 0$, which means that the optimal contract is the second-best, but not the first-best. This rent that results from the additional payments made by the principal to the agent is due to the combined effect of moral hazard and limited liability. With the continuous increase of l, the tension between moral hazard and limited liability constraints is getting smaller. When $l > \pi_0\psi/\Delta\pi$, there will be no tension between them.

Summarizing the above discussion, we have the following proposition.

Proposition 17.3.1 *With limited liability, the optimal contract inducing effort from the agent has the following properties:*

(1) For $l > \frac{\pi_0}{\Delta\pi}\psi$, the participation constraint is binding so that the set of the first-best transfers is given in Figure 17.3, and one of them that satisfies the incentive-compatibility and participant equality constraints is given by $(\underline{t}^*, \bar{t}^*) = (-\pi_0/\Delta\pi\psi, (1-\pi_0)\psi/\Delta\pi)$. The agent has no expected limited liability rent, i.e., $EU^{SB} = 0$.

(2) For $0 \leqq l \leqq \frac{\pi_0}{\Delta\pi}\psi$, the limited liability and incentive-compatibility constraints are binding. The second-best transfers are given by

$$\underline{t}^{SB} = -l, \tag{17.3.20}$$

$$\bar{t}^{SB} = -l + \frac{\psi}{\Delta\pi}. \tag{17.3.21}$$

Moreover, the agent's expected limited liability rent EU^{SB} *is non-negative:*

$$EU^{SB} = \pi_1 \bar{t}^{SB} + (1-\pi_1)\underline{t}^{SB} - \psi = -l + \frac{\pi_0}{\Delta\pi}\psi \geqq 0. \quad (17.3.22)$$

17.3.2 Optimal Contract under Risk-Aversion

We now discuss the optimal contract when the agent is risk-averse. In this case, there is a fundamental trade-off: incentive and risk, consequently, efficiency and insurance. We have seen that in the case of incomplete information, when the agent is risk-neutral, the pooling contract (i.e., compensation is independent of the outcome) does not satisfy the incentive-compatibility constraints. In this case, the agent needs to take a certain risk in order to meet incentive-compatibility constraints. However, for a risk-averse agent, taking risks will reduce his expected payoff. Since the agent's compensation must also satisfy the participation constraints, the cost for the risk will eventually be transferred to the principal. The problem of the principal is then to have a trade-off between the efficiency from incentives and the cost of risk from providing incentives.

The principal's program is then written as:

$$\max \pi_1(\bar{S} - \bar{t}) + (1-\pi_1)(\underline{S} - \underline{t}) \quad (17.3.23)$$

$$\text{s.t.} \quad \pi_1 u(\bar{t}) + (1-\pi_1)u(\underline{t}) - \psi \geqq \pi_0 u(\bar{t}) + (1-\pi_0)u(\underline{t}), \quad (17.3.24)$$

$$\pi_1 u(\bar{t}) + (1-\pi_1)u(\underline{t}) - \psi \geqq 0. \quad (17.3.25)$$

If this is a concave program, the first-order Kuhn-Tucker condition is the necessary and sufficient condition for the optimal solution. However, this is not the case since the concave function $u(\cdot)$ appears on both sides of the incentive constraint (17.3.24). Then, we use the following change of variables that can make the new program be a concave one.

Define $\bar{u} = u(\bar{t})$ and $\underline{u} = u(\underline{t})$, and thus $\bar{t} = h(\bar{u})$ and $\underline{t} = h(\underline{u})$. In this way, the principal's optimization problem becomes:

$$\max_{\{(\bar{u},\underline{u})\}} \pi_1(\bar{S} - h(\bar{u})) + (1-\pi_1)(\underline{S} - h(\underline{u})) \quad (17.3.26)$$

$$\text{s.t.} \quad \pi_1\bar{u} + (1-\pi_1)\underline{u} - \psi \geqq \pi_0\bar{u} + (1-\pi_0)\underline{u}, \quad (17.3.27)$$

$$\pi_1\bar{u} + (1-\pi_1)\underline{u} - \psi \geqq 0. \quad (17.3.28)$$

Since $h(\cdot)$ is strictly convex, the objective function of the above problem is strictly concave in $\bar{u}$ and $\underline{u}$. In addition, since the constraints given by (17.3.27) and (17.3.28) are linear, the set of constraints is convex. Thus, the program becomes a concave one, and then the first-order condition of the Lagrange function is the sufficient condition for the optimal solution of the program.

Let λ and μ be the Lagrange multipliers for the incentive-compatibility constraint (17.3.27) and participation constraint (17.3.28), respectively. The Lagrange function is

$$\begin{aligned} L(\bar{u}, \underline{u}; \lambda, \mu) &= \pi_1(\bar{S} - h(\bar{u})) + (1 - \pi_1)(\underline{S} - h(\underline{u})) \\ &\quad +\lambda[\pi_1 \bar{u} + (1 - \pi_1)\underline{u} - \psi - \pi_0 \bar{u} - (1 - \pi_0)\underline{u}] \\ &\quad +\mu[\pi_1 \bar{u} + (1 - \pi_1)\underline{u} - \psi]. \end{aligned}$$

Thus, the first-order conditions for $\bar{u}$ and $\underline{u}$ are given by

$$\frac{1}{u'(\bar{t}^{SB})} = \mu + \lambda \frac{\Delta\pi}{\pi_1}, \tag{17.3.29}$$

$$\frac{1}{u'(\underline{t}^{SB})} = \mu - \lambda \frac{\Delta\pi}{1 - \pi_1}, \tag{17.3.30}$$

where $\bar{t}^{SB}$ and $\underline{t}^{SB}$ are the transfers.

Multiplying (17.3.29) by π_1 and (17.3.30) by $1-\pi_1$, and making the summation, yields $\mu = \frac{\pi_1}{u'(\bar{t}^{SB})} + \frac{1-\pi_1}{u'(\underline{t}^{SB})} = E\left(\frac{1}{u'(\tilde{t}^{SB})}\right) > 0$. Thus, the participation constraint (17.3.28) is binding.

Multiplying (17.3.29) by $\pi_1 u(\bar{t}^{SB})$ and (17.3.30) by $(1 - \pi_1)u(\underline{t}^{SB})$, and making the summation, taking into account the expression of μ obtained above yields

$$\lambda\Delta\pi\left(u(\bar{t}^{SB}) + u(\underline{t}^{SB})\right) = E_q\left(u(\tilde{t}^{SB})\left(\frac{1}{u'(\tilde{t}^{SB})} - E\left(\frac{1}{u'(\tilde{t}^{SB})}\right)\right)\right). \tag{17.3.31}$$

Using the complementary slackness condition $\lambda\left(\Delta\pi(u(\bar{t}^{SB}) + u(\underline{t}^{SB})) - \psi\right) = 0$ to simplify the left-hand side of (17.3.31), we get

$$\lambda\psi = cov\left(u(\tilde{t}^{SB}), \frac{1}{u'(\tilde{t}^{SB})}\right). \tag{17.3.32}$$

By assumption, $u(\cdot)$ and $u'(\cdot)$ co-vary in opposite directions. In addition, the polling wage $\bar{t}^{SB} = \underline{t}^{SB}$ does not satisfy the incentive-compatibility constraint, and thus we must have $\bar{t}^{SB} \neq \underline{t}^{SB}$. Therefore, the right-hand side of (17.3.32) is positive. Thus, we have $\lambda > 0$, and then the incentive-compatibility constraint (17.3.27) is binding.

Also, since u is a concave function, we have $\bar{t}^{SB} > \underline{t}^{SB}$. The compensation when the task is successful is greater than the compensation when it fails.

Since both the incentive-compatibility constraint (17.3.27) and the participation constraint (17.3.28) are binding, we obtain the following equations by solving these two equations:

$$\bar{u} = \psi + (1 - \pi_1)\frac{\psi}{\Delta\pi}, \tag{17.3.33}$$

$$\underline{u} = \psi - \pi_1 \frac{\psi}{\Delta\pi}, \tag{17.3.34}$$

and thus we have

$$\bar{t}^{SB} = h(\psi + (1 - \pi_1)\frac{\psi}{\Delta\pi}), \tag{17.3.35}$$

$$\underline{t}^{SB} = h(\psi - \pi_1 \frac{\psi}{\Delta\pi}). \tag{17.3.36}$$

We then have the following proposition.

Proposition 17.3.2 *When the agent is risk-averse, the second-best optimal contract that induces effort makes both the agent's participation and incentive-compatibility constraints binding. This contract must be a separating contract so that it does not provide full insurance to the agent, and the second-best transfers are given by*

$$\bar{t}^{SB} = h\left(\psi + (1 - \pi_1)\frac{\psi}{\Delta\pi}\right) = h\left(\frac{1 - \pi_0}{\Delta\pi}\psi\right) \tag{17.3.37}$$

and

$$\underline{t}^{SB} = h\left(\psi - \pi_1 \frac{\psi}{\Delta\pi}\right) = h\left(-\frac{\pi_0}{\Delta\pi}\psi\right). \tag{17.3.38}$$

17.3.3 Basic Trade-off: Efficiency and Insurance

The second-best cost of inducing a high effort $e = 1$ is:

$$\begin{aligned} C^{SB} &= \pi_1 \bar{t} + (1 - \pi_1)\underline{t} \\ &= \pi_1 h\left(\psi + (1 - \pi_1)\frac{\psi}{\Delta\pi}\right) + (1 - \pi_1) h\left(\psi - \frac{\pi_1 \psi}{\Delta\pi}\right) \\ &= \pi_1 h\left(\frac{1 - \pi_0}{\Delta\pi}\psi\right) + (1 - \pi_1) h\left(-\frac{\pi_0}{\Delta\pi}\psi\right). \end{aligned} \tag{17.3.39}$$

Since $h(\cdot)$ is strictly convex, we have $C^{SB} > h(\psi) = C^{FB}$ according to Jensen's inequality. The principal's return when she induces the agent to exert $e = 1$ is still

$$B = \Delta\pi\Delta S.$$

Therefore, in the risk-averse situation, the principal will induce the agent to choose $e = 1$ if and only if $B = \Delta\pi\Delta S \geqq C^{SB} > h(\psi) = C^{FB}$. Thus, under moral hazard, inducing high-level efforts is more challenging than under complete information. Figure 17.5 compares the first-best contract and the second-best contract under complete and incomplete information.

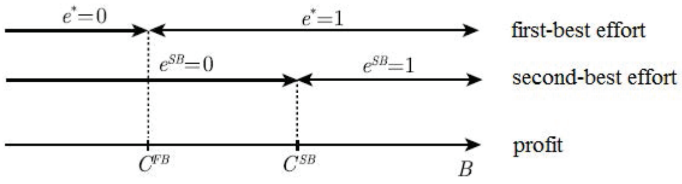

Figure 17.5: The second-best solution under risk-aversion.

When B belongs to the interval $[C^{FB}, C^{SB}]$, the second-best effort level is zero, and therefore it is strictly less than the first-best effort level. Due to the combined effect of moral hazard and the agent's risk-aversion, the agent's efforts are distorted downward.

The following proposition summarizes the basic trade-off under moral hazard.

Proposition 17.3.3 *For the optimal contract with moral hazard and risk-aversion, there is a trade-off between inducing effort and providing insurance to the agent (avoiding risk), and the principal induces a positive effort from the agent less frequently than when effort is observable.*

17.4 Contract Theory at Work

This section elaborates on the basic moral hazard model discussed above in a number of settings that have been discussed extensively in the contract theory literature. The discussions on these models are drawn from Laffont and Martimort (2002).

17.4.1 Efficiency Wage

In the labor market, a common observation is that the wages of employees are frequently higher than the competitive wages in the labor market. This wage is called the **efficiency wage**. Here, the exceeding part refers to the deduction of possible human capital factors. In this regard, a macroeconomic question is what prevents firms from adjusting their wages to market equilibrium wages. The study of these economic phenomena is sometimes called the Keynesian School, which discusses the microscopic basis of some neo-classical economic equilibrium phenomena, of which efficiency wages are one of them, and "involuntary unemployment" is closely associated with it.

Shapiro and Stiglitz (1984) first discussed the microeconomic logic behind efficient wages. Their analytical framework is to consider this problem in a dynamic environment. Here, we consider a simplified static version from Laffont and Martmort (2002).

Suppose that a firm employs a worker to perform a task, and both the owner and the employee are risk-neutral. We normalize the equilibrium wages outside of the market to 0. The firm wants the worker to work diligently. Assume that the level of effort is discrete and takes two values, i.e., $e \in \{0, 1\}$, the level of effort affects the outcomes of the task, and there are only two outcomes of performance, $y \in \{0, 1\}$. When the task succeeds, $y = 1$, the firm's added value is $\bar{V}$; when it fails, the added value is $\underline{V}$ with $0 \leqq \underline{V}$, denoting $\Delta V = \bar{V} - \underline{V}$. The probability of success of a task depends on the efforts of the employee, and let π_e be the probability of success when the effort is e with $1 > \pi_1 > \pi_0 > 0$, denoting $\Delta\Pi = \pi_1 - \pi_0$. Assume that the effort function is $\psi(1) = \psi$ and $\psi(0) = 0$, which cannot be directly observed by the firm. The firm wants the worker to work hard, but the most likely punishment is dismissal. At this point, the agent can only be rewarded for good performance and cannot be punished for a bad outcome, since they are protected by limited liability.

To induce effort, the principal (firm) must find an optimal compensation scheme $\{(\underline{t}, \bar{t})\}$ that is the solution to the program below:

$$\max_{\{(\underline{t},\bar{t})\}} \pi_1(\bar{V} - \bar{t}) + (1 - \pi_1)(\underline{V} - \underline{t}) \tag{17.4.40}$$

$$\text{s. t.} \quad \pi_1\bar{t} + (1 - \pi_1)\underline{t} - \psi \geqq \pi_0\bar{t} + (1 - \pi_0)\underline{t}, \tag{17.4.41}$$

$$\pi_1\bar{t} + (1 - \pi_1)\underline{t} - \psi \geqq 0, \tag{17.4.42}$$

$$\underline{t} \geqq 0. \tag{17.4.43}$$

The problem is isomorphic to the one analyzed earlier. The limited liability constraint for $\underline{t}$ is binding at the optimum, and the firm chooses to induce a high effort when $\Delta\pi\Delta V \geqq \frac{\pi_1\psi}{\Delta\pi}$. As such, we obtain a second-best optimal contract, at which $\underline{t}^{SB} = 0$ and $\bar{t}^{SB} = \frac{\psi}{\Delta\pi} > 0$. At this point, the expected wage is greater than the market equilibrium wage, and the reason why the efficiency wage is greater than the market equilibrium wage is that the firm needs to give the worker an incentive to work hard. The expected utility of the worker is

$$\pi_1\bar{t} + (1 - \pi_1)\underline{t} - \psi = \frac{\pi_0\psi}{\Delta\pi} > 0.$$

The incentive cost of the firm is $\frac{\pi_1\psi}{\Delta\pi}$, and only when $\Delta\pi\Delta V > \frac{\pi_1\psi}{\Delta\pi}$ does the firm have an incentive to induce the worker to work hard.

The positive wage $\bar{t}^{SB} = \frac{\psi}{\Delta\pi}$ is usually called the **efficiency wage** because it induces the agent to exert a high (efficient) level of effort. To induce the high effort, the principal must give up a positive share of the firm's profit to the agent.

17.4.2 Sharecropping

The theoretical framework of moral hazard is also widely applied to development economics. The sharing of rent in the agricultural economy is an incentive mechanism. If the landlord employs a tenant to work on a farm, the tenant with a fixed income does not have the incentive to do his best. If the landlord has full control over the work situation of the tenant, she can naturally command the tenant accordingly. However, to obtain such sufficient information, the landlord will have to spend much effort in personal monitoring, which may not be worthwhile for the landlord. Another approach is to rent the land at a fixed amount, which can give the tenant a very large incentive. However, agriculture is a high-risk form of production. The tenant may not be able to bear this uncertainty at all. The sharecropping arrangement then arises. This approach attenuates the incentives for work rather than eliminates them, and bears some of the risk. Similar ideas have also been applied to health insurance. Most health insurance policies require patients make a co-payment.

The idea of sharecropping originated from Cheung (1969) who studied the rent-sharing contract. Stiglitz (1974) extends Cheung's idea for a general setup of the moral hazard problem. In the sharecropping model examined by Stiglitz (1974), the principal is the landlord and the agent is the tenant. The tenant has only limited responsibilities. Thus, as shown below, the optimal contract provided by Stiglitz is the second-best, while the rent-sharing contract that has been widely utilized in practice is generally not a second-best contract, and is more beneficial to tenants but not to landlords.

We now discuss a simplified version of the Stiglitz (1974) model. By exerting an effort e in $\{0, 1\}$, the probability that results in a large $\bar{q}$ (resp. small $\underline{q}$) quantity of an agricultural product is $\pi(e)$ (resp. $1 - \pi(e)$). The price of the product is normalized to 1, so that the principal's stochastic return on the activity is also $\bar{q}$ or $\underline{q}$, depending on the state of nature.

We assume that the agent is risk-neutral and protected by limited liability. The principal's optimal contract must solve

$$\max_{\{(\underline{t},\bar{t})\}} \pi_1(\bar{q} - \bar{t}) + (1 - \pi_1)(\underline{q} - \underline{t}) \tag{17.4.44}$$

$$\text{s.t.} \quad \pi_1\bar{t} + (1 - \pi_1)\underline{t} - \psi \geqq \pi_0\bar{t} + (1 - \pi_0)\underline{t}, \tag{17.4.45}$$

$$\pi_1\bar{t} + (1 - \pi_1)\underline{t} - \psi \geqq 0, \tag{17.4.46}$$

$$\underline{t} \geqq 0. \tag{17.4.47}$$

The optimal contract therefore satisfies $\underline{t}^{SB} = 0$ and $\bar{t}^{SB} = \frac{\psi}{\Delta\pi}$. This is again akin to an efficiency wage. The expected utilities obtained by the principal and the agent, respectively, are given by

$$EV^{SB} = \pi_1\bar{q} + (1 - \pi_1)\underline{q} - \frac{\pi_1\psi}{\Delta\pi}. \tag{17.4.48}$$

and

$$EU^{SB} = \frac{\pi_0 \psi}{\Delta\pi}. \tag{17.4.49}$$

The flexible second-best contract described above has sometimes been criticized as not corresponding to the contractual arrangements observed in most agrarian economies. Contracts often take the form of simple linear schedules linking the tenant's production to his compensation. In particular, in practice the tenant's contract usually has the form of sharing. $t = \alpha q$, where α is the share of tenants' harvest.

Next, we will discuss the optimal share ratio under sharecropping. Since $\underline{q} > 0$ and $\underline{t} = \alpha \underline{q} > 0$, the limited liability constraint is naturally satisfied in the following discussion.

The optimal rent-sharing contract is to solve the problem:

$$\max_{\alpha}(1-\alpha)(\pi_1 \bar{q} + (1-\pi_1)\underline{q}) \tag{17.4.50}$$

$$\text{s. t.} \quad \alpha(\pi_1 \bar{q} + (1-\pi_1)\underline{q}) - \psi \geqq \alpha(\pi_0 \bar{q} + (1-\pi_0)\underline{q}), \tag{17.4.51}$$

$$\alpha(\pi_1 \bar{q} + (1-\pi_1)\underline{q}) - \psi \geqq 0, \tag{17.4.52}$$

The (17.4.51) and (17.4.52) are incentive-compatibility constraint and participation constraint, respectively, when the tenant chooses to exert effort. Since $\pi_0 > 0$, $\bar{q} > 0$ and $\underline{q} > 0$, (17.4.52) must hold if (17.4.51) holds. Therefore, only the incentive-compatibility constraint (17.4.51) is binding. From this, we obtain that the optimal contract for sharecropping is $\alpha^{SB} = \frac{\psi}{\Delta\pi\Delta q}$. Since the condition for selecting an incentive-compatibility contract is $\Delta q \Delta\pi > \psi$, the optimal sharing ratio needs to satisfy $\alpha^{SB} < 1$.

This sharing rule yields the following expected utilities to the principal and the agent, respectively,

$$EV_{\alpha} = \pi_1 \bar{q} + (1-\pi_1)\underline{q} - \left(\frac{\pi_1 \bar{q} + (1-\pi_1)\underline{q}}{\Delta q}\right)\frac{\psi}{\Delta\pi} \tag{17.4.53}$$

and

$$EU_{\alpha} = \left(\frac{\pi_1 \bar{q} + (1-\pi_1)\underline{q}}{\Delta q}\right)\frac{\psi}{\Delta\pi}. \tag{17.4.54}$$

Since the limited liability constraint has no effect, this optimal sharecropping contract is different from the previous second-best contract for the principal. Indeed, comparing (17.4.48) and (17.4.53) on the one hand and (17.4.49) and (17.4.54) on the other hand, we observe that the constant sharing rule benefits the agent, but not the principal, since the second-best contract brings more expected utility to the principal than the constant sharecropping contract. A linear contract is less powerful than the second-best contract.

Thus, the constant sharecropping contract is an inefficient way to extract rent from the agent, even if it still provides sufficient incentives to exert effort. With a linear sharing rule, the agent always benefits from a positive return on his production, even in the worst state of nature. This positive return yields to the agent more than what is requested by the second-best optimal contract in the worst state of nature, i.e., zero.

17.4.3 Financial Contracts

Information asymmetry in financial markets has two important issues: one is the adverse selection discussed in the previous chapter, and the other is the moral hazard of borrowers to be discussed here. The discussion is about credit rationing in financial markets, which was studied by Holmstrom and Tirole (1994) on credit rationing. It is also referred to in the discussion of Laffont and Martimort (2002).

In the model of Holmstrom and Tirole (1994), it is assumed that a risk-averse entrepreneur wants to start a project that requires an initial investment worth an amount I. The entrepreneur has no cash of his own and must raise money from a bank or other financial intermediary. The return on the project is random and equal to $\bar{V}$ (resp. $\underline{V}$) with probability $\pi(e)$ (resp. $1-\pi(e)$), where the effort exerted by the entrepreneur e belongs to $\{0,1\}$. Denote the spread of profits by $\Delta V = \bar{V} - \underline{V} > 0$. The financial contract consists of repayments $\{(\bar{z}, \underline{z})\}$, depending on whether or not the project is successful.

To induce effort from the borrower, the risk-neutral lender's program is written as

$$\max_{\{(\underline{z},\bar{z})\}} \pi_1 \bar{z} + (1-\pi_1)\underline{z} - I \tag{17.4.55}$$

$$\begin{aligned} \text{s.t.} \quad & \pi_1 u(\bar{V} - \bar{z}) + (1-\pi_1)u(\underline{V} - \underline{z}) - \psi && (17.4.56)\\ & \geqq \pi_0 u(\bar{V} - \bar{z}) + (1-\pi_0)u(\underline{V} - \underline{z}), \\ & \pi_1 u(\bar{V} - \bar{z}) + (1-\pi_1)u(\underline{V} - \underline{z}) - \psi \geqq 0. && (17.4.57) \end{aligned}$$

Note that the project is a valuable venture if it provides the bank with a positive expected profit.

With the change of variables, $\bar{t} = \bar{V} - \bar{z}$ and $\underline{t} = \underline{V} - \underline{z}$, the principal's program takes its usual form. Then, the second-best cost of implementing positive effort is given by

$$C^{SB} = \pi_1 h(\psi + (1-\pi_1)\frac{\psi}{\Delta\pi}) + (1-\pi_1)h(\psi - \frac{\pi_1\psi}{\Delta\pi}),$$

where $h = u^{-1}$. At the same time, we have $\Delta\pi\Delta V \geqq C^{SB}$, so that the lender wants to induce a positive effort.

The lender's expected profit is worth:

$$V_1 = \pi_1 \bar{V} + (1-\pi_1)\underline{V} - C^{SB} - I. \tag{17.4.58}$$

At this point, the initial investment of I has an important impact on whether the principal will lend to the firm, because the principal will finance the project only if the expected profit from the project is positive: $V_1 > 0$. From (17.4.58), this requires the investment not too high, and typically we must have

$$I < I^{SB} = \pi_1 \bar{V} + (1-\pi_1)\underline{V} - C^{SB}. \tag{17.4.59}$$

However, under complete information and no moral hazard, the project would instead be financed as long as

$$I < I^* = \pi_1 \bar{V} + (1-\pi_1)\underline{V} \tag{17.4.60}$$

Comparing (17.4.59) with (17.4.60), we obtain $I^{SB} < I^*$. This implies that in an asymmetric environment, the borrower's moral hazard indicates that the borrower's credit scale is smaller than that in a complete information environment, i.e., for I in $[I^{SB}, I^*]$, some projects are financed under complete information but no longer under moral hazard. This result can be seen as similar to the form of credit rationing.

Note that the optimal financial contract offered to the risk-averse and cashless entrepreneur does not satisfy the limited liability constraint $\underline{t} \geqq 0$. Indeed, we have $\underline{t}^{SB} = h\left(\psi - \frac{\pi_1\psi}{\Delta\pi}\right) < 0$. To be induced to make an effort, the agent must bear some risk, which implies a negative payoff in the bad state of nature. Adding the limited liability constraint, the optimal contract would instead entail $\underline{t}^{LL} = 0$ and $\bar{t}^{LL} = h\left(\frac{\psi}{\Delta\pi}\right)$. This contract has sometimes been interpreted in the corporate finance literature as a **debt contract**, with no money being left to the borrower in the bad state of nature and the residual being pocketed by the lender in the good state of nature.

Finally, since $h(\cdot)$ is strictly convex and $h(0) = 0$, we have[4]

$$\begin{aligned} \bar{t}^{LL} - \underline{t}^{LL} = h\left(\frac{\psi}{\Delta\pi}\right) &> \bar{t}^{SB} - \underline{t}^{SB} \\ &= h\left(\psi + (1-\pi_1)\frac{\psi}{\Delta\pi}\right) - h\left(\psi - \frac{\pi_1\psi}{\Delta\pi}\right). \end{aligned} \tag{17.4.61}$$

This inequality shows that the debt contract has a worse incentive effect than the optimal incentive contract. Under the debt contract, in order to encourage the agent to choose to work hard, it is necessary to widen the transfer gap of the agent under different states, which increases the risk faced by the agent, but will undoubtedly bring greater agency cost to the

[4]It can be easily seen that, when h is differentiable, by the Mean-Value Theorem, $h(\frac{\psi}{\Delta\pi}) - h(\psi + (1-\pi_1)\frac{\psi}{\Delta\pi}) = h'(\delta_1)(\frac{\pi_1\psi}{\Delta\pi} - \psi) > h'(\delta_2)(\frac{\pi_1\psi}{\Delta\pi} - \psi) = h(0) - h(\psi - \frac{\pi_1\psi}{\Delta\pi})$ by noting that $h'(\cdot)$ is monotonically increasing and $\delta_1 > \psi + (1-\pi_1)\frac{\psi}{\Delta\pi} > 0 > \delta_2$. It may be remarked that inequality becomes equality when $h(\cdot)$ is linear, i.e., the entrepreneur is risk-neutral.

principal.

We have discussed above the case in which the entrepreneur does not own any amount of initial capital. In practice, however, an entrepreneur shall not have access to external finance, such as bank loans, if he does not own a certain amount of capital as collateral. Indeed, the entrepreneur with a larger amount of collateral capital can have access to more bank loans, which is also called the **credit rationing**. Holmstrom and Tirole (1997) and Chapter 3 of Tirole (2006) discuss this issue. We briefly discuss it here.

Assume that there is an entrepreneur who starts out with an amount of initial capital (cash) A. There is an economically viable project which costs $I > A$. Thus, the entrepreneur needs at least $I - A$ in external funds (e.g., from a bank) to be able to invest. Assume that the investment generates a financial return equaling either 0 (failure) or R (success). In the absence of proper incentives or outside monitoring, the entrepreneur may deliberately reduce the probability of success in order to obtain a private benefit. If the entrepreneur is diligent, the probability of success is p_H; otherwise, it is p_L with a private benefit equaling B. Assume that $\Delta p = p_H - p_L$. For simplicity, the entrepreneur and the bank are assumed to be risk-neutral. Due to limited liability, the income of the entrepreneur is no less than 0. Assuming a competitive market for banks, the equilibrium market interest rate equals 0. In the credit rationing contract, if the project succeeds, the bank is paid R_I, and an entrepreneur receives $R_B = R - R_I$.

For a perfectly competitive bank, the zero-profit condition implies that $p_H R_I = I - A$. Further assume that the project generates positive returns only when the entrepreneur is diligent, i.e., $p_H R > I > p_L R + B$.

A necessary condition for the entrepreneur to be diligent is that $p_H R_B \geqq p_L R_B + B$, or $R_B \geqq B/\Delta p$. Then, this leaves at most $R - (B/\Delta p)$ to compensate the bank, and thus the pledgable expected income for the bank equals $\mathbf{p} = p_H[R - (B/\Delta p)]$.

Thus, the entrepreneur shall have access to direct finance only when $\mathbf{p} = p_H[R - (B/\Delta p)] \geqq I - A$ is fulfilled. Define $\bar{A} = p_H(B/\Delta p) - (p_H R - I)$. Then, only the entrepreneur with an initial amount of capital $A \geqq \bar{A}$ can invest using direct finance. As is obvious, $\bar{A} > 0$ when $p_H R - I < p_H(B/\Delta p)$. Moreover, the lower bound requirement for external financing, $\bar{A}$, increases as either B or I increases. As such, the entrepreneur with a small amount of initial capital, i.e., small A, may not have access to direct finance. Moreover, such a credit rationing issue will be more serious if the project scale I increases or the private benefit B increases (i.e., the moral hazard issue becomes worse).

17.5 Extensions of the Basic Model

We now extend the basic model to more general situations. In the basic model, there are two outcomes and two actions of the agent. If the outcomes or actions are finite or continuous, what will the new features of the second-best contracts be? We mainly focus on situations in which the agent is risk-averse.

17.5.1 More than Two Outcomes and Two Actions

We first extend our previous 2×2 model to allow for more than two levels of performance. We consider a production process where n possible outcomes can be realized. Those performances can be ordered in way of $q_1 < q_2 < \cdots < q_i < \ldots < q_n$. Denote the principal's return in each of those states of nature by $S_i = S(q_i)$. In this context, a contract is a n-tuple of transfers $\{(t_1, \ldots, t_n)\}$. Moreover, let π_{ik} be the probability that the production level is q_i when the effort level is e_k, so that for all k, we have $\sum_{i=1}^{n} \pi_{ik} = 1$. We still assume that only two levels of effort are feasible. i.e., e_k in $\{0, 1\}$, and let $\Delta\pi_i = \pi_{i1} - \pi_{i0}$.

We want the agent's effort $e = 1$ to result in better performance (in the random sense). Therefore, we may use the maximum likelihood ratio criterion to reward the agent.

Definition 17.5.1 The probabilities of success satisfy the **monotone likelihood ratio property (MLRP)** if $\frac{\pi_{i1}-\pi_{i0}}{\pi_{i1}}$ is nondecreasing in i.

Suppose now that the agent is risk-averse. The optimal contract that induces effort must solve the program below:

$$\max_{\{t_1,\cdots,t_n)\}} \sum_{i=1}^{n} \pi_{i1}(S_i - t_i), \tag{17.5.62}$$

$$s.t. \qquad \sum_{i=1}^{n} \pi_{i1} u(t_i) - \psi \geqq \sum_{i=1}^{n} \pi_{i0} u(t_i), \tag{17.5.63}$$

$$\sum_{i=1}^{n} \pi_{i1} u(t_i) - \psi \geqq 0, \tag{17.5.64}$$

where (17.5.62) and (17.5.64) are the agent's incentive-compatibility and participation constraints, respectively.

Using the same change of variables as previously, it should be clear that the program is again concave with respect to the new variables $u_i = u(t_i)$, so that $h_i(\cdot) = u_i^{-1}$. The optimization problem then becomes:

$$\max_{\{u_1,\cdots,u_n)\}} \sum_{i=1}^{n} \pi_{i1}(S_i - h(u_i)) \tag{17.5.65}$$

$$\text{s. t.} \qquad \sum_{i=1}^{n} \pi_{i1}u_i - \psi \geqq \sum_{i=1}^{n} \pi_{i0}u_i, \tag{17.5.66}$$

$$\sum_{i=1}^{n} \pi_{i1}u_i - \psi \geqq 0. \tag{17.5.67}$$

Let μ and λ be the Lagrange multipliers of the participation constraint (17.5.67) and the incentive-compatibility constraint (17.5.66). Then, the first-order conditions of the principal's program are written as:

$$\frac{1}{u'(t_i^{SB})} = \mu + \lambda\left(\frac{\pi_{i1} - \pi_{i0}}{\pi_{i1}}\right) \quad \forall\, i \in \{1, \cdots, n\}. \tag{17.5.68}$$

Multiplying each of these equations by π_{i1} and summing over i yields $\mu = E_q\left(\frac{1}{u'(\tilde{t}^{SB})}\right) > 0$, where E_q denotes the expectation operator with respect to the distribution of outputs induced by effort $e = 1$.

Multiplying (17.5.68) by $\pi_{i1}u(t_i^{SB})$, summing all these equations over i, and taking into account the expression of μ obtained above result in

$$\lambda\left(\sum_{i=1}^{n}(\pi_{i1} - \pi_{i0})u(t_i^{SB})\right) = E_q\left(u(\tilde{t}^{SB})\left(\frac{1}{u'(\tilde{t}^{SB})} - E\left(\frac{1}{u'(\tilde{t}^{SB})}\right)\right)\right). \tag{17.5.69}$$

Using the complementary slackness condition $\lambda\left(\sum_{i=1}^{n}(\pi_{i1} - \pi_{i0})u(t_i^{SB}) - \psi\right) = 0$ to simplify the left-hand side of (17.5.69), we finally get

$$\lambda\psi = cov\left(u(\tilde{t}^{SB}), \frac{1}{u'(\tilde{t}^{SB})}\right). \tag{17.5.70}$$

By the same argument as for the case of two outcomes and two actions, we have $\lambda > 0$, and thus the incentive-compatibility constraint is binding.

Coming back to (17.5.68), we observe that the left-hand side is increasing in t_i^{SB} since $u(\cdot)$ is concave. For t_i^{SB} to be nondecreasing with i, MLRP must hold. Then, higher outputs are also those that are the more informative ones about the realization of a large effort. Therefore, the agent should receive more rewards as output increases.

When the level of performance belongs to a continuum, we can similarly analyze the optimal contract. In this case, the level of performance $\tilde{q}$ is drawn from a continuous distribution with a cumulative function $F(\cdot|e)$ and the density function $f(\cdot|e)$ on the support $[\underline{q}, \bar{q}]$. This distribution is conditional on the agent's level of effort, which still takes two possible values in $\{0, 1\}$. By the same procedure as above for the discrete case, we can find the optimal contract $t^{SB}(\pi)$ that is monotonically increasing in π when the monotone likelihood property $\frac{d}{dq}\left(\frac{f(q|1)-f(q|0)}{f(q|1)}\right) \geqq 0$ is satisfied.

17.5.2 Two Levels of Performance and Continuous Actions

We now discuss the optimal contract with continuous actions and two levels of performance. Assume that the set of the agent's effort is a continuum, $a \in [0, \infty)$. There are two possible outcomes, $q \in \{0, 1\}$. $q = 1$ represents good performance; otherwise, it is bad performance. Performance is contingent on the effort, and $p(a) \equiv prob(q = 1|a) \in (0, 1)$ indicates the probability of a good performance occurring when the effort is a. Suppose that there is $p'(a) > 0$, which means that if the agent chooses a higher level of effort, the probability of a good performance is also higher. To simplify the discussion, we assume that the agent's effort to select a is $\Psi(a) = a$. Let w be the principal's compensation to the agent. Since actions are not observable, the transfers to the agent are based on performance q.

The utility function of the principal is $V(q-w) = q-w$, i.e., the principal is risk-neutral. The agent's utility function is $u(w) - \Psi(a)$, which satisfies $u'(\cdot) > 0$ and $u''(\cdot) < 0$. Therefore, the agent is risk-averse.

If the agent's actions can be observed by the principal and can be verified by a third party, then the principal's optimal (first-best) contract is to solve the following maximization problem:

$$\max_{a,w_0,w_1} p(a)(1 - w_1) + (1 - p(a)(-w_0) \tag{17.5.71}$$

$$\text{s.t.} \quad p(a)u(w_1) + (1 - p(a))u(w_0) - a \geqq 0, \tag{17.5.72}$$

where (17.5.72) is the agent's participation constraint. Let λ be the Lagrange multiplier of the participation constraint. We have the first-order condition:

$$\frac{1}{u'(w_1)} = \lambda = \frac{1}{u'(w_0)} \tag{17.5.73}$$

so that $w_1 = w_0$ which is a pooling contract. The agent is then fully insured and the principal bears all risks.

We now discuss the second-best contract when information is incomplete. Since the agent's actions cannot be observed by the principal, the agent's actions must satisfy the incentive-compatibility constraint. Then, the principal's problem is:

$$\max_{a,w_0,w_1} p(a)(1 - w_1) + (1 - p(a)(-w_0) \tag{17.5.74}$$

$$\text{s.t.} \quad p(a)u(w_1) + (1 - p(a))u(w_0) - a \geqq 0, \tag{17.5.75}$$

$$a = \text{argmax}_{a'} p(a')u(w_1) + (1 - p(a'))u(w_0) - a', \tag{17.5.76}$$

where (17.5.76) is the incentive-compatibility constraint in a continuous situation.

From the (17.5.76), we obtain:

$$p'(a)(u(w_1) - u(w_0)) = 1. \tag{17.5.77}$$

Let μ and λ be the Lagrange multipliers of the agent's participation constraint (17.5.72) and incentive-compatibility constraint (17.5.77), respectively. From the first-order condition regarding w_0 and w_1, we obtain:

$$\frac{1}{u'(w_1)} = \mu + \lambda \frac{p'(a)}{p(a)}, \tag{17.5.78}$$

$$\frac{1}{u'(w_0)} = \mu - \lambda \frac{p'(a)}{1 - p(a)}. \tag{17.5.79}$$

When the incentive-compatibility constraint (17.5.77) is binding, we obtain $w_1 > w_0$, which means that the agent will receive higher compensation for a higher level of effort and the agent bears some risk under the second-best contract.

17.5.3 Continuous Performance and Continuous Actions

We now discuss the situation in which performance and actions are both continuous. In this case, it is clearer to see the trade-off between efficiency and insurance in a second-best contract.

We assume that the agent's effort $a \in [0, \infty)$ (in the probability sense) determines the level of outcome $q = a + \epsilon$, where ϵ is normally distributed over $(-\infty, \infty)$. It can be understood as an external factor that influences outcome with an expectation of 0 and a variance of σ^2. The principal is risk-neutral and the agent's utility function for income is $u(w) = -e^{-rw}$, which is the constant-coefficient absolute risk-aversion utility function. Assume that the principal's compensation for the agent is a linear function of the level of output: $w(q) = \alpha + \beta q$. We examine how α and β are chosen in the optimal contract. In addition, we assume that the cost of choosing a for the agent is $c(a)$.

Under the linear contract, the agent's compensation is also normally distributed. The mean is $\bar{w} = \alpha + \beta a$, and the variance is $\sigma^2(w) = \beta^2\sigma^2$. In the constant-coefficient absolute risk-aversion utility function, the expected utility is $-e^{-(\bar{w} - \frac{r\sigma^2(w)}{2})}$.

Then, under the linear compensation contract, the agent's expected utility for compensation is:

$$-e^{-(\alpha + \beta a - \frac{r\beta^2\sigma^2}{2})}. \tag{17.5.80}$$

The agent's incentive-compatibility constraint is:

$$a = \text{argmax}_{a'} \alpha + \beta a' - \frac{r\beta^2\sigma^2}{2} - c(a'),$$

which leads to the first-order condition:

$$c'(a) = \beta. \tag{17.5.81}$$

Thus, the principal's problem is:

$$\max_{a,\alpha,\beta} (1-\beta)a - \alpha \tag{17.5.82}$$

subject to

$$\alpha + \beta a - \frac{r\beta^2\sigma^2}{2} - c(a) \geqq \bar{u}, \tag{17.5.83}$$

$$c'(a) = \beta. \tag{17.5.84}$$

Here, $-e^{(-\bar{u})}$ that corresponds to $-\bar{u}$ is the agent's reservation utility. Obviously, the participation constraint (17.5.83) must be binding; otherwise, a higher profit can be obtained by lowering α. Substituting the participation constraint and incentive-compatibility constraints into the objective function, the above problem becomes:

$$\max_a a - \frac{rc'(a)^2\sigma^2}{2} - c(a),$$

from which we have the first-order condition:

$$1 - rc'(a)c''(a)\sigma^2 - c'(a) = 0,$$

and thus

$$\beta = c'(a) = \frac{1}{1 + rc''(a)\sigma^2},$$

where β is the marginal payment of the agent's effort that has a direct effect on the agent's incentive for effort provision. Therefore, this coefficient is called the **incentive intensity**, which depends on the risk metric σ^2 and the degree of risk-aversion of the agent. We can regard $r \to 0$ as a risk-neutral case. In this case, the incentive intensity is 1. As the agent's risk-aversion increases or the risk (variance) of the outcome increases, the incentive intensity should be smaller, which is precisely the fundamental trade-off between efficiency and insurance under moral hazard.

For general continuous performance and continuous actions, optimal contracts do not necessarily have linear forms. Holmstrom and Milgrom (1987) established conditions that ensure that the linear contracts are optimal.

In practice, incentive contracts usually contain more factors. We use the results from the above continuous case to discuss an incentive contract with multiple factors. Consider the following case: A manager who is attempting to invest a will affect the firm's current profit and stock price, as well. These two ways of effects are different because the factors affecting the stock price exceed the corporate accounting statements. These factors affect the future profitability of the firm.

Let $q = a + \epsilon_q$ be the current profit level of the firm where $\epsilon_q \sim N(0, \sigma_q^2)$. $p = a + \epsilon_p$ is the stock price of the firm, where $\epsilon_p \sim N(0, \sigma_p^2)$. $Cov(\epsilon_q, \epsilon_p) = \sigma_{pq}$ is the covariance between stock price and current profit (external) influence factor. If $\sigma_{pq} = 0$, this means that these two factors are independent.

The agent's utility function for income is $u(w) = -e^{-rw}$, and the principal is risk-neutral. We consider a linear contract for manager compensation $w = \alpha + p\beta_p + q\beta_q$. We focus on how the manager's compensation is dependent on the current profits and the firm's stock price in the optimal contract under incomplete information.

To simplify the discussion, we assume that the private cost of the manager's effort is $c(a) = \frac{ca^2}{2}$. The shareholder is the principal, and her problem is:

$$\begin{aligned} &\max_{a,\alpha,\beta_p,\beta_q} q - (\alpha + p\beta_p + q\beta_q) \\ \text{s. t.} \quad & E[-e^{-(\alpha + p\beta_p + q\beta_q - c(a))}] \geqq -e^{-\bar{u}}, \\ & a \in \operatorname{argmax}_{a'} E[-e^{-(\alpha + p(a')\beta_p + q(a')\beta_q - c(a'))}], \end{aligned}$$

where $E[\cdot]$ is the expected value for ϵ_p and ϵ_q. Using the certainty equivalence principle of income, this problem can be rewritten as follows:

$$\max_{a,\alpha,\beta_p,\beta_q} (1 - \beta_p - \beta_q)a - \alpha \tag{17.5.85}$$

$$\text{s. t.} \quad (\beta_p + \beta_q)a - \frac{r[\beta_p^2\sigma_p^2 + \beta_q^2\sigma_q^2 + 2\beta_p\beta_q\sigma_{pq}]}{2} - \frac{ca^2}{2} \geqq \bar{u}; \tag{17.5.86}$$

$$a \in \operatorname{argmax}_{a'} (\beta_p + \beta_q)a' - \frac{r[\beta_p^2\sigma_p^2 + \beta_q^2\sigma_q^2 + 2\beta_p\beta_q\sigma_{pq}]}{2} - \frac{ca'^2}{2}. \tag{17.5.87}$$

Solving the incentive-compatibility constraint (17.5.87), we obtain:

$$a = \frac{\beta_p + \beta_q}{c}.$$

Substituting it into the objective function (17.5.85) and noting that the participation constraint is binding, we can simplify the principal's problem as:

$$\max_{a} (1 - \beta_p - \beta_q)\frac{\beta_p + \beta_q}{c} - \alpha \tag{17.5.88}$$

subject to

$$\begin{aligned}(\beta_p+\beta_q)\frac{\beta_p+\beta_q}{c} - \frac{r\left(\beta_p^2\sigma_p^2+\beta_q^2\sigma_q^2+2\beta_p\beta_q\sigma_{pq}\right)}{2} \\ -\left(\frac{\beta_p+\beta_q}{c}\right)^2\frac{c}{2} = \bar{u}.\end{aligned} \tag{17.5.89}$$

Simplifying the first-order conditions of β_p and β_q, we obtain:

$$\begin{aligned}\beta_p^* &= \frac{\sigma_q^2-\sigma_{pq}}{\sigma_q^2+\sigma_p^2-2\sigma_{pq}}\frac{1}{1+rc\Omega},\\ \beta_q^* &= \frac{\sigma_p^2-\sigma_{pq}}{\sigma_q^2+\sigma_p^2-2\sigma_{pq}}\frac{1}{1+rc\Omega},\end{aligned}$$

where $\Omega = \frac{\sigma_q^2\sigma_p^2-\sigma_{pq}^2}{\sigma_q^2+\sigma_p^2+2\sigma_{pq}}$. The β_p^* and β_q^* obtained above are the marginal compensations for the managers of share price and current profit in the optimal contract, respectively.

When $\sigma_{pq}=0$, β_p^* and β_q^* in the optimal contract become:

$$\begin{aligned}\beta_p^* &= \frac{\sigma_q^2}{\sigma_q^2+\sigma_p^2+rc2\sigma_q^2\sigma_p^2},\\ \beta_q^* &= \frac{\sigma_p^2}{\sigma_q^2+\sigma_p^2+rc2\sigma_q^2\sigma_p^2}.\end{aligned}$$

When the agent is risk-neutral (i.e., $r=0$), β_p^* and β_q^* in the optimal contract become:

$$\begin{aligned}\beta_p^* &= \frac{\sigma_q^2-\sigma_{pq}}{\sigma_q^2+\sigma_p^2-2\sigma_{pq}},\\ \beta_q^* &= \frac{\sigma_p^2-\sigma_{pq}}{\sigma_q^2+\sigma_p^2-2\sigma_{pq}},\end{aligned}$$

and thus $\beta_p^*+\beta_q^*=1$. Along with σ_q^2 (resp. σ_p^2) increases, β_q^* becomes smaller (resp. larger), and β_p^* becomes larger (resp. smaller). In other words, if the profit risk is greater, the compensation to the manager reduces sensitivity to profit and increases sensitivity to stocks, and vice versa.

When $\epsilon_p=\epsilon_q+\zeta$ with $\zeta\sim N(0,\sigma_\zeta^2)$, and ζ and ϵ_q are independent so that $\sigma_{pq}=\sigma_q^2$, we then have $\beta_p^*=0$ and $\beta_q^*=\frac{1}{1+rc\sigma_q^2}$. Thus, the stock will not enter the manager's compensation contract because the profit is a sufficient statistic of the stock. Introducing the stock will not increase the manager's incentives, but will result in greater risks. Holmstrom (1979) proved the conclusion of sufficient statistics in similar incentive contracts.

17.6 Performance Incentive under Multi-Task

In many organizations, agents often perform multiple tasks at the same time. For example, in the case of teachers, not only do they need to impart knowledge to students, they also need to cultivate students' imagination and creativity. These different aspects of the overall job are markedly different in terms of measurement. The imparting of knowledge can be reflected in students' achievement. However, students' imagination and creativity are not well recognized. How do we form effective incentives for teachers in these two tasks? In the literature, the study of this problem is called the multi-task principal-agent model. Holmstrom and Milgrom (1991) conducted a deep study of this issue. We now discuss the incentive problem with multiple tasks through a simple model.

Assume that an agent is engaged in two tasks, and the outcome of each task depends on respective effort:

$$q_i = a_i + \epsilon_i, i = 1, 2.$$

To simplify the discussion, assume $\epsilon_i \sim N(0, \sigma_i^2)$, $\text{cov}(\epsilon_1, \epsilon_2) = 0$, where ϵ_i is understood as a measurement error.

Assume that the cost of efforts exerted by the agent is:

$$\Psi(a_1, a_2) = \frac{1}{2}(c_1 a_1^2 + c_2 a_2^2) + \delta a_1 a_2,$$

where $0 \leqq \delta \leqq \sqrt{c_1 c_2}$. If $\delta = 0$, this means that the two efforts are perfectly independent. If $\delta = \sqrt{c_1 c_2}$, the two efforts can be substitutes.

Assume that the principal is risk-neutral, and the agent has a constant-coefficient absolute risk-aversion utility function:

$$u(w; a_1, a_2) = -e^{-r[w - \Psi(a_1, a_2)]},$$

where w is the principal's monetary compensation to the agent.

Since the principal can only observe the two outcomes of the agent, the principal formulates a linear incentive contract contingent on outcomes:

$$w = \alpha + \beta_1 q_1 + \beta_2 q_2.$$

By the certainty equivalence principle, the agent's certainty equivalent net income is:

$$\alpha + \beta_1 a_1 + \beta_2 a_2 - \frac{r}{2}(\beta_1^2 \sigma_1^2 + \beta_2^2 \sigma_2^2) - \frac{1}{2}(c_1 a_1^2 + c_2 a_2^2) - \delta a_1 a_2.$$

By the incentive-compatibility constraint, the first-order condition of a_i is obtained:

$$\beta_i = c_i a_i + \delta a_j, i \neq j, i, j = 1, 2,$$

and thus

$$a_i = \frac{\beta_i c_j - \delta\beta_j}{c_1 c_2 - \delta^2}.$$

The principal's problem is then to choose α, β_1, β_2 to solve the following problem:

$$\max a_1(1-\beta_1) + a_2(1-\beta_2) - \alpha \tag{17.6.90}$$

$$\text{s.t.} \quad a_i = \frac{\beta_i c_j - \delta\beta_j}{c_i c_j - \delta^2}, i \neq j, i, j = 1, 2, \tag{17.6.91}$$

$$\alpha + \beta_1 a_1 + \beta_2 a_2 - \frac{r}{2}(\beta_1^2\sigma_1^2 + \beta_2^2\sigma_2^2) - \frac{1}{2}(c_1 a_1^2 + c_2 a_2^2) - \delta a_1 a_2 \geqq \bar{w}, \tag{17.6.92}$$

where $\bar{w}$ is the certainty equivalence income under the agent's reservation utility. Optimization means that the participation constraint (17.6.92) must be binding. Substituting a_1 and a_2 of the incentive-compatibility constraint (17.6.91) and α of the participation constraint (17.6.92) into the objective function (17.6.90), we have the unconstrained maximization problem:

$$\max_{(\beta_1,\beta_2)} \left(\frac{\beta_1 c_2 - \delta\beta_2 + \beta_2 c_1 - \delta\beta_1}{c_1 c_2 - \delta^2}\right) - \frac{r}{2}(\beta_1^2\sigma_1^2 + \beta_2^2\sigma_2^2)$$
$$-\frac{1}{2}c_1\left(\frac{\beta_1 c_2 - \delta\beta_2}{c_1 c_2 - \delta^2}\right)^2 - \frac{1}{2}c_2\left(\frac{\beta_2 c_1 - \delta\beta_1}{c_1 c_2 - \delta^2}\right)^2$$
$$-\delta\left(\frac{\beta_1 c_2 - \delta\beta_2}{c_1 c_2 - \delta^2}\right)\left(\frac{\beta_2 c_1 - \delta\beta_1}{c_1 c_2 - \delta^2}\right).$$

The first-order condition is then given by

$$\beta_1 = \frac{c_2 - \delta + \delta\beta_2}{c_2 + r\sigma_1^2(c_1 c_2 - \delta^2)}, \tag{17.6.93}$$

$$\beta_2 = \frac{c_1 - \delta + \delta\beta_1}{c_1 + r\sigma_2^2(c_1 c_2 - \delta^2)}, \tag{17.6.94}$$

from which we obtain β_1^* and β_2^* in the optimal contract:

$$\beta_1^* = \frac{1 + (c_2 - \delta)r\sigma_2^2}{1 + rc_2\sigma_2^2 + rc_1\sigma_1^2 + r^2\sigma_1^2\sigma_2^2(c_1 c_2 - \delta^2)}, \tag{17.6.95}$$

$$\beta_2^* = \frac{1 + (c_1 - \delta)r\sigma_1^2}{1 + rc_2\sigma_2^2 + rc_1\sigma_1^2 + r^2\sigma_1^2\sigma_2^2(c_1 c_2 - \delta^2)}. \tag{17.6.96}$$

When the two tasks are independent (i.e., $\delta = 0$), we have $a_i = \frac{\beta_i}{c_i}$ according to the agent's incentive-compatibility constraint (17.6.91). Based on the conditions (17.6.95) and (17.6.96) of the second-best contract, we get the same result as one in the previous subsection $\beta_i^* = \frac{1}{1+rc_i\sigma_i^2}$. The result of this is that the principal takes a separate second-best contract for each of the agent's tasks.

In addition, as σ_2^2 increases, the incentive intensity of the second task will decrease $\frac{\partial \beta_2^*}{\partial \sigma_2^2} < 0$. Since the derivative of β_1^* with respect to σ_2 in (17.6.95) is less than 0, the incentive for the first task will also decrease. Thus, if there is a substitutability for efforts, there will be complementarity between multiple tasks. A decrease in the incentive intensity of a task means that the incentive intensity of the other task will also decrease. Holmstrom and Milgrom (1991) found that under certain conditions, if the principal cares about the performance of these two tasks, the principal's optimal contract does not provide any incentive for all tasks when the measurement error of a certain task tends to be infinite ($\sigma_i^2 \to \infty$) or cannot be measured. Such a situation is called low-powered incentive.

This again shows that the intensity of incentives that the principal needs to face is a trade-off between efficiency and insurance. Incentive performance with high-powered incentives will amplify the uncertainty resulting from noise and increase the risk that the agent must bear. If a relationship between performance indicators and effort is weak and highly random, the high-powered incentives on such performance indicators undoubtedly mean guiding the agent to engage in activities that are tantamount to gambling.

17.7 Relative Performance Incentive of Multi-Agents

In practice, there are usually multiple agents inside of an organization which will bring about some new incentive issues. The compensation of an agent depends not only on his own performance but also on the performance of others, which is usually referred to as relative performance incentives in the literature. There are numerous studies that utilize incentive theory to analyze incentives for government-level institutions. In addition to fiscal incentives, local officials are also concerned about their promotion. Indeed, a local official's promotion depends on his performance relative to other officials at the same level, which is a common incentive method in hierarchical organizations or nations, such as in China.

In this section, we discuss the logic of promotion or tournaments in organizational incentives and compare the differences between it and the incentive of sharing.

17.7.1 Tournament under Risk Neutrality and No Common Shock

We first discuss the tournament model of Lazear and Rosen (1981). They first considered a simple case: a principal allows two risk-neutral agents to complete their tasks. The performance of the task depends on their respective level of effort and some external factors. Lazear and Rosen revealed that in a risk-neutral situation, a tournament can result in the first-best compensation contract.

Suppose that the performance function of the two agents is

$$q_i = a_i + \epsilon_i, i = 1, 2,$$

where a_i is effort and ϵ_i for $i = 1, 2$ are independent and identically distributed. The distribution function is $F(\cdot)$ with mean 0 and variance σ^2. Assume that each agent's cost function of effort is $c(a_i) = \frac{ca_i^2}{2}$.

Optimal effort a^* then satisfies:

$$a^* = \frac{1}{c}.$$

Consider a tournament: if an agent's performance is higher than that of another agent, he will receive a high compensation; otherwise, he will get a low compensation. Let W_1 and W_2 be the high and low compensations respectively (i.e., $W_1 > W_2$), and $\Delta W = W_1 - W_2$. Assume that the two agents are symmetric, i.e., they have the same effort cost and the same reservation utility. Their participation constraints are:

$$W_1 P_i + W_2(1 - P_i) - \frac{ca_i^2}{2} \geqq \bar{u},$$

where P_i is the probability that agent i wins, defined as:

$$P_i = prob(q_i > q_j) = prob(\epsilon_j - \epsilon_i < a_i - a_j) = H(a_i - a_j),$$

where $H(\cdot)$ is the distribution function of $\epsilon_j - \epsilon_i$ with a mean 0, a variance $2\sigma^2$, and its density function is denoted by $h(\cdot)$. Under such a contract, the first-best choice for agent i is given by:

$$\Delta W \frac{\partial P_i}{\partial a_i} - ca_i = \Delta W h(a_i - a_j) - ca_i = 0.$$

In a symmetric equilibrium, i.e., $a_i^T = a^T$, we have

$$a^T = \frac{\Delta W h(0)}{c}.$$

Thus, when

$$\Delta W = \frac{1}{h(0)}, \tag{17.7.97}$$

we have $a^T = a^*$. Meanwhile, the agents' participation constraints are:

$$W_1 \frac{1}{2} + W_2 \frac{1}{2} - \frac{1}{2c} \geqq \bar{u}, \tag{17.7.98}$$

which is binding at the optimum, resulting in first-best outcomes.

Then, solving (17.7.97) and (17.7.98) for W_1 and W_2, we have

$$W_1 = \frac{h(0)+c}{2ch(0)} + \bar{u}$$

and

$$W_2 = \frac{h(0)-c}{2ch(0)} + \bar{u},$$

at which the tournament contract implements first-best outcomes, although the form of the contract is markedly different.

17.7.2 Tournament under Risk-Aversion and Common Shocks

Now, consider tournament contracts when the agents are risk-averse and compare them to **piece-rate contracts**, which are also called the **individual performance contracts**. The following discussion is from Green and Stockey (1983), who showed that the equivalent relation above is no longer true. We will see that the independent contract dominates the tournament in the absence of common shock, while the tournament dominates the independent contract in the presence of common shock. The model below is a simplified version, which is drawn from Bolton and Dewatripont (2005).

There are two agents who need to undertake two tasks. There are only two outcomes, i.e., $q_i \in \{0,1\}$, corresponding to "failure" and "success", respectively. Assume that the two tasks are similar, and their outcomes depend on a common state that has two possibilities. One of the possibilities is no external shock. The probability of success of the task depends on the agent's effort level a_i. The other possibility is an external shock. Once it occurs, the task will fail, irrespective of the agent's effort. The principal knows that the external shock occurs with the probability $1-\xi$. In the absence of the external shock, each agent i can increase the probability of success by exerting more effort. We model this idea by assuming that the probability of success in the absence of the external shock is given by

$$\Pr(q_i = 1|a_i) = a_i.$$

The agent's effort cost function is:

$$c(a_i) = \frac{ca_i^2}{2}.$$

The agents' utility functions are the same:

$$u(w) - c(a),$$

with $u' > 0$ and $u'' < 0$. The principal is risk-neutral.

Let us discuss the tournament first. Since the outcome of the task and the common shock are observable, the feasible contract contains five possible compensations $w = (w_{11}, w_{10}, w_{01}, w_{00}, w_c)$, where $w_{q_i q_j}$ is the compensation for agent i when his outcome is q_i and the outcome of his opponent j is q_j. w_c is the compensation obtained by the agent when external shock occurs. In the following, we discuss the optimal incentive contract for agent i.

The incentive constraint for agent i is:

$$\begin{aligned} a_i &= \operatorname{argmax}_a \max\{\xi[a(1-a_j)u(w_{10}) + aa_j u(w_{11}) + (1-a)(1-a_j)u(w_{00}) \\ &\quad +(1-a)a_j u(w_{01})] + (1-\xi)u(w_c) - \frac{1}{2}ca^2\}. \end{aligned} \tag{17.7.99}$$

The first-order condition for the above incentive-compatibility constraint (17.7.99) can be written as:

$$\xi[(1-a_j)u(w_{10}) + a_j u(w_{11}) - (1-a_j)u(w_{00}) - a_j u(w_{01})] = ca_i. \tag{17.7.100}$$

If the principal wants the agent i to choose a_i, her problem is to minimize the implementation cost:

$$\begin{aligned} &\min_w \xi[a_i(1-a_j)w_{10} + a_i a_j w_{11} + (1-a_i)(1-a_j)w_{00} + (1-a_i)a_j w_{01}] \\ &\qquad +(1-\xi)w_c \\ \text{s.t.} \quad &\xi[a_i(1-a_j)u(w_{10}) + a_i a_j u(w_{11}) + (1-a_i)(1-a_j)u(w_{00}) \\ &\qquad +(1-a_i)a_j u(w_{01})] + (1-\xi)u(w_c) - \frac{1}{2}ca_i^2 \geqq \bar{u}, \qquad (17.7.101) \\ &\xi[(1-a_j)u(w_{10}) + a_j u(w_{11}) - (1-a_j)u(w_{00}) - a_j u(w_{01})] \\ &= ca_i. \qquad (17.7.102) \end{aligned}$$

Let μ and λ be the Lagrange multipliers of the participation constraint (17.7.101) and the incentive-compatibility constraint (17.7.102), respectively. The first-order conditions for w_{10} and w_{11} are, respectively, given by:

$$\xi a_i(1-a_j) = \xi(1-a_j)[\mu a_i + \lambda]u'(w_{10}), \tag{17.7.103}$$

$$\xi a_i a_j = \xi a_j[\mu a_i + \lambda]u'(w_{11}). \tag{17.7.104}$$

From (17.7.103) and (17.7.104), we obtain $w_{10} = w_{11}$.

Similarly, we can get the first-order conditions for w_{00} and w_{01}:

$$\xi(1-a_i)(1-a_j) = \xi(1-a_j)[\lambda(1-a_i) - \lambda]u'(w_{00}), \tag{17.7.105}$$

$$\xi(1-a_i)a_j = \xi a_j[\mu(1-a_i) - \lambda]u'(w_{01}). \tag{17.7.106}$$

By (17.7.105) and (17.7.106), we obtain $w_{00} = w_{01}$.

The first-order condition for w_c is:

$$(1-\xi) = (1-\xi)\mu u'(w_c). \tag{17.7.107}$$

From (17.7.107) and (17.7.105), due to $\lambda > 0$, we have $w_c \neq w_{00}$.

Comparing (17.7.105) with (17.7.103) and (17.7.107) with (17.7.103), we have $w_{10} \neq w_{00}$ and $w_c \neq w_{10}$.

Comparison of Tournament and Individual Performance Contract

Let us compare the tournament and the individual performance contract. For the tournament, it is mainly necessary to compare the performances of agent i and agent j, i.e., q_i and q_j, respectively. If $q_i > q_j$, agent i's reward is W; if $q_i < q_j$, agent i's reward is L; if $q_i = q_j$, agent i's reward is T. According to the previous discussion of the optimal contract, this means that $W = w_{10}, L = w_{01}, T = w_{00}, T = w_{11}, T = w_c$.

For the individual performance contract, since this compensation method only depends on an agent's own performance, $W_1 = w_{10} = w_{11}, W_0 = w_{01} = w_{00} = w_c$ according to the previous optimal contracts.

We then obtain the following conclusions:

Conclusion 1: When the common shock can be separated (i.e., $w_c \neq w_{ij}$) and there is no common shock (i.e., $\xi = 1$), then any relative performance incentive, which would result in either $w_{10} \neq w_{11}$, $w_{00} \neq w_{01}$ or both, is suboptimal. Only an individual performance contract is then optimal.

Conclusion 2: When $\xi < 1$, there is a common shock, and then even an individual performance contract is only suboptimal. This is because the individual performance contract requires $w_c = w_{00}$, but the first-best contract requires $w_c \neq w_{00}$.

Conclusion 3: When $\xi < 1$, the tournament dominates the individual performance contract for the principal.

Conclusions 1 and 2 above can be derived from the characteristics of the tournament and the individual performance contract.

In the following, we discuss the possibility of Conclusion 3. Assume that the best situation to the principal is to make both agents choose $a_1 = a_2 = 1$. In the tournament, given that agent j chooses $a_j = 1$, the cost for the principal to induce the agent i to choose $a_i = 1$ is given by

$$\begin{aligned} &\min_w \xi[a_iT + (1-a_i)L] + (1-\xi)T \\ \text{s.t.} \quad &\xi[a_iu(T) + (1-a_i)u(L)] + (1-\xi)u(T) - \frac{1}{2}ca_i^2 \geqq \bar{u}, \\ &\xi[u(T) - (1-a_j)u(L)] = ca_i. \end{aligned}$$

To induce the agent i to choose $a_i = 1$, the principal sets:

$$u(T) = u(L) + \frac{c}{\xi}.$$

When $a_i = a_j = 1$, agents i and j will receive a payment of T. In this case, the agent completely avoids the risk, and there is no insurance cost.

However, for individual performance contracts, if the principal wants to induce the agents to choose $a_1 = a_2 = 1$, the agents' compensations must be: $W_1 = w_{11} = w_{10} > W_0 = w_{01} = w_{00}$; otherwise, the agents will not have the incentive to choose $a_i = 1$, which means that the agents' payment is risky. Therefore, under a common shock, the tournament contract dominates the individual performance contract for the principal.

The above is just a comparison of the utility of the two contract forms through a simple case. For more general situations, see Lazear and Rosen (1981) and Green and Stockey (1983).

17.8 Dynamic Moral Hazard: Implicit Incentive and Career Concern

In the labor market, the incentives of workers come from explicit incentives for job performance. For example, the better is the performance, the higher is the compensation. On the other hand, it also comes from the evaluation of the workers in the market, especially the market's assessment of their abilities. Indeed, it directly influences their future value in the labor market. The latter is called implicit incentive or career motivation. Holmstrom (1982, 1999) analyzed the logic of implicit incentive. Here, we discuss the implicit incentive in the principal-agent model through a simple example.

Assume that there is no incentive contract between the principal and the agent. The agent pays attention to current income and future earnings. Consider a model of two periods. The outcome of the agent in each period depends on effort, capabilities, and external uncontrollable factors:

$$q_t = \theta + a_t + \epsilon_t, t = 1, 2,$$

where θ is the intrinsic ability of the agent and does not change over time. Assume that the ability and effort of the agent cannot be observed. This model differs from the previous moral hazard model in that there is an information asymmetry about type. To simplify the discussion, assume that the principal and the agent are risk-neutral. The agent's effort cost is $\psi(a_t)$, which satisfies $\psi'(a_t) > 0$ and $\psi''(a_t) > 0$.

In the future period 2, all employers in the market will observe the agent's outcome q_i. Since in period 2, the agent does not have incentive for working, which means $a_2 = 0$, the market wage to the agent is $w_2(q_1) = E(\theta|q_1)$. The higher is the ability, the greater is the outcome. Therefore, on the posterior estimate of the agent's ability, the higher is the outcome of the first-stage agent, the higher is the estimation of his ability in the market. In a pure-strategy equilibrium, suppose that the market expects the agent to invest a^* in period 1. The ability's posterior belief is:

$$w_2(q_1) = E(\theta|q_1) = q_1 - a^* = \theta + a_1 - a^*.$$

Let δ be the agent's discount rate for the future. The longer is the future period, the greater is the discounting, and so $\delta > 1$.

The agent's goal for the first period is:

$$\max_{a_1} = w_1 + \delta w_2(q_1) - \psi(a_1) = w_1 + \delta(\theta + a_1 - a^*) - \psi(a_1).$$

Since there is no explicit incentive, w_1 is not related to q_1, and the first-order condition for a_1 is:

$$\psi'(a^*) = \delta.$$

The optimal labor input is determined by solving the problem:

$$\max_{a_1} a_1 - \psi(a_1).$$

In this way, $\psi'(a^{FB}) = 1$. When $\delta > 1$, the agent's effort will exceed the first-best effort even if there is no incentive contract between the principal and the agent under the implicit incentive.

17.9 Trade-Off between Incentive Intensity and Performance Measurement Distortion

In the previous principal-agent problem, the incentive intensity of the agent is traded off between efficiency and insurance, which is based on an implicit assumption that there is a performance measure about the principal's interest (target). However, this assumption does not necessarily hold in practice. For listed companies, the stock price can be considered as an objective measure of the interests of shareholders. But, for non-listed companies, government agencies, non-profit organizations and other institutions, there is usually no objective measure. Even if there is interest in an organization, it cannot be contracted and incorporated into the contract. Therefore, performance measurement is not an objective measure of the organization's goals.

For example, for government agencies, social welfare can be regarded as the government's goal, but it is difficult to measure. Economic outcomes, such as GDP, are easy to measure, but it is not equal to social welfare. In this situation, how can the principal-agent problem be analyzed and discussed? Baker (1992) took the lead in introducing the analytic framework of this issue. He found that a trade-off exists between incentive intensity and performance measurement distortion.

Assume that the principal's goal is $V(a, \epsilon)$, where a is the agent's effort whose cost function is assumed to be $C(a) = \frac{c}{2}a^2$, and ϵ is an external influence factor. $V(a, \epsilon)$ cannot be specified in the contract. Assume that $P(a, \epsilon)$ is a performance measure that can be contracted. For the sake of discussion, we assume that the principal and the agent are risk-neutral.

Moreover, assume that the incentive contract provided by the principal is $w(P) = \alpha + \beta P$. If $P(a, \epsilon) = V(a, \epsilon)$, the incentive intensity under the optimal contract that is first-best is $\beta = 1$ since the agent does not have a risk cost. What is the incentive intensity under the optimal (second-best) contract if $P(a, \epsilon) \neq V(a, \epsilon)$?

First, we standardize the performance measurement so that the effort's marginal contribution to the principal's goal and the performance are consistent:

$$E[P_a(a, \epsilon)] = E[V_a(a, \epsilon)].$$

Under the above incentive contract, the agent's utility is:

$$u(w, a) = w - C(a) = \alpha + \beta P(a, \epsilon) - C(a).$$

Suppose that the agent's reservation utility is $\bar{u}$. Then, the incentive-compatibility and participation constraints of the agent are:

$$\alpha + \beta P(a, \epsilon) - C(a) \geqq \bar{u}, \tag{17.9.108}$$

$$\beta P_a(a^*, \epsilon) = C'(a^*). \tag{17.9.109}$$

The principal's goal is:

$$\max_{\alpha,\beta} E[V(a^*, \epsilon) - \alpha + \beta P(a^*, , \epsilon)]$$

subject to the incentive-compatibility and participation constraints (17.9.108), and (17.9.109), which leads to:

$$\beta^* = \frac{E[V_a a^*_\beta]}{P_a a^*_\beta}. \tag{17.9.110}$$

By (17.9.109), we have:

$$a^*_\beta = \frac{P_a}{c - \beta P_{aa}}, \tag{17.9.111}$$

where $P_{aa} = \frac{\partial^2 P}{\partial a \partial a}$, i.e., the second-order partial derivative of P in a.

Substituting (17.9.111) into (17.9.110), we obtain:

$$\beta^* = \frac{E[V_a P_a]}{P_a^2}. \tag{17.9.112}$$

Without loss of generality, assume that $E[P_a] = E[V_a] = 1$ for a^*. Then, (17.9.112) can be written as:

$$\beta^* = \frac{cov(V_a, P_a) + 1}{var(P_a) + 1} = \frac{\rho \sigma_{V_a} \sigma_{P_a} + 1}{\sigma^2_{P_a} + 1}, \tag{17.9.113}$$

where $var(P_a)$ is the variance of P_a, ρ is the correlation coefficient of V_a and P_a, and σ_{V_a} and σ_{P_a} are the standard deviations of V_a and P_a, respectively.

Several important aspects of the performance measurement problem become apparent. Note first that if V_a and P_a have the same variance and are perfectly correlated, then $\beta^* = 1$. Of course, these two conditions combined with equation $E[P_a(a,\epsilon) = E[V_a(a,\epsilon]$ imply that $V_a = P_a$ in all states of the world, which results in a first-best contract. Secondly, equation (17.9.113) demonstrates that the correlation between V_a and P_a is important to the determination of the optimal piece rate. All else being equal, the higher is this correlation, the more incentive intensity that the principal gives to the agent. At the same time, if $\rho < 0$, β^* may even be negative under certain conditions.

If the marginal product of effort on a performance measure is strongly correlated with the marginal product of effort on value, then the agent, who chooses an effort level based on the value of P_a, will choose high levels of effort when V_a is high and low levels when V_a is low. If these marginal products are not strongly correlated, then the agent's effort choice will not match the principal's desired effort level in most states. Because the agent's disutility function of effort is convex, choosing the wrong level of effort is costly. In response to this cost, the principal reduces the piece rate and reduces incentives.

For a more in-depth discussion on the impact of the difference between contract performance and the value of the organization, see the discussion in Baker (1992).

17.10 A Mixed Model of Moral Hazard and Adverse Selection

So far, when we discuss asymmetric information between a principal and an agent in the previous and current chapters, we have only considered either adverse selection or moral hazard, but not both. However, in many cases, a principal may know no information about the agent's action (e.g., his effort) or the agent's characteristics (e.g., his risk-aversion or cost). In this section, we consider the case in which both actions and characteristics of an agent are unknown to the principal. In doing so, we introduce a hybrid (or mixed) model of moral hazard and adverse selection, which was studied in Meng and Tian (2013). This model investigates the optimal wage contract design when both efforts and risk-aversion or production cost of the agent are unobservable.

From the previous section on performance incentives under multi-task, we saw that using high-powered incentives amplify the uncertainty brought about by noise and increases the risk that agents have to take, but it is assumed that the noise parameters can be observed by the principal. In this section, we assume that the risk-aversion coefficient of an agent is his private information. The general theoretical model in this section reveals that in an innovative economy, the risk (insurance) effect should dominate the incentive (efficiency) effect because of the greater risk of innovative activities, and thus the low-powered incentive contract is reasonable and relatively optimal. When moral hazard and adverse selection coexist, the incentive intensity of the optimal contract will be further reduced. This provides a new explanation for the ubiquitous phenomenon of low-powered incentives, and proves the necessity of low-powered incentives for innovation-driven development.

In the previous sections, we also assumed that the effort e or the performance output $\tilde{q}$ is one-dimensional. Now, we turn to a more general situation in which effort and performance are both multidimensional and continuous. We will first consider the benchmark case in which only the agent's efforts are unobservable, and then consider the case in which either risk-aversion or cost is also unobservable.

17.10.1 Optimal Wage Contract with Unobservable Efforts

Consider a principal-agent relationship in which the agent controls n activities that influence the principal's payoff. The principal is risk-neutral, and her gross payoff is a linear function of the agent's effort vector e:

$$V(e) = \beta' e + \eta, \tag{17.10.114}$$

where the $n-$dimensional vector β characterizes the marginal effect of the agent's effort e on $V(e)$, and η is a noise term with zero mean. The agent chooses a vector of efforts $e = (e_1, \cdots, e_n)' \in \mathcal{R}^n_+$ at quadratic personal cost $\frac{e'Ce}{2}$, where C is a symmetric positive definite matrix. The diagonal element C_{ii} reflects the agent's task-specific productivities, while the sign of off-diagonal elements C_{ij} indicates the relationship between two tasks i and j, which are substitutes (resp. complements, independent) if $C_{ij} > 0$ (resp. $< 0, = 0$).

The agent's preferences are represented by the negative exponential utility function $u(x) = -e^{-rx}$, where r is the agent's constant coefficient absolute risk-aversion, and x is his compensation minus personal cost.

It is assumed that there is a linear relation between the agent's efforts and the expected levels of the performance measures:

$$P_i(e) = b_i' e + \varepsilon_i, i = 1, \cdots, m, \tag{17.10.115}$$

where $b_i \in \mathcal{R}^n$ captures the marginal effect of the agent's effort e on the performance measure $P_i(e)$; $B = (b_1, \cdots, b_m)'$ is an $m \times n$ matrix of performance parameters, and it is assumed that the matrix B has full row rank m

so that every performance measure cannot be replaced by the other measures; and $\epsilon = (\varepsilon_1, \cdots, \varepsilon_m)'$ is an $m \times 1$ vector of normally distributed variables with mean zero and variance-covariance matrix Σ.

Definition 17.10.1 (Orthogonality) A performance system is said to be orthogonal if and only if $b_i' C^{-1} b_j = 0$ and $Cov(\varepsilon_i, \varepsilon_j) = 0$, for $i \neq j$, i.e., $B'C^{-1}B$ and Σ are both diagonal matrices.

Definition 17.10.2 (Cost-adjusted Correlation) The **cost-adjusted correlation** between two performance measures i and j is the ratio of the cost-adjusted inner product of their vectors of sensitivities divided by the covariance of the error terms:

$$\rho_{ij}^c = \frac{b_i' C^{-1} b_j}{\sigma_{ij}} \tag{17.10.116}$$

which measure performance not only about the agent's task-specific abilities and interaction among tasks, but also to what degree the two performance measures are aligned with each other. If n tasks are technologically independent and identical, i.e., $C = cI$, then we use the concept of *correlation* $\rho_{ij} \equiv b_i' b_j / \sigma_{ij}$ to measure the degree of alignment between the two performance measures.

Definition 17.10.3 (Signal-to-noise ratio) The **signal-to-noise ratio of a performance measure**$P_i(\mathbf{e}) = \mathbf{b}_i'\mathbf{e} + \varepsilon_i$ is defined as

$$\gamma_i \equiv \frac{(\nabla P_i(\mathbf{e}))'(\nabla P_i(\mathbf{e}))}{\mathrm{var}(\varepsilon_i)} = \frac{\mathbf{b}_i'\mathbf{b}_i}{\sigma_i^2}. \tag{17.10.117}$$

Definition 17.10.4 (Cost-adjusted Congruence) The **cost-adjusted congruence of a performance measure** $P_i = b_i' e + \varepsilon_i$ is defined as

$$\Gamma_i = \frac{b_i' C^{-1} \beta}{\sqrt{b_i' C^{-1} b_i}\sqrt{\beta' C^{-1} \beta}}. \tag{17.10.118}$$

A performance measure with non-zero cost-adjusted congruence is called *congruent*; a performance measure with unit cost-adjusted congruence is said to be *perfectly congruent*. We assume that there exists at least one congruent measure, i.e., $BC^{-1}\beta \neq 0$.

The principal compensates the agent's effort through a linear contract:

$$W(e) = w_0 + w'P(e), \tag{17.10.119}$$

where $P(e) = (P_1(e), \cdots, P_m(e))'$, w_0 denotes the base wage, and $w = (w_1, \cdots, w_m)'$ the performance wage. Under this linear compensation rule, the principal's expected profit is:

$$\Pi_p = V(e) - W(e) = \beta' e - w_0 - w' Be,$$

and the agent's certainty equivalent is

$$CE_a = w_0 + w'Be - \frac{1}{2}e'Ce - \frac{r}{2}w'\Sigma w. \tag{17.10.120}$$

The principal's problem is to design a contract (w_0, w) that maximizes her expected profit Π_p while ensuring the agent's participation and eliciting his optimal effort. The problem of the principal is thus formulated as:

$$\begin{cases} \max\limits_{\{w_0,w,e\}} \beta'e - w_0 - w'Be \\ \text{s.t:} \qquad IR : w_0 + w'Be - \frac{1}{2}e'Ce - \frac{r}{2}w'\Sigma w \geqq 0, \\ \qquad\qquad IC : e \in arg\max\limits_{\tilde{e}} \left[w_0 + w'B\tilde{e} - \frac{1}{2}\tilde{e}'C\tilde{e} - \frac{r}{2}w'\Sigma w \right]. \end{cases}$$

The IR constraint ensures that the principal cannot force the agent into accepting the contract, and here the agent's reservation utility is normalized to zero; the IC constraint represents the rationality of the agent's effort choice.

We now consider the effort-choosing problem of the agent for a given incentive scheme (w_0, w). Since the objective is concave by noting that the second-order derivative of CE_a with respect to e is a negative-definite matrix $-C$, the maximizer can be determined by the first-order condition: $Ce = B'w$. After replacing e with $e^* = C^{-1}B'w$ and substituting the IR constraint written with equality into the principal's objective function, the principal's problem simplifies to:

$$\max_{w \in \mathcal{R}^m} \left[\beta'C^{-1}B'w - \frac{1}{2}w'\left(BC^{-1}B' + r\Sigma\right) w \right].$$

The optimal wage contract and effort to be elicited are therefore:

$$w^p = \left[BC^{-1}B' + r\Sigma\right]^{-1} BC^{-1}\beta, \tag{17.10.121}$$

$$w_0^p = \frac{rw^{p'}\Sigma w^p - w^{p'}BC^{-1}B'w^p}{2}, \tag{17.10.122}$$

$$e^p = C^{-1}B'w^p. \tag{17.10.123}$$

The resulting surplus of the principal is[5]

$$\Pi^p = \frac{1}{2}\beta'C^{-1}B'\left[BC^{-1}B' + r\Sigma\right]^{-1} BC^{-1}\beta. \tag{17.10.124}$$

If the performance evaluation system is an orthogonal system and the n tasks are technically identical and independent ($C = cI$), we then have

$$w_i = \frac{\mathbf{b}_i'\beta}{\mathbf{b}_i'\mathbf{b}_i + rc\sigma_i^2} = \frac{|\mathbf{b}_i||\beta|cos(\widehat{\mathbf{b}_i, \beta})}{|\mathbf{b}_i|^2 + rc\sigma_i^2},$$

[5]The superscript “p” denotes “pure moral hazard”.

which implies that the more sensitive is the principal's revenue to the agent's effort (the larger is the norm of vector β_i), the greater is the incentive intensity (i.e., w_i), and the greater is the uncertainty of performance indicators (σ_i^2), the lower is the incentive intensity (i.e., w_i).

Thus, a higher incentive pay could induce the agent to implement a higher effort, but it will also expose the agent to a higher risk. It therefore necessitates a premium to compensate the risk-averse agent for the risk that he bears. The optimal power of incentive is therefore determined by the trade-off between incentives and risk. Moreover, the results above show that in multi-task agency relationships, the degree of congruity of available performance measures and the agent's task-specific abilities also affect the power and distortion of the incentive contract.

17.10.2 Optimal Wage Contract with Unobservable Efforts and Risk-Aversion

In practice, the types and behaviors of agents are often unobservable, which leads to the mixed problem of moral hazard and adverse selection. In this subsection, we will consider unobservable agent risk-aversion based on the above moral hazard model.

The pure moral hazard incentive contract stated above relies crucially on the agent's attitude towards risk, which is the key factor affecting incentive intensity. In the following, we assume that risk-aversion r is also private information of the agent, and its cumulative distribution function $F(r)$ and density function $f(r)$ supported on $[\underline{r}, \overline{r}]$ are common knowledge to all parties. The principal then has to offer a contract menu $\{w_0(\hat{r}), w(\hat{r})\}$ contingent on the agent's reported "type" $\hat{r}$ to maximize her expected payoff. The timing of the problem is as follows:

- only agents know their risk-aversion coefficient;
- the principal provides the agent with a wage contract, and the agent decides whether to accept the contract and whether to report his private information truthfully;
- the agent chooses his efforts;
- the realization of the performance of the agent and the profit of the principal;
- the principal pays the agent.

A contract $\{w_0(\hat{r}), w(\hat{r})\}$ is said to be implementable if the following incentive-compatibility condition is satisfied:

$$w_0(r)+\frac{1}{2}w(r)'\left[BC^{-1}B'-r\Sigma\right]w(r) \geqq w_0(\hat{r})+\frac{1}{2}w(\hat{r})'\left[BC^{-1}B'-r\Sigma\right]w(\hat{r}). \tag{17.10.125}$$

Let $U(r,\hat r) \equiv w_0(\hat r) + \frac{1}{2}w(\hat r)'\left[BC^{-1}B' - r\Sigma\right]w(\hat r)$,[6] and $U(r) \equiv U(r,r)$. Then, the implementability condition of $\{U(r), w(r)\}$ is stated equivalently as:

$$\begin{aligned}&\exists w_0 : [\underline{r},\overline{r}] \to \mathcal{R}_+, \forall (r,\hat r) \in [\underline{r},\overline{r}]^2, \text{ we have:}\\ &U(r) = \max_{\hat r}\left\{w_0(\hat r) + \frac{1}{2}w(\hat r)'\left[BC^{-1}B' - r\Sigma\right]w(\hat r)\right\}. \quad (17.10.126)\end{aligned}$$

By the taxation principle (see Guesnerie (1981), Hammond (1979), and Rochet (1985)), the above is, in turn, equivalent to the following highly similar condition:

$$\begin{aligned}&\exists w_0 : \mathcal{R}^m \to \mathcal{R}_+, \forall r \in [\underline{r},\overline{r}], \text{ we have:}\\ &U(r) = \max_{w}\left\{w_0(w) + \frac{1}{2}w'[BC^{-1}B' - r\Sigma]w\right\}. \quad (17.10.127)\end{aligned}$$

Then, $U(\cdot)$ is continuous, convex[7], and satisfies the envelope condition:

$$U'(r) = -\frac{1}{2}w'\Sigma w. \qquad (17.10.128)$$

Conversely, if (17.10.128) holds and $U(r)$ is convex, then

$$U(r) \geqq U(\hat r) + (r-\hat r)U'(\hat r) = U(\hat r) - \frac{1}{2}(r-\hat r)w'(\hat r)\Sigma w(\hat r),$$

which implies the incentive-compatibility condition $U(r) \geqq U(r,\hat r)$. Formally, we have

Lemma 17.10.1 *The surplus function $U(r)$and performance wage function $w(r)$ are implementable if and only if:*

(1) envelope condition (17.10.128) is satisfied;

(2) $U(r)$ is convex in r.

Substituting $U(r)$ into the principal's expected payoff, we obtain

$$\begin{aligned}\Pi &= \int_{\underline{r}}^{\overline{r}}\left[\beta' e^* - w_0(r) - w(r)'Be^*\right]f(r)\mathrm{d}r\\ &= \int_{\underline{r}}^{\overline{r}}\left\{\beta' C^{-1}B'w(r) - \frac{1}{2}w(r)'\left[BC^{-1}B' + r\Sigma\right]w(r) - U(r)\right\}f(r)\mathrm{d}r.\end{aligned}$$

[6]Substituting $e^* = C^{-1}B'w$ into expression (17.10.120) yields $U = w_0 + \frac{1}{2}w'(BC^{-1}B' - r\Sigma)w$.

[7]One way to define convex functions is through representing them as the maximum of the affine functions, i.e., $s(x)$ is convex if and only if

$$s(x) = \max_{a,b\in\Omega}(a\cdot x + b)$$

for some $a \in \mathcal{R}^n$, $b \in \mathcal{R}$ and some $\Omega \subseteq \mathcal{R}^{n+1}$. In this example $a = -\frac{1}{2}w'\Sigma w, b = w_0(w) + \frac{1}{2}w'BC^{-1}B'w$, and thus $U(r) = \max_{(a,b)\in\mathcal{R}_-\times\mathcal{R}_+}(ar+b)$ is a convex function in r.

The principal's problem is therefore:

$$\max_{U(r),w(r)} \Pi, \text{s.t.: } U(r) \geqq 0, U'(r) = -\frac{1}{2}w(r)'\Sigma w(r), U(r) \text{ is convex.} \tag{17.10.129}$$

The following proposition summarizes the solution of the principal's problem.

Proposition 17.10.1 *If $\Phi(r)$ is nondecreasing,*[8] *then the optimal wage contract is given by*

$$w^h(r) = \left[BC^{-1}B' + \Phi(r)\Sigma\right]^{-1} BC^{-1}\beta \tag{17.10.130}$$

$$w_0^h(r) = \frac{1}{2}\int_r^{\overline{r}} w^h(\tilde{r})'\Sigma w^h(\tilde{r})d\tilde{r} - \frac{1}{2}w^h(r)'\left[BC^{-1}B' - r\Sigma\right] \times w^h(r), \tag{17.10.131}$$

where $\Phi(r) \equiv r + \frac{F(r)}{f(r)}$.[9]

PROOF. Using the envelope condition $U'(r) = -\frac{1}{2}w'\Sigma w$, the participation constraint $U(r) \geqq 0$ simplifies to $U(\bar{r}) \geqq 0$. Incentive-compatibility implies that only the participation constraint of the most risk-averse type can be binding, i.e., $U(\bar{r}) = 0$. We therefore get

$$U(r) = \int_r^{\bar{r}} \frac{1}{2}w(\tilde{r})'\Sigma w(\tilde{r})dr. \tag{17.10.132}$$

The principal's objective function becomes

$$\Pi = \int_{\underline{r}}^{\overline{r}} \left\{\beta' C^{-1}B'w(r) - \frac{1}{2}w(r)'\left[BC^{-1}B' + r\Sigma\right]w(r) - \int_r^{\bar{r}} \frac{1}{2}w(\tilde{r})'\Sigma w(\tilde{r})dr\right\} f(r)\mathrm{d}r$$

which, by an integration of parts, yields

$$\int_{\underline{r}}^{\overline{r}} \left\{\beta' C^{-1}B'w(r) - \frac{1}{2}w(r)'\left[BC^{-1}B' + \left(r + \frac{F(r)}{f(r)}\right)\Sigma\right]w(r)\right\} f(r)\mathrm{d}r.$$

Maximizing pointwise the above expression, we obtain

$$w^h(r) = \left[BC^{-1}B' + \Phi(r)\Sigma\right]^{-1} BC^{-1}\beta$$

and

$$w_0^h(r) = \frac{1}{2}\int_r^{\overline{r}} w^h(\tilde{r})'\Sigma w^h(\tilde{r})d\tilde{r} - \frac{1}{2}w^h(r)'\left[BC^{-1}B' - r\Sigma\right]w^h(r).$$

[8]This condition is weaker than, and could be implied by, the monotone hazard rate property: $\frac{d}{dr}\left[\frac{F(r)}{f(r)}\right] \geqq 0$.

[9]The superscript "h" denotes the "hybrid model of moral hazard and adverse selection".

The only remaining task is to verify the convexity of $U(r)$. Note that

$$U''(r) = -(D_r w^h)'\Sigma w^h = \Phi'(r) w^h(r)'\Sigma \left[BC^{-1}B' + \Phi(r)\Sigma\right]^{-1} \Sigma w^h(r).$$

The second equality comes from the fact that the derivative of w^h with respect to r is[10]

$$\begin{aligned} D_r w^h &= -\left[BC^{-1}B' + \Phi(r)\Sigma\right]^{-1} \Phi'(r)\Sigma \left[BC^{-1}B' + \Phi(r)\Sigma\right]^{-1} BC^{-1}\beta \\ &= -\Phi'(r)\left[BC^{-1}B' + \Phi(r)\Sigma\right]^{-1} \Sigma w^h. \end{aligned}$$

It is clear that $U''(r) \geqq 0$ because $\Phi'(r) \geqq 0$ and the matrix $\Sigma\left[BC^{-1}B' + \Phi(r)\Sigma\right]^{-1}\Sigma$ is positive definite. □

The following conditions prove to be sufficient for the emergence of low-powered incentives.

Condition 17.10.1 *Σ is diagonal.*

Condition 17.10.2 *Matrix $BC^{-1}B'$ is diagonal.*

Condition 17.10.3 *Matrices $BC^{-1}B'$ and Σ commute: $BC^{-1}B'\Sigma = \Sigma BC^{-1}B'$.*[11]

Condition 17.10.4 *The following inequality holds:*

$$2r\lambda_m^2 + \rho > 0 \qquad (17.10.133)$$

where

$$\begin{aligned} \rho &= \max\left\{ \min_{i=1,m} \lambda_i \mu_m \frac{(\sqrt{k_\lambda}+1)^2 - k_\mu(\sqrt{k_\lambda}-1)^2}{2\sqrt{k_\lambda}}, \min_{i=1,m} \mu_i \lambda_m \frac{(\sqrt{k_\mu}+1)^2 - k_\lambda(\sqrt{k_\mu}-1)^2}{2\sqrt{k_\mu}} \right\} \\ &= \begin{cases} \lambda_m\mu_m \frac{(\sqrt{k_\lambda}+1)^2 - k_\mu(\sqrt{k_\lambda}-1)^2}{2\sqrt{k_\lambda}} & \text{if } \sqrt{k_\mu} \leqslant \frac{\sqrt{k_\lambda}+1}{\sqrt{k_\lambda}-1}, k_\mu \geqq k_\lambda; \\ \lambda_m\mu_m \frac{(\sqrt{k_\mu}+1)^2 - k_\lambda(\sqrt{k_\mu}-1)^2}{2\sqrt{k_\mu}} & \text{if } \sqrt{k_\mu} \leqslant \frac{\sqrt{k_\lambda}+1}{\sqrt{k_\lambda}-1}, k_\mu < k_\lambda; \\ \lambda_1\mu_m \frac{(\sqrt{k_\lambda}+1)^2 - k_\mu(\sqrt{k_\lambda}-1)^2}{2\sqrt{k_\lambda}} & \text{if } \sqrt{k_\mu} > \frac{\sqrt{k_\lambda}+1}{\sqrt{k_\lambda}-1}, k_\mu \geqq k_\lambda; \\ \lambda_m\mu_1 \frac{(\sqrt{k_\mu}+1)^2 - k_\lambda(\sqrt{k_\mu}-1)^2}{2\sqrt{k_\mu}} & \text{if } \sqrt{k_\mu} > \frac{\sqrt{k_\lambda}+1}{\sqrt{k_\lambda}-1}, k_\lambda < k_\lambda. \end{cases} \end{aligned}$$

Here, λ_i and μ_i are the $i-$th eigenvalues of Σ and $BC^{-1}B'$, respectively, in a descending enumeration, $k_\lambda = \frac{\lambda_1}{\lambda_m}$, and $k_\mu = \frac{\mu_1}{\mu_m}$ denote the spectral condition number of Σ and $BC^{-1}B'$, respectively.

[10] Let A be a nonsingular, $m \times m$ matrix whose elements are functions of the scalar parameter α, then

$$\frac{\partial A^{-1}}{\partial \alpha} = -A^{-1}\frac{\partial A}{\partial \alpha}A^{-1}.$$

[11] In other words, $BC^{-1}B'\Sigma$ is symmetric.

Condition 17.10.5 *There exists a positive number λ, such that $BC^{-1}B' = \lambda\Sigma$.*

Condition 17.10.1 requires that the error terms of performance measures are stochastically independent. It also implies that different measures cannot be affected by a common stochastic factor. Condition (17.10.2) states that $b_i'C^{-1}b_j = 0$ for all $i \neq j$. Intuitively, it requires that different performance measures respond in distinct ways to the agent's effort when cost is incorporated. Condition (17.10.4) holds when the agent is sufficiently risk-averse or when either matrix $BC^{-1}B'$ or Σ is well-conditioned.[12] Several important special cases are:

(1) The performance measures system is orthogonal. In this case, conditions (17.10.1) to (17.10.3) are all satisfied.

(2) Σ is a scalar matrix, in which case (17.10.1), (17.10.3), and (17.10.4) are satisfied.

(3) $BC^{-1}B'$ is a scalar matrix, in which case conditions (17.10.2), (17.10.3), and (17.10.4) are satisfied.

Condition (17.10.5) emphasizes that the covariance matrix Σ is a transformation of the measure-cost efficiency matrix $BC^{-1}B'$. In other words, correlation between any pair of performance measures i and j is constant: $\rho_{ij} = \lambda$.

By comparing the wage contract obtained in the hybrid model to the benchmark pure moral hazard model, we find that the principal will reduce the power of incentives offered to the agent.

Theorem 17.10.1 *We have the following conclusions:*

1. *Given any one of the conditions 17.10.1 to 17.10.4, there exists an $i \in \{1 \cdots m\}$, such that $|w_i^h(r)| < |w_i^p(r)|$ for all $r \in (\underline{r}, \bar{r}]$.*

2. *If both condition 17.10.1 and condition 17.10.2 are satisfied, then $|w_i^h(r)| < |w_i^p(r)|$ for all $r \in (\underline{r}, \bar{r}]$ and all $i \in \{1 \cdots m\}$.*

3. *Let ω_i, $i \in \mathcal{K} \equiv \{1, 2, \cdots, k\}$ denote k distinct generalized eigenvalues of $BC^{-1}B'$ relative to Σ, $\mathcal{V}_i \equiv \mathcal{N}(BC^{-1}B' - \omega_i\Sigma)$ be the eigenspace corresponding to ω_i, and $\mathcal{V}_i^{\perp}$ be its orthogonal complement. Suppose that $BC^{-1}\beta \notin \bigcup_{i\in\mathcal{K}} \mathcal{V}_i^{\perp}$. Then, there exists a positive number $k \in (0, 1)$, such that $w^h = kw^p$ if and only if condition 17.10.5 is met.*

PROOF. Since the proof is long, it is omitted here and can be found in Meng and Tian (2013). □

[12]Matrices with condition numbers near 1 are said to be well-conditioned, while matrices with high condition numbers are said to be ill-conditioned.

Thus, when the risk-aversion parameter is unobservable to the principal, the less risk-averse agent gains information rent by mimicking the more risk-averse one. The amount of information rent gained by the agent depends on the performance wage of the agent with larger risk aversion, and therefore the basic trade-off between efficiency and rent extraction leads to low-powered incentive for all but the least risk-averse types. Under conditions (17.10.1) to (17.10.4), wage vector w is shortened in different quadratic-form norms compared with the pure moral hazard case. Under condition (17.10.5), the wage vector that minimizes the cost of effort $e'Ce = w'BC^{-1}B'w$ points in the same direction as the wage vector that minimizes the risk premium $rw'\Sigma w$. Consequently, the trade-off between efficiency and rent-extraction alters only the overall intensity of the wage vector, but not its relative allocation among performance measures.

17.10.3 Optimal Wage Contract with Unobservable Efforts and Cost

In this subsection, we assume that the cost parameter, rather than the degree of risk-aversion, is private information. To avoid the complicated multidimensional mechanism design issue, we assume that $C = cI$, i.e., the tasks are technologically identical and independent. $\delta = \frac{1}{c}$ is assumed to be distributed on the support $[\underline{\delta}, \overline{\delta}]$, according to a cumulative distribution function $G(\delta)$ and density $g(\delta)$. The timeline of this problem is analogous to that in Section 17.10.2, except that the agent is now required to report $\hat{\delta}$. A contract menu $\{w_0(\delta), w(\delta)\}$ is said to be implementable if the following incentive-compatibility condition is satisfied:

$$w_0(\delta)+\frac{1}{2}w(\delta)'\left[\delta BB' - r\Sigma\right]w(\delta) \geqq w_0(\hat{\delta})+\frac{1}{2}w(\hat{\delta})'\left[\delta BB' - r\Sigma\right]w(\hat{\delta}), \forall(\delta,\hat{\delta}) \in [\underline{\delta}, \overline{\delta}]^2. \tag{17.10.134}$$

Let

$$U(\delta,\hat{\delta}) \equiv w_0(\hat{\delta}) + \frac{1}{2}w(\hat{\delta})'\left[\delta BB' - r\Sigma\right]w(\hat{\delta}),$$

and

$$U(\delta) \equiv U(\delta,\delta).$$

Then, $\{U(\delta), w(\delta)\}$ is called implementable if

$$\exists w_0 : [\underline{\delta}, \overline{\delta}] \to \mathcal{R}_+, \forall(\delta,\hat{\delta}) \in [\underline{\delta}, \overline{\delta}]^2, U(\delta) = \max_{\hat{\delta}}\left\{w_0(\hat{\delta}) + \frac{1}{2}\mathbf{w}(\hat{\delta})'\left[\delta BB' - r\Sigma\right]\mathbf{w}(\hat{\delta})\right\}. \tag{17.10.135}$$

or equivalently,

$$\exists w_0 : \mathcal{R} \to \mathcal{R}_+, \forall\delta \in [\underline{\delta}, \overline{\delta}], U(\delta) = \max_{w\in\mathcal{R}^m}\left\{w_0(\mathbf{w}) + \frac{1}{2}\mathbf{w}'\left[\delta BB' - r\Sigma\right]\mathbf{w}\right\}. \tag{17.10.136}$$

$U(\delta)$ is necessarily continuous, increasing, and convex in δ[13] and satisfies the envelope condition:

$$U'(\delta) = \frac{1}{2} w' B B' w. \tag{17.10.137}$$

Conversely, similar to the case with unobservable risk-aversion, the convexity of $U(\delta)$ and the envelope condition (17.10.137) implies

$$U(\delta) \geqq U(\hat{\delta}) + (\delta - \hat{\delta}) U'(\hat{\delta}) = U(\hat{\delta}) + \frac{1}{2}(\delta - \hat{\delta}) w' B B' w = U(\delta, \hat{\delta}),$$

which, in turn, implies implementability of the contract. We summarize the above discussion in the following lemma.

Lemma 17.10.2 *The surplus function $U(\delta)$ and the wage function $w(\delta)$ are implementable if and only if*

(1) $U'(\delta) = \frac{1}{2} w' B B' w$;

(2) $U(\delta)$ is convex in δ.

The optimal $\delta-$contingent contract solves the following problem:

$$\begin{cases} \max\limits_{w(\delta), U(\delta)} \displaystyle\int_{\underline{\delta}}^{\bar{\delta}} \left\{ \delta \beta' B' w(\delta) - \frac{1}{2} w(\delta)' \left[\delta B B' + r\Sigma \right] w(\delta) - U(\delta) \right\} g(\delta) \mathrm{d}\delta \\ \quad \text{s.t: } U(\delta) \geqq 0, U'(\delta) = \dfrac{w' B B' w}{2}, U(\delta) \text{ is convex} \end{cases}.$$

Proposition 17.10.2 *With unobservable cost, if $\delta H(\delta)$ is nonincreasing,* [14] *then the optimal wage is given by*

$$w^h(\delta) = \left(H(\delta) B B' + \frac{r\Sigma}{\delta} \right)^{-1} B\beta \tag{17.10.138}$$

$$\begin{aligned} w_0^h(\delta) &= \frac{1}{2} \int_{\underline{\delta}}^{\delta} w^h(\tilde{\delta})' B B' w^h(\tilde{\delta}) d\tilde{\delta} \\ &- \frac{1}{2} w^h(\delta)' \left[\delta B B' - r\Sigma \right] w^h(\delta), \end{aligned} \tag{17.10.139}$$

where $H(\delta) \equiv 1 + \frac{1 - G(\delta)}{\delta g(\delta)}$.

Proof. Using integration by parts, we obtain

$$\int_{\underline{\delta}}^{\bar{\delta}} U(\delta) g(\delta) = \int_{\underline{\delta}}^{\bar{\delta}} \left[\frac{1 - G(\delta)}{g(\delta)} \right] \frac{w' B B' w}{2} dG(\delta).$$

---

[13] In this case, let $a = \frac{1}{2} w' B B' w, b = w_0(w) - \frac{1}{2} w' \Sigma w$, then $U(\delta) = \max\limits_{a,b}(a\delta + b)$ is convex in δ.

[14] This assumption is somewhat stronger than the usual monotone inverse hazard rate condition. It holds for any nondecreasing $g(\cdot)$.

Substituting it into the expression of the principal's expected surplus and maximizing it with respect to w, we obtain the optimal performance wage $w^h(\delta)$, and $w_0^h(\delta)$ is also easily obtained. We now check the convexity of $U(\delta)$. The first-order derivative of $w^h(\delta)$ is

$$\begin{aligned} D_\delta w^h(\delta) &= -\left[H(\delta)BB' + \frac{r}{\delta}\Sigma\right]^{-1}\left[H'(\delta)BB' - \frac{r\Sigma}{\delta^2}\right]\left[H(\delta)BB' + \frac{r\Sigma}{\delta}\right]^{-1} B\beta \\ &= -\left[H(\delta)BB' + \frac{r\Sigma}{\delta}\right]^{-1}\left[H'(\delta)BB' - \frac{r\Sigma}{\delta^2}\right] w^h(\delta) \\ &= -\left[H(\delta)BB' + \frac{r\Sigma}{\delta}\right]^{-1}\left\{-\frac{H(\delta)}{\delta}BB' - \frac{r\Sigma}{\delta^2} + \left[\frac{H(\delta)}{\delta} + H'(\delta)\right]BB'\right\} w^h(\delta) \\ &= \frac{1}{\delta}\left\{(BB')^{-1} - \left[H(\delta)BB' + \frac{r\Sigma}{\delta}\right]^{-1}[H(\delta) + \delta H'(\delta)]\right\} BB'w^h(\delta). \end{aligned}$$

It can be verified that the matrix

$$\frac{1}{\delta}\left\{(BB')^{-1} - \left[H(\delta)BB' + \frac{r\Sigma}{\delta}\right]^{-1}[H(\delta) + \delta H'(\delta)]\right\}$$

is positive-definite since $\delta + \frac{1-G(\delta)}{g(\delta)} = \delta H(\delta)$ is decreasing. Therefore

$$\begin{aligned} U''(\delta) &= D_\delta w^h(\delta)BB'w^h(\delta) \\ &= \frac{1}{\delta}w^h(\delta)'BB'\left\{(BB')^{-1} - \left[H(\delta)BB' + \frac{r\Sigma}{\delta}\right]^{-1}[H(\delta) + \delta H'(\delta)]\right\} BB'w^h(\delta) \\ &\geqq 0, \end{aligned}$$

which implies the convexity of $U(\delta)$. □

The following conditions justify the adoption of low-powered incentives in the presence of unobservable costs.

Condition 17.10.6 *BB' is a diagonal matrix.*

Condition 17.10.7 *Matrices BB' and Σ commute: $BB'\Sigma = \Sigma BB'$.*[15]

Condition 17.10.8 *The following inequality holds:*

$$2\nu_m^2 + \frac{r}{\delta}\eta > 0.$$

$$\begin{aligned} \eta &= \max\left\{\min_{i=1,m}\lambda_i\nu_m\frac{(\sqrt{k_\lambda}+1)^2 - k_\nu(\sqrt{k_\lambda}-1)^2}{2\sqrt{k_\lambda}}, \min_{i=1,m}\nu_i\lambda_m\frac{(\sqrt{k_\nu}+1)^2 - k_\lambda(\sqrt{k_\nu}-1)^2}{2\sqrt{k_\nu}}\right\}, \\ &= \begin{cases} \lambda_m\nu_m\frac{(\sqrt{k_\lambda}+1)^2 - k_\nu(\sqrt{k_\lambda}-1)^2}{2\sqrt{k_\lambda}} & \textit{if} \quad \sqrt{k_\nu} \leqslant \frac{\sqrt{k_\lambda}+1}{\sqrt{k_\lambda}-1}, k_\nu \geqq k_\lambda, \\ \lambda_m\nu_m\frac{(\sqrt{k_\nu}+1)^2 - k_\lambda(\sqrt{k_\nu}-1)^2}{2\sqrt{k_\nu}} & \textit{if} \quad \sqrt{k_\nu} \leqslant \frac{\sqrt{k_\lambda}+1}{\sqrt{k_\lambda}-1}, k_\nu < k_\lambda, \\ \lambda_1\nu_m\frac{(\sqrt{k_\lambda}+1)^2 - k_\nu(\sqrt{k_\lambda}-1)^2}{2\sqrt{k_\lambda}} & \textit{if} \quad \sqrt{k_\nu} > \frac{\sqrt{k_\lambda}+1}{\sqrt{k_\lambda}-1}, k_\nu \geqq k_\lambda, \\ \lambda_m\nu_1\frac{(\sqrt{k_\nu}+1)^2 - k_\lambda(\sqrt{k_\nu}-1)^2}{2\sqrt{k_\nu}} & \textit{if} \quad \sqrt{k_\nu} > \frac{\sqrt{k_\lambda}+1}{\sqrt{k_\lambda}-1}, k_\lambda < k_\lambda, \end{cases} \end{aligned}$$

[15] Again, it is true if $BB'\Sigma$ is symmetric.

represents the lower bound of eigenvalues of Jordan product $BB'\Sigma+\Sigma BB'$; λ_i, ν_i are the $i-th$ eigenvalues of Σ and BB', respectively, in a descending enumeration. $k_\lambda = \frac{\lambda_1}{\lambda_m}$ and $k_\nu = \frac{\nu_1}{\nu_m}$ denote the spectral condition number of Σ and BB', respectively.

Condition 17.10.9 *There exists a positive number k, such that $BB' = k\Sigma$.*

In the special case in which the performance measures system is orthogonal, conditions (17.10.1), (17.10.6), and (17.10.7) are satisfied. If Σ (resp. BB') is a scalar matrix, then conditions (17.10.1) (resp. (17.10.6)) and (17.10.8) are both satisfied. In addition, condition (17.10.8) could hold even for nondiagonal BB' and Σ, provided that either of them is well-conditioned or $\frac{r}{\delta}$ is sufficiently small.

Theorem 17.10.2 *We have the following conclusions:*

1. *Given any of conditions (17.10.1), (17.10.6), (17.10.7), and (17.10.8), there exists at least one $i \in \{1, \cdots, m\}$, such that $|w_i^h(\delta)| < |w_i^p(\delta)|$ for all $\delta \in [\underline{\delta}, \bar{\delta})$.*

2. *If both conditions (17.10.1) and (17.10.6) are satisfied, i.e., the performance measure system is orthogonal, then $|w_i^h(\delta)| < |w_i^p(\delta)|$ for all $\delta \in [\underline{\delta}, \bar{\delta})$ and all i.*

3. *Let τ_i, $i \in \mathcal{L} \equiv \{1, 2, \cdots, l\}$ denote l distinct generalized eigenvalues of BB' relative to Σ, $\mathcal{U}_i \equiv \mathcal{N}(BB' - \tau_i\Sigma)$ be the eigenspace corresponding to τ_i, $\mathcal{U}_i^\perp$ be its orthogonal complement. Suppose that $B\beta \notin \bigcup_{i\in\mathcal{L}} \mathcal{U}_i^\perp$. Then, there exists a positive number $s \in (0,1)$, such that $w^h = sw^p$ if and only if condition (17.10.9) is met.*

PROOF. Again, it is omitted here, and interested readers are referred to Meng and Tian (2013). □

When the agent possesses private information about his own cost, a more efficient type (the agent with higher δ) would acquire information rent by mimicking his less efficient counterpart. To minimize agency costs, optimality requires a downward distortion of the power of the inefficient type's incentive wage. Theorem 17.10.2 gives various conditions ensuring low-powered incentives. If the performance measure sensitivities are orthogonal to each other ($b_i'b_j = 0$ for $i \neq j$), or error terms are uncorrelated ($\sigma_{ij} = 0$ for $i \neq j$), or either BB' or Σ is well-conditioned (k_ν or k_λ is close to one), or the agent is nearly risk-neutral (r is very small), or the agent is highly efficient (δ is very large), then the power of incentives will be lowered for at least one performance measure. For an orthogonal system with all of its performance measures congruent ($b_i'\beta \neq 0$ for all i), the wage vector in the hybrid model is shorter than, but points in the same direction as, its pure moral hazard counterpart if and only if all of the measures share the same signal-to-noise ratio ($b_i' b_i / \sigma_i^2 \equiv k$ for all i).

17.11 Biographies

17.11.1 James Mirrlees

James A. Mirrlees (1936-2018) is the founder of incentive theory and has made a significant contribution to information economics. He received the 1996 Nobel Memorial Prize in Economic Sciences with William Vickrey.

Morris was born in Minnigaff, Scotland, which is also home to Adam Smith, in 1936. He received an M.S. degree in mathematics from the University of Edinburgh in 1957 and a Ph.D. in economics from the University of Cambridge in 1963. Mirrlees took up a teaching fellowship at Trinity College, Cambridge. He taught economics as an Assistant Lecturer and then as a Lecturer up to 1968. He also served as a consultant for the Pakistan Institute of Development Economics, Karachi, during this period. In 1969, he became a professor at the University of Oxford, and was the youngest professor of economics at this university. He taught at Oxford from 1969 to 1996, and served as a Professor of Economics and a Fellow at Nuffield College.

Mirrlees had been active in the economics profession since the 1960s, and he is known for his research on incentive theory. In the 1970s, together with Joseph E. Stiglitz, Alvin E. Roth, Michael Spence, and others, he initiated the study of principal-agent theory. Mirrlees published three papers in 1974, 1975, and 1976, respectively. The "Notes on Welfare Economics, Information and Uncertainty", "The Theory of Moral Hazard and Unobservable Behaviour", and "The Optimal Structure of Incentives and Authority within an Organization" laid the basic model framework of principal-agent theory. The framework pioneered by Mirrlees was further developed by Holmstrom and others, and in the literature, it is called the Mirrlees-Holmstrom approach.

Mirrlees also made remarkable achievements in economic growth and development. He co-edited the book "Models of Economic Growth" (Proceedings of the Conference of the International Economic Association, Jerusalem, 1970) with Nicholas H. Stern and co-authored "Project Appraisal and Planning for Developing Countries" with Ian Malcolm David Little. In 1975, he published the paper "A Pure Theory of Underdeveloped Economies using a Relationship between Consumption and Productivity" which conducted a utilitarian analysis of economic policies. In the area of development economics, Mirrlees proposed a cost-benefit analysis method, established a development model for low-income economies, and studied the effectiveness and consequences of international aid policies.

17.11.2 Joseph E. Stiglitz

Joseph E. Stiglitz (1943—) was born in Gary, Indiana, a small town that is well known for producing steel and gave birth to two contemporary economists, Samuelson and Stiglitz. In 2001, Stiglitz won the Nobel Memorial Prize in Economic Sciences (with George Akerlof and A. Michael Spence) for his fundamental contribution to information economics.

Stiglitz is one of the most cited economists in information economics, and the same is true of broader microeconomics and macroeconomics. Some of his theories have become standard tools for economists and policy-makers. Stiglitz is also one of the most well-known American economics educators. His book “Economics”, which was first published in 1993, has been reprinted repeatedly and translated into many languages. Indeed, it is recognized as one of the most classic textbooks on economics, and is another landmark introduction textbook of economics after Samuelson's “Economics”.

Stiglitz also focuses on the situation in developing countries, and often addresses problems from the perspective of developing countries. He advocated highlighting the role of governments in macroeconomic regulation, and asserted that the best way to achieve sustainable growth and long-term efficiency was to find an appropriate balance between government and the market. It is in this way, he stated, that the world economy could return to a path of more equitable and stable growth, which would benefit everyone. The origins of Stiglitz's ideas may be related to his experience growing up. He came from a hardworking family. His father retired as an insurance agent at the age of 95, and his mother, who retired from the position of an elementary school teacher at age 67, then got another job teaching until she was 84-years-old. When Stiglitz was in college, he was interested in social activities. In 1963, the civil rights movement in the United States was occurring. Stiglitz took part in a march led by Dr. Martin Luther King in Washington, D.C., which culminated in Dr. King's historic speech “I Have a Dream”. Social activities, such as these, had a profound influence on his efforts to promote fair and equitable market ideas after he became well known.

Professor Stiglitz is the author of hundreds of academic papers and other works, including “Economics of the Public Sector” (Norton Corp), “Readings in the Modern Theory of Economic Growth” edited with H. Uzawa (1969), “Lectures on Public Economics” written with Anthony Barnes Atkinson (1980), and “Theory of Commodity Price Stabilization: Study in the Economics of Risk” written with D. M. G. Newbery (1981). In 1987, he also founded the Journal of Economic Perspectives. Stiglitz's economics works are extensive, but they consistently focus on the role of incomplete information in the process of competition. In several pioneering papers, he showed that the common assumption that an economic agent has complete information about alternative market opportunities is not as innocuous as it may seem. These papers are summarized in the paper "Information and Competitive Price Systems" , written with Sanford J. Grossman (1976).

17.12 Exercises

Exercise 17.1 (Debit and Credit under Moral Hazard) A cashless entrepreneur wants to borrow money to run an investment project. With an investment of 1 unit, if the entrepreneur puts in an effort of $\overline{e} > 0$, he will get q units with probability $\overline{p}$; if the entrepreneur does not put any effort, he will get q with probability $\underline{p} > 0$ ($\overline{p} > \underline{p}$). Let ψ denote the entrepreneur's cost of effort $\overline{e}$. In addition, normalize the reservation utility of entrepreneur to 0, and assume $\underline{p}q < r$. For a monopolistic bank, the cost of capital is r. When the project is completed, each unit of the loan is repaid $z - x$.

1. Describe the principal-agent issue faced by the bank.
2. Find out the optimal loan contract in which the bank maximizes it-s expected profit under the entrepreneurial's incentive-compatibility constraint and participation constraint.

Exercise 17.2 (Risk-averse Principal and Moral Hazard) Suppose that the risk-averse principal delegates the task to a risk-neutral agent. The agent's effort is denoted as e, and gets $\overline{q}$ with probability e and $\underline{q}$ with probability $1 - e$, where $\underline{q} < \overline{q}$. The risk-averse principal's utility function is $v(q - t)$, where t is the transfer to the agent, and $v(\cdot)$ is a CARA VNM utility function. The cost of the agent's effort is $\psi(e)$ with $\psi' > 0$ and $\psi'' > 0$.

1. Suppose that e is not observable. Find the optimal contract.
2. With a limited liability constraint, find the second-best effort level.
3. Analyze two extreme cases: (a) the principal is infinitely risk-averse; (b) the principal is risk-neutral. Explain your answers.

Exercise 17.3 (More than Two Levels of Effort) Consider the moral hazard model in this chapter, and extend the model by allowing more than two levels of effort. Consider the more general case with n levels of production $q_1 < q_2 < \cdots < q_n$ and K levels of effort with $0 = e_0 < e_1 < \cdots < e_{K-1}$. We still make the normalization $\psi_0 = 0$, and assume that ψ_k is increasing in k, for $k = 0, 1, \cdots, K-1$. Let π_{ik} denote the probability of producing q_i when the effort level is e_k. Assume that the agent is risk-averse. Other conditions are the same as those in the standard moral hazard model.

1. Write down the participation and incentive-compatibility constraints for this problem.

2. Solve the optimization problem under complete information.

3. Solve the problem under moral hazard.

Exercise 17.4 (Continuum of Effort Levels) Extend the moral hazard model by allowing for a continuum of effort levels. We re-parameterize the model by assuming that $\pi(e) = e$ for all $e \in [0, 1]$. The disutility of effort function $\psi(e)$ is increasing and convex in e with $\psi(0) = 0$. In addition, to ensure interior solutions, we assume that the Inada condition holds $\psi'(0) = +\infty$, and also $\lim_{e\to 0} \psi'(e)e = 0$. Other conditions are the same as those in the moral hazard model in the textbook. Let us finally consider a risk-neutral agent with zero initial wealth who is protected by the limited liability constraint.

1. Write down the participation constraint and incentive-compatibility constraint for this problem.

2. Solve the optimization problem under complete information.

3. Solve the problem under moral hazard.

Exercise 17.5 (The First-Order Approach) Consider an economy in which the risk-averse agent may exert a continuous level of effort $e \in [0, \overline{e}]$ and by doing so incurs a disutility $\psi(e)$, which is increasing, convex, and $\psi(0) = 0$. To avoid corner solutions, we will also assume that the Inada condition $\psi'(0) = +\infty$, and also $\lim_{e\to 0} \psi'(e)e = 0$. Other conditions are the same as those in the moral hazard model in the textbook. The agent's performance is $q \in [\underline{q}, \overline{q}]$ with the conditional distribution $F(q|e)$ and the density function $f(q|e)$. We assume that $F(\cdot|e)$ is twice differentiable with respect to e. The principal's payment to the agent is $t(q)$.

1. Write down the participation constraint and incentive-compatibility constraint for this problem.

2. Prove that when the monotone likelihood ratio property (MLRP) holds, $t(q)$ is increasing in q.

3. Prove that when the convexity of the distribution function condition (CDFC) holds, which means that $F_{ee}(q|e) > 0$, the principal's valuation function $U(e)$ is concave in e.

4. Prove that when MLRP and CDFC both hold, the solution to the optimal contract can be obtained from the first-order conditions.

5. Verify that the moral hazard model with continuous actions in the textbook satisfies MLRP and CDFC.

6. What are your intuitive explanations for MLRP and CDFC?

Exercise 17.6 Consider a loan relationship with moral hazard. The risk-neutral borrower wants to borrow I of funds from the lender to support a risk-free project with a return of V. The project has damage to a third party with probability $1 - e$. The borrower's effort e for reducing the damage costs $\psi(e)$, and h is a compensation to the third party for the damage. A loan contract is $\{\underline{t}, \overline{t}\}$, which $\underline{t}$ ($\overline{t}$) is the repayment to the bank when the borrower generates (does not generate) environmental damages.

1. Suppose that e is observable. Find the first-best level of effort e for reducing the damage.

2. Suppose that e is unobservable, the bank is competitive, and the borrower has sufficient repayment ability. Prove that if the bank can compensate the third party h in the accident, the first-best outcome can still be implementable.

3. Suppose that in the accident, the bank must compensate the third party $c < h$. Let w represent the borrower's initial asset. Prove that when w gradually gets smaller, the first-best outcome can no longer be implemented.

4. Find the second-best effort level to maximize the borrower's expected return under the conditions of the bank's zero-profit constraint, and the borrower's incentive-compatibility and limited liability constraints.

5. Prove that raising the bank's repayment obligation c will reduce the expected welfare level.

6. Prove that when the banking industry is a monopoly industry, this result is no longer valid.

Exercise 17.7 (Risk-averse Agent with Hidden Actions) Consider the principal-agent problem of hidden actions. Suppose that $h(u) = u + \frac{ru^2}{2}$, where $r > 0$, $u > -\frac{1}{r}$. Equivalently, $u(x) = \frac{-1+\sqrt{1+2x}}{r}$, where $x > -\frac{1}{2r}$.

1. Find the second-best transfers required by the principal to induce the agent to exert a high level of effort.

2. Find the second-best cost required by the principal to induce the agent to exert a high level of effort.

3. Find the agent's optimal utilities, $\overline{u}^{SB}$ and $\underline{u}^{SB}$, of inducing a high effort for the principal's incentive-compatible optimization problem.

4. Find the optimal agency cost AC that is defined as the difference between the principal's first-best and second-best expected profits.

Exercise 17.8 A risk-averse individual has a utility function of $u(\cdot)$ and an initial wealth of w_0. The risk that he faces is the potential loss of x for an accident. In a competitive insurance market, a risk-neutral insurance firm can provide the individual with a net payout of $R(x)$ (excluding insurance fees). Assume that the distribution of x is dependent on the degree of effort e to prevent accidents and is not continuous at $x = 0$, where $f(0, e) = 1-p(e)$. When $x > 0$, $f(x, e) = p(e)g(x)$. Assume that $p''(e) > 0 > p'(e)$, and the individual's effort cost function $\psi(e)$ is an increasing convex function and is separable from the utility function. Find the first-best and second-best insurance contracts.

Exercise 17.9 (Insurance Contract) Consider the moral hazard problem in an insurance market. The consumer's VNM utility function is $u(w) = \sqrt{w}$, and the initial wealth is $w_0 = 500$. Assume that there are two possible loss levels: $l = 0$ and $l = 200$. There are also two levels of consumer's effort: $e = 0$ and $e = 1$. The cost of effort is $\psi(e)$, where $\psi(0) = 0$ and $\psi(1) = 1/3$. The corresponding probability distribution of wealth loss and effort is as follows:

	$l = 0$	$l = 200$
$e = 0$	$1/4$	$3/4$
$e = 1$	$3/4$	$1/4$

1. Prove that the above probability distribution satisfies the monotone likelihood ratio property.

2. Suppose that there is only one insurance company, and the consumer can only choose the way of self-insurance. Find the consumer's level of reservation utility.

3. If there is no insurance company, what is the level of the consumer's effort?

4. Prove that if the information is symmetric, then it is optimal for the insurance company to let the consumer choose a high level of effort.

5. If the information is asymmetric, prove that the result in the previous question will make the consumer choose a low level of effort.

6. Find out the solution to the optimal contract under asymmetric information.

Exercise 17.10 (Information Learning) Consider the following principal-agent problem: A risk-neutral principal confronts an unfunded risk-neutral agent protected by limited liability. Assume that there is a risk-free project, and that the principal can get return 0 with probability of 1. There is also a risky project, in the absence of information, and the principal gets return

$\overline{S}$ with probability of ν and return $\underline{S}$ with probability of $1-\nu$. Assume that $\nu\overline{S}+(1-\nu)\underline{S}=0$. The principal can obtain information on the quality of the risky project and make a decision on whether to invest in the risky project. By paying the cost of ψ, the agent can get a signal $\sigma \in \{\underline{\sigma}, \overline{\sigma}\}$. This signal can provide useful information for future return on the risky project. Assume that $\Pr(\overline{\sigma}|\overline{S}) = \Pr(\underline{\sigma}|\underline{S}) = \theta$, where $\theta \in [\frac{1}{2}, 1]$ is the degree of accuracy of the signal.

1. As the benchmark, suppose that the principal uses an information gathering technology by herself. Prove that this project will only be implemented when the principal observes $\overline{\sigma}$. Write down the optimal information learning conditions.

2. Suppose now that the agent decides whether to implement the risky project, and the principal adopts a contract $\{\bar{t}, \underline{t}, t_0\}$ to motivate the agent. $\bar{t}$($\underline{t}$) is the transfer received by the agent when she chooses to implement the risky project with $\overline{S}$($\underline{S}$) being realized. t_0 is the transfer received when the agent selects the risk-free project. Write down the incentive constraint that guarantees that the risky project is implemented only when $\overline{\sigma}$ is observed.

3. Write down the incentive-compatibility constraint that encourages the agent to learn information.

4. Find the contract for the agent and the $\bar{t}$ that prompted the agent to learn the information.

5. Find the second-best contract followed by the principal.

Exercise 17.11 Consider a principal-agent problem with three exogenous states of nature: $\theta_1, \theta_2, \theta_3$, and two effort levels: e_H, e_L. The level of output and the corresponding probability are as follows:

nature state	θ_1	θ_2	θ_3
probability	0.4	0.3	0.3
output of e_H	20	20	1
output of e_L	20	1	1

The principal is risk-neutral, while the agent has the utility function $\sqrt{w} - \psi(e)$. The cost of effort is normalized to 0 for e_L and 1 for e_H. The agent's reservation expected utility is 1.

1. Derive the first-best contract.

2. Derive the second-best contract when only output levels are observable.

3. Suppose that the principal can buy for a price of 1 an information system that allows the parties to verify whether or not the state of nature θ_3 occurred. Will the principal purchase this information system?

Exercise 17.12 Consider the following moral hazard model. The agent is risk-neutral, and his preference is defined in terms of the mean and variance of his income w and his own effort e. The agent's expected utility is $E(w) - \phi\text{Var}(w) - g(e)$, where $g'(0) = 0$, $g'(e), g''(e), g'''(e) > 0$ for all $e > 0$. The profit, denoted by π, generated by effort e is normally distributed with a mean of e and a variance of σ^2.

1. Consider the linear compensation scheme $w(\pi) = \alpha + \beta\pi$. Prove that given $w(\pi)$, e and σ^2, the agent's expected utility is $\alpha + \beta E - \phi\beta^2\sigma^2 - g(e)$.

2. Derive the first-best contract when e is observable.

3. Derive the optimal linear compensation scheme when e is unobservable and find the effects induced by the changes of β and σ^2, respectively.

Exercise 17.13 A principal needs to hire an agent. The work that the agent receives may be good (G) or bad (B). The principal does not know the type of work and only knows that the probability of G type is $p \in (0, 1)$. The agent knows the type of work and can choose to exert either high effort H or low effort L after being hired. Performance of work G with high effort is 4 and is 2 with low effort. Performance of work B with high effort is 2 and is 0 with low effort. The agent's high effort will cost 1 unit, and low effort has no cost. The agent's reservation utility is 0. The principal observes the performance and then gives the agent a payment.

1. Find the optimal contract.

2. How does the conclusion relate to p? Please explain.

Exercise 17.14 Consider the following moral hazard model. The agent is risk-averse, his utility function is $u(w) = \sqrt{w}$, and the reservation utility is 0. The agent's effort has three levels: $E = \{e_1, e_2, e_3\}$. There are two possible profit outcomes: $\pi_H = 10$ and $\pi_L = 0$. The probability distribution is $f(\pi_H|e_1) = 2/3$, $f(\pi_H|e_2) = 1/2$, and $f(\pi_H|e_3) = 1/3$, respectively. The agent's effort cost function satisfies $g(e_1) = 5/3$, $g(e_2) = 8/5$, and $g(e_3) = 4/3$, respectively.

1. Derive the first-best contract when levels of effort are observable.

2. Prove that if the effort levels are not observable, then e_2 will not be implementable. Find the value of $g(e_2)$, such that e_2 is implementable.

3. When the levels of effort are not observable, find the optimal contract.

Exercise 17.15 Continue with the previous question. Assume that the relationship between the principal and the agent continues for two periods. The type of work is established before the contract is signed and remains unchanged in both periods. The principal provides the agent with a contract for each period. The order is: the agent first knows the type of work, then the principal provides the contract of period 1. The agent chooses the effort level, and the performance and payment are realized. The principal provides the contract of period 2, the agent chooses the level of effort, and performance and payment are realized. Assume that, at the first period, the principal always wants to make the agent exert high effort when work is B. In period 2, the principal wants both types of the agent to exert high effort.

1. Write down the optimal contract.

2. Suppose that at the beginning of these two periods, the principal can make a commitment. For the two cases above, discuss how this commitment ability affects the principal's return. Under what conditions will the promise-keeping be optimal?

3. Some organizations regularly perform job rotations for their employees, and this practice is frequently criticized as sacrificing human capital at specific positions. According to the answers of previous questions, explain why job rotation is a good idea.

Exercise 17.16 (Debt Financing) An entrepreneur has two projects, each of which necessitates an investment of 6 at $t = 0$. The cash flow generated by the first project is $C_1 \in \{10, 80\}$ at $t = 1$. The second project generates a cash flow of $C_2 \in \{0, 90\}$. The probability of obtaining high cash flow in both cases is v, and e is the entrepreneurial effort level. The entrepreneur's effort cost is $50e^2$. He can choose three levels of effort: $e \in \{0, 1, 2\}$. The entrepreneur does not have any assets. All participants are risk-neutral, and there is no discounting.

1. If the entrepreneur has the ability to invest by himself, what level of effort will he choose for each of the two projects?

2. Suppose that the entrepreneur has financing constraints, and all project funds need to be financed through debt. In both projects, what face value of debt, denoted by D, will he choose? If the entrepreneur can get an unconditional loan, which project will he eventually choose?

3. Can the entrepreneur replace debt financing with issuing equities with a share ratio of s to improve his situation?

Exercise 17.17 (Insurance Contract) Consider a risk-averse consumer whose utility function is $u(\cdot)$. The consumer's initial wealth is W, and he faces the risk of losing L with probability θ, where $W > L > 0$. The insurance company's contract is $\{c_1, c_2\}$, where c_1 is the amount of wealth that the consumer has in the absence of a loss, and c_2 is the amount of wealth in the event of a loss. In the event that the loss does not occur, the consumer pays the insurance company a premium of $W - c_1$ and in the event of a loss, the compensation that he receives from the insurance company is $c_2 - (W - L)$.

1. Suppose that the insurance company is a risk-neutral monopolist. Find the optimal contract when the consumer's probability of loss θ is observable.

2. Suppose that the insurance company cannot observe θ that may take values from $\{\theta_H, \theta_L\}$, and $p(\theta_L) = \lambda$. Derive the insurance company's optimal contract. Does the amount of insurance purchase have the rationing property?

Exercise 17.18 (Political Economics of Regulation) Consider an enterprise that implements two projects with values of S_1 and S_2, respectively. The enterprise can exert effort e to reduce production costs. For project i, the enterprise's production cost is $C_i = \beta - e_i$. $\beta \in \left\{\underline{\beta}, \overline{\beta}\right\}$, $\nu = \Pr(\beta = \underline{\beta})$. Efforts of reducing production cost will bring the enterprise a cost $\psi(e_1, e_2) = \frac{1}{2}(e_1^2 + e_2^2) + \gamma e_1 e_2$ with $\gamma > 0$. The regulator compensates the enterprise t with observable cost C_1 and C_2. The utility of the enterprise is $U = t - \psi(e_1, e_2)$, and the social welfare is $S_1 + S_2 - (1+\lambda)(t + C_1 + C_2) + U$.

1. What is the optimal mechanism under complete information?

2. What is the optimal regulatory mechanism when β is the private information of the enterprise?

3. Suppose that the regulatory mechanism is determined by a majority vote. There are two types of individuals: shareholders and non-shareholders of the regulated enterprise. Let α denote the proportion of shareholders. If $\alpha > 1/2$, the goal of regulation is to maximize shareholders' objective function $\alpha(S_1 + S_2 - (1+\lambda)(t_1 + C_1 + t_2 + C_2)) + U$. If $\alpha < 1/2$, the goal of regulation is to maximize non-shareholders' objective function $(1-\alpha)[S_1 + S_2 - (1+\lambda)(t_1 + C_1 + t_2 + C_2)]$. Derive the optimal regulatory mechanism under incomplete information in these two cases.

Exercise 17.19 (Bolton and Dewatripont, 2005) Consider the following principal-agent problem. There is a project whose probability of success is a (a is also the effort made by the risk-neutral agent, at cost a^2). In the case of success, the return is R, and in the case of failure, the return is 0. The parameter

R can take two values, X with probability λ and 1 with probability $1-\lambda$. To undertake the project, the agent needs to borrow an amount I from the principal. The sequence of events is as follows:

- First, the principal offers the agent a debt contract, with face value D_0 (i.e., with a limited liability D_0). The agent accepts or rejects this contract.
- Second, nature determines the value R that would occur in the case of success. This value is observed by both the principal and the agent. The principal can then choose to lower the debt from D_0 to D_1.
- The agent chooses a level of effort a. This level is not observed by the principal.
- The project succeeds or does not succeed. If the project succeeds, the agent pays the minimum of R, and the face value of debt D_1.

Answer the following questions:

1. Find the subgame-perfect equilibrium of this game as a function of I, λ, and X.
2. When do we have $D_1 < D_0$? Discuss.

Exercise 17.20 (Regulation of a Risk-averse Enterprise) Consider a regulator who wants to implement a public project that is worth S. An enterprise can put in a cost of $C = \beta - e$ to implement this project, where $\beta \in \left\{\underline{\beta}, \overline{\beta}\right\}$ is a parameter of efficiency. e is the effort level that brings a negative effect of $\psi(e)$ ($\psi' > 0$, $\psi'' > 0$,$\psi''' > 0$) to the enterprise. The cost C can be observed by the regulator, and the regulator gives the enterprise transfer t with the price of $1+\lambda$. The enterprise is risk-averse, and the utility function is $u(t-\psi(e))$ ($u' > 0$, $u'' < 0$). The regulator cannot observe e or β, but $\nu = \Pr(\beta = \underline{\beta})$ is common knowledge.

1. Consider the revelation mechanism $\left\{t(\bar{t}) = \bar{t}, C(\overline{\beta}) = \overline{C}; t(\underline{\beta}) = \underline{t}, C(\underline{\beta}) = \underline{C}\right\}$. Write down the enterprise's incentive-compatibility and participation constraints.
2. The expected social welfare is defined as:

   $$W = S - (1+\lambda)[\nu(\underline{t}+\underline{C}) + (1-\nu)(\bar{t}+\overline{C})] + u^{-1}(\nu u(\underline{\pi}) + (1-\nu)u(\overline{\pi})),$$

   where $\underline{\pi} = \underline{t} - \psi(\underline{\beta} - \underline{C})$ and $\overline{\pi} = \bar{t} - \psi(\overline{\beta} - \overline{C})$. Explain the social welfare function. Suppose that it is concave for $\underline{\pi}$ and $\overline{\pi}$. Find out the optimal regulatory mechanism for the principal in the interim stage.

3. Compare the conclusion in the previous question with the case with a risk-neutral enterprise.

4. Consider a special case $v(x) = \frac{1}{\rho}(1 - e^{-\rho x})$. Prove that the effort level of type $\overline{\beta}$ increases with ρ.

Exercise 17.21 (Information Collection before Signing the Contract) Consider a principal-agent problem: An agent produces q units of output of a good at a cost of θq, where $\theta \in \left\{\underline{\theta}, \overline{\theta}\right\}$ and $\overline{\theta} > \underline{\theta}$. Let t denote the transfer to the agent. The agent's utility is $U = t - \theta q$. The principal's utility is $V = S(q) - t$, $S' > 0$, $S'' < 0$.

- On the first day, the principal proposes a menu of contracts $(\underline{t}, \underline{q}), (\overline{t}, \overline{q})$.
- On the second day, the agent decides whether to learn θ at a cost of ψ. Let e denote this decision: if the agent learns, then $e = 1$; if not, then $e = 0$. e is a moral risk variable that the principal cannot observe.
- On the third day, the agent decides whether to accept the contract.
- On the fourth day, the agent knows θ (if he decides not to study on the second day).
- On the fifth day, the contract is implemented.

Consider two contract sets: the C_1 type induces the agent to choose $e = 1$, and the C_2 type induces the agent to choose $e = 0$.

1. Write down the optimization problem for the principal to maximize her expected utility subject to the agent's incentive-compatibility constraints and budget constraints, irrespective of whether it is a contract of C_1 or C_2 type.

2. Prove that the principal can restrict the contract to $\underline{t} - \underline{\theta}\underline{q} \neq 0$ and $\overline{t} - \overline{\theta}\overline{q} \neq 0$ to achieve a utility lower bound. Prove that a meaningful contract requires $\overline{t} - \overline{\theta}\overline{q} \leqq 0$, and that the principal can always imitate a contract in C_2 with a contract in C_1.

3. Find the optimal contract in C_1. (To discuss the scope of ψ, distinguish three cases: (1) whether the ex-ante participation constraint is binding; (2) whether the incentive-compatibility constraint is binding; and (3) whether both of the constraints are binding.).

Exercise 17.22 A risk-neutral principal employs a risk-averse agent. For the principal, the agent's effort level a is not observable, and the principal's profit level has n possible outcomes which are $q_1 < q_2 < \cdots < q_n$. These n outcomes are both observable and verifiable. The probability of realization of profit q_i is $\pi_i(a)$, and the agent's corresponding income is $I = (I_1, I_2, \cdots, I_n)$. The agent's reservation utility is $\overline{u}$, and the agent's utility function is $u(I, a) = -e^{-r(I-a)}$.

1. Prove that the first-best effort level is independent of the reservation utility $\overline{u}$.

2. Prove that the second-best effort level is independent of the reservation utility $\overline{u}$.

3. Prove that the optimal income scheme of the agent can be expressed as $(I_1 + k, I_2 + k, \cdots, I_n + k)$, where k is only a function of the reservation utility $\overline{u}$.

4. Suppose that the agent's utility function is $V(I) - a$ with $V' > 0$ and $V'' < 0$. Are the results of the previous three questions still correct? Explain your answer.

5. Suppose that the agent's utility function is $V(I) - a$ with $V' > 0$, $V'' < 0$ and $n = 2$. There are only two levels of effort, which are work-hard, a_H, and lazy, a_L. In the second-best outcome, the principal wants the agent to exert the effort level a_H. Prove that the optimal incentive scheme makes the agent be indifferent between working hard and being lazy.

6. Continue with the previous question and prove that the optimal incentive scheme satisfies $I_2 > I_1,\ q_2 - q_1 > I_2 - I_1$.

Exercise 17.23 In a two-period problem, $t = 1, 2$, the agent chooses the effort level $e_t = H$ or $e_t = L$, and the profit is $q_t = 1$ or $q_t = 0$. The harder the agent works, the greater is the probability of achieving a high profit. In particular, $\Pr(q_t = 1|e_t = H) = p_H$, $\Pr(q_t = 1|e_t = L) = p_L < p_H$. The agent's effort cost is $c(e_t)$, where $c = c(H) > c(L) = 0$. The agent's utility function is $u(s, e_1, e_2) = -e^{-r(s-c(e_1)-c(e_2))}$, where s is the wage that the principal pays to the agent. The agent's reservation utility is a certainty equivalence level of zero, and the agent can observe q_1 prior to determining the action e_2. The principal is risk-neutral, and her profit is $q_1 + q_2 - s$. The wage of s depends on the profit of each period and is paid at the end of the second period.

1. Suppose that the principal wants the agent to work hard in two periods. Prove that the contract that achieves the minimum cost of (H, H) has the following form: $s(q_1, q_2) = \alpha(q_1 + q_2) + \beta$.

2. If the agent cannot observe q_1 before determining the action e_2, is the contract given in question 1 still optimal? Explain your answer.

Exercise 17.24 A monopolist sells a product at the price of p, and the only production input is labor. In order to produce q units of product, workers must work hard, and the cost of the effort is $\frac{q^2}{2}\theta$. The number of workers is standardized to 1, the skill level of the workers is represented by θ, and the principal cannot observe θ. Assume that the workers with proportion λ are highly skilled (i.e., $\theta = \theta_H$), and the rest are low-skilled (i.e., $\theta = \theta_L < \theta_H$). The principal can hire any number of workers. The reservation utility of each worker is 0. The principal's profit is the product revenue minus the wages paid to the workers.

1. Solve for the optimal contract.

2. Suppose that the output q is observable and verifiable, and that only the wage w can be written into the contract (the transfer for each worker i is the same, and $T_i = w_i q_i$). Solve for the optimal solution of w. Is w the same for high-skilled and low-skilled workers? Does the optimal w depend on the proportion of highly skilled workers?

3. Suppose that any contract (q_i, T_i) is possible. Find the optimal contract and compare it with the result of the previous question. You may assume that the proportion of high-skilled workers is quite low, and it is not optimal for firms to employ highly skilled workers only.

Exercise 17.25 Assume that the principal is risk-neutral. The agent chooses the effort levels e_H and e_L to achieve three possible outputs of $q_H > q_M > q_L > 0$. If the agent chooses e_H, the utility level is $2\sqrt{x} - 5$. If e_L is selected, the utility level is $2\sqrt{x}$. If the agent does not work, the utility level is 3. If the agent chooses to work, his minimum wage is w, $0 \leqq w \leqq 5$. The principal cannot observe the agent's effort level, but she can observe the realized output, $P(q = q_H|e_H) = \frac{1}{2}$, $P(q = q_M|e_H) = \frac{1}{2}$, $P(q = q_L|e_H) = 0$, $P(q = q_H|e_L) = \frac{1}{4}$, $P(q = q_M|e_L) = 0$, $P(q = q_L|e_L) = \frac{3}{4}$. Assume that $q_H - q_M$ and $q_M - q_L$ are sufficiently large. Derive the optimal contract for the principal.

Exercise 17.26 Define the monotone likelihood ratio and first-order stochastic dominance.

1. Prove that the above two concepts are equivalent under the case of two outcomes.

2. Give an example to show that, in the general case, the above two concepts are different. Prove that if the monotone likelihood ratio is satisfied, the first-order stochastic dominance is also satisfied.

Exercise 17.27 Consider a moral hazard model. The principal is risk-neutral, and the agent is risk-averse. The agent can choose two levels of effort, e_H

and e_L. The cost of high effort is c, and the cost of low effort is 0. There are two types of outcomes which are x_H and x_L, $p(x_L|a_L) > p(x_L|a_H)$. The agent's utility function is $u(w, c_i) = \ln w - c_i$, and the agent's reservation utility is 0.

1. Describe the principal-agent problem of implementing a high level of effort.
2. Solve for the optimal wage contract.
3. Find the second-best cost C^{SB} of inducing high effort e_H.

Exercise 17.28 (Supervision Cost) Consider a financial contract with two parties. One is a risk-neutral entrepreneur with wealth constraints, and the other is a wealthy risk-neutral investor. The investment I is required at $t = 0$. The project generates a random gain of $\pi(\theta, I) = 2\min\{\theta, I\}$ when $t = 1$, where θ is a nature state and obeys a uniform distribution on $[0, 1]$.

1. Describe the characteristics of the first-best investment level I^{FB}.
2. Suppose that at $t = 1$, the realized return can only be observed by the entrepreneur, and the investor must pay a cost of $K > 0$ to observe $\pi(\theta, I)$. Consider that the return cannot exceed the difference of the return minus the cost of supervision, and the investor's expected profit is 0. Derive the optimal contract.
3. Prove that the second-best investment level is lower than the first-best investment level I^{FB}.

Exercise 17.29 (Forged Output) Consider a risk-averse entrepreneur whose utility function is $u(\cdot)$, and the entrepreneur's output q is a random variable distributed over the interval $[0, \overline{q}]$. The entrepreneur wants to diversify the risk by signing a contract with a risk-neutral financier whose initial wealth is $w \geqq \overline{q}$. The contract provides that the payment to the entrepreneur depends on the level of output. The entrepreneur can observe the output. The financier can also observe the output, unless the entrepreneur falsifies the accounts. After observing the output q, the entrepreneur may falsify an output report R, which costs $\psi(q, R) = \frac{1}{4}(q - R) + \frac{1}{2}c(q - R)^2, c > 0$. Assume that the entrepreneur is protected by limited liability, and his reservation utility is higher than $\overline{q}/2$.

1. Characterize the optimal contract.
2. If there is no output falsification in the contract, then for all $q \in [0, \overline{q}]$, we have $R(q) = q$. Prove that the optimal contract involves falsification. According to the optimal contract, find the equilibrium solution of the entrepreneur's falsified output.

3. What is the optimal no-falsification contract? Prove that the optimal no-falsification contract is linear in the output q.

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Chapter 18

Mechanism Design with Multiple Agents: Complete Information

18.1 Introduction

In the previous two chapters, we discussed the basic model and core contents of principal-agent theory. The contracts under investigation are complete contracts that specify what is to be done in every possible contingency, and in which the principal maximizes her revenue (the issue of optimality). These contracts highlight the trade-off between allocative efficiency and information rent extraction under (exogenous) hidden information of the agents and the trade-off between efficiency and insurance (or between incentive and risk) under (endogenous) hidden actions. Since they essentially involve only one agent, the design of the principal's optimal contract can be reduced to a constrained optimization problem without having to appeal to sophisticated game theoretical settings.

In the remaining three chapters of this part, we will consider mechanism design with strategic interactions among multiple agents, in which the designer wants to maximize the aggregate welfare of all agents (the issue of efficiency) or implement a general social goal, rather than maximizing her own revenue (the issue of optimality).

In the current chapter, we discuss mechanism design under complete information in the sense that all agents know the characteristics of each other, although the designer does not know the characteristics of the agents. In such environments, asymmetric information may not only affect the relationship between the principal (often called the designer) and agents, but it may also affect the relationships among agents. To describe the strategic interactions among agents, game theoretical reasoning is thus used to model institutions as varied voting systems, auctions, bargaining protocols, and methods for deciding on public projects.

Mechanism design theory comprises implementation theory and realization theory. The former studies how the designer can find a mechanism (game form) such that individual interests are compatible with the designer's goal. The latter mainly focuses on informational efficiency of a mechanism. One of the main indicators is that the size of the message space, which represents the operation cost of the mechanism. This chapter first discusses implementation theory, and then briefly discusses realization theory.

Incentive problems arise when a social planner cannot distinguish between things that are indeed different. A typical problem is the so-called free-rider issue. A free rider can improve his well-being by not telling the truth about his own unobservable characteristic(s). Like the principal-agent model, a basic insight of the incentive mechanism with more than one agent is that incentive constraints should be considered coequally with resource constraints. One of the insights of mechanism design theory is that the free-rider problem may or may not occur, depending on the game form (mechanism) and the solution concept used.

Examples of incentive mechanism design that take strategic interactions among agents have long existed. An early example is the Biblical story of the judgement of Solomon for determining who is the real mother of a baby. Two women came before the King, disputing who was the mother of a child. The King's solution was to use a method of threatening to cut the lively baby in two and give half to each woman. One woman was willing to give up the child, but the other woman agreed to have the baby cut in two. The King then made his decision: the baby is to be given to the first woman, who is the true mother. Another example of incentive mechanism design is how to cut a pie and divide it equally among all participants.

The first major development of incentive mechanism design occurred in the 1970s. When information is private, an appealing solution concept is dominant strategy equilibrium. The fundamental conclusion of Gibbard-Hurwicz-Satterthwaite's Impossibility Theorem is that we should have a trade-off between truth-telling and Pareto efficiency. Of course, if one is willing to give up Pareto efficiency, we can have a truth-telling mechanism, such as the Vickery-Clark-Groves (VCG) mechanism.

On the other hand, we could give up the truth-telling requirement, and instead implement Pareto efficient outcomes. When information about the characteristics of the agents is shared by individuals, but not by the designer, then an appealing equilibrium concept is Nash equilibrium. In this situation, one has to give up truth-telling, and use a general message space instead. Indeed, one may design a mechanism that Nash implements Pareto efficient allocations.

In the next chapter, we will explore the case of incomplete information, in which agents do not know each other's characteristics, and thus we consider Bayesian incentive-compatible mechanisms.

18.2 Basic Settings

The basic analytical framework that is used to study mechanism design consists of the following five components: (1) economic environments; (2) a social goal to be reached; (3) an economic mechanism that specifies the rules of the game; (4) a description of solution concept on individuals' self-interested behavior; and (5) implementation of the social goal.

18.2.1 Economic Environments

Suppose that there is a designer (also known principal or a social planner) and multiple agents (also known as participants, individuals, or players). The designer determines the rules of the game, while the agents provide information and participate in games. The designer may be a concrete or an abstract entity, such as Congress, or may even be a consensus in rules or conventions that are commonly accepted by all of the participants in advance. The designer does not observe the real state of the world composed of participants' economic characteristics, and such information is dispersed among agents. Let

- $N = \{1, \cdots, n\}$: the set of agents.
- $e_i = (Z_i, w_i, \succcurlyeq_i, Y_i)$: economic characteristic of agent i, which consists of the space of outcomes/alternatives, initial endowment if any, preference relation, and production set if agent i is also a producer.
- $\mathbf{e} = (e_1, \ldots, e_n)$: an economy, also known as an economic environment, type, or state.
- E: the set of all prior admissible economic environments.
- $U = U_1 \times \ldots \times U_n$: the set of all admissible utility functions.
- $\Theta = \Theta_1 \times \Theta_2 \cdots \times \Theta_n$: the set of all admissible parameters $\theta = (\theta_1, \cdots, \theta_n) \in \Theta$ that determine the types of parametric utility functions, $u_i(\cdot, \theta)$, and thus it is called the space of types or the state of the world.

Remark 18.2.1 Thus, an economy $\mathbf{e}$ consists of three components: agents, objects, (alternatives or outcomes) and information about the characteristics of economic agents (state of the world). We assume that N and $Z = \prod_{i \in N} Z_i$ are state-independent, and thus the only factors affecting an environment involve $\succcurlyeq_i$ (U_i), Y_i or w_i. Therefore, E is a general expression of economic environments, depending on the situations faced by a designer.

The set of admissible economic environments under consideration may be just given by $E = U$, $E = \Theta$, or by the set of all possible initial endowments or production sets.

The designer does not know the true economic characteristics of individuals, but an individual may know, partially know, or not know the economic characteristics of other individuals. These three classes of timing of information structures about economic characteristics are called, respectively, ex-post, interim, and ex-ante. If each participant knows the economic characteristics of other participants, it is called a **complete information** environment;[1] otherwise, it is called an **incomplete information** environment. If e_i can only be observed by individual i, i.e., each agent only knows his own economic characteristic and does not know the characteristics of others (but knows the distributions, which may be independent or correlated), then it is the interim case.

For simplicity, we assume that preferences are given by parametric utility functions, and each agent i privately observes his type $\theta_i \in \Theta_i$, which determines agent i's preference over outcomes. The state θ is drawn randomly from a prior distribution with density $\varphi(\cdot)$. Each agent maximizes a von Neumann-Morgenstern expected utility over outcomes, given by (Bernoulli) utility function $u_i(y, \theta)$. Thus, we assume that $\varphi(\cdot)$ and $\{u_i(\cdot, \cdot)\}_{i=1}^n$ are common knowledge.

The analytical framework can be divided into two categories: (1) the **private value model**, in which the Bernoulli utility function of each individual depends only on his own value and does not depend on the type of other participants, denoted as $u_i(y, \theta_i)$. At the same time, the private value model can be further divided into the **independent distribution model** of private values and the **correlated distribution model** of private values. (2) the **interdependent value model**, in which the Bernoulli utility function of each individual depends not only on his own type, but also on the type of other participants, denoted as $u_i(y, \theta)$. It is crucial to distinguish these differences, because they often lead to markedly different results.,

This chapter discusses the case of complete information, and the next chapter explores the case of incomplete information.

[1]The term is different from the setting of the principal-agent model, in which the principal is also a participant. As such,we call it incomplete information, provided that the principal (designer) does not know the agents' information. However, in the general mechanism design discussed in this chapter, the designer is generally not a player of the mechanism. As such, although we always assume that the designer does not know the information of participants, as long as agents know information about each other, it is called complete information. It is important to understand these subtle differences in terms.

18.2.2 Social Goal

The second component of mechanism design is the **social goal** that the designer desires to reach, which is also often called the **goal** or the **social optimum**. Given economic environments, the designer wants to reach a certain goal that is considered socially optimal by some criterion.

Let

- $Z = Z_1 \times \ldots \times Z_n$: the outcome space.
- $A \subseteq Z$: the feasible set.
- $F : E \to\to A$: the social choice rule, or called social choice correspondence, in which $F(e)$ is the set of **socially desired or optimal** outcomes under some criteria of social optimality.

Thus, a social choice rule, F, is defined as a mapping from the environment space to the outcome space: If this correspondence is single-valued, it is called a **social choice function** (SCF). If random choice is allowed, Z can be substituted for the probability distribution on it, denoted by $\triangle(Z)$.

Examples of Social Choice Correspondences:

- $P(e)$: the set of Pareto efficient allocations.
- $I(e)$: the set of individually rational allocations.
- $W(e)$: the set of Walrasian allocations.
- $L(e)$: the set of Lindahl allocations.
- $FA(e)$: the set of fair allocations.

These concepts of resource allocations have been discussed in detail in the general equilibrium theory.

Examples of Social Choice Functions:

Solomon's goal.

Majority voting rule.

18.2.3 Economic Mechanism

If a social planner knows the information about economic characteristics, she can directly determine the outcomes according to a given social choice rule. However, since the designer lacks such information, she needs to design an appropriate incentive mechanism (rules of the game) to coordinate the personal interests and the collective interests determined by the social goal so that all individuals have an incentive to choose actions which result in socially optimal outcomes at equilibrium when they pursue their personal interests. To do so, the designer informs the agents about how the information that she collects from individuals will be utilized to determine outcomes, i.e., the designer first specifies the rules of the game, then uses the information or actions of agents and the rules of the game to determine the outcomes of individuals. Consequently, a **mechanism** consists of a **message space** and an **outcome function**. Let

- M_i : the message space of agent i.
- $M = M_1 \times \ldots \times M_n$: the message space in which communications take place.
- $m_i \in M_i$: a generic message reported by agent i.
- $\boldsymbol{m} = (m_1, \ldots, m_n) \in M$: a profile of messages of all agents.
- $h : M \to Z$: outcome function that translates messages into outcomes.
- $\Gamma = \langle M, h \rangle$: a mechanism

Thus, given the rules of the game, each participant is required to send a message $m_i : E \to M_i$ based on what she knows about the economic environment, and then she determines the outcome of each participant according to the information sent by the participant and the outcome function $h : \prod_{i \in I} M_i \to Z$. Assume that the state space E has a Cartesian product structure $E = \prod_{i \in N} E_i$.

If each agent's message reported is affected only by his own economic characteristic E_i (or U_i, or "type" θ_i), the message that she sends is then $m_i : E_i \to M_i$, namely, the case of **privacy-preserving**. The mechanism is called a **direct mechanism** if the participant's message space is just his type space (i.e., $E_i = M_i$).

Remark 18.2.2 Prior to the emergence of mechanism design theory, economics was mostly concerned with taking a mechanism as given, and then investigating under what economic environments it can reach a given social goal. For instance, in general equilibrium theory, we take the market mechanism as given, and then study under what economic environments a competitive market leads to an efficient allocation of resources. However, for a mechanism design issue, given economic environments and a social goal, the designer wants to design an economic mechanism to achieve the social goal through making individuals' interests be compatible with the goal. One of the main purposes of mechanism design theory is to determine what kind of social goals can be implemented and what kind of social goals cannot.

Remark 18.2.3 A mechanism is often referred to as a game form. Mechanism design theory distinguishes itself from game theory in a number of ways: (1) In contrast to game theory, which is a positive approach, mechanism design is a normative approach. Game theory is important because it predicts how a given game will be played by agents. Mechanism design goes one step further: given an economic environment and the constraints faced by the designer, what goal can be realized or implemented by designing a suitable mechanism? What mechanisms are optimal among those that are feasible? (2) The consequence of a message profile is an outcome in mechanism design theory rather than a payoff profile in game theory. Of course, once the preferences of individuals are specified, then a game form or a mechanism induces a game. (3) The preferences of individuals in the mechanism design setting vary in the admissible set E, while they are taken as given in game theory. This distinction is critical. An equilibrium (e.g., dominant strategy equilibrium) in mechanism design is much easier to exist than in a given game. (4) In designing mechanisms, one must take into account incentive constraints such that individual interests are compatible with the social goal.

Remark 18.2.4 The design of incentive mechanisms is different from the information adjustment mechanisms considered in the last section of this chapter. In the design of incentive mechanisms, participants' behavior is not described by a response function or an information feedback process, but rather is determined by participants' preferences and the way in which they adopt strategies. However, as shown in the last section, one way to regard an incentive mechanism as an information adjustment mechanism is to consider all points of the message space of the incentive mechanism as the stationary point of the information adjustment mechanism.

18.2.4 Solution Concepts of Self-Interested Behavior

When sending information to a designer, participants have strategic interactions with each other. The fourth component of mechanism design is then the equilibrium solution concept that deals with incentives of individuals. Note that the use of different equilibrium solution concepts may lead to very different conclusions.

A basic assumption in economics is that individuals are self-interested. Different economic environments and different rules of the game will result in different reactions of the individuals, and thus an agent's strategy (i.e., the message that the agent sends) will depend on his self-interested behavior which, in turn, depends on the economic environments and the mechanism at work.

Let $b(e, \Gamma)$ be the **set of equilibrium strategies** that describes the self-interested behavior of individuals, and it is a subset of the individual strategy space. $b(e, \Gamma)$ is a general notation that can represent different equilibrium solution concepts, such as dominant strategy equilibrium, Nash equilibrium, subgame perfect equilibrium, Bayesian-Nash equilibrium, and perfect Bayesian (Nash) equilibrium. The set of the resulting equilibrium outcomes is denoted by $h(b(e, \Gamma))$.

Thus, given E, M, h, and b, the resulting equilibrium outcome is a composite function of the rules of the game and the equilibrium strategy, i.e., $h(b(e, \Gamma))$.

18.2.5 Implementation and Incentive Compatibility

If the designer knew the information about individuals' economic characteristics, she could directly determine the outcomes according to a social choice rule, which is not the case in most situations. As such, the designer needs to design appropriate rules of the game, namely, incentive-compatible mechanisms, to reconcile potential conflicts between individual interests and collective interests, so that all self-interested individuals have an incentive to choose actions which result in socially optimal outcomes.

Therefore, the fifth component of mechanism design is the implementation of a given social goal so that, under a given solution concept, individual interests have no conflicts with the social interest. Given the mechanism Γ and equilibrium strategy $b(e, \Gamma)$, the implementation of a social choice rule F studies the relationship of $F(e)$ and $h(b(e, \Gamma))$, which can be illustrated by the diagram in Figure 18.1.

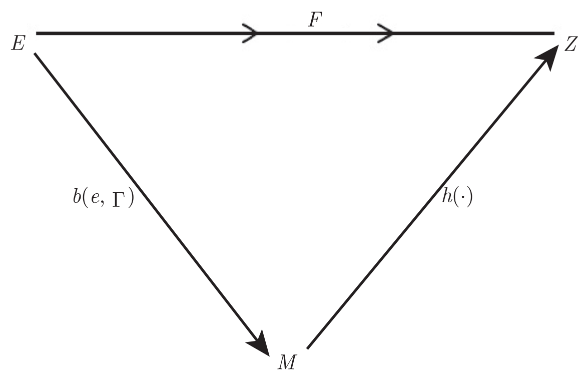

Figure 18.1: Diagrammatic illustration of mechanism design problems

In Figure 18.1, E is the set of economic environments, Z is the set of outcomes, and the social choice correspondence F specifies the social goal. The designer needs to design a mechanism which implements the social goal. Under the mechanism Γ, individuals interact strategically to reach an equilibrium strategy $b(e, \Gamma)$, and then the outcome is given by $h(b(e, \Gamma))$. In general, the mechanism does not always lead to socially desired outcomes.

If $h(b(e,\Gamma))$ is in $F(e)$, then the social goal F is implementable under this mechanism. Formally, we have the following definitions.

Definition 18.2.1 A mechanism $\langle M, h\rangle$ is said to

(i) **fully implement** a social choice correspondence F in equilibrium strategy $b(e,\Gamma)$ on E if for **every** $e \in E$

(a) $b(e,\Gamma) \neq \emptyset$ (equilibrium solution exists),
(b) $h(b(e,\Gamma)) = F(e)$ (individual interests are fully consistent with social goals);

(ii) **implement** a social choice correspondence F in equilibrium strategy $b(e,\Gamma)$ on E if for every $e \in E$

(a) $b(e,\Gamma) \neq \emptyset$,
(b) $h(b(e,\Gamma)) \subseteq F(e)$ (individual rationality is compatible with collective rationality);

(iii) **partially implement** a social choice correspondence F in equilibrium strategy $b(e,\Gamma)$ on E if for every $e \in E$

(a) $b(e,\Gamma) \neq \emptyset$,
(b) $h(b(e,\Gamma)) \cap F(e) \neq \emptyset$ (i.e., there is an equilibrium strategy $m^* \in b(e,\Gamma)$, such that $h(m^*) \in F(e)$).

Partial implementation is a weak implementation, which allows for some socially undesired equilibrium outcomes, i.e., they do not belong to $F(e)$.

Definition 18.2.2 A mechanism $\langle M, h\rangle$ is said to be (**fully or partially) incentive compatible** with a social choice correspondence F in $b(e,\Gamma)$-equilibrium if it (fully or partially) implements F in $b(e,\Gamma)$-equilibrium.

In the literature, “implementation” defined above is sometimes referred to as “strong implementation”, while “partial implementation” is called “implementation” or “weak implementation”.

As shown below, whether or not a social choice correspondence is implementable depends on the assumption regarding the solution concept of self-interested behavior. When information is complete, the solution concept can be dominant equilibrium, Nash equilibrium, strong Nash equilibrium, subgame perfect Nash equilibrium, or undominated equilibrium. For the case of incomplete information, the equilibrium strategy can be Bayesian Nash equilibrium or undominated Bayesian Nash equilibrium.

In specific issues, we want the social choice rules to be implemented to possess certain properties.

Definition 18.2.3 (Ordinality) If for $\forall (a,b) \in Z^2$ and $\forall (e,e') \in E^2$, $a \succcurlyeq_i b \Leftrightarrow a \succcurlyeq_i' b$, then $F(e) = F(e')$.

This property requires that even when economic environments are changed, but the rank between any two alternatives remains unchanged, then the set of alternatives selected in both states must be the same. In other words, for any two economies e and e', if $F(e) \neq F(e')$, then there must be $(a,b) \in Z^2$, such that $a \succcurlyeq_i b$ but $b \succ_i' a$.

Definition 18.2.4 Weak Pareto optimality: for $\forall\ e \in E$ and $a \in Z$, if there is no $b \in Z$, such that $b \succ_i a, \forall i \in N$, then $a \in F(e)$.

This property suggests that if everyone thinks that one alternative is strictly inferior to the other, the former will not be selected.

Definition 18.2.5 Pareto optimality: for $\forall\ e \in E$ and $a \in Z$, if there is no $b \in Z$, such that $b \succcurlyeq_i a, \forall i \in N$ and $b \succ_j a$ for some $j \in N$, then $a \in F(e)$.

Definition 18.2.6 Pareto indifference: for $\forall (a,e) \in Z \times E$ and $b \in F(e)$, if $a \sim_i b, \forall i \in N$, then $a \in F(e)$.

Definition 18.2.7 Dictatorship: There is $i \in I$, such that $F(e) \subseteq C_i(e,Z) \equiv \{a \in Z : a \succcurlyeq_i^e b, \forall b \in Z\}, \forall e \in E$.

If there is an individual whose optimal choice is also socially optimal, then the social choice rule is called a dictatorship, with the individual being the dictator.

Definition 18.2.8 Unanimity: for $\forall (a,e) \in Z \times E$, if $a \succcurlyeq_i^e b$, for $\forall i \in I, \forall b \in Z$, then $a \in F(e)$.

This property shows that if a scheme is considered by everyone to be optimal, then it must be selected as the social optimum.

Definition 18.2.9 Strong unanimity: $\forall (a,e) \in Z \times E$, if $a \succ_i b$, for $\forall i \in N, \forall b \neq a$, then $F(e) = \{a\}$.

This property shows that if a scheme is strictly preferred by everyone, the scheme must be the only one selected, so that the social choice correspondence becomes a social choice function.

Definition 18.2.10 No veto power: $\forall (a,j,e) \in Z \times N \times E$, if $a \succcurlyeq_i b, \forall i \neq j, \forall b \in Z$, then $a \in F(e)$.

The property of no veto power requires that if a scheme is preferred by all but one person, it must be selected. In other words, no one has the power to veto a scheme that has been unanimously agreed upon by others.

Definition 18.2.11 Maskin monotonicity: Let $L_i(a, e) = \{b \in Z : a \succcurlyeq_i b\}$. If for any two economies $e, e' \in E$ and any $a \in F(e)$, $L_i(a, e) \subseteq L_i(a, e')$ $\forall i \in N, \forall b \in Z$, then $a \in F(e')$.

This property indicates that if a scheme was originally selected by society, and the change in environment only led the scheme to be preferred by everyone, then the scheme would be re-selected by society under the new economic environment e'. [2] Maskin monotonicity plays a critical role in mechanism design theory.

18.3 Examples

Prior to discussing basic results in mechanism design theory, we first provide some examples of economic environments and social goals to illustrate the necessity of designing certain mechanisms to solve incentive compatibility problems.

Example 18.3.1 (Walrasian correspondence) Consider a pure exchange economic environment e as follows:

- Individuals: $N = \{1, 2, \cdots, n\}$ represent the set of consumers.
- Goods: There are L private goods, the endowment of consumer i is $\boldsymbol{\omega}_i \in \mathcal{R}^L_+$, X_i represents consumer i's consumption set, $\mathcal{A} = \{(x_1, \cdots, x_n) | \boldsymbol{x}_i \in X_i$, and $\sum_{i \in N} \boldsymbol{x}_i \leqq \sum_{i \in N} \boldsymbol{\omega}_i, \forall i \in N\}$ represents the set of feasible allocations.
- Type (state of the world): $\boldsymbol{\theta} = (\theta_1, \cdots, \theta_n)$, where θ_i is consumer i's preference parameter. $\succcurlyeq_i^{\theta}$ is strictly increasing on his own consumption x_i.

The Walrasian social choice rule is defined by

$$W(\theta) = \left\{ \boldsymbol{x} \in \mathcal{A} \;\middle|\; \begin{array}{l} \exists \boldsymbol{p} \in \mathcal{R}^L_+ \text{ such that } \boldsymbol{p}\boldsymbol{x}_i \leqq \boldsymbol{p}\omega_i \\ \text{and } \boldsymbol{y}_i \succ_i^{\theta} \boldsymbol{x}_i \Rightarrow \boldsymbol{p} \cdot \boldsymbol{y}_i > \boldsymbol{p}\omega_i, \ \forall i \in n \end{array} \right\}. \tag{18.3.1}$$

Example 18.3.2 (Lindahl correspondence) The economic environment e is characterized as follows:

There are n agents, L private goods, and K public goods. $\mathbf{x}_i \in \mathcal{R}^L_+$ is a vector of consumer i's private consumption goods, the endowment of

[2] Take an election as an example, in which someone, such as Barack Obama, was elected president. Four years later, the environment changed, but his position in the minds of every voter rose, and then he would be bound to be re-elected.

consumer i is $\boldsymbol{\omega}_i \in \mathcal{R}^L_+$. $\boldsymbol{y} \in \mathcal{R}^K_+$ represents the quantity of public goods. Production technology is represented by production set $\mathcal{F}$.[3]

The utility function of consumer i is $v_i(\boldsymbol{x}_i, \boldsymbol{y}, \theta_i)$ (and the corresponding preference is $\succcurlyeq_i^\theta$). Then, the Lindahl correspondence is

$$L(\theta) = \left\{ (\boldsymbol{x}, \boldsymbol{y}) \in \mathcal{R}^{nL+K}_+ \;\middle|\; \begin{array}{l} (\boldsymbol{x}, \boldsymbol{y}) \in \mathcal{F}, \exists (\boldsymbol{p}, \boldsymbol{q}_1, \cdots, \boldsymbol{q}_n) \in \mathcal{R}^{L+nK}_+ \text{ such that} \\ (\boldsymbol{x}_i, \boldsymbol{y}) \text{ maximize } v_i(\cdot, \cdot, \theta_i) \text{ s.t. } \boldsymbol{p}\boldsymbol{x}_i + \boldsymbol{q}_i\boldsymbol{y} \leqq w_i, \\ \text{where } \sum_{i \in I} w_i = \max_{(\boldsymbol{x}, \boldsymbol{y}) \in \mathcal{F}} (\boldsymbol{p}(\boldsymbol{x} - \boldsymbol{\omega}_i) + \sum_i \boldsymbol{q}_i \boldsymbol{y}) \end{array} \right\}.$$

Example 18.3.3 (The Judgment of Solomon) The Old Testament of the Bible told the story of two mothers (one of whose child died one night) who brought a baby boy to King Solomon and asked him to decide who was the real mother. When King Solomon suggested splitting the child in half and each mother getting half, one declared, “I will give up the child!” and the other one stated, “Split him up!” King Solomon declared that the mother who would give up her child was the real mother and gave her the child. We can formulate this problem in the language of mechanism design. Two mothers are Anne and Bets, denoted by A and B, respectively; the state of world is denoted by α and β, respectively; α denotes “A is the real mother”, and β denotes “B is the real mother”. There are four alternatives for the king:

- a: Anne gets the child;
- b: Bets gets the child;
- c: Cut the child in half;
- d: Death penalty for everyone (two mothers and one baby).

Preferences of the two mothers under the two states are:

$$\begin{array}{c} a \succ_A^\alpha b \succ_A^\alpha c \succ_A^\alpha d; \\ a \succ_A^\beta c \succ_A^\beta b \succ_A^\beta d; \\ b \succ_B^\alpha c \succ_B^\alpha a \succ_B^\alpha d; \\ b \succ_B^\beta a \succ_B^\beta c \succ_B^\beta d. \end{array}$$

The king wants the real mother to get the child, i.e., the social choice rule $f(\cdot)$ that he wants to implement satisfies: $f(\alpha) = a, f(\beta) = b$.

Example 18.3.4 (Allocating Public Goods) n households in a community consider whether or not to build a public project (e.g., a road or a bridge). Let y denote the size of the project (y can be continuous, such as $y \in Y =$

[3]For example, public goods can be produced by private goods with constant-returns-to-scale, and the initial endowment vector of private goods is ω_i, then $\mathcal{F} = \{(\boldsymbol{x}, \boldsymbol{y}) \in \mathcal{R}^{l+m}_+ | \boldsymbol{x} + \boldsymbol{y} \leqq \sum_{i \in N} \omega_i\}$.

$[0, \infty)$; can be discrete, such as $y \in Y = \{y_1, \cdots, y_k\}$; and in practice, $y \in Y = \{0, 1\}$ often denotes "build" if $y = 1$ or "not build" if $y = 0$). Individual preferences can be represented by a quasi-linear utility function $v_i(y, \theta) + t_i$ (if the private value model is used, the utility function is $v_i(y, \theta_i) + t_i$), where $y \in Y$ represents the scale of the project, θ_i represents the individual's preference intensity for this public project, and t_i represents the tax or compensation that an individual needs to pay or gain. This environment can be characterized as follows:

- Individuals: $N = \{1, \cdots, n\}$.
- Goods (Alternatives): An outcome consists of the scale of the public project and the transfers (tax or subsidy), and the total subsidy should not exceed the total tax. Then, the outcome space is $Z = \mathcal{R}^{n+1}$. The set of feasible outcomes is $A = \{(y, t_1, \cdots, t_n) : y \in Y, t_i \in \mathcal{R}, \sum_i t_i \leqq 0, \forall i \in N\}$.
- State of the world (Preferences or types): $\theta = (\theta_1, \cdots, \theta_n)$.

The problem is then what mechanisms can be designed to ensure efficient provision of public goods.

Example 18.3.5 (Allocating an Indivisible Private Good) An indivisible good is to be allocated to a member of a society. For instance, it can be the right to an exclusive license or a firm to be privatized. In this case, the outcome space is $Z = \{\boldsymbol{y} \in \{0, 1\}^n : \sum_{i=1}^n y_i = 1\}$, where $y_i = 1$ means that individual i obtains the object, and $y_i = 0$ means that individual i does not get the object. If individual i gets the object, his net value is v_i. If individual i does not get the object, it is 0. Thus, agent i's valuation function is

$$v_i(\boldsymbol{y}) = v_i y_i.$$

Note that we can regard $\boldsymbol{y}$ as an n-dimensional vector of public goods since $v_i(\boldsymbol{y}) = v_i y_i = \boldsymbol{v}^i \boldsymbol{y}$, where $\boldsymbol{v}^i$ is a vector with the i-th component being v^i and the others being zeros, i.e., $\boldsymbol{v}^i = (0, \cdots, 0, v_i, 0, \cdots, 0)$. So, the problem of allocating an indivisible private good also constitutes a mechanism design problem.

From these examples, a socially optimal decision depends on the individual's true valuation function $v_i(\cdot)$. For the examples of public goods and indivisible goods, let $h = (y(\cdot), t_1(\cdot), t_2(\cdot), \cdots, t_n(\cdot)) : \Theta \to Z$ be an outcome function of $\mathbf{y}$ and the transfer t_i, where $(t_1(\cdot), t_2(\cdot), \cdots, t_n(\cdot))$ is called an **allocation rule**, and $y(\cdot)$ is a **decision rule**.

We call the decision rule $y(\cdot)$ **efficient** if and only if it maximizes the social surplus (the sum of individuals' surpluses), i.e.,

$$\sum_{i \in N} v_i(y(\theta), \theta) \geqq \sum_{i \in N} v_i(y(\theta'), \theta), \quad \forall \theta' \in \Theta.$$

For the private value model, it becomes

$$\sum_{i\in N} v_i(y(\theta),\theta_i) \geqq \sum_{i\in N} v_i(y(\theta'),\theta_i), \quad \forall \theta' \in \Theta.$$

The social planner wants to implement an outcome that maximizes the sum of all individuals' surpluses. A more general utility or valuation function, instead of a parameterized utility function, can be considered. For instance, let V_i be the set of all valuation functions v_i and $V = \prod_{i\in N} V_i$. Then, $y(\cdot)$ is said to be efficient if and only if

$$\sum_{i\in N} v_i(\boldsymbol{y}(\boldsymbol{v})) \geqq \sum_{i\in N} v_i(\boldsymbol{y}(\boldsymbol{v}')), \quad \forall \boldsymbol{v}' \in V.$$

18.4 Dominant Strategy and Truthful Revelation Mechanisms

This section discusses whether there is a mechanism, such that all participants are willing to truthfully reveal their private information in equilibrium. As shall be demonstrated, dominant strategy equilibrium is equivalent to truth-telling. The dominant strategy equilibrium calls for the least information in terms of informational efficiency because the strategy chosen by each individual is optimal, irrespective of the choices of others.

For $e \in E$, a mechanism $\Gamma = \langle M, h\rangle$ is said to have a **dominant strategy equilibrium $\mathbf{m}^*$** if for all i

$$h_i(m_i^*, \mathbf{m}_{-i}) \succcurlyeq_i h_i(m_i, \mathbf{m}_{-i}), \forall \mathbf{m}_i \in M_i. \tag{18.4.2}$$

Denote by $\mathcal{D}(e,\Gamma)$ the set of dominant strategy equilibria for $\Gamma = \langle M, h\rangle$ and $e \in E$.

For a given game, dominant strategies are not likely to exist. However, since we can choose various game forms in mechanism design, the possibility of the existence of dominant strategies is much higher. Another advantage of using dominant strategy as the solution concept is that bad equilibria are usually not a problem. For instance, when agents have strict preferences, if an agent has two dominant strategies, they must be payoff-equivalent, which is, in general, not true for other equilibrium solution concepts.

Definition 18.4.1 A mechanism $\Gamma = \langle M, h\rangle$ is said to **fully implement a social choice correspondence** F in dominant strategies on E if for every $e \in E$, we have

(a) $\mathcal{D}(e,\Gamma) \neq \emptyset$;

(b) $h(\mathcal{D}(e,\Gamma)) = F(e)$.

Definition 18.4.2 A mechanism $\Gamma = \langle M, h \rangle$ is said to **implement a social choice correspondence** F in dominant strategies on E if for every $e \in E$, we have

(a) $\mathcal{D}(e, \Gamma) \neq \emptyset$;

(b) $h(\mathcal{D}(e, \Gamma)) \subseteq F(e)$.

Definition 18.4.3 A mechanism $\Gamma = \langle M, h \rangle$ is said to **partially implement a social choice correspondence** F in dominant strategies on E if for every $e \in E$, we have

(a) $\mathcal{D}(e, \Gamma) \neq \emptyset$;

(b) $h(\mathcal{D}(e, \Gamma)) \cap F(e) \neq \emptyset$ (i.e., there is a dominant strategy $\mathbf{m}^* = d(e) \in \mathcal{D}(e, \Gamma)$, such that $h(\mathbf{m}^*) \in F(e)$).

In contrast to general (indirect) mechanisms, a special class of mechanisms are revelation mechanisms, in which the message space M_i for each agent i coincides with the set of his economic characteristics E_i.

Definition 18.4.4 A mechanism $\Gamma = \langle M, h \rangle$ is said to be **a revelation or direct mechanism** if $M = E$.

Example 18.4.1 The optimal contracts that we discussed in Chapter 16 are revelation mechanisms.

Example 18.4.2 The Vickrey-Clark-Groves mechanism that we will explore below is a revelation mechanism.

The most appealing property of revelation mechanisms is that truth-telling of characteristics always turns out to be an equilibrium. The absence of such a property is called the "free-rider problem" in the theory of public goods provision.

Definition 18.4.5 (Truthful Implementation) A revelation mechanism $\langle E, h \rangle$ is said to **truthfully implement** a social choice correspondence F in dominant strategies on E if for every $e \in E$,

(a) $e \in D(e, \Gamma)$;

(b) $h(e) \subseteq F(e)$.

In other words, $F(\cdot)$ is **truthfully implementable in dominant strategies** if truth-telling is a dominant strategy equilibrium, and the resulting outcome is socially optimal.

Note that the truthful implementation in dominant strategies is just a partial implementation, and not a full implementation, since a misreport, denoted e', may also be a dominant equilibrium, i.e., $h(e') \in b(e, \Gamma)$.

Truthful implementation in dominant strategies is often called **strategy-proof**, **dominant strategy incentive compatible**, **strongly individually incentive compatible**, or **strategy straightforward**.

Although the message space of a mechanism can be general, the following revelation principle informs us that one only needs to employ the revelation mechanism in which the message space consists solely of the set of individuals' characteristics, and it is unnecessary to seek more complicated mechanisms. This significantly reduces the complexity of constructing a mechanism.

Theorem 18.4.1 (Revelation Principle) *Suppose that a mechanism $\langle M, h\rangle$ partially implements a social choice rule F in dominant strategies. Then, there is a revelation mechanism $\langle E, g\rangle$ that truthfully implements F in dominant strategies.*

PROOF. Let d be a selection of dominant strategy correspondence of the mechanism $\langle M, h\rangle$ that results in a social outcome, i.e., for every $\mathbf{e} \in E$, $\mathbf{m}^* = d(e) \in \mathcal{D}(\mathbf{e}, \Gamma)$, such that $h(d(\mathbf{e})) \in F(\mathbf{e})$. Since $\Gamma = \langle M, h\rangle$ partially implements social choice rule F, such a selection exists. In addition, since the strategy of each agent is independent of the strategies of others, each agent i's dominant strategy equilibrium can be expressed as $m_i^* = d_i(e_i)$.

Define the revelation mechanism $\langle E, g\rangle$ by $g(e) \equiv h(d(e))$ for each $e \in E$. We first show that truth-telling is always a dominant strategy equilibrium of the revelation mechanism $\langle E, g\rangle$. Suppose that this is not the case. Then, there exists a message e' and an agent i, such that

$$u_i[g(e_i', \mathbf{e}_{-i}')] > u_i[g(e_i, \mathbf{e}_{-i}')].$$

However, since $g = h \circ d$, we have

$$u_i[h(d_i(e_i'), d_{-i}(\mathbf{e}_{-i}'))] > u_i[h(d_i(e_i), d_{-i}(\mathbf{e}_{-i}'))],$$

which contradicts the assumption that $m_i^* = d_i(e_i)$ is a dominant strategy equilibrium. This is because, when the true economic environment is $(e_i, \mathbf{e}_{-i}')$, agent i has an incentive to not report $m_i^* = d_i(e_i)$ truthfully, but to report $m_i' = d_i(e_i')$, which is a contradiction.

Finally, since $\mathbf{m}^* = d(\mathbf{e}) \in \mathcal{D}(e, \Gamma)$ and $\langle M, h\rangle$ partially implements the social choice rule F in dominant strategies, we have $g(\mathbf{e}) = h(d(\mathbf{e})) = h(\mathbf{m}^*) \in F(\mathbf{e})$. Therefore, the revelation mechanism truthfully implements F in dominant strategies. □

Thus, by the revelation principle, if truthful implementation rather than full implementation or implementation is all that is required, we do not need to consider general mechanisms. In particular, when F becomes a single-valued function f, $\langle E, f\rangle$ can be regarded as a revelation mechanism. As a consequence, if a mechanism $\langle M, h\rangle$ implements f in dominant strategies, then the revelation mechanism $\langle E, f\rangle$ is incentive compatible in dominant strategies.

It should be noted that in the above proof, the difference between dominant strategy equilibrium and Nash equilibrium is that individual choice in dominant strategy equilibrium is optimal regardless of the choices of other individuals; whereas, the choice of individual strategy in Nash equilibrium depends on the optimal strategy choices of other individuals. Therefore, we can rewrite the dominant strategy equilibrium as $m_i^* = d_i(e_i)$.

Remark 18.4.1 Note that the revelation principle may be valid only for partial implementation, but not for implementation or full implementation. It specifies a mapping between a dominant strategy equilibrium of the original mechanism $\langle M, h\rangle$ and the true profile of characteristics as a dominant strategy equilibrium of the revelation mechanism. Moreover, it does not require the revelation mechanism to have a unique dominant strategy equilibrium so that there may also exist non-truthful dominant strategy equilibria under the revelation mechanism $\langle E, g\rangle$. Dasgupta, Hammond, and Maskin (1979) provided such an example (see the exercise below). Thus, shifting from a general (indirect) dominant strategy mechanism to a direct one may introduce undesired dominant strategies which are not truth-telling. Indeed, these strategies may even create a situation in which the indirect mechanism (fully) implements a social choice correspondence F, while the corresponding revelation mechanism $\langle E, g\rangle$ may only partially implement F.

However, if agents have strict preferences, i.e., $\forall a, b \in \mathcal{A}$, $a = b$ if and only if $a \sim_i^e b, \forall i \in N$, any two different outcomes cannot be indifferent. Then, the revelation principle is valid for full implementation.

The following result shows that, although (Γ, θ) may have more than one dominant strategy equilibrium, the resulting equilibrium outcome is unique under strict preferences, and thus the revelation mechanism $\langle E, g\rangle$ with $g(e) \equiv h(d(e))$ implements the social choice correspondence in dominant strategies.

Proposition 18.4.1 *Suppose that agents have strict preferences. Then, the equilibrium outcome of $\Gamma = \langle M, h\rangle$ that implements a social choice rule F in dominant strategies is unique, and thus the revelation mechanism $\langle E, g\rangle$ with $g(e) \equiv h(d(e))$ implements F in dominant strategies.*

PROOF. Suppose that $\Gamma = \langle M, h\rangle$ implements F in dominant strategies, but the equilibrium outcome is not unique. Then, there is $m_i^*, m_i^{*\prime} \in \mathcal{D}_i(\Gamma, \theta)$, such that $m_i^* \neq m_i^{*\prime}$. By the dominant strategy, for $\forall \mathbf{m}_{-i} \in M_{-i}$, we have

$$h(m_i^*, \mathbf{m}_{-i}) \succcurlyeq_i^e h(m_i^{*\prime}, \mathbf{m}_{-i}),$$

$$h(m_i^{*\prime}, \mathbf{m}_{-i}) \succcurlyeq_i^e h(m_i^*, \mathbf{m}_{-i}),$$

Then, $h(m_i^*, \mathbf{m}_{-i}) \sim_i^e h(m_i^{*\prime}, \mathbf{m}_{-i})$. By strict preferences, we have $h(m_i^*, \mathbf{m}_{-i}) = h(m_i^{*\prime}, \mathbf{m}_{-i})$. Repeating the process for each i, we have $h(\mathbf{m}^*) = h(\mathbf{m}^{*\prime})$ for all $\mathbf{m}^*, \mathbf{m}^{*\prime} \in \mathcal{D}(\Gamma, \theta)$. Therefore, the dominant strategy outcome is unique.

Thus, for all $\mathbf{m}^* \in \mathcal{D}(e, \Gamma)$, $g(e) \equiv h(\mathbf{m}^*(e))$ is a single-valued mapping. Therefore, the revelation mechanism $\langle E, g \rangle$ implements F in dominant strategies. □

As a corollary of Proposition 18.4.1, any social goal that can be fully implemented in dominant strategies must be a single-valued social choice function. Formally, we have

Corollary 18.4.1 *Suppose that agents have strict preferences. Any social goal F that can be fully implemented in dominant strategies (i.e., $h(\mathcal{D}(e, \Gamma)) = F(e)$) must be a single-valued social choice function.*

PROOF. Since $\Gamma = \langle M, h \rangle$ fully implements F in dominant strategies, we have $h(\mathcal{D}(e, \Gamma)) = F(e)$. Then, by Proposition 18.4.1, we have $h(\mathbf{m}^*) = h(\mathbf{m}^{*\prime})$, for all $\mathbf{m}^*, \mathbf{m}^{*\prime} \in \mathcal{D}(\Gamma, \theta)$. Consequently, $\forall \theta \in \Theta$, $f(\theta) = h(\mathcal{D}(\Gamma, \theta))$ is a single-valued function. □

When discussing ex-post full implementation in the next chapter, we will give a necessary and sufficient condition for a social choice correspondence under private value to be fully implemented in dominant strategies. If a participant's utility depends only on his own type, i.e., $u_i(h(m(\theta)), \theta) = u_i(h(m(\theta)), \theta_i), \forall i \in N$, the ex-post equilibrium is equivalent to the dominant equilibrium. Thus, the necessary and sufficient conditions for ex-post full implementation of the social choice rules are also necessary and sufficient conditions for full implementation in dominant strategies under private values.

However, even if the dominant equilibrium is unique, or different dominant equilibria lead to the same outcome, there may exist different equilibrium outcomes under other solution concepts. For example, there may exist a Nash equilibrium that results in a different equilibrium outcome. We will provide such an example later when we examine a special form of VCG mechanism, i.e., the second-price sealed-bid auction mechanism.

18.5 Gibbard-Satterthwaite Impossibility Theorem

It is always desired for an implementable social choice rule to have some desired properties, such as non-dictatorship; meanwhile, since the information is expected to be as little as possible, the solution concept used to implement the social choice rule is expected to be as strong as possible, such as the dominant strategy equilibrium. The revelation principle states that if one wants a social goal F to be implementable in dominant strategies, one only needs to show that the revelation mechanism is truthfully implementable.

Unfortunately, it is generally impossible to have such a result if the set of admissible economic environments is sufficiently rich. The Gibbard-Satterthwaite Impossibility Theorem in Chapter 12 informs us that such social goals can only be dictatorial in the sense that an agent's optimal choice is the social optimum when economic environments are unrestricted. We reconsider this theorem from the perspective of mechanism design.

Theorem 18.5.1 (Gibbard-Satterthwaite Impossibility Theorem) Suppose that the outcome space $\mathcal{A}$ contains at least three alternatives, $|\mathcal{A}| \geqq 3$, the set of admissible economies consists of all strict preferences, $E = E^s$, and that social choice correspondence $F : E^s \to\to \mathcal{A}$ is an onto mapping. Then, if F is truthfully implementable in dominant strategies, it must be dictatorial.

This theorem shows that if individuals' preferences are unrestricted and each individual is required to tell the truth, then such a decentralized incentive mechanism does not exist and the only social choice rule that is implementable is dictatorial. We have proved this theorem in Part IV. To make the connection to mechanism design, we provide a new proof here. Before proving the theorem, we first give the following lemmas.

Lemma 18.5.1 *If the domain only contains strict preferences, i.e., $E = E^s$, the outcome space $\mathcal{A}$ contains at least three alternatives, $|\mathcal{A}| \geqq 3$, and the social choice rule $F : E^s \to\to \mathcal{A}$ satisfies* **unanimity** *and* **Maskin monotonicity**, *then it must be* **dictatorial**.

PROOF. We prove the lemma with the assistance of Tables 18.1 to 18.6. Consider the two alternatives $a, b \in \mathcal{A}$. Suppose that $a \succ_i^e x \succ_i^e b, \forall x \in \mathcal{A}, \forall i \in I$. According to unanimity, we obtain $F(e) = a$. In the preference order of agent 1, move b forward one position at a time. According to Maskin monotonicity, as long as b does not exceed a, then alternative a is still selected. However, when b exceeds a, according to unanimity, the possible outcome is a or b. If the alternative selected is still a, the same operation is performed on the preference ordering of agent 2, and so on. There must be an agent who changes the social choice from a to b when his preference reverses from a to b, and then mark the agent as j. In addition, also mark the state before the choice between a and b reverses as e^1, and mark the state after that as e^2. In the states of e^1 and e^2, move the preference order of a to the end for each agent $i < j$, and move the preference order of a to the second-to-last position for each agent $i > j$. The preference of agent j remains unchanged, and the new states are recorded as $e^{1'}$ and $e^{2'}$.

Since $F(e^2) = b$ and $L_i(b, e^2) = L_i(b, e^{2'}), \forall i \in I$, by Maskin monotonicity, we obtain $F(e^{2'}) = b$. If we compare $e^{2'}$ with $e^{1'}$, a and b reverse only at the preference ordering of j, which means $F(e^{1'}) = a$ or b. If $F(e^{1'}) = b$, since $L_i(b, e^{1'}) = L_i(b, e^1), \forall i \in I$, we have $F(e^1) = b$ by Maskin monotonicity, an immediate contradiction. Therefore, we must have $F(e^{1'}) = a$.

Suppose that $c \neq a, b$ is another alternative, and define the state e^3 according to Table 18.5. Since $L_i(a, e^3) = L_i(a, e^{1'})$ for $\forall i \in I$, we have $F(e^3) = a$ by Maskin monotonicity.

At the state e^3, reverse the preference for a and b for each agent $i > j$, and obtain state e^4. By Maskin monotonicity again, $F(e^4) = a$ or b. Obviously, $c \succ_i^{e^4} b, \forall i \in I$, and according to unanimity, we must have $F(e^4) = a$.

Obviously, for any state $\alpha \in E^s$, as long as a is at the top of the list of agent j's preference, then $L_i(a, e^4) \subseteq L_i(a, \alpha), \forall i \in I$, and we can have $F(\alpha) = a$ according to Maskin monotonicity. It is clear that individual j is the dictator of alternative a.

individual		preference $\succ^{\theta^1}$		
1	b	a	...	*
2	b	a	...	*
...	...	...	...	...
j-1	b	a	...	*
j	a	b	...	*
j+1	a	...	...	b
...	...	...	...	...
n	a	...	...	b

Table 18.1: Preferences under state e_1

individual			preference $\succ^{\theta^{1'}}$		
1	b	*	...	*	a
2	b	*	...	*	a
...	...	...	...	...	...
j-1	b	*	...	*	a
j	a	b	...	*	*
j+1	*	*	...	a	b
...	...	...	...	...	...
n	*	*	...	a	b

Table 18.2: Preferences under state $e_1^{'}$

individual	preference $\succ^{\theta^2}$			
1	b	a	...	*
2	b	a	...	*
...	...	...	...	...
j-1	b	a	...	*
j	b	a	...	*
j+1	a	...	...	b
...	...	...	...	...
n	a	...	...	b

Table 18.3: Preferences under state e_2

individual	preference $\succ^{\theta^{2'}}$				
1	b	*	...	*	a
2	b	*	...	*	a
...	...	...	...	...	...
j-1	b	*	...	*	a
j	b	a	...	*	*
j+1	*	...	...	a	b
...	...	...	...	...	...
n	*	*	...	a	b

Table 18.4: Preferences under state $e_2^{'}$

individual	preference $\succ^{\theta^3}$						
1	*	*	*	...	c	b	a
2	*	*	*	...	c	b	a
...	...	...	...	...	...	...	...
j-1	*	*	*	...	c	b	a
j	a	c	b	...	*	*	*
j+1	*	*	*	...	c	a	b
...	...	...	...	...	...	...	...
n	*	*	*	...	c	a	b

Table 18.5: Preferences of state e_3

individual				preference $\succ^{\theta^4}$			
1	*	*	*	...	c	b	a
2	*	*	*	...	c	b	a
...	...	...	...	...	...	...	...
j-1	*	*	*	...	c	b	a
j	a	c	b	...	*	*	*
j+1	*	*	*	...	c	b	a
...	...	...	...	...	...	...	...
n	*	*	*	...	c	b	a

Table 18.6: Preferences under state e_4

From the above proof, we can see that under strict preferences, there is a dictator for any alternative a, but there can be no more than one dictator. Suppose that at the state e, j is the dictator for alternative a, then we have $F(e) = a$; and suppose that j' is another dictator for another alternative a', then we have $F(e) = a'$, which is a contradiction. □

Lemma 18.5.2 If the social choice rule $F : E^s \to\to \mathcal{A}$ is truthfully implemented in dominant strategies and is a surjective mapping, then it must satisfy unanimity and Maskin monotonicity.

PROOF.

- **Maskin monotonicity**: For states $e, e' \in E^s$, such that $F(e) = a$ and $L_i(a, e) \subseteq L_i(a, e'), \forall i \in I$, if $F(e'_1, e_{-1}) \neq a$, by truthful implementation in dominant strategies and strict preference assumption, we must have $a = F(e) \succ_i^e F(e'_1, e_{-1})$, i.e., $F(e'_1, e_{-1}) \in L_1(a, e)$. As a consequence, $F(e'_1, e_{-1}) \in L_1(a, e')$, i.e., $a = F(e_1, e_{-1}) \succ_i^{e'} F(e'_1, e_{-1})$. This contradicts the property of the truthful implementation of $F(\cdot)$ in dominant strategies, and thus $F(e'_1, e_{-1}) = a$. The rest is deduced by similar reasoning:

  $$F(e'_1, e_{-1}) = F(e'_1, e'_2, e_3, \cdots, e_n) = \cdots = F(e') = a.$$

  Therefore, we have verified that $F(\cdot)$ satisfies Maskin monotonicity.

- **Unanimity**: Since $F(\cdot)$ is an onto mapping (surjection), for any $a \in \mathcal{A}$, there exists $e \in E^s$, such that $F(e) = a$. Suppose that β shows such a situation: alternative a is at the top of the preference list for every agent. Obviously, $L_i(a, e) \subseteq L_i(a, \beta), \forall i \in I$, and we obtain $F(\beta) = a$ according to Maskin monotonicity. Thus, unanimity is verified.

□

Now, we prove the Gibbard-Satterthwaite Impossibility Theorem:

PROOF. According to the above two lemmas, there exists $j \in I$, such that $\forall e \in E^s$, $F(e) = M_j(e)$, where $M_j(e) = \{x \in \mathcal{A} | x \succ_i^e b, \forall b \in \mathcal{A}\backslash\{x\}\}$ represents the set of the most preferred alternatives for agent j. Now, we only need to prove that for any $e \in E\backslash E^s$, we have $F(e) = M_j(e)$. We prove this by the way of contradiction. Suppose that there exists $e \in E\backslash E^s$, such that $a = F(e)$ but $a \notin M_j(e)$. Then, there exists $b \in M_j(e)$, such that $b \succ_j^e a$. Consider another state $e' \in E^s$ satisfying: (i) $a \succ_i^{e'} b \succ_i^{e'} c, \forall i \neq j, c \neq a, b$; and (ii) $b \succ_j^{e'} a \succ_j^{e'} c, \forall c \neq a, b$. Then, $L_i(a, e) \subseteq L_i(a, e'), \forall i \in I$, and $F(e') = b$ by Maskin monotonocity. According to the truthful implementation in dominant strategies, $F(e_1, e_{-1}) \succcurlyeq_i^e F(e_1', e_{-1})$, and thus $F(e_1, e_{-1}) \succcurlyeq_i^{e'} F(e_1', e_{-1})$. On the other hand, by the truthful implementation in dominant strategies again, $F(e_1', e_{-1}) \succcurlyeq_i^{e'} F(e_1, e_{-1})$, so $F(e_1', e_{-1}) = F(e_1, e_{-1}) = a$. Analogous to the proof of Lemma 18.5.2., we can obtain $F(e') = a$. Therefore $F(e') = a \neq M_j(e') = b$, which contradicts the fact that there exists $j \in I$, such that $\forall e \in E^s$, $F(e) = M_j(e)$. Therefore, $\forall e \in E\backslash E^s$, we have $F(e) = M_j(e)$. □

The Gibbard-Satterthwaite Impossibility Theorem is a rather disappointing result, which is essentially equivalent to the Arrow Impossibility Theorem. However, for some special classes of economic environments, for example, by relaxing the two conditions that "there are at least three alternatives" and there is an unrestricted domain ($E^s \subseteq E$), we may have a positive result.

Example 18.5.1 (A Simple Majority Voting with only Two Candidates) Suppose that there are N voters casting their votes on two candidates in a presidential election, a and b. Each voter reports his preference by naming the candidate that she supports, and they follow the simple majority rule:

$$F(e) = \begin{cases} a & \textit{if} \quad \#\{i \in I | a \succ_i^e b\} > \#\{i \in I | b \succ_i^e a\}, \\ b & \textit{if} \quad \#\{i \in I | a \succ_i^e b\} \leqq \#\{i \in I | b \succ_i^e a\}. \end{cases}$$

It can be seen that every voter will truthfully report his preference in this process, irrespective of whether or not others are telling the truth. Therefore, $F(\cdot)$ can be truthfully implemented in dominant strategies.

Another example is the VCG mechanism to be discussed below, which is defined on the domain of quasi-linear utility functions and leads to social surplus maximization.

The following Hurwicz Impossibility Theorem gives an even stronger impossibility result: even if further restricted to the neo-classical economic environment, Pareto efficiency and truth-telling are fundamentally incompatible.

18.6 Hurwicz Impossibility Theorem

Prior to the Hurwicz Impossibility Theorem, it was generally believed that the competitive market mechanism can handle the resource allocation of private goods economies well. In particular, it was believed that efficient allocations of resources are consistent with individuals' self-interested behavior in such economies.

For economic environments with public goods, it is relatively easy to understand that the efficient allocation of resources is incompatible with the self-interested behavior of individuals, because individuals have an incentive to "free ride" and benefit from the contributions of others. Consequently, they are unwilling to reveal their true preferences for public goods in order to contribute less. If the tax paid by an individual is determined according to the degree of self-reported personal consumption, people may underreport their wants. Once a public good is produced, they can benefit from the consumption of public goods in the same way. This differs markedly from the case of private goods, in which an individual spends money on items that benefit herself only.

Such strategic manipulation of hiding their true preferences was originally proposed by Samuelson (1954, 1955) in the criticism of the Lindahl equilibrium solution for public goods provision. He further speculated that, for an economic environment with public goods, there was no decentralized economic mechanism that could lead to Pareto efficient allocations with participants reporting their true preferences. According to the direct revelation mechanism and incentive compatibility, Samuelson's assertion means that participants have no incentive to reveal their true economic characteristics. However, Hurwicz's impossibility result surprisingly shows that this assertion holds not only for the environment of public goods, but also for that of private goods.

In 1972, Hurwicz proved that, for neoclassical economic environments (i.e., commodities are divisible, preferences are continuous, monotonic and weakly convex, and the production set is closed without exhibiting increasing returns to scale) with a finite number of participants, there was no economic mechanism that could lead to individually rational and Pareto efficient allocations with individuals reporting their economic characteristics truthfully.

Theorem 18.6.1 (Hurwicz Impossibility Theorem, 1972) *For neoclassical private goods economies, there is no mechanism $\langle M, h\rangle$ that implements Pareto efficient and individually rational allocations in dominant strategies. Consequently, any revelation mechanism $\langle M, h\rangle$ that yields Pareto efficient and individually rational allocations is not strategy-proof (i.e., truth-telling about their preferences is not a dominant strategy equilibrium).*

PROOF. By the revelation principle, we only need to show that we can find a specific pure exchange economy, such that any revelation mechanism does not truthfully implement Pareto efficient and individually rational allocations in a dominant strategy equilibrium. Then, it is sufficient to show that truth-telling is not a Nash equilibrium for any revelation mechanism that yields Pareto efficient and individually rational allocations of the pure exchange economy.

Consider a private goods economy with two agents ($n = 2$) and two goods ($L = 2$):

$$\begin{aligned} w_1 &= (0,2),\ w_2 = (2,0), \\ \mathring{u}_i(x,y) &= \begin{cases} 3x_i + y_i & \text{if } x_i \leqq y_i, \\ x_i + 3y_i & \text{if } x_i > y_i. \end{cases} \end{aligned}$$

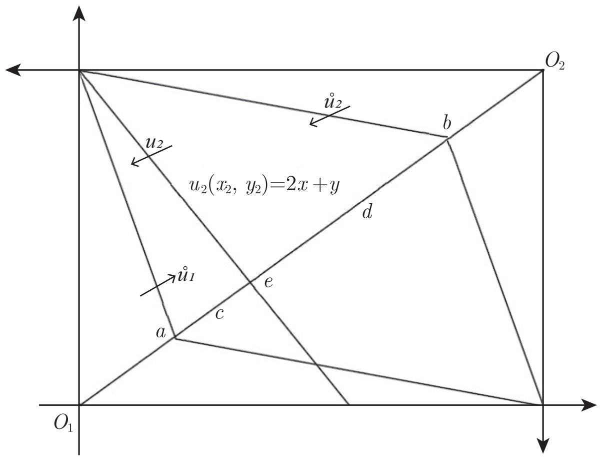

Figure 18.2: An illustration of the proof of Hurwicz's impossibility theorem

The set of feasible allocations is then given by

$$\begin{aligned} A = \Big\{&[(x_1,y_1),(x_2,y_2)] \in R^4_+ : \\ &x_1 + x_2 = 2, \\ &y_1 + y_2 = 2\}. \end{aligned}$$

Let U_i be the set of all neoclassical utility functions, i.e., they are monotonic, continuous, and quasi-concave functions that each agent i can report to the designer. Thus, the agent's true utility function, $\mathring{u}_i$, is included in U_i. Then,

$$U = U_1 \times U_2$$

and

$$h : U \to A$$

constitute a revelation mechanism. Note that if the true utility function profile $\mathring{u}$ were a Nash equilibrium, it would satisfy

$$\mathring{u}_i(h_i(\mathring{u}_i, \mathring{u}_{-i})) \geqq \mathring{u}_i(h_i(u_i, \mathring{u}_{-i})),\ i = 1, 2. \tag{18.6.3}$$

We want to demonstrate that $\mathring{u}_i$ is not a Nash equilibrium. Note from Figure 18.2 that,

(1) $P(\mathring{u}) = \overline{O_1 O_2}$ (contract curve);

(2) $IR(\mathring{u}) \cap P(\mathring{u}) = \overline{ab}$;

(3) $h(\mathring{u}_1, \mathring{u}_2) = d \in \overline{ab}$.

Now, suppose that agent 2 misrepresents his utility function as:

$$u_2(x_2, y_2) = 2x + y. \tag{18.6.4}$$

Then, with u_2, the new set of individually rational (IR) and Pareto efficient allocations is given by

$$IR(\mathring{u}_1, u_2) \cap P(\mathring{u}_1, u_2) = \overline{ae}. \tag{18.6.5}$$

Note that d is strictly inferior to any point between a and e for agent 2, and thus the allocation determined by any mechanism which is IR and Pareto efficient under $(\mathring{u}_1, u_2)$ is some point, for example, point c in the figure, on the line determined by endpoints a and e. Therefore, we have

$$\mathring{u}_2(h_2(\mathring{u}_1, u_2)) > \mathring{u}_2(h_2(\mathring{u}_1, \mathring{u}_2)) \tag{18.6.6}$$

since $h_2(\mathring{u}_1, u_2) = c \in \overline{ae}$. Similarly, if d is on $\overline{ae}$, then agent 1 has an incentive to deviate. Consequently, no mechanism that yields Pareto efficient and individually rational allocations is incentive compatible. □

The Hurwicz Impossibility Theorem tells us that, even for neoclassical economic environments with private goods only, as long as the number of participants is finite, there is no informationally decentralized economic mechanism, such that Pareto efficient and individually rational allocations can be implemented in dominant strategies. This reveals that a finite number of participants is essentially inconsistent with perfect competition. However, when the number of economic agents is a continuum, "truthful revelation of preferences" is possible, although it is also unrealistic.

Thus, if one wants a mechanism that implements Pareto optimal allocations, we must give up the requirement of truth-telling, i.e., abandon the requirement of dominant strategy implementation. For economic environments with public goods, irrespective of whether or not the number of individuals is finite, Ledyard and Roberts (1974) proved a similar impossibility result. In this sense, there is no essential difference between the two economic environments, i.e., economic environments with public goods and economic environments without public goods. Of course, as shown in the following section, a mechanism that guarantees truth-telling and social surplus maximization (not Pareto optimality, since feasibility, in general, is not satisfied) is possible.

Overall, Hurwicz's Impossibility Theorem implies that Pareto efficiency and the truthful revelation of individuals' characteristics are fundamentally incompatible. If we desire a positive result, we must give up either the requirement of Pareto efficiency or the requirement of truthful revelation. For instance, if one is willing to give up Pareto efficiency, e.g., one only requires the efficient provision of public goods, is it possible to find an incentive compatible mechanism that truthfully reveals individuals' characteristics? The answer is affirmative. For the class of quasi-linear utility functions, the so-called Vickrey-Clark-Groves mechanism can be such a mechanism.

18.7 Vickrey-Clark-Groves (VCG) Mechanisms

From Chapter 15 on public goods, we know that public goods economies may have an inefficiency issue in allocating resources. Private provision of a public good usually results in less than an efficient amount of the public good. For instance, voting may result in too much or too little of a public good. Is there any mechanism that results in the "right" amount of the public good even though privates preferences are unknown to the designer?

The Vickrey-Clark-Groves (VCG in short) mechanism can do so for private-value economic environments. Vickrey (William Vickrey, 1914-1996, his biography can be found in section 21.8.1) was the first to propose such a mechanism in the format of the second-price auction in 1961, and then Clark (1971) provided the pivotal mechanism, although they did not consider the efficient provision of indivisible private goods and public goods from the perspective of mechanism design. Groves (1973) further provided a general form of the VCG mechanism, which covers Vickrey's second-price auction and Clark's pivotal mechanism as special cases of Groves' general mechanism .

In the following, we will demonstrate that, for economic environments with quasi-linear utility functions under private value, the VCG mechanism truthfully implements the efficient provision of public goods, and is only such a mechanism when the economic environment is sufficiently "thick" .

We begin with the case of indivisible public goods.

18.7.1 VCG Mechanisms for Discrete Public Goods

Consider the provision of a discrete public good. Suppose that the economy has n agents. Let

- c: the cost of producing the public project.
- r_i: the maximum willingness to pay of i.
- g_i: the contribution by i.
- $v_i = r_i - g_i$: the net value of i.

The public project is determined according to

$$y = \begin{cases} 1 & \text{if } \sum_{i=1}^{n} v_i \geqq 0, \\ 0 & \text{otherwise.} \end{cases}$$

From the discussion in Chapter 15, it is efficient to produce the public good, $y = 1$, if and only if

$$\sum_{i=1}^{n} v_i = \sum_{i=1}^{n} (r_i - g_i) \geqq 0.$$

Since the maximum willingness to pay for each agent, r_i, is private information and so is the net value v_i, the designer needs to induce information about v_i from individuals to determine if the project is built. What mechanism should one use? One mechanism might be to simply ask each agent to report her net value and provide the public good if the sum of the reported values is positive. The problem with such a scheme is that it does not provide proper incentives for individuals to reveal their true willingness-to-pay. Indeed, individuals may have an incentive to underreport their willingness-to-pay.

Thus, a pertinent question is how can we induce agents to truthfully reveal their values for the public good. The so-called Vickrey-Clark-Groves (VCG) mechanism provides such a mechanism. Suppose that the utility functions are quasi-linear in the net demand of private goods, $x_i - w_i$:

$$\begin{aligned} \bar{u}_i(x_i - w_i, y) &= x_i - w_i + r_i y \\ s.t. \quad x_i + g_i y &= w_i + t_i, \end{aligned}$$

where t_i is the transfer to agent i. Then, we can write the original utility functions of agents as utility functions in terms of transfers and public good.

$$\begin{aligned} u_i(t_i, y) &= t_i + r_i y - g_i y \\ &= t_i + (r_i - g_i) y \\ &= t_i + v_i y. \end{aligned}$$

A Simple Form of VCG Mechanism

In a VCG mechanism, agents are required to report their net values. As a consequence, the message space of each agent i is $M_i = \mathcal{R}$. The VCG mechanism is defined as follows:

$$\Gamma = \langle M_1, \ldots, M_n, t_1(\cdot), t_2(\cdot), \ldots, t_n(\cdot), y(\cdot)\rangle \equiv \langle M, t(\cdot), y(\cdot)\rangle,$$

where the outcome function is $h = (y(\cdot), t(\cdot))$. Specifically, we have

(1) $b_i \in M_i = \mathcal{R}$: each agent i reports a "bid" for the public good, i.e., report the net value of agent i, which may or may not be his true net value v_i.

(2) The level of the public good is determined by

$$y(b) = \begin{cases} 1 & \text{if } \sum_{i=1}^n b_i \geqq 0, \\ 0 & \text{otherwise.} \end{cases}$$

(3) In order agents to have incentive in reporting their values truthfully, there transfers are determined by

$$t_i(b) = \begin{cases} \sum\limits_{j\neq i} b_j & \text{if } \sum_{i=1}^n b_i \geqq 0, \\ 0 & \text{otherwise.} \end{cases} \tag{18.7.7}$$

If $t_i < 0$, it is a tax; if $t_i > 0$, it is a lump-sum subsidy.

Then, the payoff of agent i is given by

$$\phi_i(b) = \begin{cases} v_i + t_i(b) = v_i + \sum_{j\neq i} b_j & \text{if } \sum_{i=1}^n b_i \geqq 0, \\ 0 & \text{otherwise.} \end{cases} \tag{18.7.8}$$

We want to show that it is optimal for each individual to report the true net value, i.e., $b_i = v_i$, irrespective of what the other agents report. In other words, truth-telling is a dominant strategy equilibrium.

There are two cases to be considered.

Case 1: $v_i + \sum_{j\neq i} b_j > 0$. Then, agent i desires the public good to be provided and can ensure this by reporting $b_i = v_i$. Indeed, if $b_i = v_i$, then $\sum_{j\neq i} b_j + v_i = \sum_{i=1}^n b_j > 0$, and thus $y = 1$. In this case, agent i's utility $\phi_i(v_i, \mathbf{b}_{-i}) = v_i + \sum_{j\neq i} b_j > 0$, which is greater than the utility with no public good produced. Moreover, the pursuit of individual utility maximization and the social goal of providing this public good are incentive compatible.

Case 2: $v_i + \sum_{j\neq i} b_j \leqq 0$. Agent i does not desire the public good and can ensure it not to be provided by reporting $b_i = v_i$, so that $\sum_{i=1}^n b_i \leqq 0$. In this case, $\phi_i(v_i, \mathbf{b}_{-i}) = 0 \geqq v_i + \sum_{j\neq i} b_j$. As such, the pursuit of individual utility maximization and the social goal of not providing this public good are incentive compatible.

Thus, for either case, agent i has an incentive to tell the true value of v_i, regardless of the choices of other individuals. Therefore, it is a dominant strategy for agent i to tell the truth. There is no incentive for agent i to misreport his true net value, irrespective of what other agents do.

The above VCG mechanism, however, has a major weakness: the total transfer may be very large, and so it is costly to induce the agents to tell the truth. As such, the implementation (information) cost of this mechanism is high.

Ideally, we desire to have a mechanism in which the sum of individual transfers is equal to zero, so that the feasibility condition holds, and consequently it results in Pareto efficient allocations; but, this is, in general, impossible by Proposition 18.7.4 below. Nevertheless, we can reduce the information cost by modifying the above mechanism and asking each agent to pay a tax, but not receive a subsidy. As this tax may incur a deadweight loss, the allocation of public goods is still not Pareto efficient.

General VCG Mechanism:

The basic idea is to let each individual pay an additional tax $d_i(\mathbf{b}_{-i})$ that is independent of b_i.

In this case, the transfer is given by

$$t_i(\mathbf{b}) = \begin{cases} \sum_{j\neq i} b_j + d_i(\mathbf{b}_{-i}) & \text{if } \sum_{i=1}^n b_i \geqq 0, \\ d_i(\mathbf{b}_{-i}) & \text{if } \sum_{i=1}^n b_i < 0. \end{cases}$$

The payoff to agent i now takes the following form:

$$\phi_i(\mathbf{b}) = \begin{cases} v_i + t_i(\mathbf{b}) = v_i + \sum_{j\neq i} b_j + d_i(\mathbf{b}_{-i}) & \text{if } \sum_{i=1}^n b_i \geqq 0, \\ d_i(\mathbf{b}_{-i}) & \text{otherwise.} \end{cases} \tag{18.7.9}$$

For exactly the same reason as for the mechanism above, one can prove that it is a dominant strategy for each agent i to report his true net value.

We then have the following proposition:

Proposition 18.7.1 *For discrete private-value public good economies, truth-telling is a dominant strategy for all participants under the VCG mechanism that truthfully implements the efficient decision rule (the efficient provision of public goods) in the dominant strategy.*

Remark 18.7.1 This conclusion relies on the **private value assumption**. If an individual's valuation function also depends on the types of other people, the transfer function defined above also depends on his own type, and then the VCG mechanism may not lead to truthful implementation of the social decision rule in dominant strategies. However, the generalized VCG mechanisms to be discussed in Chapter 21 concerning auction theory truthfully implement the efficient decision rule.

Remark 18.7.2 The VCG mechanism plays an important role in general mechanism design, auction theory, and dynamic mechanism design. We will provide various modifications and extensions of the VCG mechanism.

If the function $d_i(\mathbf{b}_{-i})$ is properly chosen, the size of the transfer can be significantly reduced. A typical choice was given by Clark (1971). The **Clark mechanism** is also called the **pivotal mechanism**.

The **pivotal mechanism** is a special case of the VCG mechanism, in which $d_i(\mathbf{b}_{-i})$ is given by

$$d_i(\mathbf{b}_{-i}) = \begin{cases} -\sum_{j\neq i} b_j & \text{if } \sum_{j\neq i} b_j \geqq 0, \\ 0 & \text{if } \sum_{j\neq i} b_j < 0. \end{cases}$$

In this case, it gives

$$t_i(\mathbf{b}) = \begin{cases} 0 & \text{if } \sum_{i=1}^n b_i \geqq 0 \text{ and } \sum_{j\neq i} b_j \geqq 0, \\ \sum_{j\neq i} b_j & \text{if } \sum_{i=1}^n b_i \geqq 0 \text{ and } \sum_{j\neq i} b_j < 0, \\ -\sum_{j\neq i} b_j & \text{if } \sum_{i=1}^n b_i < 0 \text{ and } \sum_{j\neq i} b_j \geqq 0, \\ 0 & \text{if } \sum_{i=1}^n b_i < 0 \text{ and } \sum_{j\neq i} b_j < 0, \end{cases} \tag{18.7.10}$$

i.e.,

$$t_i(\mathbf{b}) = \begin{cases} -|\sum_{j\neq i} b_j| & \text{if } (\sum_{i=1}^n b_i)(\sum_{j\neq i} b_i) < 0, \\ -|\sum_{j\neq i} b_j| & \text{if } \sum_{i=1}^n b_i = 0 \text{ and } \sum_{j\neq i} b_j < 0, \\ 0 & \text{otherwise.} \end{cases} \tag{18.7.11}$$

Then, the payoff of agent i is

$$\phi_i(\mathbf{b}) = \begin{cases} v_i & \text{if } \sum_{i=1}^n b_i \geqq 0 & \text{and } \sum_{j\neq i} b_j \geqq 0, \\ v_i + \sum_{j\neq i} b_j & \text{if } \sum_{i=1}^n b_i \geqq 0 & \text{and } \sum_{j\neq i} b_j < 0, \\ -\sum_{j\neq i} b_j & \text{if } \sum_{i=1}^n b_i < 0 & \text{and } \sum_{j\neq i} b_j \geqq 0, \\ 0 & \text{if } \sum_{i=1}^n b_i < 0 & \text{and } \sum_{j\neq i} b_j < 0. \end{cases} \tag{18.7.12}$$

From the transfers given in (18.7.11), one can see that a bidding strategy of agent i may change the social decision. If so, agent i is called a **pivotal agent**. The amount of the tax that agent i must pay is the amount by which agent i's bid damages the other agents. The price that agent i must pay to change the amount of public good provision is equal to the total loss that she imposes on the other agents.

18.7.2 VCG Mechanism with Continuous Public Goods

Now, we discuss the provision of continuous public goods. Consider a public goods economy with n agents, one private good, and K public goods. Denote

- x_i: the consumption of the private good by i;
- $\boldsymbol{y}$: the consumption of the public goods by all individuals;
- t_i: transfer of agent i;
- $g_i(\boldsymbol{y})$: the contribution made by i;
- $c(\boldsymbol{y})$: the cost function of producing public goods $\boldsymbol{y}$ that satisfies the condition:
$$\sum g_i(\boldsymbol{y}) = c(\boldsymbol{y}).$$

Then, agent i's budget constraint should satisfy

$$x_i + g_i(\boldsymbol{y}) = w_i + t_i, \tag{18.7.13}$$

and his utility function is given by

$$\bar{u}_i(x_i - w_i, \boldsymbol{y}) = x_i - w_i + u_i(\boldsymbol{y}). \tag{18.7.14}$$

Combining the above two equations gives

$$\begin{aligned} u_i(t_i, \boldsymbol{y}) &= t_i + (u_i(\boldsymbol{y}) - g_i(\boldsymbol{y})) \\ &\equiv t_i + v_i(\boldsymbol{y}), \end{aligned}$$

where $v_i(y)$ is called the valuation function of agent i. Summarizing the budget constraint leads to

$$\sum_{i=1}^{n}\{x_i + g_i(\boldsymbol{y})\} = \sum_{i=1}^{n} w_i + \sum_{i=1}^{n} t_i, \tag{18.7.15}$$

and thus

$$\sum_{i=1}^{n} x_i + c(\boldsymbol{y}) = \sum_{i=1}^{n} w_i + \sum_{i=1}^{n} t_i. \tag{18.7.16}$$

From the above equation, one can see that the balancedness condition is equivalent to

$$\sum_{i=1}^{n} t_i = 0. \tag{18.7.17}$$

Recall that Pareto efficient allocations are completely characterized by

$$\begin{aligned} &\max \sum a_i \bar{u}_i(x_i, \boldsymbol{y}) \\ s.t. \quad &\sum_{i=1}^{n} x_i + c(\boldsymbol{y}) = \sum_{i=1}^{n} w_i. \end{aligned}$$

For quasi-linear utility functions, it is easily seen that weights a_i must be the same for all agents since $a_i = \lambda$ for interior Pareto efficient allocations (no income effect) where λ is the Lagrange multiplier, and thus $\boldsymbol{y}$ is uniquely determined by $u_i(t_i, \boldsymbol{y}) = t_i + v_i(\boldsymbol{y})$. Then, the above characterization problem becomes

$$\max_{t_i,\mathbf{y}} \left[\sum_{i=1}^{n}(t_i + v_i(\boldsymbol{y}))\right], \tag{18.7.18}$$

or equivalently

(1) maximization of social surplus:

$$\max_{\boldsymbol{y}} \sum_{i=1}^{n} v_i(\boldsymbol{y}); \tag{18.7.19}$$

(2) balancedness condition:

$$\sum_{i=1}^{n} t_i = 0. \tag{18.7.20}$$

Then, the Lindahl-Samuelson condition is given by:

$$\sum_{i=1}^{n} \frac{\partial v_i(\boldsymbol{y})}{\partial y_k} = 0,$$

i.e.,

$$\sum_{i=1}^{n} \frac{\partial u_i(\boldsymbol{y})}{\partial y_k} = \frac{\partial c(\boldsymbol{y})}{\partial y_k}.$$

Thus, Pareto efficient allocations for quasi-linear utility functions are completely characterized by the Lindahl-Samuelson condition $\sum_{i=1}^{n} v_i'(\boldsymbol{y}) = 0$ and the balancedness condition $\sum_{i=1}^{n} t_i = 0$.[4] In general, the designer does not know the true valuation function $v_i(\cdot,\cdot)$, and then needs to design an incentive mechanism that results in the efficient provision of public goods, and under which each participant has an incentive to truthfully report his valuation function.

In a direct revelation mechanism $\langle V, h\rangle$, the designer requires each agent to report his valuation function, which may or may not be a truthful reporting, and then determine the outcome function $h(\cdot) = (\boldsymbol{y}(\cdot), t_1(\cdot), \cdots, t_n(\cdot)) : V \to \mathcal{R}_+ \times \mathcal{R}^n$, so that such a truthful revelation mechanism leads to the efficient provision of public goods, $\boldsymbol{y}^*$.

[4]In fact, they may not be fully equivalent. Since t_i may be a sufficiently large negative number, it may be that x_i is not in the consumption space. Nevertheless, the mechanism design literature often overlooks this requirement of individual feasibility when defining Pareto efficiency.

In a simple form of VCG mechanism, it is assumed that each agent is asked to report the valuation function $v_i(\boldsymbol{y})$. Denote the valuation function reported by $b_i(\boldsymbol{y})$.

To obtain the efficient level of public goods, the designer may announce that it will provide a level of public goods $\boldsymbol{y}^*$ that maximizes

$$\max_{\boldsymbol{y}} \sum_{i=1}^{n} b_i(\boldsymbol{y}). \tag{18.7.21}$$

The VCG mechanism has the following form:

$$\Gamma = \langle V, h \rangle, \tag{18.7.22}$$

where $V = V_1 \times \ldots \times V_n$ is the message space that consists of the set of all possible valuation functions with element $b_i(\boldsymbol{y}) \in V_i$, and $h(\mathbf{b}) = (\boldsymbol{y}(\mathbf{b}), t_1(\mathbf{b}), t_2(\mathbf{b}), \cdots, t_n(\mathbf{b}))$ are outcome functions. It is determined by:

(1) Ask each agent i to report his valuation function $b_i(\boldsymbol{y})$, which may or may not be the true valuation $v_i(\boldsymbol{y})$;

(2) Determine $\boldsymbol{y}^*$: the level of the public goods, $\boldsymbol{y}^* = \boldsymbol{y}(b)$, is determined by

$$\max_{\boldsymbol{y}} \sum_{i=1}^{n} b_i(\boldsymbol{y}); \tag{18.7.23}$$

(3) Determine t: the transfer of agent i, t_i, is determined by

$$t_i(\mathbf{b}) = \sum_{j \neq i} b_j(\boldsymbol{y}^*). \tag{18.7.24}$$

The payoff of agent i is then given by

$$\phi_i(\mathbf{b}(\boldsymbol{y}^*)) = v_i(\boldsymbol{y}^*) + t_i(\mathbf{b}) = v_i(\boldsymbol{y}^*) + \sum_{j \neq i} b_j(\boldsymbol{y}^*). \tag{18.7.25}$$

Under this mechanism, each participant i maximizes his utility:

$$\max_{\boldsymbol{y}} [v_i(\boldsymbol{y}) + \sum_{j \neq i} b_j(\boldsymbol{y})]. \tag{18.7.26}$$

When each individual i reports his true value function, i.e., $b_i(\boldsymbol{y}) = v_i(\boldsymbol{y})$, his maximization problem (18.7.26) and the social surplus maximization problem (18.7.21) are exactly the same. By reporting $b_i(\boldsymbol{y}) = v_i(\boldsymbol{y})$, agent i ensures that the designer will choose $\boldsymbol{y}^*$, which also maximizes agent i's payoff while the designer maximizes the social welfare. In other words, individuals' interests are compatible with the social interest that is determined by the Lindahl-Samuelson condition. Thus, truth-telling, $b_i(\boldsymbol{y}) = v_i(\boldsymbol{y}))$, is a dominant strategy equilibrium.

In general, $\sum_{i=1}^n t_i(\boldsymbol{b}(\boldsymbol{y})) \neq 0$, which means that a VCG mechanism does not result in Pareto efficient allocations even if it results in an efficient provision of public goods.

As in the discrete case, the total transfer might be very large. The VCG mechanism can then be modified to

$$t_i(\boldsymbol{b}) = \sum_{j\neq i} b_j(\boldsymbol{y}) + d_i(\boldsymbol{b}_{-i}),$$

where $d_i(\mathbf{b}_{-i})$ is an arbitrary function that is independent of $\mathbf{b}_i$.

The general form of the VCG mechanism is then $\Gamma = \langle V, t, \boldsymbol{y}(\boldsymbol{b})\rangle$, such that

(1) $\sum_{i=1}^n b_i(\boldsymbol{y}(\boldsymbol{b})) \geqq \sum_{i=1}^n b_i(\boldsymbol{y})$, for $\boldsymbol{y} \in Y$;

(2)$t_i(\mathbf{b}) = \sum_{j\neq i} b_j(\boldsymbol{y}) + d_i(\mathbf{b}_{-i})$.

We then have the following proposition:

Proposition 18.7.2 *For continuous public goods economies with private values, truth-telling is a dominant strategy under the VCG mechanism, and thus it truthfully implements the efficient decision rule $\boldsymbol{y}^*(\cdot)$ (the efficient provision of public goods) in dominant strategies.*

A special case of the VCG mechanism is independently described by Clark and is called the Clark mechanism (also called the pivotal mechanism), in which $d_i(\mathbf{b}_{-i}(\boldsymbol{y}))$ is given by

$$d_i(\mathbf{b}_{-i}) = -\max_{\boldsymbol{y}} \sum_{j\neq i} b_j(\boldsymbol{y}). \quad (18.7.27)$$

In other words, the pivotal mechanism, $\Gamma = \langle V, t, \boldsymbol{y}(\mathbf{b})\rangle$, is to choose $(\boldsymbol{y}^*, t_i^*)$, such that

(1) $\sum_{i=1}^n b_i(\boldsymbol{y}^*) \geqq \sum_{i=1}^n b_i(\boldsymbol{y}), \forall \boldsymbol{y} \in Y$;

(2) $t_i^*(\mathbf{b}) = \sum_{j\neq i} b_j(\boldsymbol{y}^*) - \max_{\boldsymbol{y}} \sum_{j\neq i} b_j(\boldsymbol{y})$.

It is interesting to point out that the Clark mechanism contains the well-known second-price auction as a special case. Under the Vickery mechanism, the highest-bid participant obtains the object, and he pays the second-highest bid price. To see this, let us explore this relationship in the case of a single good auction (Example 18.3.5 in the beginning of this chapter). In this case, the outcome space is $Z = \{\boldsymbol{y} \in \{0,1\}^n : \sum_{i=1}^n y_i = 1\}$ where $y_i = 1$ implies that agent i obtains the object, and $y_i = 0$ means that agent i does not obtain the object. Agent i's valuation function is then given by

$$v_i(\boldsymbol{y}) = v_i y_i.$$

Since we can regard y as an n-dimensional vector of public goods, by the Clark mechanism above, we know that

$$\boldsymbol{y}^* = h(\mathbf{b}) = \{\boldsymbol{y} \in Z : \max \sum_{i=1}^{n} b_i y_i\} = \{\boldsymbol{y} \in Z : \max_{i \in N} b_i\},$$

and truth-telling is a dominant strategy. Therefore, if $h_i(\boldsymbol{v}) = 1$, then $t_i(v) = \sum_{j \neq i} v_j y_j^* - \max_y \sum_{j \neq i} v_j y_j = -\max_{j \neq i} v_j$, which is the second-highest value. If $h_i(\mathbf{b}) = 0$, then $t_i(\boldsymbol{v}) = 0$. Consequently, the object is allocated to the individual with the highest valuation, and he pays an amount equal to the second-highest valuation. This is exactly the outcome predicted by the Vickrey auction format.

As mentioned previously, even if the dominant equilibrium is unique, or different dominant equilibria lead to the same equilibrium outcome, there may be other equilibrium outcomes under other solution concepts. Such an example is given below.

Example 18.7.1 In the second-price auction mechanism, besides that truth-telling is the dominant strategy equilibrium that results in an efficient outcome of the object, there is also a Nash equilibrium that results in an inefficient outcome, i.e., the agent with the highest valuation does not win the object.

To see this, without loss of generality, assume that the valuations of the object satisfy $v_1 > v_2 > \cdots > v_n > 0$. Consider the following strategy $\mathbf{b}^*$:

- $b_j^* > v_1$;
- $b_1^* < v_j$;
- $b_i^* = 0$ fot all $i \notin \{1, j\}$.

Obviously, $\mathbf{b}^*$ is a Nash equilibrium. It is easy to verify that there is no agent k who has an incentive to deviate from the strategy b_k^*.

It is worth noting that, although a second-price auction truthfully implements the efficient allocation in dominant strategies, it is generally not the full implementation in dominant strategies, nor the full implementation in Nash equilibrium, nor the full ex-post implementation. Cason, Saijo, Sjostrom and Yamato (2006) demonstrated that, for private-values models, the second-price auction does not satisfy Maskin monotonicity and ex-post monotonicity, which are the necessary conditions for full implementation in Nash equilibrium and in ex-post equilibrium, respectively.

However, for the interdependent-value model, although the second-price auction does not satisfy Maskin's monotonicity, it satisfies ex-post monotonicity and incentive compatibility, so that it is fully implementable in ex-post equilibrium (see Bergemann and Morris, 2008).

18.7.3 Uniqueness of VCG Mechanism

Do other mechanisms exist that truthfully implement the efficient decision rule $\boldsymbol{y}^*$? The answer is no if valuation functions $\boldsymbol{v}(\boldsymbol{y}, \cdot)$ are sufficiently "rich" in the sense of denseness.[5] Any mechanism that truthfully implements an efficient decision rule $y^*(\cdot)$ in dominant strategies must be a VCG mechanism. To show this result, consider the class of parametric valuation functions $v_i(\boldsymbol{y}, \cdot)$ with a continuum type space $\Theta_i = [\underline{\theta}, \bar{\theta}]$.

Proposition 18.7.3 (Laffont and Maskin (1980)) *Suppose that $Y = \mathcal{R}$, $\Theta = [\underline{\theta}, \bar{\theta}]$, and $v_i : Y \times \Theta_i \to \mathcal{R}$, $\forall i \in N$ is differentiable. Then, any mechanism that truthfully implements an efficient decision rule $y^*(\cdot)$ in dominant strategies is a VCG mechanism. In other words, if a social choice rule $f(\theta) = \{y(\theta), t_1(\theta), \cdots, t_n(\theta)\}$ is truthfully implemented in dominant strategies, and*

$$y(\theta) = \arg\max_{y \in Y} \sum_{i=1}^{n} v_i(y, \theta_i),$$

then we must have $t_i(\theta) = \sum_{j \neq i} v_j(y(\theta), \theta_j) + d_i(\theta_{-i})$, where $d_i(\theta_{-i})$ is some function that is independent of θ_i.

PROOF. Since

$$y(\theta) = arg \max_{y \in Y} \sum_{i=1}^{n} v_i(y, \theta_i),$$

we have

$$\sum_{i=1}^{n} \frac{\partial v_i}{\partial y}(y(\theta), \theta_i) = 0. \tag{18.7.28}$$

By the requirement of truthful implementation in dominant strategies,

$$v_i(y(\theta_i, \theta_{-i}), \theta_i) + t_i(\theta_i, \theta_{-i}) \geqq v_i(y(\hat{\theta}_i, \theta_{-i}), \theta_i) + t_i(\hat{\theta}_i, \theta_{-i}), \forall \theta_i, \hat{\theta}_i, \theta_{-i}.$$

Then

$$\frac{\partial v_i}{\partial y}(y(\theta), \theta_i) \frac{\partial y}{\partial \theta_i}(\theta) + \frac{\partial t_i}{\partial \theta_i}(\theta_i, \theta_{-i}) = 0, \forall (\theta_i, \theta_{-i}) \in \Theta. \tag{18.7.29}$$

Let $d_i(\theta) \triangleq t_i(\theta) - \sum_{j \neq i} v_j(y(\theta), \theta_j)$. We need to show that $d_i(\theta_i, \theta_{-i})$ is independent of θ_i. Indeed, since

$$\begin{aligned}
\frac{\partial d_i}{\partial \theta_i} =& \frac{\partial t_i}{\partial \theta_i}(\theta_i, \theta_{-i}) - \sum_{j \neq i} \frac{\partial v_i}{\partial y}(y(\theta), \theta_j) \frac{\partial y}{\partial \theta_i}(\theta_i, \theta_{-i}) \\
=& \frac{\partial t_i}{\partial \theta_i}(\theta_i, \theta_{-i}) + \frac{\partial v_i}{\partial y}(y(\theta), \theta_j) \frac{\partial y}{\partial \theta_i}(\theta_i, \theta_{-i}) - \sum_{j=1}^{n} \frac{\partial v_j}{\partial y}(y(\theta), \theta_j) \frac{\partial y}{\partial \theta_i}(\theta_i, \theta_{-i}) \\
=& 0
\end{aligned} \tag{18.7.30}$$

[5] A subset A of a topological space X is called **dense** (in X) if every point x in X either belongs to A or is a limit point of A.

by using (18.7.28) and (18.7.29), we have $d_i(\theta) = d_i(\theta_{-i})$. $\square$

The intuition behind the proof is that each agent i has the incentive to tell the truth only when he is given the residual claimant of total surplus, i.e., $U_i(\theta) = \sum_{i=1}^n v_i(y(\theta), \theta_i) + d_i(\theta_{-i})$, which is a feature of the VCG mechanism.

This result was also obtained by Green and Laffont (1979), but under less restrictive assumptions, i.e., they allowed agents to have all possible valuations over a finite decision set X, which can be described by the Euclidean type space $\Theta_i = \mathcal{R}^{|K|}$ (the valuation of agent i for decision k is represented by θ_{ik}). The proof can be seen in Mas-Colell, Whinston and Green (1995) without imposing differentiability on v_i.

18.7.4 Balanced VCG Mechanisms

Truthful implementation of a social surplus-maximizing decision rule $y^*(\cdot)$ is a necessary (but not sufficient) condition for truthful implementation of (ex-post) Pareto efficiency. To make it sufficient for Pareto efficiency, we must also ensure that there is an ex-post balanced budget (no waste of numeraire goods). Specifically, for all $\mathbf{b} \in V$,

$$\sum_{i=1}^{n} t_i(\mathbf{b}) = 0. \tag{18.7.31}$$

Substituting t_i in the VCG mechanism into (18.7.31) leads to Pareto efficient allocations if and only if

$$(n-1)\sum_{i=1}^{n} v_i(y(\boldsymbol{v})) + \sum_{i=1}^{n} d_i(\mathbf{b}_{-i}) \equiv 0. \tag{18.7.32}$$

A balanced budget implies that there is no waste of resources. For example, in order to build a public project, the government levies taxes on some residents and subsidizes some residents who are adversely affected. If the total tax revenue is equal to the total subsidies, the government's use of funds is efficient; otherwise, if $\sum_{i=1}^n t_i(\theta) < 0$, then there is a budget surplus, and the implementation of decision rule $y(\cdot)$ is not Pareto efficient. A Pareto optimal allocation must satisfy both social surplus maximization (18.7.19) and the balanced budget condition (18.7.20). A pertinent question is then: can we find a balanced VCG mechanism? Green and Laffont (1979), Holmstrom (1979), and Laffont and Maskin (1980) all gave a negative answer.

Proposition 18.7.4 (Holmstrom (1979) and Green and Laffont (1979)) *Suppose that $V(\theta) \triangleq \max_{y \in \mathcal{R}} \sum_{i=1}^n v_i(y, \theta_i)$. Then, the VCG mechanism is budget-balanced (thus resulting in a Pareto efficient allocation) if and only if*

$$\frac{\partial^n V(\theta)}{\partial \theta_1 \cdots \partial \theta_n} = 0, \forall \theta \in \Theta, \tag{18.7.33}$$

where $\frac{\partial^{n-1}}{\partial\theta_{-i}} = \frac{\partial^{n-1}}{\partial\theta_1\cdots\partial\theta_{i-1}\partial\theta_{i+1}\cdots\partial\theta_n}$.

PROOF. **Necessity**: If $\sum_{i=1}^n t_i(\theta) = 0$, by the definition of VCG mechanism, we have

$$\sum_{i=1}^n \sum_{j\neq i} v_j(y(\theta),\theta_j) + \sum_{i=1}^n d_i(\theta_{-i}) = (n-1)V(\theta) + \sum_{i=1}^n d_i(\theta_{-i}) = 0.$$

It is easy to verify that $\frac{\partial^n}{\partial\theta_1\cdots\partial\theta_n}\sum_{i=1}^n d_i(\theta_{-i}) = 0$, and thus $\frac{\partial^n V(\theta)}{\partial\theta_1\cdots\partial\theta_n} = 0$.

Sufficiency: If $\frac{\partial^n V(\theta)}{\partial\theta_1\cdots\partial\theta_n} = 0$, there must be a series of functions $d_i(\theta_{-i}), i = 1,\cdots,n$, such that $V(\theta) = \sum_{i=1}^n d_i(\theta_{-i})$. Let

$$t_i(\theta) = \sum_{j\neq i} v_i(y(\theta),\theta_j) - (n-1)d_i(\theta_{-i}).$$

Then

$$\sum_{i=1}^n t_i(\theta) = (n-1)V(\theta) - (n-1)\sum_{i=1}^n d_i(\theta_{-i}) = 0.$$

□

A trivial case for which we can achieve ex-post Pareto efficiency is given in the following example.

Example 18.7.2 If there exists some agent i, such that $\Theta_i = \{\bar{\theta}_i\}$, a singleton, then there is no incentive issue for agent i; and we can set

$$t_i(\theta) = -\sum_{j\neq i} t_j(\theta), \forall\theta\in\Theta.$$

This trivially guarantees a balanced budget.

However, in general, there is no such positive result. When the set of $v(\cdot,\cdot)$ is dense, then there may be no social choice rule $f(\cdot) = (y^*(\cdot), t_1(\cdot),\cdots,t_n(\cdot))$, where $y^*(\cdot)$ is ex post Pareto optimal, which is truthfully implementable in dominant strategies and also satisfies condition (18.7.31).

Proposition 18.7.5 *Suppose that $\mathcal{V} = \{v_i(\cdot,\theta_i) : \theta_i\in\Theta_i\}$ is the set of all admissible utility functions that are twice continuously differentiable with* $\dfrac{\partial^2 v_i}{\partial y^2} < 0$ *and* $\dfrac{\partial^2 v_i}{\partial y\partial\theta_i} \neq 0$ *at all $(y,\theta_i)\in\mathcal{R}_+\times[\underline{\theta},\bar{\theta}]$. Then, there is no such balanced* VCG *mechanism.*

PROOF. Obviously, the necessary and sufficient conditions for a balanced budget (18.7.33) cannot be satisfied, in general. For instance, if $n = 2$, by the Envelope Theorem, we have

$$\frac{\partial V(\theta)}{\partial\theta_1} = \frac{\partial v_1(y(\theta_1,\theta_2),\theta_1)}{\partial\theta_1}.$$

By the first-order condition, we have

$$\frac{\partial v_1(y(\theta_1,\theta_2),\theta_1)}{\partial y}+\frac{\partial v_2(y(\theta_1,\theta_2),\theta_1)}{\partial y}=0.$$

Then, by the Implicit Function Theorem, we obtain

$$\frac{\partial y(\theta_1,\theta_2)}{\partial \theta_2}=-\frac{\frac{\partial^2 v_2(y(\theta_1,\theta_2),\theta_1)}{\partial y\partial\theta_2}}{\frac{\partial^2 v_1(y(\theta_1,\theta_2),\theta_1)}{\partial y^2}+\frac{\partial^2 v_2(y(\theta_1,\theta_2),\theta_1)}{\partial y^2}}\neq 0$$

by noting that $\frac{\partial^2 v_i}{\partial y\partial\theta_i}\neq 0$ and $\frac{\partial^2 v_i}{\partial y^2}<0$.

Therefore,

$$\frac{\partial^2 V(\theta)}{\partial\theta_1\partial\theta_2}=\frac{\partial^2 v_1(y(\theta_1,\theta_2),\theta_1)}{\partial y\partial\theta_1}\frac{\partial y(\theta_1,\theta_2)}{\partial\theta_2}\neq 0. \tag{18.7.34}$$

□

However, when the economic environment is particularly small (nowhere dense in the functional space of utility functions), we may have a positive result, i.e., there is a balanced VCG mechanism. Groves and Loeb (1975) demonstrated this for a quadratic utility function form

$$v_i(y,\theta_i)=\theta_i y-y^2/2$$

when $n>2$.

Then,

$$V(\theta)=\max_{y\in\mathcal{R}}\sum_{i=1}^n v_i(y,\theta_i)=\frac{1}{2n}(\sum_{i=1}^n\theta_i)^2.$$

So,

$$\frac{\partial^n V(\theta)}{\partial\theta_1\cdots\partial\theta_n}=0,\forall\theta\in\Theta,\forall n\geqq 3.$$

Tian (1996a), and Liu and Tian (1999) obtained similar results for more general valuation functions

$$\{v_i(\boldsymbol{y},\theta_i)=\psi_i(\theta_i)\phi(\boldsymbol{y})-(b\phi(\boldsymbol{y})+c)^\alpha\}_{i=1}^n$$

and

$$V_i(\boldsymbol{y},\theta_i)=\psi_i(\theta_i)\phi(\boldsymbol{y})-G_i(\boldsymbol{y})+\varphi_i(\theta_i).$$

However, if we weaken the solution concept to Bayesian-Nash equilibrium, as shown in the next chapter, given the probability distribution φ on the state θ, if the transfer for every agent i, when others report truthfully, is given by

$$t_i(\hat{\boldsymbol{\theta}})=E_{\boldsymbol{\theta}_{-i}}[\sum_{j\neq i}v_j(y(\hat{\theta}_i,\boldsymbol{\theta}_{-i}),\theta_j)]+d_i(\hat{\boldsymbol{\theta}}_{-i}),$$

then we can achieve Pareto efficiency.

Remark 18.7.3 Although the VCG mechanism provides public goods efficiently, it generally does not satisfy the balanced budget requirement, and is therefore not Pareto efficient. Shao and Zhou (2016a, 2016b) proved that, when adopting certain weighted average criterion (taking expectations of the social welfare function), the VCG mechanism was not the most efficient because participants need to pay additional funds to ensure the efficient provision of public goods. If the cost of efficiently providing public goods was also taken into account, Shao and Zhou (2016a) showed that the most efficient mechanism was a certain voting mechanism. Moreover, for the indivisible goods environment with two participants, Shao and Zhou (2016b) demonstrated that the most efficient mechanism was a fixed price mechanism or an option mechanism, both of which satisfy the balanced budget requirement and incentive compatibility in dominant strategies.

18.8 Nash Implementation

18.8.1 Nash Equilibrium and Nash Implementation

Hurwicz's Impossibility Theorem yields that one must give up the dominant strategy implementation in order to have a mechanism that implements Pareto efficient and individually rational allocations. Then, we need consider more general mechanisms, $\langle M, h\rangle$, instead of using a revelation mechanism.

If we adopt Nash equilibrium as a solution concept, can we design an incentive compatible mechanism which implements Pareto efficient and individually rational allocations? The answer is yes, and this is the problem that we will consider in this section.

Recall that a Nash strategy means that each participant takes other participants' strategies as given and chooses the strategy that is most beneficial to him. A strategy is Nash equilibrium if and only if everyone's equilibrium strategy is the best response to other people's equilibrium strategy. Once such an equilibrium is reached, no one has an incentive to deviate from their choices. An important feature of Nash equilibrium is the consistency between belief and choice: A choice based on a belief is rational, and the belief supporting this choice is correct. Nash equilibrium thus has the characteristic of predictive self-enforcement.

Formally, we have

Definition 18.8.1 For $\mathbf{e} \in E$, a mechanism $\langle M, h\rangle$ is said to have a **Nash equilibrium** $\mathbf{m}^* \in M$ if

$$h_i(\mathbf{m}^*) \succcurlyeq_i h_i(m_i, \mathbf{m}^*_{-i}) \tag{18.8.35}$$

for all $m_i \in M_i$ and all i.

Denote by $\mathcal{N}(\mathbf{e}, \Gamma)$ the set of all Nash equilibria of the mechanism Γ for $e \in E$. It is obvious that every dominant strategy equilibrium is a Nash equilibrium, but the converse may not be true.

Definition 18.8.2 A mechanism $\Gamma = \langle M, h \rangle$ is said to **fully Nash implement** a social choice correspondence F on E if for every $\mathbf{e} \in E$,

(a) $\mathcal{N}(\mathbf{e}, \Gamma) \neq \emptyset$;

(b) $h(\mathcal{N}(\mathbf{e}, \Gamma)) = F(\mathbf{e})$.

We say it **Nash implements** a social choice correspondence F on E if for every $e \in E$,

(a) $\mathcal{N}(\mathbf{e}, \Gamma) \neq \emptyset$;

(b) $h(\mathcal{N}(\mathbf{e}, \Gamma)) \subseteq F(\mathbf{e})$.

The following proposition shows that if, for a revelation mechanism, truth-telling is a Nash equilibrium for all economies in the admissible set E, it must be a dominant strategy equilibrium of the mechanism. Consequently, for a revelation mechanism, the dominant equilibrium and the Nash equilibrium are equivalent.

Proposition 18.8.1 *For a revelation mechanism $\Gamma = \langle E, h \rangle$, truth-telling is a Nash equilibrium for all $\mathbf{e} \in E$ if and only if it is a dominant strategy equilibrium.*

PROOF. A dominant strategy equilibrium is clearly a Nash equilibrium, and then we only need to show that every Nash equilibrium is also a dominant strategy equilibrium. For every $\mathbf{e} \in E$ and i, by Nash equilibrium, we have

$$h(e_i, \mathbf{e}_{-i}) \succcurlyeq_i h(e'_i, \mathbf{e}_{-i}) \quad \text{for all } e'_i \in E_i.$$

Since $(e'_i, \mathbf{e}_{-i})$ is arbitrary, it is a dominant strategy equilibrium. □

Thus, we cannot obtain any new results if we are using revelation mechanisms. To obtain positive results, one has to give up the revelation mechanism and look for a more general mechanism with general message spaces. Indeed, when we consider a general message space and adopt Nash equilibrium as the solution concept, it is possible to achieve both incentive compatibility and Pareto efficiency.

Note that, when one adopts Nash equilibrium as the solution concept, partial implementation in Nash solution may be a too-weak requirement, which may result in too many Nash equilibria. To see this, consider any social choice correspondence F and the following mechanism: each individual's message space consists of the set of economic environments, i.e., it is given by $M_i = E \times Z$ with a generic element $\boldsymbol{m} = (\boldsymbol{e}, a)$. The outcome function is defined as $h(\mathbf{m}) = a$ when all agents report the same message $m_i = (\boldsymbol{e}, a)$ with $a \in A \subseteq Z$; otherwise, each agent is seriously punished by offering a worse outcome. Then, it is clear that the profile in which all agents report their true characteristics e and an outcome $a \in A$ is a Nash equilibrium for e. However, it has many, in fact infinitely many, other Nash equilibria. For instance, if they all report a false economic environment $\boldsymbol{e}'$ and an outcome $a' \in A$, so that $m_i = (\boldsymbol{e}', a')$, then this also forms a Nash equilibrium for e. Therefore, when Nash equilibrium is adopted as the solution concept, implementation or full implementation should be required.

18.8.2 Characterization of Nash Implementation

Now, we discuss what social choice rules can be (fully) Nash implementable. Maskin in 1977 provided necessary and sufficient conditions for a social choice rule to be fully Nash implementable (the paper was not published until 1999 due to some incompleteness in the proof). Maskin's result is fundamental since it not only assists us to understand what social choice correspondences can be fully Nash implementable, but also provides basic techniques and methods in characterizing the implementability of a social choice rule under many other solution concepts. Maskin used the proof of Repello (1987) when his paper was formally published in 1999. Because Maskin's work on characterization of full Nash implementation of any social choice rule had a profound influence on the development of mechanism design theory, he won the Nobel Prize in Economics with Hurwicz and Myerson in 2007.

The necessary condition for full Nash implementation, called Maskin monotonicity, is defined at the beginning of this chapter. In order to facilitate the comparison of different expressions, we provide them in slightly different forms below. It is not difficult to see that they are equivalent. The monotonic illustration is shown in Figure 18.3.

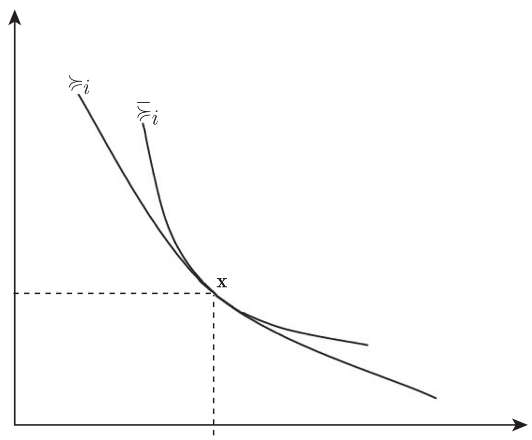

Figure 18.3: An illustration of Maskin monotonicity

Definition 18.8.3 (Maskin Monotonicity) A social choice correspondence $F : E \to\to A$ is said to be **Maskin monotonic** if for any $\mathbf{e}, \bar{\mathbf{e}} \in E$ and $\mathbf{x} \in F(\mathbf{e})$, such that for all i and all $\mathbf{y} \in A$, $\mathbf{x} \succcurlyeq_i \mathbf{y}$ implies that $\mathbf{x} \bar{\succcurlyeq}_i \mathbf{y}$, then $\mathbf{x} \in F(\bar{\mathbf{e}})$.

In other words, Maskin monotonicity requires that if an outcome $\mathbf{x}$ is socially optimal in economy $\mathbf{e}$, switching from $\mathbf{e}$ to $\bar{\mathbf{e}}$ makes all individuals even more like $\mathbf{x}$, and then $\mathbf{x}$ remains socially optimal in $\bar{\mathbf{e}}$. Below is an equivalent expression of Maskin monotonicity.

Definition 18.8.4 (A Variation of Maskin Monotonicity) An equivalent condition for a social choice correspondence $F : E \to\to A$ to be Maskin monotonic is that, if for any two economic environments $\mathbf{e}, \bar{\mathbf{e}} \in E$, $\mathbf{x} \in F(\mathbf{e})$ but $\mathbf{x} \notin F(\bar{\mathbf{e}})$, then there is an agent i and another $\mathbf{y} \in A$, such that $\mathbf{x} \succcurlyeq_i \mathbf{y}$ and $\mathbf{y} \bar{\succ}_i \mathbf{x}$.

This means that if the outcome $\mathbf{x}$ is a socially optimal choice in economy $\mathbf{e}$, but when the environment switches from $\mathbf{e}$ to $\bar{\mathbf{e}}$, x is no longer a socially optimal choice, then $\mathbf{x}$ is not the best alternative, at least for some participant.

Maskin monotonicity is a weak condition, and many well-known social choice rules satisfy Maskin monotonicity.

Example 18.8.1 (Weak Pareto Efficiency) Weak Pareto optimal correspondence $P_w : E \to\to A$ is Maskin monotonic.

PROOF. If $\mathbf{x} \in P_w(\mathbf{e})$, then for all $\mathbf{y} \in A$, there exists $i \in N$, such that $\mathbf{x} \succcurlyeq_i \mathbf{y}$. Now, if for any $j \in N$, $\mathbf{x} \succcurlyeq_j \mathbf{y}$ implies that $\mathbf{x} \bar{\succcurlyeq}_j \mathbf{y}$ for all $\mathbf{y} \in A$. Then, for particular i, we have $\mathbf{x} \bar{\succcurlyeq}_i \mathbf{y}$ for all $\mathbf{y} \in A$. Thus, we must have $\mathbf{x} \in P_w(\bar{\mathbf{e}})$. □

Example 18.8.2 (Majority Rule) Majority rule, also called the Condorcet rule, $CON : E \to\to A$ for strict preference profile, which is defined by

$$CON(\mathbf{e}) = \{\mathbf{x} \in A : \#\{i|\mathbf{x} \succ_i \mathbf{y}\} \geqq \#\{i|\mathbf{y} \succ_i \mathbf{x}\} \text{ for all } \mathbf{y} \in A\},$$

is Maskin monotonic. [6]

PROOF. If $\mathbf{x} \in CON(\mathbf{e})$, then for all $\mathbf{y} \in A$,

$$\#\{i|\mathbf{x} \succ_i \mathbf{y}\} \geqq \#\{i|\mathbf{y} \succ_i \mathbf{x}\}. \tag{18.8.36}$$

[6] It was proposed by the 18th century French mathematician and philosopher Marquis de Condorcet. The basic procedure for an election is: Each voter must make a rank for all of the candidates, in the order in which the most favorite one is ranked first. For example, if there are three candidates, denoted A, B, and C, you must mark 1, 2, and 3 under their names, respectively, not just a circle or a cross mark. After all of the voters have been marked, the two candidates are compared in this order to see who is ranked first, and the person who wins the most votes is elected. Of course, as seen in Chapter 12, there may be a Condorcet paradox.

Also, if $\bar{\mathbf{e}}$ is an economy, such that for all i, $\mathbf{x} \succ_i \mathbf{y}$ implies that $\mathbf{x} \bar{\succ}_i \mathbf{y}$, then the left-hand side of (18.8.36) cannot fall when we replace $\mathbf{e}$ by $\bar{\mathbf{e}}$. Furthermore, if the right-hand side rises, then we must have $\mathbf{x} \succ_i \mathbf{y}$ and $\mathbf{y} \bar{\succ}_i \mathbf{x}$ for some i, which is a contradiction of the relation between $\mathbf{e}$ and $\bar{\mathbf{e}}$, given the strictness of preferences. Therefore, $\mathbf{x}$ is still a majority winner with respect to $\bar{\mathbf{e}}$, i.e., $\mathbf{x} \in CON(\bar{\mathbf{e}})$. $\square$

In addition to the above two examples, constrained Walrasian correspondence, Walrasian correspondence and Lindahl correspondence with interior allocations are Maskin monotonic. The class of preferences that satisfy the "single-crossing" property and individuals' preferences over lotteries satisfying the von Neumann-Morgenstern axioms also automatically satisfy Maskin monotonicity.

The following theorem shows that Maskin monotonicity is a necessary condition for any social rule to be fully Nash implementable.

Theorem 18.8.1 *For a social choice correspondence $F : E \to\to A$, if it is fully Nash implementable, then it must be Maskin monotonic.*

PROOF. Suppose that a social choice correspondence is fully Nash implementable. Then, there exists an implementable mechanism $\Gamma = \langle M, h \rangle$, such that $h[\mathcal{N}(\mathbf{e}, \Gamma)] = F(\mathbf{e})$ for all $\mathbf{e} \in E$. For any two economies $\mathbf{e}, \bar{\mathbf{e}} \in E$, $\mathbf{x} \in F(\mathbf{e})$, then by full Nash implementability of F, there is $\mathbf{m} \in M$, such that $\mathbf{m}$ is a Nash equilibrium for e and $\mathbf{x} = h(\mathbf{m})$. This means that $h(\mathbf{m}) \succcurlyeq_i h(m_i', \mathbf{m}_{-i})$ for all i and $m_i' \in M_i$. Given $\mathbf{x} \succcurlyeq_i \mathbf{y}$ implies that $\mathbf{x} \bar{\succcurlyeq}_i \mathbf{y}$, $h(\mathbf{m}) \bar{\succcurlyeq}_i h(m_i', \mathbf{m}_{-i})$, which means that $\mathbf{m}$ is also a Nash equilibrium for $\bar{\mathbf{e}}$. Thus, by full Nash implementability again, we have $\mathbf{x} \in F(\bar{\mathbf{e}})$. $\square$

We can use another equivalent definition of Maskin monotonicity to prove the necessity of full Nash implementation. Consider the economic environment $\mathbf{e}$ and allocation outcome $\mathbf{x} \in F(\mathbf{e})$. Since F can be fully Nash implementable, there exists a Nash equilibrium $\mathbf{m} \in \mathcal{N}(e, \Gamma)$ making $h(\mathbf{m}) = \mathbf{x}$. If there exists an another economic environment $\bar{\mathbf{e}}$ making $\mathbf{x} \notin F(\bar{\mathbf{e}})$, then the fact that $\Gamma = \langle M, h \rangle$ fully Nash complements F implies that $\mathbf{m}$ is not a Nash equilibrium for $\bar{\mathbf{e}}$. Then, there must be an i and $\bar{m}_i$, making $h(\bar{m}_i, \mathbf{m}_{-i}) \bar{\succ}_i h(\mathbf{m})$. Since $\mathbf{m}$ is a Nash equilibrium in $\mathbf{e}$, we have $h(\mathbf{m}) \succcurlyeq_i h(\bar{m}_i, \mathbf{m}_{-i})$. Let $\mathbf{y} = h(\bar{m}_i, \mathbf{m}_{-i})$. As a consequence, we have $\mathbf{x} \succcurlyeq_i \mathbf{y}$, but $\mathbf{y} \bar{\succ}_i \mathbf{x}$.

It is worth noting that, despite the fact that Maskin monotonicity as a necessary condition for any social goal to be fully Nash implementable is a very weak condition, certain restrictions must be imposed on the social choice rules. For instance, Lemma 18.5.1 shows that, with the unrestricted domain, any social choice rule that satisfies Maskin monotonicity and unanimity must be dictatorial. Hurwicz and Schmeidler (1978) proved that if the domain was not restricted, the social choice rules that satisfy Maskin monotonicity were not necessarily Pareto efficient.[7] Saijo (1987) proved that on an unrestricted domain, any single-valued social choice rule must be constant, i.e., $f(\mathbf{e}) = a, \forall \mathbf{e} \in \Theta$.[8]

In addition, Maskin monotonicity per se cannot guarantee that a social choice correspondence is fully Nash implementable. However, under the so-called no-veto power, it becomes sufficient. Recall that a social choice correspondence $F : E \to\to A$ satisfies **no veto power (NVP)** if whenever for any i and $\mathbf{e}$, such that $\mathbf{x} \succcurlyeq_j \mathbf{y}$ for all $\mathbf{y} \in A$ and all $j \neq i$, $\mathbf{x} \in F(\mathbf{e})$. In other words, if $n-1$ agents regard an outcome as their best choice, then it is socially optimal.

The no veto power property is a very weak condition. It is satisfied by many "standard" social choice rules, including weak Pareto efficient and Condorcet correspondences. It is also satisfied by many social choice rules when individual preferences are restricted. For example, for private goods economies with at least three agents, if each agent's utility function is strongly monotone, then there is no other allocation, such that it is optimal for more than one agent, and thus the no veto power condition holds. The following theorem is given by Maskin (1977, 1999).

Theorem 18.8.2 *Under the no-veto power property, if the Maskin monotonicity condition is satisfied, then F is fully Nash implementable.*

PROOF. The proof is by construction. For each agent i, his message space is defined by

$$M_i = E \times A \times N,$$

where $N = \{1, 2, \ldots, \}$. Its elements are denoted by $m_i = (\mathbf{e}^i, \mathbf{a}^i, \mathbf{v}^i)$, which means that agent i announces a profile of economic characteristics of all agents including himself, an outcome, and an integer. Note that we have used $\mathbf{e}^i$ and $\mathbf{a}^i$ to denote the profiles of all individuals' economic characteristics and outcomes, but not just agent i's economic characteristic and outcome.

The outcome function is constructed in three cases:

Case (1). If $m_1 = m_2 = \ldots = m_n = (\mathbf{e}, \mathbf{a}, \mathbf{v})$ and $\mathbf{a} \in F(\mathbf{e})$, the outcome function is defined by

$$h(\mathbf{m}) = \mathbf{a}.$$

---

[7]This conclusion is proved as follows. Because the domain is unrestricted, there is $e \in E$ making $\bigcap_{i=1}^n M_i(e) = \emptyset$. Suppose that $f(e) = a$. Then, there must be some individuals $j \in I$ and alternative $b \in \mathcal{A}$, such that $b \succ_i^e a$. Suppose that under the other condition e': (i) individual j has the same preference with condition e; and (ii) for individual $i \neq j$, $a \sim_i^{e'} b$ and any other alternatives except b have the same rank. Obviously, $L_i(a,e) \subseteq L_i(a,e'), \forall i$, and by monotonicity, $f(e') = a$. However, this choice is obviously not Pareto efficient since $b \sim_i^e a, \forall i \neq j, b \succ_j^{e'} a$.

[8]Suppose that there exist $e, e' \in E$ making $f(e) = a \neq a' = f(e')$, then we can build a new environment $e'' \in E$ satisfying: $a \sim_i^{e''} a' \succcurlyeq_i^{e''} b, \forall b \notin \{a, a'\}$, and according to monotonicity $\{a, a'\} \subseteq f(e'')$. This obviously contradicts to Maskin monotonicity.

In other words, if players are unanimous in their strategy, and alterative $\mathbf{a}$ proposed is F-optimal given their economic characteristics $\mathbf{e}$, then outcome is $\mathbf{a}$.

Case (2). For all $j \neq i$, $m_j = (\mathbf{e}, \mathbf{a}, \mathbf{v})$, $m_i = (\mathbf{e}^i, \mathbf{a}^i, \mathbf{v}^i) \neq (\mathbf{e}, \mathbf{a}, \mathbf{v})$, and $\mathbf{a} \in F(\mathbf{e})$, define:

$$h(m) = \begin{cases} a^i & \text{if } a^i \in L(a, e_i), \\ a & \text{if } a^i \notin L(a, e_i), \end{cases}$$

where $L(\mathbf{a}, e_i) = \{\mathbf{b} \in A : \mathbf{a}\ R_i\ b\}$ is the lower contour set of R_i at $\mathbf{a}$. In other words, suppose that all agents but one play the same strategy and, given their characteristics, their proposed alternative $\mathbf{a}$ is F-optimal. Then, the outcome is determined by the worse one among $\mathbf{a}$ and $\mathbf{a}_i$ according to agent i's preference $\succcurlyeq_i$. As such, no one can gain by deviating from the unanimous strategy.

Case (3). If neither Case (1) nor Case (2) applies, then define

$$h(\mathbf{m}) = \mathbf{a}^{i*},$$

where $i* = \max\{i \in \mathcal{N} : \mathbf{v}^i = \max_j \mathbf{v}^j\}$. In other words, when neither Case (1) nor Case (2) applies, the outcome is the alternative proposed by the player with the highest index among those whose proposed number is maximum.

Now, we show that the mechanism $\langle M, h \rangle$ defined above fully Nash implements social choice correspondence F, i.e., $h(\mathcal{N}(\mathbf{e})) = F(\mathbf{e})$ for all $\mathbf{e} \in E$. We first demonstrate that $F(\mathbf{e}) \subseteq h(\mathcal{N}(\mathbf{e}))$ for all $\mathbf{e} \in E$, i.e., it is necessary to show that for all $\mathbf{e} \in E$ and $\mathbf{a} \in F(\mathbf{e})$, there exists a $\mathbf{m} \in M$, such that $\mathbf{a} = h(\mathbf{m})$ is a Nash equilibrium outcome. To do so, we only need to show that any $\mathbf{m}$, which is given by Case (1), is a Nash equilibrium. Note that $h(\mathbf{m}) = \mathbf{a}$ and for any given $m_i' = (e'^i, a'^i, v'^i) \neq m_i$, by Case (2), we have

$$h(m_i', \mathbf{m}_{-i}) = \begin{cases} \mathbf{a}^i & \text{if } \mathbf{a}^i \in L(\mathbf{a}, e_i), \\ \mathbf{a} & \text{if } \mathbf{a}^i \notin L(\mathbf{a}, e_i), \end{cases}$$

and thus

$$h(\mathbf{m})\ R_i\ h(m_i', \mathbf{m}_{-i}) \quad \forall m_i' \in M_i.$$

Therefore, $\mathbf{m}$ is a Nash equilibrium for e.

We now show that if $\mathbf{m}$ is a Nash equilibrium for $\mathbf{e}'$, then $h(\mathbf{m}) \in F(\mathbf{e}')$. First, consider Nash equilibrium $\mathbf{m}$ given by Case (1), so that $a \in F(\mathbf{e})$, but the true economic environment is $\mathbf{e}'$, i.e., $\mathbf{m} \in \mathcal{N}(\mathbf{e}', \Gamma)$. We need to show $h(\mathbf{m}) = \mathbf{a} \in F(\mathbf{e}')$. For any $\mathbf{b} \in L(\mathbf{a}, e_i)$ (so that $\mathbf{a} R_i \mathbf{b}$), let $m_i' = (\mathbf{e}, \mathbf{b}, \mathbf{v}^i)$. Then, $h(m_i', \mathbf{m}_{-i}) = \mathbf{b}$ by Case (2). Now, since $\mathbf{a}$ is a Nash equilibrium outcome in $\mathbf{e}'$, we have $\mathbf{a} = h(\mathbf{m})\ R_i'\ h(m_i', \mathbf{m}_{-i}) = \mathbf{b}$. Thus, we have shown that for all $i \in N$ and $\mathbf{b} \in A$, $\mathbf{a}\ R_i\ \mathbf{b}$ implies that $\mathbf{a}\ R_i'\ \mathbf{b}$. Consequently, by the Maskin monotonicity condition, we have $\mathbf{a} \in F(\mathbf{e}')$.

Next, suppose that Nash equilibrium $\mathbf{m}$ for $\mathbf{e}'$ is in the Case (2), i.e., for all $j \neq i$, $m_j = (\mathbf{e}, \mathbf{a}, \mathbf{v})$, $\mathbf{a} \in F(\mathbf{e})$, and $m_i \neq (\mathbf{e}, \mathbf{a}, \mathbf{v})$. Let $\mathbf{a}' = h(\mathbf{m})$. By Case (3), each $j \neq i$ could induce any outcome $\mathbf{b} \in A$ by choosing $(\mathbf{R}^j, \mathbf{b}, \mathbf{v}^j)$ with a sufficiently large $\mathbf{v}^j$ (which is greater than $\max_{k \neq j} v_k$), as the outcome at $(m'_j, \mathbf{m}_{-j})$, i.e., $\mathbf{b} = h(m'_j, \mathbf{m}_{-j})$. Therefore, the fact that $\mathbf{m}$ is a Nash equilibrium for $\mathbf{e}'$ implies that, for all $j \neq i$,

$$\mathbf{a}' \; R'_j \; \mathbf{b}.$$

Thus, by the no-veto power assumption, we have $\mathbf{a}' \in F(\mathbf{e}')$.

Using the same argument, we can demonstrate that if $\mathbf{m}$ is a Nash equilibrium under environment $\mathbf{e}'$ given by Case (3), then we have $\mathbf{a}' \in F(\mathbf{e}')$. □

Maskin monotonicity may be violated by a social choice rules, and thus it is not fully Nash implementable. Indeed, the second-price auction is not Maskin monotonic, and thus it is not fully Nash implementable. Walrasian allocations, Lindahl allocations, and Pareto efficient allocations that contain boundary solutions do not satisfy Maskin monotonicity, and thus they are not fully Nash implementable (if they just contain interior allocations, then they are fully Nash implementable).

In the following, we demonstrate that Solomon's mechanism also violates Maskin monotonicity. Since each woman knows who is the real mother, i.e., it is a situation of complete information, Solomon's solution can be considered with Nash implementation. However, his solution to threaten the two women to cut the child into two halves does not satisfy Maskin monotonicity, and thus is not fully Nash implementable. What decision would he make if the fake mother behaved in the same way as the real mother?

Two women: Anne and Bets,

Two economies (states): $E = \{\alpha, \beta\}$, where

> α: Anne is the real mother,
>
> β: Bets is the real mother.

Solomon has four alternatives, so that the feasible set is given by $A = \{a, b, c, d\}$, where

> a: Anne gets the baby,
>
> b: Bets gets the baby,
>
> c: Cut the baby in half,
>
> d: Death penalty for all.

King Solomon's aim (social goal) is to give the baby to the real mother,

$f(\alpha) = a$ if α happens,
$f(\beta) = b$ if β happens.

Preferences of Anne and Bets:

For Anne,

at state α, $a \succ_A^\alpha b \succ_A^\alpha, c \succ_A^\alpha d$,
at state β: $a \succ_A^\beta c \succ_A^\beta b \succ_A^\beta d$.

For Bets,

at state α, $b \succ_B^\alpha c \succ_B^\alpha a \succ_B^\alpha d$,
at state β: $b \succ_B^\beta a \succ_B^\beta c \succ_B^\beta d$.

To see why Solomon's solution does not work, we only need to show that his social choice rule is not Maskin monotonic. Note that for Anne, since

$$a \succ_A^\alpha b, c, d;$$

$$a \succ_A^\beta b, c, d,$$

and $f(\alpha) = a$, by Maskin's monotonicity, we should also have $f(\beta) = a$, but we actually have $f(\beta) = b$. Thus, Solomon's social choice rule is not fully Nash implementable. However, if one adopts subgame perfect Nash equilibrium as the solution concept so that the set of equilibria becomes smaller, it is implementable.

It is worth noting that the sufficiency conditions of the above theorems apply only to the case with $n \geqq 3$. If $n = 2$, Maskin monotonicity and NVP may not ensure full Nash implementation. Below, we provide a counterexample. In the remainder of this section, let $L_i(\mathbf{a}, \theta) = \{\mathbf{b} \in A : a \succcurlyeq_i^\theta \mathbf{b}\}$ be the lower contour set for $\succcurlyeq_i^\theta$ at $\mathbf{a}$.

Example 18.8.3 Suppose that $\mathcal{A} = \{a, b\}, N = \{1, 2\}, E = \{\theta, \phi, \xi\}$, and individual preferences and the social choice rule $F(\cdot)$ are given in the table in Figure 18.4.

	θ		ϕ		ξ	
Individual 1	b	a	a	b	b	a
Individual 2	a	b	b	a	b	a
$F(\cdot)$	$F(\theta)=\{a, b\}$		$F(\phi)=\{a, b\}$		$F(\xi)=\{b\}$	

Figure 18.4: A counterexample

It is easy to verify that $F(\cdot)$ satisfies monotonicity and NVP. Suppose that there exists a mechanism $\Gamma = \langle M_1 \times M_2, h\rangle$ that can fully Nash complement $f(\cdot)$. Then, there is $(m_1, m_2) \in \mathcal{N}(\Gamma, \theta), (m_1', m_2') \in \mathcal{N}(\Gamma, \phi)$, satisfying $h(m_1, m_2) = h(m_1', m_2') = a$. So,

$$\begin{aligned}(m_1, m_2) \in \mathcal{N}(\Gamma, \theta) \Rightarrow h(M_1, m_2) &\subseteq L_1(a, \theta) = L_1(a, \xi) = \{a\},\\ (m_1', m_2') \in \mathcal{N}(\Gamma, \phi) \Rightarrow h(m_1', M_2) &\subseteq L_2(a, \phi) = L_2(a, \xi) = \{a\}.\end{aligned}$$

Therefore, $h(m_1', m_2) = a$ and $(m_1', m_2) \in \mathcal{N}(\Gamma, \xi)$. We have $a \in F(\xi)$, which is a contradiction.

In cases with three or more participants, Maskin monotonicity and N-VP are sufficient, but not necessary, conditions for full Nash implementation. Then, a relevant question is whether there is a necessary and sufficient condition for full Nash implementation. Moore and Repullo (*Econometrica*, 1990) provided such conditions.

Definition 18.8.5 (conditions $\boldsymbol{\mu}$) We say that **condition** μ is satisfied if there is a set $\mathcal{B} \subseteq \mathcal{A}$, such that for any $i \in N, \mathbf{e} \in E$, $\mathbf{a} \in F(\mathbf{e})$, there is a set $C_i = C_i(\mathbf{a}, \mathbf{e}) \subseteq \mathcal{B}$, such that $\mathbf{a} \in C_i \subseteq L_i(\mathbf{a}, \mathbf{e}) \bigcap \mathcal{B}$ and satisfying the following conditions:

(i) if $C_i \subseteq L_i(\mathbf{a}, \mathbf{e}^*), \forall i \in N$, then $\mathbf{a} \in F(\mathbf{e}^*)$;

(ii) for some i, if $\mathbf{c} \in C_i \subseteq L_i(\mathbf{c}, \mathbf{e}^*)$ and $\mathcal{B} \subseteq L_j(\mathbf{c}, \mathbf{e}^*), \forall j \neq i$, then $\mathbf{c} \in F(\mathbf{e}^*)$;

(iii) if $\mathbf{c} \in \mathcal{B} \subseteq L_i(\mathbf{c}, \mathbf{e}^*), \forall i$, then $\mathbf{c} \in F(\mathbf{e}^*)$.

Proposition 18.8.2 *Suppose that $n \geqq 3$. Then, the necessary and sufficient condition for a social choice rule $F(\cdot)$ to be fully Nash implementable is that it satisfies the condition μ.*

PROOF.

- **Necessity**: Note that if $F(\cdot)$ is fully Nash implementable by mechanism $\Gamma = \langle M, h\rangle$, then $\forall \mathbf{e} \in E, \forall a \in F(\mathbf{e})$, there exists $\mathbf{m}(\mathbf{a}, \mathbf{e}) \in \mathcal{N}(\Gamma, \mathbf{e})$, such that $g(\mathbf{m}(\mathbf{a}, \mathbf{e})) = \mathbf{a}$. Let $\mathcal{B} = \{a = g(\mathbf{m}) \in \mathcal{A} | \exists \mathbf{m} \in M\}$, and $C_i = C_i(\mathbf{a}, \mathbf{e}) \equiv g(M_i, \mathbf{m}_{-i}(\mathbf{a}, \mathbf{e})) \equiv \{\mathbf{a} = g(\hat{m}_i, \mathbf{m}_{-i}(\mathbf{a}, \mathbf{e})) | \exists \hat{m}_i \in M_i\}$. Obviously, $\mathbf{a} \in C_i \subseteq L_i(\mathbf{a}, \mathbf{e}) \bigcap \mathcal{B}$.

  a. If $C_i \subseteq L_i(\mathbf{a}, \mathbf{e}^*) \forall i \in N$, then $\mathbf{m}(\mathbf{a}, \mathbf{e}) \in \mathcal{N}(\Gamma, \mathbf{e}^*)$. Since $F(\cdot)$ is fully Nash implementable, $\mathbf{a} = g(\mathbf{m}(\mathbf{a}, \mathbf{e})) \in g(\mathcal{N}(\Gamma, \mathbf{e}^*)) \equiv F(\mathbf{e}^*)$. The condition (i) is satisfied.

  b. Suppose that there exists $\mathbf{c}$ satisfying $\mathbf{c} \in C_i(\mathbf{a}, \mathbf{e}) \subseteq L_i(\mathbf{c}, \mathbf{e}^*)$ and $\mathcal{B} \subseteq L_j(\mathbf{c}, \mathbf{e}^*), \forall j \neq i$. Let $c_i = g(\hat{m}_i, \mathbf{m}_{-i}(\mathbf{a}, \mathbf{e}))$. Obviously, $(\hat{m}_i, \mathbf{m}_{-i}(\mathbf{a}, \mathbf{e})) \in \mathcal{N}(\Gamma, \mathbf{e}^*)$, and thus $\mathbf{c} = g(\hat{m}_i, \mathbf{m}_{-i}(\mathbf{a}, \mathbf{e})) \in g(\mathcal{N}(\Gamma, \mathbf{e}^*)) = F(\mathbf{e}^*)$. Then, the condition (ii) is satisfied.

c. If there exists $\mathbf{c} \in \mathcal{B}$ satisfying $\mathcal{B} \subseteq L_i(\mathbf{c}, \mathbf{e}^*), \forall i \in N$, then $g^{-1}(\mathbf{c}) = \{m \in M | g(\mathbf{m}) = \mathbf{c}\} \in \mathcal{N}(\Gamma, \mathbf{e}^*)$. Thus, $\mathbf{c} \in g(\mathcal{N}(\Gamma, \mathbf{e}^*)) = F(\mathbf{e}^*)$, and then the condition (iii) is satisfied.

- **Sufficiency**: We need to construct a mechanism $\Gamma = \langle M, g \rangle$, such that $F(\cdot)$ is fully Nash implementable by Γ, i.e., $F(\mathbf{e}) = g(\mathcal{N}(\Gamma, \mathbf{e})), \forall \mathbf{e} \in E$. Let

$$M_i = \{(e_i, a_i, b_i, v_i) \in E \times \mathcal{A} \times \mathcal{B} \times \mathbf{N} | a_i \in F(e_i)\},$$

and the outcome function $g(\cdot)$ is defined as follows:

Case 1: If $m_i \equiv (\mathbf{e}, \mathbf{a}, \mathbf{b}, \mathbf{v}), \forall i \in N$, then $g(\mathbf{m}) = \mathbf{a}$;

Case 2: If there exists $i \in N$, such that $m_j = (\mathbf{e}, \mathbf{a}, \mathbf{b}, \mathbf{v}), \forall j \neq i$ and $m_i \neq (\mathbf{e}, \mathbf{a}, \mathbf{b}, \mathbf{v})$, then

$$g(\mathbf{m}) = \begin{cases} b_i, & \text{if } b_i \in C_i(\mathbf{a}, \mathbf{e}) \\ \mathbf{a}, & \text{others.} \end{cases}$$

Case3: Other cases, $g(\mathbf{m}) = b_i$ with $i = \min\{i \in N : v_i = \max_{j \in N} v_j\}$.

We first show that $F(\mathbf{e}) \subseteq g(\mathcal{N}(\Gamma, \mathbf{e})), \forall \mathbf{e} \in E$. For any $\mathbf{e} \in E$, let $\mathbf{a} \in F(\mathbf{e})$ and $m_i \equiv (\mathbf{e}, \mathbf{a}, \mathbf{a}, 0), \forall i \in N$. Then, $\mathbf{m} \in \mathcal{N}(\Gamma, \mathbf{e}), g(\mathbf{m}) = \mathbf{a}$, and thus $F(\mathbf{e}) \subseteq g(\mathcal{N}(\Gamma, \mathbf{e})), \forall \mathbf{e}$. Therefore, we have $g(\mathcal{N}(\Gamma, \mathbf{e}^*)) \subseteq F(\mathbf{e}^*), \forall \mathbf{e}^*$.

If $\mathbf{m} \in \mathcal{N}(\Gamma, \mathbf{e}^*)$, then there must be one of the following three cases:

a. $m_i \equiv (\mathbf{e}^*, \mathbf{a}, \mathbf{b}, \mathbf{v}), \forall i \in N$. Then, $g(\mathbf{m}) = \mathbf{a} \in F(\mathbf{e}^*)$.

b. $m_i \neq m_j = (\mathbf{e}^*, \mathbf{a}, \mathbf{b}, \mathbf{v}), \forall j \neq i$. Then, we must have $g(\mathbf{m}) \in C_i \subseteq L_i(g(\mathbf{m}), \mathbf{e}^*), \mathcal{B} \subseteq L_j(g(\mathbf{m}), \mathbf{e}^*), j \neq i$. According to property (ii), $g(\mathbf{m}) \in F(\mathbf{e}^*)$.

c. For $m_i, i \in N$, at least two agents have different strategies. Then, anyone can get the most desired alternative among $\mathcal{B}$ by reporting the largest possible v_i. Therefore, $\mathcal{B} \subseteq L_i(g(\mathbf{m}), \mathbf{e}^*), \forall i \in N$. According to property (iii), $g(\mathbf{m}) \in F(\mathbf{e}^*)$, and thus $g(\mathcal{N}(\Gamma, \mathbf{e}^*)) \subseteq F(\mathbf{e}^*)$.

From all of the above, we have $g(\mathcal{N}(\Gamma, \mathbf{e}^*)) = F(\mathbf{e}^*)$.

□

Moore and Repullo (*Econometrica*, 1990) also provided necessary and sufficient conditions for full Nash implementation of any social choice correspondence for any economy with only two agents.

Definition 18.8.6 (Condition μ2) We say that**condition** $\mu 2$ is satisfied, if condition μ above is satisfied, and in addition, the following condition holds:

- For any $(a, e', b, \hat{e}) \in (A \times E \times A \times E)$ with $a = F(e'), b = F(\hat{e})$, there is $c = c(a, e', b, \hat{e}) \in C_1(a, e') \bigcap C_2(b, \hat{e})$, such that for any $e^* \in E$, whenever $c \in C_1(a, e') \subseteq L_1(c, e^*)$ and $c \in C_2(b, \hat{e}) \subseteq L_2(c, e^*)$, we have $c \in F(e^*)$.

We then have the following proposition whose proof is given in Moore and Repullo (*Econometrica*, 1990).

Proposition 18.8.3 *Suppose that $n = 2$. Then, a social choice rule $F(\cdot)$ is fully Nash implementable if and only if $\mu 2$ is satisfied.*

18.9 Well-Behaved Mechanism Design

Maskin's theorem and the Moore-Repullo theorem give necessary and sufficient conditions for a social choice correspondence to be fully Nash implementable. However, due to the general nature of the social choice rules, the implementing mechanisms constructed in proving the characterization theorems turn out to be quite complex and impractical. The characterization results show what is possible for the (full) implementation of a social choice rule, but not what is realistic.

Thus, like most characterization results in the literature, the mechanisms given by Maskin and Moore-Repullo are not natural in the sense that they are not continuous; small variations in an agent's strategy choice may result in large jumps in the resulting allocations, and further it has a message space of infinite dimension since it includes preference profiles as a component. Unless an a priori admissible set of preferences is sufficiently restricted, the information processing cost could be very high. As such, these mechanisms tend to be unrealistic, and will not work in practice. A natural question is then, is there a mechanism that performs well and is also easy to operate in practice? In this section, we give several mechanisms that possess some desired properties.

18.9.1 Groves-Ledyard Mechanism

Groves and Ledyard (1977, *Econometrica*) were the first to propose a specific mechanism that has certain desired properties and Nash implements Pareto efficient allocations for public goods economies.

To show the basic structure of the Groves-Ledyard mechanism, consider a simplified version here. Public goods economies under consideration have one private good x_i, one public good y, and three agents ($n = 3$). The production function is given by $y = f(v) = v$.

The mechanism is defined as follows:

- $M_i = \mathcal{R}_i$, $i = 1, 2, 3$. Its elements, m_i, can be interpreted as the proposed contribution (or tax) that agent i is willing to make.
- $t_i(\mathbf{m}) = m_i^2 + 2m_j m_k$: the actual contribution t_i determined by the mechanism with the reported m_i.
- $y(\mathbf{m}) = (m_1 + m_2 + m_3)^2$: the level of public good y.
- $x_i(\mathbf{m}) = w_i - t_i(\mathbf{m})$: the consumption of private good x_i.

Then, the mechanism is balanced since

$$\begin{aligned}
& \sum_{i=1}^{3} x_i + \sum_{i=1}^{3} t_i(\mathbf{m}) \\
= & \sum_{i=1}^{3} x_i + (m_1 + m_2 + m_3)^2 \\
= & \sum_{i=3}^{n} x_i + y = \sum_{i=1}^{3} w_i.
\end{aligned}$$

The payoff function is given by

$$\begin{aligned}
v_i(\mathbf{m}) &= u_i(x_i(\mathbf{m}), y(\mathbf{m})) \\
&= u_i(w_i - t_i(\mathbf{m}), y(\mathbf{m})).
\end{aligned}$$

To find a Nash equilibrium, we set

$$\frac{\partial v_i(\mathbf{m})}{\partial m_i} = 0. \tag{18.9.37}$$

Then,

$$\frac{\partial v_i(\mathbf{m})}{\partial m_i} = \frac{\partial u_i}{\partial x_i}(-2m_i) + \frac{\partial u_i}{\partial y} 2(m_1 + m_2 + m_3) = 0, \tag{18.9.38}$$

and thus

$$\frac{\frac{\partial u_i}{\partial y}}{\frac{\partial u_i}{\partial x_i}} = \frac{m_i}{m_1 + m_2 + m_3}. \tag{18.9.39}$$

When u_i is quasi-concave, the first-order condition turns out to be a sufficient condition for the existence of a Nash equilibrium.

Making a summation, we have at Nash equilibrium that

$$\sum_{i=1}^{3} \frac{\frac{\partial u_i}{\partial y}}{\frac{\partial u_i}{\partial x_i}} = \sum_{i=1}^{3} \frac{m_i}{m_1 + m_2 + m_3} = 1 = \frac{1}{f'(v)}, \tag{18.9.40}$$

i.e.,

$$\sum_{i=1}^{3} MRS_{yx_i} = MRTS_{yv}.$$

Thus, the Lindahl-Samuelson and budget-balanced conditions hold, yielding that every Nash equilibrium allocation is Pareto efficient.

Groves and Ledyard (1977) claimed that they solved the free-rider problem in the presence of public goods. However, the Groves-Ledyard mechanism possesses two weaknesses: (1) it is not individually rational: a payoff at Nash equilibrium may be lower than the payoff at the initial endowment, and thus they are reluctant to participate in this mechanism for resource reallocation; and (2) it is not individually feasible: for some non-equilibrium strategies, $x_i(\mathbf{m}) = w_i - t_i(\mathbf{m})$ may be negative, and the resources allocated through the mechanism are not within the individual's consumption space.

Then, one may reasonably ask: how can we design an incentive mechanism to pursue Pareto efficient and individually rational allocations?

18.9.2 Walker's Mechanism

Given that Lindahl and Walrasian allocations are Pareto efficient and individually rational, and thus the mechanisms that Nash implement these allocations will lead to both Pareto efficiency and individual rationality. Hurwicz (1979a, 1979b) provided such mechanisms under the economic environment with public goods and private goods. Walker (1981) gave a similar mechanism, which we will introduce below.

Again, consider a public goods economy with n agents, one private good, and one public good, and the production function is given by $y = f(v) = v$. We assume that $n \geqq 3$. Suppose that participant i's utility function $u_i(x_i, y)$ is continuously differentiable, quasi-concave and monotonic, and satisfies the Inada condition $\frac{\partial u}{\partial x_i}(0) = +\infty$, as well as $\lim_{x_i \to 0} \frac{\partial u}{\partial x_i} x_i = 0$, such that interior solutions exist.

Let us first recall the definition of a Lindahl equilibrium allocation:

Allocation $z = (\mathbf{x}, y) = (x_1, x_2, \cdots, x_n, y) \in \mathcal{R}^n_+ \times \mathcal{R}_+$ is called a **Lindahl equilibrium allocation** if it is feasible and there exists a personalized price vector $(q_1, \cdots, q_n) \in \mathcal{R}^n_+$, such that

(1) $x_i + q_i y = w_i, \quad i = 1, \cdots, n;$

(2) for all $i = 1, \cdots, n$, $u_i(x_i', y') > u_i(x_i, y)$ implies that $x_i' + q_i y' > w_i$;

(3) $\sum_{i=1}^{n} q_i = 1$.

We know that it is both Pareto efficient and individually rational. Walker's mechanism is defined by:

- $M_i = \mathcal{R}$: m_i is the contribution which participant i is willing to pay;

- $y(\mathbf{m}) = \sum_{i=1}^{n} m_i$: the level of public good.
- $q_i(\mathbf{m}) = \frac{1}{n} + m_{i+1} - m_{i+2}$: agent i's personalized price of public good.
- $t_i(\mathbf{m}) = q_i(\mathbf{m})y(\mathbf{m})$: the contribution (tax) made by agent i.
- $x_i(\mathbf{m}) = w_i - t_i(\mathbf{m}) = w_i - q_i(\mathbf{m})y(\mathbf{m})$: agent i's private good consumption.

Then, the budget constraint holds:

$$x_i(\mathbf{m}) + q_i(\mathbf{m})y(\mathbf{m}) = w_i, \ \forall m_i \in M_i. \tag{18.9.41}$$

Making a summation, we have

$$\sum_{i=1}^{n} x_i(\mathbf{m}) + \sum_{i=1}^{n} q_i(\mathbf{m})y(\mathbf{m}) = \sum_{i=1}^{n} w_i,$$

and thus

$$\sum_{i=1}^{n} x_i(\mathbf{m}) + y(\mathbf{m}) = \sum_{i=1}^{n} w_i,$$

which means that the mechanism is balanced.

The payoff function is

$$\begin{aligned} v_i(\mathbf{m}) &= u_i(x_i, y) \\ &= u_i(w_i - q_i(\mathbf{m})y(\mathbf{m}), y(\mathbf{m})). \end{aligned}$$

The first-order condition for interior allocations is given by

$$\begin{aligned} \frac{\partial v_i}{\partial m_i} &= -\frac{\partial u_i}{\partial x_i}\left[\frac{\partial q_i}{\partial m_i}y(\mathbf{m}) + q_i(\mathbf{m})\frac{\partial y(\mathbf{m})}{\partial m_i}\right] + \frac{\partial u_i}{\partial y}\frac{\partial y(\mathbf{m})}{\partial m_i} \\ &= -\frac{\partial u_i}{\partial x_i}q_i(\mathbf{m}) + \frac{\partial u_i}{\partial y} = 0 \\ \Rightarrow & \ \frac{\frac{\partial u_i}{\partial y}}{\frac{\partial u_i}{\partial x_i}} = q_i(\mathbf{m}) \qquad \text{(FOC) for Lindahl allocation} \\ \Rightarrow & \ N(e) \subseteq L(e). \end{aligned}$$

Since u_i is quasi-concave, it is also a sufficient condition for the existence of Lindahl equilibrium. We can also show that every Lindahl allocation is a Nash equilibrium allocation, i.e.,

$$L(\mathbf{e}) \subseteq N(\mathbf{e}). \tag{18.9.42}$$

Indeed, suppose that $[(x^*, y^*), q_i^*, \ldots, q_n^*]$ is a Lindahl equilibrium. Let $\mathbf{m}^*$ be the solution of the following equation:

$$\begin{aligned} q_i^* &= \frac{1}{n} + m_{i+1} - m_{i+2}, \\ y^* &= \sum_{i=1}^{n} m_i. \end{aligned}$$

Then, we have $x_i(\mathbf{m}^*) = x_i^*$, $y(\mathbf{m}^*) = y^*$ and $q_i(\mathbf{m}^*) = q_i^*$ for all $i \in N$. Therefore, from $(x(m_i^*, \mathbf{m}_{-i}), y(m_i^*, \mathbf{m}_{-i})) \in \mathcal{R}^2_+$ and $x_i(m_i^*, \mathbf{m}_{-i}) + q_i(\mathbf{m}^*)y(m_i^*, \mathbf{m}_{-i}) = w_i$ for all $i \in N$ and $m_i \in M_i$, we have

$$(x_i(\mathbf{m}^*), y(\mathbf{m}^*)) \ R_i \ (x_i(m_i^*, \mathbf{m}_{-i}), y(m_i^*, \mathbf{m}_{-i})),$$

which means that $\mathbf{m}^*$ is a Nash equilibrium.

Thus, Walker's mechanism fully implements Lindahl allocations which are Pareto efficient and individually rational.

The mechanisms of Hurwicz and Walker are still not satisfactory, because they cannot guarantee that individual feasibility is satisfied. If an agent claims large amounts of t_i, consumption of private good may be negative, i.e., $x_i = w_i - t_i < 0$. Moreover, the mechanism in Hurwicz (1979a) makes use of a larger dimension of message space. Hurwicz (1979b) provided an allocation mechanism that guarantees individual feasibility and results in efficient resource allocation. However, it is not continuous, and even small information processing errors can lead to large variations in resource allocation. In addition, it is not budget-balanced in the sense that resources allocated through the mechanism are more than the total resources of society.

Then, we can pose the question of whether one can design a mechanism that is both individually feasible and budget-balanced for neoclassical economies. The answer, in general, is negative. Indeed, by the Maskin theorem on Nash implementation, all outcomes at Nash equilibria and non-equilibria should be in the feasible set, and then there is no mechanism that fully Nash implements the Walrasian or Lindahl allocations that contain boundary points since they do not satisfy Maskin monotonicity. However, for a class of preferences that result in interior Lindahl allocations, Tian (1991) proposed a mechanism that overcomes the defects of the Walker mechanism. Tian's mechanism is individually feasible, balanced, continuous, and it further has a minimal dimension of message space, which fully implements Lindahl allocations on the proposed economic environments. We discuss it below.

18.9.3 Tian's Mechanism

In Tian (1991)'s mechanism, everything is the same as all other parts of the Walker mechanism, except that the outcome function for public good $y(\mathbf{m})$ is given by (see Figure 18.5):

$$y(\mathbf{m}) = \begin{cases} a(\mathbf{m}) & \text{if} \quad \sum_{i=1}^n m_i > a(\mathbf{m}); \\ \sum_{i=1}^n m_i & \text{if} \quad 0 \leqq \sum_{i=1}^n m_i \leqq a(\mathbf{m}); \\ 0 & \text{if} \quad \sum_{i=1}^n m_i < 0, \end{cases} \tag{18.9.43}$$

where $a(\mathbf{m}) = \min_{i \in N'(\mathbf{m})} \frac{w_i}{q_i(\mathbf{m})}$ can be regarded as the feasible upper bound for having a feasible allocation. Here, $N'(\mathbf{m}) = \{i \in N : q_i(\mathbf{m}) > 0\}$.

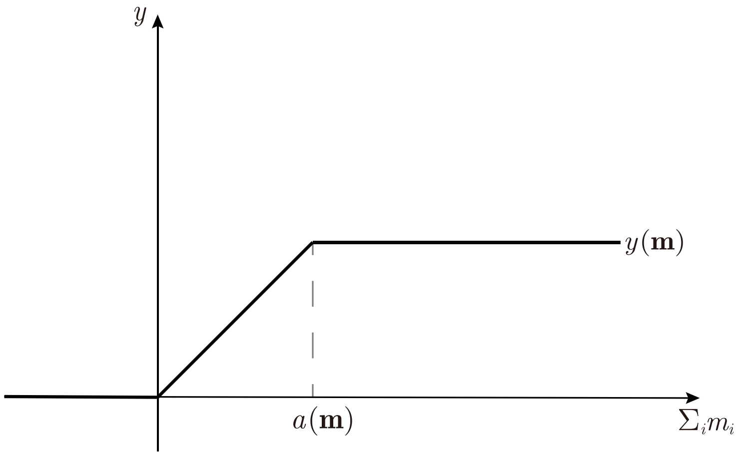

Figure 18.5: The feasible public good outcome function $y(m)$

An interpretation of this formulation is that if the total contribution that the agents are willing to pay is between zero and the feasible upper bound, the level of public good to be produced would be exactly the total contribution proposed by the agents; if the total contribution is less than zero, no public good would be produced; if the total contribution exceeds the feasible upper bound, the level of the public good would be equal to the feasible upper bound. Thus, one can easily see that Tian's mechanism possesses all of the desired properties. Specifically, it is individually feasible, balanced, continuous, and further it has a minimal dimension of message space.

To show that this mechanism fully Nash implements Lindal allocations, we assume that preferences are strongly monotonically increasing and convex, and further assume that every interior allocation is preferred to any boundary allocations. In other words, for all $i \in N$, we have $(x_i, y)\ P_i\ (x_i', y')$, for all $(x_i, y) \in \mathcal{R}^2_{++}$ and $(x_i', y') \in \partial\mathcal{R}^2_{+}$, where $\partial\mathcal{R}^2_{+}$ is the boundary of $\mathcal{R}^2_{+}$. This condition guarantees that all Lindahl equilibrium allocations be interior points; otherwise, the Lindahl correspondence with the boundary point cannot guarantee Maskin monotonicity, and thus it is impossible to be fully implemented by a feasible mechanism.

To show the equivalence between Nash allocations and Lindahl allocations, we first prove the following lemmas.

Lemma 18.9.1 *If* $(\mathbf{x}(\mathbf{m}^*), y(\mathbf{m}^*)) \in \mathcal{N}(\Gamma, \mathbf{e})$, *then* $(x_i(\mathbf{m}^*), y(\mathbf{m}^*)) \in \mathcal{R}^2_{++}$ *for all* $i \in N$.

Proof. We prove this by way of contradiction. Suppose that $(x_i(\mathbf{m}^*), y(\mathbf{m}^*)) \in \partial\mathcal{R}^2_{+}$ for some $i \in I$. Then, $x_i(\mathbf{m}^*) = 0$ or $y(\mathbf{m}^*) = 0$. Consider the quadratic equation

$$y = \frac{w^*}{2(y+c)}, \tag{18.9.44}$$

where $w^* = \min_{i \in N} w_i$; $c = b + n\sum_{i=1}^n |m_i^*|$, where $b = 1/n$. The larger root of the quadratic equation (18.9.44) is positive and denoted by $\tilde{y}$. Suppose that that player i chooses his message $m_i = \tilde{y} - \sum_{j \neq i}^n m_j^*$. Then, $\tilde{y} = m_i + \sum_{j \neq i}^n m_j^* > 0$ and

$$\begin{aligned} w_j - q_j(m_i^*, \mathbf{m}_{-i})\tilde{y} &\geqq w_j - [b + (n\sum_{s=1}^n |m_s^*| + \tilde{y})]\tilde{y} \\ &= w_j - (\tilde{y} + b + n\sum_{s=1}^n |m_s^*|)\tilde{y} \\ &= w_j - w^*/2 \geqq w_j/2 > 0 \end{aligned} \tag{18.9.45}$$

for all $j \in N$. Thus, $y(m_i^*, \mathbf{m}_{-i}) = \tilde{y} > 0$ and $x_j(m_i^*, \mathbf{m}_{-i}) = w_j - q_j(m_i^*, \mathbf{m}_{-i})y(m^*, \mathbf{m}_{-i}) = w_j - q_j(m_i^*, \mathbf{m}_{-i})\tilde{y} > 0$ for all $j \in N$. As a consequence, $(x_i(m_i^*, \mathbf{m}_{-i}), y(m_i^*, \mathbf{m}_{-i}))\ P_i\ (x_i(\mathbf{m}^*), y(\mathbf{m}^*))$ by the assumption that every interior allocation is preferred to any boundary allocations, which contradicts the hypothesis $(\mathbf{x}(\mathbf{m}^*), \mathbf{y}(\mathbf{m}^*)) \in \mathcal{N}(\Gamma, \mathbf{e})$. □

Lemma 18.9.2 *If $(\mathbf{x}(\mathbf{m}^*), y(\mathbf{m}^*)) \in \mathcal{N}(\Gamma, \mathbf{e})$, then $y(\mathbf{m}^*)$ is an interior point of $[0, a(\mathbf{m})]$ and thus $y(\mathbf{m}^*) = \sum_{i=1}^n m_i^*$.*

PROOF. By Lemma 18.9.1, $y(\mathbf{m}^*) > 0$. Therefore, we only need to show $y(\mathbf{m}^*) < a(\mathbf{m}^*)$. Suppose, by way of contradiction, that $y(\mathbf{m}^*) = a(\mathbf{m}^*)$. Then, $x_j(\mathbf{m}^*) = w_j - q_j(\mathbf{m}^*)y(\mathbf{m}^*) = w_j - q_j(\mathbf{m}^*)a(\mathbf{m}^*) = w_j - w_j = 0$ for at least some $j \in N$, which is a contradiction to $\mathbf{x}(\mathbf{m}^*) > 0$ by Lemma 18.9.1. □

Proposition 18.9.1 *If the mechanism has a Nash equilibrium $\mathbf{m}^*$, then $(\mathbf{x}(\mathbf{m}^*), y(\mathbf{m}^*))$ is a Lindahl allocation with $(q_1(\mathbf{m}^*), \ldots, q_n(\mathbf{m}^*))$ as the Lindahl price vector, i.e., $\mathcal{N}(\Gamma, \mathbf{e}) \subseteq L(\mathbf{e})$.*

PROOF. Let $\mathbf{m}^*$ be a Nash equilibrium. Now, we prove that $(\mathbf{x}(\mathbf{m}^*), y(\mathbf{m}^*))$ is a Lindahl allocation with $(q_1(\mathbf{m}^*), \ldots, q_n(\mathbf{m}^*))$ as the Lindahl price vector. Since the mechanism is feasible and $\sum_{i=1}^n q_i(\mathbf{m}^*) = 1$, as well as $x_i(\mathbf{m}^*) + q_i(\mathbf{m}^*)y(\mathbf{m}^*) = w_i$ for all $i \in N$, we only need to show that each individual is maximizing his preference. Suppose, by way of contradiction, that there is some $(x_i, y) \in \mathcal{R}_+^2$, such that $(x_i, y)\ P_i\ (x_i(\mathbf{m}^*), y(\mathbf{m}^*))$ and $x_i + q_i(\mathbf{m}^*)y \leqq w_i$. Because of monotonicity of preferences, it will be enough to confine ourselves to the case of $x_i + q_i(\mathbf{m}^*)y = w_i$. Let $(x_{i\lambda}, y_\lambda) = (\lambda x_i + (1-\lambda)x_i(\mathbf{m}^*), \lambda y + (1-\lambda)y(\mathbf{m}^*))$. Then, by convexity of preferences, we have $(x_{i\lambda}, y_\lambda)\ P_i\ (x_i(\mathbf{m}^*), y(\mathbf{m}^*))$ for any $0 < \lambda < 1$. Moreover, $(x_{i\lambda}, y_\lambda) \in \mathcal{R}_+^2$ and $x_{i\lambda} + q_i(\mathbf{m}^*)y_\lambda = w_i$.

Suppose that player i chooses his message $m_i = y_\lambda - \sum_{j\neq i}^n m_j^*$. Since $y(m^*) = \sum_{j=1}^n m_i^*$ by Lemma 18.9.2, $m_i = y_\lambda - y(\mathbf{m}^*) + m_i^*$. Consequently, as $\lambda \to 0$, $y_\lambda \to y(\mathbf{m}^*)$, and then $m_i \to m_i^*$. Since $x_j(\mathbf{m}^*) = w_j - q_j(\mathbf{m}^*)y(\mathbf{m}^*) > 0$ for all $j \in N$ by Lemma 18.9.1, we have $w_j - q_j(\mathbf{m}_i^*, \mathbf{m}_{-i})y_\lambda > 0$ for all $j \in N$ when λ is sufficiently small. Thus, $y(m_i^*, \mathbf{m}_{-i}) = y_\lambda$ and $x_i(m_i^*, \mathbf{m}_{-i}) = w_i - q_i(\mathbf{m}^*)y(m_i^*, \mathbf{m}_{-i}) = w_i - q_i(\mathbf{m}^*)y_\lambda = x_{i\lambda}$. Since $(x_{i\lambda}, y_\lambda)\ P_i\ (x_i(m^*), y(m^*))$, we have $(x_i(m_i^*, \mathbf{m}_{-i}), y(m_i^*, \mathbf{m}_{-i}))\ P_i\ (x_i(\mathbf{m}^*), y(\mathbf{m}^*))$. This contradicts $(x(\mathbf{m}^*), y(\mathbf{m}^*)) \in \mathcal{N}(\Gamma, \mathbf{e})$. □

Proposition 18.9.2 *If $(\mathbf{x}^*, y^*)$ is a Lindahl allocation with the Lindahl price vector $\mathbf{q}^* = (q_1^*, \ldots, q_n^*)$, then there is a Nash equilibrium $\mathbf{m}^*$ of the mechanism, such that $x_i(\mathbf{m}^*) = x_i^*$, $q_i(\mathbf{m}^*) = q_i^*$, for all $i \in N$, $y(\mathbf{m}^*) = y^*$, i.e., $L(\mathbf{e}) \subseteq \mathcal{N}(\Gamma, \mathbf{e})$.*

PROOF. We need to show that there is a message $\mathbf{m}^*$, such that $(\mathbf{x}^*, y^*)$ is a Nash allocation. Let $\mathbf{m}^*$ be the solution of the following linear equations system:

$$\begin{aligned} q_i^* &= \frac{1}{n} + m_{i+1} - m_{i+2}, \\ y^* &= \sum_{i=1}^n m_i \end{aligned}$$

Then, we have $x_i(\mathbf{m}^*) = x_i^*$, $y(\mathbf{m}^*) = y^*$ and $q_i(\mathbf{m}^*) = q_i^*$ for all $i \in N$. From $(x(m_i^*, \mathbf{m}_{-i}), y(m_i^*, \mathbf{m}_{-i})) \in \mathcal{R}_+^2$ and $x_i(m_i^*, \mathbf{m}_{-i}) + q_i(m^*)y(m_i^*, \mathbf{m}_{-i}) = w_i$ for all $i \in N$ and $m_i \in M_i$, we have $(x_i(\mathbf{m}^*), y(\mathbf{m}^*))\ R_i\ (x_i(m_i^*, \mathbf{m}_{-i}), y(m_i^*, \mathbf{m}_{-i}))$. □

Thus, Tian's mechanism fully Nash implements Lindahl allocations. It should be pointed out that many studies in mechanism design theory considered the issue of incentive compatibility and the issue of information efficiency separately. The incentive compatibility theory only considers the conditions under which a given goal is implementable under a solution concept of self-interested behavior, without taking the information requirements of the mechanism into account. The realization theory on information efficiency to be introduced in the last section of this chapter only considers the amount of information required to achieve a social goal (i.e., the dimension of the message space), while ignoring the incentive problem of the mechanism.

Reichelstein-Reiter (1988), however, considered both issues simultaneously. They demonstrated that, under the condition of Nash incentive compatibility, the amount of information needed to implement a social goal is at least as great as the amount of information needed to realize the same goal without considering the incentive problem. Concerning environments of public goods with only one public good, the minimum dimension of implementing the Lindahl mechanism can be the same as the minimum dimension of realizing the Lindahl allocation. Walker (1981) and Tian (1990,

1991) gave specific incentive mechanisms. However, for environments with private goods, Reichelstein-Reiter (1988) proved that the minimum dimension of the full implementation of the Walrasian mechanism is greater than the minimum dimension of realizing the Walrasian allocation.

18.10 Other Implementations of Social Choice Rules

There are many social choice rules that do not satisfy Maskin monotonicity (e.g., Solomon's judgement), and thus they are not implementable in Nash equilibrium. There are two ways to mitigate this issue. One is to refine Nash equilibrium solutions. Using Nash equilibrium as the solution concept may lead to multiple Nash equilibrium solutions. The concept of refining Nash equilibrium provides a way to eliminate those undesired Nash equilibria, thereby making the set of Nash equilibrium outcomes smaller, as well as implementable. The other is the so-called virtual implementation of a social choice rule, which only requires the equilibrium outcome to be arbitrarily close to the social choice rule. Furthermore, Nash and strong Nash double implementation can make the set of Nash equilibrium outcomes smaller.

18.10.1 Implementation in Refining Nash Equilibrium

What kind of social goals are implementable under refining Nash equilibrium? When the set of Nash equilibrium outcomes of any mechanism $Gamma$, $\mathcal{N}(\Gamma, e)$, is not a subset of the set of social goals, $F(e)$, then the social goals are not Nash implementable. However, since the set of refining Nash equilibrium outcomes can be much smaller than the set of Nash equilibrium outcomes, the former set may be a subset of the set of social goals. Thus, the social goals may be implementable in a refining Nash equilibrium solution. This is indeed the case for several concepts of refinements of Nash equilibria.

A concept of refining Nash equilibrium solutions is the strong Nash equilibrium. A strong Nash equilibrium is a Nash equilibrium in which no coalition, taking the actions of its complements as given, can cooperatively deviate in a way that benefits all of its members. Obviously, every strong Nash equilibrium is Nash equilibrium; whereas, the converse may not be true. Therefore, the set of strong Nash equilibrium is a subset of the set of Nash equilibrium. In addition, since the set of strong Nash equilibria is smaller than the set of Nash equilibria, the set of social choice rules implementable in strong Nash equilibrium may be larger. In fact, Maskin (1979) proved that any social choice rule that leads to Pareto optimal and rational allocations for appropriate preferences is implementable in strong Nash equilibrium.

Furthermore, there are many other solution concepts to characterize self-interested behaviors, such as undominated Nash equilibrium or other refinements of Nash equilibrium solutions, such as subgame perfect Nash equilibrium. Indeed, Moore-Repullo (1988) and Abreu-Sen (1990), among others, proved that almost all social goals, including Solomon's case, are implementable in subgame prefect Nash equilibrium. Palfrey-Srivastava (1991) also proved a similar result under the assumption of undominated Nash equilibrium.

18.10.2 Virtual Nash Implementation

We can also expand the scope of implementable social goals by means of virtual Nash implementation. Although the set of Nash equilibrium outcomes, $\mathcal{N}(\Gamma, e)$, cannot be a subset of social goals, $F(e)$, as long as each Nash equilibrium allocation can be arbitrarily close to some allocation in $F(e)$, we say that this social goal is virtually implementable. Matsushima (1988), Abreu-Sen (1991) and Tian (1997), among others, proved that almost all social goals are virtually implementable.

18.10.3 Double Implementation in Nash and Strong Nash Equilibrium

Although Nash equilibrium is relatively easy to achieve, it may be unstable: participants tend to respond to the designers through some forms of cooperation that may yield greater benefits. The self-interested behavior assumption in the sense of Nash equilibrium may be unrealistic, and thus it may be more reasonable to adopt the solution concept of strong Nash equilibrium. Since the set of strong Nash equilibrium is obviously a subset of Nash equilibrium, a social goal is more likely to be implemented through strong Nash equilibrium.

Although strong Nash equilibrium is more reasonable, it may not exist or be difficult to solve for. In order to achieve an equilibrium relatively easily, and simultaneously guarantee equilibrium stability, one naturally requires that a social choice be doubly implementable in Nash and strong Nash equilibrium. Suh (1997) gave the necessary and sufficient conditions for a social choice rule to be doubly implementable in Nash and strong Nash equilibrium. Since the characterization results do not consider the complexity of the mechanism in reality, Peleg (1996a, 1996b) and Tian (1999a, 2000a, 2000b, 2000c, 2000d) presented the mechanisms that have some desired properties, and that doubly implement both Walrasian and Lindahl allocations and some other social choice rules that lead to Pareto optimal and rational allocations.

18.11 Informational Efficiency in Mechanism Design

There are three important properties in designing a mechanism: allocative efficiency, incentive compatibility, and informational efficiency. It is highly desired for an economic mechanism to have these properties. Indeed: Pareto optimality requires that resources be allocated efficiently; incentive compatibility requires consistency of individuals' interests and social goals; and informational efficiency requires that an economic system have the least informational cost of operation. These properties are of fundamental importance in evaluating and choosing economic institutions.

Studies on the incentive compatibility and informational requirements of an economic system resulted in implementation theory and realization theory, respectively, which together comprise the theory of mechanism design originated by Hurwicz (1960, 1972, 1973, 1979a).

Until now, we have only considered the design of incentive mechanisms, and we now focus on the informational efficiency of economic mechanisms while ignoring the incentive compatibility issues. In this section, we examine the minimal informational size that realizes Pareto optimal allocations in terms of message space size and determine which economic institution is the most informationally efficient.

18.11.1 Framework of Information Mechanism Design

From an informational perspective, an economic mechanism can be viewed as a process of informational exchange and adjustment; it consists of a message space in which communication takes place, rules by which the agents form messages, and an outcome function that translates messages into outcomes (allocations of resources). Attention may be focused on mechanisms that possess stationary or equilibrium messages for each possible economic environment.

Decentralized decision-making is essentially a feature of incomplete information, in which information is dispersed across decision-makers of production and consumption activities. Individuals make decisions about production and consumption through the exchange and transmission of information regarding economic activities, such as demand and supply. The decentralization of information is closely associated with the most desired feature of the competitive market mechanism, as argued by Smith-Hayek-Friedman.[9] Then, certain questions naturally arise: What is the informa-

[9]The two most important theorems in microeconomics, i.e., the First Fundamental Theorem and the Second Fundamental Theorem of Welfare Economics discussed in the general equilibrium theory in Part IV, characterize the relationship between the market mechanism and Pareto efficiency of resource allocation. The First Fundamental Theorem of Welfare Economics states that a perfectly competitive market mechanism leads to a Pareto efficient allocation. It assumes that there are no externalities, and that individual preferences satis-

tion? What is the formal definition of information decentralization? What should it include? In what sense is the information cost viewed as large or small? When answering these questions, we need a unified model to study what is an economic mechanism. This model should include the informational decentralization process, informational centralization process, market-economy mechanism, planned-economy mechanism, as well as possible combinations of these mechanisms. We describe a general model of information adjustment processes below, by which we can examine the informational size of an economic mechanism.

Suppose that there are n participants in an economy. Each participant can be both a producer and a consumer, can be either a producer or a consumer, and can be a household or a government agency. The set of participants is denoted by N. The production possibility set of a producer is denoted by Y_i. Each consumer has a consumption set, denoted by X_i, and a preference ordering or a utility function (if it exists), denoted by R_i (or u_i). Each participant i has an initial endowment vector, denoted by $\boldsymbol{w}_i$. The economic characteristic of such a participant is denoted by $e_i = (X_i, \boldsymbol{w}_i, u_i, Y_i)$. As such, the economy that consists of the characteristics of all participants is called an economy or an economic environment, denoted by $\boldsymbol{e} = (e_1, e_2, \cdots, e_n)$. The set of all admissible economic environments is denoted by E. The set of all possible resource allocations is called the resource allocation space, denoted by Z.

From the perspective of information transmission, an economic mechanism transmits messages from one economic unit to another economic unit; from the perspective of informational physical forms, a message can be a letter, an email, a phone call, or an image; from the perspective of quantitative representation of a message, it can be a group of numbers, a vector, or a matrix. Mechanism design focuses on reducing the complexity of informational transmission, i.e., making a mechanism operate appropriately while paying the least amount of information cost. The message reported by agent i is denoted by m_i, all of which consist of agent i's message space, denoted by M_i. The message space of a mechanism is thus written as $M = \prod_{i \in N} M_i$.

To describe the dynamic adjustment process, we denote by $\boldsymbol{m}(t) = (m_1(t), \cdots, m_n(t))$ the vector of messages of n agents at time t. Since each agent adjusts and reports his own message according to the messages reported by the other participants, agent i's response at time $t+1$ to the message of time t can be expressed as the following first-order difference

fy local non-satiation (self-interestedness). The Second Fundamental Theorem of Welfare Economics states that any Pareto efficient allocation can be achieved by a perfectly competitive market mechanism through redistribution of endowments. However, other important assumptions also exist, such as the convexity of individual preferences and the absence of increasing returns to scale in production.

equation:

$$m_i(t+1) = \varphi_i(\boldsymbol{m}(t), \boldsymbol{e}), \;\; i \in N, \tag{18.11.46}$$

where $\varphi_i : E \to M$ is the **response function** that determines the adjustment process of informational responses. Agents shall not adjust their reported messages as long as the process arrives at the stationary state, i.e., $\boldsymbol{m}$ is the fixed point of the response function, such that $m_i = \varphi_i(\boldsymbol{m}, \boldsymbol{e})$ for $i \in N$. When arriving at the terminal time T, the allocation outcome is determined by the **outcome function** $h(\cdot) : M \to Z$, i.e., $z = h(\boldsymbol{m})$. Such an informational adjustment process and the resource allocation process determine an economic mechanism of resource allocation, which consists of a message space, a response function and an outcome function, denoted by $\langle M, \varphi, h\rangle$.

Under such an economic mechanism, the message space defines the possible messages that can be reported by each participant according to his characteristic; the response function defines the message of the next period of time, which reflects agents' responses to the messages received at this moment, depends on economic environment $\boldsymbol{e}$, and determines the stationary message state; and the allocation rule h determines resource allocation based on the messages reported by all participants. Any mechanism operates under certain constraints, which are determined by all participants from the government, the legal system, and the economic system. Under such constraints, each participant chooses messages that benefit his the most. The message set may consist of the participant's demand or supply of a certain commodity, his preference ordering over alternative commodities, or his production cost. An allocation rule determines resource allocation, i.e., it translates the process of informational transmission into the allocation process of physical resources. As such, it defines a correspondence from the message space to the resource allocation space. Specifically, it determines social output and individual consumption based on the information collected from individuals, firms, and other economic sectors.

Different from the previous model used to study incentive mechanisms, here we have an additional informational adjustment process, represented by the response function φ. Of course, we may regard the elements in the message space of an incentive mechanism as the message space composed of all stationary points determined by the response function. Then, the definitions of the two mechanisms coincide with each other.

The set of stationary points of the response function given in (18.11.46) in fact defines a correspondence from economic environment space E to message space M, denoted by $\mu_i : E \to\to M$, i.e., $\mu_i(\boldsymbol{e}) = \{\boldsymbol{m} \in M : \boldsymbol{m} = m_T$ or $m_i = \varphi_i(\boldsymbol{m}, \boldsymbol{e}), \; i \in N\}$. Thus, the (joint) equilibrium message correspondence $\mu : E \to\to M$ is defined by

$$\mu(\boldsymbol{e}) = \bigcap_{i=1}^{N} \mu_i(\boldsymbol{e})$$

which assigns to every economy $\boldsymbol{e}$ the set of stationary (equilibrium) messages determined by equation (18.11.46). As such, the informational adjustment mechanism can be equivalently defined as $\langle M, \mu, h\rangle$.

The adjustment process determined by equation (18.11.46) is an informationally-centralized adjustment process, as the message reported by agent i at the next period may depend on not only his economic characteristics, e_i, but also the characteristics of all other participants, $\boldsymbol{e}_{-i}$. If, in an economic mechanism, the message reported by each agent just depends on his own economic characteristics, the mechanism is called an **informationally decentralized mechanism**. It is a special case of equation (18.11.46), i.e.,

$$m_i(t+1) = \varphi_i(\boldsymbol{m}(t), e_i), \ \ i \in N, \tag{18.11.47}$$

which defines a **privacy-preserving mechanism** with the (joint) equilibrium message correspondence being $\mu(\boldsymbol{e}) = \bigcap_{i=1}^{N} \mu_i(e_i)$, in which $\mu_i(e_i) = \{\boldsymbol{m} \in M : \varphi_i(\boldsymbol{m}, e_i)\}$.

Readers may find such economic mechanisms too abstract. The following discussion about the informational efficiency and uniqueness of competitive market mechanisms may assist readers to understand the components of such economic mechanisms. We shall discuss how to define the market mechanism as an informationally decentralized mechanism. The message space of the market mechanism consists of prices and exchanges, and it realizes a competitive market equilibrium.

In practice, the message content of communications is usually a vector. Therefore, as long as the adjustment process and informational decentralization are defined, we can evaluate a mechanism in terms of the dimension of the corresponding message space. When considering mechanisms in practice, we find that some mechanisms need to transmit more information than do others. From the viewpoint of informational efficiency, of course, we always hope to realize a social goal using the mechanism that entails the smallest possible operation cost.

If the message space is a general topological space, the notion of informational size can be considered as a concept that characterizes the relative sizes of topological spaces that are used to convey information in the resource allocation process. It would be natural to consider that a space, say S, has more information than the other space T whenever S is topologically "larger" than T. This suggests the following definition, which was introduced by Walker (1977).

Definition 18.11.1 Let S and T be two topological spaces. The space S is said to have as much information as the space T by the Fréchet ordering, denoted by $S \geqq_F T$, if T can be embedded homeomorphically in S, i.e., if there is a subspace of S' of S that is homeomorphic to T.

Definition 18.11.2 Let $\mathcal{P}(\boldsymbol{e})$ be a subset of Pareto efficient allocations for $\boldsymbol{e} \in E$. An allocation mechanism $\langle M, \mu, h \rangle$ is said to be **non-wasteful** on E with respect to $\mathcal{P}$ if $\mu(\boldsymbol{e}) \neq \emptyset$ for all $\boldsymbol{e} \in E$, and $h(\boldsymbol{m}) \in \mathcal{P}(\boldsymbol{e})$ for all $\boldsymbol{m} \in \mu(\boldsymbol{e})$.

Definition 18.11.3 An informationally decentralized non-wasteful mechanism $\langle M, \mu, h \rangle$ is said to be **informationally efficient** on E if the size of its message space M is the smallest one among all other informationally decentralized non-wasteful mechanisms.

Definition 18.11.4 Let $\langle M, \mu, h \rangle$ be an allocation mechanism from the economic environment space E to the outcome space Z. The **performance correspondence** of the mechanism, denoted by $G : E \rightarrow\rightarrow Z$, is defined as:

$$G(\boldsymbol{e}) = \{\boldsymbol{z} \in Z : \boldsymbol{z} = h(\boldsymbol{m}), \text{ for some } \boldsymbol{m} \in \mu(\boldsymbol{e})\}.$$

Definition 18.11.5 Given a social choice correspondence $F : E \rightarrow\rightarrow Z$ and an allocation mechanism $\langle M, \mu, h \rangle$, the mechanism is said to **realize the social choice goal** F if $G(\boldsymbol{e}) \neq \emptyset$ and $G(\boldsymbol{e}) \subseteq F(\boldsymbol{e})$ for any economic environment $\boldsymbol{e} \in E$. The message mechanism is said to **fully realize the social choice rule** F if $G(\boldsymbol{e}) \neq \emptyset$ and $G(\boldsymbol{e}) = F(\boldsymbol{e})$ for all $\boldsymbol{e} \in E$.

The efficient allocation of resources is a widely accepted social goal, and we know that the competitive market mechanism results in Pareto efficient resource allocations when individuals' preferences are locally non-satiated. We may then ask: for neoclassical economic environments, are there some other informationally decentralized mechanisms (e.g., the socialist market economic mechanism) that also realize Pareto efficient allocations, but are informationally more efficient than the competitive market mechanism? Hurwicz and his coauthors proved in the 1970s that the answer to the question is no for pure-exchange neoclassical economic environments. Jordan (1982) further proved that, for pure-exchange neoclassical economic environments, the competitive market mechanism is a unique mechanism that realizes Pareto efficient allocations and simultaneously uses the least amount of information.

Since pure-exchange economies exclude the possibility of production, these theoretical results are not sufficiently appreciated. Tian (2006) then showed that, for private economies with production, the competitive market mechanism is still the unique mechanism that realizes Pareto efficient allocations and simultaneously uses the least amount of information. Since economies with production contain pure-exchange economies as special cases, in the following, we just introduce general production economies.

18.11.2 Informational Efficiency and Uniqueness of the Competitive Market Mechanism

Consider production economies with L private goods, I consumers and J firms, so that the total number of agents is $n := I + J$. Consumer i's characteristic is given by $e_i = (X_i, \boldsymbol{w}_i, R_i)$, where $X_i \subseteq \mathcal{R}^L$, $\boldsymbol{w}_i \in R^L_+$, and R_i is convex, continuous on X_i, and strongly monotonic on the set of interior points of X_i. Producer j's characteristic is given by $e_j = (\mathcal{Y}_j)$. We assume that, for $j = I+1, \ldots, n$, $\mathcal{Y}_j$ is nonempty, closed, convex, contains 0 (possibility of inaction), and $\mathcal{Y}_j - \mathcal{R}^L_+ \subseteq \mathcal{Y}_j$ (free-disposal). We also assume that the economies under consideration have no externalities or public goods.

An economy is the full vector $\mathbf{e} = (e_1, \ldots, e_I, e_{I+1}, \ldots, e_N)$, and the set of all such production economies is denoted by E and is called neoclassical production economies. E is assumed to be endowed with the product topology.

Let $\boldsymbol{x}_i$ denote the net increment in commodity holdings (net trade) by consumer i and $\boldsymbol{y}_j$ producer j's (net) output vector, by which we have $\boldsymbol{x} = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_I)$ and $\boldsymbol{y} = (\boldsymbol{y}_{I+1}, \ldots, \boldsymbol{y}_N)$. An allocation of economy e is a vector $\boldsymbol{z} := (\boldsymbol{x}, \boldsymbol{y}) \in \mathcal{R}^{NL}$. An allocation $\boldsymbol{z} = (\boldsymbol{x}, \boldsymbol{y})$ is said to be **individually feasible** if $\boldsymbol{x}_i + \boldsymbol{w}_i \in X_i$ for $i = 1, \ldots, I$, and $\boldsymbol{y}_j \in \mathcal{Y}_j$ for $j = I+1, \ldots, n$. An allocation $\boldsymbol{z} = (\boldsymbol{x}, \boldsymbol{y}) \in \mathcal{R}^{nL}$ is said to be **balanced** if $\sum_{i=1}^I \boldsymbol{x}_i = \sum_{j=I+1}^N \boldsymbol{y}_j$. An allocation $\boldsymbol{z} = (\boldsymbol{x}, \boldsymbol{y})$ is said to be **feasible** if it is balanced and individually feasible for every individual.

An allocation $\boldsymbol{z} = (\boldsymbol{x}, \boldsymbol{y})$ is said to be **Pareto efficient** if it is feasible and there does not exist another feasible allocation $\boldsymbol{z}' = (\boldsymbol{x}', \boldsymbol{y}')$, such that $(\boldsymbol{x}'_i + \boldsymbol{w}_i) R_i (\boldsymbol{x}_i + \boldsymbol{w}_i)$ for all $i = 1, \ldots, I$ $(\boldsymbol{x}'_i + \boldsymbol{w}_i) P_i (\boldsymbol{x}_i + \boldsymbol{w}_i)$ for some $i = 1, \ldots, I$. Denote by $P(\boldsymbol{e})$ the set of all such allocations. An important characterization of a Pareto optimal allocation is associated with the following concept. Let $\Delta^{L-1} = \{\boldsymbol{p} \in \mathcal{R}^L_{++} : \sum_{l=1}^L p^l = 1\}$ be the $L-1$ dimensional unit simplex.

A nonzero vector $\boldsymbol{p} \in \Delta^{L-1}$ is called a vector of **efficiency prices** for a Pareto optimal allocation $(\boldsymbol{x}, \boldsymbol{y})$ if

(a) $\boldsymbol{p} \cdot \boldsymbol{x}_i \leqq \boldsymbol{p} \cdot \boldsymbol{x}'_i$ for all $i = 1, \ldots, I$ and all $\boldsymbol{x}'_i$, such that $\boldsymbol{x}'_i + \boldsymbol{w}_i \in X_i$ and $(\boldsymbol{x}'_i + \boldsymbol{w}_i)\ R_i\ (\boldsymbol{x}_i + \boldsymbol{w}_i)$;

(b) $\boldsymbol{p} \cdot \boldsymbol{y}_j \geqq \boldsymbol{p} \cdot \boldsymbol{y}'_j$ for all $\boldsymbol{y}'_j \in \mathcal{Y}_j$, $j = I+1, \ldots, N$.

In the terminology of Debreu (1959, pp. 93), $(\boldsymbol{x}, \boldsymbol{y})$ is an equilibrium relative to the price system $\boldsymbol{p}$. It is well known that under certain regularity conditions, such as convexity, and local non-satiation, every Pareto optimal allocation $(\boldsymbol{x}, \boldsymbol{y})$ has an efficiency price correlated with it, as shown in the Second Fundamental Theorem of Welfare Economics in Chapter **??**.

We also want a mechanism to be individually rational. However, as Hurwicz (1979b) pointed out, it is non-obvious what the appropriate generalization of the individual rationality concept should be for an economy with production. The following definition of individual rationality of an allocation for a production economy was introduced by Hurwicz (1979b).

An allocation $\boldsymbol{z} = (\boldsymbol{x}, \boldsymbol{y})$ is said to be **individually rational** with respect to the fixed share guarantee structure $\gamma_i(\boldsymbol{e}; \theta)$ if $(\boldsymbol{x}_i + \boldsymbol{w}_i) R_i (\gamma_i(\boldsymbol{e}) + \boldsymbol{w}_i)$ for all $i = 1, \ldots, I$. Here, $\gamma_i(e; \theta)$ is given by

$$\gamma_i(\boldsymbol{e}; \theta) = \frac{\boldsymbol{p} \cdot \sum_{j=I+1}^{N} \theta_{ij} \boldsymbol{y}_j}{\boldsymbol{p} \cdot \boldsymbol{w}_i} \boldsymbol{w}_i, \quad i = 1, \ldots, I, \tag{18.11.48}$$

Here, $\boldsymbol{p}$ is an efficiency price vector for e and the θ_{ij} are non-negative fractions, such that $\sum_{i=1}^{n} \theta_{ij} = 1$ for $j = I+1, \ldots, N$, which can be interpreted as the profit shares of consumer i from producer j. Note that this definition of individual rationality contains pure exchange, as well as constant returns as special cases. Denote by $I_\theta(\boldsymbol{e})$ the set of all such allocations.

Now, we define the competitive equilibrium of a private ownership economy in which the i-th consumer owns the share θ_{ij} of the j-th producer, and is entitled to the corresponding fraction of its profits. Consequently, the ownership structure can be denoted by the matrix $\theta = (\theta_{ij})$. Denote by Θ the set of all such ownership structures.

An allocation $\boldsymbol{z} = (\boldsymbol{x}, \boldsymbol{y}) = (\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_I, \boldsymbol{y}_{I+1}, \boldsymbol{y}_{I+2}, ..., \boldsymbol{y}_N) \in \mathcal{R}^{IL} \times \mathcal{Y}$ is a **θ-Walrasian allocation** for an economy e if it is feasible and there is a price vector $\boldsymbol{p} \in \Delta^{L-1}$, such that

(1) $\boldsymbol{p} \cdot \boldsymbol{x}_i = \sum_{j=I+1}^{N} \theta_{ij} \boldsymbol{p} \cdot \boldsymbol{y}_j$ for all $i = 1, \ldots, I$;

(2) for all $i = 1, \ldots, I$, $(\boldsymbol{x}_i' + \boldsymbol{w}_i)\ P_i\ (\boldsymbol{x}_i + \boldsymbol{w}_i)$ implies $\boldsymbol{p} \cdot \boldsymbol{x}_i' > \sum_{j=I+1}^{N} \theta_{ij} \boldsymbol{p} \cdot \boldsymbol{y}_j$; and

(3) $\boldsymbol{p} \cdot \boldsymbol{y}_j \geqq \boldsymbol{p} \cdot \boldsymbol{y}_j'$ for all $\boldsymbol{y}_j' \in \mathcal{Y}_j$ and $j = I+1, \ldots, N$.

Denote by $W_\theta(\boldsymbol{e})$ the set of all such allocations, and by $\mathcal{W}_\theta(\boldsymbol{e})$ the set of all such price-allocation pairs $(\boldsymbol{p}, \boldsymbol{z})$.

Let $E^c \subseteq E$ be the subset of production economies on which $W(\boldsymbol{e}) \neq \emptyset$ for all $e \in E^c$, and call such a subset Walrasian production economies.

It may be remarked that every θ-Walrasian allocation is clearly individually rational with respect to $\gamma_i(\boldsymbol{e}; \theta)$, and also, by the local non-satiation of preferences, it is Pareto efficient. Therefore, we have $W_\theta(\boldsymbol{e}) \subseteq I_\theta(\boldsymbol{e}) \cap P(\boldsymbol{e})$ for all $e \in E^c$.

We now define the Walrasian process that is a privacy-preserving process and realizes the Walrasian correspondence W_θ, in which messages consist of prices and trades of all agents. In defining the Walrasian process, it is assumed that the private ownership structure matrix θ is common knowledge for all of the agents. Define the excess demand correspondence of consumer i, where $i = 1, \ldots, I$, as $D_i : \Delta^{L-1} \times \Theta \times \mathcal{R}^J_+ \times E_i \to\to \mathcal{R}^L$ by

$$D_i(\boldsymbol{p}, \theta, \pi_{I+1}, \ldots, \pi_N, e_i) = \{\boldsymbol{x}_i : \boldsymbol{x}_i + \boldsymbol{w}_i \in X_i, \boldsymbol{p} \cdot \boldsymbol{x}_i = \textstyle\sum_{j=I+1}^{N} \theta_{ij} \pi_j$$

$$(\boldsymbol{x}_i' + \boldsymbol{w}_i)\ P_i\ (\boldsymbol{x}_i + \boldsymbol{w}_i) \text{ implies } \boldsymbol{p} \cdot \boldsymbol{x}_i' > \textstyle\sum_{j=I+1}^{N} \theta_{ij} \pi_j, \tag{18.11.49}$$

where π_j is the profit of firm j, for each $j = I+1, \ldots, N$.

Define the supply correspondence of firm j, where $j = I+1, \ldots, N$, as $S_j : \Delta^{L-1} \times E_j \to\to \mathcal{R}^L$ by

$$S_i(\boldsymbol{p}, e_j) = \{\boldsymbol{y}_j : \boldsymbol{y}_j \in \mathcal{Y}_j, \boldsymbol{p} \cdot \boldsymbol{y}_j \geqq \boldsymbol{p} \cdot \boldsymbol{y}'_j \forall \boldsymbol{y}'_j \in \mathcal{Y}_j\}. \tag{18.11.50}$$

Note that $(\boldsymbol{p}, \boldsymbol{x}, \boldsymbol{y})$ is a θ-Walrasian (competitive) equilibrium for economy e with the private ownership structure θ if $\boldsymbol{p} \in \Delta^{L-1}$, $\boldsymbol{x}_i \in D_i(\boldsymbol{p}, \theta, \boldsymbol{p} \cdot \boldsymbol{y}_{I+1}, \ldots, \boldsymbol{p} \cdot \boldsymbol{y}_N)$ for $i = 1, \ldots, I$, $\boldsymbol{y}_j \in S_j(\boldsymbol{p}, e_j)$ for $j = I = 1, \ldots, N$, and the allocation $(\boldsymbol{x}, \boldsymbol{y})$ is balanced.

The Walrasian (competitive) process $\langle M_c, \mu_c, h_c \rangle$ is defined as follows:

Define $M_c = \Delta^{L-1} \times Z$.

Define $\mu_c : E \to\to M_c$ by

$$\mu_c(\boldsymbol{e}) = \cap_{i=1}^{N} \mu_{ci}(e_i), \tag{18.11.51}$$

where $\mu_{ci} : E_i \to\to M_c$ is defined as follows:

(1) For $i = 1, \ldots, I$, $\mu_{ci}(e_i) = \{(\boldsymbol{p}, \boldsymbol{x}, \boldsymbol{y}) : \boldsymbol{p} \in \Delta^{L-1}, \boldsymbol{x}_i \in D_i(\boldsymbol{p}, \theta, \boldsymbol{p} \cdot \boldsymbol{y}_{I+1}, \ldots, \boldsymbol{p} \cdot \boldsymbol{y}_N, e_i)$ and $\sum_{i=1}^{I} \boldsymbol{x}_i = \sum_{j=I+1}^{N} \boldsymbol{y}_j\}$.

(2) For $i = I+1, \ldots, N$, $\mu_{ci}(e_i) = \{(\boldsymbol{p}, \boldsymbol{x}, \boldsymbol{y}) : \boldsymbol{p} \in \Delta^{L-1}, \boldsymbol{y}_i \in S_i(\boldsymbol{p}, e_i)$ and $\sum_{i=1}^{I} \boldsymbol{x}_i = \sum_{j=I+1}^{N} \boldsymbol{y}_j\}$.

Therefore, we have $\mu_c(\boldsymbol{e}) = \mathcal{W}_\theta(\boldsymbol{e})$ for all $e \in E$.

Finally, the Walrasian outcome function $h_c : M_c \to Z$ is defined by

$$h_c(\boldsymbol{p}, \boldsymbol{x}, \boldsymbol{y}) = (\boldsymbol{x}, \boldsymbol{y}), \tag{18.11.52}$$

which is an element in $W_\theta(\boldsymbol{e})$.

The Walrasian process can be viewed as a formalization of resource allocation, simulating the competitive mechanism, which is non-wasteful and individually rational with respect to the fixed share guarantee structure $\gamma_i(e; \theta)$. The competitive message process is privacy-preserving by the construction of the Walrasian process.

Remark 18.11.1 For a given private ownership structure matrix θ, since an element, $m = (\boldsymbol{p}, \boldsymbol{x}_1, \ldots, \boldsymbol{x}_I, \boldsymbol{y}_{I+1}, \ldots, \boldsymbol{y}_N) \in \mathbb{R}^L_{++} \times \mathcal{R}^{(N)L}$, of the Walrasian message space M_c satisfies the conditions $\sum_{l=1}^{L} p^l = 1$, $\sum_{i=1}^{I} \boldsymbol{x}_i = \sum_{j=I+1}^{N} \boldsymbol{y}_j$, and $\boldsymbol{p} \cdot \boldsymbol{x}_i = \sum_{j=I+1}^{N} \theta_{ij} \boldsymbol{p} \cdot \boldsymbol{y}_j$ for each $i = 1, \ldots, I$, and one of these equations is not independent by Walras' Law, any Walrasian message is contained within a Euclidean space of dimension $(L + IL + JL) - (1 + L + I) + 1 = (L-1)I + LJ$, and thus an upper bound on the Euclidean dimension of M_c is $(L-1)I + LJ$.

To establish informational efficiency of the competitive mechanism, we consider a special class of economies, denoted by $E^{cq} = \prod_{i=1}^{N} E_i^{cq}$, where preference orderings are characterized by Cobb-Douglas utility functions, and efficient production technology is characterized by quadratic functions.

For $i = 1, \ldots, I$, consumer i's admissible economic characteristics in E_i^{cq} are given by the set of all $e_i = (X_i, \boldsymbol{w}_i, R_i)$, such that $X_i = \mathcal{R}_+^L$, $\boldsymbol{w}_i > 0$, and R_i is represented by a Cobb-Douglas utility function $u(\cdot, a_i)$ with $a_i \in \Delta^{L-1}$, such that $u(\boldsymbol{x}_i + \boldsymbol{w}_i, a_i) = \prod_{l=1}^{L}(x_i^l + w_i^l)^{a_i^l}$.

For $i = I+1, \ldots, N$, producer i's admissible economic characteristics are given by the set of all $e_i = \mathcal{Y}_i = \mathcal{Y}(b_i)$, such that

$$\begin{aligned}\mathcal{Y}(\mathbf{b}_i) \quad = \quad & \{\mathbf{y}_i \in \mathcal{R}^L : b_i^1 y_i^1 + \sum_{l=2}^{L}(\mathbf{y}^l + \frac{b_i^l}{2}(y^l)^2) \leqq 0 \\ & -\frac{1}{b_i^l} \leqq y_i^l \leqq 0 \text{ for all } l \neq 1\},\end{aligned} \tag{18.11.53}$$

where $\mathbf{b}_i = (b_i^1, \ldots, b_i^L)$ with $b_i^l > \frac{J}{\mathbf{w}_i^l}$. It is clear that any economy in E^{cq} is fully specified by the parameters $\mathbf{a} = (a_1, \ldots, a_I)$ and $\mathbf{b} = (b_{I+1}, \ldots, b_N)$. Furthermore, production sets are nonempty, closed, and convex by noting that $0 \in \mathcal{Y}(b_j)$ and their efficient points are represented by quadratic production functions, in which $(y_i^2, \ldots, y_i^L)$ are inputs and y_i^1 is possibly an output.

Given an initial endowment $\bar{w} \in \mathcal{R}_{++}^{LI}$, define a subset $\bar{E}^{cq}$ of E^{cq} by $\bar{E}^{cq} = \{\mathbf{e} \in E^{cd} : \boldsymbol{w}_i = \bar{w}_i \ \forall i = 1, \ldots, I\}$. In other words, endowments are constant over $\bar{E}^{cq}$. Let E^c be the set of all production economies in which Walrasian equilibrium exists.

We then have the following theorem that shows that the Walrasian mechanism is informationally efficient among all smooth resource allocation mechanisms that are informationally decentralized and non-wasteful over the set E^c.

Theorem 18.11.1 (Informational Efficiency Theorem, Tian, 2006) *Suppose that $\langle M, \mu, h\rangle$ provides allocation mechanisms defined on the class of production economies E^c, such that they*

(i) are informationally decentralized;

(ii) are non-wasteful with respect to $\mathcal{P}$;

(iii) have Hausdorff topological message spaces;

(iv) satisfy the local threadedness property at some point $\mathbf{e} \in \bar{E}^{cq}$.[10]

[10]A stationary message correspondence μ is said to be locally threaded at $e \in E$ if there is a neighborhood $N(\mathbf{e}) \subseteq E$ and a continuous function $f : N(\mathbf{e}) \to M$, such that $f(\mathbf{e}') \in \mu(\mathbf{e}')$ for all $\mathbf{e}' \in N(\mathbf{e})$. The stationary message correspondence μ is said to be locally threaded on E if it is locally threaded at every $\mathbf{e} \in E$.

Then, the Walrasian allocation mechanism $\langle M_c, \mu_c, h_c\rangle$ is informationally efficient among all allocation mechanisms $\langle M, \mu, h\rangle$ on E^c, i.e., $M_c =_F \mathcal{R}^{(L-1)I+LJ} \leqq_F M$.

This theorem establishes informational efficiency of the competitive mechanism within the class of all smooth resource allocation mechanisms that are informationally decentralized and non-wasteful over the class of Walrasian production economies E^c.

The following theorem further shows that the competitive allocation process is a unique, most informationally efficient decentralized mechanism for production economies among mechanisms that achieve Pareto optimal and individually rational allocations.

Theorem 18.11.2 (The Uniqueness Theorem, Tian, 2006) *Suppose that $\langle M, \mu, h\rangle$ is an allocation mechanism on the class of production economies E^{cq}, such that:*

(i) it is informationally decentralized;

(ii) it is non-wasteful with respect to $\mathcal{P}$;

(iii) it is individually rational with respect to the fixed share guarantee structure $\gamma_i(e;\theta)$;

(iv) M is a $(L-1)I + LJ$ dimensional manifold;

(v) μ is a continuous function on E^{cq}.

Then, there is a homeomorphism ϕ on $\mu(E^{cq})$ to M_c, such that

(a) $\mu_c = \phi \cdot \mu$;

(b) $h_c \cdot \phi = h$.

The conclusion of the theorem is summarized in the following commutative homeomorphism diagram:

$$\begin{array}{ccc} E & \xrightarrow{\mu_c} & M_c \\ \Big\downarrow \mu & \phi \nearrow \swarrow \phi^{-1} & \Big\downarrow h_c \\ \mu(E) & \xrightarrow{h} & Z \end{array}$$

Thus, the above theorem shows that the competitive mechanism is a unique informationally efficient process that realizes Pareto optimal and individually rational allocations over the class of production economies E^c. For a mechanism $\langle M, \mu, h\rangle$, when there exists no homeomorphism ϕ on $\mu(E^{cq})$ to M_c, such that (1) $\mu_L = \phi \cdot \mu$ and (2) $h_L \cdot \phi = h$, we may call such a mechanism $\langle M, \mu, h\rangle$ a non-Walrasian allocation mechanism. The Uniqueness Theorem then implies that any non-Walrasian allocation mechanism defined on E^{cq} must use a larger message space.

For public goods economies, Tian (2000e) obtained similar results, and showed that the Lindahl mechanism is the unique informationally efficient process that realizes Pareto optimal and individually rational allocations over the class of production economies that guarantees the existence of Lindahl equilibrium. For additional discussions about mechanism design of informational efficiency in public goods economies, readers may refer to Tian (1994a).

We thus have the following important corollary: for neoclassical economic environments, regardless of whether one considers a planned economy, state-owned economy, collective economy, mixed economy, or any other non-market economic institution, in order to realize efficient resource allocations, it must utilize more information than does the competitive mechanism, and thus all these non-market mechanisms are not informationally efficient. The implication is thus that the competitive market should be allowed to play a decisive role in resource allocation whenever it can achieve Pareto efficiency. Indeed, only when the competitive market fails will some other mechanisms be designed to complement the market. This result lays out an important theoretical foundation for explaining why a transitional economy must let the market play the fundamental and decisive role, and let the private economy rather than the state-owned economy play the dominant role in economic resource allocation. By defining the competitive mechanism as an informationally decentralized economic mechanism, the message space of the market mechanism consists of two vectors: one is the price vector; and the other is the resource allocation vector.

When adopting the command planned-economy mechanism, firms must report and transmit various information, including their production functions (technology conditions and production capabilities), to the central planners. For example, if production functions are given by polynomial functions, they may be of an arbitrarily high order, and thus the dimension of the message space reported could be arbitrarily large. The central planners also need relevant information from the demand side of consumers. As heterogeneous consumers may have different preferences and thus dissimilar utility functions, this also makes the dimension of the message space facing the planners be arbitrarily large and even infinite. As a result, in order to make production and consumption arrangements, the central planners must deal with a huge amount of information, resulting in markedly high operation cost.

On the other hand, for an economic mechanism facing a message space of a small dimension, the allocation rule might become highly complicated. It may turn out that such a mechanism entails a total operation cost that is even higher than that of those mechanisms with message spaces of much higher dimensions. Nevertheless, studies on searching for the smallest message space of a mechanism are worthwhile because they enable us to identify associated informational requirements or operation costs. In addition, they are useful for exploring other aspects of mechanisms, such as the complexity of mechanisms.

Moreover, for general non-neoclassical economic environments (e.g., indivisible commodities, and non-convex preference ordering or production set), do there exist informationally decentralized mechanisms that realize efficient resource allocations? If yes, what is the relationship between the mechanisms and the corresponding informational size? Hurwicz, among others, proved the existence of such mechanisms for quite general economic environments. However, these mechanisms entail a high information cost of operation. Calsamiglia (1977) proved that for a class of nonclassical economic environments, especially non-convex economic environments, a message space of an infinite dimension is required for realizing Pareto optimal allocations.

18.12 Biographies

18.12.1 Leonid Hurwicz

Leonid Hurwicz (1917-2008), the father of mechanism design theory, was awarded the National Medal of Science in 1990 "for his pioneering work on the theory of modern decentralized allocation mechanisms" and received the Nobel Memorial Prize in Economic Sciences with Eric Maskin and Roger Myerson for "having laid the foundations of mechanism design theory." Despite his pioneering achievements in many branches of modern economics, Hurwicz did not earn a degree in economics.

Hurwicz was born into a Jewish family in Poland in 1917 and came to the United States during World War II. The highest degree that he received is a master's degree in law in Poland. At the outbreak of World War II, Hurwicz was in Switzerland. He did not return to Poland, but went to the United States. He said, "If I stay in Poland, I could very well be a victim of the Auschwitz concentration camp." After coming to the United States, he did not pursue a doctoral degree, but instead became an assistant for Paul Samuelson, and was then promoted directly to a full professor from an assistant professor.

As early as the 1950s, Hurwicz began to think about research projects derived from general equilibrium theory, social choice theory, and game theory. In the early 1960s, his article, entitled "Optimality and Informational Efficiency in Resource Allocation Processes", brought on the prelude to mechanism design theory. Hurwicz's mechanism design theory is closely linked with major economic issues, such as market regulation, market screening and public goods provision. Subsequently, many economists have been involved in this research. Both Myerson and Maskin, who were young at the time, were active researchers in this respect.

Later, Hurwicz wrote a bunch of well-known papers, such as "Revealed Preference without Demand Continuity Assumptions" and "On the Concept and Possibility of Informational Decentralization", and gradually improved the theoretical foundation of mechanism design. In 1973, Hurwicz published the paper "The Design of Mechanisms for Resource Allocation" in the American Economic Review, which proposed the core issue in the theoretical framework of mechanism design, i.e., incentive compatibility, and established the basic framework of mechanism design theory.

Mechanism design theory can be regarded as the most crucial development in economics science in the past half century. It aims to systematically analyze resource allocation institutions and processes, and elucidate the important role of information, communication, incentives and economic agents' processing capabilities in decentralized resource allocation, thereby enabling us to identify sources of market failure. In fact, many subfields of modern economics are influenced by the mechanism design theory pioneered by Hurwicz. His "incentive compatibility" has now become a core concept in economics. The two most objective realities in the real economic world are an individual's self-interested behavior and the asymmetric nature of individual private information. In simple terms, incentive compatibility means that through the design of rules of the game, individuals wish to do what a designer wants them to do. Information asymmetry can result in market failure, and thus some incentive mechanism should be designed to induce the desired incentives for individuals.

Hurwicz had a close connection with Lange, a representative of the Lausanne school, and Hayek, a representative of the Austrian school. His work on mechanism design theory was also influenced by these two aspects of economic theory and methodology, and rigorously unifies these two seemingly very different schools of thought into mechanism design theory. Hurwicz also interacted academically with representatives of the new institutional economics, such as Douglass C. North (1920-2015, whose biography was seen in Section **??**), in an attempt to establish a channel for dialogue and connection between the two theories.

Hurwicz also carried out other pioneering work: in the late 1940s, he laid the groundwork for the identification problem of dynamic econometric models; and, as early as 1947, he first proposed and defined the concept of rational expectation, which is the most critical solution concept in neoclassical macroeconomics. Hurwicz also contributed important work on integrability, i.e., deriving utility functions from the demand function. He and Kenneth J. Arrow also conducted pioneering work on the stability of the general equilibrium of competitive markets.

In addition, Hurwicz attaches great importance to the preciseness of expressing concepts and economic problems because he believed that lacking preciseness is one of the biggest problems in numerous traditional economic theories. The greatest significance of the axiomatic approach lies in the clarity of the expression of a theory, which makes discussions and criticisms possess a universal research paradigm and analytical framework. This is also an inspiration that he got from the “socialist controversy”, i.e., the debate between Mises-Hayek and Lange-Lerner in the 1920s and 1930s, because he felt that the two sides of the discussion are incommensurable in theoretical expressions.

On April 14, 2007, the University of Minnesota hosted the 90th birthday party for professor Leonid Hurwicz. This party became a gathering of the masters of economics, including Eric Maskin and Roger Myerson, Arrow who won the 1972 Nobel Prize in Economics, his student McFadden who won the 2000 Nobel Prize in Economics, and many other accomplished economists. Six months later, Hurwicz, Eric Maskin, and Roger Myerson jointly won the Nobel Prize in Economics for their contribution to mechanism design theory. At the same time, Hurwicz became the oldest Nobel Prize winner in history at that time. After learning this news, with a sense of humor towards his later life, he stated, “I thought my time had passed. For the Nobel Prize, I am really too old. However, this bonus is indeed good for a retired old man.”

18.12.2 Eric S. Maskin

Eric S. Maskin (1950—), a professor of economics at Harvard University, has made outstanding contributions to many areas in modern economics, especially in social choice theory, game theory, incentive and information theory, and mechanism design. With his profound theoretical contributions, rigorous academic attitude and outstanding contributions to the training of advanced research talents in economics, Maskin has been regarded as the most prestigious economist in the current economics profession. His research has had a major influence on economics, political science, and law. He was awarded the Nobel Memorial Prize in Economic Sciences in 2007 for having laid the foundations of mechanism design theory.

When Maskin was a graduate student at Harvard University, he became interested in information economics after taking a course taught by Kenneth Arrow, and he later completed his doctoral dissertation under Arrow's guidance. Maskin's early research focused on implementation theory. This theory emphasizes how to design an incentive mechanism, such that individual rationality is compatible with collective rationality.

The most outstanding contribution of Maskin is the introduction of game theory into mechanism design. Prior to him, the most important scholar in mechanism design was Leonid Hurwicz. Researchers previously focused on mechanism design of central planners from the perspective of information efficiency. Maskin, however, developed an incentive mechanism design for any social goal, i.e., even if individuals aim to pursue their own benefit, the social goal will also be achieved under this mechanism. In 1977, Maskin completed the paper *Nash Equilibrium and Welfare Optimality*, but this article was not officially published until 1999. This paper is a milestone in mechanism design, in which Maskin proposed and proved the sufficient and necessary conditions of implementation in Nash equilibrium. Indeed, one of the constructed necessary conditions is called Maskin monotonicity, which was broadly circulated. The contribution of Maskin to the field of industrial organization is that he and Tirole jointly initiated the game theoretical framework for the analysis of industrial organizations.

Moreover, Maskin has also made important contributions to repeated games, political economy, and comparing different voting systems. He also introduced the soft budget constraint, a concept widely employed in the analysis of socialist economies, into the study of market economies, which has had a great influence on the study of comparing different economic systems.

18.13 Exercises

Exercise 18.1 Suppose that X is a set of alternatives, and $R = R_1 \times \cdots \times R_n$ is the set of preference profiles defined on X. Answer the given questions in response to the following statement:

A social choice function $f : R \to X$ is partially implementable in dominant strategies if and only if the social choice function is truthfully implementable by a revelation mechanism in dominant strategies.

1. Give the formal definitions of the following terms: (1) dominant strategy mechanism; (2) revelation mechanism; (3) dominant-strategy incentive compatible revelation mechanism; (4) truthful implementation in dominant strategies.

2. State whether or not the above statement is correct. If correct, prove it; otherwise, provide a counter-example.

Exercise 18.2 With a restricted domain, the Gibbard-Satterthwaite Impossibility Theorem may not be true. Justify your answer.

Exercise 18.3 (Majority Voting with Two Candidates) Suppose that N voters cast their ballots on two candidates a and b, and each voter stated his preferences on the candidates for whom he supports. The winner will be determined by the following simple majority rule:

$$F(e) = \begin{cases} a, & \text{if} \quad \#\{i \in I | a \succ_i^e b\} > \#\{i \in I | b \succ_i^e a\}, \\ b, & \text{if} \quad \#\{i \in I | a \succ_i^e b\} \leqq \#\{i \in I | b \succ_i^e a\}. \end{cases}$$

1. Prove that the simple majority rule is strategy-proof, i.e., every voter will truthfully report his preference, regardless of whether others are telling the truth, so that $F(\cdot)$ is truthfully implementable in dominant strategies.

2. Will this conclusion be true for the case with more than two candidates? Justify your answer.

Exercise 18.4 Consider pure exchange economic environments with two individuals and two goods. The consumption spaces are $X_1 = X_2 = \mathcal{R}^2_+$. For every $i = 1, 2$ and $e^i \in E^i = \{u^i_c, u^i_l\}$, the utility functions are given by

$$u_{ci}(x^1_i, x^2_i) = x^1_i x^2_i,$$
$$u_{li}(x^1_i, x^2_i) = x^1_i + 2x^2_i.$$

Then, for such economic environments $E = E^1 \times E^2$, there are four possible economies. For every economy, the endowment is fixed, i.e., $\omega^1 = (1, 0)$, $\omega^2 = (1, 2)$.

1. Find the set of all Pareto efficient allocations, and show the sets of individually rational and Pareto efficient allocations, as well as their intersection, in a graph.

2. Prove that for this economic environment, there is no incentive compatible direct mechanism that truthfully implements individually rational and Pareto efficient allocations.

Exercise 18.5 Suppose that the Clark (pivotal) mechanism is used to determine whether a public good will be provided. Individuals a, b, c report their net values of the public good as $3, 5, -7$, respectively.

1. Should the public good be provided?

2. Who is the pivotal agent under this mechanism?

3. What should be the transfer to each individual?

4. If the true value of the public good to consumer b is 3, will he still keep reporting 5? Why?

5. Is the pivotal mechanism balanced?

Exercise 18.6 Prove that for continuous public goods economies with private value, under the Vickrey-Clark-Groves mechanism, truth-telling

$$(b_1(\boldsymbol{y}), b_2(\boldsymbol{y}), \cdots, b_n(\boldsymbol{y})) = (v_1(\boldsymbol{y}), v_2(\boldsymbol{y}), \cdots, v_n(\boldsymbol{y}))$$

is a dominant strategy. As a consequence, this mechanism truthfully implements the efficient decision rule in the dominant equilibrium $y^*(\cdot)$.

Exercise 18.7 Suppose that there are two production technologies that can abate pollution, each owned by firm i, $i \in \{1, 2\}$. The production technology set is convex, and relies on private information θ_i. For simplicity, we assume that the cost function is $C_i(q_i) = \frac{1}{2}\theta_i q_i^2$, in which q_i is the amount of pollution that is abated by firm i. At the same time, we assume that marginal utility is of a linear form, $MU(q) = 1 - 2q$. The total abatement of pollution is $q = q_1 + q_2$.

1. Find the efficient amount of pollution abated by the two firms, which is a function of (θ_i, θ_j).

2. Find the efficient level of transfers in the pivotal mechanism.

Exercise 18.8 Consider the provision of public goods. Two towns, on the banks of a river, decide whether to build a bridge between the two towns at cost c with $1 > c > 0$. Let θ_i be the number of residents in town i who intend to employ bridges. Suppose that the prior distribution of θ_i is a uniform distribution on $[0, 1]$, and θ_1 is independent of θ_2. After crossing the bridge to the town on the other side of the river, the residents of town i will have an effect $\gamma\theta_i$ on the residents of town j, which can be either a positive or a negative effect. Consequently, for town i, the total value of building bridges is $\theta_i + \gamma\theta_j$.

1. For this problem, what is the VCG mechanism?

2. What is the social choice rule that is implementable in dominant strategies?

Exercise 18.9 Consider a public good economy with individuals a, b, c and public good y. The utility function of individual i is given by $u_i(t_i, y) = t_i + v_i(y)$ with $v_i(y) = \bar{\theta}_i \ln y - y$, where t_i is the transfer to individual i, and the true $\bar{\theta}_i$ is unknown to the designer.

1. Find the VCG mechanism $t_a(\theta), t_b(\theta), t_c(\theta), y(\theta)$, where θ_i is the report of the true value $\bar{\theta}_i$ by individual i.

2. Find the Clark mechanism (the pivotal mechanism).

Exercise 18.10 Suppose that $v(\cdot, \theta_i)$ is twice continuously differentiable, and for $\theta_i \in [\underline{\theta}_i, \bar{\theta}_i]$, $\partial^2 v_i(y, \theta_i)/\partial y^2 < 0$, and $\partial^2 v_i(y, \theta_i)/\partial y \partial\theta_i > 0$. Prove that a continuously differentiable social choice function $f(\cdot) = (y(\cdot), t_1(\cdot), \cdots, t_n(\cdot))$ is truthfully implementable in dominant strategies if and only if $y(\theta)$ is a nondecreasing function of θ_i and

$$t_i(\theta_i, \theta_{-i}) = t_i(\underline{\theta}_i, \theta_{-i}) - \int_{\underline{\theta}_i}^{\theta_i} \frac{\partial v_i(y(s, \theta_{-i}), s)}{\partial y} \frac{\partial y(s, \theta_{-i})}{\partial s} ds.$$

Exercise 18.11 Prove that, if the domain of agents' preferences for public goods is unrestricted (i.e., any utility function $v_i : \mathcal{D} \to \mathcal{R}$ is possible), then the VCG mechanism is the only one that can truthfully implement the efficient provision of public goods.

Exercise 18.12 Individuals 1 and 2 will decide on the ownership of an object. Consider the following mechanism: individuals 1 and 2 report prices, p_1 and p_2, separately. If $p_1 > p_2$, then individual 1 will get the object and pay p_2 to individual 2; if $p_1 < p_2$, then individual 2 will get the object and pay p_1 to individual 1; if $p_1 = p_2$, then individual 1 and 2 have half the chance to get the object and pay p_1 to the other. Answer the following questions:

1. Is this mechanism incentive compatible in dominant strategies?
2. Is the mechanism individually rational?
3. Is the mechanism budget-balanced?

Exercise 18.13 Consider quasi-linear economic environments with n participants and a public good y. Let $y^*(\cdot)$ be an ex-post Pareto efficient allocation mechanism, and define $V^*(\theta) = \Sigma_i v_i(y^*(\theta), \theta_i)$.

1. Prove that the necessary and sufficient condition for the existence of an ex-post Pareto efficient social choice function that can be truthfully implemented in dominant strategies is $V^*(\cdot) = \sum_i V_i(\theta_{-i})$, where $V_i(\cdot)$ is a function of θ_{-i} for all i.
2. Using the above conclusion, prove that when $n = 3$, $Y = \mathcal{R}$, $\Theta_i = \mathcal{R}_+$, and $v_i(y, \theta) = \theta_i y - \frac{1}{2}y^2$ for all i, there is a Pareto efficient social choice function that is truthfully implementable in dominant strategies.
3. Now, suppose that $v_i(y, \theta_i)$ are n-th continuously differentiable, $\dfrac{\partial^2 v_i}{\partial y^2} < 0$, and $\dfrac{\partial^2 v_i}{\partial y \partial \theta_i} \neq 0$ for all $(y, \theta_i) \in \mathcal{R}_+ \times \mathcal{R}_+$. Prove that a necessary and sufficient condition for the existence of an ex-post Pareto efficient social choice function is the following: for any θ,
$$\frac{\partial^n V^*(\theta)}{\partial \theta_1 \cdots \partial_n} = 0.$$
4. Using the conclusion in question (3) to verify that when $n = 2$, there is no ex-post Pareto efficient social choice function that is truthfully implementable in dominant strategies.

Exercise 18.14 Consider an economy consisting of $n \geq 3$ individuals and public good y. The utility function of individual i is $u_i(t_i, y) = t_i + v_i(y)$ with $v_i(y) = -1/2y^2 + \bar{\theta}_i y$.

1. Find the VCG mechanism.

2. Prove that if $d_i(\theta_{-1}) = \frac{1}{2n}\sum\limits_{j \neq i} \theta_j^2 + \frac{n-1}{2n(n-2)}\sum\limits_{j \neq i}\sum\limits_{k \neq i,j} \theta_j \theta_k$, then the VCG mechanism is balanced.

3. Is the VCG mechanism Pareto efficient?

Exercise 18.15 (Jackson, 2003) Consider a public good provision problem with two participants. Let $I = \{1, 2\}$, and the type of each participant be distributed as $\theta^i \in \{0, 1\}$, $\forall i = 1, 2$. The decision of public good provision is discrete, i.e., $y = \{0, 1\}$ with $y = 1$ representing that the public good will be provided. Assume that the cost for providing the public good is $c = \frac{3}{2}$. The utility functions of the two participants are quasi-linear, i.e., $u_i(y, t_i, \theta_i) = y\theta_i - t_i$, in which t_i is the transfer for participant i. $(y(\boldsymbol{\theta}), t_i(\boldsymbol{\theta}), i \in I)$ is budget-balanced if satisfying:

$$\sum_{i \in I} t_i(\boldsymbol{\theta}) = 0, \forall \boldsymbol{\theta}.$$

$(y(\boldsymbol{\theta}), t_i(\boldsymbol{\theta}), i \in I)$ is individually rational if satisfying:

$$u_i(y(\boldsymbol{\theta}), t_i(\boldsymbol{\theta})) \geqq 0, \forall \boldsymbol{\theta}, \forall i \in I.$$

1. Prove that the VCG mechanism is not budget-balanced for this economy.

2. Prove that the VCG mechanism does not satisfy individual rationality.

Exercise 18.16 (Non-Implementability by Induced Revelation Mechanism) The textbook points out the following: although the original general mechanism (fully) implements a social choice correspondence F, the revelation mechanism induced, $\langle E, g \rangle$, may only partially implement, but not fully implement nor implement, F. Now, consider the example given by Dasgupta, Hammond, and Maskin (1979). Suppose that the set of alternatives is $A = \{a, b, c, d, e, p, q, r\}$, and the sets of preferences of individuals 1 and 2 are denoted by $R_1 = \{R_1, R_1'\}$ and $R_2 = \{R_2, R_2'\}$, respectively, which are described as follows:

$$R_1 = \begin{bmatrix} q \\ a-c-e \\ d-b-p \\ r \end{bmatrix}, R_1' = \begin{bmatrix} c-b-p \\ a-d-e \\ q-r \end{bmatrix}, R_2 = \begin{bmatrix} r \\ d-a-e \\ b-c-p \\ q \end{bmatrix}, R_2' = \begin{bmatrix} d \\ b-c \\ a \\ e-p-q-r \end{bmatrix}.$$

Consider a social choice rule defined as follows:

$$F(R_1, R_2) = \{a, e\};$$
$$F(R_1', R_2) = \{c, p, b\};$$
$$F(R_1, R_2') = \{d\};$$
$$F(R_1', R_2') = \{b\}.$$

1. Prove that social choice rule F satisfies Maskin monotonicity and Pareto efficiency.

2. Prove that F can be implemented in dominant strategies by g_1, which is defined by the following table, in which the rows are the strategic choices of participant 1 and the columns are the strategic choices of participant 2.

a	d	e
c	b	p
a	b	e

3. Prove that f cannot be implemented in dominant strategies by the direct mechanism g_2 defined by the following table.

	R_2	R_2'
R_1	a	d
R_1'	c	b

4. Prove that the equilibrium outcome that cannot be implemented by a direct mechanism is not Pareto efficient.

Exercise 18.17 Prove or disprove whether the following social choice correspondences satisfy Maskin monotonicity.

1. Weakly Pareto efficient correspondence.

2. Condorcet correspondence.

3. Individually rational allocation correspondence.

4. Walrasian correspondence.

5. Lindahl correspondence.

6. Constrained Walrasian correspondence.

7. Constrained Lindahl correspondence.

8. For the above social choice correspondences that do not satisfy Maskin monotonicity, what kind of restriction on the domain can make them satisfy Maskin monotonicity? Prove your claim.

Exercise 18.18 In the King Solomon example, as the social goal does not satisfy Maskin monotonicity, it is not implementable in Nash equilibrium. Can King Solomon's goal be implementable in subgame perfect equilibrium or be virtually Nash implementable? Give your ideas about this.

Exercise 18.19 If the no-veto power property is not satisfied, Maskin monotonicity does not guarantee that the social choice rule be fully Nash implemented. Illustrate this conclusion with an example.

Exercise 18.20 Is the non-veto power condition a necessary condition for a social choice rule to be fully Nash implementable? If so, give the proof; if not, provide a counterexample.

Exercise 18.21 (Borda rule) There are N individuals in an economy. Suppose that X contains a finite number of alternatives. For any alternative x, $B^i(x)$ denotes the number of alternatives that is not as good as x in X for individual i. The Borda rule is defined as:

$$xRy \iff \sum_{i=1}^{N} B^i(x) \geqq \sum_{i=1}^{N} B^i(y).$$

1. Prove that this correspondence satisfies Maskin monotonicity.
2. Does the correspondence satisfy the no-veto power condition?

Exercise 18.22 Let $N = \{1, 2, 3\}$, $E = \{e, \bar{e}\}$ and $A = \{a, b, c\}$. The preferences of individuals are given as follows:

Under state e, $1 : b \succ a \succ c$; $2 : a \succ c \succ b$; $3 : a \succ c \succ b$.

Under state $\bar{e}$, $1 : b \succ c \succ a$; $2 : c \succ a \succ b$; $3 : c \succ a \succ b$.

For any $\theta \in E$, define the social choice rule F as follows:

(i) $a \in F(\theta)$ if and only if a is better than b by the majority rule:

(ii) $b \in F(\theta)$ if and only if b is better than a by the majority rule;

(iii) $c \in F(\theta)$ if and only if $\{x \in A | x \preceq_i^\theta c\} = A$ for any $i \in N$.

Answer the following questions:

1. Does F satisfy Maskin monotonicity? Justify your answer.
2. Does F satisfy no-veto power? Justify your answer.

Exercise 18.23 Let $N = \{1, 2, 3, 4\}$, and there be four indivisible objects, a_1, a_2, a_3, a_4. For any $i \in N$, the initial endowment is $w_i = a_i$, $E = \{e_1, e_2\}$. The preferences of individuals are given as follows:

Under state e_1,

$$1 : a_1 \succ a_2 \succ a_3 \succ a_4;$$
$$2 : a_2 \succ a_1 \succ a_3 \succ a_4;$$
$$3 : a_3 \sim a_4 \succ a_1 \succ a_2;$$
$$4 : a_3 \sim a_4 \succ a_2 \succ a_1 \circ$$

Under state e_2,

$$1 : a_1 \succ a_2 \succ a_3 \succ a_4;$$
$$2 : a_2 \succ a_1 \succ a_3 \succ a_4;$$
$$3 : a_3 \succ a_4 \succ a_1 \succ a_2;$$
$$4 : a_3 \succ a_4 \succ a_2 \succ a_1 \circ$$

The social goal is to allocate the four objects to the individuals, i.e., each one receives one object. Specifically, the social goal is a correspondence $\sigma : N \to A$. Let Z represent the set of all possible allocations. The allocation rule F satisfies individual rationality, namely: for any $e_i \in E$, we have $F(e_i) = \{\sigma \in Z | \sigma_j \succeq_j^{e_i} w_j$, for any $e_i \in E, j \in N\}$. Answer the following questions:

1. Does F satisfy Maskin monotonicity? Justify your answer.

2. Does F satisfy no-veto power? Justify your answer.

Exercise 18.24 Let $N = \{1, 2, 3\}$, $E = \{e, \bar{e}\}$ and $A = \{a, b, c\}$. The specific preferences are given as follows:

Under state e, $1 : a \succ b \succ c$; $2 : c \succ a \succ b$; $3 : b \succ c \succ a$.

Under state $\bar{e}$, $1 : a \succ b \succ c$; $2 : c \succ a \succ b$; $3 : c \succ b \succ a$.

The social choice rule F is given by the majority voting rule, with $F(e) = \{a, b, c\}$ and $F(\bar{e}) = \{c\}$. Is F Nash implementable? Justify your answer.

Exercise 18.25 (Moore and Repullo, 1990) Consider the Nash implementation problem with two participants. Let $N = \{1, 2\}$. The following condition is called $\mu 2$.

$\mu 2$: Condition μ given in Definition 18.8.6 holds. In addition, for each 4-tuple $(a, e', b, \hat{e}) \in (A \times E \times A \times E)$ with $a = F(e')$ and $b = F(\hat{e})$, there exists $c = c(a, e', b, \hat{e}) \in C_1(a, e') \bigcap C_2(b, \hat{e})$, such that for all $e^* \in E$, the following condition is satisfied:

If $c \in C_1(a, e') \subseteq L_1(c, e^*)$ and $c \in C_2(b, \hat{e}) \subseteq L_2(c, e^*)$, then $c \in F(e^*)$.

Prove: in a two-agent environment, F is Nash implementable if and only if condition $\mu 2$ is satisfied.

Exercise 18.26 (Moore and Repullo, 1990) Consider the problem of Nash implementation in two-agent environments. A social choice rule F satisfies the restricted veto power property, i.e., if for all $i \in I, e \in E, a \in A$, there exist $b \in range(F) \equiv \{\hat{a} \in A : \hat{a} = F(e) \text{ for some } e \in E\}$, such that:

$$\text{if } A \subseteq \bigcap_{j \neq i} L_j(a, e) \text{ and } a \succeq_i b, \text{ then } a \in F(e).$$

We call outcome z a bad outcome, if for all $e \in E$ and all $a \in F(e)$, we have $a \succ_i (e) z$, in which $\succ_i (e)$ is the preference of participant i under the environment e.

Prove that if a two-agent choice rule F satisfies monotonicity and restricted veto power, and if there is a bad outcome, then F can be Nash implemented.

Exercise 18.27 (Withholding Mechanism, Tian, 1993) Consider the set of economic environments E with one private good x_i, K public goods y, and $n \geqq 3$ individuals. The production function is $y = f(v) = v$. Suppose that the preference relations $\succcurlyeq_i$ of participant i are convex and strongly monotone in private goods, and for all $i \in N$, $(x_i, y)\ P_i\ (x_i', y')$, with $x_i \in \mathcal{R}_{++}$, $x_i' \in \partial \mathcal{R}_+$, and $y, y' \in \mathcal{R}_+^K$. The $\partial \mathcal{R}_+^m$ is the boundary of $\mathcal{R}_+^m$.

For each $i \in N$, define the message space as

$$M_i = (0, 1] \times (0, \mathring{w}_i] \times \mathcal{R}^K \times \mathcal{R}^K,$$

whose general elements can be represented as $(\delta_i, w_i, \phi_i, y_i)$, and let $M = \prod_{i=1}^n M_i$. The personalized price is defined as

$$q_i(\mathbf{m}) = \frac{1}{n} + m_{i+1} - m_{i+2}.$$

Define the correspondence B: $M \to 2^{\mathcal{R}_+^K}$,

$$B(\mathbf{m}) = \{y \in \mathcal{R}_+^K : (1 - \delta(\mathbf{m}))w_i - q_i(\mathbf{m}) \cdot y \geqq 0\ \forall\ i \in N\},$$

in which $\delta(\mathbf{m}) = \min\{\delta_1, \cdots, \delta_n\}$. Define the outcome function of public goods as Y: $M \to B$,

$$Y(\mathbf{m}) = \{y : \min_{y \in B(\mathbf{m})} \parallel y - \tilde{y} \parallel\},$$

and $\tilde{y} = \sum_{i=1}^n y_i$. For each i, define the tax function T_i: $M \to \mathcal{R}$,

$$T_i(\mathbf{m}) = q_i(\mathbf{m}) \cdot Y(\mathbf{m}).$$

It can be seen that

$$\sum_{i=1}^{n} T_i(\mathbf{m}) = q(\mathbf{m}) \cdot Y(\mathbf{m}).$$

The outcome function of private goods $X(\mathbf{m})$: $M \to \mathcal{R}_+$ is defined by

$$X_i(\mathbf{m}) = w_i - q_i(\mathbf{m}) \cdot Y(\mathbf{m}).$$

1. Prove that the correspondence B: $M \to 2^{\mathcal{R}_+^K}$ is a nonempty, compact, and convex continuous correspondence.
2. Prove that $Y(\mathbf{m})$ is a single-valued continuous function on M.
3. Prove that the above mechanism is a continuous and feasible mechanism.
4. Prove that the above mechanism fully Nash implements the Lindahl correspondence on E.

Exercise 18.28 (Fully Implementing Walrasian Allocations, Tian, 1992) Consider a pure exchange economy $e = (\{X_i, \boldsymbol{w}_i, \succcurlyeq_i\})$, and let E be the set of all economies in which Walrasian equilibrium exists. Define the mechanism that is individually feasible, balanced, and continuous as follows:

For each $i \in N$, define the message space as

$$M_i = \mathcal{R}^L_{++} \times \mathcal{R}^{nL},$$

whose generic elements can be represented as $m_i = (p_i, x_{i1}, \cdots, x_{in})$, and let $M = \prod_{i=1}^n M_i$. Define the price vector function as $p : M \to \mathcal{R}^L_{++}$,

$$p(\mathbf{m}) = \begin{cases} \sum_{i=1}^n \frac{a_i}{a} p_i, & \text{if } a > 0, \\ \sum_{i=1}^n \frac{1}{n} p_i, & \text{if } a = 0, \end{cases}$$

in which $a_i = \sum_{j,k \neq i} \|p_j - p_k\|$, $a = \sum_{i=1}^n a_i$ and $\|\cdot\|$ represents the Euclidean norm.

Define the correspondence that is individually feasible and budget-balanced as $B : M \to\to \mathcal{R}^{nL}_+$,

$$B(\mathbf{m}) = \{x \in \mathcal{R}^{nL}_+ : \sum_{i=1}^n x_i = \sum_{i=1}^n w_i \ \& \ p(\mathbf{m}) \cdot x_i = p(\mathbf{m}) \cdot w_i, \ \forall\, i \in N\}.$$

Let $\tilde{x}_j = \sum_{i=1}^n x_{ij}$, $\tilde{x} = (\tilde{x}_1, \tilde{x}_2, \cdots, \tilde{x}_n)$. The outcome function of private goods, X: $M \to \mathcal{R}^{nL}_+$, is given as follows:

$$X(\mathbf{m}) = \{y \in \mathcal{R}^{nL}_+ : \min_{y \in B(\mathbf{m})} \|y - \tilde{x}\|\}.$$

1. Prove that the price vector function $p : M \to \mathcal{R}^L_{++}$ is continuous.

2. Prove that the correspondence B: $M \to 2^{\mathcal{R}_+^K}$ is a continuous correspondence with nonempty, compact, and convex values.

3. Prove that $X(\mathbf{m})$ is a single-valued continuous function on M.

4. Prove: for all $e \in E$, every Nash equilibrium outcome of this mechanism is a Walrasian equilibrium allocation.

5. Prove: for all $e \in E$, every Walrasian equilibrium allocation is a Nash equilibrium outcome of the mechanism. As a consequence, the mechanism fully Nash implements the Walrasian correspondence on E.

Exercise 18.29 (Moore and Repullo, 1988) Consider an economy with two participants. Suppose that the economy has two states, $E = \{C, L\}$: when $e = C$, the preferences of two individuals are a Cobb-Douglas utility function, and the allocation is $f(C) \in A$; when $e = L$, both individual preferences are perfectly complementary utility functions (i.e., in the form of a Leontief function), and the allocation is $f(L) \in A$. Suppose that $A = \{f(C), f(L), x, y\}$. The states and allocations can be seen in Figure 18.6. Consider the following dynamic game g.

First stage: participant 1 provides report r_1. If $r_1 = L$, the game is over and the outcome is $f(L)$; if $r_1 = C$, the game goes into the second stage.

Second stage: participant 2 provides report $r_2 = C$, the game is over and the outcome is $f(C)$; otherwise, the game goes into the third stage.

Third stage: participant 1 chooses an outcome from $\{x, y\}$, and the game is over.

Prove: the above social choice rule f can be implemented in subgame perfect equilibrium by the dynamic game g.

Exercise 18.30 (Jackson,1992) Assume environments (N, A, E), in which $N = \{1, \cdots, n\}$ is the set of participants. A mechanism $\Gamma = (M, g)$ consists of the message space of participants $M = M^1 \times \cdots \times M^n$ and the outcome function $g : M \to A$. We say that m^i is an **undominated message** if there does not exist message m'^i that weakly dominates m^i. Let the set of all undominated messages of mechanism Γ under the environment e be $U(\Gamma, e)$, all of the Nash equilibria be $N(\Gamma, e)$, and the set of all of the undominated Nash equilibria be $UN(\Gamma, e) = U(\Gamma, e) \bigcap N(\Gamma, e)$. The social choice rule $F : E \to A$ is said to be **implementable in undominated Nash equilibrium** if there exists a mechanism (M, g), such that $F(e) = g(UN(\Gamma, e)), \forall e \in E$.

Suppose that $N = \{1, 2\}$, $A = \{a, b\}$, $E = \{(R^1, R^2), (\bar{R}^1, R^2)\}$, and $aP^1b, b\bar{P}^1aP^2b$. The social choice rule satisfies $F(R^1, R^2) = b$ and $F(\bar{R}^1, R^2) = a$. The following mechanism (M, g) is shown in Figure 18.7.

1. Prove that F does not satisfy Pareto efficiency and Maskin monotonicity (and thus, it is not Nash implementable).

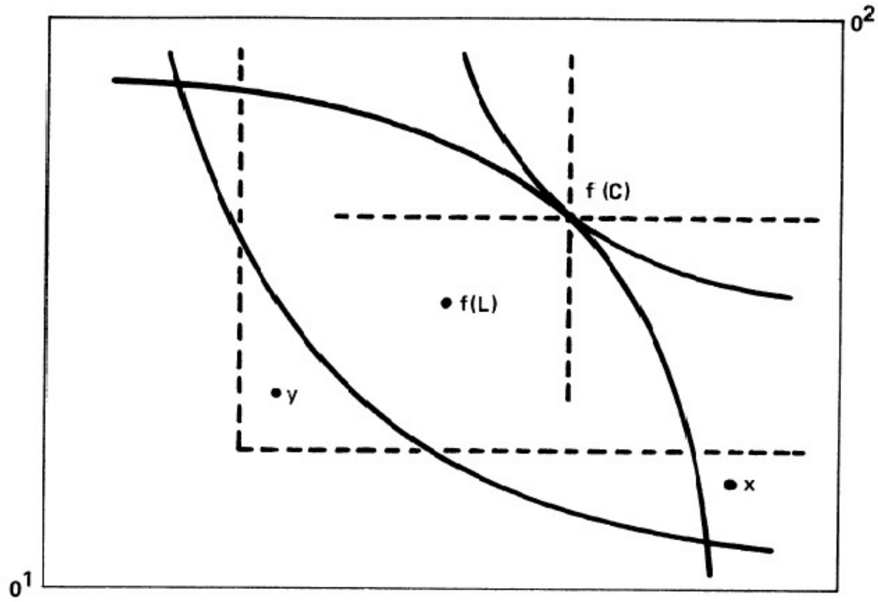

Figure 18.6: Implementation of subgame refining equilibrium

2. Prove that F can be implemented in undominated Nash equilibrium by mechanism (M, g).

Exercise 18.31 (Matsushima, 1988; Abreu and Sen, 1991; Diamantaras, 2009) For environments (N, X', E), let $A' = \{a_1, a_2, \cdots, a_K\}$ be a set of socially feasible outcomes. A is the set of all lotteries defined on A', i.e., $A = \{\mathbf{x} \in [0,1]^K : \sum_{k=1}^{K} x_k = 1\}$.

We call the environment **no-total indifference** if for every i and e, there are two outcomes $a, a' \in A'$, such that $u_i(a, e) > u_i(a', e)$, where u^i is the utility function of participant i.

We say that the two social choice rules F and H are ϵ-close, if for any $e \in E$, there is a one-to-one mapping $\tau : F(E) \to H(E)$, such that $|\mathbf{x} - \tau(\mathbf{x})| = \sqrt{\sum_{k=1}^{K}(x_k - \tau(x)_k)^2} < \epsilon$ for any $\mathbf{x} \in F(e)$.

A social choice rule F is called **virtually Nash implementable** if there is a social rule G that is Nash-implementable, as well as G and F are ϵ-close.

F is called **ordinal** if for $F(e) \neq F(e')$, there is a participant i and $\mathbf{x}, \mathbf{x}' \in A$, such that $u^i(\mathbf{x}, e) \geqq u^i(\mathbf{x}', e), u^i(\mathbf{x}', e') > u_i(\mathbf{x}, e')$.

Prove that if $|N| \geqq 3$, and the environment is no-total indifference, then any ordinal social choice rule can be virtually Nash implementable.

$$
\begin{array}{cccccccccccc}
 & & & & & & M^2 & & & & & \\
 & & m^2 & & & & & & & & & \\
 & m^1 & b & a & a & a & \ldots & a & a & a & a & \ldots \\
 & & b & a & a & a & \ldots & b & b & b & b & \ldots \\
 & & b & b & a & a & \ldots & b & b & b & b & \ldots \\
 & & b & b & b & a & \ldots & b & b & b & b & \ldots \\
M^1 & & \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & \vdots & \\
 & \bar{m}^1 & a & b & b & b & \ldots & b & b & b & b & \ldots \\
 & & a & a & a & a & \ldots & a & b & b & b & \ldots \\
 & & a & a & a & a & \ldots & a & a & b & b & \ldots \\
 & & a & a & a & a & \ldots & a & a & a & b & \ldots \\
 & & \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & \vdots &
\end{array}
$$

Figure 18.7: Undominated Nash implementation

18.14 References

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Chapter 19

Mechanism Design with Multiple Agents: Incomplete Information

19.1 Introduction

This chapter discusses the mechanism of incomplete information. The so-called incomplete information refers to that agents do not know each other's economic characteristics, such as preferences, technology, and endowments.

So far, we have only considered the implementation of social choice rules under two extreme solution concepts in terms of information requirement: dominant strategy equilibrium; and Nash equilibrium and its refinements. Dominant equilibrium is the strongest solution concept for characterizing the self-interest behavior of individuals, since it requires the least amount of information for both designers and agents, who do not need to know other participants' information. Although the mechanism with dominant equilibrium as the solution concept is able to induce participants to report their preferences truthfully, and the VCG mechanism can even implement the efficient provision of public goods or indivisible goods, the truthful revelation mechanism cannot implement Pareto optimal allocations since the balancedness condition, in general, does not hold.

The other extreme solution concept is Nash equilibrium and its refinements, which constitute much weak solution concepts, since it assumes that each participant needs to know not only his own economic characteristics but also those of all other participants. Although (weakly) Pareto optimal allocations can be implemented in Nash equilibrium, it is necessary to abandon the requirement that participants report truthfully.

Under these two solution concepts, the Hurwicz Impossibility Theorem, the VCG mechanism, and Nash implementable mechanisms all show that Pareto efficiency and truth-telling are generally impossible to be achieved simultaneously. Then, is there a solution concept between them that can achieve both properties simultaneously? The answer is affirmative if one adopts **Bayesian-Nash equilibrium** as the solution concept to study implementation of social choice rules.[1] Bayesian-Nash equilibrium assumes that, although each agent does not know the economic characteristics of the others, he does know the probability distributions of others' economic characteristics. We then consider **Bayesian incentive compatibility**, i.e., everyone has the incentive to tell the truth provided that all of the others do the same. Therefore, Bayesian incentive compatibility is a requirement of conditional truth-telling. The **expected external mechanism** introduced in this chapter shows that Bayesian incentive-compatibility and Pareto optimal allocation can be achieved simultaneously. The corresponding implementation is called **Bayesian implementation**. One can design various Bayesian incentive compatible mechanisms and characterize the full implementability of general social choice rules. We then discuss the implementability of a social choice rule with **ex-post equilibrium** as the solution concept. The advantage of adopting ex-post equilibrium is that agents are not required to know the distributions of economic characteristics. It is an equilibrium solution concept in which the economic agents have no regret.

In the following, we will first set up the basic model with incomplete information, and introduce the notations and definitions of Bayesian implementation in various senses. We will then consider Bayesian incentive compatibility and Bayesian implementation issues of social choice functions (rather than social choice correspondences). In the last two sections of this chapter, we will consider full Bayesian implementation and ex-post implementation of a social choice correspondence.

19.2 Basic Analytical Framework

19.2.1 Model

Throughout this chapter, we follow the notation introduced in the previous chapter. Let Z denote the set of outcomes, $A \subseteq Z$ be the set of feasible outcomes, and Θ_i be agent i's space of types. For simplicity, unless otherwise stated, we assume that each individual's preference is given by a parametric utility function with **private values**, denoted by $u_i(x, \theta_i)$, where $x \in Z$ and $\theta_i \in \Theta_i$. Assume that the designer and all agents know that $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_n)$ is distributed according to the probability density $\varphi(\boldsymbol{\theta})$ on $\Theta = \prod_{i \in N} \Theta_i$.

[1]The notion of Bayesian-Nash equilibrium was introduced by Harsanyi. Nash, Selten, and Harsanyi received the 1994 Nobel Prize in economics for their fundamental contributions to Nash equilibrium, sub-game perfect Nash equilibrium, and Bayesian-Nash equilibrium, respectively.

Each agent knows his own type θ_i, and update the information on other agents' types according to Bayes' rule:

$$\varphi(\boldsymbol{\theta}_{-i}|\theta_i) = \frac{\varphi(\theta_i, \boldsymbol{\theta}_{-i})}{\int_{\Theta_{-i}} \varphi(\theta_i, \boldsymbol{\theta}_{-i}) d\boldsymbol{\theta}_{-i}}.$$

Let $M = M_1 \times \cdots \times M_n$ be the message space, and $h : M \to Z$ be the outcome function. As usual, a **mechanism** is a pair of message space and outcome function, denoted by $\Gamma = \langle M, h\rangle$. Moreover, if messages of each agent consist of types, i.e., $M_i = \theta_i$ for all $i \in N$, $\Gamma = \langle \Theta, h\rangle$ is called the **revelation mechanism**.

Given $\boldsymbol{m}$ with $m_i : \Theta_i \to M_i$, agent i's interim expected utility at θ_i is given by

$$\begin{aligned}\Pi^i_{\langle M,h\rangle}(\boldsymbol{m}(\theta); \theta_i) &\equiv E_{\boldsymbol{\theta}_{-i}}[u_i(h(m_i(\boldsymbol{\theta}_i), \boldsymbol{m}_{-i}(\boldsymbol{\theta}_{-i})), \theta_i) \mid \theta_i] \\ &= \int_{\Theta_{-i}} u_i(h(\boldsymbol{m}(\theta)), \theta_i)\varphi(\boldsymbol{\theta}_{-i}|\theta_i) d\boldsymbol{\theta}_{-i}. \qquad (19.2.1)\end{aligned}$$

Definition 19.2.1 A strategy $\boldsymbol{m}^*(\cdot)$ is a **Bayesian-Nash equilibrium (BNE)** of $\langle M, h\rangle$ if for all $\theta_i \in \Theta_i$,

$$\Pi^i(\boldsymbol{m}^*(\boldsymbol{\theta}_i); \theta_i) \geqq \Pi^i(m_i, \boldsymbol{m}^*_{-i}(\boldsymbol{\theta}_{-i}); \theta_i) \quad \forall m_i \in M_i.$$

In other words, if player i believes that other players are playing strategies $\boldsymbol{m}^*_{-i}(\cdot)$, then he maximizes his interim expected utility by playing strategy $m^*_i(\cdot)$. Denote by $\mathcal{B}(\Gamma)$ the set of all Bayesian-Nash equilibria of the mechanism.

Remark 19.2.1 In a private value incomplete information setting, a message $m_i : \Theta_i \to M_i$ is a dominant strategy for agent i in mechanism $\Gamma = \langle M, h\rangle$ if for all $\boldsymbol{\theta}_i \in \Theta_i$ and all possible strategies $\boldsymbol{m}_{-i}(\boldsymbol{\theta}_{-i})$,

$$E_{\boldsymbol{\theta}_{-i}}[u_i(h(m_i(\theta_i), \boldsymbol{m}_{-i}(\boldsymbol{\theta}_{-i})), \theta_i) \mid \theta_i] \geqq E_{\boldsymbol{\theta}_{-i}}[u_i(h(m'_i, \boldsymbol{m}_{-i}(\boldsymbol{\theta}_{-i})), \theta_i) \mid \theta_i] \qquad (19.2.2)$$

for all $m'_i \in M_i$.

Since condition (19.2.2) holds for all θ_i and all $\boldsymbol{m}_{-i}(\boldsymbol{\theta}_{-i})$, it is equivalent to the condition that, for all $\theta_i \in \Theta_i$,

$$u_i(h(m_i(\theta_i), \boldsymbol{m}_{-i}), \theta_i) \geqq u_i(h(m'_i, \boldsymbol{m}_{-i}), \theta_i) \qquad (19.2.3)$$

for all $m'_i \in M_i$ and all $\boldsymbol{m}_{-i} \in M_{-i}$.[2] This leads to the following equivalent definition on dominant equilibrium.

[2](19.2.2) implies (19.2.3) by choosing $\boldsymbol{m}_{-i}(\boldsymbol{\theta}_{-i}) = \boldsymbol{m}_{-i}$ for all $\boldsymbol{\theta}_{-i} \in \boldsymbol{\Theta}_{-i}$; on the other hand, (19.2.3) implies that $u_i(h(m_i(\theta_i), \boldsymbol{m}_{-i}(\boldsymbol{\theta}_{-i})), \theta_i) \geqq u_i(h(m'_i, \boldsymbol{m}_{-i}(\boldsymbol{\theta}_{-i})), \theta_i)$ for each $\boldsymbol{\theta}_{-i} \in \boldsymbol{\Theta}_{-i}$, which further implies (19.2.2) by taking expectation with respect to $\boldsymbol{\theta}_{-i}$.

dominant strategy equilibrium

Definition 19.2.2 For a private value model, a message $\boldsymbol{m}^*$ is a **dominant strategy equilibrium** of mechanism $\Gamma = \langle M, h \rangle$ if for all i and all $\theta_i \in \Theta_i$,

$$u_i(h(m_i^*(\theta_i), \boldsymbol{m}_{-i}), \theta_i) \geqq u_i(h(m_i', \boldsymbol{m}_{-i}), \theta_i)$$

for all $m_i' \in M_i$ and all $\boldsymbol{m}_{-i} \in M_{-i}$, which is the same as that in the case of complete information.

Remark 19.2.2 It is clear that every dominant strategy equilibrium is a Bayesian-Nash equilibrium, but the converse may not be true. Bayesian-Nash equilibrium requires more sophisticated belief structures from the agents than does dominant strategy equilibrium. Each agent, in order to find his optimal strategy, must have a correct prior, $\varphi(\cdot)$, over type profiles, and must correctly predict the equilibrium strategies used by other agents.

19.2.2 Bayesian Incentive Compatibility and Bayesian Implementation

We now focus on the issues of Bayesian incentive compatibility and Bayesian implementation of a social choice function. By the revelation principle, we can further focus on direct revelation mechanisms.

Let $f : \Theta \to A$ be a social choice function. Since, at Bayesian-Nash equilibrium, every agent reaches his interim expected utility maximization, Bayesian implementation is also sometimes called the **interim implementation**. It includes (full) Bayesian implementation and partial Bayesian implementation. Since the social choice rule is a function, Bayesian-Nash implementation coincides with full Bayesian-Nash implementation.

Definition 19.2.3 A mechanism $\Gamma = \langle M, h \rangle$ is said to **Bayesian implement** a social choice function f on Θ if for *every* Bayesian-Nash equilibrium $\boldsymbol{m}^*$, we have $h(\boldsymbol{m}^*(\boldsymbol{\theta})) = f(\boldsymbol{\theta}),\ \forall \boldsymbol{\theta} \in \Theta$. If such a mechanism exists, we say that f is **Bayesian implementable**.

When there are multiple Bayesian-Nash equilibria, it may result in some undesired equilibrium outcome that is not equal to f (we will provide such an example). Then, we have the following weaker concept of Bayesian implementation.

Definition 19.2.4 A mechanism $\Gamma = \langle M, h \rangle$ is said to **partially Bayesian implement** a social choice function f on Θ if there exists a Bayesian-Nash equilibrium $\boldsymbol{m}^*$, such that $h(\boldsymbol{m}^*(\boldsymbol{\theta})) = f(\boldsymbol{\theta}),\ \forall \boldsymbol{\theta} \in \Theta$. If such a mechanism exists, we say that f is **partially Bayesian implementable**.

We will see that a social choice function is partially Bayesian implementable if and only if it is truthfully Bayesian implementable.

Definition 19.2.5 We say that a social choice function f is **truthfully Bayesian implementable or Bayesian incentive compatible** if truth-telling, i.e., $m^*(\boldsymbol{\theta}) = \boldsymbol{\theta}, \forall \boldsymbol{\theta} \in \Theta$, is a Bayesian-Nash equilibrium of the revelation mechanism $\Gamma = (\Theta, f)$, i.e.,

$$E_{\boldsymbol{\theta}_{-i}}[u_i(f(\theta_i, \boldsymbol{\theta}_{-i})), \theta_i) \mid \theta_i] \geqq E_{\boldsymbol{\theta}_{-i}}[u_i(f(\theta'_i, \boldsymbol{\theta}_{-i})), \theta_i) \mid \theta_i], \quad \forall i, \forall \theta_i, \theta'_i \in \Theta_i.$$

Bayesian incentive compatibility means that for social choice rule f, every agent will report his type truthfully *provided that all of the other agents report their types truthfully*, and thus every truthful strategy profile is a Bayesian-Nash equilibrium of the direct mechanism $\langle \boldsymbol{\theta}, f \rangle$. Note that Bayesian incentive-compatibility is only conditional truth-telling. It does not determine what is the best response of an agent when other agents are not reporting truthfully. When a mechanism has multiple equilibria, non-truth-telling may be a Bayesian-Nash equilibrium resulting in an undesired outcome that is not equal to f. When Bayesian incentive compatibility is just a basic requirement, it in general only ensures that a social choice rule is truthfully Bayesian implementable but not (fully) Bayesian implementable, i.e., it may also have some undesired equilibrium outcomes (they are not socially optimal) when a mechanism has multiple equilibria.

Similarly, we have the following revelation principle.

Proposition 19.2.1 (Revelation Principle) *A social choice rule $f(\cdot)$ is partially implementable in Bayesian-Nash equilibrium if and only if it is truthfully implementable in Bayesian-Nash Equilibrium.*

PROOF. The proof is the same as previously: Suppose that there exists a mechanism $\Gamma = (M_1, \cdots, M_n, g(\cdot))$ and an equilibrium strategy profile $\boldsymbol{m}^*(\cdot) = (m_1^*(\cdot), \cdots, m_n^*(\cdot))$ such that $g(\boldsymbol{m}^*(\cdot)) = f(\cdot)$ and $\forall i, \forall \theta_i \in \Theta_i$. We then have

$$E_{\boldsymbol{\theta}_{-i}}[u_i(g(m_i^*(\theta_i), \boldsymbol{m}_{-i}^*(\boldsymbol{\theta}_{-i})), \theta_i) \mid \theta_i] \geqq E_{\boldsymbol{\theta}_{-i}}[u_i(g(m'_i, \boldsymbol{m}_{-i}^*(\boldsymbol{\theta}_{-i})), \theta_i) \mid \theta_i], \forall m'_i \in M_i.$$

One way to deviate for agent i is by pretending that his type is $\hat{\theta}_i$, rather than θ_i, i.e., sending message $m'_i = m_i^*(\hat{\theta}_i)$. This gives

$$E_{\boldsymbol{\theta}_{-i}}[u_i(g(m_i^*(\theta_i), \boldsymbol{m}_{-i}^*(\boldsymbol{\theta}_{-i})), \theta_i) \mid \theta_i] \geqq E_{\boldsymbol{\theta}_{-i}}[u_i(g(m_i^*(\hat{\theta}_i), \boldsymbol{m}_{-i}^*(\boldsymbol{\theta}_{-i})), \theta_i) \mid \theta_i], \forall \hat{\theta}_i \in \Theta_i.$$

However, since $g(\boldsymbol{m}^*(\boldsymbol{\theta})) = f(\boldsymbol{\theta})$, $\forall \boldsymbol{\theta} \in \Theta$, we must have $\forall i, \forall \theta_i \in \Theta_i$ that

$$E_{\boldsymbol{\theta}_{-i}}[u_i(f(\theta_i, \boldsymbol{\theta}_{-i}), \theta_i) \mid \theta_i] \geqq E_{\boldsymbol{\theta}_{-i}}[u_i(f(\hat{\theta}_i, \boldsymbol{\theta}_{-i}), \theta_i) \mid \theta_i], \forall \hat{\theta}_i \in \Theta_i.$$

□

Although Bayesian incentive compatibility is a necessary and sufficient condition for f to be truthfully Bayesian implementable, it is not a sufficient condition for f to be (fully) Bayesian implementable. Bayesian implementability requires the set of all BNE outcome(s) to be equal to $f(\boldsymbol{\theta})$.

The goal for designing a mechanism is to reach a desired equilibrium outcome, but it may also result in undesired equilibrium outcomes. The severity of this multiple equilibrium problem has been exemplified by Demski and Sappington (1984), Postlewaite and Schmeidler (1986), Repullo (1988), among others. The implementation problem involves designing mechanisms to ensure that all equilibria result in desired outcomes.

Under what conditions does there exist a mechanism that has a unique Bayesian incentive-compatible equilibrium outcome? If such a mechanism exists, by definition, we know that a social choice function f is Bayesian implementable. One such condition is the existence of undominated Bayesian equilibrium.

Definition 19.2.6 A message $\boldsymbol{m} \in M$ is weakly dominated if there is an agent i, θ_i, and another Bayesian message $m_i' \neq m_i$, such that for all $\boldsymbol{m}_{-i} \in M_{-i}$,

$$\Pi^i(m_i(\theta_i), \boldsymbol{m}_{-i}; \theta_i) \leqq \Pi^i(m_i'(\theta_i), \boldsymbol{m}_{-i}; \theta_i)$$

with strict inequality for some $\boldsymbol{m}_{-i} \in M_{-i}$.

Definition 19.2.7 $\boldsymbol{m} \in M$ is an **undominated Bayesian equilibrium** of $\Gamma = \langle M, h \rangle$, if it is a Bayesian-Nash equilibrium that is not weakly dominated.

Palfrey and Srivastava (1989) proved the following proposition:

Proposition 19.2.2 *For the class of private value economic environments, if there exists a mechanism such that all agents do not have weakly dominated strategies, then any Bayesian incentive compatible social choice function is Bayesian implementable.*

In the remainder of the chapter, we will first consider truthful Bayesian implementation of a social choice rule. We then discuss the necessary and sufficient conditions for full Bayesian implementation of a general social choice rule—the set of social choice functions with interdependent values.

19.3 Truthful Implementation of Pareto Efficient Outcomes

Once again, let us return to the quasilinear setting. From the discussion on dominant strategy implementation in the last chapter, we know that the VCG mechanism truthfully implements efficient outcomes in dominant strategies (and then, of course, in Bayesian-Nash equilibrium), but it does not truthfully implement Pareto efficient outcomes in dominant strategies. In fact, there is, in general, no ex-post Pareto efficient implementation in dominant strategies. However, in quasilinear environments, changing the equilibrium concept from DE to BNE allows us to truthfully Bayesian implement ex-post Pareto efficient choice rule $f(\cdot) = (y(\cdot), t_1(\cdot), \cdots, t_i(\cdot))$, where $\forall \boldsymbol{\theta} \in \Theta$,

$$y(\boldsymbol{\theta}) \in \text{argmax}_{y \in Y} \sum_{i=1}^{n} v_i(y, \theta_i), \tag{19.3.4}$$

and there is a balanced budget:

$$\sum_{i=1}^{n} t_i(\boldsymbol{\theta}) = 0. \tag{19.3.5}$$

The **expected externality mechanism** described below was suggested independently by D' Aspermont and Gerard-Varet (1979) and Arrow (1979), and thus is called the **AGV mechanism** in the literature. This mechanism enables us to have ex-post Pareto efficient implementation under the following additional assumption.

Assumption 19.3.1 Types are distributed independently: $\varphi(\boldsymbol{\theta}) = \Pi_i \varphi_i(\theta_i), \forall \boldsymbol{\theta} \in \Theta$.

Consider the VCG transfer for agent i. Instead of using other agents' announced types, take the interim expectation of other agents' valuation functions over their types and let

$$t_i(\hat{\boldsymbol{\theta}}) = E_{\boldsymbol{\theta}_{-i}}[\sum_{j \neq i} v_j(y(\hat{\theta}_i, \boldsymbol{\theta}_{-i}), \theta_j)] + d_i(\hat{\boldsymbol{\theta}}_{-i}).$$

By Assumption 19.3.1, the expectation over $\boldsymbol{\theta}_{-i}$ does not need to be taken conditionally on θ_i. Note that, unlike VCG mechanisms, the first term depends only on agent i's announcement $\hat{\theta}_i$, and not on other agents' announcements. This is because it sums the expected valuations of agents $j \neq i$, assuming that they tell the truth and given that i announces $\hat{\theta}_i$, and does not depend on the actual announcements of agents $j \neq i$. This means that $t_i(\cdot)$ is less "variable", but on average it will cause i's incentives to be lined up with the social surplus.

To see that agent i's Bayesian incentive compatibility is satisfied given that agents $j \neq i$ announce truthfully, observe that agent i solves

$$\begin{aligned}
&\max_{\hat{\theta}_i \in \Theta_i} E_{\boldsymbol{\theta}_{-i}}[v_i(y(\hat{\theta}_i, \boldsymbol{\theta}_{-i}), \theta_i) + t_i(\hat{\theta}_i, \boldsymbol{\theta}_{-i})] \\
&\quad = \max_{\hat{\theta}_i \in \Theta_i} E_{\boldsymbol{\theta}_{-i}}[\sum_{j=1}^{n} v_j(y(\hat{\theta}_i, \boldsymbol{\theta}_{-i}), \theta_j)] + E_{\boldsymbol{\theta}_{-i}} d_i(\boldsymbol{\theta}_{-i}). \quad (19.3.6)
\end{aligned}$$

Again, agent i's announcement only matters through the decision $y(\hat{\theta}_i, \boldsymbol{\theta}_{-i})$. In addition, by the definition of the efficient decision rule $y(\cdot)$, the designer always chooses $y(\cdot)$ to maximize social surplus (19.3.4) for each realization of $\boldsymbol{\theta}_{-i} \in \Theta_{-i}$, and then agent i's interim expected utility (19.3.6) is also maximized as long as he announces truthfully, i.e., $\hat{\theta}_i = \theta_i$. As such, individuals' goals that maximize (19.3.6) are consistent with the social goal (19.3.4).

Remark 19.3.1 The argument relies on the assumption that other agents announce truthfully. Consequently, it is in general not a dominant strategy for agent i to announce the truth. Indeed, if agent i expects the other agents to misrepresent their types, i.e., they announce $\hat{\boldsymbol{\theta}}_{-i}(\boldsymbol{\theta}_{-i}) \neq \boldsymbol{\theta}_{-i}$, then agent i's interim expected utility is

$$E_{\boldsymbol{\theta}_{-i}}[v_i(y(\hat{\theta}_i, \hat{\boldsymbol{\theta}}_{-i}(\boldsymbol{\theta}_{-i})), \theta_i) + \sum_{j \neq i}^{n} v_j(y(\hat{\theta}_i, \boldsymbol{\theta}_{-i}), \theta_j)] + E_{\boldsymbol{\theta}_{-i}} d_i(\boldsymbol{\theta}_{-i}),$$

which may not be maximized, even though agent i has announced truthfully.

To have specific transfers that achieve ex-post Pareto efficiency, we now choose functions $d_i(\cdot)$, so that the budget is balanced. To see this, let

$$\xi_i(\theta_i) = E_{\boldsymbol{\theta}_{-i}}[\sum_{j \neq i} v_j(y(\theta_i, \boldsymbol{\theta}_{-i}), \theta_j)],$$

so that the transfers in the expected externality mechanism are $t_i(\boldsymbol{\theta}) = \xi_i(\theta_i) + d_i(\boldsymbol{\theta}_{-i})$. Let $d(\cdot)$ "compensate" $\xi(\cdot)$ in the following way:

$$d_j(\boldsymbol{\theta}_{-j}) = -\sum_{i \neq j} \frac{1}{n-1} \xi_i(\theta_i).$$

Then,

$$\begin{aligned}
\sum_{j=1}^{n} d_j(\boldsymbol{\theta}_{-j}) &= -\frac{1}{n-1} \sum_{j=1}^{n} \sum_{i \neq j} \xi_i(\theta_i) = -\frac{1}{n-1} \sum_{j \neq i} \sum_{i=1}^{n} \xi_i(\theta_i) \\
&= -\frac{1}{n-1} (n-1 \sum_{i=1}^{n}) \xi_i(\theta_i) = -\sum_{i=1}^{n} \xi_i(\theta_i),
\end{aligned}$$

and therefore

$$\sum_{i=1}^{n} t_i(\boldsymbol{\theta}) = \sum_{i=1}^{n} \xi_i(\theta_i) + \sum_{i=1}^{n} d_i(\boldsymbol{\theta}_{-i}) = 0.$$

Thus, we have shown that when agents' Bernoulli utility functions are quasilinear and agents' types are statistically independent, the ex-post Pareto efficient social choice function is truthfully Bayesian implementable.

19.4 Characterizations of DIC and BIC to a Linear Model

In this section, we shall give a full characterization of dominant incentive compatibility (DIC) and Bayesian incentive compatibility (BIC) when participants' utility functions are of linear forms. The following BIC Characterization Theorem shall be highly useful. As a corollary, this theorem implies that for any two Bayesian incentive compatible mechanisms that truthfully implement the same decision rule $y(\cdot)$, their interim expected utilities $E_{\boldsymbol{\theta}_{-i}} U_i(\boldsymbol{\theta})$ and transfers $E_{\boldsymbol{\theta}_{-i}} t_i(\boldsymbol{\theta})$ coincide up to a constant. We will apply this theorem to prove the well-known **Revenue Equivalence Theorem** in auction theory in Chapter 21. Consider the quasi-linear environment with a decision set $Y \subseteq \mathcal{R}^n$, and the type spaces $\Theta_i = [\underline{\theta}_i; \overline{\theta}_i] \subseteq \mathcal{R}$ for all i. Each agent i's utility takes the form

$$\theta_i y_i + t_i.$$

Note that these payoffs satisfy the single-crossing property (SCP).

The decision set Y may have different forms, depending on the application under consideration.

Three examples:

1. Allocating an indivisible private good: $Z = \{y \in \{0,1\}^n : \sum_i y_i = 1\}$.
2. Provision of a public good: $Y = \{(q, \cdots, q) \in \mathcal{R}^n : q \in \{0,1\}\}$.
3. Agent i's valuation function is given by $\theta_i v_i(q_i)$ (as assumed in the basic principal-agent model in Chapter 16), where q_i is the level of consumption. Letting $y_i = v_i(q_i)$, it reduces to the linear model.

We can fully characterize both dominant incentive compatible and Bayesian incentive compatible social choice rules in this environment.

Let $U_i(\boldsymbol{\theta}) = \theta_i y_i(\boldsymbol{\theta}) + t_i(\boldsymbol{\theta})$. For any given $\boldsymbol{\theta}_{-i}$, we first have the following proposition:

Proposition 19.4.1 (DIC Characterization Theorem) *In the linear model, a social choice rule $(\boldsymbol{y}(\cdot), t_1(\cdot), \cdots, t_n(\cdot))$ is dominant incentive compatible if and only if for all $i \in N$,*

*(1) (**dominant monotonicity (DM)**): $y_i(\theta_i, \boldsymbol{\theta}_{-i})$ is nondecreasing in θ_i for all $\boldsymbol{\theta}_{-i} \in \boldsymbol{\Theta}_{-i}$;*

*(2) (**dominant incentive-compatibility first-order condition (DIC-FOC)**):*

$$U_i(\theta_i, \boldsymbol{\theta}_{-i}) = U_i(\underline{\theta}_i, \boldsymbol{\theta}_{-i}) + \int_{\underline{\theta}_i}^{\theta_i} y_i(\tau, \boldsymbol{\theta}_{-i}) d\tau, \forall \boldsymbol{\theta} \in \Theta,$$

or

$$\frac{\partial U_i(\boldsymbol{\theta})}{\partial \theta_i} = y_i(\boldsymbol{\theta}).$$

PROOF. Necessity: Dominant incentive compatibility implies that for all $\theta_i' > \theta_i$, we have

$$U_i(\theta_i, \boldsymbol{\theta}_{-i}) \geqq \theta_i y_i(\theta_i', \boldsymbol{\theta}_{-i}) + t_i(\theta_i', \boldsymbol{\theta}_{-i}) = U_i(\theta_i', \boldsymbol{\theta}_{-i}) + (\theta_i - \theta_i') y_i(\theta_i', \boldsymbol{\theta}_{-i})$$

and

$$U_i(\theta_i', \boldsymbol{\theta}_{-i}) \geqq \theta_i' y_i(\theta_i, \boldsymbol{\theta}_{-i}) + t_i(\theta_i, \boldsymbol{\theta}_{-i}) = U_i(\theta_i, \boldsymbol{\theta}_{-i}) + (\theta_i' - \theta_i) y_i(\theta_i, \boldsymbol{\theta}_{-i}).$$

Thus

$$y_i(\theta_i', \boldsymbol{\theta}_{-i}) \geqq \frac{U_i(\theta_i', \boldsymbol{\theta}_{-i}) - U_i(\theta_i, \boldsymbol{\theta}_{-i})}{\theta_i' - \theta_i} \geqq y_i(\theta_i, \boldsymbol{\theta}_{-i}). \tag{19.4.7}$$

Expression (19.4.7) implies that $y_i(\theta_i, \boldsymbol{\theta}_{-i})$ must be nondecreasing in θ_i by noting that $\theta_i' > \theta_i$. In addition, letting $\theta_i' \to \theta_i$ in (19.4.7) implies that for all θ_i, we have

$$\frac{\partial U_i(\boldsymbol{\theta})}{\partial \theta_i} = y_i(\boldsymbol{\theta}),$$

and thus

$$U_i(\theta_i, \boldsymbol{\theta}_{-i}) = U_i(\underline{\theta}_i, \boldsymbol{\theta}_{-i}) + \int_{\underline{\theta}_i}^{\theta_i} y_i(\tau, \boldsymbol{\theta}_{-i}) d\tau, \forall \boldsymbol{\theta} \in \Theta.$$

Sufficiency: Consider any two types θ_i' and θ_i. Without loss of generality, suppose $\theta_i > \theta_i'$. If **DM** and **DICFOC** hold, then for all $\boldsymbol{\theta}_{-i} \in \boldsymbol{\Theta}_{-i}$:

$$\begin{aligned} U_i(\theta_i, \boldsymbol{\theta}_{-i}) - U_i(\theta_i', \boldsymbol{\theta}_{-i}) &= \int_{\theta_i'}^{\theta_i} y_i(\tau, \boldsymbol{\theta}_{-i}) d\tau \\ &\geqq \int_{\theta_i'}^{\theta_i} y_i(\theta_i', \boldsymbol{\theta}_{-i}) d\tau \\ &= (\theta_i - \theta_i') y_i(\theta_i', \boldsymbol{\theta}_{-i}). \end{aligned}$$

Thus, we have

$$U_i(\theta_i, \boldsymbol{\theta}_{-i}) \geqq U_i(\theta_i', \boldsymbol{\theta}_{-i}) + (\theta_i - \theta_i') y_i(\theta_i', \boldsymbol{\theta}_{-i}) = \theta_i y_i(\theta_i', \boldsymbol{\theta}_{-i}) + t_i(\theta_i', \boldsymbol{\theta}_{-i}).$$

Similarly, we have

$$U_i(\theta_i', \boldsymbol{\theta}_{-i}) \geqq U_i(\theta_i, \boldsymbol{\theta}_{-i}) + (\theta_i' - \theta_i) y_i(\theta_i, \boldsymbol{\theta}_{-i}) = \theta_i' y_i(\theta_i, \boldsymbol{\theta}_{-i}) + t_i(\theta_i, \boldsymbol{\theta}_{-i}).$$

Therefore, the decision rule $(y(\cdot), t_1(\cdot), \cdots, t_n(\cdot))$ is dominant-strategy incentive compatible. $\square$

Similarly, we can provide the necessary and sufficient conditions for a social choice rule to be Bayesian incentive compatible. When agent i announces $\hat{\theta}_i$ and the others announce $\boldsymbol{\theta}_{-i}$ truthfully, we have the interim expected consumption $\bar{y}_i(\hat{\theta}_i) \equiv E_{\boldsymbol{\theta}_{-i}} y_i(\hat{\theta}_i, \boldsymbol{\theta}_{-i})$ and transfer $\bar{t}_i(\hat{\theta}_i) \equiv E_{\boldsymbol{\theta}_{-i}} t_i(\hat{\theta}_i, \boldsymbol{\theta}_{-i})$. Then, agent i' s interim expected utility is given by $\bar{U}_i(\theta_i, \hat{\theta}_i) = \theta_i \bar{y}_i(\hat{\theta}_i) + \bar{t}_i(\hat{\theta}_i)$.

Define

$$\bar{U}_i(\theta_i) \equiv \bar{U}_i(\theta_i, \hat{\theta}_i) \mid_{\hat{\theta}_i = \theta_i} = E_{\boldsymbol{\theta}_{-i}} U_i(\boldsymbol{\theta}) = \theta_i \bar{y}_i(\theta_i) + \bar{t}_i(\theta_i),$$

which is agent i's interim expected utility when he announces θ_i truthfully. Then, BIC means that truth-telling is interim-optimal. By the same way used in the proof of the above proposition for $\bar{U}(\theta_i)$, we have the following proposition:

Proposition 19.4.2 (BIC Characterization Theorem) *In the linear model, a social choice rule $(y(\cdot), t_1(\cdot), \cdots, t_n(\cdot))$ is Bayesian incentive-compatible if and only if for all $i \in N$,*

*(1) (**Bayesian monotonicity (BM)**): $E_{\boldsymbol{\theta}_{-i}} y_i(\theta_i, \boldsymbol{\theta}_{-i})$ is nondecreasing in θ_i for all $\boldsymbol{\theta}_{-i} \in \boldsymbol{\Theta}_{-i}$;*

*(2) (**Bayesian incentive-compatibility first-order condition (BIC-FOC)**):*

$$E_{\boldsymbol{\theta}_{-i}} U_i(\theta_i, \boldsymbol{\theta}_{-i}) = E_{\boldsymbol{\theta}_{-i}} U_i(\underline{\theta}_i, \boldsymbol{\theta}_{-i}) + \int_{\underline{\theta}_i}^{\theta_i} E_{\boldsymbol{\theta}_{-i}} y_i(\tau, \boldsymbol{\theta}_{-i}) d\tau, \forall \theta \in \Theta.$$

The above proposition implies that for any two BIC mechanisms that truthfully implement the same decision rule $y(\cdot)$ in Bayesian-Nash equilibrium, the interim expected utilities, $E_{\boldsymbol{\theta}_{-i}} U_i(\boldsymbol{\theta})$, and thus the transfers, $E_{\boldsymbol{\theta}_{-i}} t_i(\boldsymbol{\theta})$, coincide up to a constant. We then have the following important corollary.

Corollary 19.4.1 *In the linear model, for any BIC mechanism that implements the ex-post Pareto efficient decision rule $y(\cdot)$, there exists a VCG mechanism with the same interim expected transfers and utilities.*

Proof. Suppose that $(y(\cdot), t(\cdot))$ is a BIC mechanism, and $(y(\cdot), \tilde{t}(\cdot))$ is a VCG mechanism. By the BIC Characterization Theorem (Proposition 19.4.2), $E_{\boldsymbol{\theta}_{-i}} t_i(\boldsymbol{\theta}) = E_{\boldsymbol{\theta}_{-i}} \tilde{t}_i(\boldsymbol{\theta}) + c_i, \forall \theta_i \in \Theta_i$. Letting $\bar{t}_i(\boldsymbol{\theta}) = \tilde{t}_i(\boldsymbol{\theta}) + c_i$, then $(y(\cdot), \bar{t}(\cdot))$ is also a VCG mechanism, and $E_{\boldsymbol{\theta}_{-i}} \bar{t}_i(\boldsymbol{\theta}) = E_{\boldsymbol{\theta}_{-i}} t_i(\boldsymbol{\theta})$. □

For example, the expected externality mechanism is interim-equivalent to a VCG mechanism. Its only advantage is that it allows balancing of the budget ex-post for each state of the world. More generally, if a decision rule $y(\cdot)$ is truthfully implementable in dominant strategies, the only reason to have a BIC mechanism is that we care about ex-post transfers/utilities rather than just their interim or ex-ante expectations.

19.5 Myerson-Satterthwaite Impossibility Theorem

For mechanism design considered in the preceding chapter and this chapter, we have not yet been thinking of the mechanism as a contract. If a mechanism is a contract, it must be voluntary, i.e., it should also satisfy participation constraints. As a consequence, if we would put the principal-agent theory into the framework of general mechanism design, a mechanism should also be individually rational. As such, a social choice function that can be successfully implemented should satisfy not only the incentive compatibility in dominant strategies or Bayesian Nash dominant strategies, depending on the equilibrium concept used, but also the relevant participation constraints.

We know that the expected externality mechanism truthfully Bayesian implements Pareto efficient allocations under quasi-linear economic environments. Then, a pertinent question is, if an individual rationality constraint is also required, does such a mechanism that implements Pareto efficient allocations still exist? The answer turns out, unfortunately, to be negative.

If the mechanism is viewed as a contract, the following issues are raised: (1) Will the agents accept it voluntarily, i.e., are their participation constraints satisfied? (2) If one of the agents designs a contract, he will attempt to maximize his own payoff subject to the other agents' participation constraints. What will the optimal contract look like?

To answer these questions, it is necessary to impose additional restrictions on the social choice rule in the form of participation constraints. These constraints depend on when agents can withdraw from the mechanism, and what they get when they do so. Let $\bar{u}_i(\theta_i)$ be the reservation utility of agent i. According to the timing of information revelation, we have three types of participation constraints.

Definition 19.5.1 The social choice rule $f(\cdot)$ is

1. **ex-post individually rational** if for all i,

$$U_i(\boldsymbol{\theta}) \equiv u_i(f(\boldsymbol{\theta}), \boldsymbol{\theta}) \geqq \bar{u}_i(\theta_i), \forall \boldsymbol{\theta} \in \Theta;$$

2. **interim individually rational** if for all i,

$$E_{\boldsymbol{\theta}_{-i}}[U_i(\theta_i, \boldsymbol{\theta}_{-i})] \geqq \bar{u}_i(\theta_i), \forall \boldsymbol{\theta}_i \in \Theta_i;$$

3. **ex-ante individually rational** if for all i,

$$E_{\boldsymbol{\theta}}[U_i(\boldsymbol{\theta})] \geqq E_{\theta_i}[\bar{u}_i(\theta_i)].$$

Note that ex-post IR imply interim IR, which, in turn, imply ex-ante IR, but the reverse may not be true. Consequently, the ex-post participation constraint is the strongest.

Ex-post IR arises when the agent can withdraw in any state after the announcement of the outcome, so that there is no regret when the ex-post IR is satisfied. For instance, it is satisfied by any decentralized bargaining procedure. Indeed, it is the strongest constraint to satisfy.

The interim IR arises when the agent can withdraw after learning his type θ_i, but before learning anything about other agents' types. Once an agent decides to participate, he has to accept the outcome assigned to him. This constraint is easier to satisfy than ex-post IR.

With the ex-ante participation constraint, an agent can commit to participating before his type is realized. Indeed, it is the easiest constraint to satisfy. For example, in a quasi-linear environment, whenever the mechanism generates a positive expected total surplus, i.e.,

$$E_{\boldsymbol{\theta}}[\sum_i U_i(\boldsymbol{\theta})] \geqq E_{\boldsymbol{\theta}}[\sum_i \bar{u}_i(\theta_i)],$$

all agents' ex-ante IR can be satisfied by reallocating the expected total surplus among agents through lump-sum transfers, which will not disturb agents' incentive constraints or the balanced budget.

For this reason, we will focus mainly on interim IR. In the following, we will further illustrate the limitations on the set of implementable social choice functions that may be caused by participation constraints according to the important theorem given by Myerson-Satterthwaite (1983).

Even though Pareto efficient mechanisms do exist, such as the expected externality mechanism, it remains unclear whether such a mechanism results from private contracting among the parties, i.e., whether it satisfies interim participation constrains. Under the principal-agent framework, the principal offers an optimal contract to extract the agent's information rent.

This raises a question of whether there is some contracting or bargaining procedure that would yield Pareto efficiency. In the principal-agent situation, if the agent makes an offer to the principal who has no private information on his own, the agent would extract all of the surplus and truthfully implement efficient allocation. This is the case in which the type of one of the participants is known, and thus the budget balance of transfers can be satisfied. Consequently, the VCG mechanism truthfully implements Pareto efficient outcomes. Therefore, to make the question more interesting, we shall focus on a bilateral situation in which both parties have private information. In this situation, it turns out that, in general, there does not exist any efficient mechanism that satisfies both agents' participation and balanced constraints.

Consider the setting of allocating an indivisible object with two agents - a seller and a buyer: $N = \{S, B\}$. Each agent's type is $\theta_i \in \Theta_i = [\underline{\theta}_i, \overline{\theta}_i] \subseteq \mathcal{R}$, where $\theta_i \sim \varphi_i(\cdot)$ are independent, and $\varphi_i(\cdot) > 0$ for all $\theta_i \in \Theta_i$. Let $y \in \{0, 1\}$ indicate whether B receives the good. A social choice rule is then $f(\boldsymbol{\theta}) = (y(\boldsymbol{\theta}), t_1(\boldsymbol{\theta}), t_2(\boldsymbol{\theta}))$. The agents' utilities can thus be written as

$$\begin{aligned} u_B(y, \theta_B) &= \theta_B y + t_B, \\ u_S(y, \theta_S) &= -\theta_S y + t_S. \end{aligned}$$

It is easy to see that an efficient decision rule $y(\boldsymbol{\theta})$ must have

$$y(\theta_B, \theta_S) = \begin{cases} 1 & \text{if } \theta_B > \theta_S, \\ 0 & \text{if } \theta_B \leqq \theta_S. \end{cases}$$

We know that one can use the expected externality mechanism to truthfully implement the efficient decision rule in BNE with ex-post budget balance. Now suppose that interim IR is imposed:

$$\begin{aligned} E_{\theta_S}[\theta_B y(\theta_B, \theta_S) + t_B(\theta_B, \theta_S)] &\geqq 0, \\ E_{\theta_B}[-\theta_S y(\theta_B, \theta_S) + t_S(\theta_B, \theta_S)] &\geqq 0. \end{aligned}$$

Concerning budget balance, let us relax this requirement by requiring only ex-ante budget balance:

$$E_\theta[t_B(\theta_B, \theta_S) + t_S(\theta_B, \theta_S)] \leqq 0.$$

Unlike the ex-post budget balance considered previously, the ex-ante budget balance allows the borrowing of money as long as it breaks even, on average. It also allows a surplus of funds. The only constraint is that there is no ex-ante expected shortage of funds.

We then have the following impossibility theorem:

Theorem 19.5.1 (Myerson-Satterthwaite Theorem, 1983) *In the two-party trade setting above, suppose that each agent's type is $\theta_i \in \Theta_i = [\underline{\theta}_i, \overline{\theta}_i] \subseteq \mathcal{R}$, where $\theta_i \sim \varphi_i(\cdot)$ are independent and $(\underline{\theta}_B, \bar{\theta}_B) \cap (\underline{\theta}_S, \bar{\theta}_S) \neq \emptyset$ (gains from trade are possible, but types are not certain). Then, there is no Bayesian incentive-compatible mechanism that implements the efficient decision rule and satisfies ex-ante budget balance and interim IR.*

PROOF. Consider first the case in which $[\underline{\theta}_B, \bar{\theta}_B] = [\underline{\theta}_S, \bar{\theta}_S] = [\underline{\theta}, \bar{\theta}]$.

By Corollary 19.4.1, we know that for any BIC mechanism that implements the ex-post Pareto efficient decision rule $y(\cdot)$ in the linear environment, there exists a VCG mechanism with the same interim expected transfers and utilities. As such, we can restrict our attention to VCG mechanisms, while preserving ex-ante budget balance and interim IR. Such mechanisms take the form

$$\begin{aligned} t_B(\theta_B, \theta_S) &= -\theta_S y(\theta_B, \theta_S) + d_B(\theta_S), \\ t_S(\theta_B, \theta_S) &= \theta_B y(\theta_B, \theta_S) + d_S(\theta_B). \end{aligned}$$

By interim IR of B's type $\underline{\theta}$, $E_{\theta_S}[\underline{\theta}_B y(\underline{\theta}_B, \theta_S) + t_B(\underline{\theta}_B, \theta_S)] \geqq 0$, using the fact that $y(\underline{\theta}, \theta_S) = 0$ with probability 1, we must have $E_{\theta_S} d_B(\theta_S) \geqq 0$. Similarly, by interim IR of S's type $\bar{\theta}$, $E_{\theta_B}[-\bar{\theta}_S y(\theta_B, \bar{\theta}_S) + t_S(\theta_B, \bar{\theta}_S)] \geqq 0$, using the fact that $y(\theta_B, \bar{\theta}) = 0$ with probability 1, we must have $E_{\theta_B} d_S(\theta_B) \geqq 0$. Thus, adding the transfers, we have

$$\begin{aligned} E_\theta[t_B(\theta_B, \theta_S) + t_S(\theta_B, \theta_S)] &= E_\theta[(\theta_B - \theta_S)y(\theta_B, \theta_S)] + E_{\theta_S}[d_B(\theta_S)] + E_{\theta_B}[d_S(\theta_B)] \\ &\geqq E_\theta[\max\{\theta_B - \theta_S, 0\}] > 0 \end{aligned}$$

since $\Pr(\theta_B > \theta_S) > 0$. Therefore, ex-ante budget balance cannot be satisfied.

Now, in the general case, let $(\underline{\theta}, \bar{\theta}) = (\underline{\theta}_B, \bar{\theta}_B) \cap (\underline{\theta}_S, \bar{\theta}_S)$, and observe that any type of either agent above $\bar{\theta}$ [or below $\underline{\theta}$] has the same decision rule, and therefore must have the same transfer, as this agent's type $\bar{\theta}$ (resp. $\underline{\theta}$). Therefore, the transfers are the same as if both agents had valuations distributed on $[\underline{\theta}, \bar{\theta}]$; with possible atoms on $\underline{\theta}$ and $\bar{\theta}$. The argument thus still applies. □

The intuition for the proof is simple: In a VCG mechanism, in order to induce truthful revelation, each agent must become the residual claimant for the total surplus (since it is given by the sum of individuals' values plus $d_i(\hat{\boldsymbol{\theta}}_{-i})$). This means that, in the case in which trade is implemented, the buyer pays the seller's cost for the object, and the seller receives the buyer's valuation for the object. Any additional transfers to the agents must be nonnegative, in order to satisfy the interim IR of the lowest-valuation buyer and the highest-cost seller. Thus, each agent's utility must be at least equal to the total surplus. In BNE implementation, agents receive the same expected utilities as in the VGC mechanism, and thus again each agent's expected utility must equal at least the total expected surplus. This cannot be done without having an expected infusion of funds equal to the expected surplus, which breaks the budget balance.

When agents face ex-ante rather than interim IR, we can easily have a BIC Pareto efficient optimal contract. For instance, in a quasi-linear environment, whenever a mechanism, such as the AVG mechanism, generates a positive expected total surplus $E_{\boldsymbol{\theta}}[\sum_i U_i(\boldsymbol{\theta})] \geqq E_{\boldsymbol{\theta}}[\sum_i \hat{u}_i(\theta_i)]$, all agents' ex-ante IR can be satisfied by reallocating expected surplus among agents through lump-sum transfers, which will not disturb agents' incentive constraints or budget balance.

The above impossibility theorem shows that when an agent is at the interim stage, he would be interested not only in efficiency, but also in maximizing his own payoff. As such, it cannot form incentive compatibility. A better interpretation of the result is then as an upper bound on the efficiency of decentralized bargaining procedures. Indeed, any such procedure can be thought of as a mechanism, and must satisfy interim IR (even ex-post IR), and ex-ante budget balance (indeed, even ex-post budget balance). The theorem shows that decentralized bargaining in this case cannot be Pareto efficient. In the terminology of the Coase Theorem, private information creates a “transaction cost”.

19.6 Optimal Contract Design with Multiple Agents

In Chapter 16, we discussed optimal contract design with one agent. Now we discuss optimal contract design with multiple agents who have continuous types, and can obtain parallel results that reveal that one in general cannot obtain first best outcomes.

Consider a situation in which a principal (seller) contracts with n agents (buyers) who possess private information. The seller typically screens buyers of different types, i.e., induces them to select different consumption bundles. Suppose the production cost per unit is c. For each agent i, suppose that θ_i is in the interval $\Theta_i = [\underline{\theta}_i, \overline{\theta}_i]$, which is an independent random variable with density function f_i (distribution function F_i). By the revelation principle, we can restrict the analysis to direct revelation mechanisms $\{(q_i(\boldsymbol{\theta}), T(\boldsymbol{\theta})_i)\}$.

Assume that buyer i's utility is

$$U_i(\boldsymbol{\theta}) = \theta_i v_i(q_i(\boldsymbol{\theta})) + t_i(\boldsymbol{\theta}).$$

The principal's problem is to maximize the total expected revenue,

$$\begin{aligned}
\Pi &= \sum_{i=1}^{n} E_{\boldsymbol{\theta}}(-t_i(\boldsymbol{\theta}) - cq_i(\boldsymbol{\theta}))f_i(\theta_i) \\
&= \sum_{i=1}^{n} E_{\boldsymbol{\theta}}[\theta_i v_i(q_i(\boldsymbol{\theta})) - cq_i(\boldsymbol{\theta}) - U_i(\boldsymbol{\theta})] \\
&= \sum_{i=1}^{n} E_{\boldsymbol{\theta}}[\theta_i v_i(q_i(\boldsymbol{\theta})) - cq_i(\boldsymbol{\theta})] - \sum_{i} E_{\theta_i} E_{\boldsymbol{\theta}_{-i}}[U_i(\boldsymbol{\theta})], \qquad (19.6.8)
\end{aligned}$$

subject to the interim incentive compatibility and interim individual rationality constrains:

$$\begin{aligned}
E_{\theta_i} U_i(\boldsymbol{\theta}) &\geqq E_{\theta_i} U_i(\hat{\theta}_i, \boldsymbol{\theta}_{-i}),\ \forall \boldsymbol{\theta} \in \Theta, \forall \theta_i \in \Theta_i \text{ for all } i; \qquad (19.6.9)\\
E_{\theta_i} U_i(\boldsymbol{\theta}) &\geqq 0, \quad \forall \theta_i \in \Theta_i, \forall i. \qquad (19.6.10)
\end{aligned}$$

Here, the principal's total expected payoff (19.6.8) equals the difference between the total surplus and the expected information rent of agents.

By Proposition 19.4.2 (the Bayesian Incentive Compatibility Characterization Theorem), the interim expected utility of every buyer must satisfy

$$E_{\boldsymbol{\theta}_{-i}}[U_i(\theta_i, \boldsymbol{\theta}_{-i})] = E_{\boldsymbol{\theta}_{-i}}[U_i(\underline{\theta}_i, \boldsymbol{\theta}_{-i})] + \int_{\underline{\theta}_i}^{\theta_i} E_{\boldsymbol{\theta}_{-i}} v_i(q_i(\tau, \boldsymbol{\theta}_{-i}))d\tau, \forall \theta_i \in \Theta_i. \qquad (19.6.11)$$

By the same reason, the principal will optimally choose transfers to set the interim expected utilities of buyers with the lowest value as zero, i.e.,

$$E_{\boldsymbol{\theta}_{-i}}[U_i(\underline{\theta}_i, \boldsymbol{\theta}_{-i})] = 0, \forall i.$$

Integrating by parts for (19.6.11), buyer i's expected information rent can be rewritten as

$$E_{\theta_i} E_{\boldsymbol{\theta}_{-i}} \left[\frac{1 - F_i(\theta_i)}{f_i(\theta_i)} v_i(q_i(\boldsymbol{\theta})) \right].$$

Substituting into the seller's revenue formula, we have

$$E_{\boldsymbol{\theta}} \sum_{i=1}^{n} \left[\left(\theta_i - \frac{1 - F_i(\theta_i)}{f_i(\theta_i)} \right) v_i(q_i(\boldsymbol{\theta})) - c q_i(\boldsymbol{\theta}) \right].$$

The interim individual rationality constraint of the buyer requires

$$E_{\boldsymbol{\theta}_{-i}}[U_i(\underline{\theta}_i, \boldsymbol{\theta}_{-i})] \geqq 0, \forall i.$$

Let

$$\nu_i(\theta_i) = \theta_i - \frac{1 - F(\theta_i)}{f(\theta_i)},$$

which is agent i's **virtual valuation function**. Provided the hazard rate $h_i(\theta_i) = \frac{f(\theta_i)}{1-F(\theta_i)}$ is nondecreasing, the monotonicity condition is satisfied.

The principal chooses the contract that solves the following optimization problem:

$$\max_{\{q_i(\boldsymbol{\theta})\}} E_{\boldsymbol{\theta}} \sum_{i=1}^{n} \left[\nu_i(\theta_i) v_i(q_i(\boldsymbol{\theta})) - c q_i(\boldsymbol{\theta}) \right], \tag{19.6.12}$$

$$s.t. \quad E_{\boldsymbol{\theta}_{-i}} v_i(q_i(\theta_i, \boldsymbol{\theta}_{-i})) \text{ is non-decreasing in } \theta_i, \forall i \geqq 1 \ (BM). \tag{19.6.13}$$

The same as the optimal contract design with one agent, the Bayesian monotonicity constraint is satisfied provided virtual valuation function $\nu_i(\theta_i)$, for every agent i, is non-decreasing in θ_i, which is true when the hazard rate

$$h_i(\theta_i) = \frac{f_i(\theta_i)}{1 - F_i(\theta_i)}$$

is a non-decreasing function.

Therefore, the optimal contract is fully characterized by the following first-order condition:

$$\nu_i(\theta_i) v_i'(q_i(\boldsymbol{\theta})) = c, \ \forall i. \tag{19.6.14}$$

Again, we obtain the basic conclusion: For the highest type $\theta_i = \bar{\theta}_i$, there is no distortion in consumption, while for other types of $\theta_i < \bar{\theta}_i$, there is a downward distortion in consumption, resulting in second-best outcomes. Intuitively, since the seller (principal) cannot extract all of the buyers' information rents, the principal will adopt the virtual valuation rather than the true value of the participant. An agent's virtual valuation is lower than his true valuation because it accounts for the buyers' information rents which cannot be extracted by the principal.

Also, by the dominant Incentive Compatibility Characterization Theorem (Proposition 19.4.1) and Corollary 19.4.1, an optimal allocation rule that is Bayesian implementable can also be implementable in dominant strategies. Thus, we have DIC (dominant-strategy incentive compatible) transfers by integrating the DICFOC (first-order condition for dominant-strategy incentive compatibility), and the resulting transfers can take the form of VCG mechanism.

Similarly, if virtual valuation functions are not increasing, Bayesian monotonicity may not hold, and an ironing procedure should be used to identify the regions on which it binds. We can do so by the ironing procedure discussed in Chapter 16.

19.7 Cremer-McLean Full Surplus Extraction Theorem

The above result on optimal contracting with multiple agents shows that we generally cannot obtain first-best (consequently, not socially efficient) outcomes through the optimal contract design. However, when individuals' types are correlated, we can have first best outcomes.

A central theme of optimal contract design under incomplete information concerns the surplus extraction ability of agents. Both realistic observation and economic intuition inform us that the presence of private information makes the principal give up some positive information rent. However, when private types are correlated, Cremer and McLean (1988) show that the principal can extract all information rents through verifying the information reported, and thus first-best outcomes are attainable. This conclusion is called the Cremer-McLean Theorem and has wide applications. When applied in auction theory, a seller can design an efficient auction for economic environments with interdependent types.

Consider an economy with n agents. Each agent's type is discrete, $\theta_i \in \Theta_i \equiv \{\theta_i^1, \theta_i^2, \ldots, \theta_i^{k_i}\}, \forall i \in N = \{1, \ldots, n\}$, and the utility function is quasi-linear with private values, i.e., $U_i(\boldsymbol{y}, \boldsymbol{t}, \theta_i) = v_i(\boldsymbol{y}, \theta_i) + t_i$. Assume that agents' types are correlated and have density function $\pi(\boldsymbol{\theta})$. When agent i's type is θ_i, his beliefs about other agents types are given by

$$\pi_i(\boldsymbol{\theta}_{-i}|\theta_i) = \frac{\pi(\boldsymbol{\theta})}{\sum_{\boldsymbol{\theta}'_{-i} \in \boldsymbol{\theta}_{-i}} \pi(\boldsymbol{\theta}'_{-i}, \theta_i)}.$$

For each agent i, his beliefs form a matrix Π_i with element $\pi_i(\boldsymbol{\theta}_{-i}|\theta_i)$. Thus, matrix Π_i has k_i rows and $\prod_{j \neq i} k_j$ columns. Each row represents agent i's belief distribution about other agents' types when his type is θ_i. If types are independent, all rows are the same, and thus the rank of matrix Π_i is 1. If types are correlated, then different rows represent different beliefs. We assume that Π_i is of full row rank that is k_i, and hence an agent's belief distribution about others' beliefs is different.

The following well-known Cremer-McLean Full Surplus Extraction Theorem shows that, even under the requirement of DIC, the principal can extract all surplus. The idea of the proof is fairly simple. We first use any efficient and dominant-strategy incentive-compatible mechanism, such as the VCG mechanism, and then form a corresponding efficient and dominant-strategy incentive-compatible mechanism with binding individual rationality constraints such that information rent is zero. We thus obtain a dominant-strategy incentive-compatible mechanism that implements first-best outcomes.

Theorem 19.7.1 (Cremer-McLean Full Surplus Extraction Theorem) *Suppose that agents' types under the private value model are correlated, and the information matrix Π_i has full row rank. Then, for any efficient and dominant-strategy incentive-compatible mechanism $(y(\cdot), t_1(\cdot), \dots, t_n(\cdot))$, there exists another dominant-strategy incentive-compatible and efficient mechanism $(y(\cdot), \tilde{t}_1(\cdot), \dots, \tilde{t}_n(\cdot))$ with the same decision rule $y(\cdot)$, in which interim individual rationality constraints are binding, and thus the first-best is implementable.*

PROOF. For any dominant-strategy incentive-compatible mechanism $(y(\cdot), t_1(\cdot), t_2(\cdot))$, since agents have quasi-linear utility functions, agent i's interim expected utility with θ_i at equilibrium can be written as

$$\bar{U}_i(\theta_i) = \sum_{\boldsymbol{\theta}_{-i}} \pi_i(\boldsymbol{\theta}_{-i}|\theta_i)[u_i(\boldsymbol{y}(\boldsymbol{\theta}), \theta_i)) + t_i(\boldsymbol{\theta})].$$

Let $\bar{u}_i^* = [\bar{U}(\theta_i^1), \bar{U}(\theta_i^2), \dots, \bar{U}(\theta_i^{k_i})]'$. Since Π_i is of full row rank, there is a vector $c_i = (c_i(\boldsymbol{\theta}_{-i}))_{\boldsymbol{\theta}_{-i} \in \boldsymbol{\theta}_{-i}}$ with $\prod_{j \neq i} k_j$ columns, such that

$$\Pi_i c_i' = -\bar{u}_i^*;$$

i.e., for $\forall \theta_i$, we have

$$\bar{U}_i(\theta_i) + \sum_{\boldsymbol{\theta}_{-i}} \pi_i(\boldsymbol{\theta}_{-i}|\theta_i) c_i(\boldsymbol{\theta}_{-i}) = 0.$$

To ensure that i's interim individual rationality constraint is binding, let

$$\tilde{t}_i(\boldsymbol{\theta}) = t_i(\boldsymbol{\theta}) + c_i(\boldsymbol{\theta}_{-i}).$$

Construct a new mechanism, called the Cremer-McLean mechanism:

$$(\boldsymbol{y}(\boldsymbol{\theta}), \tilde{t}_i(\cdot)) = (\boldsymbol{y}(\boldsymbol{\theta}), t_i(\boldsymbol{\theta}) + c_i(\boldsymbol{\theta}_{-i}), \forall i \in N).$$

Since $c_i(\boldsymbol{\theta}_{-i})$ is independent of θ_i, the Cremer-McLean mechanism is still dominant-strategy incentive-compatible and efficient. However, under the Cremer-McLean mechanism, agent i's interim expected utility at equilibrium becomes

$$\begin{aligned}\bar{V}_i(\theta_i) &= \sum_{\boldsymbol{\theta}_{-i}} \pi_i(\boldsymbol{\theta}_{-i}|\theta_i)[u_i(\boldsymbol{y}(\boldsymbol{\theta}), \theta_i)) + \tilde{t}_i(\boldsymbol{\theta}))] \\ &= \sum_{\boldsymbol{\theta}_{-i}} \pi_i(\boldsymbol{\theta}_{-i}|\theta_i)[u_i(\boldsymbol{y}(\boldsymbol{\theta}), \theta_i)) + t_i(\boldsymbol{\theta}) + c_i(\boldsymbol{\theta}_{-i})] \\ &= \bar{U}_i(\theta_i) + \sum_{\boldsymbol{\theta}_{-i}} \pi_i(\boldsymbol{\theta}_{-i}|\theta_i) c_i(\boldsymbol{\theta}_{-i}) \\ &= 0.\end{aligned}$$

Thus, all agents' interim individual-rationality constraints are binding, and the designer extracts all of the surplus; in particular, the designer can extract agents' surplus through the VCG mechanism to achieve the first-best outcome. □

In fact, the Cremer-McLean Theorem still holds for interdependent environments. Although the VCG mechanism with interdependent values is not dominant-strategy incentive-compatible, the generalized VCG mechanism is. Such a mechanism will be discussed in auction theory with interdependent values in Chapter 21.

The Cremer-McLean Theorem relies on the assumptions of quasi-linear utility functions and unlimited liability, as well as on an implicit assumption that there is no collusion. A major criticism of this result comes from its vulnerability to collusion among agents. In the FSE mechanism, transfers to and from agents depend on the reports of other agents. Therefore, it is highly susceptible to collusion among agents, especially in nearly independent environments in which these transfers are very large.

The pioneering work that studies collusion in principal-multi agent settings is Laffont and Martimort (1997, 2000). In procurement/public good settings with two agents, they showed that if the types are correlated, preventing collusion entails a strict cost to the principal, and thus only the second-best outcome is implementable. Che and Kim (2006) showed that the results obtained by Laffont and Mortimort (1997, 2000) depend on the two-agent assumption. For $n > 2$, they showed that agents' collusion, including both reporting manipulation and arbitrage, is harmless to the principal, so that the first-best is still implementable in a broad class of circumstances.

For a two-agent nonlinear pricing private goods environment, when types are correlated and arbitrage is allowed, Meng and Tian (2017) demonstrated that a major difference arises if types are positively or negatively correlated. Under negative correlation, the principal can exploit the conflict of interest inside of the coalition to prevent collusion at no cost, i.e., the Cremer-McLean Theorem is still true. However, under positive correlation, the threat of collusion forces the principal to distort the allocation away from the first-best level obtained without collusion.

19.8 Characterization of Bayesian Implementation

This section discusses the necessary and sufficient conditions for full Bayesian implementation of a set of social choice functions in general economic environments.

Assume that economic environments under consideration allow for interdependent types, and agent i's parametric utility function is denoted by $u_i(x, \boldsymbol{\theta})$, where $x \in Z$ and $\boldsymbol{\theta} \in \boldsymbol{\Theta}$. The designer and agents know that $\boldsymbol{\theta} = (\theta_1, \cdots, \theta_n)$ has density function $\varphi(\boldsymbol{\theta})$ on $\Theta = \prod_{i \in N} \Theta_i$.

Let $X = \{x : \Theta \to A\}$ be the set of all feasible outcomes and $\hat{F} = \{f_1, f_2, \cdots\} \subseteq X$ the set of social choice functions. When a social choice set $\hat{F}$ contains only one element, it is called a social choice function, denoted by f. We will see below that a set of social choice functions $\hat{F}$, in general, differs from social choice correspondence, unless every status $\boldsymbol{\theta} \in \Theta$ is an event of common knowledge (every information set is singleton for all agents, reducing to an economic environment with complete information) and the set satisfies closure to be defined.

Given a mechanism $\langle M, h \rangle$, like Nash implementation, Bayesian-implementation also involves the relationship between $\hat{F}$ and $\mathcal{B}(\Gamma)$.

Definition 19.8.1 A mechanism $\Gamma = \langle M, h \rangle$ is said to **fully Bayesian implement** $\hat{F}$, if

(i) for every social choice function $f \in \hat{F}$, there is a Bayesian-Nash equilibrium $\mathbf{m}^*$, such that $h(\mathbf{m}^*) = f(\cdot)$;

(ii) If $\mathbf{m}^*$ is a Bayesian-Nash equilibrium, then $h(\mathbf{m}^*) \in \hat{F}$.

If such a mechanism exists, we say that $\hat{F}$ is **fully Bayesian implementable**.

Definition 19.8.2 A mechanism $\Gamma = \langle M, h \rangle$ is said to **Bayesian implement** $\hat{F}$ if whenever $\mathbf{m}^*$ is a Bayesian-Nash equilibrium, then $h(\mathbf{m}^*) \in \hat{F}$, i.e., condition (ii) above holds.

If such a mechanism exists, we say that $\hat{F}$ is **Bayesian implementable**. When $\hat{F}$ contains only one social choice function, Bayesian implementation and full Bayesian implementation are the same.

Definition 19.8.3 A mechanism $\Gamma = \langle M, h \rangle$ **partially Bayesian implements social choice set** $\hat{F}$ if for every social choice function $f \in \hat{F}$, there exists a Bayesian-Nash equilibrium $\mathbf{m}^*$, such that $h(\mathbf{m}^*) = f(\cdot)$, i.e., condition (i) above holds.

If such a mechanism exists, we say that $\hat{F}$ is **partially Bayesian implementable**. The reason that we call this partial Bayesian implementation is that there may exist another Bayesian-Nash equilibrium at which the resulting outcome is not in the social choice set.

Like implementation of a social goal under other solution concepts, full Bayesian implementation needs to deal with two issues: Condition (i) requires to seek a mechanism such that any social choice function in the social choice set $\hat{F}$ is a Bayesian-Nash equilibrium outcome of the mechanism; and Condition (ii) requires that every Bayesian-Nash equilibrium outcome of the mechanism is a social choice function in $\hat{F}$. While the incentive compatibility requirement is central, it may not be sufficient for a mechanism to give all of the desired outcomes due to the existence of multiple equilibria. The severity of this multiple-equilibrium problem is a critical issue to be solved in (full) Bayesian implementation. Then, we need to find conditions, such that there exists a mechanism in which all Bayesian-Nash equilibrium outcomes are social choice functions in the social choice set $\hat{F}$.

We first consider an example given by Palfrey and Srivastava (1989), which shows how the issue of multiple Bayesian-Nash equilibria affects full Bayesian implementation even for a social choice function.

Example 19.8.1 (Palfrey and Srivastava, 1989) Consider an exchange economy with two agents and two goods. Agent 1 has two possible preferences (types) θ_1 and θ'_1, and agent 2 only has one preference θ_2. Agent 1's preference is private information. Under preference profile $\boldsymbol{\theta} = (\theta_1, \theta_2)$, Pareto efficient allocation is $x(\boldsymbol{\theta})$; and under $\boldsymbol{\theta}' = (\theta'_1, \theta_2)$, Pareto efficient allocation is $x(\boldsymbol{\theta}')$, see Figure 19.1. For these two allocations $x(\boldsymbol{\theta})$ and $x(\boldsymbol{\theta}')$, agent 1 with type θ_1 likes $x(\boldsymbol{\theta})$ more, while agent 1 with θ'_1 feels indifferent about the two allocations, but agent 2 likes $x(\boldsymbol{\theta}')$ more. Consider the social choice rule satisfying $f(\boldsymbol{\theta}) = x(\boldsymbol{\theta})$ and $f(\boldsymbol{\theta}') = x(\boldsymbol{\theta}')$, which is Pareto efficient.

In the above exchange economy, consider the following mechanism $\Gamma = (\Theta, \boldsymbol{h}(\cdot), \boldsymbol{m}(\cdot))$: Each agent (mainly agent 1) reports his types $M_1 = \Theta_1 = \{\theta_1, \theta'_1\}, M_2 = \{\theta_2\}$. If the message reported is $\boldsymbol{m} = (\theta_1, \theta_2)$, then $\boldsymbol{h}(\theta_1, \theta_2) = \boldsymbol{x}(\boldsymbol{\theta})$; if the message reported is $\boldsymbol{m} = (\theta'_1, \theta_2)$, then $\boldsymbol{h}(\theta'_1, \theta_2) = \boldsymbol{x}(\boldsymbol{\theta}')$. This mechanism has two Bayesian-Nash equilibria: One is truth-telling $m_1(\tilde{\theta}_1) = \tilde{\theta}_1, \forall \tilde{\theta}_1 \in \Theta_1; m_2(\theta_2) = \theta_2$; and the other is not truth-telling $m_1(\tilde{\theta}_1) = \theta_1, \forall \tilde{\theta}_1 \in \Theta_1; m_2(\theta_2) = \theta_2$. However, these two Bayesian-Nash equilibria are not equally desired from the Pareto optimality criterion. When the true type profile is $\boldsymbol{\theta}' = (\theta'_1, \theta_2)$, the second Bayesian-Nash equilibrium outcome is not Pareto efficient. This example shows that we have a multiple-equilibria problem in Bayesian implementation.

To eliminate undesired Bayesian-Nash equilibria, we consider the following indirect mechanism, see Table 19.1.

In this mechanism, agent 2 has two different reports, θ_2 and ρ. If agent 2 reports ρ, then $\boldsymbol{h}(\theta_1, \rho) = \boldsymbol{x}(\boldsymbol{\theta}')$ and $\boldsymbol{h}(\theta'_1, \rho) = \boldsymbol{x}(\boldsymbol{\theta})$. Under this indirect

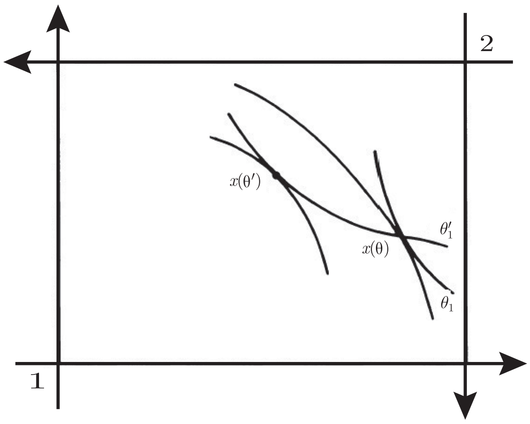

Figure 19.1: Bayesian implementation with two-agent and two-good economies

		Agent 2	
		θ_2	ρ
Agent 1	θ_1	$x(\boldsymbol{\theta})$	$x(\boldsymbol{\theta}')$
	θ_1'	$x(\boldsymbol{\theta}')$	$x(\boldsymbol{\theta})$

Table 19.1: Fully Bayesian implementable social choice rule

mechanism, $m_1(\tilde{\theta}_1) = \theta_1, \forall \tilde{\theta}_1 \in \Theta_1$ is no longer agent 1's Bayesian-Nash equilibrium strategy; otherwise, agent 2 should play ρ, resulting in outcome $\boldsymbol{x}(\boldsymbol{\theta}')$ when agent 1 reports θ_1. This is because $\boldsymbol{x}(\boldsymbol{\theta}')$ is strictly preferred to $\boldsymbol{x}(\boldsymbol{\theta})$ for agent 2. When agent 2 chooses ρ, agent 1 should play θ_1' when he is of type θ_1. Thus the bad equilibrium of the direct mechanism has been eliminated.

One can easily see that the mechanism has two Bayesian-Nash equilibria:

Equilibrium 1: $m(\theta_1) = \theta_1, m(\theta_1') = \theta_1', m_2(\theta_2) = \theta_2$, and its equilibrium outcome is $\boldsymbol{h}(\theta_1, \theta_2) = \boldsymbol{x}(\boldsymbol{\theta})$ and $\boldsymbol{h}(\theta_1', \theta_2) = \boldsymbol{x}(\boldsymbol{\theta}')$;

Equilibrium 2: $m(\theta_1) = \theta_1', m(\theta_1') = \theta_1, m_2(\theta_2) = \rho$, resulting in equilibrium outcome $\boldsymbol{h}(\theta_1, \rho) = \boldsymbol{x}(\boldsymbol{\theta})$ and $\boldsymbol{h}(\theta_1', \rho) = \boldsymbol{x}(\boldsymbol{\theta}')$.

Thus, although this mechanism has two Bayesian-Nash equilibria, the outcome at these two equilibria is identical to the outcome under the social choice rule, i.e., this mechanism fully Bayesian implements f.

In the following, we show that under some technical conditions, a social choice set $\hat{F}$ is fully Bayesian implementable if and only if $\hat{F}$ is Bayesian incentive compatible (it deals with the first problem that any outcome determined by a social choice rule is a Bayesian-Nash equilibrium outcome of the mechanism), and it is Bayesian monotonic (it solves the second problem: all Bayesian-Nash equilibrium outcomes of the mechanism are outcomes under the social choice rule).

We first discuss these conditions. It will be seen that Bayesian monotonicity is similar to Maskin monotonicity when the expected utility function replaces the Bernoulli utility function.

The following important concept of Bayesian incentive compatibility solves the first problem of full Bayesian implementation.

Definition 19.8.4 A social choice set $\hat{F}$ is said to be **Bayesian incentive-compatible** or **truthfully Bayesian implementable** if for every social choice function $f \in \hat{F}$, truth-telling, i.e., $\mathbf{m}^*(\boldsymbol{\theta}) = \boldsymbol{\theta}\ \forall \boldsymbol{\theta} \in \Theta$, is a Bayesian-Nash equilibrium of the revelation mechanism $\Gamma = (\Theta, f)$, i.e., for any $f \in \hat{F}$, we have

$$E_{\theta_{-i}}[u_i(f(\theta_i, \boldsymbol{\theta}_{-i})), \boldsymbol{\theta}) \mid \theta_i] \geqq E_{\theta_{-i}}[u_i(f(\theta'{}_i, \boldsymbol{\theta}_{-i})), \boldsymbol{\theta}) \mid \theta_i], \quad \forall i, \forall \theta_i, \theta'{}_i \in \Theta_i.$$

We then have the first necessary condition for full Bayesian implementation of the social choice set $\hat{F}$.

Proposition 19.8.1 *If a social choice set $\hat{F}$ is partially Bayesian implementable, then it satisfies Bayesian incentive compatibility.*

PROOF. Suppose that mechanism $\Gamma = \langle M, h\rangle$ partially Bayesian implements the social choice set $\hat{F}$. If some social choice function $f \in \hat{F}$ is not Bayesian incentive-compatible, then there are i and $\theta_i, \theta_i' \in \Theta_i$, such that

$$E_{\theta_{-i}}[u_i(f(\theta_i, \boldsymbol{\theta}_{-i})), \boldsymbol{\theta}) \mid \theta_i] < E_{\theta_{-i}}[u_i(f(\theta'{}_i, \boldsymbol{\theta}_{-i})), \boldsymbol{\theta}) \mid \theta_i]. \qquad (19.8.15)$$

Let $\mathbf{m} \in \mathcal{B}(\Gamma)$, such that $h \circ \mathbf{m} = f$. When agent i's type is θ_i, if he chooses $m_i(\theta_i)$, then his interim expected utility is given by

$$E_{\theta_{-i}}[u_i(h(m_i(\theta_i), \mathbf{m}_{-i}(\boldsymbol{\theta}_{-i}))), \boldsymbol{\theta}) \mid \theta_i] = E_{\theta_{-i}}[u_i(f(\theta_i, \boldsymbol{\theta}_{-i})), \boldsymbol{\theta}) \mid \theta_i].$$

If agent i chooses a message $m_i' = m_i(\theta_i')$, his interim expected utility is

$$E_{\theta_{-i}}[u_i(h(m_i(\theta_i'), \mathbf{m}_{-i}(\boldsymbol{\theta}_{-i}))), \boldsymbol{\theta}) \mid \theta_i] = E_{\theta_{-i}}[u_i(f(\theta_i', \boldsymbol{\theta}_{-i})), \boldsymbol{\theta}) \mid \theta_i].$$

By (19.8.15), we know that the agent has an incentive to deviate from $m_i(\theta_i)$, which contradicts the fact that $\mathbf{m} \in \mathcal{B}(\Gamma)$. □

We now examine the second necessary condition for a social choice set to be fully Bayesian implementable, i.e., the Bayesian monotonicity condition. Consider a revelation mechanism, an agent i, and a strategy $\alpha_i : \Theta_i \to \Theta_i$. If agent i tells the truth, yielding that $\alpha_i(\theta_i) = \theta_i, \forall \theta_i \in \Theta_i$; otherwise, α_i is called the **deception strategy** of agent i. We call $\boldsymbol{\alpha}(\boldsymbol{\theta}) =$

$(\alpha_1(\theta_1), \cdots, \alpha_n(\theta_n))$ a deception, if at least one agent has a deception strategy.

Let

$$\boldsymbol{\alpha}_{-i}(\boldsymbol{\theta}_{-i}) = (\alpha_1(\theta_1), \cdots, \alpha_{i-1}(\theta_{i-1}), \alpha_{i+1}(\theta_{i+1}), \cdots, \alpha_n(\theta_n)).$$

For a social choice function f and deception $\boldsymbol{\alpha}$, $f \circ \boldsymbol{\alpha}$ denotes the resulting social choice function by deception. When social status is $\boldsymbol{\theta}$, social choice function is $f \circ \boldsymbol{\alpha}(\boldsymbol{\theta}) = f(\boldsymbol{\alpha}(\boldsymbol{\theta}))$. For every $\boldsymbol{\theta}' \in \Theta$, define $f_{\alpha_i(\theta_i)}(\boldsymbol{\theta}') \equiv f(\alpha_i(\theta_i), \boldsymbol{\theta}'_{-i})$, i.e., it is the resulting social choice function when only agent i chooses the deception strategy.

Similar to Maskin monotonicity, we have the following Bayesian monotonicity condition.

Definition 19.8.5 (Bayesian monotonicity) A social choice set $\hat{F}$ is said to satisfy **Bayesian monotonicity** if for any social choice function $f \in \hat{F}$ and deception $\boldsymbol{\alpha}$ that results in $f \circ \boldsymbol{\alpha} \notin \hat{F}$, there exists an agent i and a function $y : \Theta_{-i} \to A$, such that

$$E[u_i(f(\theta_i, \boldsymbol{\theta}_{-i}), \boldsymbol{\theta})|\theta_i] \geqq E[u_i(y(\boldsymbol{\theta}_{-i}), \boldsymbol{\theta})|\theta_i] \tag{19.8.16}$$

for all $\theta_i \in \Theta_i$, and for some θ'_i, we have

$$E[u_i(f(\theta'_i, \boldsymbol{\theta}_{-i}), \boldsymbol{\theta}')|\theta'_i] < E[u_i(y(\boldsymbol{\theta}_{-i}), \boldsymbol{\theta}')|\theta'_i]. \tag{19.8.17}$$

Bayesian monotonicity is a variation of Maskin monotonicity under incomplete information. It is used to prevent undesired outcomes from Bayesian-Nash equilibria. Consider a mechanism $\Gamma = \langle M, h \rangle$ that fully Bayesian implements a social choice set $\hat{F}$, and a social choice function $f \in \hat{F}$ that can be Bayesian implemented by Bayesian-Nash equilibrium $\mathbf{m}$, i.e., for any $\boldsymbol{\theta} \in \Theta$, $h(\mathbf{m}(\boldsymbol{\theta})) = f(\boldsymbol{\theta})$. Suppose that agent i adopts a deception $\boldsymbol{\alpha}$. Then, strategy $\mathbf{m} \circ \boldsymbol{\alpha}$ results in an outcome $f \circ \boldsymbol{\alpha}$. If $f \circ \boldsymbol{\alpha} \notin \hat{F}$, then $\mathbf{m} \circ \boldsymbol{\alpha}$ is not a Bayesian-Nash equilibrium. The inequality in the Bayesian monotonicity condition (19.8.17) avoids the possibility that $\mathbf{m} \circ \boldsymbol{\alpha}$ becomes a Bayesian-Nash equilibrium, while another inequality (19.8.16) ensures that no one has an incentive to misreport his type.

We then have another necessary condition for full Bayesian implementation, i.e., Bayesian monotonicity.

Proposition 19.8.2 *If a social choice set $\hat{F}$ is fully Bayesian implementable, then $\hat{F}$ must satisfy Bayesian monotonicity.*

PROOF. Suppose that $\Gamma = \langle M, h \rangle$ fully Bayesian implements a social choice set $\hat{F}$. Then, for any social choice function $f \in \hat{F}$, there is a Bayesian-Nash equilibrium $\mathbf{m}^*$, such that $f(\boldsymbol{\theta}) = h(\mathbf{m}^*(\boldsymbol{\theta})), \forall \boldsymbol{\theta} \in \Theta$. Let $\boldsymbol{\alpha}$ be a deception, such that $f \circ \boldsymbol{\alpha} \notin \hat{F}$. Consider strategy $\mathbf{m}^* \circ \boldsymbol{\alpha}$. Under this strategy, for any

$\boldsymbol{\theta} \in \Theta$, agents' strategy profile is $\mathbf{m}^* \circ \boldsymbol{\alpha}(\boldsymbol{\theta}) = \mathbf{m}^*(\boldsymbol{\alpha}(\boldsymbol{\theta}))$. Since $f \circ \boldsymbol{\alpha} \notin \hat{F}$, full Bayesian implementation implies that $\mathbf{m}^* \circ \boldsymbol{\alpha}$ is not a Bayesian-Nash equilibrium, yielding that there is $\theta_i' \in \Theta_i$, such that agent i has an incentive to choose some $m_i' \neq m_i^*(\alpha_i(\theta_i'))$, such that

$$E[u_i(h(\mathbf{m}^* \circ \boldsymbol{\alpha}(\theta_i', \boldsymbol{\theta}_{-i})), (\theta_i', \boldsymbol{\theta}_{-i}))|\theta_i'] < E[u_i(h(m_i', \mathbf{m}^*_{-i}(\boldsymbol{\alpha}_{-i}(\boldsymbol{\theta}_{-i})), (\theta_i', \boldsymbol{\theta}_{-i}))|\theta_i]. \tag{19.8.18}$$

Define function $y : \Theta_{-i} \to A$: $y(\boldsymbol{\theta}_{-i}) = h(m_i', \mathbf{m}^*_{-i}(\boldsymbol{\theta}_{-i}))$. We have

$$y(\boldsymbol{\alpha}_{-i}(\boldsymbol{\theta}_{-i})) = h(\mathbf{m}^*_{-i}(m_i', \boldsymbol{\alpha}_{-i}(\boldsymbol{\theta}_{-i})).$$

Thus, from the above inequality (19.8.18), we can obtain (19.8.17), while (19.8.16) comes from the fact that f can be Bayesian implemented by $h \circ \mathbf{m}^*$. Therefore, Bayesian monotonicity is satisfied. In this case, for any θ_i, all agents have an incentive to tell the truth. □

In addition to Bayesian incentive compatibility and Bayesian monotonicity, to ensure that they are sufficient conditions, some technical conditions are required. For fully implementable social choice set $\hat{F}$, they first need to satisfy the closure condition. We call a subset of a type space $\Theta' \subseteq \Theta$ a **common knowledge event** if for any $\boldsymbol{\theta}' = ({\theta'}_i, {\boldsymbol{\theta}'}_{-i}) \in \Theta', \boldsymbol{\theta} = (\theta_i, \boldsymbol{\theta}_{-i}) \notin \Theta'$, we have $\varphi({\boldsymbol{\theta}'}_{-i}|\theta_i) = 0, \ \forall i$.

If an agent does not know the true status for all possible states, then the agent needs to predict what messages the other agents will report. All such possible states consist of common knowledge, which are the bases for reporting messages.

Definition 19.8.6 (Closure of Social Choice Set) Let Θ_1 and Θ_2 be a partition of Θ, i.e., $\Theta_1 \bigcap \Theta_2 = \emptyset$ and $\Theta_1 \bigcup \Theta_2 = \Theta$. A social choice set $\hat{F}$ is said to satisfy **closure** if for any $f_1, f_2 \in \hat{F}$, there is $f \in \hat{F}$, such that $f(\boldsymbol{\theta}) = f_1(\boldsymbol{\theta}), \forall \boldsymbol{\theta} \in \Theta_1; f(\boldsymbol{\theta}) = f_2(\boldsymbol{\theta}), \forall \boldsymbol{\theta} \in \Theta_2$.

If every state $\boldsymbol{\theta} \in \Theta$ is common knowledge, it reduces to an economic environment with complete information discussed in the previous chapter, and thus a social choice set that satisfies closure becomes a social choice correspondence. If a social choice set $\hat{F}$ does not satisfy closure, say, $\Theta = \{\boldsymbol{\theta}, \boldsymbol{\theta}'\}$, and every state is common knowledge, $\hat{F} = \{f_1, f_2\}$ satisfies $f_1(\boldsymbol{\theta}) = f_2(\boldsymbol{\theta}') = a; f_1(\boldsymbol{\theta}') = f_2(\boldsymbol{\theta}) = b, a \neq b$, then the social choice set $\hat{F}$ is not fully Bayesian implementable. This is because, if $\hat{F}$ is fully Bayesian implementable, we need them to be Bayesian-Nash equilibrium outcomes under two states a and b, and then by the definitions of f_1 and f_2, there is no way that ensures that the implementable outcomes are different under these two different states. It is worth noting that, in this case, the social choice set is not the same as the social choice correspondence, i.e., $\hat{F} \neq F$, where $F(\boldsymbol{\theta}) = F(\boldsymbol{\theta}') = \{a, b\}$.

Jackson (1991) characterized the sufficient condition for full Bayesian implementation of a social choice set in incomplete information environments. We first give the definition of an economic environment under incomplete information.

Definition 19.8.7 An environment is called an **economic environment under incomplete information** if for any state $\boldsymbol{\theta} \in \Theta$ and any outcome $y \in Y$, there are two agents i and j, and two outcomes y_i and y_j, such that

$$u_i(y_i, \boldsymbol{\theta}) > u_i(y, \boldsymbol{\theta})$$

and

$$u_j(y_j, \boldsymbol{\theta}) > u_j(y, \boldsymbol{\theta}).$$

Intuitively, for incomplete information economic environments, for any social choice function and state, there are at least two agents who hope to change the social choice outcome under the state, which implies that a social choice outcome cannot make all agents reach their satiated points. If every utility function is monotonic in private goods economies, this condition is satisfied. Such a condition is also satisfied for public good economies and economies with externalities.

We now state the following theorem given by Jackson (1991) without proof.

Proposition 19.8.3 (Necessity and Sufficiency for Full Bayesian Implementation) *For economic environments under incomplete information with $N \geqq 3$ agents, suppose that a social choice set $\hat{F}$ satisfies the closure condition. Then, $\hat{F}$ is fully Bayesian implementable if and only if it satisfies Bayesian incentive compatibility and Bayesian monotonicity.*

Under more general environments of incomplete information, Jackson (1991) strengthened Bayesian monotonicity by introducing the monotonicity-no-veto-power condition and showed that in economic environments with $N \geqq 3$ agents, if a social choice function satisfies Bayesian incentive compatibility and the monotonicity-no-veto-power condition, it is fully Bayesian implementable. For environments with two agents, Dutta and Sen (1994) provided sufficient conditions for a social choice function to be fully Bayesian implementable.

One can similarly investigate implementation of a social choice set in refinements of Bayesian-Nash equilibria and virtual Bayesian implementation. Palfrey-Srivastava (1989) showed that for economic environments with $N \geqq 3$ agents, a necessary and sufficient condition for a social choice set $\hat{F}$ to be fully Bayesian implementable in undominated Bayesian equilibria is that it satisfies Bayesian incentive compatibility. Abreu-Matsushima (1990), Matsushima (1993), Duggan (1993), and Tian (1997) showed under various technical conditions that a social choice set is virtual Bayesian implementable if and only if the social goal is Bayesian incentive compatible.

19.9 Characterization of Ex-post Implementation

We have so far only discussed implementation using Bayesian equilibrium as the solution concept, which assumes that agents have common knowledge about the distribution of types. However, this assumption constitutes a serious limitation on applications since it is highly difficult to achieve under incomplete information, including for the mechanism designer, if not completely unachievable. Wilson (1987) summarized the validity and limitations of game theory on the analysis of practical problems:

> 'Game Theory has a great advantage in explicitly analyzing the consequences of trading rules that presumably are really common knowledge; it is deficient to the extent it assumes other features to be common knowledge, such as one player's probability assessment about another's preferences or information. I foresee the progress of game theory as depending on successive reductions in the base of common knowledge required to conduct useful analysis of practical problems. Only by repeated weakening of common knowledge assumptions will the theory approximate reality.

In the development of mechanism design theory, especially in the practice of auction design, many studies have focused on a more robust equilibrium concept, i.e., **ex-post equilibrium**. Generally, the **ex-post equilibrium** means that the participant's interim equilibrium strategy will not have the incentive to change or repent afterwards (i.e., after knowing the true types of other people). The dominant equilibrium under private value that we discussed previously is a typical ex-post equilibrium. This is because, under the dominant strategy, participants do not have to consider the probability distribution of other participants' types, i.e., the strategy is independent of belief. If a participant's utility function depends on the types of other participants, then the concept of ex-post equilibrium needs to be introduced. In this section, we mainly discuss the issue of the social choice set (or function) that can be implemented in ex-post equilibrium.

Similar to the Nash implementation and Bayesian implementation discussed previously, ex-post implementation also has its own relevant incentive compatibility conditions and monotonicity conditions. This section discusses the necessary and sufficient conditions for ex-post implementation. The discussion is mainly based on Bergemann and Morris (2005, 2008).

Assume that the set of participants is $N = \{1, \cdots, n\}$, a type of participant i is $\theta_i \in \Theta_i$, and a profile of types for all participants is $\boldsymbol{\theta} \in \Theta$. Assume that the set of feasible outcomes is A, and the utility function of participant i is $u_i : A \times \Theta \to \mathcal{R}$, which depends on the types of other people, i.e., it is an interdependent value model. A social choice function is denoted as f. The set of all social choice functions is denoted as F, and a subset is denoted as $\hat{F} \subseteq F$. $\Gamma = \langle M_1, \cdots, M_n, h(.)\rangle$ is a mechanism, and a participant's pure strategy of sending message is $m_i : \Theta_i \to M_i$.

Definition 19.9.1 (Ex-post equilibrium) A message profile $\boldsymbol{m}^* = (m_1^*, \cdots, m_n^*)$ is an **ex-post equilibrium** of mechanism $\Gamma = \langle M_1, \cdots, M_n, h(.)\rangle$ if for any $i \in N$, $\boldsymbol{\theta} \in \Theta$ and $m_i \in M_i$, we have

$$u_i(h(m_i^*, \boldsymbol{m}_{-i}^*), \boldsymbol{\theta}) \geqq u_i(h(m_i, \boldsymbol{m}_{-i}^*), \boldsymbol{\theta}).$$

Obviously, an ex-post equilibrium is a Nash equilibrium for each type profile $\boldsymbol{\theta}$. Also, in a private value environment where the participants' utilities depend only on their own types, i.e., $u_i(h(\mathbf{m}(\boldsymbol{\theta})), \boldsymbol{\theta}) = u_i(h(\mathbf{m}(\boldsymbol{\theta})), \theta_i), \forall i \in N$, ex-post equilibrium is equivalent to dominant equilibrium. As a corollary, here the necessary and sufficient conditions for full ex-post implementation of a social choice rule are also necessary and sufficient conditions for a social choice correspondence to be fully implementable in dominant strategies under private values. In games of incomplete information, ex-post equilibrium has the property of ex-post no regret, i.e., even if the participant knows the types of other participants ex-post, he will not change his strategy.

Definition 19.9.2 (Full Ex-post Implementation) A social choice set $\hat{F}$ is (pure-strategy) **fully ex-post implementable or fully implementable in ex-post strategies** if there is a mechanism $\Gamma = (M, h(.))$, such that:

(i) For any social choice function $f \in \hat{F}$, there is an ex-post equilibrium, $\boldsymbol{m}^*$, satisfying:

$$h(\boldsymbol{m}^*(\boldsymbol{\theta})) = f(\boldsymbol{\theta}), \quad \forall \boldsymbol{\theta} \in \Theta;$$

(ii) For any ex-post equilibrium, $\boldsymbol{m}^*$, there is a social choice function, $f \in \hat{F}$, satisfying:

$$h(\boldsymbol{m}^*(\boldsymbol{\theta})) = f(\boldsymbol{\theta}), \quad \forall \boldsymbol{\theta} \in \Theta.$$

The above implementation requires that the equilibrium of the mechanism be exactly the same as the outcome of the social choice set, i.e., to achieve full implementation. Similarly, we can define ex-post implementation and partial ex-post implementation.

Like full Nash implementation with complete information and Bayesian implementation with incomplete information, a social choice set usually needs to satisfy the conditions of incentive compatibility and monotonicity to be ex-post implementable.

Definition 19.9.3 (Ex-post Incentive Compatibility) A social choice set $\hat{F}$ is **ex-post incentive compatible** if for any social choice function $f \in \hat{F}$, $i \in N$, $\boldsymbol{\theta} \in \Theta$ and $\theta'_i \in \Theta_i$, we have

$$u_i(f(\boldsymbol{\theta}), \boldsymbol{\theta}) \geqq u_i(f(\theta'_i, \boldsymbol{\theta}_{-i}), \boldsymbol{\theta}).$$

Social choice set $\hat{F}$ is **strictly ex-post incentive compatible** if the above inequality holds strictly, i.e., for all $i \in N$, $\boldsymbol{\theta} \in \Theta$ and $\theta'_i \neq \theta_i$, we have

$$u_i(f(\boldsymbol{\theta}), \boldsymbol{\theta}) > u_i(f(\theta'_i, \boldsymbol{\theta}_{-i}), \boldsymbol{\theta}).$$

In a direct mechanism, consider participant i's manipulation of information disclosure. Let $\alpha_i : \Theta_i \to \Theta_i$ be a deception of participant i, denoted as $\alpha_i(\theta_i) \neq \theta_i$, and $\boldsymbol{\alpha}(\boldsymbol{\theta})$ be a profile of participants' deceptions. Under the social choice function f, this deception profile shall result in an outcome with $f(\boldsymbol{\alpha}(\boldsymbol{\theta})) \neq f(\boldsymbol{\theta})$. The ex-post monotonicity condition guarantees that, for each deception, there must exist an alert provided by one of the participants, and the alert satisfies incentive compatibility.

Definition 19.9.4 (Ex-post Monotonicity) A social choice set $\hat{F}$ satisfies **ex-post monotonicity** if for any social choice function $f \in \hat{F}$ and a deception $\boldsymbol{\alpha}$ with $f \circ \boldsymbol{\alpha} \notin \hat{F}$, there is a participant $i \in N$ and an outcome y, such that:

$$u_i(y, \boldsymbol{\theta}) > u_i(f(\boldsymbol{\alpha}(\boldsymbol{\theta})), \boldsymbol{\theta}), \tag{19.9.19}$$

and

$$u_i(f(\theta'_i, \boldsymbol{\alpha}_{-i}(\boldsymbol{\theta}_{-i})), (\theta'_i, \boldsymbol{\alpha}_{-i}(\boldsymbol{\theta}_{-i}))) \geqq u_i(y, (\theta'_i, \boldsymbol{\alpha}_{-i}(\boldsymbol{\theta}_{-i}))), \quad \forall \theta'_i \in \Theta_i. \tag{19.9.20}$$

Condition (19.9.19) means that when a deception $\boldsymbol{\alpha}$ appears, participant i has an incentive to provide the alert y; but the inequality (19.9.20) means that the alert will not appear in the absence of deception. For convenience of discussion, we define a set in which, given that other participants' types are $\boldsymbol{\theta}_{-i}$, participant i does not have an incentive to provide an alert for all $\theta_i \in \Theta_i$:

$$Y_i^f(\boldsymbol{\theta}_{-i}) \equiv \{y | u_i(y, (\theta'_i, \boldsymbol{\theta}_{-i})) \leqq u_i(f(\theta'_i, \boldsymbol{\theta}_{-i}), (\theta'_i, \boldsymbol{\theta}_{-i})), \forall \theta'_i \in \Theta_i\}. \tag{19.9.21}$$

Condition (19.9.20) implies that $y \in Y_i^f(\boldsymbol{\theta}_{-i})$. $Y_i^f(\boldsymbol{\theta}_{-i})$ is called a **reward set**, i.e., participant i who reports information truthfully receives this reward, which depends on the social choice function f. At the same time, define a **successful reward set**: $Y_i^{f*}(\boldsymbol{\theta}_{-i}) = \{y : u_i(y, \boldsymbol{\theta}) > u_i(f(\boldsymbol{\alpha}(\boldsymbol{\theta})), \boldsymbol{\theta})\} \bigcap Y_i^f(\boldsymbol{\theta}_{-i})$. In this set,participant i only sends out an alert on deception, thus avoiding some poor choices.

When comparing ex-post monotonicity to Maskin monotonicity, one may feel that the former is stronger than the latter. In the definition of ex-post monotonicity, the truth-telling constraint is satisfied at $(\theta_i', \boldsymbol{\alpha}_{-i}(\boldsymbol{\theta}_{-i}))$ for all $\theta_i' \in \Theta_i$, while Maskin monotonicity only requires it to hold on $\boldsymbol{\alpha}(\boldsymbol{\theta})$. However, this difference has no effect on ex-post implementation and Nash implementation. Indeed, as proven by Bergemann and Morris (2008), in general, ex-post monotonicity does not lead to Maskin monotonicity, nor does Maskin monotonicity mean ex-post monotonicity. For example, the weakly Pareto efficient allocation satisfies Maskin monotonicity, but does not satisfy ex-post monotonicity; the single unit auction with interdependent valuations satisfies ex-post monotonicity, but does not satisfy Maskin monotonicity. However, for environments that possess the single-crossing property, it can be shown that these two are equivalent.

In the following, we explore the necessary and sufficient conditions for ex-post full implementation.

Theorem 19.9.1 (Necessary Conditions for Full Ex-post Implementation) *If a social choice set $\hat{F}$ is fully ex-post implementable, then it must satisfy ex-post incentive compatibility and ex-post monotonicity.*

PROOF. Let $\Gamma = (M_1, \cdots, M_n; h(.))$ be a mechanism that fully ex-post implements the social choice set $\hat{F}$. Given $f \in \hat{F}$, by full ex-post implementation, there is an ex-post equilibrium $\mathbf{m}^*$, such that $f = h \circ \boldsymbol{m}^*$. Since $\boldsymbol{m}^*$ is an ex-post equilibrium, for any $i \in N, \theta'_i \in \Theta_i, \boldsymbol{\theta} \in \Theta$, we have:

$$u_i(h(\mathbf{m}^*(\boldsymbol{\theta})), \boldsymbol{\theta}) \geqq u_i(h(m_i^*(\theta'_i), \mathbf{m}^*_{-i}(\boldsymbol{\theta}_{-i})), \boldsymbol{\theta}).$$

As $f(\boldsymbol{\theta}) = h(\mathbf{m}^*(\boldsymbol{\theta})), f(\theta'_i, \boldsymbol{\theta}_{-i}) = h(m_i^*(\theta'_i), \mathbf{m}^*_{-i}(\boldsymbol{\theta}_{-i}))$, the social choice set $\hat{F}$ satisfies ex-post incentive compatibility.

Consider any deception $\boldsymbol{\alpha}$ that satisfies $f \circ \boldsymbol{\alpha} \notin \hat{F}$. Then $\mathbf{m}^* \circ \boldsymbol{\alpha}$ must not be an equilibrium under some $\boldsymbol{\theta}$. Then, there are $i \in N$ and $m_i \in M_i$, such that:

$$u_i(h(m_i, \boldsymbol{\alpha}_{-i}(\boldsymbol{\theta}_{-i})), \boldsymbol{\theta}) > u_i(h(\mathbf{m}^*(\boldsymbol{\alpha}(\boldsymbol{\theta}))), \boldsymbol{\theta}).$$

Letting $y \equiv h(m_i, \boldsymbol{\alpha}_{-i}(\boldsymbol{\theta}_{-i}))$, we have

$$u_i(y, \boldsymbol{\theta}) > u_i(h(\mathbf{m}^*(\boldsymbol{\alpha}(\boldsymbol{\theta})))), \boldsymbol{\theta}).$$

Since $\mathbf{m}^*$ is an ex-post equilibrium and $f = h \circ \mathbf{m}^*$, we have:

$$\begin{aligned} u_i(f(\theta'_i, \boldsymbol{\alpha}_{-i}(\boldsymbol{\theta}_{-i})), (\theta'_i, \boldsymbol{\alpha}_{-i}(\boldsymbol{\theta}_{-i}))) &= u_i(h(\mathbf{m}^*(\theta'_i, \boldsymbol{\alpha}_{-i}(\boldsymbol{\theta}_{-i}))), (\theta'_i, \boldsymbol{\alpha}_{-i}(\boldsymbol{\theta}_{-i}))) \\ &\geqq u_i(h(m_i, \mathbf{m}^*_{-i}(\alpha_{-i}(\boldsymbol{\theta}_{-i}))), (\theta'_i, \boldsymbol{\alpha}_{-i}(\boldsymbol{\theta}_{-i}))) \\ &= u_i(y, (\theta'_i, \boldsymbol{\alpha}_{-i}(\boldsymbol{\theta}_{-i}))). \end{aligned}$$

Therefore, the alert y satisfies the incentive compatibility constraint of participant i or $y \in Y_i^f(\boldsymbol{\alpha}_{-i}(\boldsymbol{\theta}_{-i}))$. We thus have shown the ex-post monotonicity of $\hat{F}$. □

For the economic environments specified in Definition 19.8.7, Bergemann and Morris (2008) also obtained sufficient conditions that are similar to those for Bayesian implementation.

Theorem 19.9.2 (Sufficient Conditions of Ex-post Implementation) *In economic environments with three or more participants, if a social choice set $\hat{F}$ satisfies ex-post incentive compatibility and ex-post monotonicity, then it is ex-post implementable.*

Proof. For the social choice set $\hat{F}$, Bergemann and Morris (2008) constructed the following mechanism.

First of all, the message space of each participant contains message $m_i = (\theta_i, f_i, z_i, y_i)$, in which: θ_i is the type of participant i himself; f_i is a social choice function suggested by participant i; z_i is similar to the positive integer in the Maskin implementation mechanism, but the scope of z_i here is $N = \{1, \cdots, n\}$; and y_i is the reward (outcome) that participant i proposed. The message space of i is $M_i = \Theta_i \times \hat{F} \times N \times Y$, in which Y is the set of all possible outcomes. The mechanism is restricted by the following rules:

Rule 1: If $\forall i, f_i = f$, then $h(\mathbf{m}) = f(\boldsymbol{\theta})$.

Rule 2: If there is a participant j and a social choice rule $f \in \hat{F}$, such that for $\forall i \neq j$ we have $f_i = f$ and $f_j \neq f$, then we choose y_j when $y_j \in Y_j^f(\boldsymbol{\theta}_{-j})$; otherwise, we choose $f(\boldsymbol{\theta})$.

Rule 3: In addition to the two cases above, if $j(z) = \sum_{i=1}^{n} z_i (mod\ n)$, then we choose outcome $y_{j(z)}$.

Now, we prove the sufficiency of the theorem.

First, we rewrite the message of participant i as:

$$m_i(\theta_i) = (m_i^1(\theta_i), m_i^2(\theta_i), m_i^3(\theta_i), m_i^4(\theta_i)) \in \Theta_i \times \hat{F} \times N \times Y.$$

We shall complete the proof in three steps.

Step 1: For a given $f \in \hat{F}$, there is an ex-post equilibrium $\mathbf{m}^*$, such that $h(\mathbf{m}^*(\boldsymbol{\theta})) = f(\boldsymbol{\theta}), \forall \boldsymbol{\theta} \in \Theta$.

Consider the message profile $(m_i^*(\theta_i) = (\theta_i, f, ., .), i \in N)$. By Rule 1, $h(\mathbf{m}^*(\boldsymbol{\theta})) = f(\boldsymbol{\theta})$. We now prove that this message profile is an ex-post equilibrium. Given other participants' messages $(m_j^*(\theta_i), j \neq i)$, consider that participant i deviates from message $m_i^*(\theta_i)$, choosing to send a message $m_i(\theta_i) = (\theta'_i, f_i, ., .)$. When $y_i \notin Y_i^f(\boldsymbol{\theta}_{-i})$, by Rule 2, society will choose $f(\theta'_i, \boldsymbol{\theta}_{-i})$, and then for participant i, the payoff of deviation is:

$$u_i(f(\theta'_i, \boldsymbol{\theta}_{-i}), (\theta_i, \boldsymbol{\theta}_{-i})) - u_i(f(\theta_i, \boldsymbol{\theta}_{-i}), (\theta_i, \boldsymbol{\theta}_{-i})) \leqq 0.$$

The above inequality is derived by the ex-post incentive compatibility, and the participants do not have an incentive to deviate. When $f_i \neq f, y_i \in$

$Y_i^f(\boldsymbol{\theta}_{-i})$, by Rule 2, society will choose y_i, and then for participant i, the payoff of deviation is:

$$u_i(y, (\theta_i, \boldsymbol{\theta}_{-i})) - u_i(f(\theta_i, \boldsymbol{\theta}_{-i}), (\theta_i, \boldsymbol{\theta}_{-i})) \leqq 0.$$

The inequality above is derived by the definition (19.9.21) of $Y_i^f(\boldsymbol{\theta}_{-i})$ or the ex-post monotonicity condition (19.9.20). When $f_i = f$, according to Rule 1, the other message choices of the participant will not change the choice of the whole society.

Step 2: For any ex-post equilibrium $\mathbf{m}^*$, there is a social choice function $f \in \hat{F}$, such that $m_i^{*2}(\theta_i) = f, \forall i \in N, \boldsymbol{\theta} \in \Theta$. Then, by Rule 1, $h \circ \mathbf{m}^* = f$.

We prove this by way of contradiction. Suppose that, for ex-post equilibrium $\mathbf{m}^*$, and for any $f \in \hat{F}$, there is i and θ_i, such that $m_i^{*2}(\theta_i) \neq f$. Then, there is a state $\boldsymbol{\theta}$, such that Rule 1 is not applicable.

We first consider the case in which Rule 2 can be applied to state $\boldsymbol{\theta}$, which means that there is a j and an f, such that $f_i = f, i \neq j$. Now, for any participant $i \neq j$ whose state is θ_i, he sends a message $m_i(., f_i, z_i, y_i)$ with taking the other participants' states as $\boldsymbol{\theta}_{-i}$, in which $f_i \neq f, i = \sum_{k=1}^n z_k (mod\ n)$. Therefore, the social rule will choose y_i under Rule 3, and the utility is $u_i(y_i, \boldsymbol{\theta})$. Therefore, if $\mathbf{m}^*$ is an ex-post equilibrium, for any $y \in Y, i \neq j$, it must satisfy $u_i(h(\mathbf{m}^*(\boldsymbol{\theta})), \boldsymbol{\theta}) \geqq u_i(y, \boldsymbol{\theta})$, which contradicts the condition of the economic environment.

We now consider the case in which Rule 3 can be applied in state $\boldsymbol{\theta}$, which means that for any participant with state θ_i, he will send the message $m_i(., f_i, z_i, y_i)$ when taking the other participants' states as $\boldsymbol{\theta}_{-i}$, in which $i = \sum_{k=1}^n z_k (mod\ n)$. Then, by Rule 3, the social rule will choose y_i, and the utility level is $u_i(y_i, \boldsymbol{\theta})$. Therefore, if $\mathbf{m}^*$ is an ex-post equilibrium, for any $y \in Y, i \neq j$, there must be $u_i(h(\mathbf{m}^*(\boldsymbol{\theta})), \boldsymbol{\theta}) \geqq u_i(y, \boldsymbol{\theta})$. This contradicts the condition of the economic environment.

Step 3: For any social choice function $f \in \hat{F}$ and any ex-post equilibrium $\mathbf{m}^*$ that satisfies $m_i^{*2}(\theta_i) = f, \forall i, \theta_i$, we must have $f \circ m^{*1} \in \hat{F}$.

Suppose by way of contradiction that $f \circ m^{*1} \notin \hat{F}$. According to ex-post monotonicity, there is an i, $\boldsymbol{\theta}$, $y \in Y_i^f(m_{-i}^{*1}(\boldsymbol{\theta}_{-i}))$ satisfying:

$$u_i(y, \boldsymbol{\theta}) > u_i(f(m^{*1}(\boldsymbol{\theta})), \boldsymbol{\theta}).$$

Consider a participant i with state θ_i, believing other participants' states to be $\boldsymbol{\theta}_{-i}$, he will send the message $m_i = (., f_i, ., y)$ that satisfies $f_i \neq f$. Given that other participants choose their equilibrium strategies, by Rule 2, the social choice outcome is $h(m_i, \mathbf{m}_{-i}^*(\boldsymbol{\theta}_{-i})) = y$. The participant i's utility under m_i is:

$$u_i(h(m_i, \mathbf{m}^*(\boldsymbol{\theta}_{-i})), \boldsymbol{\theta}) = u_i(y, \boldsymbol{\theta}) > u_i(f(\mathbf{m}^{*1}(\boldsymbol{\theta})), \boldsymbol{\theta}) = u_i(h(\mathbf{m}^*(\boldsymbol{\theta})), \boldsymbol{\theta}),$$

which contradicts the fact that $\mathbf{m}^*$ is an ex-post equilibrium.

We have thus proved that under economic environments, ex-post incentive compatibility and ex-post monotonicity are sufficient conditions for full ex-post implementation. □

For non-economic environments, Bergemann and Morris (2008) showed that with an additional condition of "no veto power", ex-post incentive compatibility and ex-post monotonicity are sufficient conditions for full ex-post implementation. For an economic environment that satisfies the single-crossing property, they also proved that any social choice set, with strict ex-post incentive compatibility and consisting of interior points, satisfies the ex-post monotonicity condition and is therefore fully ex-post implementable. When the number of participants is greater than 2, Chapter 21 will introduce the generalized direct VCG mechanism under interdependent valuation environments, which also satisfies ex-post monotonicity and has a unique ex-post equilibrium.

Ohashi (2012) also provided similar sufficient and necessary conditions for full ex-post implementation in two-participant environments. Moreover, using the concept of ex-post equilibrium, Bergemann and Morris (2005) established an analytical framework for robust mechanism design.

19.10 Biographies

19.10.1 A. Michael Spence

A. Michael Spence (1943—) was born in New Jersey in the United States, and he received his Ph.D. degree from Harvard University in 1972. He is currently a professor at New York University and a Senior Fellow at Stanford University's Hoover Institution. In 2001, he won the Nobel Memorial Prize in Economic Sciences, along with George Akerlof and Joseph Stiglitz, for their pioneering analyses of markets with asymmetric information.

Spence's main contribution to information economics is on how individuals with informational advantages transmit "signals" reliably to individuals with informational disadvantages, in order to avoid such problems as adverse selection. Sending signals requires individuals to take observational and costly measures to convince other individuals of their abilities or, more generally, of the value or quality of their products. Spence formalized this idea and analyzed its economic effect. For example, Spence interpreted education as a signal of productivity in the labor market in his 1973 groundbreaking study. The basic assertion was that the signal of education would not generate a positive effect for the signal sender unless signaling cost was significantly different among job seekers. Employers could not distinguish competent job seekers from those with low abilities, unless when the latter chose a lower level of education the former found that their education investment paid off; otherwise, it would lead the labor market to be occupied by low-ability workers with low wages, forming the phenomenon of “bad money drives out good” in the labor market. Spence also pointed out the possibility of different expectation equilibria concerning the impact of education on labor market returns, featuring that men and Caucasians earn more than women and African-Americans, respectively, when they have the same level of labor productivity.

Spence’s subsequent research applied this theory, demonstrated the importance of market signaling, and analyzed a large number of economic phenomena, such as expensive advertising and full warranty as productivity signals; the active price reduction as a signal of market power; the strategy of delaying wage quotation as a signal of bargaining power; debt financing as a sign of profitability rather than the financing method of issuing new shares; and monetary policy at the expense of recession as a signal of a strong commitment to lowering high inflation.

19.10.2 Roger B. Myerson

Roger Bruce Myerson (1951—) is an economics professor at the University of Chicago, and was awarded the Nobel Prize in Economics with Leonid Hurwicz and Eric Maskin in 2007 for his contribution to the creation and development of mechanism design theory and auction theory, such as the Revenue Equivalence Theorem.

Myerson was born in Boston, MA in the United States on March 29, 1951. In 1976, he received his Ph.D. in applied mathematics from Harvard University. His doctoral thesis was entitled “A Theory of Cooperative Games”. He has been at the University of Chicago since 2007, and his research expertise includes mechanism design theory, game theory, and the analysis of voting systems in political science. Myerson’s “Optimal Auction Design”, published in 1981, was the cornerstone of optimal mechanism design. He also published other influential books, including “Game Theory: Analysis of Conflict” and “Probability Models for Economic Decisions”.

After Vickrey, economists began to work on auction theory. Among them, Myerson deserves special mention. He used the newly developed mechanism design theory to revisit auction theory and promoted Vickrey’s theory on this basis. Myerson came to the following conclusion through rigorous mathematical analysis: Under a series of assumptions (e.g., symmetric independent private values), all standard auction mechanisms will bring the same expected revenue to the auctioneer. Obviously, this conclusion surpassed previous ideas of Vickrey and other scholars, who only focused on specific auction forms, thereby greatly advancing auction theory.

Myerson is an economist who succeeds in making abstract economic theory practical. For example, he solved economic problems for California's power crisis. In the 1980s, electricity market reform in California was implemented to remove the disadvantages of power monopoly, but it was impossible for the power industry to implement perfect competition. Myerson thus applied mechanism design theory to design a plan for California's electricity market reform. In addition, he solved the problem of recruiting students to medical school in the U.S. Indeed, Myerson's contribution to solving real economic problems has impressed the economics profession, making "economic engineering" possible, in which economics could be as pragmatic as engineering.

19.11 Exercises

Exercise 19.1 Consider economic environments with one seller and one buyer. The value of buyer is v_b, and the value of seller is v_s. Both v_b and v_s follow the uniform distribution on $[0, 1]$, and are independent. The trading mechanism is given as follows: the buyer and the seller simultaneously make offers at the purchasing price p_b and the selling price p_s, respectively. If $p_b \geqq p_s$, the transaction is conducted with the price $p = (p_b + p_s)/2$, and the profits of buyers and sellers are, respectively, given by:

$$\begin{aligned} &v_b - p, \\ &p - v_s. \end{aligned}$$

If $p_b < p_s$, there will be no transaction, and the profits of buyers and sellers are 0. It is known that, under this mechanism, there are multiple-equilibrium bidding strategies. For example, there is a linear equilibrium bidding strategy: $p_i(v_i) = a_i + c_i v_i$, $i \in b, s$. Find the linear equilibrium strategy for buyers and sellers, and prove whether or not it can truthfully implement ex-post Pareto efficient outcomes.

Exercise 19.2 (Arrow,1979; d'Aspremont and Gerard-Varet, 1979) Suppose that the set of participants is N, with $|N| = n$, and the set of types for participant i is Θ_i. The utility functions are $u_i(d, \theta_i, t_i) = v_i(d, \theta_i) + t_i$. AVG mechanism $(d(\boldsymbol{\theta}), t_i(\boldsymbol{\theta}))$ satisfies:

$$d(\boldsymbol{\theta}) \in \mathrm{argmax}_d \sum_{i \in N} v_i(d, \theta_i);$$

$$t_i(\boldsymbol{\theta}) = E_{\boldsymbol{\theta}_{-i}}[\sum_{j \neq i} v_j(d(\boldsymbol{\theta}), \theta_j)|\theta_i] - \frac{1}{n} \sum_{k \neq i} E_{\boldsymbol{\theta}_{-i}}[\sum_{j \neq k} v_j(d(\boldsymbol{\theta}), \theta_j)|\theta_k].$$

Prove that if participants' type distributions are independent, the above AVG mechanism satisfies Bayesian incentive compatibility and is interim budget-balanced.

Exercise 19.3 (Diamantaras, 2009) Consider economic environments with two participants and one public good. The set of types for each participant is $\Theta_i = \{0, 1\}, i = 1, 2$. The two types have equal probability, and the types of two participants are independently distributed. Participant i's evaluation of the public good is θ_i. Assume that the cost of the public good is $c = \frac{3}{4}$

Prove that, in such environments, there is no mechanism that simultaneously satisfies the following features: (i) interim participation constraints; (ii) Bayesian incentive compatibility; (iii) the decision-making mechanism of public good provision is efficient; and (iv) and mechanism is budget-balanced, i.e., it does not require extra resources from outside of the economy under consideration.

Exercise 19.4 (Palfrey and Srivastava, 1989) Consider economies in which the set of participants is $I = \{1, 2, 3\}$, the set of alternatives is $A = \{a, b\}$, and the set of each participant is $T^i = \{t_a, t_b\}$. Suppose that the types of participants are all independently and identically distributed, i.e., $Prob(t^i = t_b) = q > 0.5^{1/2}$, and each participant's utility function is the same and satisfies:

$$u^i(a, t_a) = 1 > 0 = u^i(b, t_a);$$
$$u^i(b, t_b) = 1 > 0 = u^i(a, t_b).$$

Suppose that the social choice function f is given as follows:

$$f(t^1, t^2, t^3) = \begin{cases} t_a, & \text{If } t^1 = t^2 = t_a, \text{or } t^1 = t^3 = t_a, \text{or } t^2 = t^3 = t_a; \\ t_b, & \text{if } t^1 = t^2 = t_b, \text{or } t^1 = t^3 = t_b, \text{of } t^2 = t^3 = t_b. \end{cases}$$

1. Prove that $f(\cdot)$ is the only allocation rule that satisfies the following five properties: (i) it is Bayesian incentive compatible; (ii) it is ex-ante, interim, and ex-post efficient; (iii) $f(\boldsymbol{t})$ is the majority voting decision rule; (iv) it maximizes the Arrow social welfare function; and (v) it can be implemented in dominant strategies by a direct revelation mechanism.

2. Prove that $f(\cdot)$ is not fully Bayesian implementable (hint: prove that it does not satisfy Bayesian monotonicity).

Exercise 19.5 (Palfrey and Srivastava, 1989) Suppose that the set of participants is $I = \{1, 2, 3\}$, and the set of alternatives is $A = \{a, b\}$. Participants' type spaces are $T_i = \{t^a, t^b\}$, which is independently and identically distributed, with $prob(t_i = t^b) = q > 0.5^{1/2}$, and their utility functions are the same and depend on the type of other participants, and satisfy:

$$u_i(a, \boldsymbol{t}) = \begin{cases} 1, & \text{if at least two participants are of the type } t^a, \\ 0, & \text{otherwise.} \end{cases}$$

$$u_i(b, \boldsymbol{t}) = \begin{cases} 1, & \text{if at least two participants are of the type } t^b, \\ 0, & \text{otherwise.} \end{cases}$$

Suppose that the social choice function f is given as follows:

$$f(t_1, t_2, t_3) = \begin{cases} t^a, & \text{if } t_1 = t_2 = t^a, \text{or } t_1 = t_3 = t^a, \text{or } t_2 = t_3 = t^a, \\ t^b, & \text{if } t_1 = t_2 = t^b, \text{or } t_1 = t_3 = t^b, \text{or } t_2 = t_3 = t^b. \end{cases}$$

Prove that this social choice function cannot be implemented in undominated Bayesian equilibrium. This so-called undominance means that, for each participant, there is no other strategy that weakly dominates this s-trategy.

Exercise 19.6 (Characterization of Bayesian Incentive Compatibility) Consider the linear model of independent distribution of private values. Prove that social choice rule $(\boldsymbol{y}(\cdot), t_1(\cdot), \cdots, t_n(\cdot))$ is Bayesian incentive compatible if and only if for all $i \in N$,

1. $E_{\boldsymbol{\theta}_{-i}} y_i(\theta_i, \boldsymbol{\theta}_{-i})$ is non-decreasing in θ_i;

2. $E_{\boldsymbol{\theta}_{-i}} U_i(\theta_i) = E_{\boldsymbol{\theta}_{-i}}[U_i(\underline{\theta}_i, \boldsymbol{\theta}_{-i})] + \int_{\underline{\theta}_i}^{\theta_i} E_{\boldsymbol{\theta}_{-i}} y_i(\tau, \boldsymbol{\theta}_{-i}) d\tau, \ \forall \theta_i \in \Theta_i$.

Exercise 19.7 A seller and a buyer are bargaining over an indivisible item. The valuation of the buyer is $\theta_b = 10$. There are two possible valuations for the seller, i.e., $\theta_s \in \{0, 9\}$. Let t be the period when the transaction occurs ($t = 1, 2, \cdots$), and P represent the agreed-upon price. The discount factor for both the buyer and the seller is δ.

1. In the current economic environment, what is the set of feasible alternatives?

2. Imagine that under a Bayesian Nash equilibrium of the bargaining process, when the valuation of the seller is 0, the transaction occurs immediately; and when the valuation of the seller is θ_s, the agreed-upon transaction price is $(10 + \theta_s)/2$. When the valuation of the seller is 9, what is the earliest possible time for the transaction to take place?

Exercise 19.8 Consider the Myerson-Satterthwaite bilateral trade model under discrete conditions. It is given that the valuation of the buyer is v and the cost of the seller is c, which is uniformly distributed over $1, 2, 3, 4$.

1. Prove that the efficient trade is Bayesian incentive-compatible.

2. What does this example tell us about the Myerson-Satterthwaite Impossibility Theorem?

Exercise 19.9 Consider a direct incentive-compatible mechanism $(q, \mathbf{t})$, in which $\mathbf{t}$ is ex-ante budget-balanced, i.e., $E_0 \sum_{i=1}^{N} t_i(\boldsymbol{\theta}) = 0$. Suppose that the type is independently distributed.

1. Prove that there is another Bayesian incentive-incapable mechanism $(q, \mathbf{t}')$, such that $\mathbf{t}'$ is ex-post budget-balanced, i.e., $\sum_{i=1}^{N} t_i'(\boldsymbol{\theta}) = 0$.

2. Prove that $E_{\boldsymbol{\theta}_{-i}} t_i(\boldsymbol{\theta}) = E_{\boldsymbol{\theta}_{-i}} t_i'(\boldsymbol{\theta})$.

3. Prove that if there is an individually rational VCG mechanism $(q^*, \mathbf{t})$, such that $E_{\boldsymbol{\theta}} \sum_{i=1}^{N} t_i(\boldsymbol{\theta}) \geqq 0$, then there is a Bayesian incentive compatible, budget-balanced, individually rational, and efficient mechanism denoted $(q^*, \mathbf{t}')$.

Exercise 19.10 Consider a bilateral trading economy in which two participants initially own a unit of a commodity. Each participant's valuation for every unit of the product consumed is $\theta_i (i = 1, 2)$. Suppose that θ_i follows a uniform distribution over $[0, 1]$ and is independent of each other.

1. Characterize a trading rule in an ex-post efficient social choice function.

2. Consider the following mechanism: Each participant submits a bid, and the highest bidder obtains the unit of good from the opponent participant and pays the offer. Give a symmetric Bayesian-Nash equilibrium for this mechanism.

3. What is the social choice function that can be implemented by this mechanism? Verify that it is Bayesian incentive compatible. Is it efficient? Does it satisfy individual rationality? Intuitively, why does it differ from the conclusion of the Myerson-Satterthwaite Theorem?

Exercise 19.11 Consider the problem of bilateral trading between a seller and a consumer. The seller's production cost is determined by its cost type, and the cost type is the seller's private information. When the seller's type is θ, if the consumer's total payment is x, then the former's profit is $x - \theta q$. At the same time, for q units of goods, the value of the consumer depends on the seller's type θ, which is expressed as $\pi(q|\theta) = (4+2\theta)q^{1/2}$. Therefore, the consumer obtains q units of goods by paying x from the seller whose type is θ, and the net surplus is $(4 + 2\theta)\sqrt{q} - x$. Any participant can refuse to participate in the transaction, and then $q = 0$ and $x = 0$.

1. When the seller's type θ is common knowledge, find the expression of the quantity of goods, denoted by $q^*(\theta)$, that maximizes the social surplus.

2. Suppose that the seller's type is either $\theta_L = 2$ or $\theta_H = 3$. The probability of being a low-cost θ_L-type is p_L, and the probability of being a high-cost θ_H-type is $p_H = 1 - p_L$. For the consumer's expected net return maximization problem that takes into account incentive constraints and participation constraints, find a trading plan and give the constrained optimization problem. (Hint: $\pi(q|\theta_L) = 8\sqrt{q}$ and $\pi(q|\theta_H) = 10\sqrt{q}$.)

3. Write down the problem for finding the consumer's optimal Bayesian incentive compatible trading plan.

4. Consider the following case: the consumer believes that the seller's cost type is uniformly distributed over $[2, 3]$. Write down the problem for finding the consumer's optimal Bayesian incentive compatible trading plan, which maximizes the consumer's expected surplus subject to the participation constraint and incentive compatibility constraint.

5. Consider another case: the seller's cost type is either $\theta_L = 2$ or $\theta_H = 3$. In all incentive-compatible plans in which the consumer's surplus is non-negative for any type of seller, find the optimal trading plan for both types of sellers.

6. For the solution to the problem (5), prove that if p_L is sufficiently close to 0 and p_H is sufficiently close to 1, then there is a pooling strategy that can give the consumer a non-negative expected surplus, which is better than the trading plan in question (5).

Exercise 19.12 There are two construction firms that bid for the construction of a government building. Each firm is one of two types: an H-type firm, which can use the cost C_H to construct a high-quality project that values v_H; and an L-type firm, which can use the cost C_L to construct a low-quality project that values v_L. Here, $C_H > C_L$ and $v_H > v_L$. Suppose that $v_H - C_H > v_L - C_L > 0$. The types are independent of each other and are the private information of firms. Each firm has a probability of v to be a high-quality firm. Here, we consider the direct revelation mechanism. When one firm declares its type as θ and the other firm declares its type as θ', we use $q(\theta, \theta'), \theta, \theta' \in \{H, L\}$ to represent the probability that the previous firm wins the bid; similarly, we use $T(\theta, \theta')$ to represent the government's payment. Given that both the government and the two firms are risk-neutral, the government's goal is to maximize its expected surplus, denoted as $E(v - \omega)$.

1. Write down the government's optimal bidding design problem that satisfies participation constraints and incentive compatibility constraints.

2. Derive the government's optimal bidding mechanism.

Exercise 19.13 (Dana and Spier, 1994) Suppose that there are two firms, $j = 1, 2$, who shall obtain the production rights in a given market through competition. A social planner designs an optimal production-rights auction to maximize the expectation of a social surplus function. The social surplus function is defined by

$$W = \Sigma_j \pi_j + S + (\lambda - 1)\Sigma_j t_j,$$

in which π_j is the firm j's total (pre-transfer) profit, S is the consumer's surplus, $\lambda > 1$ is the shadow cost of public funds, and t_j denotes the transfer from firm j to the planner. The auction determines each firm's transfer and the structure of a market, i.e., either two firms do not obtain production rights, or one firm obtains production rights, or both have the right to production.

Every firm j privately observes its fixed production cost, denoted by θ_j. The fixed costs θ_1 and θ_2 follow an independent and identical distribution over $[\theta^1, \theta^2]$. The density function $\phi(\cdot)$ and the distribution function $\Phi(\cdot)$ are all continuously differentiable. Suppose that $\frac{\Phi(\cdot)}{\phi(\cdot)}$ is increasing in $\boldsymbol{\theta}$. The firms have a common marginal cost $c < 1$ and produce homogeneous products. The inverse market demand is $p(x) = 1 - x$. If both firms obtain production rights, they become Cournot competitors.

1. Write the problem of the optimal auction mechanism for production rights.

2. Characterize the optimal auction mechanism.

Exercise 19.14 There is a seller ($i = 0$) who owns two identical indivisible goods. At the same time, there are two buyers ($i = 1, 2$). For a unit of each good, buyers and sellers each have a willingness-to-pay, denoted by θ_i, which are private information. The three participants' willingness-to-pay follows a uniform distribution over $[0, 1]$ and is independent of each other. This is a common knowledge. For the portion of the product that exceeds one unit, everyone's willingness-to-pay is 0.

1. Describe the efficient allocation function $f(\theta_0, \theta_1, \theta_2)$ for these two units of goods.

2. For an incentive-compatible mechanism that truthfully Bayesian implements the efficient allocation of these two units of goods, what is the interim expected utility of each participant?

3. What is the interim participation constraint for each participant?

4. When there is no external transfer, is there a mechanism that satisfies the participation constraint in question (3), as well as incentive compatibility constraints, and can efficiently allocate these two units of goods?

Exercise 19.15 There are two types of students, denoted by L and H, searching for jobs in the market. There are at least two potential employers. The profit of hiring type-i students is π_i, and $\pi_H > \pi_L$. Recruitment is conducted as follows: (1) student observes his own type (not observed by the employer); (2) student chooses an unproductive education level e, which is costly and can be observed by the employer; (3) the employer offers a salary w to future workers (current students) based on the observed level of education (Bertrand competition); and (4) student chooses an employer and becomes a worker. The utility function for i-type worker who accepts w is $U(w, e) = w - c_i(e)$. The corresponding employer's profit is $\pi_i - w$. Suppose $C'_H(e) > C'_L(e)$ for all $e > 0$. Only consider pure strategy equilibrium.

1. Does this model satisfy the Spence-Mirrlees single-crossing property? Explain your answer.

2. What is the level of education obtained by a student of type L in the separating equilibrium? Derive the expressions of the maximum and minimum education levels of workers of type H in the separating equilibrium.

3. Derive the expressions of the maximum and minimum education levels of all of the students in the pooling equilibrium.

4. Which equilibrium outcome is more dominant? Explain your answer.

5. Suppose that the government taxes t for education level (i.e., a person who receives e years of education will pay a tax of te), and the government holds all of the tax revenue. How does this tax change the separating and pooling equilibrium conditions that you derived in questions (2) and (3)?

6. Only consider efficient separating equilibrium. Based on question (5), derive the partial derivatives of H- and L-type workers' utility functions with respect to tax t. Give the sign of the partial derivatives. How does this tax affect the welfare of workers?

Exercise 19.16 Answer the following questions:

1. Under conditions of complete information, can the single-crossing property imply Maskin monotonicity? If yes, prove it; if no, provide a counterexample.

2. Under conditions of incomplete information, can the single-crossing property imply Bayesian monotonicity? If yes, prove it; if no, provide a counterexample.

Exercise 19.17 Give an example to illustrate independence between Maskin monotonicity and ex-post monotonicity.

Exercise 19.18 (Bergemann and Morris, 2008) Consider a modified VCG mechanism under interdependent values. We call $(\Theta, \boldsymbol{y}^*, \boldsymbol{t}^*)$ a generalized VCG mechanism if the highest bidder gets the good and the price is $\max_{j \neq i} u_j(\vartheta_i(\boldsymbol{\theta}_{-i}), \boldsymbol{\theta}_{-i})$ (not directly dependent on θ_i).

1. Prove that the generalized VCG mechanism satisfies ex-post monotonicity.
2. Prove that the generalized VCG mechanism satisfies ex-post incentive compatibility.
3. Prove that the generalized VCG mechanism does not satisfy Maskin monotonicity, so that it is not Nash implementable.

Exercise 19.19 Prove that for economic environments with two participants, if the social choice function f satisfies ex-post incentive compatibility and ex-post monotonicity, then it is ex-post implementable.

Exercise 19.20 For economic environments with two participants, suppose that a social choice set $\hat{F}$ contains more than one element. Explain whether ex-post incentive compatibility and ex-post monotonicity are sufficient conditions for $\hat{F}$ to be ex-post implementable.

Exercise 19.21 (Bergemann and Morris, 2008) They obtained the following conclusion: for a general case with the number of participants being $n \geqq 3$, if social choice function f satisfies ex-post incentive compatibility, ex-post monotonicity and the no-veto-power condition, then it is ex-post implementable. Explain whether this conclusion can be generalized to the case with $n = 2$.

Exercise 19.22 (Bergemann and Morris, 2005) Consider two participants, 1 and 2. The sets of their types are $\Theta_1 = \{\theta_1, \theta_1'\}$ and $\Theta_2 = \{\theta_2, \theta_2'\}$, respectively. The set of socially feasible outcomes is $A = \{a, b, c\}$. With different outcomes and types of participants, their utilities can be described by the following table: (the two numbers in each box represent, respectively, the utilities of participant 1 and 2):

a	θ_2	θ_2'
θ_1	(1,0)	(-1,2)
θ_1'	(0,0)	(0,0)

b	θ_2	θ_2'
θ_1	(-1,2)	(1,0)
θ_1'	(0,0)	(0,0)

c	θ_2	θ_2'
θ_1	(0,0)	(0,0)
θ_1'	(1,1)	(1,1)

A social planner considers the following social choice set, $\hat{F}$, to maximize the sum of the utilities of these two participants, and it is described by the following table:

F	θ_2	θ_2'
θ_1	{a,b}	{a,b}
θ_1'	{c}	{c}

Prove that social choice set $\hat{F}$ cannot be ex-post implemented.

Exercise 19.23 (Bergemann and Morris, 2005) Consider two participants, 1 and 2. Their type sets are $\Theta_1 = \{\theta_1, \theta_1', \theta''_1\}$ and $\Theta_2 = \{\theta_2, \theta_2'\}$, respectively. Suppose that the socially feasible outcome set is $A = \{a, b, c, d\}$. With different outcomes and types of participants, their utilities can be described by the following table: (the two numbers in each box represent, respectively, the utilities of participants 1 and 2):

a	θ_2	θ_2'
θ_1	(0,2)	(0,2)
θ_1'	(-4,0)	(1,0)
θ''_1	(-4,0))	(-4,0)

b	θ_2	θ_2'
θ_1	(0,0)	(0,0)
θ_1'	(0,2)	(0,0)
θ''_1	(-4,0))	(0,0)

c	θ_2	θ_2'
θ_1	(0,0)	(-4,0)
θ_1'	(0,0)	(0,2)
θ''_1	(0,0))	(0,0)

d	θ_2	θ_2'
θ_1	(-4,0))	(-4,0)
θ_1'	(1,0)	(-4,0)
θ''_1	(0,2)	(0,2)

A social planner considers the following social choice set, $\hat{F}$, to maximize the sum of the utilities of these two participants, and it is described by the following table:

F	θ_2	θ_2'
θ_1	a	a
θ_1'	b	c
θ''_1	d	d

Suppose that the social planner can use transfers among the participants. Each participant has a quasi-linear utility function. Prove that when the social choice set $\hat{F}$ is ex-post budget-balanced, it cannot be ex-post implemented.

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Chapter 20

Dynamic Mechanism Design

20.1 Introduction

Chapters 16-19 discussed optimal static mechanism design problems. Most practical contractual relationships, however, have a dynamic nature, involving long-term, non-anonymous interactions between the designer and agent(s). Examples include income taxation, regulation, managerial compensation, or a monopolist repeatedly selling a nondurable good to a buyer. In these environments, contracts can be made contingent on past realizations of the agent's type, allowing the principal to use the agent's revealed information to screen future type realizations. This may be particularly useful in mitigating asymmetric information and agency problems when the type is persistent over time.

The last two decades have witnessed significant progress in extending the theory of mechanism design from static to dynamic environments. Two objectives in dynamic mechanism design are maximizing the revenue of the designer (optimality) and maximizing the aggregate welfare of all parties (efficiency). The first strand of literature focuses on the design of revenue-maximizing mechanisms in various dynamic settings. The second strand of studies investigates how to implement dynamically efficient (surplus-maximizing) allocations in settings in which the agents' types change over time, which extends the Vickrey-Clarke-Groves (VCG) and d'Aspremont-Gérard-Varet (AGV) results from static to dynamic settings.

In this chapter, we shall discuss dynamic mechanism design problems from both *optimality* and *efficiency* perspectives. To this end, we first consider optimal contract design in a simple principal-agent framework with repeated interactions, concerning the changes of incentive constraints and participation constraints, and the corresponding tradeoff. We then explore the efficient dynamic mechanism design, especially Pareto efficient dynamic mechanisms.

20.2 Dynamic Optimal Contracts under Full Commitment

In this section, we discuss a simple dynamic contracting problem, in which the principal has full commitment power. We derive the optimal dynamic contracts in three cases regarding the agent's types: (1) constant (time-invariant) types; (2) independent types over time; and (3) correlated types over time.

20.2.1 Constant Types

For simplicity, consider a two-period situation that consists of a monopolist (or principal) and a consumer (or agent). We examine the second-best contract of the principal. Let us start with the simplest case in which the agent's type is constant over time: $\theta_1 = \theta_2 \in \{\theta_L, \theta_H\}$, $\Delta\theta \equiv \theta_H - \theta_L$. As in Chapter 16, we assume that the probability of type θ_L is β.

The timing of contracting is described in Figure (20.1). At time 0, the agent learns his type θ_1; at time 0.25, the principal offers a long-term contract $\{q_1(\theta_1), T_1(\theta_1); q_2(\theta_1, \theta_2), T_2(\theta_1, \theta_2)\}$ to the agent for both periods; at time 0.5, the agent decides whether or not to accept the contract, and the interaction continues if he chooses to accept and is over otherwise; in period 1, the agent buys q_1 units, and pays T_1 to the principal; in period 2, the agent buys q_2 units, and pays T_2 to the principal, where the transfer depends on the current and the past reports.

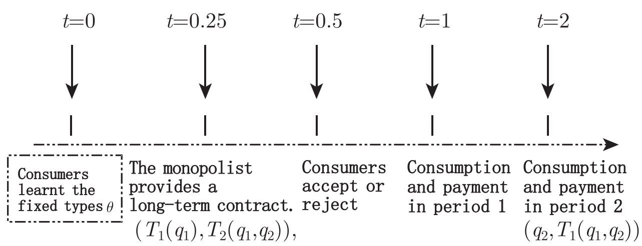

Figure 20.1: Timing of contracting with constant types

Since type is constant, only $\theta_1 = \theta_2 = \theta$ may happen on the truthful reports path. To economize the notation, we write $q_2(\theta)$ and $T_2(\theta)$, instead of $q_2(\theta, \theta)$ and $T_2(\theta, \theta)$, respectively. The monopolist's expected profit is

$$\begin{aligned}
\Pi &\equiv \mathbb{E}_\theta \left\{T_1(\theta) - cq_1(\theta) + \delta\left[T_2(\theta, \theta) - cq_2(\theta, \theta)\right]\right\} \\
&= \mathbb{E}_\theta \left\{\theta v(q_1(\theta)) - cq_1(\theta) - u_1(\theta) + \delta\left[\theta v(q_2(\theta)) - cq_2(\theta) - u_2(\theta)\right]\right\} \\
&= \mathbb{E}_\theta \left\{\theta v(q_1(\theta)) - cq_1(\theta) + \delta\left[\theta v(q_2(\theta)) - cq_2(\theta)\right] - U(\theta)\right\}
\end{aligned}$$

where $u_i(\theta) \equiv \theta v(q_i(\theta)) - T_i(\theta), i = 1, 2$ is the instantaneous information rents obtained by the consumer with the status quo utility that is normalized to 0; $U(\theta) \equiv u_1(\theta) + \delta u_2(\theta)$ denotes the discounted lift-long rent; and δ is the discount factor.

As in the previous discussion of static adverse selection, if there is only one period, we know that the second-best contract is described as (q_i^{SB}, T_i^{SB}), $i \in \{L, H\}$, where outputs $q_H^{SB} = q_H^{FB}$ and q_L^{SB} satisfy

$$\begin{aligned} \theta_H v'(q_H^{SB}) &= c, \\ \theta_L v'(q_L^{SB}) &= \frac{c}{1 - \frac{1-\beta}{\beta}\frac{\Delta\theta}{\theta_L}}. \end{aligned}$$

The transfers T_H^{SB} and T_L^{SB} satisfy

$$\begin{aligned} \theta_H v(q_H^{SB}) - T_H^{SB} &= U_H = \Delta\theta v(q_L^{SB}), \\ \theta_L v(q_L^{SB}) - T_L^{SB} &= U_L = 0. \end{aligned}$$

If the principal has full commitment power, the optimal contract for two periods is twice the repetition of the optimal static contract, as shown below. Since the principal can commit intertemporally, he can omit the information learned in period one, and strictly enforce the optimal static contract. In this case, the revelation principle remains valid. In the following, we discuss this optimal long-term contract, denoted by (q_1, q_2, U). The complete form of the contract is $(q_{1H}, q_{2H}, U_H; q_{1L}, q_{2L}, U_L)$. If the contract is incentive-feasible, then the four conditions below are satisfied:

$$U_H \geqq U_L + \Delta\theta[v(q_{1L}) + \delta v(q_{2L})], \tag{20.2.1}$$

$$U_L \geqq U_H - \Delta\theta[v(q_{1H}) + \delta v(q_{2H})]; \tag{20.2.2}$$

$$U_H \geqq 0, \tag{20.2.3}$$

$$U_L \geqq 0. \tag{20.2.4}$$

The principal solves the following constrained maximization problem:

$$\max_{\left\{\begin{smallmatrix}(q_{1H}, q_{2H}, U_H); \\ (q_{1L}, q_{2L}, U_L)\end{smallmatrix}\right\}} \Pi, s.t. : (20.2.1), (20.2.2), (20.2.3), (20.2.4), \tag{20.2.5}$$

where

$$\Pi = \begin{bmatrix} (1-\beta)[\theta_H(v(q_{1H}) + \delta v(q_{2H})) - U_H - c(q_{1H} + \delta q_{2H})] \\ +\beta[\theta_L(v(q_{1L}) + \delta v(q_{2L})) - U_L - c(q_{1L} + \delta q_{2L})] \end{bmatrix}.$$

As we discussed previously, we only consider the binding (20.2.1) and (20.2.4) and implementability (monotonicity) condition

$$v(q_{1H}) + \delta v(q_{1H}) \geqq v(q_{1L}) + \delta v(q_{1L}). \tag{20.2.6}$$

Substituting $U_L = 0$ and $U_H = \Delta\theta(v(q_{1L}) + \delta v(q_{2L}))$ into the principal's objective (20.2.5), we obtain the first-order conditions for $q_{1H}, q_{2H}; q_{1L}, q_{2L}$, respectively:

$$\begin{aligned}\theta_H v'(q_{1H}^{SB}) &= \theta_H v'(q_{2H}^{SB}) = c,\\ \theta_L v'(q_{1L}^{SB}) &= \theta_L v'(q_{2L}^{SB}) = \frac{c}{1-(\frac{1-\beta}{\beta}\frac{\theta_H-\theta_L}{\theta_L})}.\end{aligned}$$

(20.2.6) is obviously satisfied. Therefore, the second-best long-term contract with full commitment for two periods is twice the repetition of the static second-best contract.

20.2.2 Independent Types Over Time

Now, we consider the other case in which types are independent over time.

The timing of contracting when the agent's type is independent over time is described in Figure (20.2). Compared to Figure (20.1), at time 0, the agent only learns his type in period 1; and only at time 1.5, the agent can learn his type in period 2. The type distribution of the consumer is i.i.d., i.e., the probability of being θ_L is β in each period. The other assumptions are the same as those in the case of constant types.

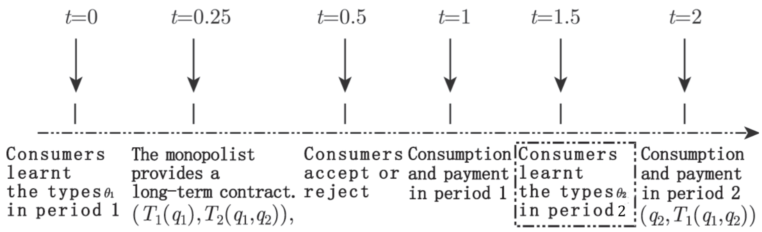

Figure 20.2: Timing of contracting with independent types

The principal's optimization problem at date-0 is:

$$\max_{\{q_i, U_i\}_{i=1}^2} \Pi, s.t. : IR_1(\theta_1), IC_1(\theta_1), IR_2(\theta_2|\theta_1), IC_2(\theta_2|\theta_1),$$

where

$$\begin{aligned}\Pi &\equiv \mathbb{E}_{\theta_1,\theta_2}\left\{\theta_1 v(q_1(\theta_1)) - cq_1(\theta_1) - U_1(\theta_1) + \delta\left[\theta_2 v(q_2(\theta_1,\theta_2)) - cq_2(\theta_1,\theta_2)\right]\right\}\\ &= \beta[\theta_L v(q_{1L}) - cq_{1L} - U_{1L}] + (1-\beta)[\theta_H v(q_{1H}) - cq_{1H} - U_{1H}]\\ &\quad +\delta\begin{bmatrix}\beta^2[\theta_L v(q_{2L}(\theta_L)) - cq_{2L}(\theta_L)]\\ +\beta(1-\beta)[\theta_H v(q_{2H}(\theta_L)) - cq_{2H}(\theta_L)]\\ +\beta(1-\beta)[\theta_L v(q_{2L}(\theta_H)) - cq_{2L}(\theta_H)]\\ +(1-\beta)^2[\theta_H v(q_{2H}(\theta_H)) - cq_{2H}(\theta_H)]\end{bmatrix},\end{aligned} \tag{20.2.7}$$

$$IR_2(\theta_2|\theta_1): \quad U_2(\theta_1,\theta_2) \equiv \theta_2 v(q_2(\theta_1,\theta_2)) - T_2(\theta_1,\theta_2) \geqq 0, \forall \theta_1, \theta_2,$$

$$IR_1(\theta_1): \quad U_1(\theta_1) \equiv \left[\begin{array}{c} \theta_1 v(q_1(\theta_1)) - T_1(\theta_1) \\ +\delta \mathbb{E}_{\theta_2}\left[\theta_2 v(q_2(\theta_1,\theta_2)) - T_2(\theta_1,\theta_2)\right] \end{array}\right] \geqq 0, \forall \theta_1,$$

$$IC_2(\theta_2|\theta_1): \quad U_2(\theta_1,\theta_2) \geqq \theta_2 v(q_2(\theta_1,\hat{\theta}_2)) - T_2(\theta_1,\hat{\theta}_2), \forall \theta_1, \theta_2, \hat{\theta}_2,$$

$$IC_1(\theta_1): \quad U_1(\theta_1) \geqq \left[\begin{array}{c} \theta_1 v(q_1(\hat{\theta}_1)) - T_1(\hat{\theta}_1) \\ +\delta \mathbb{E}_{\theta_2}\left[\theta_2 v(q_2(\hat{\theta}_1,\theta_2)) - T_2(\hat{\theta}_1,\theta_2)\right] \end{array}\right], \forall \theta_1, \hat{\theta}_1.$$

Let $\tilde{T}_1(\theta_1) \equiv T_1(\theta_1) + \delta \mathbb{E}_{\theta_2}\left[\theta_2 v(q_2(\theta_1,\theta_2)) - T_2(\theta_1,\theta_2)\right]$. Then, $IR(\theta_1)$ and $IC(\theta_1)$ take the following forms:

$$IC_{1L}: U_{1L} = \theta_L v(q_{1L}) - \tilde{T}_{1L} \geqq 0 \quad (20.2.8)$$

$$IC_{1H}: U_{1H} = \theta_H v(q_{1H}) - \tilde{T}_{1H} \geqq 0 \quad (20.2.9)$$

$$IR_{1L}: U_{1L} \geqq U_{1H} - \Delta v(q_{1H}) \quad (20.2.10)$$

$$IR_{1H}: U_{1H} \geqq U_{1L} + \Delta v(q_{1L}) \quad (20.2.11)$$

Analogous to the static screening model, we have binding IR_{1L} and IC_{1H}, and will check the ex-post implementability condition $q_{1H} \geqq q_{1L}$. Since $U_{2L}(\theta_1)$ and $U_{2H}(\theta_1)$ do not enter the principal's objective, they can then be chosen arbitrarily to meet $IC_{2L}(\theta_1)$ and $IC_{2H}(\theta_1)$, respectively, for any θ_1.

Substituting $U_{1H} = \Delta\theta v(q_{1L})$, $U_{1L} = 0$ into (20.2.7), then optimizing with respect to quantities yields:

$$q_{1H} = q_H^{SB} = q_H^{FB},$$

$$q_{1L} = q_L^{SB} < q_L^{FB},$$

$$q_{2H}(\theta_1) = q_H^{FB},$$

and

$$q_{2L}(\theta_1) = q_L^{FB}$$

for all θ_1, with

$$\theta_H v'(q_H^{FB}) = \theta_L v'(q_L^{FB}) = \left(\theta_L - \frac{1-\beta}{\beta}\Delta\theta\right) v'(q_L^{SB}) = c. \quad (20.2.12)$$

The life-long information rents are therefore: $U_{1L} = 0$ and $U_{1H} = \Delta\theta v(q_{1L}^{SB})$. The information rents of the second period $U_{2L}(\theta_1)$ and $U_{2H}(\theta_1)$ are, however, not uniquely identified. Without loss of generality, we can put $U_{2H}(\theta_1) = \Delta\theta v(q_L^{FB})$ and $U_{2L}(\theta_1) = 0, \forall \theta_1$.

Since types are independent across periods, the agent has no private information about his own future type beyond the initial period when the contract is signed. The quantities offered at $t = 2$ are therefore first-best because there is no asymmetric information between the seller and buyers.

This amounts to a screening problem with an ex-ante participation constraint. However, the quantities at period $t = 1$ are not first-best because at that point, the agent is privately informed about his own type. The traditional rent extraction versus efficiency tradeoff thus leads to an inefficient allocation.

20.2.3 Correlated Types Over Time

Now, we consider the more general case in which the agents' types are imperfectly correlated over time. In this case, there are some new features of the second-best dynamic contract. The problem was initially studied by Baron and Besanko (1984), in which they derived the second-best contracts with correlated types over time and full commitment of the principal.

We still keep the assumption of full commitment, and study the long-term contract in the monopoly economy of intertemporal price discrimination. In this case, the agent learns his first-period type $\theta_1 \in \{\theta_H, \theta_L\}$, $prob(\theta_1 = \theta_L) = \beta$. The principal only knows the distribution of types. The agent's second-period type is imperfectly correlated to the first-period one, and we assume that their correlation is:

$$\beta_i = prob(\theta_2 = \theta_H | \theta_1 = \theta_i) \text{ for } i \in \{H, L\}.$$

We also assume that $\beta_H \geqq \beta_L$, which means that the intertemporal correlation is positive, i.e., when type at period 1 is a high-type, the probability that it is a high-type at period 2 is higher than the probability that it is a low-type. When $\beta_H = 1 > \beta_L = 0$, it is equivalent to the case of constant types; when $\beta_H = \beta_L = \beta$, it is equivalent to the case of independent types.

The timing of the contract is the same as in Figure (20.2). In this framework, a direct revelation mechanism requires that the agent report the new information in each period he has learned about his current type. Typically, at date 0, the agent learns his period-1 type as θ_1; at date 0.25, the principal offers a full-commitment contract, denoted by $\{(T_1(\tilde{\theta}_1), q_1(\tilde{\theta}_1)), (T_2(\tilde{\theta}_1, \tilde{\theta}_2), q_2(\tilde{\theta}_1, \tilde{\theta}_2))\}$; at date 0.5, the agent decides whether or not to accept the contract; and if the agent accepts, at date 1, he reports his period-1 type, denoted by $\tilde{\theta}_1$, to the principal and the first period transaction is implemented; at date 1.5, the agent learns his period-2 type as θ_2; the agent reports his period-2 type, denoted by $\tilde{\theta}_2$, to the principal and the second period transaction is implemented.

The first-period report by the agent regarding his type can now be used by the principal to update his beliefs about the agent's second period type. This report can be viewed as an informative message that is useful for improving the second-period contract. The difference is that now the message utilized by the principal to improve the second-period contract is not exogenously-given by nature, but rather comes from the first-period report

$\tilde{\theta}_1$ of the agent on his type θ_1. Therefore, this message can be strategically manipulated by the agent in the first period in order to increase his second-period rent.

We assume that $v(\cdot)$ is increasing and concave. The agent has a quasi-linear per-period utility function $v(q_t) - T_t$. The agent's utility of the second period is $U_2(\theta_1, \theta_2) \equiv \theta_2 v(q_2(\theta_1, \theta_2)) - T_2(\theta_1, \theta_2)$; his continuation utility of the first period is $U_1(\theta_1) \equiv \theta_1 v(q_1(\theta_1)) - T_1(\theta_1) + \delta \mathbb{E}[U_2(\theta_1, \theta_2)|\theta_1]$. Assume that the principal's unit cost is c, and the common discount rate of the principal and agent is δ. The principal's problem is to maximize his two-period discounted profits subject to IR and IC constraints.

The principal's discount profit is represented as

$$\begin{aligned}
\Pi &\equiv \mathbb{E}\left\{T_1(\theta_1) - cq_1(\theta_1) + \delta\left[T_2(\theta_1,\theta_2) - cq_1(\theta_1,\theta_2)\right]\right\} \\
&= \mathbb{E}\left\{\theta_1 v(q_1(\theta_1)) - cq_1(\theta_1) + \delta\left[\theta_2 v(q_2(\theta_1,\theta_2)) - cq_2(\theta_1,\theta_2)\right] - U_1(\theta_1)\right\} \\
&= \beta[\theta_L v(q_1(\theta_L)) - cq_1(\theta_L) - U_1(\theta_L)] + (1-\beta)[\theta_H v(q_1(\theta_H)) - cq_1(\theta_H) - U_1(\theta_H)] \\
&\quad +\delta\begin{bmatrix} \beta(1-\beta_L)[\theta_L v(q_2(\theta_L,\theta_L)) - cq_2(\theta_L,\theta_L)] \\ +\beta\beta_L[\theta_H v(q_2(\theta_L,\theta_H)) - cq_2(\theta_L,\theta_H)] \\ +(1-\beta)(1-\beta_H)[\theta_L v(q_2(\theta_H,\theta_L)) - cq_2(\theta_H,\theta_L)] \\ +(1-\beta)\beta_H[\theta_L v(q_2(\theta_H,\theta_H)) - cq_2(\theta_H,\theta_H)] \end{bmatrix}. \qquad (20.2.13)
\end{aligned}$$

The following IR and IC constraints need to be satisfied:

$$\begin{aligned}
&IC_R(\theta_2|\theta_1) : U_2(\theta_1,\theta_2) \equiv \theta_2 v(q_2(\theta_1,\theta_2)) - T_2(\theta_1,\theta_2) \geqq 0, \\
&IC_2(\theta_2|\theta_1) : U_2(\theta_1,\theta_2) \geqq \theta_2 v(q_2(\theta_1,\hat{\theta}_2)) - T_2(\theta_1,\hat{\theta}_2), \\
&IR_1(\theta_L) : U_1(\theta_L) \equiv \begin{bmatrix} \theta_L v(q_1(\theta_L)) - T_1(\theta_L) \\ +\delta\left[\beta_L U_2(\theta_L,\theta_H) + (1-\beta_L)U_2(\theta_L,\theta_L)\right] \end{bmatrix} \geqq 0, \\
&IR_1(\theta_H) : U_1(\theta_H) \equiv \begin{bmatrix} \theta_H v(q_1(\theta_H)) - T_1(\theta_H) \\ +\delta\left[\beta_H U_2(\theta_H,\theta_H) + (1-\beta_H)U_2(\theta_H,\theta_L)\right] \end{bmatrix} \geqq 0, \\
&IC_1(\theta_L) : U_1(\theta_L) \geqq \begin{bmatrix} \theta_L v(q_1(\theta_H)) - T_1(\theta_H) \\ +\delta\left[\beta_L U_2(\theta_H,\theta_H) + (1-\beta_L)U_2(\theta_H,\theta_L)\right] \end{bmatrix}, \\
&IC_1(\theta_H) : U_1(\theta_H) \geqq \begin{bmatrix} \theta_H v(q_1(\theta_L)) - T_1(\theta_L) \\ +\delta\left[\beta_H U_2(\theta_L,\theta_H) + (1-\beta_H)U_2(\theta_L,\theta_L)\right] \end{bmatrix}.
\end{aligned}$$

Applying the standard technique, we can identify that $IR_2(\theta_L|\theta_1)$, $IC_2(\theta_H|\theta_1)$, $IR_1(\theta_L)$ and $IC_1(\theta_H)$ are binding $\forall \theta_1 \in \{\theta_L, \theta_H\}$; and the implementability conditions

$$IM_2 : \quad q_2(\theta_1,\theta_H) \geqq q_2(\theta_1,\theta_L), \theta_1 \in \{\theta_L,\theta_H\} \qquad (20.2.14)$$

$$IM_1 : \quad \begin{bmatrix} [v(q_1(\theta_H)) - v(q_1(\theta_L))]+ \\ \delta\Delta\beta[v(q_2(\theta_H,\theta_L)) - v(q_2(\theta_L,\theta_L))] \end{bmatrix} \geqq 0. \qquad (20.2.15)$$

All incentive compatibility constraints can be represented in Figure 20.3.

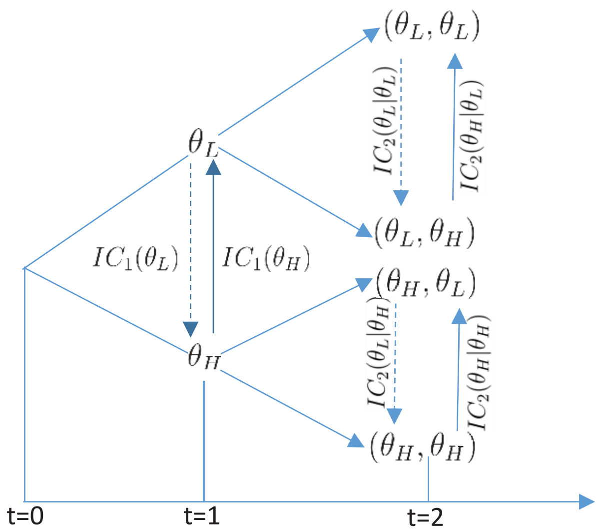

Figure 20.3: Incentive constraints with correlated types (solid lines denote binding constraints, and dashed lines denote slack constraints)

We thus have

$$\begin{aligned} U_2(\theta_1, \theta_L) &= 0, \forall \theta_1 \in \{\theta_L, \theta_H\}, && (20.2.16) \\ U_2(\theta_1, \theta_H) &= \Delta\theta v(q_2(\theta_1, \theta_L)), \forall \theta_1 \in \{\theta_L, \theta_H\}, && (20.2.17) \\ U_1(\theta_L) &= 0, && (20.2.18) \\ U_1(\theta_H) &= \Delta\theta v(q_1(\theta_L)) + \delta\Delta\theta\Delta\beta v(q_2(\theta_L, \theta_L)). && (20.2.19) \end{aligned}$$

Inserting (20.2.16) to (20.2.19) into (20.2.13), and then optimizing with respect to $q_1(\theta_1), q_2(\theta_1, \theta_2)$, yields:

$$\theta_H v'(q_1(\theta_H)) = c \quad (20.2.20)$$

$$\left[\theta_L - \frac{(1-\beta)\Delta\theta}{\beta}\right] v'(q_1(\theta_L)) = c \quad (20.2.21)$$

$$\left[\theta_L - \frac{\Delta\theta(1-\beta)(\beta_H - \beta_L)}{\beta(1-\beta_L)}\right] v'(q_2(\theta_L, \theta_L)) = c \quad (20.2.22)$$

$$\theta_L v'(q_2(\theta_H, \theta_L)) = c \quad (20.2.23)$$

$$\theta_H v'(q_2(\theta_L, \theta_H)) = c \quad (20.2.24)$$

$$\theta_H v'(q_2(\theta_H, \theta_H)) = c, \quad (20.2.25)$$

which means that the agent's consumptions with θ_H-type in both periods or with θ_H-type in the first period and θ_L-type in the second period or with θ_L-type in the first period and θ_H-type in the second period are efficient (first-best); whereas, the agent's consumptions with θ_L-type in both periods are inefficient and have a downward distortion.

It is easy to verify that implementability conditions (20.2.14) and (20.2.15) are satisfied.

Note that when $\beta_H = 1 > \beta_L = 0$, which is the case of constant types, (20.2.22) reduces to

$$\left(1 - \frac{\Delta\theta}{\theta_L}\frac{1-\beta}{\beta})\theta_L v'(q_2(\theta_L, \theta_L)\right) = c,$$

and then we obtain the second-best solution for $q_2(\theta_L, \theta_L) = q_1(\theta_L) = q_L^{SB}$, which is the same as the case of constant types.

While $\beta_H = \beta_L = \beta$, it is equivalent to the case of independent types, and (20.2.22) reduces to

$$\theta_L v'(q_2(\theta_L, \theta_L)) = c,$$

and then we obtain the first-best solution for $q_2(\theta_L, \theta_L) = q_L^{FB}$, which is the same as the case of independent types.

20.3 Dynamic Contracts under Different Degrees of Commitment Power

Now, we explore how dynamic interaction affects the agent's incentives when the principal exhibits different degrees of commitment power. If the principal does not have full commitment power, the agent may have no incentive to reveal his type at an early time, since he may use the information to increase his information rent at a later date. As such, the agent's incentive constraints in a dynamic setting may be different from those in a static setting.

There are manifold such examples in real-world economies. In the regulation case, the regulated price is dependent on the cost reported by a regulated monopolist. If the cost information is to be employed in the future regulation, a phenomenon of **rachet effect** arises, which destroys the monopolist's incentive to report the true information in the first place. This problem is analyzed by Freixas, Guesnerie and Tirole (1985).

For optimal contract design under adverse selection, there is a basic tradeoff between information rent extraction and allocative efficiency, i.e., the principal can distort the allocation of one type to reduce the information rent accruing to other types. In the dynamic setting, if information of the agent's type is completely revealed, then the principal and the agent of some types have the incentive to reduce, and even eliminate, the allocative distortion mentioned above, which, in turn, will change the incentive constraints confronted by the other types of agent. This situation is called *dynamic renegotiation* in the literature.

There are two types of dynamic contracts, i.e., spot contract and long-term contract. The implementability of a long-term contract depends on the commitment power of the principal, which can be divided into three cases. First, the principal has full commitment power, and as long as the contract is signed, it shall not be changed in the future. Second, the principal has no commitment power, and he can unilaterally change the contract in the future. Third, the principal has partial commitment power, which means that any future change of the contract must be agreed upon by the agent. In practice, all of the three cases occur, and the principal's commitment power relies on the underlying environment, such as legal constraints and political institutions.

If the principal has no commitment power, the dynamic principal-agent problem is, in general, very complicated. For simplicity, we utilize examples to characterize the new incentive issues facing the principal in designing long-term contracts. We discuss dynamic contracting in a monopoly selling. Hart and Tirole (1988) analyzed this problem in detail, and we adopt the simplified version from Segal (2010).

For this purpose, we consider a principal-agent relationship that is repeated over time, over a finite or an infinite horizon. First, as a benchmark, we assume that the agent's type θ is realized ex-ante and remains constant over time. For simplicity, we focus on a two-period principal-agent model with a constant type and stationary payoffs. We let $t = 1$ represent today, $t = 2$ represent the future, and the common discount rate be δ. We will allow δ to be smaller or greater than one, and the latter case can be interpreted as capturing situations in which the second period is very long.

The agent (consumer) has his private evaluation about the product value, denoted by θ. The agent's type θ is realized before the relationship starts and is the same in both periods. Assume that the type space is $\{\theta_H, \theta_L\}, \theta_H > \theta_L > 0$, and the probability of θ_H is β.

For simplicity, we assume that the consumer has a unit demand of the good. Denote $x_{it} \in \{0, 1\}$ as the purchasing decision of type θ_i at time t. Suppose that the production cost is normalized to 0, and p_t is the uniform price at time t.

The total discounted utility of the type-i agent is:

$$U(x_{i1}, x_{i2}) = \sum_{t=1}^{2} \delta^{t-1} x_{it}(\theta_i - p_t).$$

The total discounted profit of the principal is:

$$\pi(p_1, p_2) = \sum_{t=1}^{2} \delta^{t-1} p_t[\beta x_{H,t} + (1 - \beta) x_{L,t}],$$

where $x_{H,t}$ and $x_{L,t}$ are the purchasing decisions of types θ_H and θ_L, respectively, at time t.

20.3.1 Contracting with Full Commitment

As a benchmark, we first discuss the static case, i.e., the one-period optimal contract. The monopolist (principal) is assumed to interact with a single consumer (agent).

Note that if $p \leqq \theta_L$, then $x_H = x_L = 1$ and the profit is p; if $\theta_L < p \leq \theta_H$, then $x_H = 1, x_L = 0$ and the profit is βp; if $p > \theta_H$, then $x_H = x_L = 0$ and the profit is 0. Let $\bar{\beta} = \frac{\theta_L}{\theta_H}$. Then, the principal's profit-maximizing price is given as follows:

$$p^*(\beta) = \begin{cases} \theta_H & \text{if } \beta > \bar{\beta}, \\ \theta_L & \text{if } \beta < \bar{\beta}, \\ \theta_H \text{ or } \theta_L & \text{if } \beta = \bar{\beta}. \end{cases}$$

The agent's optimal consumption choice is given as follows:

$$(x_H^*, x_L^*) = \begin{cases} (1,0) & \text{if } \beta > \bar{\beta}, \\ (1,1) & \text{if } \beta < \bar{\beta}, \\ (1,0) \text{ or } (1,1) & \text{if } \beta = \bar{\beta}. \end{cases}$$

The principal's one-period profit is:

$$\pi_1^*(\beta) = \max\{\beta\theta_H, \theta_L\}.$$

If $\beta < \bar{\beta}$, the utility of type θ_H is $U(\theta_H) = \theta_H - \theta_L$, which is also the information rent of type θ_H.

Now, suppose that there are two periods. We already know that, for constant types over time, any two-period contract can be replaced with a repetition of the same static contract. Therefore, the optimal commitment contract sets the same price, i.e., $p_t^* = p^*, t = 1, 2$, and the agent's consumption choice is $x_i^*(t) = x_i^*$ for $i \in \{H, L\}$. The principal's profit is thus $\pi^*(\beta) = (1 + \delta)\pi_1^*(\beta)$, which is the biggest profit that he can earn.

20.3.2 Dynamic Contract without Commitment

If the principal has no commitment power, i.e., he can change the contract unilaterally, will the contract above be implemented? Or, will he want to modify the price at the second period? These answers depend on $p^*(\beta)$.

If $\beta \leqq \bar{\beta}$, $p_1^*(\beta) = \theta_L$ and then both types will purchase. Thus monopolists cannot distinguish the types of the consumers, so there is no belief updating on consumer′ types. Therefore, in the second period, the monopolist will still abide by the contract, which is the price of profit maximization under the initial belief.

However, when $\beta > \bar{\beta}$, the monopolist may have an incentive to revise the contract unilaterally. Indeed, if the consumer does not purchase the good at the first period, i.e., $x_1 = 0$, the monopolist knows the consumer is of type θ_L, and then he will modify the contract so that $p_2^*(\beta) = \theta_L$. Once the principal knows following a purchase that he deals with a high-type and following no purchase that he deals with a low type, he wants to restore efficiency for the low type by setting the price to θ_L in the second period. Expecting this price reduction following no purchase, the high-type will not buy in the first period, and then the contract is not implementable. Thereby, when the principal deals optimally with the renegotiation problem, the ratchet problem analyzed by Freixas, Guesnerie and Tirole (1985) may arise and further harm the principal. As result, there are only short-term contracts in this situation.

In the following, we suppose that $\beta > \bar{\beta}$. Let the parties play the following two-period game: at the first period, the principal offers price p_1 and the agent chooses $x_1 \in \{0, 1\}$; at the second period, the principal offers price p_2 and the agent chooses $x_2 \in \{0, 1\}$.

We assume that the solution concept used is **(weak) perfect Bayesian equilibrium** (PBE), i.e., Bayes' rule is used to update the belief on the equilibrium path, and the posterior belief is arbitrary outside of the equilibrium path. So, we do this by considering Bayesian-Nash equilibria for every continuation game at different first-period price p_1, from which we need to determine one that is sequentially rational. As such, the principal's equilibrium price p_1^* can be determined. The principal's strategy in the continuation game is a function $p_2(x_1)$, which sets the second-period price following the agent's first-period choice x_1. Let $\hat{\beta}(x_1) = prob(\theta_H | x_1)$ be the principal's posterior belief that the agent is a high-type after x_1 is chosen.

In the following, we analyze three possible equilibrium types of PBE: "**full revelation**" equilibrium; "**no revelation**" equilibrium; and "**partial revelation**" equilibrium.

"Full Revelation" Equilibrium without Commitment

Under the first-period price p_1, there is only one possible "full revelation" (or separating) equilibrium: $x_H = 1$ and $x_L = 0$. The principal's posterior belief is given by

$$\hat{\beta}(x_1) = \begin{cases} 1 & \text{if } x_1 = 1, \\ 0 & \text{if } x_1 = 0. \end{cases}$$

According to sequential rationality, the principal's second-period price is:

$$p_2(x_1) = \begin{cases} \theta_H & \text{if } x_1 = 1, \\ \theta_L & \text{if } x_1 = 0. \end{cases}$$

Therefore, if such equilibrium exists, the incentive compatibility constraints for the types $\{\theta_H, \theta_L\}$ should hold:

$$IC_H: \quad \underbrace{\theta_H - p_1}_{\text{buy at } t=1} + \underbrace{\delta(\theta_H - \theta_H)}_{\text{no rent at } t=2} \geqq \underbrace{0}_{\text{don't buy at } t=1} + \underbrace{\delta(\theta_H - \theta_L)}_{\text{rent at } t=2}, \quad (20.3.26)$$

$$IC_L: \quad \underbrace{0}_{\text{don't buy at } t=1} + \underbrace{\delta(\theta_L - \theta_L)}_{\text{no rent at } t=2} \geqq \underbrace{\theta_L - p_1}_{\text{buy at } t=1} + \underbrace{0}_{\text{don't buy at } t=2}. \quad (20.3.27)$$

In the incentive compatibility constraint (20.3.26) for the type θ_H, the left hand is his utility of first-period consumption, and the right hand is his utility of second-period consumption (with second-period price $p_2 = \theta_L$). Without consumption in the first period, the inequality (20.3.26) can be rewritten as:

$$p_1 \leqq (1-\delta)\theta_H + \delta\theta_L. \quad (20.3.28)$$

In the incentive compatibility inequality constraint (20.3.27) for the type θ_L, the right hand is his utility of first-period consumption and no consumption in the second period, and the left hand is his utility of no consumption in the first period (with second-period price $p_2 = \theta_L$). This is because, in a short-term contract, there is no such constraint that the consumer must have consumption in the first period. The inequality (20.3.27) can be rewritten as

$$p_1 \geqq \theta_L. \quad (20.3.29)$$

From the inequalities (20.3.28) and (20.3.29), we have $\delta \leqq 1$. Consequently, *the necessary condition for the existence of "full revelation" equilibrium is that $\delta \leqq 1$.* If $\delta > 1$, then the incentive compatibility inequality (20.3.28) for type θ_H means $p_1 \leqq \theta_L$. Then, both types of θ_H and θ_L choose to buy at the first-period, and for the type θ_L, he will not buy at the second-period (since $p_2 = \theta_H$) and is of the type of "take the money and run".

Therefore, when $\delta \leqq 1$ holds, "full revelation" equilibrium exists. The principal will set the highest possible first-period price $p_1^R = (1-\delta)\theta_H + \delta\theta_L$, and the principal's profit is

$$\pi^R = \beta(p_1^R + \delta\theta_H) + (1-\beta)\delta\theta_L = \beta\theta_H + \delta\theta_L,$$

and the information rent for type θ_H is $\delta(\theta_H - \theta_L)$.

"No Revelation" Equilibrium without Commitment

In the "no revelation" (or pooling) equilibrium, $x_{H1} = x_{L1}$. Then, $\hat{\beta}(x_1) = \beta > \bar{\beta}$, and $p_2 = \theta_H$. Although there are two possible "no revelation" equilibria, i.e., $x_{H1} = x_{L1} = 0$ and $x_{H1} = x_{L1} = 1$, by sequential rationality, for the principal, the optimal (profit maximizing) "no revelation" equilibrium is $x_{H1} = x_{L1} = 1$. Therefore, *the necessary condition for the existence of such* *equilibrium is that the participation constraint of type θ_L should be binding, and the first-period price is $p_1 = \theta_L$.*

The principal's profit is

$$\pi^p = \theta_L + \delta\beta\theta_H,$$

and the information rent for type θ_H is $\theta_H - \theta_L$.

Now, we compare π^R and π^p. For $\delta \leqq 1$, by

$$\pi^R - \pi^p = (\beta\theta_H - \theta_L)(1-\delta) \geqq 0,$$

we have $\pi^R \geqq \pi^p$.

"Partial Revelation" Equilibrium without Commitment

Let ρ_i be the probability of the purchase for type θ_i under the first-period price p_1. Since type θ_H has more of an incentive to purchase than does type θ_L, it must be true that $\rho_H > \rho_L$, and then the posterior belief is

$$\hat{\beta}(x_1 = 1) = \frac{\beta\rho_H}{\beta\rho_H + (1-\beta)\rho_L} \geqq \beta > \bar{\beta}.$$

Therefore, $p_2(x_1 = 1) = \theta_H$. There are two possibilities for the second-period price:

1. $\hat{\beta}(x_1 = 0) \leqq \bar{\beta}$. Then, $p_2(1) = \theta_H$ and $p_2(0) = \theta_L$. By $\rho_H > \rho_L \geqq 0$, we have

$$\theta_H - p_1 \geqq \delta(\theta_H - \theta_L),$$

which implies that $p_1 \leqq (1-\delta)\theta_H + \delta\theta_L$.

If $\delta > 1$, both types have an incentive to buy in period 1, and then $\rho_H = \rho_L = 1$, which is not "partial revelation".

If $\delta \leqq 1$, the principal has two options:

(a) $p_1 = (1-\delta)\theta_H + \delta\theta_L$. Then, $\rho_H \in (0, 1]$, $\rho_L = 0$, and his profit is

$$\pi^S = \rho_H\beta[(1-\delta)\theta_H + \delta\theta_L] + \delta[\beta\theta_H + (1-\beta)\theta_L],$$

which is the same as the "full revelation" case, i.e., $\pi^S = \pi^R$;

(b) $p_1 = \theta_L$. Then, $\rho_H = 1, \rho_L \in (0, 1)$, and his profit is $\pi^S = \pi^p$.

Since $\pi^R > \pi^p$, the principal chooses $p_1 = (1-\delta)\theta_H + \delta\theta_L$, which is the same as that in the "full revelation" case.

2. $\hat{\beta}(x_1 = 0) > \bar{\beta}$. Then, $p_2(x_1 = 1) = p_2(x_1 = 0) = \theta_H$. Since $\hat{\beta}(x_1 = 0) = \frac{\beta(1-\rho_H)}{\beta(1-\rho_H)+(1-\beta)(1-\rho_L)} > 0$, then $\rho_H < 1$, and thus $p_1 \geqq \theta_H$. Indeed, if

$p_1 < \theta_H$, then $\rho_H = 1$, and thus $\rho_L = 0$. The first-period price must be $p_1 = \theta_H$. Due to

$$\hat{\beta}(x_1 = 0) = \frac{\beta(1-\rho_H)}{\beta(1-\rho_H) + (1-\beta)} \geqq \bar{\beta},$$

we have

$$\rho_H \leqq \hat{\rho}_H = \frac{\beta\theta_H - \theta_L}{\beta(\theta_H - \theta_L)},$$

and thus the principal's profit is

$$\pi^S = \beta\theta_H[\hat{\rho}_H + \delta].$$

If $\delta \leqq 1$, then $\pi^R \geqq \pi^p$. We now compare π^R and π^S.

$$\begin{aligned}
\pi^S - \pi^R &= \beta\theta_H\hat{\rho}_H + \beta\theta_H\delta - \beta\theta_H - \delta\theta_L] \\
&= \beta\theta_H\hat{\rho}_H + \delta(\beta\theta_H - \theta_L) - \beta\theta_H \\
&= \theta_H\frac{\beta\theta_H - \theta_L}{\theta_H - \theta_L} + \delta(\beta\theta_H - \theta_L) - \beta\theta_H \\
&= (\beta\theta_H - \theta_L)[\frac{\theta_H}{\theta_H - \theta_L} + \delta] - \beta\theta_H \\
&= \beta\theta_H[\frac{\theta_H}{\theta_H - \theta_L} + \delta - 1] - \theta_L[\frac{\theta_H}{\theta_H - \theta_L} + \delta].
\end{aligned}$$

Therefore, $\pi^S \geqq \pi^R$ is equivalent to

$$\beta \geqq \beta^{RS} \equiv \frac{\theta_L}{\theta_H}\frac{\theta_H + \delta(\theta_H - \theta_L)}{\theta_L + \delta(\theta_H - \theta_L)}.$$

Now, we examine the principal's contract choice under $\delta > 1$. Since there is no "full revelation" equilibrium, it is necessary to compare π^p and π^S.

$$\begin{aligned}
\pi^S - \pi^p &= \beta\theta_H\hat{\rho}_H + \beta\theta_H\delta - \theta_L - \delta\beta\theta_H] \\
&= \beta\theta_H\hat{\rho}_H - \theta_L \\
&= \frac{\beta\theta_H - \theta_L}{\theta_H - \theta_L}\theta_H - \theta_L \\
&= \frac{\theta_H}{\theta_H - \theta_L}[\beta - \frac{\theta_L}{\theta_H}(2 - \frac{\theta_L}{\theta_H}).]
\end{aligned}$$

Thus, $\pi^S \geqq \pi^p$ is equivalent to

$$\beta \geqq \beta^{PS} \equiv \frac{\theta_L}{\theta_H}(2 - \frac{\theta_L}{\theta_H}).$$

Summarizing the above discussion, we have the following proposition:

Proposition 20.3.1 *In the lack of commitment, under the solution concept of perfect Bayesian equilibrium, the optimal contract for the principal entails:*

If $\beta \leqq \bar{\beta}$, the principal chooses the same contract as under the full commitment, and it can be implemented;

If $\delta \leqq 1$ and $\beta \in (\bar{\beta}, \beta^{RS}]$, the principal chooses the "full revelation" equilibrium, and the optimal contract is $p_1 = (1-\delta)\theta_H + \delta\theta_L$, $p_2(x_1 = 1) = \theta_H$, and $p_2(x_1 = 0) = \theta_L$;

If $\delta \leqq 1$ and $\beta > \beta^{RS}$, the principal chooses the "partial revelation" equilibrium, and the optimal contract is $p_1 = \theta_H$ and $p_2(x_1 = 1) = p_2(x_1 = 0) = \theta_H$;

If $\delta > 1$ and $\beta \in (\bar{\beta}, \beta^{PS}]$, the principal chooses the "no revelation" equilibrium, and the optimal contract is $p_1 = \theta_L$ and $p_2(x_1 = 1) = p_2(x_1 = 0) = \theta_H$;

If $\delta > 1$ and $\beta > \beta^{PS}$, the principal chooses the "partial revelation" equilibrium, and the optimal contract is $p_1 = \theta_H$ and $p_2(x_1 = 1) = p_2(x_1 = 0) = \theta_H$.

20.3.3 Dynamic Contracts with Partial Commitment

Next, we turn to the case of partial commitment, in which the principal can revise the contract only with the consent of the agent. The timing of this dynamic contracting is the following: (1) in the first period, the principal offers a contract $(p_1; p_2(x_1))$; (2) the agent then chooses x_1; (3) the principal offers a new contract $p_2'(x_1)$ in the second period; (4) the agent then accepts $p_2'(x_1)$, or rejects it and sticks to the original contract; (5) the agent chooses the second-period decision x_2. If $p_2' = p_2$, the principal adopts the original contract.

We will look for the principal's PBE of this contract. Note that a renegotiation offer $p_2'(x_1)$ following the agent's choice x_1 will be accepted by the agent if and only if $p_2'(x_1) < p_2(x_1)$. Therefore, a long-term renegotiable contract commits the principal against raising the price, but does not commit his against lowering the price. Analysis is simplified by observing that the principal can, without loss of generality, offer a contract that is not renegotiated in equilibrium.

We first define the renegotiation-proof contract.

Definition 20.3.1 A **renegotiation-proof (RNP) contract** is one that is not renegotiated in PBE.

The following principle can simplify the analysis of the above problem.

Proposition 20.3.2 (Renegotiation-Proof Principle) *In the above problem, for any PBE in the contract with renegotiation, there exists another PBE that implements the same outcome and in which the principal offers a RNP contract.*

PROOF. In the PBE, let $(p_1; p_2(1), p_2(0))$ be the initial contract. If the principal offers $(p_1; p_2(1), p_2(0))$ and in equilibrium renegotiates to $p_2'(x_1) < p_2(x_1)$ after the agent's choice of $x_1 \in \{0, 1\}$, then the contract $(p_1; p_2'(1), p_2'(0))$ is RNP. □

Dewatripont (1989) discussed the Renegotiation-Proof Principle in detail. The RNP Principle is similar to the Revelation Principle, i.e., for any equilibrium, one can always find a new equilibrium that has the same equilibrium outcome. By the RNP Principle, the optimal dynamic contract can be implemented by a RNP contract. As such, we can restrict our attention to RNP mechanisms.

When $\beta \leqq \bar{\beta}$, with full commitment power, all consumers will purchase at date 1, and thus the consumer's action will not modify the principal's belief. As such, a contract with full commitment is always implementable even though the principal may modify the contract without the consent of the agent, and of course it is implementable with the consent of the agent. As such, we only need to consider the case of $\beta > \bar{\beta}$.

Firstly, the contract with full commitment is not a RNP contract. This is because, under such a contract, the principal offers $p_1 = p_2(1) = p_2(0) = \theta_H$, and the agent chooses $x_1(\theta_H) = x_2(\theta_H) = 1, x_1(\theta_L) = x_2(\theta_L) = 0$. However, with $x_1 = 0$ being observed, the consumer can be identified as θ_L type. Then, $p_2(0) = \theta_L$ will be accepted by the consumer of type θ_L, and this will increase the principal's profit. As such, the principal has an incentive to modify the contract. However, he cannot do so without the consent of the agent.

Secondly, any short-term contract without commitment is implementable with the consent of the agent. This is because, if p_1^N and $(p_2(1)^N, p_2(0)^N)$ are two short-term contracts, it is clear that with x_1 being observed, the contract $(p_1^N; p_2(1)^N, p_2(0)^N)$ in the second period can be implemented. If the principal provides a new contract $(p_2'(1), p_2'(0))$ that can be accepted by the agent, we then must have $p_2'(1) < p_1^N(1)$ or $p_2'(0) < p_1^N(0)$. However, for the profit-maximizing dynamic contract, given x_1 and the principal's belief, $p_1^N(x_1)$ is an optimal choice. As such, the principal does not have an incentive to modify the original contract.

Thirdly, do long-term renegotiable contracts offer an advantage relative to short-term contracts? The answer is affirmative when the ability to commit to a low price is useful. Consider the following contract: $p_1 = \theta_H + \delta\theta_L; p_2(1) = 0, p_2(0) = \theta_L$. This contract provides a separating equilibrium under the consent of the agent. Indeed, θ_H-consumer's utility with purchase in the first period is

$$\theta_H - (\theta_H + \delta\theta_L) + \delta\theta_H = \delta(\theta_H - \theta_L) > 0,$$

and his utility without purchase in the first period is

$$\delta(\theta_H - \theta_L).$$

Then, the incentive compatibility and individual rationality constraints are satisfied for the type-θ_H agent.

As for type-θ_L agent, his utility with purchase in the first period is

$$\theta_L - (\theta_H + \delta\theta_L) + \delta\theta_L = -(\theta_H - \theta_L) < 0,$$

and his utility without purchase in the first period is

$$\delta(\theta_L - \theta_L) = 0.$$

Then, the incentive compatibility and individual-rationality constraints are satisfied for the type-θ_L agent.

In addition, the principal will not provide a new contract ($p_2'(1)$, $p_2'(0)$) for the second period. This is because, if the agent agrees, then $p_2'(1) < 0$ or $p_2'(0) < \theta_L$, which makes the principal worse off. Therefore, the contract with the consent of the agent is RNP.

However, the contract without the consent of the agent is not implementable. Indeed, once $x_1 = 1$ is observed, the consumer must be θ_H-type, and then the principal will choose $p_2(1) = \theta_H$. As such, the contract without commitment power is not implementable.

20.4 Sequential Screening

From the above discussion, in dynamic mechanism design, the agent may possess an information advantage of different degrees over time. This information advantage will turn into the information rent of the agent. In fact, the agent may need some time to learn his future consumption type, and has some private information of different degrees over time. Then, the principal can employ some mechanism to elicit the private information over time, such as dynamic contracting to screen agents over time. Courty and Li (2000) introduced sequential screening mechanisms to analyze dynamic principal-agent problems.

20.4.1 An Example of Sequential Screening

Consider the demand for airplane tickets. Travelers typically do not know their valuations for tickets until just prior to departure, but they know their likelihood to have high or low valuations in advance. A monopolist can wait until the travelers learn their valuations and charge the monopoly price, and thus more consumer surplus can be extracted by requiring them to reveal their private information sequentially. An illustration of such monopoly practice is a menu of refund contracts, each consisting of an advance payment and a refund amount in case the traveler decides not to use the ticket. By selecting a refund contract from the menu, travelers reveal their private information about the distribution of their valuations, and by deciding later whether they want the ticket or the specified refund, they reveal what they have learned about their actual valuations.

Suppose that at time $t = 0$, one-third of all potential buyers are leisure travelers (type L) whose valuation is uniformly distributed on $\theta_L \in [1, 2]$, and two-thirds are business travelers (type B) whose valuation is uniformly distributed on $\theta_B \in [0, 1] \cup [2, 3]$. Intuitively, business travelers face greater valuation uncertainty than do leisure travelers. Suppose that the cost of flying an additional traveler is 1. Suppose that the monopolist and travelers are all risk-neutral. At time $t = 1$, all travelers know their true valuations. The monopolist designs a contract at time $t = 0$.

We first consider the monopolist's pricing at $t = 1$. If the seller waits until travelers have privately learned their valuations, he faces a valuation distribution that is uniform on $[0, 3]$, i.e., the type distribution function is $F(x) = \frac{x}{3}$, and the problem of the monopolist is:

$$\max_p (p - 1)(1 - F(p)),$$

solving which gives the monopoly price as $p^m = 2$ and his profit as $\pi^m = \frac{1}{3}$.

Now, we consider the monopolist's contract at $t = 0$. Travelers only know their general types, i.e., leisure type or business type. Suppose that instead that the seller offers two contracts before the travelers learn their valuations: one with an advance payment of $p_l = 1.5$ and no refund (unchangeable), i.e., $r_l = 0$; and the other with an advance payment of $p_b = 1.75$ and a partial refund of $r_b = 1$ (changeable, but with a 0.75 cancellation fee). Leisure travelers strictly prefer the contract with no refund, (p_l, r_l). Business travelers are indifferent between the two contracts, and thus we assume that they choose the contract with the refund, (p_b, r_b).

For the leisure type, if choosing the contract (p_l, r_l), his expected utility is:

$$\int_1^2 (\theta_L - 1.5) d\theta_L = 0 > \int_1^2 (\theta_L - 1.75) d\theta_L.$$

For the business type, if choosing the contract (p_b, r_b), his expected utility is:

$$\begin{aligned}\frac{1}{2}\int_0^1 (1 - 1.75) d\theta_B + \frac{1}{2}\int_2^3 (\theta_B - 1.75) d\theta_B = 0 = \\ \frac{1}{2}\int_0^1 (\theta_B - 1.5) d\theta_B + \frac{1}{2}\int_2^3 (\theta_B - 1.5) d\theta_B.\end{aligned}$$

Therefore, such screening contracts satisfy the incentive compatibility and participation constraints, and the expected information rents for both types are 0. In such contracts, the monopolist's profit is:

$$1/3(1.5 - 1) + 2/3[\int_0^1 (1.75 - 1) d\theta_B + \int_2^3 (1.75 - 1) d\theta_B] = \frac{2}{3} > \pi^m.$$

At time $t = 1$, for the business type, if $\theta_B \in [0, 1]$, he will choose the refund, and if $\theta_B \in [2, 3]$, he will travel. Consequently, the monopolist can obtain 0.75 profit from each business traveler.

The above example reveals the basic idea of sequential screening. On the one hand, with time passing, the agent has more of an advantage in terms of information. To screen such information, the principal needs to give up more information rent, and the earlier screening is, the more reduction of the agent's information rent would be. On the other hand, with time passing, more information causes an increase in the efficiency of the outcome, and earlier screening results in a loss of efficiency. Therefore, in sequential screening, a tradeoff exists between allocative efficiency and information rent extraction over time.

There are numerous other examples of sequential mechanisms that take different forms, such as hotel reservations (cancellation fees), car rentals (free mileage vs. fixed allowance), telephone pricing (calling plans), public transportation (day passes), and utility pricing (optional tariffs). Sequential price discrimination can also play a role in contracting problems, such as taxation and procurement, in which the agent's private information is revealed sequentially.

Although sequential mechanisms can take various forms, we restrict our attention to situations in which consumers have unit demand, as in the airplane ticket pricing problem. In these situations, the optimal ex-post pricing scheme (after consumers have complete private information about their demands) degenerates to standard monopolist pricing, and the monopolist sequentially discriminates only by choosing the probability of delivering the good.

20.4.2 Sequential Screening under Incomplete Information

Is the result of the above example the optimal choice of the monopolist under incomplete information? In order to discuss the sequential screening mechanism under incomplete information, consider a simple model with two periods and two types.

Consider a monopoly seller of airplane tickets facing two types of travelers, B and L, with proportions being β_B and β_L, respectively. We can think of type B as the "business traveler" and type L as the "leisure traveler". There are two periods. At the beginning of period one, the traveler privately learns his type. The seller and the traveler contract at the end of period one. At the beginning of period two, the traveler privately learns his actual valuation $v \in [\underline{v}, \bar{v}]$ for the ticket, and then decides whether to travel. Each ticket costs the seller c. The seller and the traveler are risk-neutral, and the reservation utility of each type of traveler is normalized to zero.

The business type may value the ticket more in the sense of first-order stochastic dominance (FSD): type B's distribution of valuation G_B first-order stochastically dominates the leisure type's distribution G_L if $G_B(v) \leqq G_L(v)$ for all v in the range of valuations $[\underline{v}, \bar{v}]$. Or, alternatively, assume that the business type may face greater valuation uncertainty in the sense of mean-preserving-spread (MPS). In other words, G_B is a mean-preserving-spread (MPS) of G_L: for G_B and G_L and their random variables v_B and v_L, respectively, there exists v_ϵ so that $v_B = v_L + v_\epsilon$ with v_ϵ being independent of v_L, or equivalently, they have the same mean and $\int_{\underline{v}}^{v}(G_B(u) - G_L(u))du \geqq 0$ for all $v \in [\underline{v}, \bar{v}]$, which means that G_L second-order stochastically dominates G_B.

Example 20.4.1 Let us consider again the example above. G_B and G_L are described as follows:

$$G_B(v) = \begin{cases} \frac{v}{2}, & \text{if } v \in [0,1], \\ \frac{1}{2}, & \text{if } v \in [1,2], \\ \frac{v-1}{2}, & \text{if } v \in [2,3]; \end{cases}$$

and

$$G_L(v) = \begin{cases} 0, & \text{if } v \in [0,1], \\ v-1, & \text{if } v \in [1,2], \\ 1, & \text{if } v \in [2,3]. \end{cases}$$

The distribution function of v_ϵ is:

$$G_\epsilon(v) = \frac{v+2}{4}, v \in [-2,2].$$

Thus,

$$\int_{\underline{v}}^{v}(G_B(u) - G_L(u))du = \begin{cases} \frac{v^2}{4} \geqq 0, & \text{if } v \in [0,1], \\ \frac{1}{4} - \frac{(v-1)(v-2)}{2} \geqq 0, & \text{if } v \in [1,2], \\ \frac{(v-3)^2}{4} \geqq 0, & \text{if } v \in [2,3]. \end{cases}$$

Therefore, G_L second-order stochastically dominates G_B.

A refund contract consists of an advance payment a at the end of period one, and a refund k that can be claimed at the end of period two after the traveler learns his valuation. Clearly, regardless of the payment a, the traveler uses it only if he values the ticket more than k. The seller offers two refund contracts (a_B, k_B, a_L, k_L). The profit maximization problem can be written as follows:

$$\begin{aligned} \max_{(a_B,k_B;a_L,k_L)} \ & \beta_B[a_B - k_B G_B(k_B) - c(1 - G_B(k_B))] \\ & +\beta_L[a_L - k_L G_L(k_L) - c(1 - G_L(k_L))] \end{aligned} \tag{20.4.30}$$

subject to

$$-a_B + k_B G_B(k_B) + \int_{\underline{v}}^{\bar{v}} \max\{v, k_B\} dG_B(v)$$
$$\geqq -a_L + k_L G_B(k_L) + \int_{\underline{v}}^{\bar{v}} \max\{v, k_L\} dG_B(v); \quad (20.4.31)$$
$$-a_L + k_L G_L(k_L) + \int_{\underline{v}}^{\bar{v}} \max\{v, k_L\} dG_L(v)$$
$$\geqq -a_B + k_B G_L(k_B) + \int_{\underline{v}}^{\bar{v}} \max\{v, k_B\} dG_L(v); \quad (20.4.32)$$
$$-a_B + k_B G_B(k_B) + \int_{\underline{v}}^{\bar{v}} \max\{v, k_B\} dG_B(v) \geqq 0; \quad (20.4.33)$$
$$-a_L + k_L G_L(k_L) + \int_{\underline{v}}^{\bar{v}} \max\{v, k_L\} dG_L(v) \geqq 0. \quad (20.4.34)$$

The constraints of (20.4.33) and (20.4.34) are the participation constraints at period one, and (20.4.31) and (20.4.32) are the incentive compatibility constraints at period one.

We now verify that the constraints of (20.4.31) and (20.4.34) are binding. First, note that since $\max\{v, k_L\}$ is an nondecreasing function of v, when G_B dominates G_L by FSD, we have:

$$\int_{\underline{v}}^{\bar{v}} \max\{v, k_L\} dG_B(v) \geqq \int_{\underline{v}}^{\bar{v}} \max\{v, k_L\} dG_L(v).$$

Or, since $\max\{v, k_L\}$ is a convex function of v, the above inequality also holds by the properties of the MPS. Therefore, the constraint of (20.4.31) holds, and thus we have

$$\begin{aligned} -a_B + k_B G_B(k_B) + \int_{\underline{v}}^{\bar{v}} \max\{v, k_B\} dG_B(v) &\geqq -a_L + k_L G_B(k_L) \\ &+ \int_{\underline{v}}^{\bar{v}} \max\{v, k_L\} dG_L(v) \geqq 0. \end{aligned}$$

Therefore, we obtain the participation constraint (20.4.33) for type B.

Then, by the same arguments, the constraints of (20.4.31) and (20.4.34) must be binding at the optimal contract; otherwise, profits could be increased. At the moment, we omit the incentive compatibility constraint (20.4.32) for type L (we show below that it is loosely satisfied). Substituting the equations of (20.4.31) and (20.4.34) into (20.4.30), the profit maximization problem can be rewritten as:

$$\max_{(k_B, k_L)} \int_{k_B}^{\bar{v}} \beta_B (v-c) dv + \int_{k_L}^{\bar{v}} \beta_L (v-c) g_L(v) - \beta_B (G_L(v) - G_B(v)) dv. \quad (20.4.35)$$

In the objective function (20.4.35), define

$$S_t(k_t) = \int_{k_t}^{\bar{v}} (v-c) g_L(v) dv$$

as the consumer surplus for type $t \in \{L, B\}$, and

$$R_B(k_L) = \int_{k_L}^{\bar{v}} (G_L(v) - G_B(v)) dv$$

as the information rent for type B .

The solution for the problem is thus:

$$\begin{aligned} k_B &= c, && (20.4.36) \\ k_L &= arg \max_k [f_L S(k) - f_B R(k)]. && (20.4.37) \end{aligned}$$

The second-best solution reflects the tradeoff between allocative efficiency and information rent extraction, according to the same logic of static principal-agent problems under adverse selection.

Next, we demonstrate that under the solutions (20.4.36) and (20.4.37), the incentive compatibility constraint (20.4.32) for type L is satisfied.

As binding (20.4.31) implies that $a_L - a_B = \int_{k_B}^{k_L} G_B(v)dv$, we have

$$\begin{aligned} -a_L + k_L G(k_L) + \int_{k_L}^{\bar{v}} v dG_L(v) &= -a_B + k_B G(k_L) + \int_{k_B}^{\bar{v}} v dG_L(v) \\ &\quad - \int_{k_B}^{k_L} (G_B(v) - G_L(v)) dv. \end{aligned}$$

Then, (20.4.32) is equivalent to $\int_{k_B}^{k_L} (G_B(v) - G_L(v))dv \leqq 0$. Therefore, we only need to verify that $\int_{k_B}^{k_L} (G_B(v) - G_L(v))dv \leqq 0$. When $k_L = c$, this is obviously true. When $k_L \neq c$, suppose by way of contradiction that $\int_c^{k_L} (G_B(v) - G_L(v))dv > 0$, and consider a new contract in which $k'_B = k'_L = c$. We have $S_L(k'_L) = S_L(c) > S_L(k_L)$, and the information rent is:

$$\begin{aligned} R_B(k'_L) &= \int_c^{\bar{v}} (G_L(v) - G_B(v)) dv \\ &= \int_{k_L}^{\bar{v}} (G_L(v) - G_B(v)) dv + \int_c^{k_L} (G_L(v) - G_B(v)) dv \\ &\leqq \int_{k_L}^{\bar{v}} (G_L(v) - G_B(v)) dv = R_B(k_L), \end{aligned}$$

which is a contradiction to the second-best contract (k_B, k_L). Consequently, it must be that $\int_c^{k_L} (G_B(v) - G_L(v))dv \leqq 0$. Thus, the incentive compatibility constraint (20.4.32) holds under the binding (20.4.31) and (20.4.34).

We now show that, when $k_L \neq c$, we must have $k_L < c$. When $k_L > c$, it means that the type L is rationed, and when $k_L < c$, it means that the type L is subsided.

For the buyer's surplus of type L, $S_L(k_L) = \int_{k_L}^{\bar{v}}(v-c)g_L(v)dv$, $\forall k_L \in [\underline{v}, \bar{v}], S_L(c) \geqq S_L(k_L)$; thus, if $k_L = c$, the surplus is the biggest. However, for the information rent of type L,

$$R_B(k_L) = \int_{k_L}^{\bar{v}}(G_L(v) - G_B(v))dv = \int_{\underline{v}}^{k_L}(G_B(v) - G_L(v))dv \geqq 0.$$

When $k_l = \underline{v}$ or $k_l = \bar{v}$, $R_B(k_L) = 0$, we thus know that the rent R_B has an extreme point in the interior of $[\underline{v}, \bar{v}]$.

Consider the following special case in which $R_B(k_L)$ is a single-peaked function, which means that there exists z, such that $\frac{dR_B(k_L)}{dk_L} > 0 \; \forall k_L < z$ and $\frac{dR_B(k_L)}{dk_L} < 0 \; \forall k_L > z$. If the density functions g_B and g_L are symmetric at point z, we must have $k_L < c$ (or $k_L > c$) if $c < z$ (or $c > z$). To see this, we only explore the case of $c < z$ (the case of $c > z$ is similar). Let k_L^{SB} be the second-best solution.

(1) $k_L^{SB} \notin (c, z]$. If $k_L \in (c, z]$, then $\frac{dS_L(k_L)}{dk_L} < 0$ and $\frac{dR_B(k_L)}{dk_L} > 0$. As a consequence, if k_L decreases, then $S_L(k_L) - R_B(k_L)$ increases.

(2) $k_L^{SB} \notin (z, 2z - c]$. If $k_L \in (z, 2z - c]$, consider a new refund $\tilde{k}_L = 2z - k_L$, so that $R_B(\tilde{k}_L) = R_B(k_L)$. Therefore, for $c \leqq \tilde{k}_L < k_L$, we have $S_L(\tilde{k}_L) > S_L(k_L)$.

(3) $k_L^{SB} \not> 2z - c$. If $k_L > 2z - c$, consider a new refund $\tilde{k}_L = 2z - k_L$, so that $R_B(\tilde{k}_L) = R_B(k_L)$. Since $k > 2z - c$, for $\tilde{k} = 2z - k$, we obtain:

$$-\frac{dS_L(k)}{dk} = (k-c)g_L(k) = (k-c)g_L(\tilde{k}) > (c-\tilde{k})g_L(\tilde{k}) = \frac{dS_L\tilde{k}}{dk},$$

in which the second equality is from the symmetric assumption of g_L at z, and the third equality is from $k + \tilde{k} = 2z > 2c$. We then have

$$\begin{aligned} S_L(c) - S_L(k_L) &= \int_c^{k_L} -\frac{dS_L(k)}{dk}dk \\ &> \int_{\tilde{k}_L}^{c} \frac{dS_L(\tilde{k})}{dk}dk \\ &= S_L(c) - S_L(\tilde{k}_L). \end{aligned}$$

Therefore, $S_L(\tilde{k}_L) > S_L(k_L)$, which contradicts the fact that k_L is a second-best solution.

From the above discussion, we obtain $k_L^{SB} < c$.

Example 20.4.2 Return to the example from the last subsection and let $c = 1$. We show that the contract given in this example is, in fact, the second-best screening contract to the principal. Indeed, $S_L(k_L)$ and $R_B(k_L)$ are

$$S_L(k_L) = \int_{k_L}^{\bar{v}} (v - c) g_L(v) dv \begin{cases} \frac{1}{2}, & \text{if } k_L \in [0,1], \\ k_L - \frac{k_L^2}{2}, & \text{if } k_L \in [1,2], \\ 0, & \text{if } k_L \in [2,3]; \end{cases}$$

and

$$R_B(k_L) = \int_{\underline{v}}^{k_L} (G_B(u) - G_L(u)) du = \begin{cases} \frac{k_L^2}{4} \geqq 0, & \text{if } k_L \in [0,1], \\ \frac{1}{4} - \frac{(k_L-1)(k_L-2)}{2} \geqq 0, & \text{if } k_L \in [1,2], \\ \frac{(k_L-3)^2}{4} \geqq 0, & \text{if } k_L \in [2,3]. \end{cases}$$

So, $z = 1.5$, $k_L^{SB} = arg \max_k [\frac{1}{3} S_L(k) - \frac{2}{3} R_B(k)]$, and therefore $k_L^{SB} = 0$. Its graphic illustration is shown in Figure (20.4).

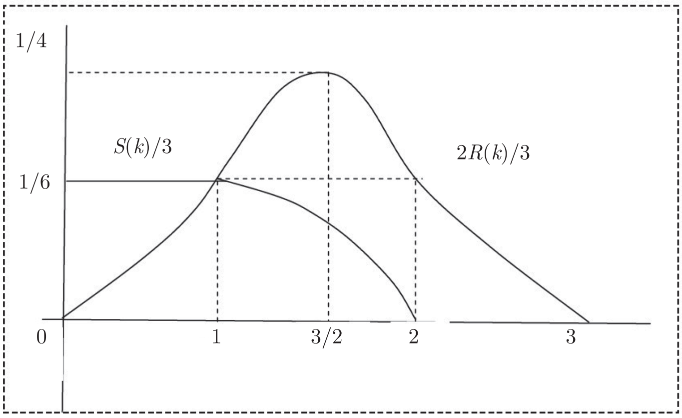

Figure 20.4: Sequential screening for two types

Since the participation constraint (20.4.34) is binding for type L, if $k_L = 0$, then $a_L = 1.5$; and since $k_B = c = 1$, for type B, the participation constraint (20.4.31) is binding, as well. Consequently, we obtain $a_B = 1.75$. Thus, $(a_L = 1.5, k_L = 0; a_B = 1.75, k_B = 1)$ is the second-best screening contract to the principal.

Courty and Li (2000) further discussed the case of continuous types, and interested readers can refer to their paper for more details.

20.5 Efficient Budget-Balanced Dynamic Mechanism

We now turn to the design of efficient dynamic mechanisms in multi-agent environments with strategic interactions. In the previous sections, we discussed that, in a dynamic environment without strategic interactions among agents, there are intertemporal constraints in the interaction between the principal and agent(s), such as how to balance information rent between different periods, and the trade-off between information rent extraction and allocative efficiency. However, in a general multi-agent dynamic mechanism, because the behavior of participants in early periods may reveal their types, there will be some strategic dynamic considerations in the choice of cross-period behavior, which will change the incentive constraints. Athey and Segal (2013) provided an analytic framework of dynamic mechanism design and established an efficient dynamic mechanism.

The Vickery-Clark-Groves (VCG) mechanism revealed the existence of incentive-compatible and efficient mechanisms for a class of static mechanism design problems. The VCG mechanism provides incentives using monetary transfers for truthful reporting of private information under the assumptions of private values and quasi-linear utility functions. The VCG mechanism is, nevertheless, not ex-post budget-balanced, so that it is not Pareto efficient. Subsequently, Arrow (1979) and d'Aspremont and Gerard-Varet (1979) (AGV) constructed an efficient and incentive-compatible mechanism, called the expected externality mechanism, in which the transfers are budget-balanced and the solution concept is Bayesian-Nash equilibrium, thus resulting in ex-post Pareto efficient outcomes under the additional assumption that private information is independent across agents. However, Myerson and Satterthwaite (1983) found that such mechanisms generally do not satisfy participation constraints.

In this section, we discuss efficient dynamic mechanism design by omitting the requirement of ex-post participation constraints. In a static setting, having a transfer equal to the "expected externality" imposed on the other agents, the AGV mechanism makes every agent have an incentive to truthfully report his type given his belief about opponents' types. As a consequence, an agent's current belief about opponents' types play an important role in determining the transfer that he will receive. However, in a dynamic setting, these beliefs evolve over time as a function of opponents' reports and the decisions that these reports induce. If the transfers are constructed using the agents' prior beliefs at the beginning of the mechanism, the transfers will no longer induce truthful reporting after agents have learned some information about every other's type.

Athey and Segal (2013) constructed a balanced team mechanism that satisfies the budget-balanced property. Such mechanism sustains an equilibrium in truthful strategies by giving each agent in each period an incentive payment equal to the change in the expected present value (EPV) of the other agents' utilities that are induced by his current report. On the one hand, these incentive payments cause each agent to internalize the expected externality imposed on the other agents by his report. On the other hand, *the expected incentive payment to an agent is zero when the agent reports truthfully regardless of what reporting strategies the other agents use*. Constructing a transfer to each agent as a function of others' reports in a way without affecting their incentives for truthful reporting, the latter property can make the mechanism budget-balanced.

Prior to introducing the dynamic Pareto efficient mechanism proposed by Athey and Segal, we first consider the static case and discuss the difficulties in dynamic mechanism design.

Consider a seller (agent 1) and a buyer (agent 2) who engage in a two-period relationship. In each period $t = 1, 2$, they can trade a contractible quantity $x_t \in \mathcal{R}_+$.

Before the first period, the seller privately observes a random type $\tilde{\theta}_1 \in [1, 2]$, whose realization θ_1 determines his cost. The cost function is given by

$$C(\theta_1, x_t) = \frac{1}{2}\theta_1 {x_t}^2,\ t = 1, 2,$$

where x_t is output at time t.

The buyer's value per unit of the good in period 1 is equal to 1, and in period 2 it is given by a random type $\tilde{\theta}_2 \in [0, 1]$ whose realization is privately observed between the periods.

When information is complete, the trading problem at period 1 is:

$$\max_{x_1} x_1 - \frac{1}{2}\theta_1 {x_1}^2,$$

giving us $x_1(\theta_1) = \frac{1}{\theta_1}$, and the trading problem at period 2 is:

$$\max_{x_2} \theta_2 x_2 - \frac{1}{2}\theta_1 {x_2}^2,$$

giving us $x_2(\theta_1, \theta_2) = \frac{\theta_2}{\theta_1}$. Therefore, an efficient (surplus-maximizing) mechanism must have trading decisions x_1 and x_2 determined by the decision rules: $x_1(\theta_1) = \frac{1}{\theta_1}$ and $x_2(\theta_1, \theta_2) = \frac{\theta_2}{\theta_1}$.

When information is incomplete, if the agents can observe their own types at period 1, it is a static interaction since they report simultaneously even although there are trades for two periods. In this situation, the problem of designing an efficient mechanism can be solved by designing transfers to each agent as a function of their reports. In this simple setting, each agent makes only one report. Let us first consider the AGV mechanism for this problem, where we assume that the seller and buyers observe their types at the same time. Taking this as a static benchmark, we examine the incentive change under asynchronous observations and reports.

20.5.1 Efficient Budget-Balanced Static Mechanism

Since the AGV mechanism can make each agent internalize the externality induced by his action to the others, the agents have an incentive to reveal their information truthfully in the efficient decision rule. In the following, we demonstrate how the AGV mechanism can truthfully implement efficient allocations in the above example, i.e., $x_1(\theta_1) = \frac{1}{\theta_1}$ and $x_2(\theta_1, \theta_2) = \frac{\theta_2}{\theta_1}$.

Let $\gamma_i(\theta_i)$ be the payoff to agent i. To encourage the buyer to reveal his type truthfully, the transfer to his must be the expected externality to the seller:

$$\begin{aligned}\gamma_2(\theta_2) &= -E_{\tilde{\theta}_1}[\frac{1}{2}\tilde{\theta}_1(x_1(\tilde{\theta}_1))^2 + \frac{1}{2}\tilde{\theta}_1(x_2(\tilde{\theta}_1, \theta_2))^2] \\ &= -\frac{1}{2}E_{\tilde{\theta}_1}[\frac{1}{\tilde{\theta}_1}](1 + (\theta_2)^2).\end{aligned}$$

We verify that under such transfer, the buyer has the incentive to truthfully reveal his true type. Let the buyer's report type as $\hat{\theta}_2$, and then his expected utility is:

$$\gamma_2(\hat{\theta}_2) + E_{\tilde{\theta}_1}[x_1\tilde{\theta}_1 + \theta_2 x_2(\tilde{\theta}_1, \hat{\theta}_2)] = -\frac{1}{2}E_{\tilde{\theta}_1}[\frac{1}{\tilde{\theta}_1}](1 + (\hat{\theta}_2)^2) + E_{\tilde{\theta}_1}[\frac{1}{\tilde{\theta}_1}](1 + \theta_2\hat{\theta}_2,]$$

and then the first-order condition for $\hat{\theta}_2$ is:

$$-E_{\tilde{\theta}_1}[\frac{1}{\tilde{\theta}_1}](\hat{\theta}_2) + E_{\tilde{\theta}_1}[\frac{1}{\tilde{\theta}_1}]\theta_2 = 0,$$

which gives us

$$\hat{\theta}_2 = \theta_2.$$

By a similar logic, the transfer to the seller is equal to the externality caused to the buyer:

$$\begin{aligned}\gamma_1(\theta_1) &= E_{\tilde{\theta}_2}[x_1(\theta_1) + \tilde{\theta}_2 x_2(\theta_1, \tilde{\theta}_2)] \\ &= \frac{1}{\theta_1}[1 + E_{\tilde{\theta}_2}(\tilde{\theta}_2)^2].\end{aligned}$$

We can easily check that, under such a transfer, the seller has an incentive to reveal his type.

The transfer above is not budget-balanced. However, due to the independence between $\gamma_i(\theta_i)$ and $\theta_j, j \neq i$, when the transfer to agent θ_i is

$$\psi_i(\theta_i, \theta_j) = \gamma_i(\theta_i) - \gamma_j(\theta_j), i \neq j, i, j \in \{1, 2\},$$

it is a budget-balanced AGV mechanism. Therefore, in the static mechanism, the efficient rule is Bayesian implementable and budget-balanced, and so it is Pareto efficient. Actually, we know this result in Chapter 19.

20.5.2 Incentive Problems in Dynamic Environments

Now, we turn to the dynamic model, in which the buyer (agent 2) observes his type $\tilde{\theta}_2$ between period 1 and period 2, and the seller (agent 1) reports his type at period 1. The above AGV mechanism cannot truthfully implement the efficient rule $x_1(\theta_1) = \frac{1}{\theta_1}$ and $x_2(\theta_1, \theta_2) = \frac{\theta_2}{\theta_1}$. This is because, under the dynamic setting, when the seller reports his true type θ_1, the buyer can learn the seller's type θ_1 from the first-period trade $x_1(\theta_i)$, and then the transfer $\gamma_2(\theta_2)$ cannot induce true revelation for the buyer. Indeed, if the buyer announces his type to be $\hat{\theta}_2$, his expected utility is

$$\gamma_2(\hat{\theta}_2) + \theta_1[x_1\theta_1 + \theta_2 x_2(\theta_1, \hat{\theta}_2)] = -\frac{1}{2}E_{\tilde{\theta}_1}\frac{1}{\tilde{\theta}_1}(1 + (\hat{\theta}_2)^2) + \frac{1}{\theta_1}(1 + \theta_2\hat{\theta}_2). \quad (20.5.38)$$

In this dynamic setting, the buyer can learn the seller's private information from the first-period trade. From the buyer's expected utility (20.5.38), the first-order condition for $\hat{\theta}_2$ is

$$-E_{\tilde{\theta}_1}[\frac{1}{\tilde{\theta}_1}](\hat{\theta}_2) + \frac{\theta_2}{\theta_1} = 0,$$

and so

$$\frac{\hat{\theta}_2}{\theta_2} = \frac{1}{\theta_1}(E_{\tilde{\theta}_1}[\frac{1}{\tilde{\theta}_1}])^{-1}.$$

Therefore, when $\frac{1}{\theta_1} > E_{\tilde{\theta}_1}[\frac{1}{\tilde{\theta}_1}]$, the buyer has an incentive to over-report his value; otherwise, he has an incentive to underreport his value. The buyer has an incentive to misreport his type because he does not need to fully internalize the externality induced by his action.

However, if we let the buyer bear externality, then it is given by:

$$\tilde{\gamma}_2(\theta_1, \theta_2) = -\frac{1}{2}\theta_1(1 + (\theta_2)^2).$$

Under this amount of transfer, the buyer's incentive can be restored. In this case, we have

$$\tilde{\gamma}_2(\theta_1, \hat{\theta}_2) + \theta_1[x_1\theta_1 + \theta_2 x_2(\theta_1, \hat{\theta}_2)] = -\frac{1}{2}\theta_1[\frac{1}{\theta_1}](1 + (\hat{\theta}_2)^2) + \frac{1}{\theta_1}(1 + \theta_2\hat{\theta}_2). \quad (20.5.39)$$

The first-order condition for $\hat{\theta}_2$ is:

$$-\frac{1}{\theta_1}(\hat{\theta}_2) + \frac{\theta_2}{\theta_1} = 0,$$

and thus

$$\hat{\theta}_2 = \theta_2.$$

Although $\gamma_1(\theta_1)$ and $\tilde{\gamma}_2(\theta_1, \theta_2)$ are incentive compatible for truthful revelation, they are not budget-balanced. In order to restore the budget-balanced property, the transfers to the seller and buyer must be:

$$\psi_1(\theta_1, \theta_2) = \gamma_1(\theta_1) - \tilde{\gamma}_2(\theta_1, \theta_2),$$

$$\psi_2(\theta_1, \theta_2) = \tilde{\gamma}_2(\theta_1, \theta_2) - \gamma_1(\theta_1).$$

However, as $\tilde{\gamma}_2(\theta_1, \theta_2)$ depends on θ_1, such transfer $\psi_1(\theta_1, \theta_2)$ is not incentive compatible for θ_1.

20.5.3 Efficient Budget-Balanced Dynamic Mechanism

The difficulty mentioned above is called the problem of contingent deviation. Athey and Segal (2013) proposed a mechanism that overcomes this difficulty. Similar to the AGV mechanism, their construction proceeds in two steps: (1) construct incentive compatible transfers $\gamma_1(\theta)$ and $\gamma_2(\theta)$ to make each agent report truthfully if he expected the other to do so, where $\theta = (\theta_1, \theta_2)$; and (2) charge each agent's incentive compatible transfer to the other agent, making the total transfer to agent i equal to $\psi_i(\theta) = \gamma_i(\theta) - \gamma_j(\theta)$. However, in contrast to AGV transfers, the incentive transfer $\gamma_i(\theta)$ to agent i will now depend not just on agent i's announcement θ_i, but also on those of the other agents.

How do we then make sure that step (2) does not destroy incentives? For this purpose, we ensure that even though agent i can affect the other's incentive payment $\gamma_i(\theta_i, \theta_{-i})$, he cannot manipulate the expectation of the payment, given that agent i reports truthfully. We achieve this by letting $\gamma_i(\theta_i, \theta_{-i})$ be the change in the expectation of agent i's utility, conditional on all of the previous announcements, which is brought about by the report of agent i (in the general model, in which an agent reports in many periods, these incentive transfers would be calculated in each period for the latest report). Irrespective of what reporting strategy agent i adopts, if he believes that agent i reports truthfully, his expectation of the change in his expected utility due to agent i's future announcements is zero by the law of iterated expectations. Therefore, agent i can be charged $\gamma_i(\theta_i, \theta_{-i})$ without affecting his incentives.

For the above situation, our construction entails giving the buyer an incentive transfer of

$$\gamma_2(\theta_1, \theta_2) = -\frac{1}{2\theta_1}[(\theta_2)^2 - E_{\tilde{\theta}_2}(\tilde{\theta}_2)^2].$$

Such transfer is incentive compatible for the buyer. This is because, upon the announcement $\hat{\theta}_2$, the buyer's utility is

$$\gamma_2(\theta_1, \hat{\theta}_2) + \theta_1[x_1\theta_1 + \theta_2 x_2(\theta_1, \hat{\theta}_2)] = -\frac{1}{2\theta_1}[(\hat{\theta}_2)^2 - E_{\tilde{\theta}_2}(\tilde{\theta}_2)^2] + \frac{1}{\theta_1}(1 + \theta_2(\hat{\theta}_2)).$$

The first-order condition for $\hat{\theta}_2$ is:

$$-\frac{1}{\theta_1}(\hat{\theta}_2) + [\frac{1}{\theta_1}]\theta_2 = 0,$$

giving us $\hat{\theta}_2 = \theta_2$, which means that it satisfies incentive compatibility.

Although $\gamma_2(\theta_1, \hat{\theta}_2)$ depends on θ_1,

$$E_{\tilde{\theta}_2}[\gamma_2(\theta_1, \tilde{\theta}_2)] = 0, \forall \theta_1.$$

Therefore, the seller's report does not affect the interim expected transfer to the buyer. As we know, $\gamma_1(\theta_1)$ is the incentive transfer for the seller, and thus the total transfers for the agents are:

$$\begin{aligned} \psi_1(\theta_1, \theta_2) &= \gamma_1(\theta_1) - \gamma_2(\theta_1, \theta_2), \\ \psi_2(\theta_1, \theta_2) &= -\psi_1(\theta_1, \theta_2). \end{aligned}$$

Therefore, these transfer schemes are budget-balanced and incentive compatible for the seller, because the seller reports earlier than does the buyer, and the interim expected transfer to the seller is:

$$E_{\tilde{\theta}_2}\psi_1(\theta_1, \tilde{\theta}_2) = \gamma_1(\theta_1).$$

For a general dynamic model, Athey and Segal (2013) designed an efficient mechanism by generalizing the idea of using incentive transfers that give an agent the change in the expected present value of opponent utilities induced by his report, and interested readers can find more details in their paper.

Dynamic mechanism design is a popular theoretical topic. Bergemann and Valimaki (2010) proposed an alternative efficient dynamic mechanism, called the dynamic pivot mechanism, which satisfies the ex-post participation and incentive compatibility constraints, as well as "efficient exit" conditions, but it may not be balanced. Pavan, Segal, and Toikka (2014) developed a more general allocation model, and derived the revenue-maximizing and incentive compatible mechanism in a dynamic setting. For interested readers, the last chapter of the monograph by Borgers (2015) is also a useful reference on dynamic mechanism design.

20.6 Biographies

20.6.1 Jean-Jacques Laffont

Jean-Jacques Laffont (1947-2004) is the founder of Toulouse's Industrial Economics Institute (Institut D'Economie Industrielle, IDEI) and one of the founders of the new regulatory economics, information economics, and incentive theory. He made pioneering contributions in public economics, development economics, and the theory of imperfect information, incentives, and regulation. He was also a pioneer of the econometrics of auctions and structural estimations of industrial organization models.

Jean-Jacques Laffont, born in Toulouse, southeastern France, was a member of the post-war generation in France. This generation grew up under the influence of General de Gaulle. They had strong patriotism and were committed to revitalization of the French nation. In 1968, Laffont graduated from the University of Toulouse, which has an almost legendary mathematics education tradition. Later, he completed a doctorate at the National School for Statistics and Economic Administration in France. Together with Maskin and Elhanan Helpman, he became part of a world-renowned trio in the economics community – the so-called Three Musketeers at the Harvard Economics Department. Laffont received a Ph.D. in Economics (1975) from Harvard University in just one and a half years, which is very rare in the history of Harvard. He taught at the École Polytechnique from 1975-1987, and was a professor of economics at Ecole des hautes études en sciences sociales from 1980-2004, and at the University of Toulouse from 1991-2001. In 1991, he founded Toulouse's Industrial Economics Institute (Institut D'Economie Industrielle, IDEI), which has become one of the most prominent European research centres in economics.

In the 1970s, general equilibrium theory was still a dominant field in economics, but Laffont was deeply aware of the importance of incentive theory in economics. Laffont chose incentive theory as his main research field. At the end of the 1970s, information economics was emerging. The main subject of information economics is incentive problems under incomplete information and asymmetric information. Integrated with game theory methodologically, information economics has successfully solved many problems for which the framework of general equilibrium theory might not work well. Laffont took information economics as the basic framework for his research on incentives, and began to explore the systematic integration of incentive theory.

In 1979, Laffont's book "Incentives in Public Decision Making" (co-authored with Jerry Green), established his authoritative position in the field of public economics. Laffont and Tirole created a general framework for incentive regulation, which led to the birth of new regulatory economics which combines the basic ideas of public economics and industrial organization, as well as the basic methods of information economics and mechanism design theory. Laffont and Tirole published their 1993 book "A Theory of Incentives in Procurement and Regulation", thus establishing their academic leadership in the field.

Laffont was an extremely diligent and productive scholar. He is the author of 17 books and 200 high-level academic papers. His academic contributions have earned him a high reputation in the economics profession. Even in his fight against cancer, he still completed the new book, "Regulation and Development" (December 2003), applying his knowledge to understand the challenges faced by developing countries.

Laffont was elected as the President of the European Economic Association (1998), Honorary Member of the American Economic Association (1991), and Foreign Honorary Member of the American Academy of Arts and Sciences (1993); jointly with Jean Tirole, he was awarded the Yrjo-Jahnsson Award from the European Economic Association (1993), awarded each year to the best European economist under the age of 45. He might well have shared the Nobel Prize in Economics with Tirole if he were alive.

Jean-Jacques Laffont passed away at his home in Colomiers in the Haute Garonne region of southern France on May 1, 2004. He was 57.

20.6.2 Jean Tirole

Jean Tirole (1953-) is a French professor of economics. His academic contribution has been awe-inspiring: more than 300 high-quality papers and 11 monographs, covering many important fields of economics – macroeconomics, the theory of industrial organization, game theory, incentive theory, corporate finance, international finance, and the interdisciplinary studies of economics and psychology.

In 2000, as a summary of regulation theory and policy research of monopoly industry in the past 10 years at that time, he and Laffont co-authored the book, *Competition in Telecommunications*, which provides a hallmark theoretical framework for economic analysis and policy formulation of competition and regulation of the telecom and network industries. Tirole is the Director of the Industrial Economics Institute at the University of Toulouse in France, and is a visiting professor at the Massachusetts Institute of Technology. He was the President of the Econometric Society and the President of the European Economic Association, and was awarded the Nobel Prize in Economics in 2014 for his “analysis of market power and regulation”.

Tirole was born on August 9, 1953 in Troyes, located on the Seine River. His father was a gynecologist and his mother was a teacher who taught French, Latin, and Greek. Tirole’s mother attached great importance to family education and taught him a lot since he was very young. In 1978, after receiving a doctorate in applied mathematics at the Paris Dauphine University, he went on to study economics at the Massachusetts Institute of Technology and received his Ph.D. in economics there in 1981.

Inheriting the tradition of French scholars’ emphasis on humanities and coupled with a deep mathematical foundation, Tirole has an extraordinary talent to generalize and synthesize. He is always able to put essential research questions and important results into simple economic models, and collate them into a systematic theoretical framework. For example, in 1982 and 1985, he published two classic papers in *Econometrica*, “On the Possibility of Speculation under Rational Expectations” and “Asset Bubbles and Overlapping Generations”, establishing his authority in the field.

Since the revival of economics in the 1980s, the Institute of Industrial Economics (IDEI) at the University of Toulouse in France has been the most successful such institute on the European continent. In 1988, Tirole returned from the United States to France, and founded the world-renowned IDEI along with Jean-Jacques Laffont. He has made remarkable contributions to the revitalization of French and European economics.

20.7 Exercises

Exercise 20.1 Consider a two-period principal-agent model under full commitment. The firm as the principal produces a certain product, and the value of q units of product to the principal is $S(q)$, satisfying $S' > 0$, $S'' < 0$ and $S(0) = 0$. The firm intends to delegate the production of q_1 and q_2 units of product to the agent in the two periods, and to provide transfers $t(q_1)$ and $t(q_1, q_2)$, respectively. The agent has two types: $\underline{\theta}$ and $\overline{\theta}$, where $\underline{\theta}$ means that the marginal cost of production is low, whereas $\overline{\theta}$ means that the marginal cost of production is high, and the agent knows his type. However, the principal only knows the probability distribution of types, i.e., $\beta(\overline{\theta}) = 1 - \beta$ and $\beta(\underline{\theta}) = \beta$. The goal of the firm is to maximize $S(q_1) - t(q_1) + S(q_2) - t(q_1, q_2)$, subject to the agent's incentive compatibility constraint and participation constraint.

1. Under the assumption of constant types, solve the principal-agent problem.

2. Under the assumption of independent types, solve the model and compare the result with the above one.

Exercise 20.2 Under the model of the previous problem, we add type correlation over the two periods. At the first period, the agent observes his type of the first period as $\theta_1 \in \{\underline{\theta}, \overline{\theta}\}$, and the principal just knows that the probability of $\theta_1 = \underline{\theta}$ is β. For the information structure of the second period, the agent cannot observe his type, and there is only one correlation between the two periods, i.e., the probability of the marginal cost of the second period is

$$\underline{\beta} = prob(\underline{\theta}|\underline{\theta});$$

$$\overline{\beta} = prob(\underline{\theta}|\overline{\theta}).$$

When $\underline{\beta} > \overline{\beta}$, there is a positive correlation; when $\underline{\beta} = 1 > \overline{\beta} = 0$, it is the case of constant types; when $\underline{\beta} = \overline{\beta} = \beta$, it is the case of independent types over time. Using the ways of analyzing consumers and monopolists provided in this book, analyze the principal-agent problem under positively correlated types over time.

Exercise 20.3 (Laffont and Tirole, 1993) For dynamic contracts without any commitment power, consider the following two-period model: for each period, the firm must complete a project at a cost

$$C_\tau = \beta - e_\tau, \tau = 1, 2, \tag{20.7.40}$$

where β is the fixed parameter and its value is only observable by the firm; and e_τ reflects the cost reduction in period τ, or the level of effort that the manager has expended. At time τ, social welfare is

$$W_\tau = S - (1+\lambda)(C_\tau + t_\tau) + U_\tau, \tag{20.7.41}$$

where $U_\tau - \psi(e_\tau)$ is the utility of the manager; S is the social value of the project itself; λ is the shadow cost of capital; and $(1+\lambda)(C_\tau + t_\tau)$ is the total cost. The discount factor of the social planner and the manager is δ.

Consider a continuum situation of $\beta \in [\underline{\beta}, \overline{\beta}]$, whose prior probability distribution is $F_1(\cdot)$, and the density function $f_1(\cdot)$ is strictly positive over $[\underline{\beta}, \overline{\beta}]$, with $\frac{d(F_1(\beta)/f_1(\beta))}{d\beta} > 0$. Under this assumption, the optimal static mechanism is a separating contract (full revelation).

1. Prove that for any scheme $t_1(\cdot)$ of the first period, there is a perfect Bayesian equilibrium that is not of full revelation (i.e., not a separating equilibrium).

2. Consider any incentive scheme $t_1(\cdot)$ of the first period, and suppose that the lower bound of ψ' is positive. Prove that for any ϵ, there is $\beta_\epsilon < \overline{\beta}$, such that when $\underline{\beta}_n \geq \beta_\epsilon$ for any n, there does not exist a perfect Bayesian equilibrium, such that the social planner benefits more than from an optimal pooling (no-revelation) contract.

Exercise 20.4 Consider a bilateral economy between a buyer and a seller. From the ex-ante perspective, the buyer only has a private signal about the value θ of the commodity, i.e., the buyer does not know θ until accepting the mechanism designed by the seller. The ex-ante signal τ is discrete: $\tau \in \{\tau_L, \tau_H\}$. Each ex-ante type is equally likely, for $\theta \in [0,1]$. The conditional distributions of the two types are $F(\theta|\tau_H) = \theta$ and $F(\theta|\tau_L) = \sqrt{\theta}$, respectively. The corresponding direct revelation mechanism, (q,t), can be written as $(q_L, q_H, t_L, t_H) : [0,1] \to [0,1]^2 \times \mathcal{R}^2$.

1. For $i \in \{L, H\}$, define a random variable $\gamma_i = F(\theta|\tau_i)$. Prove that γ_i is randomly independent of τ and is uniformly distributed over $[0,1]$.

2. Let $(\tilde{q}^*, \tilde{t}^*)$ represent the optimal mechanism that is incentive compatible with the publicly observable γ. Prove that

$$\tilde{q}^*_H(\gamma) = 1, \quad \forall \gamma \in [0,1];$$

$$\tilde{q}_L^*(\gamma) = \begin{cases} 0, & \text{if } \gamma < 1/2, \\ 1, & \text{others.} \end{cases}$$

3. Prove that the optimal mechanism under publicly observable γ brings expected utility of $1/12$ to the buyer of ex-ante type τ_H.

4. For the following questions, assume that only the buyer privately observes γ. Suppose that the mechanism $(\tilde{q}_L^*, \tilde{t}_L)$ leads to that the buyer of ex-ante type τ_L reports truthfully γ after reporting τ_L. Prove in this case that there is $\overline{\tau}_L \in R$ satisfying

$$\tilde{t}_L(\gamma) = \begin{cases} \bar{t}_L, & \text{if } \gamma < 1/2, \\ \bar{t}_L + 1/4, & \text{otherwise.} \end{cases}$$

5. Prove that if $(\tilde{q}_L^*, \tilde{t}_L)$ leads to that ex-ante type τ_L reports γ truthfully, for any $\gamma \in (1/4, 1/2)$, ex-ante type τ_H will not report γ truthfully and will report γ to be over $1/2$ after reporting τ_L.

6. Prove that if $(\tilde{q}_L^*, \tilde{t}_L)$ leads to that ex-ante type τ_L reports γ truthfully, the expected utility of ex-ante type τ_H exceeds the expected utility of ex-ante type τ_L by $11/96 > 1/12$.

7. Explain using problems (3) and (4) that, compared with the case in which γ is publicly observable, in order to realize $(\tilde{q}^*, \tilde{t}^*)$, the seller has to provide a higher information rent to the buyer of ex-ante type τ_H when only the buyers observe γ privately.

Exercise 20.5 (Borgers, 2015) Consider an economic environment of sequential screening. There is a seller and a buyer whose ex-ante and ex-post types are discrete. Before the buyer meets the seller, he has a discrete signal $\tau \in \{1, 3\}$. The seller offers a mechanism. The buyer then accepts the mechanism and receives a second discrete signal $\sigma \in \{1, 3\}$. The value of the buyer is $\theta = \theta_{\tau\sigma} = \tau + \sigma$. Each signal value is equally likely. The opportunity cost for the seller to sell the product is $c = 1/2$. For each $(\tau, \sigma) \in \{1, 3\}^2$, a direct revelation mechanism (q, t) specifies the probability of a sale to be $q_{\tau,\sigma} \in [0, 1]$ and the transfer to be $t_{\tau,\sigma} \in R$.

1. Prove that any direct revelation mechanism will result in the following lies along the off-equilibrium path. The best choice of the buyer of ex-ante type $\tau = 3$ is to always report $\sigma = 3$ after reporting $\tau = 1$, but the best choice of the buyer of ex-ante type $\tau = 1$ is to always report $\sigma = 1$ after reporting $\tau = 3$.

2. Prove that the optimal mechanism (q, t) satisfies $q_{11} = 1, q_{13} = q_{31} = q_{33} = 1$.

3. Suppose that the buyer could not obtain the ex-post signal σ. Thus, the expected value of the buyer of ex-ante type $\tau = 1$ to the product is $\theta_1 = 3$, but the expected value of the buyer of ex-ante type $\tau = 3$ to the product is $\theta_3 = 5$. Prove that $q_1 = q_3 = 1$ is optimal in a (static) direct revelation mechanism $(q_\tau, t_\tau)_{\tau=1,3}$.

4. Prove that a buyer's situation after receiving an ex-post private signal is worse than the ones that are not able to receive any ex-post signal.

Exercise 20.6 Consider the problem in which a principal delegates an agent to produce q units of product. The value of q units of product to the principal is $S(q)$, with $S' > 0$, $S'' < 0$ and $S(0) = 0$. The principal cannot observe the cost of production of the agent, and the marginal cost satisfies $\theta \in \{\underline{\theta}, \overline{\theta}\}$. The probabilities of the agent being the high-efficient type $\underline{\theta}$ and low-efficient type $\overline{\theta}$ are, respectively, v and $1 - v$. The cost function is $C(q, \underline{\theta}) = \underline{\theta} q$ with probability v, or is $C(q, \overline{\theta}) = \overline{\theta} q$ with probability $1 - v$, and the agent gets transfer t.

1. Suppose that this is a static case. Solve for the optimal contract.

2. Suppose that there are two periods, and the objective function of the principal is $V = S(q_1) - t_1 + \delta(S(q_2) - t_2)$. The risk-neutral agent has the same discount factor, and the objective function is $U = t_1 - \theta q_1 + \delta(t_2 - \theta q_2)$. Solve the dynamic principal-agent problem under constant types.

Exercise 20.7 (Independent types and risk-neutrality) Consider the same setting as the previous problem, but suppose that the distributions of the agent's marginal cost are independent across the two periods. Solve for the optimal dynamic contract under full commitment.

Exercise 20.8 (The moral hazard problem of repeating two periods) In the moral hazard model of Chapter 17, we now assume that the relationship between the principal and the agent is repeated for two periods, and the utility function of the risk-averse agent is $U = u(t_1) - \psi(e_1) + \delta(u(t_2) - \psi(e_2))$, for $e_i \in \{0, 1\}$, $\psi(1) = \psi$, and $\psi(0) = 0$. The effort level of the agent at each period will generate a random return q_i with the probability $\pi(e_i)$, for $\pi_0 = \pi(0)$ and $\pi_1 = \pi(1)$. The principal is risk-neutral, and the utility function is $V = S(q_1) - t_1 + \delta(S(q_2) - t_2)$. In this two-stage problem, solve for the optimal contract.

Exercise 20.9 (Renegotiation-proof contract) Suppose that the relationship between the principal and the agent is repeated twice, and the utility function of the risk-averse agent is $U = u(t_1) - \psi(e_1) + \delta(u(t_2) - \psi(e_2))$, for $e_i \in \{0, 1\}$, $\psi(1) = \psi$, and $\psi(0) = 0$. The effort level of the agent at each period will generate a random return q_i with the probability $\pi(e_i)$, satisfying $\pi_0 = \pi(0)$ and $\pi_1 = \pi(1)$. The principal is risk-neutral, and the utility function is $V = S(q_1) - t_1 + \delta(S(q_2) - t_2)$. All conditions are consistent with the previous one, given that $u_2(q_1)$ is the utility level that the agent can obtain in the second period, guaranteed by the principal if the output in the first period is q_1. The renegotiation-proof condition is $u_2(q_1) \geqq 0$.

1. Solve the optimal contract problem when the condition of renegotiation-proof is satisfied.

2. What happens to the optimal contract problem when the agent is allowed to borrow?

Exercise 20.10 (The moral hazard problem of repeating infinite periods) Now, extend the two-period problem in the previous ones to infinite periods, i.e., assuming that the relationship between the principal and the agent is repeated infinitely, and the utility function of the risk-averse agent is $U = u(t_1) - \psi(e_1) + \delta(u(t_2) - \psi(e_2))$, for $e_i \in \{0, 1\}$, $\psi(1) = \psi$ and $\psi(0) = 0$. The effort level of the agent at each period will generate a random return q_i with the probability $\pi(e_i)$, satisfying $\pi_0 = \pi(0)$ and $\pi_1 = \pi(1)$. The principal is risk-neutral, and the utility function is $V = S(q_1) - t_1 + \delta(S(q_2) - t_2)$.

1. Characterize the optimal contract problem recursively and find the state variable of this dynamic programming problem.

2. Solve for the optimal contract and prove that it exhibits the Markov property.

Exercise 20.11 (Bolton and Dewatripont, 2005) Consider a two-period durable-good monopoly problem. When a seller faces a buyer, the buyer's reservation utility of the durable-good is $v \in \{v_L, v_H\}$, for $v_H > v_L > 0$. The initial belief of the seller regarding the reservation utility of the buyer is $\Pr(v = v_H) = 0.5$. The cost of production by the seller takes two values for the same probability: $c \in \{c_L, c_H\}$ and $c_H > c_L \geqq 0$. The seller's cost and the value of the buyer's reservation utility are independently distributed and are their private information. Suppose that $v_L - c_L \geqq \frac{v_H - c_L}{2}$ and $v_L - c_H \geqq \frac{v_H - c_H}{2}$. The common discount factor is $\delta > 0$.

1. For the following situations, what are the conditions for the existence of a pooling equilibrium?

   (a) The seller of both types sets the same first-period price: $p_1 = v_H - \delta(v_H - v_L)/2$.

   (b) The buyer of the type v_H accepts the price with the probability $\gamma = \frac{v_H + c_H - 2v_L}{v_H - v_L}$, but the buyer of the type v_L rejects the price with probability 1.

(c) After the price is rejected in the first period, the seller of the type c_L will set the price $p_2^L = v_L$ in the second period, but the seller of the type c_H will set the price $p_2^H = v_H$ in the second period.

2. Explain why the seller of the type c_L can benefit from his private information of the cost, while the type c_H cannot.

Exercise 20.12 (Two-period regulation model) Consider a two-period regulation model where the type of firm is endogenous. Firstly, the regulator proposes an income function $R_1(q)$, which defines the transfer for an output level q. Next, the firm invests a sunk cost I, which is only observable to the firm. The production cost of each period is $c(q, I)$, where $c(0, I) = c_q(0, I) = 0$, $c_{qq}(q, I) > 0$ and $c_I(q, I) < 0$. The firm selects the output of q_1, thus generating a first-period return of $R_1(q_1) - c(q_1, I) - I$. If the firm gives up, it gets $-I$ and the game ends. If the firm chooses to produce q, then the first-period return of the regulator is $q_1 - R_1(q_1)$. In the second period, the regulator observes q_1 and then proposes $R_2(q)$, and thus the firm either gives up or selects an output q_2. Accordingly, the second-period returns of the two sides are, respectively, $R_2(q_2) - c(q_2, I)$ and $q_2 - R_2(q_2)$.

1. What is the regulator's full-commitment strategy?

2. Prove that if there is no commitment, when $I > 0$, there is no pure-strategy perfect Bayesian equilibrium.

Exercise 20.13 (Repeated moral hazard, Rogerson, 1985) Consider a two-stage moral hazard problem. At the time 1, the agent selects the effort level of a, thereby gaining an independently and identically distributed profit in each stage: $q_1, q_2 \in \{q_L, q_H\}$. The probability of the occurrence of q_H is $p(a)$, which is strictly increasing with a; the probability of the occurrence of q_L is $1 - p(a)$. For all $a \in [0, \overline{a}]$, $a < \infty, 1 > p(a) > 0$. The utility function of the agent is $u(w) - a$, satisfying $u' > 0$ and $u'' < 0$. The agent could not borrow, and thus his income in each period would be consumed entirely. The principal is risk-neutral, and can deposit and loan at a zero interest rate.

1. The transfers of the agent, depending on output in the first stage and the second stage, are expressed by $\{w_L, w_H, w_{LL}, w_{LH}, w_{HL}, w_{HH}\}$. Prove that the optimal contract satisfies:

$$\frac{1}{u'(w_i)} = \frac{p(a)}{u(w_{iH})} + \frac{1 - p(a)}{u'(w_{iL})}, i = H, L.$$

2. Prove: under the optimal contract, when $1/u'$ is a concave function, $w_i \leqq p(a)w_{iH} + [1 - p(a)]w_{iL}, i = H, L.$

3. Suppose that, under the above optimal contract, the agent is allowed to deposit a loan after the q_1 is realized in stage 1. Explain why he is willing to do so.

Exercise 20.14 (Bolton and Dewatripont, 2005) Consider an investment insurance problem under private information. A risk-averse agent invests $p/2$ in a project, and there is a random shock to incomes in two periods $t = 1, 2$, where w_i is equal to1 with a probability of p, and is equal to 0 with a probability of $1 - p$. In addition, $\Pr(w_2 = 1|w_1 = 1) = \gamma \leqq p$, Pr $(w_2 = 1|w_1 = 0) = \mu \geqq p$, $\gamma > 0.5$, and $p < 1$. The utility function of the agent is time-separable, $U(c_1, c_2) = u(c_1) + u(c_2)$. $u(c)$ is the following piecewise linear function: when $c \geqq 1/2$, $u(c) = c/2 + 1/4$; when $c < 1/2$, $u(c) = c$. At the beginning of each period, the agent can obtain insurance against income shocks at a fair rate.

1. Characterize the optimal consumption allocation when the agent cannot make private deposits.

2. Suppose that the income shock is private information. Prove that the agent cannot get any insurance coverage when only the temporary (short-term) contract is truthfully implementable.

3. Suppose that the agent can deposit and borrow money at a zero interest rate from a bank. Characterize the optimal lending margins of the agent.

4. When is the insurance of the form of deposit and loan the optimal contract?

Exercise 20.15 (International Lending, Atkeson, 1991) There are two types of economic agents in an infinite economy: the risk-averse agent (borrower) and the risk-neutral principal (lender). After the agent borrows money, he can invest it and generate a random investment return. The distribution of returns depends on the number of investments. The agent has an initial endowment of $Y_0 - d_0$ at period 0. At the beginning of period t, the agent borrows b_t from the principal; at the end of period t, the agent returns d_{t+1} to the principal, and the utility of the agent depends on the consumption level in each period. Set $(b_t, d_{t+1}(Y_{t+1}))$ as a loan contract, and let $Q_t = Y_t - d_t$.

1. Define the feasible allocations of the agent.

2. Write down the participation constraints of the principal and the agent of the dynamic problem.

3. Write down the incentive compatibility constraint of the agent of the dynamic problem, i.e., the agent does choose the optimal level of investment.

4. Suppose that once the agent defaults, the principal will terminate the loan agreement, i.e., the agent will return to autarky. Write down an incentive constraint to prevent the agent from default.

5. Solve the optimal contract problem. Show that when observing that the output level is low, the agent faces capital outflow.

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Part III

Market Design

The last part of this textbook discusses the development and extension of mechanism design: market design of auctions and matchings, which concerns allocative efficiency, stability, fairness, strategic interactions, and computational complexity.

The theory of market design helps us to understand price and non-price mechanisms, and it focuses on the design of practical auction and matching procedures. The essence of the market design process is to find prices or obtain stable matchings in an effective manner, while ensuring that complex constraints can be met. It is regarded as microeconomic engineering because it uses laboratory research, game theory, algorithms, and simulations, and has wide practical applications. The recent development of auction design has moved from single-unit auction to multi-units auction and combinational auction, while matching theory has been applied to solving the problems of school choice, organ transplantation, office allocation, and numerous other fields.

The theory of market design does not take the market as given, but rather applies the empirical or experimental results of microeconomics and, in particular, game theory to the design of new mechanisms. Auction theory primarily considers auction design, in which the price mechanism characterized by **monetary transfers** plays an essential role — agents who are willing to pay the highest win objects; whereas, the price mechanism usually does not play a dominant role in matching theory, and there might be no **monetary transfers** in many cases. Auction theory and matching theory have established their respective analytical frameworks, which have increasingly become independent subfields of modern microeconomic theory.

Chapter 21 discusses basic contents in auction theory. We will see that many results in auction theory can be obtained by applying the results in mechanism design theory introduced in the previous part. We will discuss the single item auction with private values, the single item auction with interdependent values, the simultaneous and sequential auction of multiple items, and combinational auction.

Chapter **??**, the last chapter of the textbook, discusses how to allocate indivisible objects generally without transfers playing a dominant role in matching issues. In many situations (e.g., human organ transplantation, school choice, and office allocation), transfers are not allowed or it does not play a substantial role (e.g., the admission of students is not determined by who pays the highest tuition fees, and the recruitment in the labor market is not mainly determined by who is willing to ask for the least remuneration to hire who, more importantly, it depends on their potential, quality or whether they are suitable for the job they are applying for), so that it constitutes a non-price mechanism. We will first explore matchings in a two-sided market and a one-sided market, respectively, and then analyze the applications in school choice and organ transplantation.

Chapter 21

Auction Theory

21.1 Introduction

Auction theory has developed systematic theoretical results and also provided useful insights for practical operations. Since auction theory can be considered as a branch or an extension of the theory of mechanism design, applying the results established in the previous part can obtain some fundamental results in auction theory.

As an effective trading mechanism of objects, the **auction** or **tender** is being increasingly widely applied, and has penetrated into all aspects of daily economic activities. In practice, many major objects or projects are auctioned or tendered. Objects that are often auctioned include tangible assets, such as antiques, paintings, jewelry, used cars, buildings and agricultural products, as well as intangible assets, such as land use rights, oilfield exploitation rights, and even some special telephone numbers and license plate numbers. In addition, the central banks of many countries often sell government bonds by auction, and the Ministry of the Interior conducts regular auctions of oil exploration rights. The Federal Communications Commission of the United States invited Paul Milgrom and Robert Wilson, the 2020 Nobel Laureates in Economics, to design the radio spectrum auction, which generated more than $200 billion in revenue in the 20 years from 1994 to 2014. The auction mechanism has also been applied to other scenarios, such as auctions of electricity and gas sales. With the explosive growth of demand for wireless Internet service, the wireless spectrum used to transmit data rapidly became in short supply. In 2012, the Federal Communications Commission decided to reconfigure the idle radio spectrum of television companies, and the incentive auction implemented in 2016 was also led by Milgrom, which generated $20 billion in revenue for the Federal Communications Commission. European countries have also adopted similar auctions, and generated enormous revenue.

In fact, the history of the auction can be traced back to at least the ancient Babylonian period of 700 BC. The first recorded auction activity occurred in 500 BC in Greece, where women were auctioned off by their families as brides. The Romans, too, were advocates of the auction process, regularly auctioning off the spoils of war, slaves, the assets of debtors, and even influenced the whole course of history. In AD 193, Roman Emperor Pertinax was killed by his guards because he wanted to rectify military discipline. Subsequently, the controlling Praetorian Guard auctioned the throne in the barracks. The two candidates were Senator Didius Julianus and Pertinax's father-in-law, Titus Flavius Claudius Sulpicianus. Didius Julianu offered 25,000 sesterces (almost two kilos of gold) to each guardsman and Sulpicianus only 20,000, and thus the guardsmen chose him as the next emperor. This, however, did not last long. Two months later, an insurgent army stormed into Rome, and this politician throned by an auction was eventually killed.

Although the practice of auctions has lasted for thousands of years, some of which even produced a huge historical impact, scientific research on auctions based on rigorous economic theory did not begin until the 1960s. The first work was Vickrey's 1961 paper, "Counterspeculation, Auctions, and Competitive Sealed Tenders" published in *Journal of Finance* (for William Vickrey' biography, see Section 21.7.1). In this classic paper, he discussed the four basic auction formats for a single item. He provided a landmark conclusion on the development of the auction theory, so-called the Revenue Equivalence Theorem, and proved that all of the bidders will bid truthfully in the second-price sealed-bid auction.

Four Basic Auctions

There are many kinds of auction formats, among which four auction formats are basic and studied, while others constitute their variations.

The first is the **English auction** (also called the **ascending-price auction**). Under this format, the price of an object is *publicly* called, and it shall increase until there is a player who is willing to pay the highest price. Then, the bidder wins the object and pays the bid. One variation of this auction is that all of the bidders bid, and the price increases until only one bidder bids and pays her bid. This format is frequently used in live auctions, especially regarding cultural relics, calligraphy and paintings, second-hand commodities, and commercial land owned by the government.

The second is the **Dutch auction** (also called the **descending-price auction**). The Dutch auction is opposite to the English auction, where the price goes from high to low. The auctioneer calls the price at a high level; if there is no bidder who wants to buy the object, the auctioneer reduces the price in accordance with the predetermined range until some bidder wants to accept it. When there is a bidder who accepts the price, she wins the object at the price. Although the price of this auction goes from high to low, the winner is still the bidder whose bid is the highest. In Europe, and especially the Netherlands, auctions for flowers always take this form.

The third is the **first-price sealed-bid auction** or simply called the **first-price auction**. Under this rule, each bidder submits her bid document to the auctioneer in sealed form in the specified time independently, and indicates the price that she wishes to pay. As a consequence, the bids of other bidders cannot be seen. The auctioneer then invites all bidders to open their bids on-site at a specified time. Then, the bidder with the highest bid wins the object by her bid. Therefore, the first-price auction is also called the **high price auction, bidding auction** or the **mail auction**.

The fourth is the **second-price sealed-bid auction** or simply called the **second-price auction**, which is also called the **Vickrey auction**. Under this format, the bid is also sealed, but after opening the bid, the bidder whose bid is the highest wins the object and pays the second-highest bid. This form is not often used in practice, but it has a good theoretical basis, which is the unique truthful-telling auction format. This auction has been discussed in Chapter 18 as a special case of the VCG mechanism. It was first proposed by Vickrey in 1961, and economists began to make an in-depth study of auctions since that time.

In each format of these auctions, if several bidders bid the same with the highest price, then the ties shall be decided by lot. The introduction of these auctions is only for a single-unit item auction, and we will also discuss auctions of multi-unit items. Their auction formats are different, but are essentially variants of the above four basic auction formats.

It can be seen from the above introduction of the four basic auctions that the auction mainly adopts two major formats: the first two are open outcry formats, such as for antiques, calligraphy and paintings, and second-hand goods (e.g., used cars), which require that the bidders assemble in the same place; whereas, the latter two are sealed auctions, which may be submitted by mail. Thus a bidder may observe the behavior of other bidders in one format and not in the other. In many auctions or bidding activities, the confidentiality of participants' business secrets should be ensured, and thus the auction or bidding formats adopted should avoid commercial leaks. Compared with open outcry formats, sealed price formats possess an advantage in this respect, and thus the use of sealed price bidding is necessary.

However, for rational decision-makers, some of these differences are superficial. The Dutch open descending price auction is strategically equivalent to the first-price sealed-bid auction, in the sense that they have the same equilibrium strategies. When values are private, the English open ascending auction is also outcome-equivalent to the second-price sealed-bid auction in equilibrium outcome, in a weak sense that they have the same equilibrium outcomes. This is because in an English auction, a winner should obtain a positive payoff only if the bid is less than her valuation of the object, and thus is slightly higher than the value of the second-highest bidder.

In addition, sellers often impose two restrictions for each auction format. One is to set a base (minimal) price, which is called the **reserve price**; the other is the charge for bidding. Under first-price and second-price sealed-bid auction formats, a bidder must bid higher than, or equal to, the reserve price; otherwise, no transaction will be made. Concerning the second-price auction, if there is only one bidder bidding and bids above the reserve price, she shall get the item and pays the reserve price. The reserve price has a similar effect in both the ascending-price and the descending-price auctions.

There are two basic issues to be discussed in auction theory:

(1) **Efficiency**: can the resources be efficiently allocated using these auctions?

(2) **Optimality**: which auction format can make the highest revenue for the auctioneer?

To answer these two questions, it is necessary to set up the basic analytical framework, including a description of bidders' preferences and the information structure – **private value**, **interdependent value**, or **common value** (a special case of interdependent value). Then, we will analyze their bidding strategies to see if they lead to efficient outcomes, and which one is optimal by comparing the auctioneer's expected revenue among the auction formats under consideration.

In the following, we will first examine auctions with private values and focus primarily on the symmetric, risk-neutral, unlimited liability, private value economic environment with a single object. We then discuss auctions under the context of interdependent values for a single object and auctions for multi-unit objects. Subsequently, we will examine simultaneous and sequential auction of multiple items, and finally we will explore combinational auctions. As an auction format is a special case of incomplete information mechanism design discussed in Chapter 19, the solution concept adopted is mainly the Bayesian-Nash equilibrium. In order to make it easier to understand, we attempt to employ the same terms and notations as those in Chapters 18-20. Discussions in this chapter are significantly drawn from the classic monograph by Krishna (2010).

21.2 Single Object Auctions with Private Values

21.2.1 Basic Analytical Framework

We consider the private value benchmark model of auction theory, which usually includes the following six assumptions:

1. **Private Value**: The values of the item by bidders is private information;
2. **Independence**: The values of bidders are independent;
3. **Symmetry**: The values of bidders have the same probability distribution;
4. **Risk Neutrality**: Bidders are risk-neutral;
5. **Unlimited liability**: Bidders have no budget constraints, and have the ability to pay the bidding price;
6. **Non-collusion**: All buyers decide on their own bidding strategy independently, and there is no binding cooperative agreement.

We thus get the **Symmetric Independent Private Value** (in short, **SIPV) model**. We focus on the solution concept of symmetric Bayesian-Nash equilibrium and discuss relevant properties, although there may be asymmetric equilibrium solutions for general utility functions.

Consider an auction with an indivisible object owned by the auctioneer (seller) and n risk-neutral bidders (buyers). Every bidder i has a private value of the object, denoted by θ_i, the type of the bidder. The auction model can then be considered as a special case of the indivisible object model that we studied in Chapters 18-19: $Y = \{y \in \{0,1\}^n : \sum_i y_i = 1\}$, and the payoff of bidder i can be written as

$$\theta_i y_i + t_i,$$

where t_i is agent i's transfer, or in terms of payment to the auctioneer, it is

$$m_i = -t_i.$$

The value θ_i for buyer i is a random variable distributed on $[\underline{\theta}_i, \bar{\theta}_i]$ with independent density $\varphi_i(\cdot) > 0$, where $\underline{\theta}_i < \bar{\theta}_i$, and the cumulative distribution function is denoted by $\Phi_i(\cdot)$.

We first examine equilibrium outcomes of the four typical auctions under symmetric scenarios, and discuss the forms and properties of the Bayesian-Nash equilibrium, such as validity. Then, we compare the seller's expected revenue under these auctions.

In private value auctions, first-price (sealed-bid) auction and Dutch auction have the highest price called or accepted as the auction price, and then they are strategically equivalent. In an English auction, a bidder will obtain a positive payoff from participation if and only if she can win the auction with a bid that is less than her valuation of the object. Therefore, the bidder with the highest value wins the auction and pays an amount slightly higher than the value of the second-highest bidder. This is why the English auction is sometimes referred to as an **open second-price auction**. Therefore, in both the second-price (sealed-bid) auction and the English auction, the second-highest price called or accepted is the auction price, and thus the two auctions are outcome-equivalent. However, this equivalence applies only for private values, or if there are only two bidders.

Therefore, without loss of generality, we only need to discuss the first-price (sealed-bid) auction and the second-price (sealed-bid) auction for the private value environment.

Below, we assume that all bidders' private values are symmetric, i.e., for any i, there is $\theta_i \sim \varphi(\cdot)$ distributed on $[0, w]$ with distribution function $\Phi(\cdot)$. We consider symmetric Bayesian-Nash equilibrium, in which we focus on bidder 1's strategic choices.

21.2.2 First-Price (Sealed-Bid) Auctions

In a first-price auction, given bidder i's value θ_i, if her bid is b_i and other bidders' bids are $b_j, j \neq i$, then bidder i's ex-post utility is of the following form:

$$U_i = \theta_i y_i(b_i, \boldsymbol{b}_{-i}) + t_i(b_i, \boldsymbol{b}_{-i}),$$

or in terms of payment to the auctioneer,

$$U_i = \theta_i y_i(b_i, \boldsymbol{b}_{-i}) - m_i(b_i, \boldsymbol{b}_{-i}),$$

where the allocation rule is given by

$$y_i(b_i, \boldsymbol{b}_{-i}) = \begin{cases} 1 & \text{if } b_i > \max_{j\neq i} b_j \\ 0 & \text{if } b_i < \max_{j\neq i} b_j, \end{cases} \tag{21.2.1}$$

and the payment is given by

$$m_i(b_i, \boldsymbol{b}_{-i}) = \begin{cases} b_i & \text{if } b_i > \max_{j\neq i} b_j \\ 0 & \text{if } b_i < \max_{j\neq i} b_j. \end{cases} \tag{21.2.2}$$

If $b_i = \max_{j\neq i} b_j$, the ties shall be decided by lot, i.e., the object is randomly assigned by the same probability.

Consider a symmetric equilibrium bidding function, denoted by $\beta^I(\cdot)$, which is a mapping $\beta^I : [0, \omega] \to [0, \infty)$. Without loss of generality, suppose $i = 1$. Let $\vartheta_1, \ldots, \vartheta_{n-1}$ be the order from the largest to the smallest of $\theta_2, \ldots, \theta_n$, or be called the 1st, ..., $n-1$th sequential order statistics of $\theta_2, \ldots, \theta_n$. Suppose that $\beta^I(\cdot)$ is an increasing function, as we will show later.

When other bidders choose strategy $\beta^I(\cdot)$, then only if $b_1 > \beta^I(\vartheta_1)$, bidder 1 can get the object; otherwise, she would not. Since ϑ_1 is a continuous random variable, the probability of $b_1 = \beta^I(\vartheta_1)$ is zero, and thus we can omit the case. As $\theta_j, j \neq 1$ are all distributed according to the density function φ (the corresponding distribution function is $\Phi(\cdot)$), it is easy to obtain that ϑ_1's distribution function is $\Psi(\cdot) = \Phi^{n-1}(\cdot)$ (the corresponding density function is ψ).

Given that other bidders choose strategies $\beta(\cdot) = \beta^I(\cdot)$, the interim expected utility of bidder 1 with value θ and bid b is then:

$$E_{\boldsymbol{\theta}_{-1}} U_1 = \Psi(\beta^{-1}(b))(\theta - b),$$

where $\beta^{-1}(\cdot)$ is the inverse function of $\beta(\cdot)$.

The first-order condition for b is:

$$\frac{\psi(\beta^{-1}(b))(\theta - b)}{\beta'(\beta^{-1}(b))} - \Psi(\beta^{-1}(b)) = 0.$$

At equilibrium, $b = \beta(\theta)$, we then have:

$$\beta'(\theta) = \frac{\psi(\theta)}{\Psi(\theta)}(\theta - \beta(\theta)), \tag{21.2.3}$$

or equivalently,

$$\frac{d}{d\theta}(\Psi(\theta)\beta(\theta)) = \theta\psi(\theta). \tag{21.2.4}$$

Since $\beta(0) = 0$, the solution of first-order differential equation (21.2.3) is:

$$\beta(\theta) = \frac{1}{\Psi(\theta)} \int_0^\theta \vartheta_1 \psi(\vartheta_1) d\vartheta_1 = E[\vartheta_1 | \vartheta_1 < \theta].$$

Obviously $\beta(\theta) < \theta$. From the differential equation (21.2.3), we obtain $\beta'(\cdot) > 0$, i.e., the bidding function $\beta(\cdot)$ is an increasing function. We then have the following proposition:

Proposition 21.2.1 *With a symmetric independent private value, the symmetric Bayesian-Nash equilibrium bidding function of the first-price auction is*

$$\beta^I(\theta) = \frac{1}{\Psi(\theta)} \int_0^\theta \vartheta_1 \psi(\vartheta_1) d\vartheta_1 = E[\vartheta_1 | \vartheta_1 < \theta].$$

PROOF. We obtain the first-order necessary condition for a bidding strategy to be a Bayesian-Nash equilibrium from the above argument, and in the following we only need to show the sufficiency of the proposition.

If other bidders all choose $\beta = \beta^I$, under θ, bidder i chooses bid $\beta(\tilde{\theta})$, and then her interim expected utility is:

$$\begin{aligned}
\bar{U}_i(\theta, \tilde{\theta}) &\equiv E_{\boldsymbol{\theta}_{-1}}[\theta y_i(\beta(\tilde{\theta}), \beta_{-i}(\boldsymbol{\theta}_{-i})) + t_i(\beta(\tilde{\theta}), \beta_{-i}(\boldsymbol{\theta}_{-i}))] \\
&= E_{\boldsymbol{\theta}_{-1}}[\theta y_i(\beta(\tilde{\theta}), \beta_{-i}(\boldsymbol{\theta}_{-i})) - m_i(\beta(\tilde{\theta}), \beta_{-i}(\boldsymbol{\theta}_{-i}))] \\
&= \Psi(\tilde{\theta})[\theta - \beta(\tilde{\theta})] \\
&= \Psi(\tilde{\theta})\theta - \Psi(\tilde{\theta})E[\vartheta_1 | \vartheta_1 < \tilde{\theta}] \\
&= \Psi(\tilde{\theta})\theta - \int_0^{\tilde{\theta}} \vartheta_1 \psi(\vartheta_1) d\vartheta_1 \\
&= \Psi(\tilde{\theta})\theta - \Psi(\tilde{\theta})\tilde{\theta} + \int_0^{\tilde{\theta}} \Psi(\vartheta_1) d\vartheta_1 \text{ (integration by parts)} \\
&= \Psi(\tilde{\theta})(\theta - \tilde{\theta}) + \int_0^{\tilde{\theta}} \Psi(\vartheta_1) d\vartheta_1.
\end{aligned}$$

Then, irrespective of whether $\theta \geqq \tilde{\theta}$ or $\theta \leqq \tilde{\theta}$, we have:

$$\bar{U}_i(\theta, \theta) - \bar{U}_i(\theta, \tilde{\theta}) = \Psi(\tilde{\theta})(\tilde{\theta} - \theta) + \int_{\theta}^{\tilde{\theta}} \Psi(\vartheta_1) d\vartheta_1 \geqq 0.$$

□

Using the formula of integration by parts, the Bayesian-Nash equilibrium bidding function can be also written as:

$$\beta^I(\theta) = \theta - \frac{1}{\Psi(\theta)} \int_0^{\theta} \Psi(\vartheta_1) d\vartheta_1 < \theta, \tag{21.2.5}$$

which again means that the bidder bids less than her true value. In other words, no bidders have an incentive to reveal their true values.

Since $\beta^I(\theta)$ is a monotonically increasing function, the equilibrium bid has a positive correlation with the private value (i.e., for high private value, the equilibrium bid is also high), but it is always less than the private value. However, because

$$\frac{\Psi(\vartheta)}{\Psi(\theta)} = \left[\frac{\Phi(\vartheta)}{\Phi(\theta)}\right]^{N-1},$$

when the number of bidders N increases, which means it is more competitive, the Bayesian-Nash equilibrium solution $\beta^I(\theta)$ converges to θ.

Example 21.2.1 Suppose that that values are exponentially distributed on $[0, \infty)$, and there are only two bidders. If $\Phi(\theta) = 1 - \exp(-\lambda\theta),\ \lambda > 0$, then

$$\begin{aligned}
\beta^I(\theta) &= \theta - \int_0^{\theta} \frac{\Phi(\vartheta)}{\Phi(\theta)} d\vartheta \\
&= \frac{1}{\lambda} - \frac{\theta \exp(-\lambda\theta)}{1 - \exp(-\lambda\theta)}.
\end{aligned}$$

21.2.3 Second-Price (Sealed-Bid) Auctions

Now, we explore the equilibrium solution of second-price (sealed-bid) auction. Denote the highest bid for other bidders by $\bar{b}_{(i)} \equiv \max_{j \neq i} b_j$, where b_j is bidder j's bid. For bidder i with private value θ_i, the ex-post utility of the bid b_i is

$$U_i = \theta_i y_i(b_i, \bar{b}_{(i)}) - m_i(b_i, \bar{b}_{(i)}),$$

where

$$y_i(b_i, \bar{b}_{(i)}) = \begin{cases} 1 & \text{if } b_i > \bar{b}_{(i)} \\ 0 & \text{if } b_i < \bar{b}_{(i)}, \end{cases} \tag{21.2.6}$$

$$m_i(b_i, \bar{b}_{(i)}) = \begin{cases} \bar{b}_{(i)} & \text{if } b_i > \bar{b}_{(i)} \\ 0 & \text{if } b_i < \bar{b}_{(i)}. \end{cases} \tag{21.2.7}$$

The following proposition gives the properties of the Bayesian-Nash equilibrium bidding function β^{II} for a second-price sealed-bid auction.

Proposition 21.2.2 *For a second-price (sealed-bid) auction, every bidder' Bayesian-Nash equilibrium bidding function $\beta^{II}(\theta) = \theta$ is a weakly dominant strategy.*

PROOF. This result can be obtained by applying the Vickrey-Clarke-Groves mechanism directly. We present a direct proof of this.

When $\theta > \bar{b}_{(i)}$, bidding $b_i > \bar{b}_{(i)}$ and the true value θ_i bring the same revenue $\theta_i - \bar{b}_{(i)} > 0$; but when $b_i < \bar{b}_{(i)}$, the opportunity to win is lost, so that the revenue is less than that brought by bidding the true value θ_i. As a consequence, when $\theta_i > \bar{b}_{(i)}$, $\beta^{II}(\theta_i) = \theta_i$ is a weakly dominant strategy.

When $\theta_i \leqq \bar{b}_{(i)}$, bidding $b_i \leqq \bar{b}_{(i)}$ and the true value θ_i bring the same revenue 0; but when the bidding price $b_i > \bar{b}_{(i)}$, the revenue $\theta_i - \bar{b}_{(i)} < 0$ is smaller than the payoff of bidding θ_i. Therefore, when $\theta_i \leqq \bar{b}_{(i)}$, $\beta^{II}(\theta_i) = \theta_i$ is also a weakly dominant strategy.

Therefore, for any $\bar{b}_{(i)}$, to truthfully reveal θ_i is a weakly dominant strategy. □

Consequently, in a second-price sealed-bid auction, all of the bidders bid truthfully; whereas, in a first-price sealed-bid auction, bidders' bids are lower than their true values. The intuition of this conclusion is simple. When using a first-price sealed-bid auction, if a bidder bids her true value of the object, then winning the auction is unprofitable. In order to obtain potential profits, bidders have an incentive to submit prices lower than their true values. On the other hand, , this issue can be well solved in the second-price sealed-bid auction.

21.2.4 Revenue Comparison of First- and Second-Price Auctions

As mentioned above, Vickrey (1961) compared the four auction forms and obtained the landmark theory of auction - the “Revenue Equivalence Theorem”. We now compare the expected revenue of auctioneers with the first-price auction and the second-price auction under symmetric conditions.

For the first-price auction, the interim expected expenditure of bidder 1 with private value θ paid to the auctioneer is:

$$\begin{aligned} m^I(\theta) &\equiv E_{\boldsymbol{\theta}_{-1}} m_1^I(\boldsymbol{\theta}) \\ &= \Psi(\theta)\beta^I(\theta) = \Psi(\theta)E[\vartheta_1|\vartheta_1 < \theta], \end{aligned}$$

where $\Psi(\theta)$ is the probability that bidder 1 wins the auction.

Therefore, the auctioneer's ex-ante expected revenue from bidder 1 is:

$$\begin{aligned} Em^I(\theta) &= \int_0^w \Psi(\theta)\beta^I(\theta)d\theta \\ &= \int_0^w (\int_0^\theta \vartheta_1\psi(\vartheta_1)d\vartheta_1)\varphi(\theta)d\theta \\ &= \int_0^w (\int_{\vartheta_1}^w \varphi(\theta)d\theta)\vartheta_1\psi(\vartheta_1)d\vartheta_1 \\ &= \int_0^w (1-\Phi(\vartheta_1))\vartheta_1\psi(\vartheta_1)d\vartheta_1, \end{aligned}$$

where the third equality is obtained by interchanging the order of integration.

Thus, the ex-ante expected revenue that an auctioneer receives from n bidders is:

$$\begin{aligned} E_{\boldsymbol{\theta}} R^I &= nEm^I(\theta) \\ &= \int_0^w n(1-\Phi(\vartheta_1))\vartheta_1\psi(\vartheta_1)d\vartheta_1. \end{aligned}$$

For the second-price auction, the revenue of the auctioneer is the second-highest private value of n bidders, denoted by $\vartheta_2^{(n)}$, and its distribution function is

$$\Phi^n(\vartheta_2^{(n)}) + n\Phi^{n-1}(\vartheta_2^{(n)})(1-\Phi(\vartheta_2^{(n)})).$$

The density function is:

$$n(n-1)(1-\Phi(\vartheta_2^{(n)}))\Phi^{n-2}(\vartheta_2^{(n)})\varphi(\vartheta_2^{(n)}) = n(1-\Phi(\vartheta_2^{(n)}))\psi(\vartheta_2^{(n)}).$$

Therefore, in the second-price auction, the auctioneer's ex-ante expected revenue is:

$$\begin{aligned} E_{\boldsymbol{\theta}} R^{II} &= \int_0^w n\vartheta_2^{(n)}(1-\Phi(\vartheta_2^{(n)}))\psi(\vartheta_2^{(n)})d\vartheta_2^{(n)} \\ &= E_{\boldsymbol{\theta}} R^I. \end{aligned}$$

Therefore, in a symmetric environment with private value, the ex-ante expected revenues of the auctioneer in the first-price auction and the second-price auction are identical. In fact, we will obtain a more general revenue equivalence result that all auctions, which satisfy the same allocation rule (i.e., the highest bid wins the object) and have the same interim expected utility of the bidder at the lowest private value, generate the same expected revenue for the auctioneer.

In order to obtain the specific expression of the auctioneer's expected revenue, from (21.2.5), we have the first-price auction bid:

$$\beta^I(\theta) = \theta - \frac{1}{\Psi(\theta)} \int_0^{\theta} \Psi(\vartheta_1) d\vartheta_1. \tag{21.2.8}$$

By applying integration by parts to the second part of the above equation, we have

$$E\beta^I(\theta) = \int_0^w \nu(\theta') d\Phi^n(\theta'),$$

where

$$\nu(\theta) = \theta - \frac{1 - \Phi(\theta)}{\varphi(\theta)},$$

which can be interpreted as the bidder's **virtual valuation function** that we first discussed in Chapter 16 and then Chapter 19.

Thus, we obtain the specific expression of the auctioneer's expected revenue:

$$E_{\boldsymbol{\theta}}(m^I) = E_{\boldsymbol{\theta}}(m^{II}) = \int_0^w \nu(\theta') d\Phi^n(\theta').$$

In auction theory and mechanism design theory, $\nu(\theta)$ given above plays an important role. Since θ is the buyer's private information, she would tend to under-report this information when dealing with the seller to have information rent. How much the buyer can under-report depends on the seller's knowledge about θ, which is the distribution function of θ. From the seller's point of view, $\nu(\theta)$ is the highest price that the buyer is willing to pay due to information asymmetry and adjustment. In the subsection on the design of the optimal auction below, one will see that $\nu(\theta)$ can be regarded as the seller's (or auctioneer's) marginal revenue.

21.2.5 Reserve Price in First- and Second-Price Auctions

The above discussion assumes that the auctioneer has no value on the object and can sell it at any price. In many situations, the auctioneer may have a value on the object, denoted by $r > 0$ and called **reserve price**. Since a winning bid can never be lower than r, no bidder with a value $\theta < r$ can win.

In a second-price auction with reserve price r, the reserve price makes no difference to the bidding behavior of bidders, and thus all bidders' Bayesian-Nash equilibrium bidding strategy is still

$$\beta^{II}(\theta) = \theta, \tag{21.2.9}$$

which is a weakly dominant strategy, and then the expected payment is

$$m^{II}(\theta, r) = \begin{cases} r\Psi(r) & \text{if } \theta = r, \\ r\Psi + \int_r^\theta \vartheta_1 \psi(\vartheta_1')d\vartheta_1' & \text{if } \theta > r. \end{cases} \tag{21.2.10}$$

In a first-price auction with reserve price r, a bidder with value $\theta = r$ wins the object only if all other bidders have values less than r and pays r. When $\theta > r$, the analysis is the same as previously. Then, for a first-price auction with reserve price r, the symmetric equilibrium bidding strategy for any bidder is given by

$$\beta^{I}(\theta) = \begin{cases} r & \text{if } \theta = r, \\ r\frac{\Psi(r)}{\Psi(\theta)} + \frac{1}{\Psi(\theta)}\int_r^\theta \vartheta_1 \psi(\vartheta_1)d\vartheta_1 & \text{if } \theta > r. \end{cases} \tag{21.2.11}$$

and the expected payment of a bidder with a reserve price r is

$$m^{I}(\theta, r) = \begin{cases} r\Psi(r) & \text{if } \theta = r, \\ r\Psi + \int_r^\theta \vartheta_1 \psi(\vartheta_1')d\vartheta_1' & \text{if } \theta > r, \end{cases} \tag{21.2.12}$$

which is the same as in the second-price auctions with a reserve price. Therefore, the revenue equivalence between the first- and second-price auctions remains true even with a reserve price.

21.2.6 Efficient Allocation and Revenue-Equivalence Principle

From the previous discussion, we know that if all of the bidders' bids are given independently in all of the four auction formats, the auctioneer can obtain the same expected revenue, regardless of auction form. The four auction formats all share some common features, in that bidders are asked to give their bidding prices that they are willing to pay, which determines who will receive the object and how much they will pay. Since bidders' equilibrium bidding functions are both monotonic in the first-price and second-price auctions, and both are auctions in which the highest bids win, the resulting outcomes are efficient, and we have the following theorem:

Theorem 21.2.1 (Allocative Efficiency Theorem) *In symmetric independent private value (SIPV) environments, the first-price and second-price auctions result in efficient allocations of an object (the one with the highest value gets the object).*

We will see that when symmetry is not satisfied, the second-price auction is still efficient, but the first-price auction may not be efficient.

Another notable feature is that when the distribution functions are different, the bids are varied and even not strategically-equivalent, the auctioneer receives the same expected revenue. As such, to what extent are the results still valid?

Myerson (for his biography, see Section 19.10.2) utilized the mechanism design approach to investigate this issue. On this basis, he extended Vickrey's theory and proved that: under the assumption that the bidders' values of the object are independent, and that the bidders only care about their expected payments, all standard auctions will bring the same expected revenue to the auctioneer. This result is significant, making auction theory extend one great step further.

We call an auction **standard** if the bidder with the highest bid gets the object. An example of a non-standard form of auction is to buy a lottery ticket. The chance of a bidder's winning is related to the ratio of the bid amount to the total bid amount of all of the bidders. It is non-standard because it does not necessarily guarantee obtaining the object despite the highest bidding.

In addition to the four common standard auctions, another example of a standard auction is the **all-pay auction**, where every bidder pays according to her bid, but only the bidder with the highest bid wins the object. For example, lobbying for the government to adopt a policy belongs to such auction formats. We will see any all-pay auction is also efficient. For all of these standard auctions, do they still result in the same revenue to the auctioneer? Myerson's Revenue Equivalence Theorem provides an affirmative answer.

We now examine the question of whether the auctioneer's expected revenues for all standard auctions are the same. Still consider a linear model that allocates an indivisible object among n risk-neutral buyers:

$$Y = \{y \in \{0,1\}^n : \sum_i y_i = 1\},$$

buyer i's revenue is

$$\theta_i y_i + t_i,$$

and the value is a random variable with individually independent density $\varphi_i(\cdot) > 0$ defined on $[\underline{\theta}_i, \bar{\theta}_i]$.

Buyer i's utility then is

$$U_i(\boldsymbol{\theta}) = \theta_i y_i(\boldsymbol{\theta}) + t_i(\boldsymbol{\theta}).$$

Under an auction rule (or more general social choice in a linear environment) $(y(\cdot), t_1, \cdots, t_n)$, the auctioneer's expected payoff is can be written as

$$-\sum_i E_{\boldsymbol{\theta}} t_i(\boldsymbol{\theta}) = \sum_i E_{\boldsymbol{\theta}}[\theta_i y_i(\boldsymbol{\theta}) - U_i(\boldsymbol{\theta})] = \sum_i E_{\boldsymbol{\theta}}[\theta_i y_i(\boldsymbol{\theta})] - \sum_i E_{\theta_i} E_{\boldsymbol{\theta}_{-i}}[U_i(\boldsymbol{\theta})].$$

In the above equation, the first item on the right side is the expected total surplus for all bidders, while the second item is the expected utility for all bidders. According to the Bayesian Incentive Compatibility Characterization Theorem (Proposition 19.4.2) provided in Chapter 19, the second item is given by

$$E_{\boldsymbol{\theta}_{-i}} U_i(\boldsymbol{\theta}) = E_{\boldsymbol{\theta_{-i}}}[U_i(\underline{\theta}_i, \boldsymbol{\theta}_{-i})] + \int_{\underline{\theta}_i}^{\theta_i} E_{\boldsymbol{\theta}_{-i}} y(\tau, \boldsymbol{\theta}_{-i}) d\tau,$$

and thus it can be completely determined by allocation rule $y(\cdot)$ and the lowest type of interim expected utility $E_{\boldsymbol{\theta}_{-i}}[U_i(\underline{\theta}_i, \boldsymbol{\theta}_{-i})]$. Since the total surplus remains unchanged (all mechanisms implement the same decision rule), we have the following **Revenue-Equivalence Theorem**.

Theorem 21.2.2 (Myerson's Revenue-Equivalence Theorem) *Suppose that in SIPV environments, two different auctions both have symmetric Bayesian-Nash equilibrium, and at the equilibrium: (1) the same decision (object allocation) rule $y(\cdot)$ is implemented; and (2) all buyers have the same interim expected utilities at the lowest value $\underline{\theta}_i$. Then, the Bayesian-Nash equilibrium outcomes of these two auctions result in the same revenue to the auctioneer.*

This theorem shows that there are many ways to obtain interim expected transfers $\bar{t}_i(\theta_i)$. Even though the decision rule and the utility at the lowest-type is fixed, the seller possesses great degree of freedom in designing an auction scheme.

For example, suppose that the buyers are symmetric (i.e., they have the same distribution function). The seller wants to implement efficient decision rule $y(\cdot)$, and the interim expected utilities of buyers with the lowest value are zero. We can use the second-price sealed-bid auction to truthfully implement the efficient decision rule or the first-price auction to Bayesian implement the efficient decision rule. More generally, consider the k-th price auction, where $1 \leqq k \leqq n$, and the highest bidder gets the object and pays the k-th highest bid. Suppose that the value of every buyer on the object is an independently and identically distributed random variable. Then, we can prove that the auction has an equilibrium that is unique and symmetric, and the bid of each participant $b(\theta_i)$ is an increasing function of her value (see Fudenberg-Tirole's *Game Theory*, pp: 223-225.) Since the bidder with the highest bid gets the object, the k-th price auction implements the efficient outcome, and thus is an efficient decision rule. Meanwhile, the probability of the buyer with the lowest value $\underline{\theta}$ getting the object is zero, and thus her interim expected utility is zero. As a consequence, by the Revenue Equivalence Theorem, for any k, k-th price auction brings the same revenue to sellers. All-pay auction and k-th price auction also bring the same revenue to sellers.

For a given bidding scheme, it seems that the auctioneer will get a higher revenue when k is smaller. How do the above results hold? The answer is that the winning bid of a bidder at equilibrium would be smaller when k is small, but the price paid may not be. For example, we know that the bidder's bid is the true value under a second-price auction, but the price paid is the second-highest bid. In a first-price auction, a buyer's bid is less than her true value because the bid equal to her true value would bring zero expected utility. Myerson's Revenue-Equivalence Theorem shows that, for both auctions, the seller's expected revenue is the same. In particular, we obtain this equivalence without solving the auction equilibrium. When $k > 2$, the Revenue Equivalence Theorem shows that a bidder's bid is greater than her true value (but since the price paid is the k-th ($1 \leqq k \leqq n$) highest price, it is not greater than her true value).

21.2.7 Applications of Revenue-Equivalence Principle

There are many auction formats in theory and practice. In addition to the four common auction formats discussed above, there are also all-pay auctions, k-th price auction, auctions with an uncertain number of bidders, and other auction formats. These auctions have numerous applications in practice. For example, an all-pay auction can be applied to attrition competition in industry, in which an enterprise that lasts until the end of the auction will win. However, these enterprises, including enterprises that quit midway, have paid certain costs in the process. In addition, an all-pay auction can also be applied to lobbying activities of interest groups in politics. Although the $k \geqq 3$-th price auction is rarely used in practice, such auctions have theoretical significance. We find that bidders may bid more than their own value in a third-price auction. Indeed, there are some fine balances of interests. In addition, for online auctions in which the number of bidders is uncertain, how will each bidder choose her own strategy? Therefore, these auction formats have great theoretical and practical significance. However, in these auctions, the solution to bidding equilibrium tends to be relatively complicated; whereas, applying Myerson's Revenue Equivalence Theorem provides some shortcuts to the solution process. The following are such examples drawn from Krishna (2002).

All-Pay Auctions

An all-pay auction is similar to first-price and second-price auctions in which a bidder with the highest bid gets the object. The difference, however, lies in that each bidder, regardless of whether she obtains the object eventually, pays the auctioneer at her own bidding price. Let $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_n)$ be n types of bidders independently distributed according to $\varphi(\cdot)$ over $[0, w]$, $\boldsymbol{b} = (b_1, \ldots, b_n)$ be the bidding vector, and bidder i's utility be $U_i = \theta_i y_i(b_i, \mathbf{b}_{-i}) - m_i(b_i, \mathbf{b}_{-i})$, where:

$$y_i(b_i, b_{-i}) = \begin{cases} 1 & \text{if } b_i > \max_{j \neq i} b_j \\ 0 & \text{if } b_i < \max_{j \neq i} b_j, \end{cases} \tag{21.2.13}$$

where, if $b_i = \max_{j \neq i} b_j$, with the same probability the object is randomly allocated and

$$m_i(b_i, b_{-i}) = b_i. \tag{21.2.14}$$

As previously, $\Psi(\cdot)$ is the distribution of ϑ_1 (the first-order statistics of other participant types). We focus on the case in which symmetric equilibrium $\beta(\cdot) = \beta^{AP}(\cdot)$ is a monotonic function (which will be further proved). When $\theta = 0$, it is obvious that bidders will choose the bidding price 0, and at the same time the bidder with the highest bids wins the object. According to the Revenue Equivalence Theorem, at equilibrium, this auction has the same expected utility as the first-price auction and the second-price auction, which can be used to easily find the equilibrium bidding strategy of the all-pay auction.

Indeed, when bidder i's value is θ, her interim expected utility under the all-pay auction at equilibrium is:

$$\Pi_i^{AP}(\beta(\tilde{\theta}), \theta) = \Psi(\tilde{\theta})\theta - \beta^{AP}(\theta),$$

and the interim expected utility at equilibrium under the second-price auction is:

$$\Pi_i^{II}(\beta(\tilde{\theta}), \theta) = \Psi(\tilde{\theta})\theta - \Psi(\tilde{\theta})\frac{\int_0^\theta \vartheta_1 \psi(\vartheta_1) d\vartheta_1}{\Psi(\tilde{\theta})}.$$

Therefore, making the above two equations equal, we obtain a symmetric equilibrium bidding strategy for an all-pay auction:

$$\beta^{AP}(\theta) = \int_0^\theta \vartheta_1 \psi(\vartheta_1) d\vartheta_1,$$

which is obviously a strictly increasing function, and therefore it is efficient.

Third-price Auctions

Consider a third-price auction. We use the previous approach to find the equilibrium bidding strategy. Let $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_n)$ be n types of bidders independently distributed according to $\varphi(\cdot)$ on $[0, w]$, and $\boldsymbol{b} = (b_1, \ldots, b_n)$ be the bidding vector. Then, bidder i's payoff is

$$U_i = \theta_i y_i(b_i, \mathbf{b}_{-i}) - m_i(b_i, \mathbf{b}_{-i}),$$

which satisfies:

$$y_i(b_i, b_{-i}) = \begin{cases} 1 & \text{if } b_i > \max_{j \neq i} b_j \\ 0 & \text{if } b_i < \max_{j \neq i} b_j, \end{cases} \tag{21.2.15}$$

$$m_i(b_i, b_{-i}) = \begin{cases} b_{(3)} & \text{if } b_i > \max_{j \neq i} b_j \\ 0 & \text{if } b_i < \max_{j \neq i} b_j. \end{cases} \tag{21.2.16}$$

where $b_{(3)}$ is the third-highest bidding price.

In a symmetric Bayesian-Nash equilibrium under the third-price auction, like first-price and second-price auctions, the highest bidder obtains the object; meanwhile, the bid of the bidder with $\theta = 0$ is zero, and the payoff is zero.

Let $\beta^{III}(\cdot)$ be the third-price Bayesian-Nash equilibrium bidding function, which is assumed to be an increasing function. In a symmetric environment, consider bidder 1's strategy choice. It is obvious that only when $\theta_1 = \theta > \vartheta_1$, bidder 1 can win the object. Consider ϑ_2 to be the second-order statistics of $\{\theta_2, \ldots, \theta_n\}$. At equilibrium, bidder 1's interim expected payment for winning the object is $E_{\vartheta_2}[\beta^{III}(\vartheta_2)|\vartheta_1 \leqq \theta]$. By the Revenue Equivalence Theorem, we have:

$$\Psi_1(\theta)^{(n-1)} E_{\vartheta_2}[\beta^{III}(\vartheta_2)|\vartheta_1 \leqq \theta] = \int_0^\theta \vartheta_1 \psi(\vartheta_1) d\vartheta_1.$$

Let $\Psi_k^{(m)}$ be the distribution function of the k-th order statistics in m i.i.d. random variables, and $\psi_k^{(m)}$ be the corresponding density function. $(\vartheta_2|\vartheta_1 \leqq \theta)$'s density function is:

$$\psi_2^{(n-1)}(\vartheta_2 = \tau|\vartheta_1 \leqq \theta) = \frac{1}{\Psi_1(\theta)^{(n-1)}}(n-1)[\Phi(\theta) - \Phi(\tau)]\psi_1(\tau)^{(n-2)},$$

where $(n-1)[\Phi(\theta) - \Phi(\tau)]$ is the probability of ϑ_1's value in $[\tau, \theta]$, and $\psi_1(\tau)^{(n-2)}$ is the density function of the first-order statistics in $n-2$ i.i.d. random variables.

Therefore, for the bidder with $\theta_1 = \theta$, the interim expected payment is:

$$\begin{aligned} &\Psi_1(\theta)^{(n-1)} \int_0^\theta \beta^{III}(\tau)\psi_2^{(n-1)}(\vartheta_2 = \tau|\vartheta_1 < \theta)d\tau \\ = &\int_0^\theta \beta^{III}(\tau)(n-1)[\Phi(\theta) - \Phi(\tau)]\psi_1(\tau)^{(n-2)}d\tau. \end{aligned}$$

We then have by the Revenue Equivalence Theorem:

$$\int_0^\theta \beta^{III}(\tau)(n-1)[\Phi(\theta) - \Phi(\tau)]\psi_1(\tau)^{(n-2)}d\tau = \int_0^\theta \vartheta_1 \psi(\vartheta_1) d\vartheta_1,$$

where $\psi(\cdot)$ is ϑ_1's density function.

Taking differentiation on both sides with respect to θ leads to

$$(n-1)\varphi(\theta) \int_0^\theta \beta^{III}(\tau)\psi_1(\tau)^{(n-2)}d\tau = \theta\psi(\theta).$$

Since $\psi(\theta) = (n-1)\Phi(\theta)^{n-2}\varphi(\theta)$, we then have

$$\int_0^\theta \beta^{III}(\tau)\psi_1(\tau)^{(n-2)}d\tau = \theta\Phi(\theta)^{n-2}.$$

Taking differentiation on both sides with respect to θ again, we obtain:

$$\beta^{III}(\theta)\psi_1(\theta)^{(n-2)} = \Phi(\theta)^{n-2} + \theta(n-2)\Phi(\theta)^{n-3}\varphi(\theta).$$

Since $\Psi_1^{(n-2)}(\tau) = \Phi(\tau)^{n-2}$, we have

$$\beta^{III}(\theta) = \theta + \frac{\Phi(\theta)}{(n-2)\varphi(\theta)},$$

which is an increasing function when $\frac{\Phi(\theta)}{\varphi(\theta)}$ is increasing or, equivalently, $\Phi(\theta)$ is log-concave.

Comparing first-price, second-price and third-price auctions, we have:

$$\beta^I(\theta) < \beta^{II}(\theta) = \theta < \beta^{III}(\theta).$$

In other words, bidders under the first-price auction will bid lower than their true value (otherwise, the interim expected utility is less than or equal to zero), bidders under the second-price auction will bid their true value, and bidders under the third-price auction would bid more than their true value.

Uncertain Number of Bidders

We now consider auctions with an uncertain number of bidders. Suppose that $\mathcal{N} = \{1, 2, \ldots, N\}$ denote the set of potential bidders. Let $A \subseteq \mathcal{N}$ be the set of actual bidders. Assume that for each potential bidder, the values of the object are i.i.d. with distribution function $\Phi(\cdot)$. For participant $i \in A$, assume that the probability of facing n other bidders is p_n, and the probability does not depend on the identity of bidders nor on their values. As long as a bidder is consistent with her type and the number of opponents she faces, then for a symmetric equilibrium, we can also use the Revenue Equivalence Theorem to solve for the Bayesian-Nash equilibrium bidding function.

Suppose that the symmetric Bayesian-Nash equilibrium bidding strategy β is a monotonically increasing function. Let $\vartheta_1^{(n)}$ be the first-order statistics of types in n other bidders, and the distribution function is $\Psi^{(n)}(\cdot) = \Phi^n(\cdot)$. When bidder 1's value is θ and the bidding price is $\beta(\tilde{\theta})$, the probability of winning the auction is:

$$\Psi(\tilde{\theta}) = \sum_{n=0}^{N-1} p_n \Psi^{(n)}(\tilde{\theta}),$$

and the interim expected utility/payoff is:

$$\bar{U}_1(\boldsymbol{\theta}, \tilde{\theta}) = \Psi(\tilde{\theta})\theta - \bar{m}_1(\tilde{\theta}),$$

where $\bar{m}_1(\tilde{\theta})$ is the interim expected payment to the auctioneer when choosing $b_1 = \beta(\tilde{\theta})$.

Consider the first-price and second-price auctions with an uncertain number of bidders. First, for the second-price auction, despite the number of opponents, $\beta(\theta) = \theta$ would always be a weakly dominant strategy for bidder 1. Then, the interim payment to the auctioneer under the second-price auction is

$$\bar{m}_1^{II}(\tilde{\theta}) = \sum_{n=0}^{N-1} p_n \Psi^{(n)}(\theta) E[\beta^{II}(\vartheta_1^{(n)}) | \vartheta_1^{(n)} < \theta].$$

For the first-price auction, the interim expected payment is:

$$\bar{m}_1^I(\tilde{\theta}) = \Psi(\theta)\beta^I(\theta).$$

According to the Revenue Equivalence Theorem, we obtain: $\bar{m}_1^{II}(\tilde{\theta}) = \bar{m}_1^I(\tilde{\theta})$, that is:

$$\begin{aligned} \beta^I(\theta) &= \sum_{n=0}^{N-1} \frac{p_n \Psi^{(n)}(\theta)}{\Psi(\theta)} E[\beta^{II}(\vartheta_1^{(n)}) | \vartheta_1^{(n)} < \theta] \\ &= \sum_{n=0}^{N-1} \frac{p_n \Psi^{(n)}(\theta)}{\Psi(\theta)} \beta^{I,(n)}(\theta), \end{aligned}$$

where $\beta^{I,(n)}(\theta)$ denotes the equilibrium bidding price under the first-price auction with her own type θ when the number of opponents is n.

Thus, for a first-price auction with an uncertain number of bidders, the bidding strategy of a symmetric equilibrium is the weighted average of first-price bids under a given number of bidders, whose weight is equal to the probability distribution of the number of opponent bidders.

21.2.8 Optimal Auction Design with Private Values

Now, we examine the seller's optimal auction design, which means to consider the optimal auction that satisfies Bayesian incentive compatibility and participation constraints, and allows the seller to also have the option of keeping the object to herself. The discussion is essentially the application of the optimal contract design with multiple agents.

We call the seller with the "reserve object" "participant 0", the value of the object is denoted by θ_0, and the decision about whether to retain the object is written as $y_0 \in \{0,1\}$. Then, we must have $\sum_{i=0}^{n} y_i = 1$, and the seller's total expected revenue can be written as

$$\theta_0 E_{\boldsymbol{\theta}} y_0(\boldsymbol{\theta}) + \sum_{i=1}^{n} E_{\boldsymbol{\theta}}[\theta_i y_i(\boldsymbol{\theta})] - \sum_{i=1}^{n} E_{\theta_i} E_{\boldsymbol{\theta}_{-i}}[U_i(\theta_i, \boldsymbol{\theta}_{-i})].$$

As such, the total expected payoff of the seller equals the difference between the total surplus and the expected information rent of the participants.

By the Bayesian Incentive Compatibility Characterization Theorem in Chapter 19, the interim expected utility of the buyer must satisfy

$$E_{\boldsymbol{\theta}_{-i}}[U_i(\theta_i, \boldsymbol{\theta}_{-i})] = E_{\boldsymbol{\theta}_{-i}}[U_i(\underline{\theta}_i, \boldsymbol{\theta}_{-i})] + \int_{\underline{\theta}_i}^{\theta_i} E_{\boldsymbol{\theta}_{-i}} y_i(\tau, \boldsymbol{\theta}_{-i}) d\tau, \forall \theta_i \in \Theta_i. \quad (21.2.17)$$

The interim individual rationality constraint of the buyer requires

$$E_{\boldsymbol{\theta}_{-i}}[U_i(\underline{\theta}_i, \boldsymbol{\theta}_{-i})] \geqq 0, \forall i.$$

Given the decision rule $y(\boldsymbol{\theta})$, the seller will optimally choose transfers to set the interim expected utilities of buyers with the lowest value as zero, i.e.,

$$E_{\boldsymbol{\theta}_{-i}}[U_i(\underline{\theta}_i, \boldsymbol{\theta}_{-i})] = 0, \forall i.$$

Integrating by parts for (21.2.17), we can write buyer i's expected information rent as

$$E_{\theta_i} E_{\boldsymbol{\theta}_{-i}} \left[\frac{1 - \Phi_i(\theta_i)}{\varphi_i(\theta_i)} y_i(\boldsymbol{\theta}) \right].$$

Substituting into the seller's revenue formula, we can write it as expected virtual surplus:

$$E_{\boldsymbol{\theta}} \left[\theta_0 y_0(\boldsymbol{\theta}) + \sum_{i=1}^{n} \left(\theta_i - \frac{1 - \Phi_i(\theta_i)}{\varphi_i(\theta_i)} \right) y_i(\boldsymbol{\theta}) \right].$$

For $i \geqq 1$, let

$$\nu_i(\theta_i) = \theta_i - \frac{1 - \Phi_i(\theta_i)}{\varphi_i(\theta_i)},$$

which is participant i's "virtual valuation", and let

$$\nu_0(\theta_0) = \theta_0.$$

The seller's optimization problem then can be written as

$$\max_{x(\cdot)} E_{\theta} \left[\sum_{i=0}^{n} \nu_i(\theta_i) y_i(\theta_i) \right]$$

$$s.t. \sum_{i=0}^{n} y_i(\boldsymbol{\theta}) = 1,$$

$E_{\boldsymbol{\theta}_{-i}} y_i(\theta_i, \boldsymbol{\theta}_{-i})$ is non-decreasing in $\theta_i, \forall i \geqq 1 \ (BM)$.

Notice that the Bayesian monotonicity constraint is satisfied provided virtual valuation function $\nu_i(\theta_i)$, for every agent i, is non-decreasing in θ_i, which is true when the hazard rate

$$h_i(\theta_i) = \frac{\varphi_i(\theta_i)}{1 - \Phi_i(\theta_i)}$$

is a non-decreasing function. For each state $\boldsymbol{\theta}$, we maximize the above expectation. The solution then is to give the object to the bidder who has the highest virtual valuation, i.e., $y_i(\theta) = 1$ when $\nu_i(\theta_i) > \nu_j(\theta_j)$, and $y_j(\boldsymbol{\theta}) = 0$ for all individuals $j \neq i$. The corresponding decision rule is

$$y(\boldsymbol{\theta}) = y(\nu_0(\theta_0), \nu_1(\theta_1), \cdots, \nu_n(\theta_n)),$$

where $y(\cdot)$ is the efficient decision rule. Thus, the seller should allocate the object to the participant with the highest virtual valuation.

Again, by by the dominant Incentive Compatibility Characterization Theorem (Proposition 19.4.1) and Corollary 19.4.1, an optimal allocation rule that is Bayesian implementable can also be implementable in dominant strategies. Thus, we have DIC (dominant-strategy incentive compatible) transfers by integrating the DICFOC (first-order condition for dominant-strategy incentive compatibility), and the resulting transfers take the form of:

$$t_i(\boldsymbol{\theta}) = -p_i(\boldsymbol{\theta}_{-i}) y_i(\boldsymbol{\theta}),$$

where

$$p_i(\boldsymbol{\theta}_{-i}) = \inf\{\hat{\theta}_i \in [\underline{\theta}_i, \bar{\theta}_i] : y_i(\hat{\theta}_i, \boldsymbol{\theta}_{-i}) = 1\}.$$

Therefore, for each buyer i, there exists a pricing rule $p_i(\boldsymbol{\theta}_{-i})$, which is a function of other bidders' true values. If the participant's bid is greater than $p_i(\boldsymbol{\theta}_{-i})$, the bidder will get the object, and pays price $p_i(\boldsymbol{\theta}_{-i})$. This means that telling the truth is the dominant strategy. The logic of the argument is the same as that of the second-price auction: misreporting does not affect the price paid, but only affects the acquisition of the object. A bidder wants to get the object just when her valuation θ_i is higher than the price $p_i(\boldsymbol{\theta}_{-i})$.

Suppose that for all i, $\psi_i = \psi$ and $\nu_i = \nu$. In other words, the distributions of bidders' types are the same, and then their virtual valuation functions are the same, denoted by $\nu(\cdot)$. Assume that the function is increasing. When the principal sells the object, since bidder i with the highest valuation has the highest virtual valuation $\nu(\theta_i)$, she obtains the object. When $\nu(\max_{i \geqq 1} \theta_i) > \theta_0$, the principal sells the object. Then, we have

$$p_i(\boldsymbol{\theta}_{-i}) = \max\{\nu^{-1}(\theta_0), \max_{j \neq i} \theta_j\}.$$

Thus, the optimal auction that is truthfully implementable in dominant strategies is a second-price auction with reserve price $r^* = \nu^{-1}(\theta_0)$.

The optimal reserve price has two characteristics. First, if it is greater than the value of the object for the principal, it is equivalent to monopoly pricing, the price of which is higher than the marginal cost. In this sense, if the seller can determine the optimal reserve price, the standard auction does not achieve Pareto efficiency, and the object cannot be sold in a socially optimal manner because when the highest value of buyers is between θ_0 and r^*, the seller will not resell the object to this buyer. The intuition here is that the principal can reduce information rent by reducing individual spending. This is similar to the case of only one individual.

Why is it optimal to allocate the object based on the optimal reserve price (virtual valuation)? It is not hard to understand. We give the following explanation. Suppose that the seller adopts a take-it-or-leave-it offer to sell the object to a buyer at price p. If the seller decides on a reserve price r, then only when the buyer's private value θ is higher than or equal to r, she would buy the object. The probability that the buyer will accept this price level is $1 - \Phi(p)$, which is also the probability of the value of buyer exceeding p. If we consider the probability of purchasing as the buyer i's demand, then the demand function is $q(p) \equiv 1 - \Phi(p)$, where the inverse demand function is $p(q) \equiv \Phi^{-1}(1 - q)$. Then, the seller's revenue function is

$$p(q) \times q = q\Phi^{-1}(1 - q),$$

where the derivative with respect to q is

$$\frac{d}{dq}(p(q) \times q) = \Phi^{-1}(1 - q) - \frac{q}{\Phi'(\Phi^{-1}(1 - q))}.$$

Since $\Phi^{-1}(1 - q) = p$, we have

$$MR(p) \equiv p - \frac{1 - \Phi(p)}{\phi(p)} = \nu(p),$$

which means marginal revenue equals buyer's virtual valuation at $p(q) = p$. As a consequence, buyer's virtual valuation $\nu(p)$ can be interpreted as marginal revenue. Since ψ is strictly increasing, the seller can set monopoly price r^* according to the marginal cost equaling the marginal revenue $MR(p) = MC$. Since the latter is assumed to be θ_0, $MR(r^*) = \nu(r^*) = \theta_0$ or $r^* = \nu^{-1}(\theta_0)$.

When the seller faces different types of buyers, the optimal mechanism is to adopt discriminatory reserve price $r_i^* = \nu_i^{-1}(\theta_0)$. If no buyer's value of the object exceeds the reserve price r^*, the seller retains this object; otherwise, the seller sells the object at marginal revenue, and the buyer who wins the object is asked to pay $p_i = p_i(\boldsymbol{\theta}_{-i})$, which is the lowest value of winning the object for the buyer.

The second characteristic of the optimal reserve price is that it is independent of the number of bidders, but depends on the distribution of bidders' private values. This conclusion does not hold when bidders' private values are not independent or when bidders are not risk-neutral.

21.2.9 Factors Affecting Revenue Equivalence and Efficiency

Although Myerson's Revenue Equivalence Theorem is theoretically significant, the conditions behind the theorem are too strong to hold in practice. If any of the following situations occurs, the theorem may not be true. In this sense, Myerson's conclusion does not end the auction theory, but provides a benchmark theory and a new starting point for its development of more realistic auctions. Indeed, most of the subsequent study on auction theory was based on the work of Myerson.

Here, we do not provide a detailed discussion of cases in which Myerson's Revenue Equivalence Theorem may not hold; instead, we provide the basic conclusions.

Number of Bidders

The bidding price of an object is determined by competition, and thus the more competitive an auction is, the higher the bidder's bid and the greater payoff for the auctioneer. In the symmetric independent private value (SIPV) model, this conclusion can easily be proved. The two previous auctions bring the following maximum average net profit to the seller:

$$E\beta^I(\theta) = \int_0^{\theta} I(\theta) d\Phi^n(\theta).$$

By integration by parts we obtain:

$$E\beta^I(\theta) = \int_0^{\theta} (1 - \Phi^n(\theta)) I'(\theta) d\theta > 0.$$

Therefore, the auctioneer's maximum payoff increases with the number of bidders. In a real auction, to attract more bidders to participate in the auction is absolutely essential.

Limited Liability

In the previous discussions, we assume that a bidder is able to pay up to her highest value $\bar{\theta}_i$. Suppose that the assumption of unlimited liability does not hold, and the bidder's budget $\tilde{w}$ is a random variable distributed on $[0, \bar{w}]$. w is a realization of $\tilde{w}$.

It can be shown that a second-price auction with limited liability still has a dominant equilibrium, whose equilibrium bidding function is given by $B^{II}(\theta, w) = \min\{\theta, w\}$, while the equilibrium bidding function of a first-price auction is given by $B^I(\theta, w) = \min\{\beta^I(\theta), w\}$. The auctioneer's expected payoff under the first-price auction is greater than the expected payoff under the second-price auction, so that Myerson's Revenue Equivalence Theorem is no longer true.

Asymmetry between Bidders

Symmetry is an important assumption in the SIPV model. The distribution functions of all bidders are the same. It is difficult, however, to imagine that this symmetry is satisfied in practice. What impact will asymmetry between bidders have on the auction?

First, since truth-telling in the second-price auction is the dominant strategy equilibrium, it is ex-post implementable. Therefore, it is irrespective of whether or not the auctioneer's distribution function is symmetric, the highest bidder gets the object, so that the second-price auction is still efficient. However, as truth-telling may not be a dominant strategy equilibrium in other formats of auctions, they are likely inefficient. For example, the first-price auction is actually not efficient. To see this, suppose that there are two bidders, and the equilibrium bidding functions are continuously and monotonically increasing, denoted by β_1 and β_2. Assume that for some θ, $\beta_1(\theta) < \beta_2(\theta)$. Therefore, for sufficiently small $\epsilon > 0$, we still have $\beta_1(\theta + \epsilon) < \beta_2(\theta - \epsilon)$. This means that bidder 2 gets the object despite that her true value is smaller.

Secondly, for the auctioneer, the profits of different auctions may not be the same. From the previous discussion on optimal auction design, $\nu(\theta_i)$ is the virtual valuation function of bidder i, which may be regarded as the seller's marginal revenue. As such, the seller's optimal mechanism should sell the object to the bidder with the highest marginal revenue $\nu(\theta_i)$, provided that her marginal revenue is not less than the seller's true value or marginal cost. Consequently, when bidders are not symmetric, the optimal auction is discriminatory and the bidder who wins the object is not necessarily the one with the highest private value. This also means that the seller's optimal mechanism is not efficient. Moreover, the seller will also set different reserve prices for different bidders. This mechanism is similar to monopoly price discrimination, where the monopolist sets different prices according to different needs of consumers.

Which of first-price and second-price auctions is beneficial to the seller under asymmetric conditions? First, the asymmetry does not affect the bidder's strategy in the second-price auction, and bidding her true value is still a dominant strategy. However, in a first-price auction, the weaker bidders may bid more aggressively. Therefore, for the seller, a first-price auction may be better than an ascending-price auction or a second-price auction. For a detailed discussion, see Krishna (2010).

Non-neutral Risk Attitudes

For bidders, an auction may bring risk. When a bidder wins, she gains profits; when she loses, her payoff is zero. If the bidder dislikes the risk or has dissimilar degrees of risk-aversion, her bidding strategy is likely to be different.

Under the first-price auction, if a bidder increases her bid, it will increase the probability of winning, but at the same time her profit is reduced after winning. A risk-averse bidder will be more inclined to raise her bid than a risk-neutral bidder. Therefore, if all bidders are risk-averse, a first-price auction may bring higher profits to the seller. See Krishna (2010) for the corresponding proof.

Under a second-price auction, regardless of whether a bidder is risk-neutral or risk-averse, she always bids her true value. Therefore, if other assumptions in the SIPV model remain unchanged, but all bidders are risk-averse, then the Revenue Equivalence Theorem no longer holds. Indeed, a first-price auction may yield a higher expected revenue than a second-price auction or an ascending-price auction.

Collusive Bid

In many auction situations, bidders often bid together to reduce competition and lower the price paid to the seller. This collusive bidding behavior exists in several ways, depending on auction formats and information structures. For example, some bidders may first conduct pre-auctions together to choose the best bidders for the formal auction; sometimes, bidders take turns participating in multiple auctions; in some cases, while all bidders participate in the auction, they may make an agreement in advance on low bids, allowing one bidder to win and later (or in advance) sharing the revenue.

The analysis of collusive bidding strategy is complicated. The biggest difficulty is that bidders may change their strategy. Thus, in order to prevent participants from changing their strategy, the collusive bidding mechanism must satisfy the **collusion-proof incentive-compatible constraint**. In order to bring more bidders into collusion, but not be mandatory, this mechanism must meet the participation constraint. In addition, collusive bidding is best worked out by letting the bidder who is willing to bid the most win the object.

In the SIPV model, McAfee and McMillan (AER, 1992) demonstrated that there exists a direct collusion mechanism that includes all bidders, satisfying both efficiency and collusion-proof incentive compatibility. If the seller uses a standard auction, such as the first-price or the second-price auction, every bidder is willing to participate in this collusion mechanism that satisfies the collusion-proof incentive-compatible constraint. The basic procedure for such a mechanism is that, prior to participating in the formal auction, all of the participants hold their own auction to select only one bidder for the formal auction. The rule is that each bidder will bid separately, and the highest bidder will be selected and she must pay her own bid, whereby the profit (i.e., the difference of the winner's bid and seller's reserve price) is equally divided among all other bidders. The selected bidder participates in the formal auction, and her bid is equal to the seller's reserve price, thus winning the auction and paying her bid. This collusive bidding mechanism is equivalent to the case in which every bidder wants to bribe others to quit the competition, and the bidder who is willing to pay the highest bribe wins and shares the profits of that bribery with others.

Participation Costs

When bidders have participation costs, low-value bidders are more motivated to participate in the collusion mechanism than are high-value bidders. High-value bidders send out a signal of higher value by denying the collusion mechanism, which is more credible. Tan and Yilankaya (2007) showed that when the auctioneer uses the second-price auction, such information shall reduce participation in the bidding mechanism for other bidders, making it harder to form a collusion mechanism that satisfies the collusion-proof incentive-compatible constraint. Therefore, McAfee-McMillan's conclusion no longer holds. Cao, Hsueh and Tian (2020) proved that when the auctioneer adopts the first-price auction, McAfee-McMillan's conclusion also fails. In addition, Tan and Yilankaya (2006), Cao and Tian (2010, 2013), and Cao, Tan, Tian and Yilankaya (2018) discussed the existence of equilibrium in first-price and second-price auctions with participation costs. In general, in the presence of participation costs, the Revenue Equivalence Theorem may not be true.

In addition to the above factors, the interdependent value that will be discussed below also affects the validity of the Revenue Equivalence Theorem.

21.3 Single Object Auctions with Interdependent Values

In the previous section, we obtained the expected revenue equivalence result: For any two auctions, if the bidders' expected payments at the lowest private value are the same and the allocation rules are also the same, then the expected auction revenue generated by them is the same. In a symmetric economic environment with independent private values, auctions, such as first-price sealed-bid auction, second-price sealed-bid auction, English auction (with ascending price) and Dutch auction (with descending price), all have the same allocation rule (i.e., the highest bidder gets the object) and the same expected payoff (usually zero) at the minimum private value, and therefore the expected revenue generated by them is the same. As such, they are indifferent to the auctioneer.

However, Myerson's Revenue Equivalence Theorem relies on a series of assumptions, the most crucial of which is that all bidders' values of the object are independent. In practice, this assumption is apparently not true, and a bidder's valuation depends not only on his own valuation of the object, but also on those of other bidders. For example, in an auction of artworks, bidders take into account not only their own preferences for the artwork, but also the potential profits from resale, which is obviously affected by others' valuations. After considering the behavior of other bidders, the value that a bidder think of the object is called the **interdependent value**. When interdependence values are present, Myerson's theorem is no longer true, and the auctioneer may be able to improve his expected revenue through choosing different auction formats.

Milgrom and Weber (1982) were the first to investigate auctions with interdependent values, which has a significant influence on the development of auction theory. Based on the observation of auction practice, they found that bidders' valuations may be **affiliated**: a higher value (thus a higher bid) of a bidder can easily increase the rating of other participants. Therefore, an auction with interdependent values can be understood as that any bidder's bid will not only reveal her own information about the value of the object, but also partially reveal the private information of other bidders. In this sense, the payoff of a bidder will depend on the degree of privacy of the information. Once the information is revealed in the auction, bidders can guess each other's possible bid, and they have to bid a higher price in order to win the auction. Therefore, for an auctioneer, the auction that will bring him the highest expected revenue must be the one that most efficiently reveals the private information of all bidders.

In practice, auctioneers attempt to designing various revenue-maximizing auctions. By applying auction theory, many well-known auction theorists, such as Larry Ausubel, Ken Binmore, Paul Klemperer, Preston McAfee, John McMillan, Paul Milgrom, and Robert Wilson, have brought their talents into the practice of auction designs, such as the spectrum auction design for the Federal Communications Commission, and 3G license in the United Kingdom. These objects are not private values, but rather with interdependent values. Another typical example is oilfields. The amount of oilfield storage, though unknown until it is mined, has an impact on the value of the oilfield for all bidders. Overall, the assumption of private value is not applicable for auctions of such objects, and thus auction design for economic environments with interdependent values will have a large impact on auction outcomes, which shall be discussed in this section.

21.3.1 Basic Analytical Framework

In the following, we use the mechanism design approach to discuss auctions in the presence of interdependent values, and conduct revenue comparisons of different auction formats.

We assume that there are n bidders competing for an object. Bidder i's private information is summarized by signal θ_i, and her value for the object is V_i. We assume that V_i is a nondecreasing function of all bidders' signals, and write it as

$$V_i = u_i(\theta_1, \ldots, \theta_n),$$

and such values are known as the **interdependent values**.

If $V_i = u_i(\theta_i)$ (or equivalently writing $V_i = \theta_i$), it is the private value of the bidder i, as discussed in the previous sections. If

$$V_i = u(\theta_1, \ldots, \theta_n) \equiv V,$$

then the object is of **common value**. Thus, private value and common value can be regarded as special cases of interdependent values.

Suppose that bidder i's bidding strategy is b_i, the probability of getting the object is $y_i(b_i, \boldsymbol{b}_{-i})$, and the payment is $m_i(b_i, \boldsymbol{b}_{-i})$. As previously, bidder i's ex-post utility in term of bids can be written as:

$$U(\theta_i, \boldsymbol{\theta}_{-i}, b_i, \boldsymbol{b}_{-i}) = u_i(\theta_1, \ldots, \theta_n) y_i(b_i, \boldsymbol{b}_{-i}) - m_i(b_i, \boldsymbol{b}_{-i}).$$

For the four common auctions with interdependent values, bidder i's probability to get the object, $y_i(b_i, \boldsymbol{b}_{-i})$, and interim expected payment after winning, $m_i(b_i, \boldsymbol{b}_{-i})$, are determined in the same way as in the case of private value. For the first-price sealed-bid auction and the Dutch auction, the allocation rule $y_i(b_i, \boldsymbol{b}_{-i})$ is determined by equation (21.2.1), and the ex-post payment $m_i(b_i, \boldsymbol{b}_{-i})$ is given by equation (21.2.2); for the second-price sealed-bid auction and the English auction, the allocation rule $y_i(b_i, \boldsymbol{b}_{-i})$ is determined (21.2.6), and the ex-post transfer $m_i(b_i, \boldsymbol{b}_{-i})$ is given by (21.2.7).

For an object with common value V, we assume that the private signal θ_i under given common value V is independently distributed, which is also an unbiased estimate of value V, i.e.,

$$E(\theta_i | V = v) = v.$$

.

Consider bidder 1 with signal $\theta_1 = \theta$, and then she might over estimate and thought the value of the object is $E(V|\theta_1 = \theta)$. Let $\vartheta_1, \vartheta_2, \ldots, \vartheta_{n-1}$ be the largest, second-largest, ... , and smallest signals of $\theta_2, \theta_3, \ldots, \theta_n$, or be the first-order, second-order, ..., $n-1$th order statistics of $\{\theta_2, \ldots, \theta_n\}$. Assume that the bidders are symmetric, and they choose the same bidding strategy.

Milgrom and Weber also showed the possibility of the so-called "winner's curse", in which a bidder realizes that winning the auction was not "worth it" only after the winning. If all bidders are symmetric and follow the same strategy β, then this fact reveals to bidder 1 that the highest of the other $n-1$ signals is less than θ. As such, after bidder 1 wins, her estimate of the value should be $E(V|\theta_1 = \theta, \vartheta_1 < \theta)$ rather than $E(V|\theta_1 = \theta)$, which is less than the initial estimate $E(V|\theta_1 = \theta)$, leading to a decrease in the estimated value and is lower than her initial estimate, so that winning brings "bad news" . To put this succinctly, if the successful bid comes from an overly optimistic estimate of the object, a bidder would regret winning the object, which is termed the winner's curse.

In practice, it is often seen that some bidders will boldly raise their bids to a certain level when they recognize other competitors' intention of raising prices. When they find that the competitors are not going to follow anymore, they realize that they just bid too much; however, at this moment, such regret comes too late. This is the "curse" of winning (e.g., Didius Julianus paid nearly two kilos of gold to each Praetorian Guard soldier, winning their support and gaining the emperor's throne, which, however, resulted not only in too high of a price, but also cost his life; for another example, corruption brings promotion, but also leads to prison). In fact, this is caused by the bidder's incorrect calculation of her expected payoff according to $E(V|\theta_1 = \theta)$, instead of using $E(V|\theta_1 = \theta, \vartheta_1 < \theta)$.

Obviously, the traditional assumption of independent values of the auction object cannot explain the emergence of the winner's curse, while this becomes easy to understand with the introduction of interdependent values. The winning bidder also receives private information about other bidders' valuations, which causes the winning bidder to lower her rating of the object that she has just acquired. If a bidder is rational, in order to avoid the winner's curse, she should lower her estimation on the value of the object, and the extent of the reduction will be influenced by the auction rules. Therefore, an appropriate auction needs to minimize the impact of the winner's curse on bidders. By correct estimation according to $E(V|\theta_1 = \theta, \vartheta_1 < \theta)$, this curse will not occur at equilibrium.

Basic Assumptions

Bidders have the profile of private signals about their valuations on the object, denoted by $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_n)$. If the density function of $\boldsymbol{\theta}$ does not satisfy $\varphi(\boldsymbol{\theta}) = \Pi_i^n \varphi_i(\theta_i)$, where $\varphi_i(\cdot)$ is the density function of θ_i, then the distribution of private signals is not independent. We hereby introduce the concept of "affiliation" between signals.

Definition 21.3.1 The random vector $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_n)$ on $\Theta \subseteq \mathcal{R}^n$ is **affili-**

ated if for any $\boldsymbol{\theta}, \boldsymbol{\theta'} \in \Theta$,

$$\varphi(\boldsymbol{\theta} \vee \boldsymbol{\theta'})\varphi(\boldsymbol{\theta} \wedge \boldsymbol{\theta'}) \geqq \varphi(\boldsymbol{\theta})\varphi(\boldsymbol{\theta'}),$$

where

$$\boldsymbol{\theta} \vee \boldsymbol{\theta'} = (\max\{\theta_1, \theta_1'\}, \ldots, \max\{\theta_n, \theta_n'\})$$

and

$$\boldsymbol{\theta} \wedge \boldsymbol{\theta'} = (\min\{\theta_1, \theta_1'\}, \ldots, \min\{\theta_n, \theta_n'\}).$$

Let $\vartheta_1, \ldots, \vartheta_{n-1}$ be the first-, second-, ..., and $n-1$th- order statistics of $\{\theta_2, \ldots, \theta_n\}$. By Proposition 2.11.1 (see the proof in Milgrom and Weber(1982)), we know that if $\{\theta_1, \ldots, \theta_n\}$ is affiliated, the variables in any of their subsets are affiliated, and $\{\theta_1, \vartheta_1 \ldots, \vartheta_{n-1}\}$ are also affiliated. If $\boldsymbol{\theta'}$s density function $\varphi(\cdot)$ is positive on Θ, and twice continuously differentiable, then $\{\theta_1, \ldots, \theta_n\}$ are affiliated if and only if

$$\frac{\partial^2 \ln(\varphi)}{\partial\theta_i \partial\theta_j} \geqq 0, \forall i \neq j.$$

In simple terms, the affiliation between variables means that if the value of a random variable is larger, the probability that any other random variables associated with it have a larger value is also higher.

Let $\Psi(\cdot|\theta)$ be the conditional distribution function of ϑ_1 given $\theta_1 = \theta$, and $\psi(\cdot|\theta)$ as the corresponding conditional density function. The affiliation between θ_1 and ϑ_1 implies that, if $\theta' > \theta$, then $\Psi(\cdot|\theta')$ dominates $\Psi(\cdot|\theta)$ in terms of the reverse hazard rate, i.e.:

$$\frac{\psi(\vartheta|\theta')}{\Psi(\vartheta|\theta')} \geqq \frac{\psi(\vartheta|\theta)}{\Psi(\vartheta|\theta)}.$$

It is easy to verify that, for an increasing function $h(\cdot)$, we have $E(h(\vartheta_1)|\theta') \geqq E(h(\vartheta_1)|\theta)$ for $\theta' > \theta$.

Applying the concept of affiliation, Milgrom and Weber analyzed four basic standard auctions.

Differences of the Four Basic Auctions under Interdependent Values

For the case of private value, the second-price sealed-bid auction and the English auction are outcome-equivalent, as private information of other participants does not affect a bidder's valuation of an object. This is because a bidder's valuation relies solely on her own signal, while the exit decisions of others will not affect the bidder's expected valuation revision, and the rules which govern the winning bidder and the winner's payment (the second-highest price) are also the same. Similarly, the first-price sealed-bid auction and the Dutch auction are also outcome-equivalent and, in fact, strategically equivalent.

However, for the case of interdependent value and affiliated information, if the number of bidders is three or more, there is no equivalence between the second-price sealed-bid auction and the English auction. Intuitively, this is because in English auctions, the exit of bidders at different prices provides additional information that can be used by the remaining bidders, especially the winning bidder, to revise the estimated values of the object. Since the winning bid is affiliated to the signals of all losing bidders, it results in a higher profit to the seller. On the other hand, in second-price sealed-bid auctions, the winning bid is only affiliated to the bidder who bids the second-highest price, resulting in lower profits to the sellter. Therefore, the outcomes may be different under these two auction formats. However, if there are only two bidders, once one bidder exits, the other would win the auction. In this case, the exit of a bidder under a bidding price does not change the winner's payment, whereby the English auction and the second-price sealed-bid auction are also outcome equivalent.

Regarding the Dutch auction and the first-price sealed-bid auction, because there is no affiliation among bids of bidders, the value and payment to the winner are the same in both auctions, and thus it does affect their equivalence. Since the English auction reveals more information than the others, its revenue is the highest. This finding provides a good explanation for the prevalence of English auctions in practice.

Thus, we only need to discuss equilibrium solutions of the second-price auction, the English auction, and the first-price auction.

Symmetric Model

Symmetry here has two meanings: one is that bidders' valuation of the object is symmetric; and the other is that bidders' signal distribution is symmetric, so that we can focus on the discussion of the symmetric equilibrium bidding strategy.

We then assume that all bidders' signals are drawn from the same distribution, $\theta_i \in [0, w]$, and their valuations of the object are symmetric. Bidder i's valuation function can be written as

$$V_i = u_i(\boldsymbol{\theta}) = u(\theta_i, \boldsymbol{\theta}_{-i}),$$

which satisfies: $u_i(\theta_i, \boldsymbol{\theta}_{-i}) = u(\theta_i, \tilde{\boldsymbol{\theta}}_{-i})$, where $\tilde{\boldsymbol{\theta}}_{-i}$ is an arbitrary rearrangement of $\boldsymbol{\theta}_{-i}$, such as $u(\theta_i, \theta_j, \theta_k) = u(\theta_i, \theta_k, \theta_j), i \neq j \neq k \neq i$. Of course, an object with symmetric values does not necessarily have a common value. The following example illustrates this point.

Example 21.3.1 (Symmetric Valuation Function) Consider that three bidders' value signals for the object are $\theta_1, \theta_2, \theta_3$. Suppose that, for every bidder i, the valuation function is

$$u(\theta_i, \theta_j, \theta_k) = a\theta_i + b\theta_j + c\theta_k.$$

Bidders' valuations are symmetric if and only if $b = c$. When $a \neq b = c$, the valuations of bidders are symmetric, but the object does not have a common value. In this example, the object has a common value if and only if $a = b = c$.

We further assume that $u(\theta_i, \boldsymbol{\theta}_{-i})$ is a continuous function, which is strictly increasing in θ_i, non-decreasing in $\theta_j, \forall j \neq i$, and satisfies $u(0, \dots, 0) = 0$. In the symmetric case, we only need to consider bidder 1's choice.

Define

$$v(\theta, \vartheta) = E(V_1|\theta_1 = \theta, \vartheta_1 = \vartheta) = E(u(\theta_1, \vartheta_1, \dots, \vartheta_{n-1}))|\theta_1 = \theta, \vartheta_1 = \vartheta),$$

which characterizes bidder 1's interim expected value of the object when her private signal is θ, and the first-order statistic for other bidders' private signals is ϑ. Meanwhile, we assume that all bidders are risk-neutral, and $v(0, 0) = 0$. $\psi(\vartheta|\theta)$ is the density function of $\vartheta_1 = \vartheta$ under $\theta_1 = \theta$.

21.3.2 Second-Price Sealed-Bid Auctions

In a second-price sealed-bid auction, the bidder with the highest bid gets the object and pays the second-highest price. We want to derive the symmetric Bayesian-Nash equilibrium.

Proposition 21.3.1 *For economic environments with symmetric, interdependent values and affiliated signals, symmetric Bayesian-Nash equilibrium for the second-price sealed-bid auction has the form of* $\beta^{II}(\theta) = v(\theta, \theta)$.

PROOF. Suppose that bidders $j \neq 1$ all choose the bidding strategy $\beta = \beta^{II}$. When bidder 1's signal is θ and bids b, then her interim expected payoff is:

$$\begin{aligned}
\bar{U}(b, \theta) &= \int_0^{\beta^{-1}(b)} [v(\theta, \vartheta) - \beta(\vartheta)]\psi(\vartheta|\theta)d\vartheta \\
&= \int_0^{\beta^{-1}(b)} [v(\theta, \vartheta) - v(\vartheta, \vartheta)]\psi(\vartheta|\theta)d\vartheta.
\end{aligned}$$

Since $v(\theta, \vartheta)$ increases in θ, we have $v(\theta, \vartheta) > v(\vartheta, \vartheta)$ for $\vartheta < \theta$ and $v(\theta, \vartheta) < v(\vartheta, \vartheta)$ for $\vartheta > \theta$. Thus, by the first-derivative test, when $\beta^{-1}(b) = \theta$, $\bar{U}(b, \theta)$ reaches the maximum, which means that β^{II} is a symmetric Bayesian-Nash equilibrium of the second-price sealed-bid auction. □

Note that, unlike the second-price sealed-bid price auction with private values, β^{II} is not a weakly dominant strategy strategy equilibrium.

Example 21.3.2 Suppose that there are three bidders with a common value V for an object that is uniformly distributed on $[0, 1]$. Given V, bidders' signals θ_i are uniformly and independently distributed on $[0, 2V]$, i.e., the density of θ_i conditional on V is

$$\varphi(\theta_i|V) = 1/2V$$

on the interval $[0, 2V]$.

Let $\boldsymbol{\theta} = (\theta_1, \theta_2, \theta_3)$ and $\bar{\theta} \equiv \max\{\theta_1, \theta_2, \theta_3\}$. Then, the joint density of $(V, \boldsymbol{\theta})$ on the set $\{(V, \boldsymbol{\theta})|\theta_i \leqq 2V, \forall i = 1, 2, 3\}$ is

$$\varphi(V, \boldsymbol{\theta}) = \varphi(V)\varphi(\boldsymbol{\theta}|V) = 1/8V^3.$$

Note that V and $\boldsymbol{\theta}$ are related by $V \geqq \bar{\theta}/2$. The joint density of $\boldsymbol{\theta}$ is then

$$\begin{aligned}\varphi(\boldsymbol{\theta}) &= \int_{\bar{\theta}/2}^{1} \varphi(V, \boldsymbol{\theta}) dV = \int_{\bar{\theta}/2}^{1} \frac{1}{8V^3} dV \\ &= \frac{4 - \bar{\theta}^2}{16\bar{\theta}^2}.\end{aligned}$$

Therefore, the density of V conditional on θ is the same as the density of V conditional on $\bar{\theta}$, and then we have

$$\begin{aligned}\varphi(V|\theta) = \varphi(V|\bar{\theta}) &= \frac{\varphi(V, \boldsymbol{\theta})}{\varphi(\boldsymbol{\theta})} \\ &= \frac{1}{\varphi(\boldsymbol{\theta})} \times \frac{1}{8V^3} \\ &= \frac{16\bar{\theta}^2}{4 - \bar{\theta}^2} \times \frac{1}{8V^3}\end{aligned}$$

on the interval $[\bar{\theta}/2, 1]$. Thus, we obtain

$$\begin{aligned}E(V|\theta) &= E(V|\bar{\theta}) \\ &= \int_{1/2\bar{\theta}}^{1} V\varphi(V|\theta) dV \\ &= \frac{2\bar{\theta}}{2 + \bar{\theta}}.\end{aligned}$$

By noting $\vartheta_1 = \max\{\theta_2, \theta_3\}$ and $\bar{\theta} = \max\{\theta_1, \theta_2, \theta_3\}$, we have $\bar{\theta} = \max\{\theta_1, \vartheta_1\}$, and then

$$\begin{aligned}v(\theta, \vartheta) &= E(V|(\theta = \theta_1, \vartheta = \vartheta_1)) \\ &= E(V|\bar{\theta}) \\ &= \frac{2\bar{\theta}}{2 + \bar{\theta}}.\end{aligned}$$

Thus, we obtain

$$\beta^{II}(\theta) = v(\theta, \theta) = \frac{2\theta}{2 + \theta}.$$

21.3.3 English Auctions

Under an English auction, as the bidding price increases, bidders successively exit from the bidding (they can no longer participate in the bidding after exiting). The bids from bidders who exit early reveal the information about their values of the object, and the remaining bidders can use such information to infer (is affiliated to) the signals of all exiting bidders. The last remaining bidder is the "winner", and the *winner's bidding price is the price at the time of the last bidder's exit*.

Consider bidder 1's choice: when there are $k \geqq 1$ bidders remaining, let $p_n, p_{n-1}, \ldots, p_{k+1}$ be the prices (random variables) when the first bidder, second bidder, ..., the $n-k$ bidder exit; obviously when $j > i$, then $p_j \leqq p_i$. If bidder 1's private signal is θ, the minimum price at which she will exit is

$$\beta^k(\theta, p_{k+1}, p_{k+2}, \ldots, p_n),$$

which depends on the remaining number of bidders and on the price at the time of the bidder's exit. This minimum price is equivalent to the bidding price in this case. Due to different information and dissimilar beliefs on the expected value of the auction under different numbers of remaining bidders, the bidding strategy of bidders in the English auction is a series of bidding functions

$$\boldsymbol{\beta} = (\beta^n, \beta^{n-1}, \ldots, \beta^2).$$

Consider the bidding strategy below: when no bidder has yet exited, and the bidder's private signal is θ, the bidding strategy is

$$\beta^n(\theta) = u(\theta, \theta, \ldots, \theta). \tag{21.3.18}$$

Suppose that bidder n firstly exits after observing signal θ_n, and the price at which she exits is p_n, which means

$$\beta^n(\theta_n) = u(\theta_n, \theta_n, \ldots, \theta_n) = p_n.$$

As $\beta^n(\theta)$ is a continuous and strictly increasing function in θ, θ_n in the equation is unique. When bidder n exits, there are $n-1$ bidders in the auction.

Consider the following bidding function:

$$\beta^{n-1}(\theta, p_n) = u(\theta, \ldots, \theta, \theta_n).$$

Moreover, since $\beta^{n-1}(., p_n)$ is a continuous and monotonically increasing function, if a bidder, say the $n-1$-th bidder, exits secondly at bidding price $p_{n-1} > p_n$, then there is a unique private signal θ_{n-1} satisfying

$$\beta^{n-1}(\theta_{n-1}, p_n) = u(\theta_{n-1}, \ldots, \theta_{n-1}, \theta_{n-1}, \theta_n) = p_{n-1}.$$

Proceeding this recursively, at prices $p_n, p_{n-1}, \ldots, p_{k+1}$, there are $n-k$ bidders exiting the auction in order, and the bidding function of the remaining k bidders is:

$$\beta^k(\theta, p_{k+1}, \ldots, p_n) = u(\theta, \theta, \ldots, \theta, \theta_{k+1}, \ldots, \theta_n) = p_k,\ k > 1, \qquad (21.3.19)$$

where $\theta_j, j \geqq k+1$, satisfies:

$$\beta^j(\theta_j, p_{j+1}, \ldots, p_n) = u(\theta_j, \theta_j, \ldots, \theta_j, \theta_{j+1}, \ldots, \theta_n) = p_j.$$

Since $\beta^j(., p_{j+1}, \ldots, p_n)$ is a continuously increasing function, the θ_j satisfying the above formula is unique.

For $k = 2$, when the $n-1$-th bidder (bidder 2) exits, there is a unique signal $\theta = \theta_2$ that satisfies

$$\beta^2(\theta_2, p_3, \ldots, p_n) = u(\theta_2, \theta_2, \theta_3, \ldots, \theta_n) = p_2, \qquad (21.3.20)$$

which can be used to determine the winner (bidder 1)'s payment to the auctioneer. In this case, bidder 1 wins the object if and only if $\theta_1 > \theta_2$, i.e., if and only if

$$u(\theta_1, \theta_2, \theta_3, \ldots, \theta_n) > p_2,$$

and pays $\beta^2(\vartheta_1, \vartheta_1, \vartheta_2, \ldots, \vartheta_{n-1})$ at equilibrium.

We then have the following proposition that characterizes the symmetric Bayesian-Nash equilibrium of the English auction.

Proposition 21.3.2 *For economic environments with symmetric, interdependent values and affiliated signals, the symmetric Bayesian-Nash equilibrium in English auction $\boldsymbol{\beta} = (\beta^n, \beta^{n-1}, \ldots, \beta^2)$ is fully characterized by (21.3.18) and (21.3.19) .*

PROOF. Suppose that all bidders, except bidder 1, follow the bidding strategy above. Consider bidder 1's strategy choice. Let $\vartheta_1, \ldots, \vartheta_{n-1}$ be the order statistics of $\theta_2, \ldots, \theta_n$. Given any realized values of order statistics $\vartheta_1 \geqq \vartheta_2 \geqq \ldots \geqq \vartheta_{n-1}$, let θ be the private signal observed by bidder 1. If bidder 1 also follows the bidding strategy above, then only when $\theta > \vartheta_1$, the bidder can win the auction, and the price that the bidder shall pay is the exit price when bidders with signal ϑ_1 exit, denoted by p_2. Using (21.3.20), we can get $p_2 = u(\vartheta_1, \vartheta_1, \vartheta_2, \ldots, \vartheta_{n-1})$. Following the bidding strategy $\boldsymbol{\beta}$ above, when $\theta > \vartheta_1$, bidder 1's utility is:

$$u(\theta, \vartheta_1, \vartheta_2, \ldots, \vartheta_{n-1}) - p_2 = u(\theta, \vartheta_1, \vartheta_2, \ldots, \vartheta_{n-1}) - u(\vartheta_1, \vartheta_1, \vartheta_2, \ldots, \vartheta_{n-1}) > 0.$$

Obviously, when $\theta > \vartheta_1$, other bidding strategies cannot bring bidder 1 a utility higher than does $\boldsymbol{\beta}$.

When $\theta < \vartheta_1$, and following $\boldsymbol{\beta}$, bidder 1 cannot win the auction. Indeed, if under other bidding strategies, bidder 1 wins the auction, then her payment is $p_2 = u(\vartheta_1, \vartheta_1, \vartheta_2, \ldots, \vartheta_{n-1})$. However, if bidder ϑ_1 exits at price p_2, then bidder 1's utility is:

$$u(\theta, \vartheta_1, \vartheta_2, \ldots, \vartheta_{n-1}) - u(\vartheta_1, \vartheta_1, \vartheta_2, \ldots, \vartheta_{n-1}) < 0.$$

So, when $\theta < \vartheta_1$, there are no other bidding strategies that can bring bidder 1 a utility higher than does $\boldsymbol{\beta}$. Since θ_1 is continuously distributed, the probability of $\theta_1 = \vartheta_1$ is zero, which has no real impact on the bidder's expected utility. In sum, there are no other bidding strategies that can bring bidder 1 a utility higher than does $\boldsymbol{\beta}$. □

It should be remarked that the English auction has an advantageous property, in which its symmetric equilibrium given by (21.3.18) and (21.3.19) does not depend on distribution function $\varphi(\cdot)$. As such, it forms an ex-post equilibrium, which means that the Bayesian-Nash equilibrium strategy $\boldsymbol{\beta}$ has an important property of no regret: irrespective of realization of the signals, the bidders shall not regret the outcome. This characteristic makes English auctions different from second-price auctions, which rely on the signal distribution function, and also the first-price auction discussed below, both of which do not generate ex-post equilibrium.

21.3.4 First-Price Sealed-Bid Auctions

In the first-price sealed-bid auction, the highest bidder wins the object and pays the bidding price. Since it is a sealed auction, the bidding information of one bidder can not affect the bidding decisions of other bidders, which is an important difference from the English auction. Suppose that β is a symmetric Bayesian-Nash equilibrium, which is an increasing function (to be verified later). Given that other bidders all choose this strategy, for bidder 1, her private signal is θ. If her bid is $\beta(\theta')$, her interim expected utility is:

$$\bar{U}(\theta', \theta) = \int_0^{\theta'} [v(\theta, \vartheta) - \beta(\theta')]\psi(\vartheta|\theta) d\vartheta,$$

where

$$v(\theta, \vartheta) = E(V_1|\theta_1 = \theta, \vartheta_1 = \vartheta) = E(u(\theta_1, \vartheta_1, \ldots, \vartheta_{n-1})|\theta_1 = \theta, \vartheta_1 = \vartheta),$$

and $\psi(\cdot|\theta_1 = \theta)$ is the density function of the first-order statistics of other participants' signals given the bidder's private signal θ, and $\Psi(\cdot|\theta)$ is the corresponding distribution function.

The first-order condition on θ' satisfies:

$$[v(\theta, \theta') - \beta(\theta')]\psi(\theta'|\theta) - \beta'(\theta')\Psi(\theta'|\theta) = 0.$$

Since β is a symmetric Bayesian-Nash equilibrium, $\theta' = \theta$ is optimal, and then we have the differentiation equation:

$$\beta'(\theta) = v(\theta, \theta) - \beta(\theta)\frac{\psi(\theta|\theta)}{\Psi(\theta|\theta)}.$$

Since $v(0,0) = 0$, then $\beta(0) = 0$. Solving the above first-order differential equation yields

$$\beta^I(\theta) = \int_0^\theta v(\vartheta, \vartheta) dF(\vartheta|\theta), \tag{21.3.21}$$

where $F(\vartheta|\theta) = \exp(-\int_\vartheta^\theta \frac{\psi(\tau|\tau)}{\Psi(\tau|\tau)} d\tau)$.

The following proposition characterizes the symmetric Bayesian-Nash equilibrium of the first-price sealed-bid auction.

Proposition 21.3.3 *For economic environments with symmetric, interdependent values and affiliated signals, the symmetric Bayesian-Nash equilibrium of the first-price sealed-bid auction $\beta^I(\theta)$ is given by (21.3.21).*

PROOF. We first demonstrate that $F(\vartheta|\theta)$ is a distribution function on $[0, \theta]$. By the assumption of affiliations, for any $\tau > 0$, we have:

$$\frac{\psi(\tau|\tau)}{\Psi(\tau|\tau)} > \frac{\psi(\tau|0)}{\Psi(\tau|0)}.$$

So,

$$-\int_0^\theta \frac{\psi(\tau|\tau)}{\Psi(\tau|\tau)} d\tau \quad \leqq \quad -\int_0^\theta \frac{\psi(\tau|0)}{\Psi(\tau|0)} d\tau = \int_0^\theta \frac{d}{d\tau}(\ln \Psi(\tau|0)) d\tau = -\infty,$$

Then, $F(0|\theta) = 0$. Moreover, $F(\theta|\theta) = 1$ and $F(\cdot|\theta)$ is nondecreasing. Thus, $F(\vartheta|\theta)$ is a distribution function.

Moreover, $F(\vartheta|\theta') \leqq F(\vartheta|\theta)$ for $\theta' > \theta$, which implies that $F(\vartheta|\theta')$ first-order stochastically dominates $F(\vartheta|\theta)$, and thus for all $\theta' > \theta$, we have

$$\beta^I(\theta') = \int_0^{\theta'} v(\vartheta, \vartheta) dF(\vartheta|\theta') \geqq \int_0^\theta v(\vartheta, \vartheta) dF(\vartheta|\theta) = \beta^I(\theta).$$

Thus, $\beta^I(\theta)$ is a monotonically increasing function.

We now show that $\beta^I(\theta)$ is a symmetric Bayesian-Nash equilibrium. Suppose that bidder 1 bids $\beta(\theta') = \beta^I(\theta')$ when her signal is θ. Then, her interim expected utility is:

$$U(\theta', \theta) = \int_0^{\theta'} [v(\theta, \vartheta) - \beta(\theta')] \psi(\vartheta|\theta) d\vartheta.$$

The first-order condition on θ' is:

$$\begin{aligned} \frac{\partial U}{\partial \theta'} &= [v(\theta, \theta') - \beta(\theta')] \psi(\theta'|\theta) - \beta'(\theta') \Psi(\theta'|\theta) \\ &= \Psi(\theta'|\theta)[(v(\theta, \theta') - \beta(\theta')) \frac{\psi(\theta'|\theta)}{\Psi(\theta'|\theta)} - \beta'(\theta')]. \end{aligned}$$

If $\theta' < \theta$, then $v(\theta, \theta') > v(\theta', \theta')$. Meanwhile, by affiliation, we have:

$$\frac{\psi(\theta'|\theta)}{\Psi(\theta'|\theta)} > \frac{\psi(\theta'|\theta')}{\Psi(\theta'|\theta')},$$

we obtain:

$$\frac{\partial U}{\partial \theta'} > \Psi(\theta'|\theta)[(v(\theta', \theta') - \beta(\theta'))\frac{\psi(\theta'|\theta')}{\Psi(\theta'|\theta')} - \beta'(\theta')] = 0.$$

Similarly, we can get: if $\theta' > \theta$, then $\frac{\partial U}{\partial \theta'} < 0$. Therefore, by the first-derivative test, when $\theta' = \theta$, $U(\theta', \theta)$ reaches the maximum. $\square$

The above bidding function is an extension of the first-price auction with private values. Indeed, under private value, $v(\vartheta, \vartheta) = \vartheta$. If the signal is distributed independently, and $\Psi(\cdot|\theta)$ does not depend on θ, i.e., it is just a distribution function of ϑ, then $F(\vartheta|\theta) = \frac{\psi(\vartheta)}{\Psi(\theta)}$. Therefore, the Bayesian-Nash equilibrium for the symmetric private value first-price (sealed-bid) auction is:

$$\beta^{I,p}(\theta) = \int_0^\theta \vartheta \frac{\psi(\vartheta)}{\Psi(\theta)} d\vartheta.$$

21.3.5 Revenue Comparison of Alternative Auctions

We now compare the seller's expected revenues under the three auction forms discussed above. The basic finding will be that under symmetric equilibrium, the English auction dominates the second-price sealed-bid auction, while the seller's expected revenue under the second-price sealed-bid auction is not less than that under the first-price sealed-bid auction.

We first compare the expected revenues of the seller under the English auction and the second-price sealed-bid auction.

The equilibrium strategy under the second-price sealed-bid auction with symmetric equilibrium is given by $\beta^{II}(\theta) = v(\theta, \theta)$, where

$$v(\theta, \vartheta) = E(V_1|\theta_1 = \theta, \vartheta_1 = \vartheta) = E(u(\theta_1, \vartheta_1, \ldots, \vartheta_{n-1}))|\theta_1 = \theta, \vartheta_1 = \vartheta).$$

If $\theta > \vartheta$, the price paid by bidder 1 (the income of the auctioneer) is:

$$\begin{aligned} v(\vartheta, \vartheta) &= E(u(\theta_1, \vartheta_1, \ldots, \vartheta_{n-1})|\theta_1 = \vartheta, \vartheta_1 = \vartheta) \\ &= E(u(\vartheta_1, \vartheta_1, \ldots, \vartheta_{n-1})|\theta_1 = \vartheta, \vartheta_1 = \vartheta) \\ &\leqq E(u(\vartheta_1, \vartheta_1, \ldots, \vartheta_{n-1})|\theta_1 = \theta, \vartheta_1 = \vartheta). \end{aligned}$$

The expected revenue of the seller is then:

$$\begin{aligned} E(R^{II}) &= E(\beta^{II}(\vartheta_1)|\theta_1 > \vartheta_1) \\ &= E(v(\vartheta_1, \vartheta_1)|\theta_1 > \vartheta_1) \\ &\leqq E(E(u(\vartheta_1, \vartheta_1, \ldots, \vartheta_{n-1})|\theta_1 = \theta, \vartheta_1 = \vartheta)|\theta_1 > \vartheta_1) \\ &= E(u(\vartheta_1, \vartheta_1, \ldots, \vartheta_{n-1})|\theta_1 > \vartheta_1) \\ &= E(\beta^2(\vartheta_1, \vartheta_1, \ldots, \vartheta_{n-1})) \\ &= E(R^{Eng}), \end{aligned}$$

where the second-last equation is obtained by the definition of β^2 by (21.3.19); and R^{II} and R^{Eng} are the prices paid by the winning bidders of the second-price sealed-bid auction and the English auction, respectively.

Now, we compare the payments paid by the winning bidder of the second-price sealed-bid auction and of the first-price sealed-bid auction.

We can also suppose that signal of bidder 1 is the highest, and thus we only have to consider the expected payment paid by bidder 1 under the symmetric Bayesian-Nash equilibrium.

$$\begin{aligned} E(R^{II}(\vartheta_1)|\theta_1 = \theta, \theta_1 > \vartheta_1) &= E(v(\vartheta_1, \vartheta_1)|\theta_1 = \theta, \theta_1 > \vartheta_1) \\ &= \int_0^\theta v(\vartheta, \vartheta) dG(\vartheta|\theta), \end{aligned}$$

where $G(\vartheta|\theta) \equiv \frac{\Psi(\vartheta|\theta)}{\Psi(\theta|\theta)}$ is a distribution function with support $[0, \theta]$.

However, in the first-price sealed-bid auction, $\beta^I = \int_0^\theta v(\vartheta, \vartheta) dF(\vartheta|\theta)$. Next, we will verify that $F(\vartheta|\theta)$ is first-order stochastically dominated by $G(\vartheta|\theta)$.

By affiliation, if $\tau < \theta$, then we have

$$\frac{\psi(\tau|\tau)}{\Psi(\tau|\tau)} \leqq \frac{\psi(\tau|\theta)}{\Psi(\tau|\theta)}.$$

So, for any $\vartheta < \theta$, we have:

$$\begin{aligned} -\int_\vartheta^\theta \frac{\psi(\tau|\tau)}{\Psi(\tau|\tau)} d\tau &\geqq -\int_\vartheta^\theta \frac{\psi(\tau|\theta)}{\Psi(\tau|\theta)} d\tau \\ &= -\int_\vartheta^\theta \frac{d}{d\tau} \ln \Psi(\tau|\theta) d\tau \\ &= \ln \left(\frac{\Psi(\vartheta|\theta)}{\Psi(\theta|\theta)} \right). \end{aligned}$$

We thus have

$$F(\vartheta|\theta) = \exp\left(-\int_\vartheta^\theta \frac{\psi(\tau|\tau)}{\Psi(\tau|\tau)} d\tau \right) \geqq \frac{\Psi(\vartheta|\theta)}{\Psi(\theta|\theta)} = G(\vartheta|\theta),$$

which means that $G(\vartheta|\theta)$ first-order stochastically dominates $F(\vartheta|\theta)$. Since $v(\vartheta, \vartheta)$ is an increasing function, by the equivalence condition of first-order dominance, we have

$$E(R^{II}(\vartheta_1)|\theta_1 = \theta, \theta_1 > \vartheta_1) \geqq E(\beta^I(\vartheta_1)|\theta_1 = \theta, \theta_1 > \vartheta_1).$$

The proposition below summarizes the expected revenue comparisons of different auction formats.

Proposition 21.3.4 *For economic environments with symmetric, interdependent values and affiliated signals, the expected revenue of the seller at symmetric equilibrium under the English auction is the highest, followed by the second-price sealed-bid auction, while the expected revenue of the first-price sealed-bid auction is the lowest.*

21.3.6 Efficiency of Auctions

Under the symmetric economic environment and symmetric Bayesian-Nash equilibria of all of the three auction formats above, the bidder with the highest bid wins the object. Can we say that the allocation is efficient? Efficiency means that the bidder with the highest valuation wins the object (of course, it should not be less than auctioneer's reserve price). The example below shows that the allocations of all of the auction formats above may not be efficient.

Example 21.3.3 Consider a two-bidder auction in which their valuation functions are symmetric, and θ_1 and θ_2 are private signals of the two bidders. Suppose that their valuation functions are:

$$u_1(\theta_1, \theta_2) = \frac{1}{2}\theta_1 + \theta_2,$$
$$u_2(\theta_1, \theta_2) = \theta_1 + \frac{1}{2}\theta_2.$$

Obviously, if $\theta_1 > \theta_2$, then $u_1 < u_2$. Therefore, bidder 1 wins regardless of whether the auction is an English auction, a second-price auction or a first-price auction, although her valuation is lower. So, the outcomes of all of the three forms of auctions are inefficient.

In this example, the private signal θ_i of each bidder i has less impact on herself than from the other bidder, and then the winner of the auction is not the bidder with the highest valuation. Thus, we need to impose some additional condition to guarantee efficiency.

Definition 21.3.2 (Single Crossing Property) For the valuation system of auctions, we will say that $u_i, i \in \{1, \ldots, n\}$, satisfies the **single crossing property** if for any $j \neq i$, and for all $\boldsymbol{\theta}$,

$$\frac{\partial u_i}{\partial \theta_i}(\boldsymbol{\theta}) \geqq \frac{\partial u_j}{\partial \theta_i}(\boldsymbol{\theta}).$$

Under symmetric equilibrium, if the single crossing condition is satisfied, then the higher is the signal of a bidder, the higher is the valuation of the bidder, and thus the three auction formats result in efficient outcomes.

Proposition 21.3.5 *For economic environments with symmetric, interdependent values and affiliated signals, if the single crossing property is satisfied, then all symmetric Bayesian-Nash equilibria of the second-price, English, and first-price auctions result in efficient outcomes.*

PROOF. In the symmetric model with interdependent values, the valuation of bidder i becomes

$$u_i(\boldsymbol{\theta}) = u(\theta_i, \boldsymbol{\theta}_{-i}).$$

Suppose that $\theta_i > \theta_j$, and define $\theta(\lambda) = (1-\lambda)(\theta_j, \theta_i, \boldsymbol{\theta}_{-ij}) + \lambda(\theta_i, \theta_j, \boldsymbol{\theta}_{-ij})$. According to the Mean Value Theorem of Integrals:

$$u(\theta_i, \theta_j, \boldsymbol{\theta}_{-ij}) - u(\theta_j, \theta_i, \boldsymbol{\theta}_{-ij}) = \int_0^1 \nabla u(\theta(\lambda)) \cdot \theta'(\lambda) d\lambda,$$

where

$$\nabla u(\theta(\lambda)) \cdot \theta'(\lambda) = \frac{\partial}{\partial \theta_i} u(\theta(\lambda))(\theta_i - \theta_j) + \frac{\partial}{\partial \theta_j} u(\theta(\lambda))(\theta_j - \theta_i) \geqq 0$$

since $\theta_i > \theta_j$ and by single crossing property, $\frac{\partial}{\partial \theta_i} u(\theta(\lambda)) \geqq \frac{\partial}{\partial \theta_j} u(\theta(\lambda))$. As a consequence, we have

$$u(\theta_i, \theta_j, \boldsymbol{\theta}_{-ij}) \geqq u(\theta_j, \theta_i, \boldsymbol{\theta}_{-ij}).$$

□

For a general Bayesian-Nash equilibrium, such as non-symmetric equilibrium (if any), does it also result in an efficient allocation? The following example given by Krishna (2002) illustrates that there is no efficient allocation in an auction if the single crossing condition is not satisfied.

Example 21.3.4 Suppose that there are two bidders, whose signal are θ_1 and θ_2, and their valuation functions for an indivisible object are:

$$\begin{aligned} u_1(\theta_1, \theta_2) &= \theta_1, \\ u_2(\theta_1, \theta_2) &= \theta_1^2, \end{aligned}$$

where $\theta_1 \in [0, 2]$; and the signal of bidder 2 does not affect their valuations of the object. Clearly, the valuations do not satisfy the single crossing property, such as $\frac{\partial u_1}{\partial \theta_1}(1, \theta_2) < \frac{\partial u_2}{\partial \theta_1}(1, \theta_2)$.

It can be easily seen that $u_1 > u_2$ if and only if $\theta_1 < 1$. Let $\boldsymbol{y}(\theta_1) = (y_1(\theta_1), y_2(\theta_1))$ be the probabilities that bidder 1 and bidder 2 get the object, respectively; and $\boldsymbol{t}(\theta_1) = (t_1(\theta_1), t_2(\theta_1))$ the transfers of bidder 1 and bidder 2. If $(\boldsymbol{y}(\theta_1), \boldsymbol{t}(\theta_1))$ is efficient, it requires that $y_1(\theta_1) = 1$ if $\theta_1 < 1$, and $y_2(\theta_1) = 1$ if $\theta_1 > 1$.

Let θ_1 and θ_1' be agent 1's two signals with $\theta_1 < 1 < \theta_1'$. Then, efficiency and incentive compatibility together require that

$$0 + t_1(\theta_1') \geqq \theta_1' + t_1(\theta_1)$$

when agent 1's private signal is θ_1', and

$$\theta_1 + t_1(\theta_1) \geqq 0 + t_1(\theta_1')$$

when her private signal is θ_1. The two inequalities above imply $\theta_1 \geqq \theta_1'$, which is a contradiction. Therefore, there is no incentive-compatible auction that is also efficient.

In fact, we can show that the single crossing property is a necessary condition for the efficiency of an auction. In general, if there exists an efficient mechanism, then by the revelation principle, there must exist an efficient direct mechanism in which "truth-telling" is weakly dominant strategy. Suppose that all of the bidders other than bidder i have observed a signal $\boldsymbol{\theta}_{-i}$, and bidder i's private signal is θ_i. If, regardless of what the value of her signal θ_i is, bidder i either always wins or always loses, then the single crossing property means nothing for θ_i.

Therefore, we are only concerned about the situation in which whether or not she gets the object if buyer i's signal can be determined. We shall say that buyer i is **pivotal** if there exist signals θ_i and θ_i', such that $u_i(\theta_i, \boldsymbol{\theta}_{-i}) > \max_{j \neq i} u_j(\theta_i, \boldsymbol{\theta}_{-i})$ and $u_i(\theta_i', \boldsymbol{\theta}_{-i}) < \max_{j \neq i} u_j(\theta_i', \boldsymbol{\theta}_{-i})$. Incentive compatibility requires that when bidder i's signal is θ_i, she would not report θ_i', which means that:

$$u_i(\theta_i, \boldsymbol{\theta}_{-i}) + t_i(\theta_i, \boldsymbol{\theta}_{-i}) \geqq 0 + t_i(\theta_i', \boldsymbol{\theta}_{-i}).$$

Furthermore, when she observes θ_i', she would not report θ_i, which means that:

$$0 + t_i(\theta_i', \boldsymbol{\theta}_{-i}) \geqq u_i(\theta_i', \boldsymbol{\theta}_{-i}) + t_i(\theta_i, \boldsymbol{\theta}_{-i}).$$

From the two inequalities above, we have:

$$u_i(\theta_i, \boldsymbol{\theta}_{-i}) \geqq u_i(\theta_i', \boldsymbol{\theta}_{-i})$$

In other words, bidder i's value when she wins the object must be at least as high as when she does not. Overall, keeping others' signals unchanged, an increase in bidder i's private signal will not reduce the probability of winning the object.

So, by incentive compatibility, the efficient mechanism must satisfy the single-crossing property, which requires that, for any θ_i, we must have:

$$\frac{\partial u_i}{\partial \theta_i}(\theta_i, \boldsymbol{\theta}_{-i}) \geqq \frac{\partial u_j}{\partial \theta_i}(\theta_i, \boldsymbol{\theta}_{-i}).$$

Thus, if an allocation that satisfies incentive compatibility is efficient, it must satisfy the single-crossing property. In fact, we already know this conclusion in Chapter 16 when dealing with the principal-agent issues.

21.3.7 The Generalized VCG Mechanism

For economic environments with symmetric and interdependent values, a symmetric equilibrium of an auction is efficient if the single crossing condition is satisfied. Can we have an efficient mechanism under general environments with interdependent values, but without symmetry? In the previous chapters on mechanism design, we have shown that VCG is an efficient mechanism if the players' valuation functions only depend on their own types.

However, it will lose efficiency when a bidder's valuation function is interdependent on others' signals. Indeed, if bidder i who gets the object is asked to pay the second-highest value at all of the reported signals, $\max_{j\neq i} u_j(\theta_i, \boldsymbol{\theta}_{-i})$, which depends on θ_i, then player i would have an incentive to report a lower signal in order to reduce the payment. As such, the usual VCG mechanism is not "truth-telling". Nevertheless, the so-called **generalized VCG mechanism**, which is a variation of VCG, restores the incentive to tell truth and is efficient. We now describe the generalized VCG mechanism.

Suppose that there are n bidders. $(\Theta, \boldsymbol{y}(\boldsymbol{\theta}), \boldsymbol{t}(\boldsymbol{\theta}))$ is a direct mechanism, where $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_n) \in \Theta$, $\boldsymbol{y}(\boldsymbol{\theta}) = (y_1(\boldsymbol{\theta}), \ldots, y_n(\boldsymbol{\theta}))$ are the probabilities of bidders allocated to the object, and $\boldsymbol{t}(\boldsymbol{\theta}) = (t_1(\boldsymbol{\theta}), \ldots, t_n(\boldsymbol{\theta}))$ are the transfers of the players.

If the mechanism is efficient, we have:

$$y_i^*(\boldsymbol{\theta}) = \begin{cases} 1 & \text{if } u_i(\boldsymbol{\theta}) > \max_{j\neq i} u_j(\boldsymbol{\theta}) \\ 0 & \text{if } u_i(\boldsymbol{\theta}) < \max_{j\neq i} u_j(\boldsymbol{\theta}). \end{cases}$$

If there is more than one bidder who has the highest value, the object is allocated to each of these players with an equal probability. However, with a continuous distribution, the probability of this situation equals 0.

Let the transfer of the bidder who gets the object be:

$$t_i^* = -u_i(\vartheta_i(\boldsymbol{\theta}_{-i}), \boldsymbol{\theta}_{-i}),$$

where

$$\vartheta_i(\boldsymbol{\theta}_{-i}) = \inf\{\bar{\theta}_i : u_i(\bar{\theta}_i, \boldsymbol{\theta}_{-i}) \geqq \max_{j\neq i} u_j(\bar{\theta}_i, \boldsymbol{\theta}_{-i})\}.$$

This implies that, given other bidders' signals, $\vartheta_i(\boldsymbol{\theta}_{-i})$ is the lower bound of the signal of bidder i that still wins the object. The bidder who does not win the object pays nothing. Since such a defined transfer does not depend on the signal of bidder i directly, just as in the private value model, it can result in efficient outcomes.

We call the direct mechanism $(\Theta, \boldsymbol{y}^*, \boldsymbol{t}^*)$ defined above a **generalized VCG mechanism**, because it modifies the VCG mechanism for an interdependent values setting. As mentioned above, the usual VCG mechanism under interdependent values is not a "truth-telling" mechanism. To guarantee that the winning player i has an incentive to tell the truth, agent i shall be asked to pay only $\max_{j\neq i} u_j(\vartheta_i(\boldsymbol{\theta}_{-i}), \boldsymbol{\theta}_{-i})$ (which does not depend on θ_i), rather than $\max_{j\neq i} u_j(\theta_i, \boldsymbol{\theta}_{-i})$. When $u_i(\theta_i, \boldsymbol{\theta}_{-i}) > \max_{j\neq i} u_j(\theta_i, \boldsymbol{\theta}_{-i})$, we have $\vartheta_i(\boldsymbol{\theta}_{-i}) \leqq \theta_i$, so $\max_{j\neq i} u_j(\vartheta_i(\boldsymbol{\theta}_{-i}), \boldsymbol{\theta}_{-i}) \leqq \max_{j\neq i} u_j(\theta_i, \boldsymbol{\theta}_{-i})$.

The following proposition shows that the generalized VCG mechanism is an efficient mechanism under the affiliated valuation system.

Proposition 21.3.6 *Suppose that the valuations of bidders of the object satisfy the single crossing property. The generalized VCG mechanism is an efficient and incentive-compatible (truth telling) mechanism in a dominant strategy equilibrium.*

PROOF. Obviously, if everyone tells the truth, the generalized VCG mechanism is efficient. Now, we need to prove that it satisfies incentive compatibility in dominant strategies. When $u_i(\theta_i, \boldsymbol{\theta}_{-i}) > \max_{j\neq i} u_j(\theta_i, \boldsymbol{\theta}_{-i})$, bidder i who gets the object pays $u_i(\vartheta_i(\boldsymbol{\theta}_{-i}), \boldsymbol{\theta}_{-i})$. As the payment does not depend on the signal θ_i that the bidder reports, and $\theta_i \geqq \vartheta_i(\boldsymbol{\theta}_{-i})$, the expected utility of reporting the true signal is: $u_i(\theta_i, \boldsymbol{\theta}_{-i}) - u_i(\vartheta_i(\boldsymbol{\theta}_{-i}), \boldsymbol{\theta}_{-i}) \geqq 0$. When the bidder reports other signal θ_i', if $\theta_i' > \vartheta_i(\boldsymbol{\theta}_{-i})$, her expected utility does not change; if $\theta_i' < \vartheta_i(\boldsymbol{\theta}_{-i})$, she would not win the object, and her expected utility will be 0. As a consequence, bidder i gets the object, and there is no other strategy that can bring a higher expected utility than to tell the truth.

When $u_i(\theta_i, \boldsymbol{\theta}_{-i}) < \max_{j\neq i} u_j(\theta_i, \boldsymbol{\theta}_{-i})$, if she tells the truth, bidder i's expected utility equals 0; if she reports θ_i' to alter the outcome, by the single crossing property: $\theta_i' > \vartheta_i(\boldsymbol{\theta}_{-i}) > \theta_i$, bidder i wins the object, but her expected utility is $u_i(\theta_i, \boldsymbol{\theta}_{-i}) - u_i(\vartheta_i(\boldsymbol{\theta}_{-i}), \boldsymbol{\theta}_{-i}) < 0$. Thus, we have shown that the generalized VCG mechanism is an efficient mechanism that is also incentive compatible in dominant strategies. $\square$

For private values, the generalized VCG mechanism above returns to the Clark mechanism, a special form of the VCG mechanism. However, when values are interdependent, the generalized VCG mechanism is different from the Clark mechanism since the transfer $u_i(\vartheta_i(\boldsymbol{\theta}_{-i}), \boldsymbol{\theta}_{-i})$ depends on the valuation functions of all other bidders.

21.3.8 Optimal Auction Design with Interdependent Values

We now examine the optimal mechanism design when values are interdependent. Under the optimal mechanism of independent private values, in order to provide incentives to bidders to tell the truth, the bidder with the highest value is able to obtain certain positive information rent. As such, the optimal mechanism when values are independent only results in second-best outcomes. However, as long as signals of bidders are correlated, we can obtain the same result under interdependent values as the Full Surplus Extraction Theorem under private values discussed in Chapter 16: there exists an optimal mechanism, such that the auctioneer can extract all of the surplus even in the presence of interdependent values.

In order to understand the Full Surplus Extraction Theorem easily, we set the space of signals to be discrete. Define $\Theta_i = \{\theta_i^1, \theta_i^2, \ldots, \theta_i^{m_i}\}$ as the discrete message space of bidders, where $\theta_i^1 < \theta_i^2 < \ldots < \theta_i^{m_i}$. Suppose that $\theta_i \in \Theta_i$, the signals of all bidders are written as $\boldsymbol{\theta}$, and the valuation function of bidder i is $u_i = u_i(\theta_i, \boldsymbol{\theta}_{-i})$. For the discrete values, we can define the single crossing condition as: for $j \neq i$, if $u_i(\theta_i, \boldsymbol{\theta}_{-i}) \geqq u_j(\theta_i, \boldsymbol{\theta}_{-i})$, then $u_i(\theta_i', \boldsymbol{\theta}_{-i}) \geqq u_j(\theta_i', \boldsymbol{\theta}_{-i})$ holds for any $\theta_i < \theta_i' \in \Theta$; at the same time, the conclusion does not change when the inequality is strict.

For any bidder i, let Π_i be a matrix with m_i rows and $\prod_{j \neq i} m_j$ columns, and its element be denoted by $\pi_i(\boldsymbol{\theta}_{-i}|\theta_i)$. Each row corresponds to bidder i's beliefs regarding the signal distribution of other bidders given θ_i. If the signals are independent, then all of the rows are the same, which means that the rank of Π_i is 1. If the signals are correlated, the beliefs represented by different rows are not the same. In the proposition below, we suppose that the Π_i is of rank m_i, which means that the bidder's beliefs regarding the signal distribution of other bidders will change according to different θ_i.

We then have the following Full Surplus Extraction Theorem under interdependent values.

Theorem 21.3.1 (Full Surplus Extraction Theorem for Interdependent Vaules) *Suppose that signals are discrete, and the valuations satisfy the single crossing condition. If for any bidder i, Π_i, the matrix of beliefs about other players, is of full rank (that is m_i), then there exists a direct mechanism in which truth-telling is an efficient ex-post equilibrium, and the expected information rent (that is expected utility in the event) of every bidder is equal to zero. As a result, we have first-best outcomes, i.e., all of the surplus is extracted by the auctioneer.*

PROOF. For the generalized VCG mechanism $(\Theta, \boldsymbol{y}(\boldsymbol{\theta}), \boldsymbol{t}(\boldsymbol{\theta}))$ defined above, we know that it constitutes an efficient truth-telling mechanism. In this mechanism, the expected utility of bidder i with signal θ_i at equilibrium is:

$$\bar{U}_i^*(\theta_i) = \sum_{\boldsymbol{\theta}_{-i}} \pi_i(\boldsymbol{\theta}_{-i}|\theta_i)[y_i(\boldsymbol{\theta})u_i(\boldsymbol{\theta}) + t_i(\boldsymbol{\theta})].$$

Let $\bar{u}_i^* = [\bar{U}^*(\theta_i^1), \bar{U}^*(\theta_i^2), \ldots, \bar{U}^*(\theta_i^{m_i})]'$. As the matrix Π_i has full row rank, there is a column vector $c_i = (c_i(\boldsymbol{\theta}_{-i}))_{\boldsymbol{\theta}_{-i} \in \boldsymbol{\Theta}_{-i}}$ of size $\sum_{j \neq i} m_j$, such that:

$$\Pi_i c_i = -\bar{u}_i^*,$$

for $\forall \theta_i$, and then we have:

$$\sum_{\boldsymbol{\theta}_{-i}} \pi_i(\boldsymbol{\theta}_{-i}|\theta_i) c_i(\boldsymbol{\theta}_{-i}) = \bar{U}_i^*(\theta_i).$$

Comparing the generalized VCG mechanism $(\Theta, \boldsymbol{y}(\boldsymbol{\theta}), \boldsymbol{t}(\boldsymbol{\theta}))$ with the Cremer-McLean mechanism $(\Theta, \boldsymbol{y}(\boldsymbol{\theta}), \boldsymbol{t}(\boldsymbol{\theta}) + c_i(\boldsymbol{\theta}_{-i}))$, we find that the allocation rule is the same, and the transfer of every bidder does not depend on bidder i's report. Therefore, the Cremer-McLean mechanism is still a truth-telling mechanism, and it is efficient. At the same time, every bidder's interim expected utility is zero. Consequently the participation constraints of bidders are binding, and the auctioneer extracts all of the surplus. □

This conclusion is robust. As long as signals of individuals are correlated, irrespective of what the distribution functions are, the information rent is reduced to zero. This conclusion does not depend on whether or not values are interdependent, but rather depends on the correlations of signals.

21.4 Multi-Unit Simultaneous Auctions with Private Values

The remainder of this chapter discusses multiple object auctions under private values. In practice, most auctions are multiple objects auctions, which exist in many different forms that depend on the relationship of objects, as well as the timing and formats of auctions.

Objects can be either identical or heterogenous. When the objects are **identical**, a buyer may only need one or some of them. Under this situation, the marginal willingness to pay will decrease with the increase in the number of objects bought. When the objects are **heterogenous**, the objects are dissimilar but may affect each other. Then, the willingness to pay of the buyer will change with the relationship of these objects. For instance, the value of consuming two objects simultaneously may be higher than the sum of value from consuming them separately, which means that the objects are somehow **complements**. Of course, they may also be **substitutes**. The complementary and substitutability aspects are worthy of special attention for designing an auction.

Multiple object auctions also depend on the timing and formats of auctions. When multiple objects are to be sold, numerous options can be chosen by the seller. The seller can sell the objects jointly in a single auction or separately in multiple auctions. The former case is called a **simultaneous auction**, in which all of the objects are sold in a single auction at one time, but not necessarily all to the same bidder, and the bids on the various objects collectively influence the overall allocation. The latter case is called a **sequential auction**, in which the objects are sold one at a time in separate auctions in a way in which the bids in the auction for one of the objects do not directly influence the outcome of the auction for another object.

The seller can also choose different auction formats. Formats can be sealed-bid, such as a **discriminatory auction**, a **uniform-price auction** and a **Vickrey auction**, or open-bid actions such as a **Dutch auction**, an **English auction**, and an **Ausubel auction**.

We consider multiple identical-object simultaneous auctions in this section, examine multiple identical-object sequential auctions in the next section, and investigate heterogeneous multiple-object combinational auctions in private values in the last section of this chapter.

21.4.1 Basic Model

Suppose that there are K units of identical objects for sale and n bidders. The set of bidders is $N = \{1, \ldots, n\}$. The marginal value vector of bidder i to different units of goods is $\boldsymbol{V}_i(\boldsymbol{\theta}) = (V_i^1(\boldsymbol{\theta}), \ldots, V_i^K(\boldsymbol{\theta}))$, where V_i^k is bidder i's marginal value to item k. Assume that $V_i^k \leqq V_i^l, \forall k > l$, which means diminishing marginal utility, i.e., the marginal value decreases with increases in the number of identical objects; θ_i is the signal observed by bidder i, and $\boldsymbol{\theta}$ is the profile of signals observed by all of the bidders. The bidding function of bidder i is $\boldsymbol{b}_i = (b_i^1(\boldsymbol{V}_i), \ldots, b_i^K(\boldsymbol{V}_i))$, where b_i^k is the bid of bidder i to the k-th unit of objects.

As multi-unit auctions are relatively complex, we focus on private value environments:

$$\boldsymbol{V}_i(\boldsymbol{\theta}) = \boldsymbol{V}_i(\theta_i) \equiv \boldsymbol{V}_i.$$

We assume that $\boldsymbol{V}_i$ of bidder i is independently and identically distributed on $\{\boldsymbol{v}_i \in [0, w]^K : v_i^k \geqq v_i^{k+1}, \forall k < K\}$ according to the density function $\varphi(\boldsymbol{v}_i)$.

We will discuss the allocation of objects and transfers under different auction formats. All of the formats of auctions are standard auctions, which obey the rule that objects are given to the K highest of these bids.

In a simultaneous auction environment with K identical objects to be auctioned, each bidder i is asked to submit K bids $\boldsymbol{b}_i = (b_i^1, \ldots, b_i^K)$, satisfying $b_i^1 \geqq b_i^2 \geqq \ldots, \geqq b_i^K$, to indicate how much bidder i is willing to pay for each additional unit.

Given the bid vector $\boldsymbol{b}_i$ of each bidder i, we will obtain her **demand function** d_i:

$$d_i(p) = \max\{k : b_i^k \geqq p\},$$

where $d_i(p)$ is non-increasing. If the price p is between b_i^k and b_i^{k+1}, bidder i will buy k units. In turn, if we know the demand for the objects, we can get the bid vector of the bidders. Indeed, these two methods are equivalent.

Given a bidding set $\{b_i^k, i = 1, \ldots, n; k = 1, \ldots, K\}$, assume that K units of objects are given to the K highest bids among all bids by bidders. For the discriminatory auction, uniform-price auction and Vickrey auction that we discuss below, the allocation of objects is the same, but the transfers to win the objects are different in these three auctions.

We will illustrate the allocation and pricing rules of the three auctions through a simple example below.

Example 21.4.1 Suppose that there are six units of identical objects and three bidders in an auction. The bid vectors of the three bidders are:

$$\begin{aligned}\mathbf{b}_1 &= (54, 46, 43, 40, 33, 15),\\ \mathbf{b}_2 &= (60, 55, 47, 32, 27, 13),\\ \mathbf{b}_3 &= (48, 45, 35, 24, 14, 9).\end{aligned}$$

The six highest bids are then:

$$(b_2^1, b_2^2, b_1^1, b_3^1, b_2^3, b_1^2) = (60, 55, 54, 48, 47, 46).$$

As such, bidder 1 gets two units, bidder 2 gets three units, and bidder 3 gets one unit.

21.4.2 Multi-Unit Simultaneous Sealed-Bid Auctions

For multi-unit simultaneous sealed-bid auctions, there are three basic auction formats: the discriminatory auction, the uniform-price auction, and the Vickrey auction. We define these here one by one.

Discriminatory Auctions

In a **discriminatory auction**, each bidder pays an amount equal to the sum of her bids that are among the K highest of the $N \times K$ bids submitted in all. In other words, if bidder i is allocated to k_i units of objects, her total payment is $b_i^1 + \ldots + b_i^{k_i}$, which amounts to perfect price discrimination relative to the demand submitted.

In Example 21.4.1, bidder 1 gets two units of the object, and her payment equals $54 + 46 = 100$. Bidder 2 and bidder 3 get three units and one unit of the object, respectively, and their payments equal $60 + 55 + 47 = 162$ and 48, respectively. Thus, the total revenue received by the auctioneer is $100 + 162 + 48 = 310$.

The discriminatory pricing rule can also be described by the **residual supply function**. At price p, the residual supply faced by bidder i, denoted as $s_{-i}(p)$, is equal to the total units of objects subtracting the sum of the amounts demanded by other bidders, so:

$$s_{-i}(p) \equiv \max\{K - \sum_{j\neq i} d_j(p), 0\}.$$

Obviously, as $d_i(p)$ is non-increasing, $s_{-i}(p)$ is non-decreasing.

In Example 21.4.1, the demand of each bidder is:

$$d_1(p) = \begin{cases} 0 & \text{if } p > 54, \\ 1 & \text{if } 54 \geqq p > 46, \\ 2 & \text{if } 46 \geqq p > 43, \\ 3 & \text{if } 43 \geqq p > 40, \\ 4 & \text{if } 40 \geqq p > 33, \\ 5 & \text{if } 33 \geqq p > 15, \\ 6 & \text{if } 15 \geqq p > 0, \end{cases}$$

$$d_2(p) = \begin{cases} 0 & \text{if } p > 60, \\ 1 & \text{if } 60 \geqq p > 55, \\ 2 & \text{if } 55 \geqq p > 47, \\ 3 & \text{if } 47 \geqq p > 32, \\ 4 & \text{if } 32 \geqq p > 27, \\ 5 & \text{if } 27 \geqq p > 13, \\ 6 & \text{if } 13 \geqq p > 0, \end{cases}$$

$$d_3(p) = \begin{cases} 0 & \text{if } p > 48; \\ 1 & \text{if } 48 \geqq p > 45, \\ 2 & \text{if } 45 \geqq p > 35, \\ 3 & \text{if } 35 \geqq p > 24, \\ 4 & \text{if } 24 \geqq p > 14, \\ 5 & \text{if } 14 \geqq p > 9, \\ 6 & \text{if } 9 \geqq p > 0. \end{cases}$$

The residual supply of bidder 1 is:

$$s_{-1}(p) = \begin{cases} 6 & \text{if } p > 60, \\ 5 & \text{if } 60 \geqq p > 55, \\ 4 & \text{if } 55 \geqq p > 48, \\ 3 & \text{if } 48 \geqq p > 47, \\ 2 & \text{if } 47 \geqq p > 45, \\ 1 & \text{if } 45 \geqq p > 35, \\ 0 & \text{if } 35 \geqq p \geqq 0. \end{cases}$$

When $K = 1$, the discriminatory auction turns out to be the first-price sealed-bid auction. Therefore, the discriminatory price is just an extension of the first-price sealed-bid auction with multiple identical objects.

Uniform-Price Auctions

In a **uniform-price auction**, all of the objects are sold to the bidders at a market-clearing price. In the discrete model discussed here, we take the highest losing bid that is *the* $K+1$ *highest of all of the bids as the market-clearing price*. In Example 21.4.1, the seventh-highest bid is 45.

In a general situation, if the bidding set of n bidders is $\{b_i^k, i = 1, \ldots, n; k = 1, \ldots, K\}$, and bidder i gets k_i units of objects from the K highest of these bids, then the market-clearing price is:

$$p = \max_i \{b_i^{k_i+1}\}.$$

For the above example, $p = \max_i\{b_i^{k_i+1}\} = \max\{43, 32, 45\} = 45$. The total revenue received by the auctioneer is $6 \times 45 = 270$.

There is another alternative way to determine the uniform price p. Let $\boldsymbol{c}_{-i}$ be the K-vector of **competing bids** of bidder i facing other bidders, which is constructed by the K highest bids of $\{b_j^k, j \neq i, k = 1, \ldots, K\}$, as these bids will affect whether or not bidder i can get the objects. $\boldsymbol{c}_{-i} = (c_{-i}^1, \ldots, c_{-i}^K)$ is a price vector for rearranging the component in a decreasing order. As such, bidder i wins exactly $k^i > 0$ units if and only if

$$b_i^{k^i} > \mathbf{c}_{-i}^{K-k^i+1} \text{ and } b_i^{k^i+1} < \mathbf{c}_{-i}^{K-k^i}.$$

Then, the residual supply of bidder i can be also written as:

$$s_{-i}(p) = K - \max\{k : c_{-i}^k \geqq p\},$$

which is the inverse function of $\boldsymbol{c}_{-i}$. Then, the market-clearing price can also be determined by

$$p = \max\{b_i^{k_i+1}, c_{-i}^{K-k_i+1}\},$$

which can be obtained from any bidder i, resulting in the same uniform price p.

In Example 21.4.1, since

$$\begin{aligned}
\mathbf{b}_1 &= (54, 46, 43, 40, 33, 15) \\
\mathbf{b}_2 &= (60, 55, 47, 32, 27, 13) \\
\mathbf{b}_3 &= (48, 45, 35, 24, 14, 9)
\end{aligned}$$

and

$$\begin{aligned}
\mathbf{c}_{-1} &= (60, 55, 48, 47, 45, 35) \\
\mathbf{c}_{-2} &= (54, 48, 46, 45, 43, 40) \\
\mathbf{c}_{-3} &= (60, 55, 54, 47, 46, 43),
\end{aligned}$$

the market-cleaning price can be obtained from bidder 1: $p = \max\{b_1^3, c_{-1}^5\} = \max\{43, 45\} = 45$. We can also obtain the same price $p = \max\{b_2^4, c_{-2}^4\} = \max\{32, 45\} = 45$ from bidder 2 or $p = \max\{b_3^2, c_{-3}^6\} = \max\{45, 43\} = 45$ from bidder 3.

When $K = 1$, the uniform-price auction reduces to the second-price sealed-bid auction. When $K > 1$, the uniform-price auction is different from the second-price sealed-bid auction in several ways, especially in the bidding strategy, which may not be truth-telling. Therefore, we cannot extend the second-price sealed-bid auction to the uniform-price auction directly. The Vickrey auction discussed below is more like an extension of the second-price auction with a single object.

Vickrey Auctions

In second-price auction with a single object, the winner's price is the highest losing bid of the other bidders. In a **Vickrey auction** with multi-unit objects, a bidder who wins k^i units pays the k^i highest losing bids of the other bidders. The basic principle of the Vickrey auction is the same as the VCG mechanism: Each bidder is asked to pay an amount equal to the externality that she exerts on other competing bidders.

Suppose that bidder i wins k_i units, to get her first unit of objects, it must be true that $b_i^1 > c_{-i}^K$. The bidder is then asked to pay an amount equal to c_{-i}^K units of externality, where c_{-i}^K is the K-highest bid among other bidders. As such, the bids of bidders only affect the allocation of the objects, but do not affect the payment of the bidders when they win. When the bidder gets the second unit, it must be true that $b_i^2 > c_{-i}^{K-1}$, and thus she pays c_{-i}^{K-1} units of externality to other bidders. Therefore, if the bidder wins k_i units of objects, she will bring $\sum_{k=1}^{k_i} c_{-i}^{K-k+1}$ units of externality to other bidders.

Consequently, under the Vickrey auction, if bidder i wins k_i units, then the amount that the bidder pays is equal to

$$\sum_{k=1}^{k_i} c_{-i}^{K-k+1}.$$

In Example 21.4.1, bidder 1 wins two units, $\boldsymbol{c}_{-1} = (60, 55, 48, 47, 45, 35)$, and her payment is $45 + 35 = 80$; bidder 2 gets three units of objects, $\boldsymbol{c}_{-2} = (54, 48, 46, 45, 43, 40)$, and thus bidder 2's payment is $45 + 43 + 40 = 128$; and bidder 3 gets one unit of objects, $\boldsymbol{c}_{-3} = (60, 55, 54, 47, 46, 43)$, and thus her payment is 43. The total revenue of the auctioneer is 251.

When $K = 1$, the Vickrey auction reduces to the second-price sealed-bid auction, just like the uniform-price auction. However, when $K > 1$, the Vickrey auction is an appropriate extension of the second-price sealed-bid auction to the case of multiple units. Indeed, as we will show, unlike the uniform price auction, it shares numerous important properties with the second-price sealed-bid auction while the other auction formats do not. For instance, in a uniform-price auction, bidder i's payment may depend on her own bid because the market-clearing price is $p = \max\{b_i^{k_i+1}, c_{-i}^{K-k_i+1}\}$, which may be determined by $b_i^{k_i+1}$.

21.4.3 Simultaneous Open-Bid Auctions

In a single-object auction, under private value, each sealed-bid auction has a corresponding open format. For example, the Dutch auction is strategically equivalent to the first-price auction, and the English auction is outcome-equivalent to the second-price auction. We will have similar, but not the same, features in multi-unit auctions. In other words, the Dutch auction is outcome-, but not strategically-, equivalent to the discriminatory auction, the English auction and the Ausubel auction are outcome-equivalent to the uniform-price auction and the Vickrey auction, respectively.

Dutch Auctions

In a **multi-unit Dutch auction** (or **open descending price auction**), the auctioneer starts the auction by calling out a price that is high enough so that no bidder wants to accept it. The price then keeps going down until there is a bidder who wants to buy a unit at the current price. The bidder then gets one unit, and the auction continues. Then, the price continues to go down until there is another bidder who wants to buy a unit at the bidding price. The bidder then gets the second unit, and the auction continues until the price is reduced to a level at which some bidder wants to buy the K-th unit.

The multi-unit Dutch auction is outcome-equivalent to the discriminatory auction. If bidder i accepts the price according to her bid vector $\boldsymbol{b}_i$ in the Dutch auction, she will purchase one unit when the price is reduced to $p = b_i^1$ and buy the second unit when the price is $p = b_i^2$, and so on. However, the Dutch auction is not strategically equivalent to the multi-unit extension of the discriminatory auction. This is because after observing one bidder's willingness to get one unit, other bidders can calculate their willingness to buy other units of objects. However, in the discriminatory auction, as bidders have to report their bids to different amounts of objects at the same time, there is no opportunity for other bidders to update their beliefs. Therefore, they are not strategically equivalent.

English Auctions

The process of a multi-unit **English auction** (or **open ascending-price auction**) is opposite to that of the Dutch auction. The auctioneer changes the price from low to high, and each bidder reports the units that she wants to buy at each reported price. Usually, the total amount of the objects that the bidders want to buy is larger than K at the initially low price. With the price goes up, the bidders will adjust the amounts that they want to buy. At a certain price when the amount that all of the bidders want to buy is equal to K, this price will be the market-clearing price. All of the bidders will buy their desired units at this price.

Just like the relationship of the Dutch auction and the discriminatory auction, the multi-unit English auction is outcome-equivalent to the uniform-price auction, but they are not strategically equivalent.

Ausubel Auctions

The **Ausubel auction** is an alternative ascending-price format that is outcome-equivalent to the Vickrey auction. Similar to the English auction, the price also changes from low to high. The bidders need to report their demand $d_i(p)$ at price p, and the demand will decrease as the price goes up. However, being different from the English auction, in the Ausubel auction, the price of the objects is determined by the following procedure.

At the beginning price p, if each bidder i's demand $d_i(p)$ is sufficiently large, the residual supply of each bidder is

$$s_{-i}(p) \equiv \max\{K - \sum_{j \neq i} d_j(p), 0\} = 0.$$

With the price going up, the demand of each bidder goes down. The price continues to increase until it reaches a level p', such that the residual supply of at least some bidder, say bidder i, satisfies $s_{-i}(p') > 0$. Then, $s_{-i}(p')$ units are sold to bidder i for whom $s_{-i}(p') > 0$ at price p'. The price continues to increase as long as the sale is less than K.

When the price rises to p'', such that the residual supply of at least one bidder, say bidder i, satisfies $s_{-i}(p'') - s_{-i}(p') > 0$. Then, $s_{-i}(p'') - s_{-i}(p') > 0$ units are sold to bidder i for whom $s_{-i}(p'') - s_{-i}(p') > 0$ at price p''. The price continues to go up until the market clears, and then the auction ends.

We use Example 21.4.1 again to describe the rule of the Ausubel auction. Given the strategy of each bidder, i.e.,

$$\begin{aligned}
\mathbf{b}_1 &= (54, 46, 43, 40, 33, 15), \\
\mathbf{b}_2 &= (60, 55, 47, 32, 27, 13), \\
\mathbf{b}_3 &= (48, 45, 35, 24, 14, 9).
\end{aligned}$$

the residual supply of each bidder is:

$$s_{-1}(p) = \begin{cases} 6 & \text{if } p > 60; \\ 5 & \text{if } 60 \geqq p > 55, \\ 4 & \text{if } 55 \geqq p > 48, \\ 3 & \text{if } 48 \geqq p > 47, \\ 2 & \text{if } 47 \geqq p > 45, \\ 1 & \text{if } 45 \geqq p > 35, \\ 0 & \text{if } 35 \geqq p \geqq 0. \end{cases}$$

$$s_{-2}(p) = \begin{cases} 6 & \text{if } p > 54, \\ 5 & \text{if } 54 \geqq p > 48, \\ 4 & \text{if } 48 \geqq p > 46, \\ 3 & \text{if } 46 \geqq p > 45, \\ 2 & \text{if } 45 \geqq p > 43, \\ 1 & \text{if } 43 \geqq p > 40, \\ 0 & \text{if } 40 \geqq p \geqq 0. \end{cases}$$

$$s_{-3}(p) = \begin{cases} 6 & \text{if } p > 60, \\ 5 & \text{if } 60 \geqq p > 55, \\ 4 & \text{if } 55 \geqq p > 54, \\ 3 & \text{if } 54 \geqq p > 47, \\ 2 & \text{if } 47 \geqq p > 46, \\ 1 & \text{if } 46 \geqq p > 43, \\ 0 & \text{if } 43 \geqq p \geqq 0. \end{cases}$$

Suppose that the initial price is $p^0 \leqq 35$, $s_{-i}(p^0) = 0, \forall i \in N = \{1, 2, 3\}$. When the price rises to $p^1 = 35+\epsilon$, $s_{-1}(35+\epsilon) = 1, s_{-j}(35+\epsilon) = 0, \forall j \neq 1$. When $\epsilon \to 0$, bidder 1 buys the first unit of commodity at price 35.

When the price rises to $p^2 = 40+\epsilon$, $s_{-1}(40+\epsilon)-s_{-1}(35+\epsilon) = 0, s_{-2}(40+\epsilon) - s_{-2}(35+\epsilon) = 1, s_{-3}(40+\epsilon) - s_{-3}(35+\epsilon) = 0$. When $\epsilon \to 0$, bidder 2 buys the second unit of the commodity at price 40.

When the price rises to $p^3 = 43+\epsilon$, $s_{-1}(43+\epsilon)-s_{-1}(40+\epsilon) = 0, s_{-2}(43+\epsilon) - s_{-2}(40+\epsilon) = 1$, $s_{-3}(45+\epsilon) - s_{-3}(40+\epsilon) = 1$. When $\epsilon \to 0$, bidder 2 buys the third unit and bidder 3 buys the fourth unit at price 43. Total sales add up to 4.

When the price further rises to $p^4 = 45+\epsilon$, $s_{-1}(45+\epsilon)-s_{-1}(43+\epsilon) = 1$, $s_{-2}(45+\epsilon) - s_{-2}(43+\epsilon) = 1$, $s_{-3}(45+\epsilon) - s_{-3}(43+\epsilon) = 0$. When $\epsilon \to 0$, bidder 1 buys the fifth unit and bidder 2 buys the sixth unit at price 45. Total sales amount to 6, and the auction ends.

At the end of the Ausubel auction: bidder 1 gets two units of objects with the payment equal to $35 + 45 = 80$; bidder 2 gets three units of objects with the payment equal to $40 + 43 + 45 = 128$; and bidder 3 gets one unit of objects with the payment equal to 43. The revenue of the auctioneer is $80 + 128 + 43 = 251$, which is the same as that in the Vickrey auction.

Similarly, the Ausubel auction is outcome-equivalent, but not strategically equivalent, to the Vickrey auction.

In the following, we will explore equilibria of these auction formats with identical objects and private values. As each sealed-bid auction has an equivalent open form, we will focus on the three sealed-bid auctions. In the single-object auction, when the bidders' values are private and independently identically distributed and bidders are risk-neutral and have no budget constraints, the symmetric equilibria of both the first-price auction and the second-price auction are efficient. As we will show, this conclusion in general is not true for multi-unit auctions. Only Vickrey auction is efficient, while the discriminatory auction and the uniform-price auction in general are not.

21.4.4 Equilibrium of Vickrey Auctions

Suppose that an auctioneer has K units of identical objects, and the marginal value vector of bidder i to different units of objects is $\boldsymbol{V}_i = (V_i^1, \dots, V_i^K)$, where V_i^k is the marginal value of bidder i to the kth unit of object that satisfies the law of the diminishing marginal value, i.e., $V_i^k \geqq V_i^{k+1}, \forall k < K$.

Given the valuation vector of each bidder i, her **true demand function** is $\mathring{d}_i$, and is defined as:

$$\mathring{d}_i(p) = \max\{k : v_i^k \geqq p\}.$$

From the previous subsection, we know that the demand reported by bidder i is $d_i(p) = \max\{k : b_i^k \geqq p\}$. As a consequence, the reported demand equals the true demand if and only if every bidder bids according to her true value.

We also know that, in a multi-unit Vickrey auction, each bidder simultaneously submits a K-vector $\boldsymbol{b}_i$ that satisfies $b_i^k \geqq b_i^{k+1}$. All of the bid vectors of the bidders construct the bidding set $\{b_i^k, i \in N, k = 1, \dots, K\}$, in which the K highest bids win the objects and bidder i wins k_i units of objects, so that $\sum_i k_i = K$. The K-vector c_{-i} of competing bids faced by bidder i is obtained by rearranging in decreasing order the bids of the bidding set $\{b_j^k, j \neq i, j \in N, k = 1, \dots, K\}$ and choosing the first K of these. In the Vickrey auction, bidder i getting k_i units of objects means $b_i^{k_i} > c_{-i}^{K-k_i+1}$ and $b_i^{k_i+1} < c_{-i}^{K-k_i}$. At the same time, the total payment of bidder i to k_i units of objects is equal to $\sum_{k=1}^{k_i} c^{K+1-k}$. Now, we show that truth-telling, i.e., $\boldsymbol{b}_i(\boldsymbol{v}_i) = \boldsymbol{v}_i$, is a weakly dominant strategy in the Vickrey auction.

Proposition 21.4.1 *For every bidder i, it is a weakly dominant strategy to bid according to $\boldsymbol{b}_i(\boldsymbol{v}_i) = \boldsymbol{v}_i$ in a multi-unit Vickrey auction.*

PROOF. Let $\boldsymbol{b}_i = \boldsymbol{v}_i$ be the truthful pricing strategy of bidder i and c_{-i} be the vector of competing bids of other bidders faced by bidder i. k_i is the units of objects that bidder i wins when she chooses to report truthfully, which means $v_i^{k_i} > c_{-i}^{K-k_i+1}, v_i^{k_i+1} < c_{-i}^{K-k_i}$, and the total payment is $\sum_{k=1}^{k_i} c_{-i}^{K-k+1}$. Then, bidder i's utility is

$$\sum_{k=1}^{k_i}[v_i^k - c_{-i}^{K-k+1}].$$

If bidder i were to submit a different bid profile $\boldsymbol{b}'$ and still got k_i units of objects, her utility would be the same. Suppose that bidder i submitted a different bid profile $\boldsymbol{b}'$, but the units of the objects she gets were $k_i' \neq k_i$.

Then, if $k_i' > k_i$, bidder i gets the k-th units of goods with $k_i' \geqq k \geqq k_i+1$, and her marginal net utility is

$$v_i^k - c_{-i}^{K-k+1} \leqq v_i^{k_i+1} - c_{-i}^{K-k_i+1} < 0.$$

Thus, bidder i's utility of bidding $\boldsymbol{b}'$ is lower than the utility of bidding $\boldsymbol{b}_i$.

If $k_i' < k_i$, bidder i's utility is not maximized. Indeed, if she gets one more unit, her marginal utility of getting the $k_i' + 1$-th unit of goods would be

$$v_i^{k_i'+1} - c_{-i}^{K-k_i'+2} \geqq v_i^{k_i+1} - c_{-i}^{K-k_i+1} > 0.$$

Therefore, bidder i's expected utility of bidding $\boldsymbol{b}'$ is lower than the interim expected utility of bidding $\boldsymbol{b}_i$.

Thus, bidder i's truthful pricing strategy $\boldsymbol{b}_i$ is a weakly dominant strategy. □

In a Vickrey auction, the payment that bidder i makes to get each unit of the object is equal to the opportunity cost resultant from this unit of goods to other bidders, which is the externality caused by bidder i to other bidders.

Ausubel and Milgrom (2006) defined the payments of participants in a Vickrey auction by the VCG mechanism. Let $\boldsymbol{v}_i$ be the vector of true values of bidder i to multiple objects. Let $\hat{\boldsymbol{v}}_i$ be the vector of reported values of bidder i. Let $\boldsymbol{k} = (k_1, \ldots, k_n)$ be an allocation of the objects and $\boldsymbol{k}^*$ be the optimal allocation given bidders' reported values, so that

$$\boldsymbol{k}^* = \arg\max_{\boldsymbol{k}} \sum_i \hat{\boldsymbol{v}}_i(k_i), s.t. \quad \sum_i k_i \leqq K.$$

The transfer of the bidder who gets k_i units of objects is

$$t_i(k_i, \hat{\boldsymbol{v}}) = \sum_{j\neq i} \hat{\boldsymbol{v}}_j(k_j^*) - \bar{\boldsymbol{v}}_{-i},$$

where $\bar{\boldsymbol{v}}_{-i} = \max\{\sum_{j\neq i} \hat{\boldsymbol{v}}_j(k_j) : \sum_{j\neq i} k_j \leqq K\}$. Therefore, the Vickrey auction is a form of the VCG mechanism with multiple objects.

In a Vickrey auction, as each bidder reports her true value and the objects are given to the K highest bids, it is an efficient auction. We then have

Proposition 21.4.2 *The Vickrey auction results in efficient allocation of the objects.*

Ausubel and Milgrom (2006) also showed that among all auctions for multiple identical objects, *the Vickrey auction is the only one in which truth-telling is a dominant strategy equilibrium, the allocation is efficient, and the losing bidders' payments are zero.*

The conclusions above reveal that the Vickrey auction possesses desired properties. However, if objects are not identical, especially in the presence of complementarity, there will be a number of shortcomings in the Vickrey auction, which we will examine later. Even with identical multiple objects, one shortcoming of the Vickrey auction is that the prices of identical objects are not the same. This will cause arbitrage and other problems. The example below shows that identical objects do not have the same price in the Vickrey auction.

Example 21.4.2 Suppose that there are two units of identical objects, and the values of two bidders are $\boldsymbol{v}_1 = (12, 2)$ and $\boldsymbol{v}_2 = (8, 1)$, respectively. In the Vickrey auction, each bidder obtains one unit of a commodity, and their payments are 1 and 2, respectively. So, they pay different prices for the same object. If the bidders are two listed companies, such bids and payments will cause external pressure to the company that pays a higher price.

In the following, we discuss the uniform-price auction and the discriminatory auction. In these auctions, bidders often bid untruthfully and the allocations, in general, are not efficient.

21.4.5 Equilibrium of Uniform Price Auctions

We now discuss equilibrium of the uniform-price auction in which there is the feature of underreported demand.

The marginal value of the bidder to the object is $\boldsymbol{v}_i$. Let $\boldsymbol{c}_{-i}$ be the K-vector of competing bids of the other bidders faced by bidder i. If bidder i wins k_i units of objects, we must have:

$$b_i^{k_i} > c_{-i}^{K-k_i+1} \text{ and } b_i^{k_i+1} < c_{-i}^{K-k_i}.$$

The price is then given by:

$$p = \max\{b_i^{k_i+1}, c_{-i}^{K-k_i+1}\}.$$

In general, we do not have an explicit solution of equilibrium strategy in the uniform-price auction. As a consequence, we just provide some characterizations of the equilibrium strategy of the auction.

The equilibrium strategy of a uniform-price auction has two important features: one is that the bids will not exceed the marginal values; and the other is that the bid for the first unit is equal to its marginal value.

Fact 21.4.1 *$\forall i, k$, $b_i^k \leqq v_i^k$; i.e., the bids cannot exceed marginal values.*

PROOF. Suppose by way of contradiction that bid $\boldsymbol{b}_i$ has $b_i^k > v_i^k$ for some i and some k. We show that it must be weakly dominated by a bidding strategy that satisfies ${b'}_i^k = v_i^k, {b'}_i^{k'} = b_i^{k'}, k' \neq k$. We discuss this in four cases. (1) $b_i^k > p = v_i^k$. Bidder i can get $k-1$ units of objects under both bidding strategies. However, when she chooses ${b'}_i^k = v_i^k$, p may decrease. (2) $b_i^k < p$. The two bidding strategies make no difference to bidder i. (3) $b_i^k \geqq p > v_i^k$. The bidder will suffer a loss from winning the k-th unit as the price exceeds the marginal value. (4) $b_i^k > v_i^k > p$. The two bidding strategies make no difference. Thus, in all four cases, the bidder's utility cannot be increased when $b_i^k > v_i^k$. Therefore, in the uniform-price auction, the bidder's bid will not exceed the marginal value. □

Fact 21.4.2 *For all i, $b_i^1 = v_i^1$; i.e., the bid on the first unit must be the same as its true value.*

PROOF. We want to show that, if the bidder bids on the first unit according to her marginal value, i.e., $b_i^1 = v_i^1$, then this bid is a weakly dominant strategy, or any ${b'}_i^1 < v_i^1$ is a weakly dominated strategy. We will examine this in three cases: (1) $p > v_i^1 > {b'}_i^1$. The bidder gets nothing under the two strategies, and thus they make no difference to him. (2) $v_i^1 \geqq p > {b'}_i^1$. If she bids on the first unit truthfully, bidder i will win it and have positive revenue. However, if she chooses ${b'}_i^1$, she gets nothing and no revenue. (3) $v_i^1 > {b'}_i^1 > p$. The two bidding strategies make no difference to bidder i. □

Thus, in a uniform-price auction, every bidder's strategy satisfies the condition that the bid will not exceed the true value. Despite that the bid on the first unit is equal to its value, the bids on the other units may be less than the true value. This phenomenon is called **strategic demand reduction**.

To see the logic behind the strategic demand reduction of the uniform-price auction, we consider an economy with two units of identical objects, and two bidders whose marginal value vector V_i is distributed on $[0, w]^2$ with $v_i^1 \geqq v_i^2$ according to the density function $\varphi(\cdot)$ and the distribution function $\Phi(\cdot)$. Let $\Phi^1(\cdot)$ and $\Phi^2(\cdot)$ be the marginal distribution functions of V_i^1 and V_i^2, respectively. We will focus on bidder 1's strategies of the uniform-price auction in symmetric equilibrium (subscript omitted).

Suppose that the marginal values of bidder 1 are $(v_1^1, v_1^2) \equiv (v^1, v^2)$, bids are $(b_1^1, b_1^2) \equiv (b^1, b^2)$, and the competing bids faced by her are $(b_2^1, b_2^2) = (c_{-1}^1, c_{-1}^2) \equiv (c^1, c^2)$. In symmetric equilibrium, as $\boldsymbol{b}_i(\boldsymbol{v}_i) = \beta(\boldsymbol{v}_i)$, $\boldsymbol{b}_i$ is random and its distribution function is $H(\cdot)$ (its density function is $h(\cdot)$).

Let $H^1(\cdot)$ and $H^2(\cdot)$ be the marginal distribution functions of b_2^1 and b_2^2, respectively, and thus they are marginal distribution functions of c^1 and c^2. Thus, $H^1(b^2) = Prob[C^1 < b^2]$ is the probability that bidder 1 will win two units of objects. Let h_1 and h_2 denote the corresponding densities.

$H^2(b^1) = Prob[C^2 < b^1]$ is the probability that bidder 1 will defeat the lower competing bid, and win at least one unit. The probability that bidder 1 will win exactly one unit is then the difference $H^2(b^1) - H^1(b^2) > 0$. Moreover, $H^2(b^2) - H^1(b^2) = Prob[C^2 < b^2 < C^1]$ is the probability that the highest losing bid at which the units are sold is b^2.

Then, bidder 1's interim expected utility when she bids $\boldsymbol{b}_1 = (b^1, b^2)$ is:

$$\begin{aligned}
Eu_1(\boldsymbol{b}, \boldsymbol{v}) &= \int_{\{\boldsymbol{c}:c^1<b^2\}} (v^1 + v^2 - 2c^1)h(\boldsymbol{c})d\boldsymbol{c} \\
&+ \int_{\{\boldsymbol{c}:c^1>b^2, c^2<b^1\}} (v^1 - \max\{c^2, b^2\})h(\boldsymbol{c})d\boldsymbol{c} \\
&= H^1(b^2)(v^1 + v^2) - 2\int_0^{b^2} c^1 h^1(c^1)dc^1 \\
&+ [H^2(b^1) - H^1(b^2)]v^1 - [H^2(b^1) - H^1(b^2)]b^2 \\
&- \int_{b^2}^{b^1} c^2 h^2(c^2)dc^2.
\end{aligned}$$

We know that the bid for the first unit of object satisfies $b^1 = v^1$. The first-order condition for maximizing b^2 results in:

$$\frac{dEu_1}{db^2} = h^1(b^2)(v^2 - b^2) - [H^2(b^2) - H^1(b^2)] = 0.$$

Since $H^2(b^2) - H^1(b^2) > 0$, we have

$$b^2 = v^2 - \frac{H^2(b^2) - H^1(b^2)}{h^1(b^2)} < v^2.$$

Therefore, in a symmetric equilibrium of this uniform-price auction, there is a strategic demand reduction, except for the first unit. In a more general case, Ausubel and Cramton (2002) showed that as long as $K > 1$, at equilibrium of the uniform-price auction, the bid of each bidder satisfies $\beta_i^1(v_i^1) = v_i^1, \beta_i^k(v_i^k) < v_i^k, \forall k > 1$. Since the bidding strategies associated with different units are different, it will be shown below the unform price auction is inefficient.

Milgrom (2004) related the strategic demand reduction of the uniform-price auction with the properties of the demand of the buyer monopoly. The reason for strategic demand reduction is similar to the logic behind the buyer monopoly market. In a buyer monopoly market, the monopolist pays $TE(q) = qp(q)$ to buy q units of objects, where $p(q)$ is the market supply function. The marginal expenditure of the monopolist buying the

qth unit is $TE'(q) = p(q) + qp'(q) > p(q)$ (as the market supply function satisfies $p'(q) > 0$).

Thus, there are two kinds of effects of buying one more unit of commodity, one is the expenditure $p(q)$ of buying the object, the other is the increased marginal expenditure $qp'(q)$. The latter reduces the demand incentive of the monopolist. Suppose that the value of the monopolist to the object is $v(q)$, the amount of goods purchased to achieve maximum utility is:

$$v'(q) - TE'(q) = v'(q) - p(q) - qp'(q) = 0,$$

which implies the demand q of the monopolist satisfies $v'(q) - p(q) > 0$. This is the logic behind the strategic demand reduction.

However, if every bidder has a *unit demand*, by Fact 21.4.2, all of the bidders report truthfully, and the uniform-price auction is efficient. Consider a simple example. Suppose that $K = 2$ and bidder 1's demand is unit demand, i.e., $V_1^2 = 0$. Suppose that bidder 1's value to one unit of commodity is V^1, which is distributed on [0,1] according to the uniform distribution. By the features of the uniform-price auction, we have $b_1(v^1) = v^1$.

Bidder 2's marginal value vectors are v^1 and v^2 that satisfy $1 > v^1 \geqq v^2 > 0$, each of which is distributed on [0,1] according to the uniform distribution. In bidding strategy, she takes $b_2^1 = v^1$ and $b_2^2 = b^2$. If $b^2 > v^1$, bidder 2 will win two units of objects. If $0 \leqq b^2 < v^1$, she will win one unit. Then, the interim expected utility of bidder 2 choosing b^2 is:

$$\begin{aligned} Eu_2(b^2, v^1, v^2) &= \int_{v^1<b^2} (v^1 + v^2 - 2v^1) dv^1 + \int_{v^1>b^2} (v^1 - b^2) dv^1 \\ &= v^1 - b^2(1 - v^2). \end{aligned}$$

Therefore, the optimal bidding strategy is $b^2 = 0$. This means that regardless of what value v^2 takes, bidder 2's optimal bid for the second unit is equal to zero. Thus, when bid $b^2 = 0$, bidder 2 will get one unit of goods for free, and then each bidder is bidding truthfully. Consequently, there is not possibility that a winning bidder will influence the price paid, and thus efficient.

Summarizing our discussions, we have the following propositions.

Proposition 21.4.3 *When $K > 1$, at an undominated equilibrium of the uniform-price auction, the bid on the first unit is equal to the value of the first unit, but bids on other units are lower than the respective marginal values.*

Proposition 21.4.4 *When $K > 1$, every undominated equilibrium outcome of a uniform price auction, in general, is inefficient. However, when every bidder has a unit demand, it is efficient.*

21.4.6 Equilibrium of Discriminatory Auctions

In this auction, a bidder pays her bids when she wins some units of the objects. Let $\boldsymbol{b}_i$ be the bid vector of bidder i. When she wins k_i units of objects, her total payment is $\sum_{k=1}^{k_i} b_i^k$. As the general close-form solution is impossible to obtain, like the situation of the uniform-price auction, we focus here on the features of symmetric equilibrium.

To have the features of equilibrium bids, consider a simple economy with two units of objects and two bidders. Suppose that the vector of bidders' marginal values is $\boldsymbol{v}_i = (v_i^1, v_i^2)$. Consider symmetric equilibrium (β^1, β^2), i.e., $b_i^1 = \beta^1(\boldsymbol{v}_i)$ and $b_i^2 = \beta^2(\boldsymbol{v}_i)$.

Firstly, each bidder's bid is less than, or equal to, $\max_{\boldsymbol{v}} \beta^2(\boldsymbol{v})$ on the first unit. This is because if b_i^1 on the first unit is greater than $\max_{\boldsymbol{v}} \beta^2(\boldsymbol{v})$, the bidder wins with probability one, and she could be better off by reducing it slightly. Indeed, let $\bar{b} = \max_{\boldsymbol{v}} \beta^2(\boldsymbol{v})$. For bidder i, $b_i^1 > \bar{b}$ is strictly dominated by $b'^1_i = b_i^1 - \epsilon > \bar{b}$, where $\epsilon > 0$. As $b_i^2 \leqq \bar{b}$, for bidder i, b_i^1 and b'^1_i are both winning bids, but paying b'^1_i will lower the acquisition cost.

Thus, we have the first feature that, for any symmetric equilibrium bidding strategy:

$$\max_{\boldsymbol{v}} \beta^1(\boldsymbol{v}) = \max_{\boldsymbol{v}} \beta^2(\boldsymbol{v}).$$

Secondly, for a bidder, say bidder 1, let $\boldsymbol{c}_{-1} = (b_2^1, b_2^2) = \boldsymbol{c}_{-1} = (c^1, c^2)$ denote the competing bids of bidder 1, which is the bids of bidder 2. Similarly, let $H^1(\cdot)$ and $H^2(\cdot)$ be the marginal distribution functions of c^1 and c^2, respectively. Let h^1 and h^2 denote the corresponding densities.

Then we have

$$H^k(c) = Prob[\beta^k(v) \leqq c].$$

Since $\beta^1(v) \geqq \beta^2(v)$ for all v, H^1 first-order stochastically dominates H^2.

When bidder 1's marginal value vector is $(v_1^1, v_1^2) = (v^1, v^2)$, and the bidder bids $(b_1^1, b_1^2) = (b^1, b^2)$, and will win two units if $c^1 < b^2$ with the probability $H^1(b^2)$; if $c^2 < b^1$ and $c^1 > b^2$, the bidder wins one unit with the probability $H^2(b^1) - H^1(b^2)$.

Then, the bidder's problem is to maximize the expected utility

$$\begin{aligned} Eu_1(\boldsymbol{b}_1, \boldsymbol{v}_1) &= H^1(b^2)(v^1 + v^2 - b^1 - b^2) + [H^2(b^1) - H^1(b^2)](v^1 - b^1) \\ &= H^2(b^1)(v^1 - b^1) + H^1(b^2)(v^2 - b^2). \end{aligned}$$

subject to the constraint:

$$b^1 \geqq b^2.$$

When optimal bids are $b^1 > b^2$, the first-order conditions are:

$$h^2(b^1)(v^1 - b^1) = H^2(b^1) \tag{21.4.22}$$

$$h^1(b^1)(v^2 - b^2) = H^1(b^2). \tag{21.4.23}$$

Thus, *the bids are independent*. In other words, β^1 is independent of v^2, and β^2 is independent of v^1 when $b^1 > b^2$.

When optimal bids are $b^1 = b^2 = b$, the first-order conditions are:

$$h^2(b)(v^1 - b) + h^1(b)(v^2 - b) = H^2(b) + H^1(b).$$

Then, the bidder has a flat demand, which means that the bid for each unit is the same, i.e., $\beta^1(\boldsymbol{v}) = \beta^2(\boldsymbol{v})$. Moreover, if the bidder submits the bid b for $\boldsymbol{v} = (v^1, v^2)$ and $\boldsymbol{v'} = (v'_1, v'_2)$, then for any $\lambda \in [0, 1]$, she will submit the same bid b for the value $\lambda\boldsymbol{v} + (1 - \lambda)\boldsymbol{v'}$.

Summarizing the above discussions, we have the following proposition:

Proposition 21.4.5 *When $K = 2$, at equilibrium of the discriminatory auction, the bidding function β has the following features:*

1. *The greatest bids of the first unit and second unit are the same:*

$$\max_{\boldsymbol{v}} \beta^1(\boldsymbol{v}) = \max_{\boldsymbol{v}} \beta^2(\boldsymbol{v}).$$

2. *If $\beta^1(\boldsymbol{v}) > \beta^2(\boldsymbol{v})$ for some $\boldsymbol{v}$, then β^1 is independent of v^2, and β^2 is independent of v^1, i.e.,*

$$\beta^1(\boldsymbol{v}) = \beta^1(v^1) \text{ and } \beta^2(\boldsymbol{v}) = \beta^2(v^2).$$

3. *If it is optimal to submit the flat demand bid $\beta^1(\boldsymbol{v}) = \beta^2(\boldsymbol{v}) = b$ for $\boldsymbol{v}$ and $\boldsymbol{v'}$, then it is also optimal to submit the same flat demand bid b for any value $\boldsymbol{v}_\lambda$ between $\boldsymbol{v}$ and $\boldsymbol{v'}$, i.e., for any $\lambda \in [0, 1]$, we have*

$$b = \beta^1(\lambda\boldsymbol{v} + (1 - \lambda)\boldsymbol{v'}) = \beta^2(\lambda\boldsymbol{v} + (1 - \lambda)\boldsymbol{v'}).$$

21.4.7 Efficiency in Multi-Unit Auctions

For the uniform-price auction, we know that it may not be efficient, except that all of the bidders' demand is unit demand. Is a discriminatory auction efficient? Here, we provide a result on how to determine if a multi-unit auction is efficient prior to discussing the efficiency of the discriminatory auction.

As there are different forms of auctions, we will focus on standard auctions, in which the objects are given to the bidders whose bids are the K highest. For a standard auction, there are K units of objects. Let $\boldsymbol{\beta} = (\boldsymbol{\beta}_1, \ldots, \boldsymbol{\beta}_n)$ be the profile of bidding functions of bidders, where $\boldsymbol{\beta}_i = (\beta_i^1, \ldots, \beta_i^K)$ is the vector of bidding functions of bidder i. Suppose that bidder i's marginal value vector is $\boldsymbol{v}_i$. If $\beta_i^k(\boldsymbol{v}_i)$ is a winning bid, it must belong to the K highest bids. If v_i^k belongs to the K highest marginal values, bidder i at least wins the k-th unit of objects. If the auction is efficient, the equilibrium of this auction must satisfy the condition below:

$$\beta_i^k(\boldsymbol{v}_i) > \beta_j^l(\boldsymbol{v}_j) \text{ if and only if } v_i^k > v_j^l,\ \forall i,j,k,l.$$

Then, we have the following features for efficient allocation:

(1) *To have an efficient auction, bidders' bidding strategies for different units must be independent (separable)*, i.e., the bid of any bidder i on the k-th unit only depends on v_i^k to the k-th unit:

$$\beta_i^k(\boldsymbol{v}_i) = \beta_i^k(v_i^k).$$

If the bid on the k-th unit depends on the marginal value of the other units, say $v_i^{k'}$, then $v_i^k = v_j^l$ and $v_i^{k'} \neq v_j^{l'}$, which implies that $\beta_i^k \neq \beta_j^l$. This means that if v_i^k and v_j^l belong to the K highest marginal values of bidders, but β_i^k and β_j^l might not belong to K highest bids, then this auction is not efficient.

(2) *To have an efficient auction, the bidding strategy function must be the symmetric across both bidders and objects*, i.e.,

$$\beta_i^k(\cdot) = \beta_j^l(\cdot),\ \forall i,j,k,l.$$

Otherwise, we may have $b_i^k \neq b_j^l$ when $v_i^k = v_j^l$, which will make the allocation of the auction inefficient.

In review of the three multi-unit auctions above, the Vickrey auction is efficient, as all of the bidders report their true marginal values, i.e., $b_i^k = \beta_i^k(\boldsymbol{v}_i) = v_i^k$. For the uniform-price auction, if at least one bidder's demand is not unit demand, then each bidder's bid on the first unit is the true marginal value, but their bids on the other units may be less than the true marginal values, and thus the uniform-price auction, in general, is not efficient.

For the discriminatory multi-unit auction, we consider the bidding strategy equilibrium of two units of object and two bidders. In the case of $b^1 > b^2$, by the first-order conditions (21.4.22) and (21.4.23), despite that the bids on different units are independent, they are not the same bidding strategies, unless $\frac{H^2}{h^2} = \frac{H^1}{h^1}$. Furthermore, if $\beta^1(\cdot) = \beta^2(\cdot)$, $H^1(\cdot)$ first-order stochastically dominates $H^2(\cdot)$, where $\frac{H^2}{h^2} \neq \frac{H^1}{h^1}$. However, if $b^1 = b^2$, then $v_i^1 > v_i^2$, which is also not efficient.

The above two features imply that the bidding functions $\boldsymbol{\beta}$ are separable and symmetric across both bidders and objects. Consequently, if an equilibrium outcome is efficient, the bidding function must be a single-valued increasing function. The converse is also true. We then have the following proposition:

Proposition 21.4.6 *An equilibrium allocation of a standard auction is efficient if and only if the bidding functions $\boldsymbol{\beta}$ are separable and symmetric across both bidders and objects, i.e.,*

$$\beta_i^k(\boldsymbol{v}_i) = \beta(v_i^k).$$

As a corollary, one can see that the Vickrey auction is efficient, while the uniform-price and discriminatory auctions, in general, are not efficient.

21.4.8 Revenue Equivalence Principle in Multi-Unit Auctions

For the single-object auction, if it is private value, independent, symmetric, without external (budget) constraints and with risk neutral preferences, standard auctions, such as first-price auctions, second-price auctions, English auctions, Dutch auctions and all-pay auctions, are not only efficient but also revenue-equivalent to the auctioneer. When we extend the single-object auction to the multi-unit auction, as discussed above, even if all of these assumptions are satisfied, not all of these auctions are efficient. The question then is whether the revenue equivalence principle also holds in a multi-unit auction. We now discuss this issue.

In the multi-unit auction, the discussion below also applies to auctions resulting in the same decision rule. The outcomes of these auctions, and consequently the expected revenue of the seller, may be the same in some economic environments. When two multi-unit auctions allocate the objects in the same way, the revenue equivalence principle will become a useful tool.

Suppose that there are K units of identical objects to be sold to $n > K$ bidders. The set of bidders is $N = \{1\ldots,n\}$. Each bidder's marginal value (random) vector is $\boldsymbol{V}_i$ with $\boldsymbol{V}_i \in \{\boldsymbol{v} \in [0,w]^K : v^k \geqq v^{k+1}, \forall k\}$. They are independent, but do not need to be identically distributed. Let $\Phi_i(\cdot)$ be the distribution function, and $\varphi_i(\cdot)$ be the density function.

We can utilize the mechanism design approach to discuss the revenue equivalence principle of the multi-unit auction, and especially apply the Bayesian Incentive Compatibility Characterization Theorem (Proposition 19.4.2) in Chapter 19 to obtain the revenue equivalence principle. Here, instead, we give a direct proof of the principle. Let $(\boldsymbol{\beta}_1,\ldots,\boldsymbol{\beta}_n)$ be the equilibrium bidding strategy of an auction. In equilibrium, the bidder with $\boldsymbol{v}_i$ will choose $\boldsymbol{\beta}_i(\boldsymbol{v}_i)$ as strategy. When bidder i with $\boldsymbol{v}_i$ reports $\hat{\boldsymbol{v}}_i$, which means that the bidder chooses $\boldsymbol{\beta}_i(\hat{\boldsymbol{v}}_i)$ as a strategy, the probability of getting the k-th unit of objects is $y_i^k(\hat{\boldsymbol{v}}_i)$, and the transfer is $t_i(\hat{\boldsymbol{v}}_i)$. Therefore, bidder i's interim expected utility is:

$$u_i(\hat{\boldsymbol{v}}_i, \boldsymbol{v}_i) = \boldsymbol{y}_i(\hat{\boldsymbol{v}}_i)\boldsymbol{v}_i + t_i(\hat{\boldsymbol{v}}_i),$$

where $\boldsymbol{y}_i = (y_i^1,\ldots,y_i^K)$ is the probability vector of bidder i getting the objects.

In equilibrium, for any $\hat{\boldsymbol{v}}_i$, we have

$$\boldsymbol{y}_i(\boldsymbol{v}_i)\boldsymbol{v}_i + t_i(\boldsymbol{v}_i) \geqq \boldsymbol{y}_i(\hat{\boldsymbol{v}}_i)\boldsymbol{v}_i + t_i(\hat{\boldsymbol{v}}_i). \tag{21.4.24}$$

Let $U_i(\boldsymbol{v}_i) \equiv u_i(\boldsymbol{v}_i, \boldsymbol{v}_i) = \max_{\hat{\boldsymbol{v}}_i}[\boldsymbol{y}_i(\hat{\boldsymbol{v}}_i)\boldsymbol{v}_i + t_i(\hat{\boldsymbol{v}}_i)]$.
(21.4.24) can be written as:

$$U_i(\boldsymbol{v}_i) \geqq U_i(\hat{\boldsymbol{v}}_i) + \boldsymbol{y}_i(\hat{\boldsymbol{v}}_i)(\boldsymbol{v}_i - \hat{\boldsymbol{v}}_i). \tag{21.4.25}$$

In (21.4.25), $\boldsymbol{y}_i(\hat{\boldsymbol{v}}_i)$ can be seen as the subgradient of the convex function $U_i(\cdot)$ at the point $\hat{\boldsymbol{v}}_i$.

Compared to the single-object auction, there are multidimensional private values in the multi-unit auction, i.e., there is a private marginal value vector. In the single-object auction, we are able to integrate to obtain U_i. By reducing the multidimensional problem to a single dimension, we can also express U_i as a form of integration.

For any point $\boldsymbol{v}_i$ of the K dimensional space, define a one-dimensional function $W_i : [0, 1] \to \mathcal{R}$:

$$W_i(s) = U_i(s\boldsymbol{v}_i).$$

Thus, $W_i(0) = U_i(0)$ and $W_i(1) = U_i(\boldsymbol{v}_i)$. As $U_i(\cdot)$ is a continuous convex function, $W_i(\cdot)$ is also continuous and convex.

Since a convex function of one variable is absolutely continuous, $W_i(\cdot)$ is differentiable, and can be written as the integral of its derivative:

$$W_i(1) = W_i(0) + \int_0^1 \frac{dW_i(s)}{ds} ds. \tag{21.4.26}$$

By (21.4.25), we have

$$W_i(s + \Delta) - W_i(s) = U_i((s + \Delta)\boldsymbol{v}_i) - U_i(s\boldsymbol{v}_i)) \geqq \boldsymbol{y}_i(s\boldsymbol{v}_i)\Delta\boldsymbol{v}_i.$$

When $\Delta > 0$, we get:

$$\frac{W_i(s + \Delta) - W_i(s)}{\Delta} \geqq \boldsymbol{y}_i(s\boldsymbol{v}_i)\boldsymbol{v}_i.$$

As $\Delta \to 0$, we obtain:

$$\frac{dW_i(s)}{ds} \geqq \boldsymbol{y}_i(s\boldsymbol{v}_i)\boldsymbol{v}_i.$$

When $\Delta < 0$, we similarly have:

$$\frac{dW_i(s)}{ds} \leqq \boldsymbol{y}_i(s\boldsymbol{v}_i)\boldsymbol{v}_i.$$

Since $W_i(s)$ is differentiable, we must have:

$$\frac{dW_i(s)}{ds} = \boldsymbol{y}_i(s\boldsymbol{v}_i)\boldsymbol{v}_i.$$

Thus, by (21.4.26), we have:

$$U_i(\boldsymbol{v}_i) = U_i(0) + \int_0^1 \boldsymbol{y}_i(s\boldsymbol{v}_i)\boldsymbol{v}_i ds. \tag{21.4.27}$$

Therefore, at $\boldsymbol{v}_i$, the interim expected utility $U_i(.)$ can be determined by $\boldsymbol{y}_i$, and thus

$$t_i(\boldsymbol{y}_i) = U_i(\boldsymbol{y}_i) - \boldsymbol{y}_i(\boldsymbol{v}_i)\boldsymbol{v}_i$$

which is the same, provided that $\boldsymbol{y}_i$ is the same. Then, we have the following proposition that summarizes revenue equivalence in the multi-unit auction.

Theorem 21.4.1 *For any two multi-unit auctions, suppose that the interim expected utilities $U_i(0)$ are the same and the allocation rules are the same. Then the interim expected utility and transfers are also the same under the two auctions, and consequently the expected revenue of the auctioneer is the same under these two auctions.*

21.4.9 Application of Revenue Equivalence Principle

For multi-unit auctions, although the uniform-price auction and the discriminatory auction, in general, are not efficient, they may be efficient in some special cases. For instance, when the demands of bidders are unit demand, or the uniform-price auction reduces to the Vickrey auction, every bidder will bid truthfully, and consequently it is efficient.

Usually, solving for the equilibrium of a uniform-price auction or a discriminatory auction is challenging. However, the revenue equivalence principle can be employed to derive their equilibrium bidding strategies in some situations. The following example is drawn from Krishna (2010).

Consider an economic environment in which there are three units of identical objects and two bidders, each of whom wants at most two units, which means $V_i^3 = 0, i = 1, 2$. The vector of bidders' values is $\boldsymbol{V}_i = (V_i^1, V_i^2), i = 1, 2$. Suppose that, on $\{\boldsymbol{v} \in [0,1]^2 : v^1 \geqq v^2\}$, the marginal values are identically and independently distributed according to the density function $\varphi(\cdot)$ and the distribution function $\Phi(\cdot)$. Let $\varphi^1(\cdot)$ and $\varphi^2(\cdot)$ be the marginal density functions of V_i^1 and V_i^2, respectively, and the marginal distribution functions be $\Phi^1(\cdot)$ and $\Phi^2(\cdot)$, respectively.

We demonstrate that the allocations of the uniform-price auction and the discriminatory auction for this economy are efficient. We first show that the allocation of the uniform-price auction is efficient. Recall that, for a uniform-price auction, each bidder bids truthfully on the first unit, i.e., $b_i^1 = v_i^1, i = 1, 2$. As there are three units of identical objects and two bidders, each of whom wants at most two units, each bidder wins at least one unit of object. If there exists a symmetric and increasing equilibrium bidding strategy $\beta^2(\cdot)$ on the second unit so that $\beta^2(v_i^2) > \beta^2(v_j^2), i \neq j$, which means $v_i^2 > v_j^2$, bidder i will win the second unit. Suppose that the equilibrium bidding strategy is symmetric and increasing. Then, for any (v_1^1, v_1^2) and (v_2^1, v_2^2), if $v_1^2 > v_2^2$, bidder 1 will win two units, and bidder 2 will win one unit. The opposite is similar. Obviously, the uniform-price function is efficient.

We next show that the allocation of the discriminatory auction is also efficient. Once again, each bidder is assured of winning at least one unit. Suppose that the biding profile of one bidder is (b^1, b^2), such that $b^1 > b^2$. Reducing the bid on the first unit to $b^1 - \epsilon > b^2$ does not affect a bidder winning the first unit (as each bidder will get at least one unit). This also does not affect the bidder winning the second unit, but will reduce her payoff. As a result, there must be $b^1 = b^2$ at equilibrium, which means that each bidder submits a flat demand. Moreover, the equilibrium bidding strategy is determined merely by the marginal value v_i^2 of the second unit.

In the discriminatory auction, if the bid on the second unit is an increasing function $\beta^2 : [0, 1] \to \mathcal{R}_+$, each bidder will choose $(\beta^2(v_i^2), \beta^2(v_i^2))$ in a symmetric equilibrium. As each bidder gets at least one unit, $\beta^2(v_i^2) > \beta^2(v_j^2)$ implies that $v_i^2 > v_j^2$, and bidder i will get two units. Obviously, the discriminatory auction is efficient.

Next, we will employ the revenue equivalence principle to derive the symmetric equilibrium strategy on the second unit for the uniform-price auction and the discriminatory auction.

In the three auction formats above, the bidder with $\boldsymbol{v}_i = 0$ will not win the object, and $U_i(0) = 0$. The conditions of the revenue equivalence principle are satisfied.

First, we calculate the expected revenue in the Vickrey auction. As each bidder will choose the true bids $b_i^k = v_i^k, i = 1, 2, k = 1, 2$ in the Vickrey auction, the competing bid vector faced by bidder i is $c_{-i} = (v_j^1, v_j^2, 0)$. Then, the bidder is sure to win at least one unit: If $v_i^2 < v_j^2$, bidder i will win one unit and the transfer is $t_i = 0$. If $v_i^2 > v_j^2$, bidder i will win two units, and the transfer is $t_i = -(0 + v_j^2)$ or the payment to the seller is $m_i = (0 + v_j^2)$.

Therefore, the interim expected payment of bidder i with value $\boldsymbol{v}_i$ is:

$$Em_i^v(\boldsymbol{v}_i) = \int_0^{v_i^2} v^2 \varphi^2(v^2) dv^2, \tag{21.4.28}$$

which is independent of v^1.

We then want to solve the symmetric equilibrium bidding strategy $\beta^{u2}(v_i^2)$ of the uniform-price auction for the second unit. In the equilibrium, if $v_i^2 > v_j^2$, bidder i wins two units and pays $2\beta^{u2}(v_j^2)$. If $v_i^2 < v_j^2$, bidder i wins one unit and pays $\beta^{u2}(v_i^2)$. Therefore, in the uniform-price auction, the interim expected payment of bidder i with value $\boldsymbol{v}_i$ is:

$$Em_i^u(\boldsymbol{v}_i) = \int_0^{v_i^2} 2\beta^{u2}(v^2)\varphi^2(v^2)dv^2 + (1 - \Phi^2(v_i^2))\beta^{u2}(v_i^2). \tag{21.4.29}$$

According to the revenue equivalence principle, $Em_i^v(\boldsymbol{v}_i) = Em_i^u(\boldsymbol{v}_i)$, that is:

$$\int_0^{v_i^2} v^2\varphi^2(v^2)dv^2 = \int_0^{v_i^2} 2\beta^{u2}(v^2)\varphi^2(v^2)dv^2 + (1-\Phi^2(v_i^2))\beta^{u2}(v_i^2).$$

Differentiating with respect to v_i^2, we have:

$$v_i^2\varphi^2(v_i^2) = \beta^2(v_i^2)\varphi^2(v_i^2) + (1-\Phi^2(v_i^2))\beta'^{u2}(v_i^2),$$

and thus we obtain the differential equation:

$$\beta'^{u2}(v_i^2) = (v_i^2 - \beta^2(v_i^2))\lambda_2(v_i^2),$$

where

$$\lambda_2(v_i^2) \equiv \frac{\varphi^2(v_i^2)}{1-\Phi^2(v_i^2)}.$$

Then, by the initial condition $\beta^2(0) = 0$, the symmetric equilibrium bidding strategy on the second unit is:

$$\beta^{u2}(v_i^2) = \int_0^{v_i^2} v^2\lambda_2(v^2)dL(v^2|v_i^2),$$

where $L(v^2|v_i^2) = \exp(-\int_{v^2}^{v_i^2}\lambda_2(s)ds)$.

Obviously, $\beta^{u2}(\cdot)$ is an increasing function. In the uniform-price auction, the equilibrium bidding strategy of the bidder is $(v_i^1, \beta^{u2}(v_i^2))$ at $\boldsymbol{v}_i$.

Finally, we now solve for the symmetric equilibrium bidding strategy of the discriminatory auction for the second unit. We already know that $b_i^1 = b_i^2$. Let $\beta^{d2}(\cdot)$ be the symmetric equilibrium strategy. Then, at the equilibrium, if $v_i^2 > v_j^2$, bidder i will win two units and pay $2\beta^{d2}(v_i^2)$. If $v_i^2 < v_j^2$, bidder i will win one unit and pay $\beta^{d2}(v_i^2)$, and thus the interim expected payment of the bidder whose value is $\boldsymbol{v}_i$ is:

$$Em_i^d(\boldsymbol{v}_i) = \beta^{d2}(v_i^2) + \Phi^2(v_i^2)\beta^{d2}(v_i^2). \tag{21.4.30}$$

According to the revenue equivalence principle, we have

$$\int_0^{v_i^2} v^2\varphi^2(v^2)dv^2 = \beta^{d2}(v_i^2) + \Phi^2(v_i^2)\beta^{d2}(v_i^2).$$

Then, the equilibrium bidding strategy on the second unit is:

$$\beta^{d2}(v_i^2) = \frac{1}{1+\Phi^2(v_i^2)}\int_0^{v_i^2} v^2\varphi^2(v^2)dv^2.$$

It is easy to verify that $\beta^{d2}(\cdot)$ is an increasing function. Thus, in the discriminatory auction, the equilibrium bidding strategy of the bidder is $(\beta^{d2}(v_i^2), \beta^{d2}(v_i^2))$ at $\boldsymbol{v}_i$.

Through the example above, we find that, by using the revenue equivalence principle, we can easily obtain the equilibrium bidding strategy of some auctions that are difficult to solve for directly. However, we have to ensure that the decision rules (allocations) of the two auctions are the same when we adopt this approach.

21.5 Multi-Unit Sequential Auctions with Private Values

We now explore sequential auctions of multiple identical objects. A **sequential auction** is an auction in which the units are sold one at a time in separate auctions that are conducted sequentially until all units are sold. In particular, we mainly consider two formats of auctions: the sequential first-price sealed-bid auction; and the sequential second-price sealed-bid auction. In these two auctions, at each unit of objects, all bidders simultaneously submit bids, and the bidder with the highest bid wins.

Sequential auctions of multiple objects are markedly different from simultaneous multi-unit auctions discussed previously. Sequential auctions bring new strategic issues in which bidders may choose different bidding strategies at different stages, and thus this dynamic consideration can make the analysis of the bidder's strategy quite complicated. For the sake of discussion, we focus our attention on the simplest situations in which each bidder has a *unit demand*. In other words, for any bidder i, $V_i^k = 0, \forall k \geqq 2$.

We first discuss sequential first-price sealed-bid auctions, and then sequential second-price sealed-bid auctions.

21.5.1 Equilibrium of Sequential First-Price Auctions

Consider an economic environment in which K identical items are sold to n bidders ($n > K$), and each bidder has a unit demand. We assume that bidder i has private value $\mathbf{V}_i$, and that each bidder's value $\mathbf{V}_i$ is drawn independently from the same distribution $\Phi(\cdot)$ on $[0, \omega]$. The corresponding density is denoted by $\varphi(\cdot)$. A particular bidder may be still active in round K auction (if the bidder wins an object, she will stop bidding). Therefore, a bidding strategy for a bidder consists of K functions, denoted by $\beta^{I1}(\cdot), \cdots, \beta^{IK}(\cdot)$, with the superscript I standing for the first-price auction, and the superscript k representing the strategy of the k-th round auction.

We solve for a symmetric equilibrium. As in each round, there is different information, in $k \leqq K$-th round, the bidding strategy is

$$\beta^{Ik}(v; p^1, \cdots, p^{k-1}),$$

where $p^1, \cdots, p^{k-1}$ are the prices in the previous $k-1$ auctions, respectively. Moreover, the sequential auctions are assumed to be held in a short period of time, and thus we disregard time discounting.

If the equilibrium strategies $\beta^{Ik}(v;\cdot)$ for $k \leqq K$ are increasing functions of v, the first item will go to the bidder with the highest value, and the k-th item to the bidder with the k-th highest value. In this sense, *the sequential first-price auction is efficient*. Prior to discussing the sequential auctions of K units of objects, we begin by considering the simplest situation in which only two units are sold in order to determine some basic characteristics of equilibrium bidding strategies in sequential first-price auctions.

Sequential Auctions for Two Units of Identical Objects

Consider an economy in which two units are sold. Let (β^1, β^2) (the superscript I is omitted here) be the symmetric equilibrium bidding strategy profile. In the first round, the bidding strategy is $\beta^1(\cdot) : [0, w] \to \mathcal{R}_+$. If it is a strictly increasing function, there is an inverse function, denoted by $(\beta^1)^{-1}$, which is the corresponding value of the item at the price p^1 paid in the first-price auction.

Before the start of the second-round bidding, the bidder can infer the private value of the bidder who won the auction in the first round according to the inverse function $(\beta^1)^{-1}$ at p^1. As it is a symmetric equilibrium, we can simply consider only the bidding strategy of bidder 1. Denote by $W_1 \equiv W_1^{(n-1)}$ and $W_2 \equiv W_2^{(n-1)}$ the first-order statistics and the second-order statistics of $n-1$ stochastic variables $V_2, \cdots, V_n$, and by w_1 and w_2 their realizations, respectively.

Let ψ_1 and ψ_2 be the corresponding densities, and Ψ_1 and Ψ_2 be the distributions of $W_1 \equiv W_1^{(n-1)}$ and $W_2 \equiv W_2^{(n-1)}$, respectively, where $\Psi_1 = \Phi^{(n-1)}(\cdot)$. We assume that the considered concept of bidding equilibrium is the perfect Bayesian equilibrium that requires sequential rationality, and thus the equilibrium bidding strategies starting from the first bidding price and then the second round both should be Baysian-Nash equilibria. Below, we start with the analysis of the second round of the auction.

In the second round of the auction, suppose that the auction price of the first round is p^1. If bidder 1 does not get the object in the first round, she can infer that $w_1 = (\beta^1)^{-1}(p^1)$. Suppose that all of the other bidders follow the equilibrium strategy $\beta^2(\cdot; w_1)$. Since $W_2 < w_1$ and $\beta^2(\cdot; w_1)$ is an increasing function, it makes no sense for bidder 1 to bid an amount greater than $\beta^2(w_1; w_1)$. Noting that the private value of bidder 1 is v_1, if she bids $\beta^2(\hat{v}_1; w_1)$ where $\hat{v}_1 \leqq w_1$, her interim expected utility in the second auction is:

$$Eu_1(\hat{v}_1, v_1; w_1) = \Psi_2(\hat{v}_1|w_1)[v_1 - \beta^2(\hat{v}_1; w_1)].$$

Differentiating with respect to $\hat{v}_1$, we obtain the first-order condition in equilibrium:

$$\psi_2(v_1|w_1)[v_1 - \beta^2(v_1; w_1)] - \Psi_2(v_1|w_1)\beta^{2\prime}(v_1; w_1) = 0.$$

Rearranging this results in the following differential equation:

$$\beta^{2\prime}(v_1; w_1) = \frac{\psi_2(v_1|w_1)}{\Psi_2(v_1|w_1)}[v_1 - \beta^2(v_1; w_1)], \tag{21.5.31}$$

where the initial value condition satisfies $\beta^2(0; w_1) = 0$.

Given the winner of the first round, the highest of $n-1$ values is w_1. The probability that bidder 1 will win the second auction, if she chooses $\hat{v}_1$, is $prob(W_2^{(n-1)} < \hat{v}_1)$, which is equivalent to $prob(W_1^{(n-2)} < \hat{v}_1)$, where the $W_1^{(n-2)}$ is the private value of the second-highest of $n-2$ values, not including that of bidder 1 and the winner of the first round. Actually, $W_2^{(n-1)} \equiv W_1^{(n-2)}$, and then we have

$$\begin{aligned}\Psi_2(v_1|w_1) &= \Psi_1^{(n-2)}(v_1|W_1^{(n-2)} < w_1)\\ &= \frac{\Phi(v_1)^{n-2}}{\Phi(w^1)^{n-2}},\end{aligned} \tag{21.5.32}$$

and thus we have

$$\psi_2(v_1|w_1) = \frac{(n-2)\Phi(v_1)^{n-1}\varphi(v_1)}{\Phi(w^1)^{n-2}}. \tag{21.5.33}$$

Using (21.5.32) and (21.5.33), the differential equation (21.5.31) can be written as

$$\beta^{2\prime}(v_1; w_1) = \frac{(n-2)\phi(v_1)}{\Phi(v_1)}[v_1 - \beta^2(v_1; w_1)],$$

or equivalently,

$$\frac{\partial}{\partial v_1}(\Phi(v_1)^{n-2}\beta^2(v_1; w_1)) = (n-2)\Phi(v_1)^{n-3}\varphi(v_1)v_1.$$

Together with the initial conditions, the solution to the differential equation is:

$$\begin{aligned}\beta^2(v_1) &= \frac{1}{\Phi(v_1)^{n-2}}\int_0^{v_1} vd(\Phi(v)^{n-2})\\ &= E[W_1^{(n-2)}|W_1^{(n-2)} < v_1]\\ &= E[W_2|W_2 < v_1 < W_1].\end{aligned} \tag{21.5.34}$$

Thus, bidder 1's bidding strategy for the second period is to bid $\beta^2(v_1)$ following (21.5.34) if $v_1 \leqq w_1$, and to bid $\beta^2(w_1)$ if $v_1 > w_1$. The latter may occur if bidder 1 underbid in the first auction, say by mistake, causing someone else, with a lower value to wine. Even though this represents off-equilibrium behavior on the part of bidder 1 himself, as per the concept of a strategy, we must count in this possibility.

Now, we examine the first-round equilibrium strategy $\beta^1(\cdot)$. Again, consider the bidding strategy of bidder 1 with v_1. Suppose that all of the other bidders follow the first-period strategy $\beta^1(\cdot)$. Also assume that all bidders, including bidder 1, will follow the equilibrium bidding strategy (21.5.34) in the second period. We now discuss the conditions such that the equilibrium calls on bidder 1 to bid $\beta^1(\cdot)$ in the first stage.

In equilibrium, bidder 1 should bid $\beta^1(v_1)$ in the first stage, but consider what bidder 1's interim expected utility will be if she bids $\beta^1(\hat{v}_1)$, rather than $\beta^1(v_1)$.

Case 1: $\hat{v}_1 \geqq v_1$. When $\hat{v}_1 > W_1$, bidder 1 wins the first auction. However, when $\hat{v}_1 < W_1$, bidder 1 cannot win the item in the first round, and she will enter the second round of bidding. When $W_2 < v_1 \leqq \hat{v}_1 < W_1$, bidder 1 wins the second auction. Then, the interim expected utility of choosing $\beta^1(\hat{v}_1)$ is:

$$\begin{aligned} Eu_1(\hat{v}_1, v_1) =& \Psi_1(\hat{v}_1)[v_1 - \beta^1(\hat{v}_1)] \\ &+(n-1)(1-\Phi(\hat{v}_1))\Phi(v_1)^{n-2}[v_1 - \beta^2(v_1)]. \end{aligned} \tag{21.5.35}$$

Case 2: $\hat{v}_1 < v_1$. When $\hat{v}_1 > W_1$, bidder 1 wins the first auction with a bid. When $\hat{v}_1 < W_1$, if $W_2 < v_1 < W_1$, bidder 1 loses the first auction, but wins the second at the price of $\beta^2(v_1)$. If $\hat{v}_1 < W_1 < v_1$, bidder 1 will pay $\beta(W_1)$, and then her interim expected utility is:

$$\begin{aligned} Eu_1(\hat{v}_1, v_1) =& \Psi_1(\hat{v}_1)[v_1 - \beta^1(\hat{v}_1)] + [\Psi_2(v_1) - \Psi_1(v_1)][v_1 - \beta^2(v_1)] \\ &+ \int_{\hat{v}_1}^{v_1} [v_1 - \beta^2(w_1)]\psi_1(w_1)dw_1. \end{aligned} \tag{21.5.36}$$

Differentiating equations (21.5.35) and (21.5.36) with respect to $\hat{v}_1$, we obtain the first-order conditions in the two cases:

$$\begin{aligned} 0 =& \psi_1(\hat{v}_1)[v_1 - \beta^1(\hat{v}_1)] - \Psi_1(\hat{v}_1)\beta^{1\prime}(\hat{v}_1) \\ &- (n-1)\varphi(\hat{v}_1)\Phi(v_1)^{n-2}[v_1 - \beta^2(v_1)], \end{aligned} \tag{21.5.37}$$

$$\begin{aligned} 0 =& \psi_1(\hat{v}_1)[v_1 - \beta^1(\hat{v}_1)] - \Psi_1(\hat{v}_1)\beta^{1\prime}(\hat{v}_1) \\ &- \psi_1(\hat{v}_1)[v_1 - \beta^2(v_1)]. \end{aligned} \tag{21.5.38}$$

Since $\Psi_1(v) = \Phi(v)^{n-1}$, $\psi_1(v) = (n-1)\varphi(v)\Phi(v)^{n-2}$. Therefore, the two equations (21.5.37) and (21.5.38) are the same. Therefore, in equilibrium it is optimal to bid $\beta^1(\cdot)$ and setting $\hat{v}_1 = v_1$ in either first-order condition, which results in the differential equations ((21.5.37) and (21.5.38)).

From equation (21.5.37) or equation (21.5.38), we have

$$\beta^{1\prime}(v_1) = \frac{\psi_1(v_1)}{\Psi_1(v_1)}[\beta^2(v_1) - \beta^1(v_1)]. \tag{21.5.39}$$

Together with the initial value condition $\beta^1(0) = 0$, we can obtain the solution to the above differential equation:

$$\begin{aligned}\beta^1(v_1) &= \frac{1}{\Psi_1}\int_0^{v_i} \beta^2(w_1)\psi_1(w_1)dw_1 \\ &= E[\beta^2(W_1)|W_1 < v_1] \\ &= E[E[W_2|W_2 < W_1]|W_1 < v_1] \\ &= E[W_2|W_1 < v_1]. \end{aligned} \tag{21.5.40}$$

Thus, together with equations (21.5.40) and (21.5.34), we obtain the following proposition:

Proposition 21.5.1 *Suppose that all of the bidders have a unit demand. Then, the symmetric equilibrium strategies for the sequential first-price sealed-bid auction with two units are*

$$\beta^{I1}(v) = E[W_2|W_1 < v],$$
$$\beta^{I2}(v) = E[W_2|W_2 < v < W_1],$$

where $W_1 \equiv W_1^{(n-1)}$ *and* $W_2 \equiv W_2^{(n-1)}$ *are the highest and the second-highest of* $n-1$ *independently drawn values, respectively.*

Obviously, $\beta^{I1}(v_1)$ and $\beta^{I2}(v_1)$ are strictly increasing functions in v_1. In the following, we provide two features of dynamic equilibrium bidding behavior for the case of $K = 2$.

First, from (21.5.40), we have

$$\beta^1(v_1) = E[\beta^2(W_1)|W_1 < v_1],$$

which means that if bidder 1 with v_1 is the bidder with the highest value, irrespective of whether she chooses to win the item in the first round or in the second round, her interim expected payments are the same. To be precise, *the (random) dynamic payment price is martingale* (i.e., the conditional expected value of the next observation, given all of the past observations, is equal to the most recent observation). At this point, the bidder with the highest value has no incentive to delay winning the item, and thus equilibrium outcomes can prevent strategic delays.

Secondly, from the (21.5.39) and the fact that $\beta^1(\cdot)$ is a strictly increasing function, we have:

$$\beta^2(v_1) - \beta^1(v_1) > 0,$$

which means that the bidding strategy in the second round will be more active than that in the first round. This is because, after the second round, the bidder will no longer have the opportunity to win the auction, and bidding in the latter stage is more intense. If we extend $K = 2$ to a more general case, the above two features remain true.

Sequential Auctions for K Units of Identical Objects

Consider a more general economy in which $K > 1$ identical items are sold to $n > K$ bidders. Assume that K units of identical objects are sold in a sequential first-price sealed-bid auction. At round k, the prices in the first $k-1$ auctions are $p^1, \cdots, p^{k-1}$. Let $W_k = W_k^{(n-1)}$ be the k-th highest independent and stochastically drawn variables from $n-1$ identical distributions, $\Psi_k(\cdot)$ be the distribution of W_k, and $\psi_k(\cdot)$ be the corresponding density. We will derive symmetric perfect Bayesian equilibrium bidding strategies $(\beta^1, \cdots, \beta^K)$, and similarly, only the strategies of bidder 1 are considered.

We now derive symmetric perfect Bayesian equilibrium bidding strategies backwards from the last round auction. So, first consider the K-th auction conducted in the last period. Using a similar inference process as previously, the equilibrium bidding strategy β^K in the last period is

$$\beta^K(v_1) = E[W_K | W_K < v_1 < W_{K-1}]. \tag{21.5.41}$$

We then consider the auction for the $k < K$-th round. Suppose that all bidders follow symmetric equilibrium bidding strategies $\beta^{k+1}, \cdots, \beta^K$ in the sequential auctions. When all other bidders choose the bidding strategy β^k in round k, we examine the strategy choice of bidder 1. Suppose that all of the other bidders choose the equilibrium bidding strategy.

If $v_1 > W_1^{(n-1)}$, bidder 1 wins the first round auction. If $v_1 < W_K^{(n-1)}$, she cannot get any item in equilibrium. If $W_k^{(n-1)} < v_1 < W_{k+1}^{(n-1)}, k \leqq K$, bidder 1 wins the k-th round auction with a bid of $W_k^{(n-1)}$. If $W_{k+1}^{(n-1)} < v_1, k \leqq K-1$, bidder 1 wins the $k+1$-st round auction with a bid of $W_{k+1}^{(n-1)}$, and the other bidders cannot obtain the item.

Therefore, if the symmetric equilibrium bidding strategy $(\beta^1, \cdots, \beta^K)$ is a strictly increasing function, the sequential auction is efficient. Furthermore, in the first $k-1$ round auctions, if the prices are $p^1, \cdots, p^{k-1}$, the bidders can infer that the bidders' private values of winning the first $k-1$ round auctions are:

$$w_1 = (\beta^1)^{-1}(p^1), \cdots, w_{k-1} = (\beta^{k-1})^{-1}(p^{k-1}).$$

The equilibrium calls on bidder 1 with private value v_1 to bid $\beta^k(v_1)$ in the k-th stage, but consider what happens if she decides to bid slightly higher, say, $\beta^k(v_1 + \Delta)$. (Bidder 1's choice of a slightly lower price $\beta^k(v_1 - \Delta)$ is similar. As for the previous case of $K = 2$, in the analysis of the first round of bidding, we know that the first-order conditions are the same regardless of whether the bidding price is higher or lower). If $v_1 > W_k \equiv W_k^{(n-1)}$, bidder 1 obtains the item in the k-th round, and the interim expected payment increases by

$$\Psi_k(v_1|w_{k-1})[\beta^k(v_1+\Delta) - \beta^k(v_1)]. \tag{21.5.42}$$

If $v_1 < W_k < v_1 + \Delta$, bidder 1 would have lost the k-th auction with her equilibrium bidding strategy; whereas, bidding higher $\beta^k(v_1+\Delta)$ results in her winning. Now, there are two subcases.

Case 1. When $W_{k+1} < v_1 < W_k < v_1 + \Delta$, in equilibrium, bidder 1 would have won the $(k+1)$st round auction.

Case 2. When $v_1 < W_{k+1} < W_k < v_1 + \Delta$, however, in equilibrium, bidder 1 would have lost both the k-th and the $k+1$-st auctions, and possibly won a later auction, for $l > k+1$.

When Δ is small, however, the probability of $v_1 < W_{k+1} < W_k < v_1 + \Delta$ is sufficiently small, which is of second-order in magnitude (proportional to Δ^2). The probability of $W_{k+1} < v_1 < W_k < v_1 + \Delta$ is sufficiently small, which is of first-order in magnitude (proportional to Δ). Therefore, when Δ is sufficiently small, the main effect is Case 1, and then, when bidder 1's bidding strategy changes from $\beta^k(v_1)$ to $\beta^k(v_1+\Delta)$, her interim expected utility increases by approximately:

$$[\Psi_k(v_1+\Delta|w_{k-1}) - \Psi_k(v_1|w_{k-1})][(v_1 - \beta^k(v_1)) - (v_1 - \beta^{k+1}(v_1))]. \tag{21.5.43}$$

Equating (21.5.42) and (21.5.43), we have

$$\begin{aligned}\Psi_k(v_1|w_{k-1})[\beta^k(v_1+\Delta) - \beta^k(v_1)] &= [\Psi_k(v_1+\Delta|w_{k-1}) - \Psi_k(v_1|w_{k-1})] \\ &\quad \times [(v_1 - \beta^k(v_1)) - (v_1 - \beta^{k+1}(v_1))]. \end{aligned} \tag{21.5.44}$$

Dividing equation (21.5.44) by Δ and taking the limit as $\Delta \to 0$, we obtain the differential equation:

$$\beta^{k\prime}(v_1) = \frac{\psi_k(v_1|w_{k-1})}{\Psi_k(v_1|w_{k-1})}[\beta^{k+1}(v_1) - \beta^k(v_1)], \tag{21.5.45}$$

where

$$\begin{aligned}\Psi_k(v_1|w_{k-1}) &= \Psi_1^{n-k}(v_1|w_{k-1}) \\ &= \frac{\Phi(v_1)^{n-k}}{\Phi(w_{k-1})^{n-k}},\end{aligned}$$

because of $W_k^{(n-1)} = W_1^{(n-k)}$.

Together with the initial condition $\beta^k(0) = 0$, we obtain the solution to the differential equation (21.5.45):

$$\begin{aligned}\beta^k(v_1) &= \frac{1}{\Phi^{n-k}(v_1)} \int_0^{v_1} \beta^{k+1}(v) d(\Phi^{n-k}(v)) \\ &= E[\beta^{k+1}(W_1^{(n-k)})|W_1^{(n-k)} < v_1] \\ &= E[\beta^{k+1}(W_k)|W_k < v_1 < W_{k-1}]. \end{aligned} \quad (21.5.46)$$

The equation (21.5.46) is a first-order recursive equation, and given the equilibrium bidding equation of the last round (21.5.41), we can obtain the equilibrium bidding strategy in the $K-1$-th auction:

$$\begin{aligned}\beta^{K-1}(v_1) &= E[\beta^K(W_{K-1})|W_{K-1} < v_1 < W_{K-2}] \\ &= E[E[W_K|W_K < W_{K-1}]|W_{K-1} < v_1 < W_{K-2}] \\ &= E[W_K|W_{K-1} < v_1 < W_{K-2}]. \end{aligned} \quad (21.5.47)$$

Going backwards from the last around proceeding inductively in this fashion results in the equilibrium bidding strategy for all $k < K$ rounds:

$$\begin{aligned}\beta^k(v_1) &= E[\beta^{k+1}(W_k)|W_k < v_1 < W_{k-1}] \\ &= E[E[W_K|W_{k+1} < W_k]|W_k < v_1 < W_{k-1}] \\ &= E[W_K|W_k < v_1 < W_{k-1}]. \end{aligned} \quad (21.5.48)$$

Let $W_0 = \infty$, and we obtain a symmetric equilibrium of a sequential first-price sealed-bid auction.

Proposition 21.5.2 . *Suppose that K identical items are sold to $n > K$ bidders by sequential first-price sealed-bid auctions, and each bidder has a unit demand and private values. Then, symmetric equilibrium strategies $(\beta^{I1}(v), \cdots, \beta^{IK}(v))$ are given by*

$$\beta^{Ik}(v) = E[W_K|W_k < v < W_{k-1}], k = 1, \cdots, K,$$

where $W_k \equiv W_k^{(n-1)}$ is the k-th highest of independently and stochastically drawn variables from $n-1$ identical distributions.

Again, two features of sequential auctions should be mentioned. First, as the auction goes on, the bidding prices become increasingly active, i.e., $\beta^k(v) > \beta^m(v)$ with $k > m$. It can be shown that $\beta^{Ik}(v)$ is an increasing function, and can also be induced by equation (21.5.45).

Secondly, the equilibrium price path is a martingale. Suppose that, in equilibrium, bidder 1 with value v_1 wins the $k < K$-th round auction. Then, there must be

$$W_K < \cdots < W_k < v_1 < W_{k-1} < \cdots < W_1,$$

and the (equilibrium) realized price in period k is $p^k = \beta^k(v_1)$. Moreover, the price in the $k+1$-st round is a random variable $P^{k+1} = \beta^{k+1}W_k$ and from (21.5.48), we have:

$$\begin{aligned} E(P^{k+1}|p^k) &= E[\beta^{k+1}(W_k)|W_k < v_1 < W_{k-1}] \\ &= \beta^k(v_1) \\ &= p^k. \end{aligned}$$

This establishes that the price path in sequential auctions is a martingale. The logic behind this conclusion is very simple: if $E(P^{k+1}|p^k) \neq p^k$, for example, $E(P^{k+1}|p^k) < p^k$, the bidder who wins the auction in the k-th round would be unwilling to use the equilibrium strategy of β^k, as the expected auction price will be lower and the expected utility for her winning the auction in the next round will be greater.

In the model, we have assumed that the bidder is risk neutral, but the conclusion will be similar even for the case of risk aversion.

21.5.2 Equilibrium of Sequential Second-Price Auctions

We now discuss sequential second-price sealed-bid auctions. If there is an increasing symmetric bidding strategy $(\beta^{II1}, \cdots, \beta^{IIK})$ in the sequential second-price sealed-bid auction, the bidder with the highest value at each round wins one unit, and then such second-price sealed-bid auction will also be efficient. Then, the sequential first-price and second-price auctions are revenue equivalent. We will simply find the equilibrium strategies by making use of this revenue equivalence principle.

Specifically, let $m_I(v)$ and $m_{II}(v)$ denote the interim expected payments by a bidder with value v in sequential first- and second-price auctions, respectively. Let $m_I^k(v)$ be the expected payment made in the k-th auction by a bidder when the items are sold by means of first-price sealed-bid auctions. Then,

$$m_I(v) = \sum_{k=1}^{K} m_I^k(v).$$

Similarity, define

$$m_{II}(v) = \sum_{k=1}^{K} m_{II}^k(v)$$

for sequential second-price sealed-bid auctions. We first use the revenue equivalence principle to obtain a strong version of revenue equivalence:

$$m_I^k(v) = m_{II}^k(v), \forall k \in \{1, \cdots, K\}. \tag{21.5.49}$$

Equation (21.5.49) implies that each round of auction satisfies the revenue equivalence principle. We verify this by induction, starting with the

K-th auction. Prior to the last round auction, as the equilibrium strategies of the two auctions are monotonic, the remaining $n - K + 1$ bidders can infer the private value of the previous winning bidders. For instance, bidder 1 knows her own value $v_1 = v$, while her competitors have values $W_K, \cdots, W_{n-1}$. At round $K - 1$, the winner's value is $W_{k-1} = w_{k-1}$. The revenue equivalence principle implies that $m_I(v) = m_{II}(v)$. Since bidders with a private value of v do not win the auction in the previous $K - 1$ rounds, $m_I^k(v) = m_{II}^k(v) = 0, \forall k \leqq K - 1$, and thus we have:

$$m_I^K(v) = m_{II}^K(v).$$

Now, consider the start of auction $K-1$. The remaining $n-K+2$ bidders bid on the remaining items. Suppose that bidder 1 knows her own value $v_1 = v$, and her competitors have values $W_{K-1}, \cdots, W_{n-1}$. At round $K-2$, the winner's value is $W_{k-2} = w_{k-2}$. Once again, the revenue equivalence principle implies that

$$m_I^{K-1}(v) + m_I^K(v) = m_{II}^{K-1}(v) + m_{II}^K(v).$$

Moreover, since $m_I^K(v) = m_{II}^K(v)$, we have $m_I^{K-1}(v) = m_{II}^{K-1}(v)$. Proceeding inductively in this way, we will obtain:

$$m_I^k(v) = m_{II}^k(v), \ \forall k \in \{1, \cdots, K\}.$$

With this strong version of the revenue equivalence principle, we are ready to find the equilibrium bidding strategies of sequential second-price auctions.

Note that for the previous $K - 1$ rounds, bidder 1 does not win the auction. She wins the K-th auction with a private value of $v_1 = v$, and then apparently in the K-th auction her (weakly dominant) bidding strategy must be:

$$\beta^{IIK}(v) = v.$$

Suppose that, in the $k < K$-th round, the bidder wins the auction, and then there must be:

$$W_K < \cdots < W_k < v < W_{k-1} < \cdots < W_1.$$

In the second-price auction, her payment is $\beta^{IIk}(W_k)$, and for the bidder with a private value of v, her interim expected payment in the k-th round is:

$$m_{II}^k(v) = Prob[W_k < v < W_{k-1}] \times E[\beta^{IIk}(W_k)|W_k < v < W_{k-1}].$$

On the other hand, in the sequential first-price auction, the payment for a winning bidder with private value v in the k-th round is:

$$\begin{aligned} m_I^k(v) &= Prob[W_k < v < W_{k-1}] \times \beta^{Ik}(v) \\ &= Prob[W_k < v < W_{k-1}] \times E[\beta^{I(k+1)}(W_k)|W_k < v < W_{k-1}], \end{aligned} \quad (21.5.50)$$

where the second line of equation (21.5.50) comes from

$$\beta^{Ik}(v) = E[\beta^{I(k+1)}(W_k|W_k < v < W_{k-1})].$$

By the revenue equivalence principle for each period mentioned above, i.e., $m_I^k(v) = m_{II}^k(v)$, we have

$$E[\beta^{IIk}(W_k)|W_k < v < W_{k-1}] = E[\beta^{I(k+1)}(W_k)|W_k < v < W_{k-1}].$$

Differentiating both sides of the equation with respect to v results in the identity:

$$\beta^{IIk}(v) = \beta^{I(k+1)}(v).$$

Thus, we have

$$\beta^{IIk}(v) = E[W_K|W_{k+1} < v < W_k].$$

Obviously, $\beta^{IIk}(\cdot)$ is an increasing function. Summarizing the above discussion, we have the following proposition:

Proposition 21.5.3 *Suppose that K identical items are sold to $n > K$ bidders by sequential second-price sealed-bid auctions, and that each bidder has a unit demand and private values. Then, symmetric equilibrium bidding strategies are given by*

$$\beta^{IIK}(v) = v$$

and for all $k < K$,

$$\beta^{IIk}(v) = \beta^{I(k+1)}(v)$$

where $\beta^{I(k+1)}(v)$ is the $(k+1)$st round equilibrium bidding strategy in the sequential first-price sealed-bid auction format, derived in Proposition 21.5.2.

Comparing the strategies for the sequential first-price auctions with the strategies for the sequential second-price auctions, we find that, due to $\beta^{I(k+1)}(v) > \beta^{Ik}(v)$, $\beta^{IIk}(v) > \beta^{Ik}(v)$. In other words, every bidder bids more actively in sequential second-price auctions than in sequential first-price auctions. In addition, as with sequential first-price auctions, the equilibrium price process in a sequential second-price auction is also a martingale. Suppose that the equilibrium price process in a sequential second-price auction is $P^{II1}, \cdots, P^{IIK}$. Since $\beta^{IIk}(v) = \beta^{I(k+1)}(v)$, we only need to verify that the prices between the last two periods have the property of a martingale.

Suppose that bidder 1 with a value of v wins in round $K-1$, and thus we have $W_{k-1} < v < W_{k-2}$. As a consequence, the auction (random) price of round $K-1$ is $P^{K-1} = \beta^{II(K-1)}(W_{K-1})$. Let the realized price be $p^{K-1} = \beta^{II(K-1)}(w_{K-1})$, where w_{K-1} is the realized value of W_{K-1}. In the last round, the bidder with a value of w_{K-1} will win, and her payment will be $P^K = \beta^{IIK}(W_K) = W_K$. We then have:

$$
\begin{aligned}
E[P^{IIK}|p^K] &= E[W_K|W_K < w_{K-1}] \\
&= \beta^{II(K-1)}(w_{K-1}) \\
&= p^{K-1}.
\end{aligned}
$$

Thus, the equilibrium price process in a sequential second-price auction is also a martingale. No participant would like to win the auction through a strategic delay.

21.6 Combinatorial Auctions of Heterogeneous Objects

We have so far only discussed homogeneous objects in considering multi-term auctions. In practice, objects are often heterogeneous: they can be substitutes, complements, or sometimes unrelated; the relevance of the objects may also vary with different bidders. All of these factors create new complications. For instance, if there is complementarity between two objects, the bidder would be very cautious if it was anticipated that the complementary items would not be obtained in the auction, which is known as the **exposure problem**. Indeed, many auctions are dedicated to solving such incentive issues in heterogeneous multi-term auctions. As the issue of combinatorial auctions of heterogeneous objects will be more complex, we mainly introduce some basic conclusions in this section. We again focus on the case of private values.

21.6.1 The Basic Model

Let $K = \{a, b, c, \cdots\}$ be a finite set of distinct objects to be auctioned, and $N = \{1, 2, \cdots, n\}$ be the set of bidders. Bidder i's private values of combination $S \subseteq K$ is $v_i(S)$, and bidder i's value vector of all combinations is $\boldsymbol{v}_i = (v_i(S))_{S \subseteq K}$. Suppose that $v_i(\emptyset) = 0$ and $v_i(S) \leqq v_i(T), \forall S \subseteq T \subseteq K$. The set of all value vectors for bidder i is χ_i, which is a nonnegative closed convex set with $\mathbf{0} \in \chi_i$.

For bidder i, there may be some specific relationship between objects, such as substitutability. Any two of multiple identical objects are perfect substitutes, and generally have a diminishing marginal value, i.e., with one object already in hand, the marginal value of the other item will decrease. For heterogeneous multi-objects, there is a similar substitution: if a set of combinational objects in hand is larger than another, the marginal value of an additional object is smaller. We then have the following definition.

Definition 21.6.1 (Substitutable Multiple Items) Objects are said to be **substitutable** for bidder i if for any $a \in K, a \notin T, S \subseteq T$,

$$v_i(S \cup a) - v_i(S) \geqq v_i(T \cup a) - v_i(T), \forall \boldsymbol{v}_i \in \chi_i. \quad (21.6.51)$$

It can be shown that (21.6.51) is equivalent to:

$$v_i(S) + v_i(T) \geqq v_i(S \cup T) + v_i(S \cap T), \forall \boldsymbol{v}_i \in \chi_i. \quad (21.6.52)$$

If (21.6.52) holds, it implies that bidder i's value has **submodular**. If $S \cap T = \emptyset$, the inequality (21.6.52) is reduced to

$$v_i(S) + v_i(T) \geqq v_i(S \cup T), \forall \boldsymbol{v}_i \in \chi_i. \quad (21.6.53)$$

In this case, we say that bidder i's value is **subadditive**.

The following defines the complementarity of objects.

Definition 21.6.2 (Complementary Multiple Items) Objects are said to be **complementary** for bidder i if for any $a \in K, a \notin T, S \subseteq T$, we have

$$v_i(S \cup a) - v_i(S) \leqq v_i(T \cup a) - v_i(T), \forall \boldsymbol{v}_i \in \chi_i. \quad (21.6.54)$$

Then, analogously, bidder i's value is **supermodular** if it satisfies

$$v_i(S) + v_i(T) \leqq v_i(S \cup T) + v_i(S \cap T), \forall \boldsymbol{v}_i \in \chi_i. \quad (21.6.55)$$

Moreover, bidder i's value is said to be **superadditive** if it satisfies:

$$v_i(S) + v_i(T) \leqq v_i(S \cup T), \forall \boldsymbol{v}_i \in \chi_i, S \cap T = \emptyset. \quad (21.6.56)$$

In multi-item auctions, let $y_i \in K$ denote the set of items obtained by bidder i in allocation $\boldsymbol{y}$. The allocation $\boldsymbol{y} = (y_1, \cdots, y_n)$ is **feasible**, if $y_i \cap y_j = \emptyset, \forall i \neq j$, and $\bigcup_i y_i = K$. As such, a feasible allocation is actually a partition of the set of all objects K. Let Y denote the set of all feasible allocations.

Next, we utilize the mechanism design approach to discuss multi-item auctions. Suppose that the value profile of all bidders is $\boldsymbol{v} = (\boldsymbol{v}^1, \cdots, \boldsymbol{v}^n)$. Define the allocation rule as $y(\boldsymbol{v}) = (y_1(\boldsymbol{v}), \cdots, y_n(\boldsymbol{v}))$, where $y_i(\boldsymbol{v})$ is the set of items obtained by participant i at a value of $\boldsymbol{v}$. We consider the direct mechanism, in which each bidder reports her value $\hat{\boldsymbol{v}}_i, i \in N$. Let $y_i(\hat{\boldsymbol{v}}_i) = (y_i(S, \hat{\boldsymbol{v}}_i)_{S \subseteq K})$, where $y_i(S, \hat{\boldsymbol{v}}_i)$ is the probability that bidder i wins the combination S when reporting $\hat{\boldsymbol{v}}_i$. $y_i(\hat{\boldsymbol{v}}_i)$ is the probability vector that bidder i wins the item combination, the payment of which is $m_i(\hat{\boldsymbol{v}}_i)$, or in terms of transfer, it is $t_i(\hat{\boldsymbol{v}}_i) = -m_i(\hat{\boldsymbol{v}}_i)$.

A direct mechanism $(\boldsymbol{y}(\cdot), \boldsymbol{t}(\cdot))$ is said to be **incentive compatible** if it satisfies

$$\begin{aligned} U_i(\boldsymbol{v}^1) &= y_i(\boldsymbol{v}_i).\boldsymbol{v}_i + t_i(\boldsymbol{v}_i) \\ &= \max_{\hat{\boldsymbol{v}}_i} y_i(\hat{\boldsymbol{v}}_i).\boldsymbol{v}^1 + t_i(\hat{\boldsymbol{v}}_i). \end{aligned}$$

Similarly, we have the following revenue equivalence theorem for combinatorial auctions.

Theorem 21.6.1 *Suppose that the allocation rule $y(\cdot)$ of two auctions is the same, and every bidder's expected utility is zero at the lowest value. Then, the auctioneer's revenues from the two auctions are the same.*

The proof is similar to that of Proposition 21.4.1. The only difference is that the vectors of reported values are of size $2^{|K|}$, instead of size K, and thus we have the following equation:

$$U_i(\boldsymbol{v}_i) = U_i(0) + \int_0^1 \boldsymbol{y}_i(s\boldsymbol{v}_i)\boldsymbol{v}_i ds. \tag{21.6.57}$$

21.6.2 A Benchmark of Combinatorial Auction: VCG Mechanism

In the following, we discuss a benchmark mechanism of combinatorial auctions, i.e., the Vickrey-Clarke-Groves (VCG) mechanism. In private value settings, similar to the cases of single-item and homogeneous multiple-object auctions, the heterogeneous multiple-object combinatorial VCG auction is also an efficient mechanism.

An allocation rule is **efficient** if for any value vector $\boldsymbol{v}$ of any combination of participants, the allocation $\boldsymbol{y}(\boldsymbol{v})$ maximizes social welfare, i.e.,

$$\boldsymbol{y}(\boldsymbol{v}) \in \mathrm{argmax}_{y_1,\cdots,y_n} \sum_{i\in N} v_i(y_i). \tag{21.6.58}$$

Given the value vector $\boldsymbol{v}$, the social welfare (social surplus) from an allocation $\boldsymbol{y}(\boldsymbol{v})$ is

$$W(\boldsymbol{v}) = \sum_{i\in N} \mathbf{v}_i(y_i(\boldsymbol{v})). \tag{21.6.59}$$

The total welfare of individuals other than i from an allocation $\boldsymbol{y}(\boldsymbol{v})$ is

$$W_{-i}(\boldsymbol{v}) = \sum_{j\neq i} \mathbf{v}_j(y_j(\boldsymbol{v})). \tag{21.6.60}$$

Now, we define the VGG mechanism of combinatorial auctions, which is an efficient mechanism. Given the value vector $\boldsymbol{v}$, and the allocation rule $\boldsymbol{y}(\boldsymbol{v})$ defined by (21.6.58), the transfer rule is

$$\begin{aligned} t_i(\mathbf{v}_i, \boldsymbol{v}_{-i}) =& \sum_{j\neq i} \mathbf{v}_j(\mathbf{y}(\boldsymbol{v})) - \sum_{j\neq i} \mathbf{v}_j(\mathbf{y}(\boldsymbol{v}_{-i})) \\ =& W_{-i}(\boldsymbol{v}) - W_{-i}(\boldsymbol{v}_{-i}). \end{aligned} \tag{21.6.61}$$

In fact, from Chapter 18, this mechanism is also called the pivotal or Clarke mechanism, which is a special case of a general VCG mechanism.

We can interpret the transfers of the VCG mechanism defined by the formula (21.6.61) as the externality that bidder i exerts on the other bidders. When bidder i does not bid, the social allocation rule is $\mathbf{y}(\boldsymbol{v}_{-i})$. When bidder i bids, the social allocation rule is $\mathbf{y}(\boldsymbol{v})$, while the change in others' welfare is characterized by (21.6.61). Like the usual VCG mechanism, the transfer $t_i(v_i, \boldsymbol{v}_{-i})$ of bidder i is independent of v_i. Therefore, truth-telling is a weakly dominant strategy in the VCG mechanism. The following proposition describes the VCG mechanism under the combinatorial auctions.

Proposition 21.6.1 *Under the combinatorial auction VCG mechanism, truth-telling is a weakly dominant strategy for every bidder. The VCG mechanism is efficient when each bidder truthfully reports her own value.*

21.6.3 Properties of VCG Mechanism

The VCG mechanism is an efficient combinatorial auction, under which it is a weakly dominant strategy for each bidder to report her true value. The efficiency of the VCG mechanism does not depend on the distribution of the value types of participants, so that it is ex-post implementable. As mentioned in Chapter 18, Green and Laffont (1979) and Holmstrom (1979) showed that, under relatively weak assumptions, the VCG mechanism is a unique weakly dominant mechanism with truthful reporting, efficient outcomes, and zero utility of losing bidders who did not win the object (see Proposition 18.7.3).

However, this conclusion needs one extra assumption on the bidder's value type, i.e., a smooth path connection. This assumption that is weaker than differentiability used in the proof of Proposition 18.7.3 has been used in the proof of revenue equivalence earlier. Here is a formal definition.

Definition 21.6.3 The set of bidder's valuation functions is said to be **s-moothly path connected** if, for any two functions $\boldsymbol{v}_i(\cdot), \hat{\boldsymbol{v}}_i(\cdot) \in \chi_i, \forall i \in N$, there is a family of valuation functions $\boldsymbol{v}_i(\cdot, s) \in \chi_i$ for $s \in [0,1]$, such that $\boldsymbol{v}_i(\cdot, 0) = \boldsymbol{v}_i(\cdot)$, $\boldsymbol{v}_i(\cdot, 1) = \hat{\boldsymbol{v}}_i(\cdot)$, $\boldsymbol{v}_i(\cdot, s)$ is differentiable in s, and

$$\int_0^1 \sup_{S \subseteq K} |\frac{\partial \mathbf{V}_i(S, s)}{\partial s}| < \infty.$$

Under the above conditions, Ausubel and Milgrom (2006) showed the uniqueness of the VCG mechanism.

Proposition 21.6.2 *Suppose that the set of all valuation functions is smoothly path connected and* $\mathbf{0} \in \chi_i, \forall i \in N$. *Then, the VCG mechanism is the unique direct revelation mechanism in which truthful reporting is a weakly dominant s-trategy, the resulting equilibrium outcomes are efficient, and there are no transfers to losing bidders.*

PROOF. Given a value combination $\boldsymbol{v}_{-i}$ of the other bidders, and consider any mechanism satisfying the above assumptions. If bidder i reports $\hat{\boldsymbol{v}}_i = 0$, then her outcome is zero, and the transfer is also zero. Suppose that bidder i reports valuation function $v_i(\cdot, 1) = \hat{\boldsymbol{v}}_i(\cdot) \neq 0$ and let $v_i(\cdot, 0) = \mathbf{0}$, $\{v_i(\cdot, s)|s \in [0,1]\}$ be a family of smooth valuation functions. Denote by $\boldsymbol{y}(s)$ the socially optimal allocation when bidder i reports $v_i(\cdot, s)$, and let

$$U_i(s) = \max_{s'}\{v_i(y_i(s'), s) + t_i(s')\},$$

where $y_i(s') = y(\mathbf{v}_i(\cdot, s'), \boldsymbol{v}_{-i})$ and $t_i(s') = t_i(\mathbf{v}_i(\cdot, s'), \boldsymbol{v}_{-i})$ is the allocation and transfer of bidder i under the VCG mechanism. By the Envelope Theorem in integral form (Milgrom and Segal (2002)),

$$U_i(1) - U_i(0) = \int_0^1 \frac{\partial v_i(\boldsymbol{y}(s), s)}{\partial s} ds.$$

Let $\hat{t}(s)$ be the transfer under any direct revelation mechanism for which truthful reporting is a dominant strategy, the resulting equilibrium outcomes are always efficient, and there are no transfers by, or to, losing bidders. Let

$$\hat{U}_i(s) = \max_{s'}\{v_i(y_i(s'), s) + \hat{t}_i(s')\}.$$

By the Envelope Theorem,

$$\hat{U}_i(1) - \hat{U}_i(0) = \int_0^1 \frac{\partial v_i(\boldsymbol{y}(s), s)}{\partial s} ds = U_i(1) - U_i(0).$$

Since $U_i(0) = \hat{U}_i(0) = 0$, we have

$$v_i(\boldsymbol{y}(1), 1) + t_i(1) = U_i(1) = \hat{U}_i(1) = v_i(\boldsymbol{y}(1), 1) + \hat{t}_i(1).$$

Therefore, $t_i(1) = \hat{t}_i(1)$, and thus the VCG mechanism is a unique direct revelation mechanism for which truthful reporting is a dominant strategy, the outcomes are efficient, and there are no payments by, or to, losing bidders. □

Another important property of the VCG mechanism is that when the solution concept of implementation is Bayesian-Nash equilibrium, combined with the previous Revenue Equivalence Theorem (21.6.1), we can obtain that the expected revenue under the VCG mechanism is not less than that of any other efficient and Bayesian incentive compatible combinatorial auction.

21.6.4 Defects of Combinatorial Auction VCG Mechanism

Despite these desired properties of the VCG mechanism, it is seldom adopted in practical multiple objects auctions. There are a number of reasons for this.

First, the VCG mechanism is complicated. For instance, the U.S. Federal Communications Commission's auction of spectrum No. 31 in 1994 contained 12 different frequency spectrums, and bidders were required to report a value of possible combinations of $2^{12} = 4,096$. This will not only entail a significant computational burden to the auctioneer, but as the number of items to be auctioned increases, the calculations required for the auction exhibit exponential growth.

Second, the VCG mechanism involves a considerable amount of private information, and such information disclosure may affect future bidding. Bidders may rationally be reluctant to report their true values, fearing that the information that they revealed will later be used against them.

Third, the VCG mechanism may trigger controversy because two bidders may pay different prices for identical items. In addition to the above-mentioned issues, we will also explore the VCG mechanism from the perspective of auction revenues: excessively low seller's revenues; non-monotonicity of the seller's revenues in the set of bidders and the bids; vulnerability to collusion by a coalition of bidders; and vulnerability to shill bidding. This subsection mainly refers to Ausubel and Milgrom (2002, 2006).

We now examine these defects of the combinatorial VCG mechanism by a series of examples.

Example 21.6.1 Consider an auction of two items $\{a, b\}$ to three bidders. The values of the three bidders are:

$$\mathbf{v}_1 = (\mathrm{v}_1^{\mathrm{a}}, \mathrm{v}_1^{\mathrm{b}}, \mathrm{v}_1^{\{\mathrm{a,b}\}}) = (0, 0, 2),$$
$$\mathbf{v}_2 = (\mathrm{v}_2^{\mathrm{a}}, \mathrm{v}_2^{\mathrm{b}}, \mathrm{v}_2^{\{\mathrm{a,b}\}}) = (2, 2, 2),$$
$$\mathbf{v}_3 = (\mathrm{v}_3^{\mathrm{a}}, \mathrm{v}_3^{\mathrm{b}}, \mathrm{v}_3^{\{\mathrm{a,b}\}}) = (2, 2, 2),$$

and $v_i^{\emptyset} = 0, \forall i$. It is easy to see that these two items are complements to bidder 1, but substitutes to bidders 2 and 3. At this point, there are two efficient allocations: one is $(y_1, y_2, y_3) = (\emptyset, \{a\}, \{b\})$; and the other is $(y_1, y_2, y_3) = (\emptyset, \{b\}, \{a\})$.

Under the VCG mechanism, the transfers of bidders 2 and 3 are both zero, irrespective of the allocation. The reason for this is that:

$$t_2(\boldsymbol{v}) = W_{-2}(\boldsymbol{v}) - W_{-2}(\boldsymbol{v}_{-2}) = 2 - 2 = 0.$$

The same conclusion applies symmetrically to $t_3(\boldsymbol{v}) = 0$. Thus, the total auctioneer's revenue is zero in the multi-item VCG mechanism.

Note that, in this example, if the auctioneer sells the two items as a bound sale, and every bidder values the combined items $\{a, b\}$ as 2, then the seller's revenue is 2. The multi-item VCG mechanism brings excessively low seller revenue.

In this example, we also find that, for the multi-item VCG mechanism, the seller's revenue is not an increasing function of the number of bidders.

Example 21.6.2 The following example is a variant of Example 21.6.1. Suppose that bidder 3 does not participate in bidding, i.e., only bidders 1 and 2 participate. Also consider the VCG mechanism in which there are four possible efficient allocations, and the corresponding VCG transfers are:

(1) $(y_1, y_2) = (\{a, b\}, \emptyset)$, and $t_1 = -2, t_2 = 0$;
(2) $(y_1, y_2) = (\{a\}, \{b\})$, and $t_1 = 0, t_2 = -2$;
(3) $(y_1, y_2) = (\{b\}, \{a\})$, and $t_1 = 0, t_2 = -2$;
(4) $(y_1, y_2) = (\emptyset, \{a, b\})$, and $t_1 = 0, t_2 = -2$.

Under the VCG mechanism with only bidder 1 and bidder 2, the seller's revenue is 2; whereas, the more bidders in Example 21.6.1 bring a smaller auction revenue, which is zero.

In addition, the multi-item VCG mechanism is prone to bidder collusion. The following example shows the collusion incentive.

Example 21.6.3 Consider a variant of Example 21.6.1. Suppose that the value of bidder 1 is unchanged, but bidders 2 and 3's value of the item is only 0.5. In other words, the values of the three bidders of the item are:

$$\boldsymbol{v}_1 = (v_1^a, v_1^b, v_1^{\{a,b\}}) = (0, 0, 2),$$
$$\boldsymbol{v}_2 = (v_2^a, v_2^b, v_2^{\{a,b\}}) = (0.5, 0.5, 0.5),$$
$$\boldsymbol{v}_3 = (v_3^a, v_3^b, v_3^{\{a,b\}}) = (0.5, 0.5, 0.5).$$

Then, the only efficient allocation is $(y_1, y_2, y_3) = (\{a, b\}, \emptyset, \emptyset)$. In the VCG mechanism, $t_1 = -1, t_2 = t_3 = 0$, and the auction revenue is 1.

However, if bidders 2 and 3 collude, and the values that they report are $\hat{\boldsymbol{v}}_2 = \hat{\boldsymbol{v}}_3 = (2, 2, 2)$, according to the analysis of Example 21.6.1, the auction outcome becomes $(\emptyset, \{a\}, \{b\})$ or $(\emptyset, \{b\}, \{a\})$, and $t_2 = t_3 = 0$. Thus, the VCG mechanism is not efficient under collusion, and the auctioneer's revenue again becomes zero.

A multi-item VCG mechanism may not prevent shill bidding. The so-called "shill" in the auction means that the bidder participates in bidding by introducing a dummy bidder to obtain higher profits. Here, the dummy bidder is called the shill of the bidder. The example below shows that, in a multi-item VCG mechanism, there is a dummy bidder.

Example 21.6.4 Consider a variant of Example 21.6.3. Suppose that there are two bidders for the auction of two items $\{a, b\}$. Bidder 1 has the same value as that in Example 21.6.3, and bidder 2 can be considered as the sum of bidder 2 and bidder 3 in Example 21.6.3. In other words, the values of bidders 1 and 2 are

$$\boldsymbol{v}_1 = (v_1^a, v_1^b, v_1^{\{a,b\}}) = (0, 0, 2),$$
$$\boldsymbol{v}_2 = (v_2^a, v_2^b, v_2^{\{a,b\}}) = (1, 1, 1).$$

In the VCG mechanism, the allocation rule is $(y_1, y_2) = (\{a, b\}, \emptyset)$, $t_1 = -1, t_2 = 0$, and the auctioneer's revenue is 1.

With a dummy bidder 3, suppose that the values of bidder 2 and dummy bidder 3 are reported as $\hat{\boldsymbol{v}}_2 = \hat{\boldsymbol{v}}_3 = (2, 2, 2)$. As in the case of Example 21.6.3, the auction allocation outcome becomes $(\emptyset, \{a\}, \{b\})$ or $(\emptyset, \{b\}, \{a\})$ under the VCG mechanism, and $t_2 = t_3 = 0$. Bidders can obtain higher profits through dummy bidder 3, but the auction allocation is no longer efficient, and at the same time the auctioneer's revenue has been reduced to zero.

In addition, the VCG mechanism may prevent some efficient decisions. The following example reveals that the VCG mechanism inhibits the adjustment of efficient organizational structure.

Example 21.6.5 This example is a variant of Example 21.6.1. The values of three bidders are the same as those in Example 21.6.1. We can consider bidders 2 and 3 as two firms. If the two firms merge and let the merged firm be bidder 4, and then suppose bidder 4 has the value of $\boldsymbol{v}_4 = (v_4^a, v_4^b, v_4^{\{a,b\}}) = (4+x, 4+x, 4+x)$. x can be regarded as the benefits of the merger, and it is clear that as long as $x > 0$, the merger can produce efficiency improvement (to the value of the item). However, if the two firms merge, the merged firm gets the item in the VCG mechanism and needs to pay 2 at the same time. The merged firm's net income is $2 + x$. If the two firms do not merge, bidders 2 and 3 can get a total of 4 net gains. Therefore, only when $x \geqq 2$, the two firms have an incentive to merge. This shows that the VCG mechanism may distort the merger decision prior to the auction.

Note that, in the above five examples, bidder 1's value of the item is not substitutable. If the value of bidder 1 is also substitutable, as that of bidders 2 and 3, e.g., the value of bidder 1 is changed to :

$$\boldsymbol{v}_1 = (v_1^a, v_1^b, v_1^{\{a,b\}}) = (1, 1, 2),$$

the defects of VCG may disappear. In Example 21.6.1, we can verify that the auctioneer can get revenue 2 under the VCG mechanism. Meanwhile, we can show that the auctioneer's revenue is an increasing function of the number of bidders in the cases from Examples 21.6.2$\sim$ to 21.6.5, and the bidders have no conspiracy nor incentive to find a dummy bidder and will not distort the pre-auction merger decision. Therefore, the substitutability of auction items is very important and affects the operational efficiency of the auction. When the assumption of substitutability of bidder's value of items is no longer satisfied, it is necessary to introduce other multi-item auctions. Ausubel and Milgrom (2002) devised a series of ascending combinatorial auctions. When substitutability is satisfied, the ascending combinatorial auctions are reduced to the VCG mechanism, and when the substitutability assumption is not satisfied, they have better properties.

21.6.5 Ascending Combinatorial Auction Mechanism

We now focus on the ascending combinatorial auction of Ausubel and Milgrom (2002). We first introduce the mechanism: **ascending combinatorial bidding** proceeds in multiple rounds, and in each round, bidders place bids in terms of a certain amount of money, say ϵ (i.e., the bids are discrete); in any round, say s, each bidder can place a bid $b_i^s(S)$ on a package $S \subseteq K$, $\boldsymbol{b}_i^s$) is the bid vector of bidder i in round s, and within a round, all bids are placed simultaneously. At the end of a round, the auctioneer designates a set of provisionally winning bids that maximize the total revenue and the allocation corresponding to the set of winning bids: given that the bids of all bidders in round s are $\boldsymbol{b}$, the allocation rule is $\boldsymbol{y}^s = (y_1^s, \cdots, y_n^s)$, which corresponds to the winning bids designated by the auctioneer, so that $\boldsymbol{y}^s$ is feasible and

$$\boldsymbol{y}^s = \text{argmax}_{\boldsymbol{y}} \sum_{i \in N} b_i^s(y_i). \tag{21.6.62}$$

Then, if $y_i^s \neq \emptyset$, bidder i will be the provisionally winning bidder in round s. As long as there is a higher bid on a certain package in round $s+1$, i.e., $b_i^{s+1}(S) > \max_{j \in N} b_j^s(S)$, the auctioneer will re-select a new set of provisionally winning bids and its corresponding allocation in round $s+1$. If no higher bids on any package appear in round s^*+1, the auction comes to an end in round s^*. Let $\boldsymbol{y}_{s^*}$ denote the allocation that corresponds to the set of winning bids. Then, if i is the winner, i.e., $y_i^{s^*} \neq \emptyset$, her expected utility is

$$u_i^{s^*} = \mathbf{v}_i(y_i^{s^*}) - b_i^{s^*}(y_i^{s^*}).$$

If i is not the winner, her utility is zero.

The ascending combinatorial auction of Ausubel and Milgrom (2002) used proxy bidding, being a direct mechanism in which each bidder submits a value vector to a proxy agent who then bids in the bidder's interest. The goal of the ascending proxy auction of Ausubel and Milgrom (2002) is to ensure that each bidder has no incentive to deceive the proxy agent. The following is an illustration of the operation of the ascending proxy auction mechanism.

Suppose that the true value of bidder i is $\boldsymbol{v}_i$, and the value reported to the proxy agent is $\hat{\boldsymbol{v}}_i$.

(1) In round 0, the auctioneer sets an initial price vector as the bidding price vector $\boldsymbol{b}_0$; at the same time, let $b_i^0(S) = b_0^0(S), \forall i \in N, S \subseteq K$, the provisional winner of this round be the seller, and $y_0^0 = K, y_i^0 = \emptyset, \forall i \in N$. Here, superscript 0 denotes the seller. Define $\tilde{N} = N \cup \{0\}$, i.e., the set of all of the bidders and the seller.

(2) Let $\boldsymbol{b}_i^{s-1}$ be bidder i's bidding price vector in round $s-1$. The minimum bid that i can place in round s is $\underline{b}_i^s(S) = b_i^{s-1}(S) + \epsilon$ if $S \neq S_i^{s-1}$; otherwise, $\underline{b}_i^s(S) = b_i^{s-1}(S)$. The ϵ here is the minimum bidding increment required if the bidder wants to change the provisional allocation of the

previous rounds.

The proxy i determines the optimal package $\hat{y}_i$ based on $\hat{y}_i$ given by

$$\hat{y}_i = \text{argmax}_S[v_i(S) - \underline{b}_i^s(S)].$$

The proxy i then places a bid of $\beta_i^s(S|\hat{\boldsymbol{v}}_i)$, that is

$$\beta_i^s(S|\hat{\boldsymbol{v}}_i) = \begin{cases} \underline{b}_i^s(S), & \text{if } S = \hat{y}_i \text{ and } \hat{\boldsymbol{v}}_i(S) \geqq \underline{b}_i^s(S); \\ b_i^{s-1}(S), & \text{otherwise} \end{cases}$$

At the end of the round, the auctioneer chooses a set of provisionally winning bids that maximize the total revenue, i.e., (21.6.62), and the allocation $\boldsymbol{y}_t$ that corresponds to the set of winning bids.

If no proxy places a new bid in the next round following s^*, the winner of round s^* will pay for the package according to the bidding price placed by her proxy. If a proxy places a new bid in the next round, then repeat (2) until the auction ends. As the bid price is ascending, the auction will end in finite rounds. It is noted that, in some cases, some auction items are held by the seller unless the auctioneer sets the initial bid price vector as zero.

When the auction items are substitutable for each other, the ascending proxy auction has the same result as the VCG mechanism. Prior to the formal proof of this conclusion, it is necessary to introduce a definition and a lemma.

Given a value vector $\boldsymbol{v}$ and a coalition of some buyers and seller $I \subseteq N$, define the coalitional valuation function

$$w(I) = \max_{\boldsymbol{y}} \sum_{i \in I} \mathbf{v}_i(y_i). \tag{21.6.63}$$

to be the maximum surplus attainable by the coalition $\mathcal{I}$ from an optimal allocation of all of the objects.

Definition 21.6.4 The coalitional valuation function is **bidder-submodular** if for all $\{0\} \in I \subseteq J \subseteq \tilde{N}$ and $i \notin J$, and all coalitions satisfy

$$w(I \cup i) - w(I) \geqq w(J \cup i) - w(J).$$

Describing the coalitional valuation function as bidder-submodular means that the surplus marginal contribution of each individual to the coalition decreases as the set of the coalition becomes larger. In the definition, symbol $\{0\}$ denotes the auctioneer. If the coalition does not include the buyer, it is clear that the value of the coalition is zero.

The property of the coalitional valuation function in the auction is closely related to the relationship between the goods, and the following is a lemma proven by Ausubel and Milgrom (2002).

Lemma 21.6.1 *If the items are substitutes for all of the bidders, the valuation function of the coalition is submodular.*

The proof of the lemma is relatively complicated, and interested readers can refer to Ausubel and Milgrom (2002).

Next, we employ the coalitional valuation function to characterize the interim expected utility of the bidder in a VCG auction. Given the bidder's value $\boldsymbol{v}$ of items, based on the definition of $W(\cdot)$, i.e., equation (21.6.59), we have $w(\tilde{N}) = W(\boldsymbol{v})$ and $w(\tilde{N}\backslash i) = W(\boldsymbol{v}_{-i})$. By the definition of transfers in the VCG mechanism, i.e., equation (21.6.61), the interim expected utility of bidder $i \in N$ under the VCG mechanism can be written as

$$\bar{u}_i = w(\tilde{N}) - w(\tilde{N}\backslash i). \tag{21.6.64}$$

Ausubel and Milgrom (2002) showed that the result of the ascending proxy auction is equivalent to that of the VCG mechanism when the items are substitutes.

As discussed earlier, the so-called ex-post equilibrium is such a strategic combination in which, even though all participants in the game know the private information of other participants' types, the participants do not want to change their strategic choices. Compared with Bayesian-Nash equilibrium, the ex-post equilibrium is more robust because it does not depend on the distribution of bidders' types. Obviously, the ex-post equilibrium is Bayesian-Nash equilibrium, but the reverse is not necessarily true.

We then have the following proposition. To prove it, we first need to establish that truthful reporting in the ascending proxy auction will result in VCG outcomes, and then show that truthful reporting is indeed an ex-post equilibrium.

Proposition 21.6.3 *Suppose that objects are substitutes for all bidders. Then, truthful reporting to the proxy is an ex-post equilibrium of the ascending proxy auction, in which the equilibrium outcome is the same as that in the VCG mechanism.*

PROOF. Suppose that all bidders report their values truthfully to the proxy. We want to demonstrate that the utility of the ascending proxy auction is the same as in the VCG mechanism. Let s^* be the round in which the ascending proxy auction ends, and the utility obtained by each bidder is $u_i^{s^*}$. We claim that, for any i, there is $u_i^{s^*} \geqq \bar{u}_i$ (up to ϵ). Suppose by way of contradiction that in some round $s \leqq s^*$, $u_i^s < \bar{u}_i$. Then, bidder i must be a provisional winner of round s.

Let $\hat{I} = I\backslash 0$ be the set of provisional winners in round s. If $i \notin I$, then the seller's provisional revenue in round s is

$$\begin{aligned} w(I) - \sum_{j \in \hat{I}} u_j^s &< w(I) - \sum_{j \in \hat{I}} u_j^s + \bar{u}_i - u_i^s \\ &= w(I) - \sum_{j \in \hat{I} \cup i} u_j^s + w(\tilde{N}) - w(\tilde{N} \backslash i) \\ &\leqq w(I) - \sum_{j \in \hat{I} \cup i} u_j^s + w(I \cup i) - w(I) \\ &= w(I \cup i) - \sum_{j \in \hat{I} \cup i} u_j^s. \end{aligned}$$

Here, the inequality in the third line follows from applying Lemma 21.6.1. Thus, for the auctioneer, including bidder i in the set of provisional winners in round s will generate a larger provisional revenue. However, in round s, the utility of bidder $u_i^s < \bar{u}_i$ shows that the proxy of bidder i does not bid in a way that maximizes bidder i's interest, unless $\bar{u}_i - u_i^s < \epsilon$. Therefore, for all i, we have $u_i^{s^*} \geqq \bar{u}_i$ (up to ϵ).

Now, suppose that there is a bidder $i \in N$, such that $u_i^{s^*} > \bar{u}_i$. Then

$$\begin{aligned} w(\tilde{N}) &= u_0^{s^*} + u_i^{s^*} + \sum_{j \neq i} u_j^{s^*} \\ &> u_0^{s^*} + w(\tilde{N}) - w(\tilde{N} \backslash i) + \sum_{j \neq i} u_j^{s^*}, \end{aligned}$$

which implies that

$$u_0^{s^*} + \sum_{j \neq i} u_j^{s^*} < w(\tilde{N} \backslash i). \tag{21.6.65}$$

On the other hand, the seller's revenue

$$\begin{aligned} u_0^{s^*} &= \max_{\boldsymbol{y}} \sum_{j \in N} b_j^{s^*}(y_j | v_j) \\ &= \max_{\boldsymbol{y}} \sum_{j \in N} \max(v_j(y_j) - u_j^{s^*}, 0) \\ &= \max_{\boldsymbol{y}} \max_{\hat{I} \subseteq N} \sum_{j \in \hat{I}} (v_j(y_j) - u_j^{s^*}) \\ &= \max_{\hat{I} \subseteq N} \max_{\boldsymbol{y}} \sum_{j \in \hat{I}} (v_j(y_j) - u_j^{s^*}) \\ &= \max_{\hat{I} \subseteq N} (w(I) - \sum_{j \in \hat{I}} u_j^{s^*}). \end{aligned}$$

and thus, in particular, for $I = \tilde{N} \backslash i$, this implies that

$$u_0^{s^*} + \sum_{j \neq i} u_j^{s^*} \geqq w(\tilde{N} \backslash i), \tag{21.6.66}$$

which contradicts (21.6.65). Thus, we have argued that with truthful reporting, the ascending proxy mechanism will have the same outcome as the VCG mechanism.

In order to show that truthful reporting constitutes an ex-post equilibrium, consider bidder i, and suppose that all bidders $j \neq i$ report truthfully. Then, we discuss the reporting incentives of bidder i. We know from (21.6.66) that, independent of what bidder i submits as her value vector, the seller's revenue satisfies

$$u_0^{s^*} > w(\tilde{N}\backslash i) - \sum_{j \neq i} u_j^{s^*}.$$

The right-hand side is a lower bound on the seller's revenue because it can be obtained by ignoring bidder i and including all other bidders in the set of provisionally winning bidders. At the same time, the total payoff of all of the participants (bidders and the seller) can never exceed $w(\tilde{N})$, which means

$$u_0^{s^*} \leqq w(\tilde{N}) - \sum_{j \in N} u_j^{s^*}.$$

For bidder i, her arbitrary report does not make her utility exceed $w(\tilde{N}) - w(\tilde{N}\backslash i) = \bar{u}_i$. The right-hand side of the equation is the expected utility obtained in the ascending proxy auction if bidder i chooses to report truthfully, and thus bidder i will have the incentive to report truthfully. We, therefore, showed that truthful reporting constitutes an ex-post equilibrium. □

21.6.6 Sun-Yang Auction Mechanism for Multiple Complements

Ausubel and Milgrom (2002, 2006) further discussed that, when the auction items are not substitutes for each other, the ascending proxy auction will be better than the VCG mechanism, and can overcome the drawbacks and problems faced by the VCG mechanism mentioned previously.

The ascending combinatorial auction is similar to the dynamic adjustment mechanism in general equilibrium theory. When the goods are overdemanded, the price will rise. The slight difference is that, in ascending combinatorial auctions, there is only a one-way price change (i.e., increment), the equilibrium price of the ascending combinatorial auction is actually the general equilibrium price (of a discrete commodity), and the demand corresponding to the aggregate bidding prices is consistent with the supply corresponding to the auctioneer's optimal income. However, when the items are complements, the ascending proxy mechanism will face an exposure problem, i.e., there may be no general equilibrium price. The following example about the exposure problem is from Milgrom (2000, p. 257, Footnote 12)) .

Example 21.6.6 (Exposure Problem in Auction for Complements) Suppose that the set of objects is $K = \{a, b, c\}$, there are three bidders, $N = \{1, 2, 3\}$, the objects are mutual complements for all bidders (the valuation function is of superadditivity), and the values attached by the bidders to these objects (package) are

$$\begin{array}{llll}
v_1(a) = 1; & v_1(b) = 1; & v_1(c) = 0; & v_1(\{a, b\}) = 3; \\
v_2(a) = 0; & v_2(b) = 0.9; & v_2(c) = 1; & v_2(\{a, b\}) = 1; \\
v_3(a) = 1, & v_3(b) = 0; & v_3(c) = 1; & v_3(\{a, b\}) = 1; \\
v_1(\{a, c\}) = 1; & v_1(\{b, c\}) = 1; & v_1(\{a, b, c\}) = 3; & \\
v_2(\{a, c\}) = 1; & v_2(\{b, c\}) = 3; & v_2(\{a, b, c\}) = 3; & \\
v_3(\{a, c\}) = 3.5; & v_3(\{b, c\}) = 1; & v_3(\{a, b, c\}) = 3.5. &
\end{array}$$

If a competitive equilibrium does exist, its allocation would be efficient. In this example, there is an efficient allocation

$$y_1 = \{b\}, y_2 = \emptyset, y_3 = \{a, c\}.$$

To support the efficient allocation, the prices must satisfy

$$p^b \leqq 1, p^a + p^b \geqq 3, p^b + p^c \geqq 3, p^a + p^c \leqq 3.5.$$

However, these together imply that $p^a + 2p^b + p^c \geqq 6$ and $p^b \leqq 1$, which, in turn, implies that $p^a + p^c \geqq 4$, which contradicts $p^a + p^c \leqq 3.5$. Therefore, there is no competitive equilibrium.

The problem associated with the existence of general equilibrium of multi-item auctions is called the exposure problem. Indeed, new problems appear when auction items are complements. If bidders submit their bids according to actual needs, they may be exposed to a possible risk. When the bidder is included in the list of provisional winners by the auctioneer, she will face the risk of not getting some complementary items, and she will not be willing to pay for the other complementary items according to the bidding price. It is quite challenging to design an efficient auction for complements.

In the following, we briefly discusses an auction introduced by Sun and Yang (2014) for multiple complements. Sun and Yang constructed a dynamic auction to solve the design of an efficient auction for multiple complements. This dynamic mechanism is called the Sun−Yang auction.

The Sun−Yang auction introduces the concept of nonlinear general equilibrium, in which the general equilibrium of nonlinearity exists when the items are complements to all of the bidders, although the general linear equilibrium may not exist.

Suppose that there are a set of objects $K = \{a, b, \cdots\}$ and a group of bidders $N = \{1, 2, \cdots, n\}$. Bidder $i \in N$ attaches a monetary value to the objects, which is $u_i : 2^K \to \mathbb{Z}_+$, where 2^K denotes the set of all subsets of items K, and $\mathbb{Z}_+$ is the set of all nonnegative integers. The seller (denoted by superscript 0) has a reserve price function $u_0 : 2^K \to \mathbb{Z}_+$ with $u_i(\emptyset) = 0, \forall i \in N \cup \{0\}$. Let $\tilde{N} = N \cup 0$ represent the set of all agents.

Suppose that the bidder's utility function is quasi-linear and satisfies superadditivity for the valuation function of the item. We first introduce the concept of non-linear general equilibrium.

A price function $p : 2^K \to \mathcal{R}_+$, with $p(\emptyset) = 0$, is **feasible** if $p(S) \geqq u_0(S), \forall S \subseteq K$. For a given pricing function p, the bidder's demand correspondence (which may be multi-valued) is

$$D_i(p) = \mathrm{argmax}_{S \subseteq K}[u_i(S) - p(S)].$$

Let $\boldsymbol{\pi} = \{\pi^1, \cdots, \pi^m\}$, $m \leqq n$, be a partition of K, and $\boldsymbol{y} = (y_0, y_1, \cdots, y_n)$ be a feasible allocation, i.e., for any y_i, or $y_i = \emptyset$, or $y_i \in \boldsymbol{\pi}$, $y_i \cap y_j = \emptyset, i \neq j, \bigcup_{i \in \tilde{N}} y_i = K$. The corresponding partition of allocation $\mathbf{y}$ is $\boldsymbol{\pi} = \{y_i, i \in \tilde{N} : y_i \neq \emptyset\}$. Given the price function p, the seller's supply correspondence is

$$\boldsymbol{Y}(p) = \mathrm{argmax}_{\boldsymbol{\pi}} \sum_{S \in \boldsymbol{\pi}} p(S).$$

An allocation $\boldsymbol{y}^* = (y_0^*, y_1^*, \cdots, y_n^*)$ is **efficient** if for every allocation $\boldsymbol{y}$, we have

$$\sum_{i \in \tilde{N}} u_i(y_i^*)) \geqq \sum_{i \in \tilde{N}} u_i(y_i)).$$

Definition 21.6.5 (Nonlinear General Equilibrium) A **nonlinear general equilibrium** consists of a price function p^* and an allocation y^*, such that

(1) for the seller, $y^* \in \boldsymbol{Y}(p^*)$;

(2) for every bidder $i \in N$, $y_i^* \in D_i(p^*)$.

In the above definition, similar to general equilibrium, condition (1) characterizes supply under the nonlinear equilibrium price function; and condition (2) depicts demand under the nonlinear equilibrium price function, in which at the equilibrium price p^*, the quantity supplied equals the quantity demanded.

The following example on complements (from Sun and Yang) shows that a nonlinear general equilibrium exists, but the general equilibrium does not exist.

Example 21.6.7 Suppose that objects $K = \{a, b, c\}$ are complements for three bidders $N = \{1, 2, 3\}$. Bidders' values and seller's reserve prices are given as follows:

$$\begin{array}{llll}
v_1(a)=2; & v_1(b)=2; & v_1(c)=0; & v_1(\{a,b\})=7; \\
v_2(a)=2; & v_2(b)=0; & v_2(c)=2; & v_2(\{a,b\})=3; \\
v_3(a)=0; & v_3(b)=2; & v_3(c)=2; & v_3(\{a,b\})=4; \\
u_0(a)=1; & u^0(b)=2; & u_0(c)=1; & u_0(\{a,b\})=3; \\
v_1(\{a,c\})=4; & v_1(\{b,c\})=4; & v_1(\{a,b,c\})=7; & \\
v_2(\{a,c\})=6; & v_2(\{b,c\})=3; & v_2(\{a,b,c\})=6; & \\
v_3(\{a,c\})=3; & v_3(\{b,c\})=6; & v_3(\{a,b,c\})=7; & \\
u_0(\{a,c\})=3; & u_0(\{b,c\})=4; & u_0(\{a,b,c\})=5. &
\end{array}$$

In this example, there are two efficient allocations: $y_1 = \{a, b\}$, $y_2 = \{c\}$, $y_3 = \emptyset$, and $y_1' = \{a, b\}$, $y_2' = \emptyset$, $y_3' = \{c\}$. At the same time, there is a nonlinear general equilibrium price: $p^*(a) = p^*(b) = p^*(c) = 2$, $p^*(\{a, b\}) = p^*(\{a, c\}) = p^*(\{b, c\}) = 6$, and $p^*(\{a, b, c\}) = 7$.

On the demand side, $D_1(p^*) = \{a, b\}$, $\{c\} \in D_2(p^*)$, $\emptyset \in D_3(p^*)$; on the supply side, $(\{a, b\}, \{c\}, \emptyset) \in \boldsymbol{Y}(p^*)$. In addition, price p^* and $\boldsymbol{y}'$ also constitute a nonlinear general equilibrium.

However, there is no general equilibrium under the linear price. Suppose that there is a price vector $p(a)$, $p(b)$, and $p(c)$. If the price p^* and the allocation $\boldsymbol{y}'$ comprise an equilibrium, then:

for the seller, we must have $p(a) \geqq 1$, $p(b) \geqq 2$, $p(c) \geqq 1$;

for bidder 1, we must have: $p(a) + p(b) \leqq 7$;

for bidder 2, we must have: $p(c) \leqq 2$, $p(a) + p(c) \geqq 6$;

for bidder 3, we must have: $p(c) \geqq 2$, $p(b) + p(c) \geqq 6$.

From $p(c) = 2$, combining the inequalities $p(a)+p(c) \geqq 6$ and $p(b)+p(c) \geqq 6$, we have $p(a) \geqq 4$ and $p(b) \geqq 4$, which contradicts $p(a) + p(b) \leqq 7$.

Sun and Yang (2014) showed that, if objects are complements to all bidders and the bidders have quasi-linear utility functions, there is always a nonlinear general equilibrium. For readers who are interested in the proof of the theorem, see Sun and Yang (2012).

Sun and Yang (2014) constructed an ascending auction in which bidders choose to bid truthfully (which will be defined below). An elaborate dynamic auction is then constructed to ensure that each bidder has an incentive to bid truthfully, and more precisely, truthful bidding was a refined ex-post equilibrium.

Definition 21.6.6 In a dynamic auction, we say that bidder i bids **truthfully** if at every round and for any price function $p^t(\cdot)$, bidder i reports demand

$$d_i^t \in D_i(p^t) = \mathrm{argmax}_{S \subseteq K}\{u_i(S) - p^t(S)\}$$

with $d_i^t = \emptyset$ when $\emptyset \in D_i(p^t)$.

When bidders choose to bid truthfully, Sun and Yang constructed the following ascending auction, by which the final result of the auction is a nonlinear general equilibrium.

Definition 21.6.7 (Ascending Auction under Truthful Reporting) There are several steps:

(1) The seller announces her reserve price u_0 and sets the initial pricing function $p^0 = u_0$.
(2) In each round $t = 0, 1, 2, \cdots$, given the pricing function p^t at round t, the auctioneer chooses a supply set $\boldsymbol{y}_t \in Y(p^t)$, and every bidder i reports her demand $d_i^t \in D_i^t(p(t))$ against prices p^t. When there is at least a pair of i, j with $i \neq j$, such that $d_i^t = d_j^t = S, S \subseteq K$ or there is only one i, such that $d_i^t = S$ but no k that makes $S \neq y_k^t$, then we call this bundle S over-demanded. For any over-demanded bundle S, the auctioneer adjusts the price function through the formula $p^{t+1}(S) = p^t(S) + 1$, and repeats (2) into round $t+1$. If no bundle at the t-th round is overly-demanded, it is written as t^*.
(3) In round t^*, the auctioneer allocates $d_i^{t^*} = y_i^{t^*}$ to bidder i, and asks bidder i to pay the price $p^{t^*}(y_i^{t^*})$. If there is no i of bundle S, such that $S = d_i^{t^*}$ and $p^{t^*}(S) = u_0(S)$, the bundle S is left to the seller. If $p^{t^*}(S) > u_0(S)$, the bundle S is left to the bidder who finally gives up the choice of bundle S. Then, the auction ends.

Here, we show that the results of the ascending auction under truthful reporting are non-linear general equilibria.

Example 21.6.8 Given

$$p^t = (p^t(a), p^t(b), p^t(c), p^t(\{a,b\}), p^t(\{a,c\}), p^t(\{b,c\}), p^t(\{a,b,c\})),$$

in the initial round, the price, supply, and demand are as follows:

$$p^0 = (1,2,1,3,3,4,5), \boldsymbol{y}_0 = \{\{b,c\}, a\}, d_1^0 = \{a,b\}, d_2^0 = \{a,c\}, d_3^0 = \{a,b,c\}.$$

In rounds 1$\sim$6, the price, supply, and demand are as follows:

$$\begin{array}{lll}
p^1 = (1,2,1,4,4,4,6); & \boldsymbol{\pi}^1 = \{\{a,c\},b\}; & d_1^1 = \{a,b\}, d_2^1 = \{a,c\}, d_3^1 = \{b,c\}; \\
p^2 = (1,2,1,5,4,5,6), & \boldsymbol{\pi}^2 = \{\{a,b\},c\}; & d_1^2 = \{a,b\}, d_2^2 = \{a,c\}, d_3^2 = \{b,c\}; \\
p^3 = (1,2,1,5,5,6,6); & \boldsymbol{\pi}^3 = \{\{a,c\},b\}; & d_1^3 = \{a,b\}, d_2^3 = \{a,c\}, d_3^3 = \{a,b,c\}; \\
p^4 = (1,2,1,6,5,6,7); & \boldsymbol{\pi}^4 = \{\{a,b\},c\}; & d_1^4 = \{a,b\}, d_2^4 = \{a,c\}, d_3^4 = \{c\}; \\
p^5 = (1,2,2,6,6,6,7); & \boldsymbol{\pi}^5 = \{\{a,c\},b\}; & d_1^5 = \{a\}, d_2^5 = \{a\}, d_3^5 = \emptyset; \\
p^6 = (2,2,2,6,6,6,7); & \boldsymbol{\pi}^6 = \{\{a,b\},c\}; & d_1^6 = \{a,b\}, d_2^6 = \emptyset, d_3^6 = \{c\}.
\end{array}$$

Then, no bundle is over-demanded, $t^* = 6$, and the equilibrium prices are $p^* = (2, 2, 2, 6, 6, 6, 7)$. Bidder 1 gets $\{a, b\}$ and pays 6. Bidder 3 gets $\{c\}$ and pays 2.

Compared with Example 21.6.7, the outcomes of an ascending auction are non-linear general equilibria.

Sun and Yang (2014) established a dynamic Sun-Yang auction that induces each bidder to report truthfully. Since this mechanism involves substantial technical details and symbols, here we only briefly discuss its basic structure.

In the incentive-compatible dynamic auction, Sun and Yang constructed $n + 1$ markets, each of which corresponds to two price functions: the first price function; and the second price function. All participants, including the seller, have different prices for each market at each round. Let p^t_{-0} and p^t_0 denote the seller's first price function and second price function in each round t of market M_{-0} with the set N of bidders. Let p^t_{-i} and p^t_i denote bidder i's first price function and second price function in each round t of market M_{-i}, which stands for the market without bidder i.

Each bidder i reports a demand bundle against the second price function p^t_i, and the auctioneer announces a revenue-maximizing supply set of each market $n + 1$ at the first price function $p_{-k}(k \in \tilde{N})$. For bidders, because of different prices in each market, the prices faced in making demand decisions are also dissimilar. When the market overly-demands a certain bundle, the seller adjusts the price depending on how much the bidders influence the demand for the bundle, and thus the adjustment of each market and each bidder is different. If bidder i is the crucial demander of a bundle (i.e., the bidder who reports the highest price), bidder i will face a price increase of 1 unit, while the price of the others will be unchanged. This manner of price adjustment can internalize the externality of price changing by bidder's demand report.

Sun and Yang (2014) showed that under this dynamic mechanism, the bidder's payment is a generalized VCG payment. The Sun−Yang auction has many elaborate designs: the over-demand is divided into first over-demand and second over-demand; the price adjustment process is bidirectional, raising the price when overly-demanded and decreasing the price when over-supplied; the bidder is allowed to withdraw some of her past bids, but will be punished if she nullifies her bids too many times; the Sun −Yang auction can always end within a finite period of time, etc. Readers interested in additional details can refer to their paper.

The Sun−Yang auction possesses numerous advantageous features. For example, it requires the bidder to report only one demand against the market price, and thus it is privacy-preserving and more informationally efficient; it can solve the exposure problem easily, which is caused by complementarity; and it tolerates mistakes made by bidders in the bidding process.

For the exposure problem, a more complex situation is that a bundle of objects can be complements to one bidder, but substitutes to another. In this case, the nonlinear general equilibrium may also not exist. Sun and Yang (2009, 2015) put forward some other efficient auctions according to the substitutability and complementarity of items.

21.7 Biographies

21.7.1 William Vickrey

William Vickrey (1914-1996) was the pioneer of auction theory, and made profound contributions in numerous areas, including information economics and incentive theory. In 1996, Vickrey and James Mirrlees were awarded the Nobel Memorial Prize in Economic Sciences for their great contributions to the economic theory of incentives under asymmetric information. Unfortunately, he passed away three days after winning the Nobel Prize, when he was on the way to a conference.

At the end of the 1940s, Vickrey began to become well-known in academia, especially in the field of optimal taxation. His doctoral dissertation, "Agenda for Progressive Taxation", was reprinted as an "economic classic" in 1972. He was not only outstanding in taxation, transportation, public services and pricing, but also well-known for his pioneering research on incentive theory and auction theory. His penetrating insights into incentive theory in his early papers did not obtain the attention of the economics community until the 1970s, which greatly stimulated the development of auction theory, information economics, incentive theory, and other fields.

Although the practice of auctions has persisted for thousands of years, and some auctions have even produced an enormous historical impact, scientific research on auctions did not begin until the 1960s. In 1960, Vickrey published an article in *The Quarterly Journal of Economics* that discussed public pricing and sealed bidding strategies. In 1961, his classic paper, entitled "Counter speculation, auctions, and competitive sealed tenders" and published in the *The Journal of Finance*, explored the relationship between auction rules and public pricing. In this paper, he also analyzed such issues as private information and strategic pricing. Vickrey further developed the landmark auction theory – the Revenue Equivalence Theorem, which asserts that if all bidders have independent private values for the auctioned items, then all four of the standard single-item auction formats generate the same expected revenue for the seller. In addition, Vickrey invented the second-price sealed-bid auction, which provides incentives for all bidders to bid truthfully, i.e., bidders will bid the price at which they value the item. Indeed, the second-price auction constitutes a special case of the general Vickrey-Clarke-Groves (VCG) mechanism. Although Vickrey did not consider the efficient provision of items from the mechanism design approach, the VCG mechanism is the only mechanism that truthfully implements socially-efficient decision rules (the maximization of the sum of individual surpluses) in dominant strategy equilibrium.

These two papers are path-breaking works concerning auctions, and established the foundation for this research area. Vickrey's research about bidding methods solved the problem of how to allocate resources most efficiently under incomplete information or asymmetric information, which is also not necessarily limited to auction problems. In addition, his research created a precedent for the study of information economics. Under incomplete information or asymmetric information, those with more information can strategically use their information to obtain benefits. Information economics thus needs to explore how to design contracts or mechanisms to deal with various incentive and regulatory issues. Vickrey's research about bidding and pricing has brought about innumerable relevant studies to solve such problems as observed in the insurance market, the credit market, internal organizations of firms, salary structures, tax systems, social insurance, and political organizations.

Vickrey also made great contributions to the optimal pricing theory of public utilities and transportation. His research interests include marginal cost pricing, responsive pricing, urban congestion pricing, simulated futures markets, inflation effects on utility adjustment and pricing methods, etc. Although Vickrey focused on specific market mechanisms, his research has great value for us to comprehend general market mechanisms.

21.7.2 Paul Milgrom

Paul Milgrom (1948-) is a professor at Stanford University, a Fellow of the American Academy of Arts and Sciences, and a Fellow of the National Academy of Sciences. He made pioneering contributions to auction theory, pricing strategy, and mechanism design theory, and won the 2020 Nobel Memorial Prize in Economic Sciences, together with Robert B. Wilson, for improvements to auction theory and inventions of new auction formats.

Paul Milgrom was born in Detroit, Michigan, the United States in 1948. He received a bachelor's degree in mathematics from the University of Michigan in 1970, and a master's degree in statistics and a Ph.D. degree in business from Stanford University in 1978 and 1979, respectively. After graduation, he first taught at the Kellogg School of Management at Northwestern University from 1979 to 1983, and then taught at Yale University from 1983 to 1987. Milgrom has been a professor of economics at Stanford University since 1987, and the Shirley R. and Leonard W. Ely, Jr. Professor of Humanities and Sciences since 1993. While studying at Stanford University, Milgrom studied under the guidance of Wilson, and completed his Ph.D. dissertation on auction theory. His proposals of interdependent value and the design of simultaneous ascending auctions have markedly enriched auction theory. His books, entitled *The Structure of Information in Competitive Bidding* and *Putting Auction Theory to Work*, have become must-read works in this field.

In addition to his deep research on auction theory, Milgrom was also devoted to applying theory to solving practical problems. One of his great real-world accomplishments was to successfully design an auction for the U.S. telecommunications market. In addition to auction theory, he has published numerous papers in the areas of incentives in organizations, mathematical economics, game theory, network economy, pricing strategy, and actuarial science.

After Vickrey's landmark conclusion on auction theory—the Revenue Equivalence Theorem, a large number of scholars began to focus on auction theory, including Roger B. Myerson, who was awarded the Nobel Memorial Prize in Economic Sciences in 2007. Myerson used the newly-developed mechanism design theory to generalize auction theory. Under a series of assumptions that bidders' values of an item are independent and that bidders only care about their own expected returns, he proved that all possible standard auctions would bring the auctioneer the same expected revenue. Obviously, this conclusion surpassed that of previous research, such as those of Vickrey and others who focused on specific auction forms, which thus enabled the study of all possible auctions and advanced auction theory a great step further.

For an auctioneer, since all possible auction formats bring her the same expected revenue, can she arbitrarily choose a specific auction format? Unfortunately, although Myerson's conclusion is highly elegant in theory, the preconditions for his conclusion to hold are very strict, so that it is almost impossible to meet these assumptions in practice. This makes Myerson's Revenue Equivalence Theorem no longer valid, especially when the bidder's value of the auction item depends not only on herself, but also on the values of other bidders. Milgrom took the lead in the research of auctions with interdependent values. In 1982, Milgrom and Weber constructed an analytical framework that deals with information, prices, and auctioneer revenue when there are interdependent values. According to their observations of auction practices, they proposed that bidders' values may be interdependent. They also asserted that a bidder's higher value of the auction item may also easily increase the values of other participants.

Milgrom has a wide range of research interests, including auction design for the real world, organizational economics, bounded rationality, and economic history. His book, entitled *Economics, Organization, and Management* published in 1992 with John Roberts, is a well-known textbook. In 1993, Milgrom was commissioned by the former U.S. President Bill Clinton to design the auction protocol for the Federal Communications Commission (FCC) and completed the major design of the auction. Because of his contribution, the FCC auction achieved great success. As such, Milgrom has become one of the most well-known figures in the world of auctions and industrial economics. His team also designed channel and public auctions for Germany, Mexico, Canada, and many other countries.

21.8 Exercises

Exercise 21.1 (All-Pay Auctions under Uniform Distribution) An all-pay auction is similar to the first-price sealed bid auction. The difference is that bidders are required to pay their bids to the seller, regardless of winning or losing. Suppose that there are n bidders. The distribution of each bidder's value is independent and identical on $[0, 1]$.

1. For only two bidders whose values obey the uniform distribution over the unit interval, characterize the symmetric equilibrium under the all-pay auction in a symmetric environment. (Hint: Guess the equilibrium price function as an increasing quadratic function.)

2. When there are n bidders, derive the corresponding symmetric equilibrium bidding function. (Hint: The equilibrium bidding function must have the form of $\beta(\theta_i) = \alpha(\theta_i)^n$ in this case.)

3. Analyze the relationship between the equilibrium bidding function under the all-pay auction and the equilibrium bidding function under the first-price sealed-bid auction.

Exercise 21.2 (First-Price Auctions under Binary Distribution) Consider a first-price auction in a symmetric environment in which prices are distributed according to a binary distribution. In other words, bidder i's value satisfies $\theta_i \in \{\theta_l, \theta_h\}$, where $0 \leqq \theta_l < \theta_h < \infty$. For any bidder i, the ex-ante probability satisfies $\Pr(\theta_i = \theta_h) = \alpha$. Consider the case with only two bidders.

1. Characterize the Bayesian-Nash equilibrium.

2. When the values obey the binary distribution, does the Revenue-Equivalence Theorem between the first-price auction and the second-price auction still hold true?

Exercise 21.3 Assume that there are n buyers in the first-price auction, whose private values are uniformly distributed on $[0, 1]$. We further assume that all buyers' utility function is $U(\theta) = \theta^c$, for $c < 1$.

1. Under what values of c is the buyer risk-averse, risk-neutral, or risk-seeking?

2. Prove that the symmetric Bayesian-Nash equilibrium γ must satisfy the following differential equation:

$$\gamma'(\theta) = \frac{n-1}{c}\frac{f(\theta)}{F(\theta)}(\theta - \gamma(\theta)).$$

3. Prove that the following strategy is a symmetric equilibrium:

$$\gamma(\theta) = \theta - \frac{\int_0^\theta F^{\frac{n-1}{c}}(x)dx}{F^{\frac{n-1}{\theta}}}.$$

4. When F is a uniform distribution on $[0, 1]$, solve for the symmetric Bayesian-Nash equilibrium γ.

5. What is the functional form of the seller's expected revenue in c? Is it an increasing function?

Exercise 21.4 Consider a first-price sealed-bid auction with two bidders. In order to avoid the case in which bidder 1 bids the price $\frac{1}{4} + \epsilon$, where $\epsilon > 0$ is small, we define the function p (who shall get the item) as follows: if both bidders' prices are $\frac{1}{4}$, then bidder 1 is the winner; if both bidders' prices are the same and lower than $\frac{1}{4}$, then each would win with the same probability. Answer the following questions and prove your conclusions.

1. Verify whether the following strategy combination (β_1, β_2) constitutes an Bayesian-Nash equilibrium:

$$\beta_1(\theta_1) = \frac{\theta_1}{2},\ \beta_2(\theta_2) = \min\left\{\frac{\theta_2}{2}, \frac{1}{4}\right\}.$$

2. Denote by (β_1^*, β_2^*) a non-decreasing equilibrium. What are the values of $\beta_1^*(0)$ and $\beta_2^*(0)$? Is it true that $\beta_1^*(1) = \beta_2^*(1)$?

3. Verify whether the Revenue-Equivalence Theorem holds.

4. Is it possible to verify whether the direct revelation mechanism that maximizes the seller's expected revenue under individual rationality and incentive compatibility is a second-price auction with a reserve price?

Exercise 21.5 Suppose that there are two buyers with independent private values distributed uniformly on $[0, 1]$. Buyer 2 has a budget constraint, and the maximum value that she can bid is $\frac{1}{4}$. Buyer 1 does not have a budget constraint. Answer the following questions:

1. If buyers are involved in a second-price sealed-bid auction, find an equilibrium strategy.

2. Derive the seller's expected revenue under the above equilibrium s-trategy.

Exercise 21.6 Consider a variant of the following second-price sealed-bid auction: there is a single indivisible item for sale, and the values of n bidders are given by an independent random variable θ_i distributed over the same interval. The number of bidders is greater than 3. If bidder i wins the item, her utility is $\theta_i + t_i$; if bidder i loses the item, her utility is t_i. Here, t_i refers to the auctioneer's money transfer to the bidder, and if $t_i < 0$, it means that the bidder pays to the auctioneer. Note that the money that the auctioneer receives from the bidder will not be refunded

The rule of the auction is as follows: all bid prices must not be negative; and the item will be allocated to the bidder of the highest price. The money is paid as follows: the highest price bidder pays the second-highest price to the auctioneer; at the same time, the highest price bidder and second-highest price bidder will receive the $\frac{1}{n}$ rebate of the third-highest bid price from the auctioneer. In addition, the highest bidder also needs to pay the auctioneer the difference between the second-highest bid and the third-highest bid; and the third-highest bidder and the lower bidders will receive a rebate that is equal to the second-highest bid of $\frac{1}{n}$ from the auctioneer.

1. Prove that there is a dominant strategy for each bidder.

2. Prove that the auctioneer does not lose money.

3. Prove that any bidder will participate in the auction.

4. The auction seems to have all of the good features of the second-price sealed-bid auction, and the bidders receive more surplus. Why do people pay more attention to the second-price sealed-bid auction instead of this one?

Exercise 21.7 (Uncertain Number of Buyers) Suppose that there are N potential buyers, whose private values $\theta_1, \theta_2, \cdots, \theta_n$ are independent and have the same distribution on $[0, 1]$. The cumulative distribution function is Φ.

Each bidder participating in the auction does not know the number of participants in the auction. Each bidder thinks that, except for herself, there are n individuals who participate in the auction with a probability p_n, satisfying $\sum_{n=0}^{N-1} p_n = 1$. Note that everyone has the same belief about the distribution of participants in the auction.

1. When using a second-price sealed-bid auction, find the symmetric Bayesian-Nash equilibrium. Explain why it constitutes an equilibrium.

2. Let $\Psi^{(n)}(\theta) = (\Phi(\theta))^n$. Prove that the expected return of a buyer who participates in this auction with private value θ is:

$$\sum_{n=0}^{N-1} p_n \Psi^{(n)}(\theta) E[\vartheta_1^{(n)} | \vartheta_1^{(n)} < \theta],$$

where $\vartheta_1^{(n)}$ is the highest of n random variables with the same cumulative distribution function Φ.

3. Prove the Revenue Equivalence Theorem in the following situation: Consider a symmetric sealed-bid auction with independent private values, and denote by β a monotonically increasing symmetric equilibrium that satisfies the following assumptions: (a) The winner is the highest bidder; (b) When the private value is 0, the expected return of the buyer is 0. Then, the expected return of a buyer with private value θ is given by the formula in question 2.

4. When using a first-price sealed-bid auction, solve for the symmetric Bayesian-Nash equilibrium strategy.

Exercise 21.8 (Risk-Averse Bidders) Consider a single-item auction. There are n bidders, whose value is θ_i uniformly distributed on the interval $[0, 1]$. Suppose that the participants are risk-averse, and the utility of bidder i to win the item on price p is given by $\sqrt{\theta_i - p}$, where $p \leqq \theta_i$. We assume that no bidder's bid or payment exceeds its value.

1. Write down the bidder's expected payment function at the first-price and second-price sealed-bid auctions.

2. Prove that the bidder with θ_i in the first-price sealed-bid auction chooses the bidding strategy $\beta(\theta_i) = \frac{(n-1)}{(n-1/2)}\theta_i$ that forms a symmetric Bayesian-Nash equilibrium.

3. Prove that, in the second-price sealed-bid auction, reporting the true value is a (weakly) dominant strategy for each bidder.

4. Compare the seller's expected revenues derived from the first-price sealed-bid auction and the second-price sealed-bid auction. In what way is this example different from the case with risk-neutral bidders?

Exercise 21.9 (Auction with Participants Cost) There are n potential bidders, whose private values are independently distributed on $[0, \overline{\theta}]$. The cumulative distribution function is Φ, and the probability density function is φ. In order to participate in the auction, a bidder must pay a nonrefundable participation cost of c. All bidders make decisions at the same time and do not know about others' decisions when they are bidding. Assume that everyone adopts the same strategy.

1. If the seller adopts a first-price auction, characterize the bidder's participation decision and derive the equilibrium bidding price.

2. If the seller adopts a second-price auction, characterize the bidder's participation decision and derive the equilibrium bidding price.

3. Under a symmetric Bayesian-Nash equilibrium, prove that the first-price auction and the second-price auction produce the same expected revenue.

Exercise 21.10 (Auction with Participants Cost, Cao and Tian (2010)) Consider a first-price auction in which there are n potential bidders whose private values are independently distributed on $[0, 1]$, and the cumulative distribution function is Φ. The probability density function is φ. In order to participate in the auction, a bidder must pay a non-refundable participation cost of c (for example, some preparations for participating in the auction). All bidders make participation decisions at the same time, and they are able to observe the number of other people participating in the auction when bidding. We focus on the strategy of threshold value participation, i.e., one participant participates in the auction if and only if her valuation θ exceeds some critical value θ^*.

1. What is the critical value if everyone chooses the same critical value and adopts the same bidding strategy? What is the corresponding bidding price function?

2. For symmetric Bayesian-Nash equilibrium, write down the seller's expected revenue and compare it with that when the number of participants in the above question is unobservable.

3. If potential bidders are divided into two groups, and each group uses a different threshold strategy, write down the equation that solves for the critical value and make a judgment about when such a critical value equilibrium exists.

Exercise 21.11 Use the Revenue-Equivalence Theorem to find the following equilibrium strategies for the auction. Consider a symmetric and independent private value auction with n bidders whose values are drawn from the distribution function $\Phi(\theta_i)$ on $[0, 1]$. Assume that the density function corresponding to Φ is continuously differentiable. Answer the following questions:

1. Find the Bayesian-Nash equilibrium strategy for the all-pay auction.

2. Find the Bayesian-Nash equilibrium strategy for the second-price all-pay auction.

Exercise 21.12 (An Application of Revenue Equivalence Theorem) In 1991, the U.S. Vice President Dan Quayle proposed that the losing party in a lawsuit should give the prevailing party a fee equal to the losing party's legal expenses. Quayle claimed that this could reduce expenditures on legal services (current U.S. law requires each party to pay its own legal fees). Suppose that the two parties $i = 1, 2$ have private values θ_i for winning. θ_i obeys an independent and identical distribution, and the cumulative distribution function is $\Phi(v)$ (we assume that the lowest value is $\underline{\theta}$, i.e., $\Phi(\underline{\theta}) = 0$). Both parties decided at the same time on how much they would spend on legal fees, and the party who spent more legal fees would win the case.

1. By using the Revenue Equivalence Theorem, find the expression of the equilibrium cost of the two parties under the current U.S. law. What other assumptions might you need in order to apply the Revenue-Equivalence Theorem?

2. Use our model to evaluate Quayle's statement without performing more calculations.

3. In the European legal system, the losing party is usually responsible for paying a part of the legal costs of the winning party. Without additional calculations, do you think this rule will increase or decrease the expected legal expenses?

Exercise 21.13 Use the distribution characteristics given in Table 4 below to characterize the corresponding revenue-maximizing mechanism under the Bayesian-Nash equilibrium solution concept. From this characterization, we can summarize what kind of conclusions and explain.

	t_2^1	t_2^2	t_2^3	t_2^4	
t_1^1	$\frac{1}{6}$	0	0	0	θ_1
t_1^2	0	$\frac{1}{6}$	$\frac{1}{6}$	0	θ_1
t_1^3	0	$\frac{1}{6}$	$\frac{1}{6}$	0	θ_2
t_1^4	0	0	0	$\frac{1}{6}$	θ_2
	θ_1	θ_1	θ_2	θ_2	

Table 4 Type Space IV

Exercise 21.14 (Auction with Reserve Price) Consider a first-price auction with $n \geqq 2$ bidders and a reserve price r. Suppose that each bidder i's value θ_i of the auction item is drawn from the i.i.d. distribution $\Phi(\theta_i)$ (not necessarily a uniform distribution) on the interval $[0, 1]$.

1. Explain why bidding with values higher than r is strictly better than not bidding for all bidders, i.e., the equilibrium bidding strategy $\beta_i(\theta_i)$ is in $[r, 1]$.

2. Prove that the equilibrium bidding function is

$$\beta_i(\theta_i) = \frac{r\Phi(r)^{n-1} + \int_r^{\theta_i} x(n-1)\varphi(x)\Phi(x)^{n-2}dx}{\Phi(\theta_i)^{n-1}}.$$

3. Use the formula of integration by parts to prove that the above equilibrium bidding function can be expressed in the following form:

$$\beta_i(\theta_i) = \theta_i - \int_r^{\theta_i} \left(\frac{\Phi(x)}{\Phi(\theta_i)}\right)^{n-1} dx.$$

Exercise 21.15 Under the first-price auction setting with the reserve price r in the above question, use the results of the above question to answer the following two questions:

1. If the cumulative distribution function Φ_A stochastically dominates Φ_B, is the equilibrium bidding price under Φ_A higher than that under Φ_B? Explain your answer.

2. Does the equilibrium bidding price increase in the number of bidders n? Explain your answer.

Exercise 21.16 Consider the first-price auction with two bidders. The utility of bidder i is $u(\theta_i - b_i)$, which is assumed to be concave, i.e., bidders are risk-averse. The valuation of bidder i follows the i.i.d. distribution $\Phi(v_i)$ on the interval $[0, 1]$.

1. If utility function $u(\cdot) : R \to R$ is increasing and strictly concave, and satisfies $u(0) = 0$, prove that the following formula must hold:

$$\frac{u(x)}{u'(x)} > x, \forall x > 0.$$

2. Let $b_i^{RN}(v_i)$ and $b_i^{RA}(v_i)$ represent equilibrium bidding functions for risk-neutral and risk-averse bidders, respectively. Prove that they satisfy the following differential equations:

$$\frac{db_i^{RN}(\theta_i)}{d\theta_i} = \frac{\varphi(\theta_i)}{\Phi(\theta_i)}(\theta_i - b_i^{RN});$$

$$\frac{db_i^{RA}(\theta_i)}{d\theta_i} = \frac{\varphi(\theta_i)}{\Phi(\theta_i)}\frac{u(\theta_i - b_i^{RA})}{u'(\theta_i - b_i^{RA})}.$$

3. Apply the concavity expression in question 1 to prove the following formula:

$$\frac{db_i^{RA}(\theta_i)}{d\theta_i} = \frac{\varphi(\theta_i)}{\Phi(\theta_i)}(\theta_i - b_i^{RA}).$$

4. Prove for all θ, $b_i^{RA}(\theta) \geqq b_i^{RN}(\theta)$.

Exercise 21.17 (Complete Information First-Price Auctions) Consider the first-price auction with bidders 1 and 2. The values of an item are $\theta_1 = 5$ and $\theta_2 = 3$, respectively. Suppose that bidders know the values of each other, but they can only submit one of the following three bids: $b_i = 0$, $b_i = 2$, or $b_i = 4$.

1. Write down the payoff matrix.
2. Which strategy combinations can survive the iterated elimination of strictly dominated strategies?
3. Which strategy combinations can form a Nash equilibrium?

Exercise 21.18 (Expected Payoffs of Two-Bidder Auctions) Consider an auctioneer auctioning an item to two bidders 1 and 2. The value θ_i of bidder i is drawn from a uniform distribution on $[0, 10]$, which is common knowledge to the auctioneer and all bidders. Assume that bidder 1's value is $\theta_1 = 8$. Find the expected utility of bidder 1 under the following two auctions:

1. English auction.
2. First-price auction. The bidder's equilibrium strategy is a linear function of valuation, i.e., $\beta_i(\theta_i) = \alpha_i\theta_i + \beta_i$, where $\alpha_i, \beta_i > 0$.

Exercise 21.19 (Private Value First-Price Auctions) Consider a first-price auction between two risk-neutral bidders. Each bidder i ($i = 1, 2$) chooses to bid price $b_i \geqq 0$ at the same time. The highest bidder gets the item and pays her bidding price. If the bidding prices are the same, then they get the item with the same probability. Prior to the auction, each bidder i privately observed a random variable θ_i, which is uniformly distributed on $[0, 2]$. The actual value of bidder i is equal to $\theta_i + 1$, so that bidder i's payoff function is:

$$u_i = \begin{cases} \theta_i + 1 - b_i, & \text{if } b_i > b_j; \\ \frac{1}{2}(\theta_i + 1 - b_i), & \text{if } b_i = b_j; \\ 0, & \text{if } b_i < b_j. \end{cases}$$

1. Derive the symmetric linear Bayesian-Nash Equilibrium of this game (i.e., each bidder uses the same equilibrium strategy of the form $\beta_i = \alpha\theta_i + \beta$).
2. In this equilibrium, what is the conditional expected payoff for bidder i of type θ_i?

Exercise 21.20 (Common Value First-Price Auctions) Consider a similar first-price auction as in the previous question. The difference now is that the true value of bidder i is $\theta_i + \theta_j$ ($j \neq i$), so that bidder i's payoff function is:

$$u_i = \begin{cases} \theta_i + \theta_j - b_i, & \text{if } b_i > b_j; \\ \frac{1}{2}(\theta_i + \theta_j - b_i), & \text{if } b_i = b_j; \\ 0, & \text{if } b_i < b_j. \end{cases}$$

Note that given bidder i's private signal θ_i, her expected value for an object is $\theta_i + 1$, which is the same as in the above question.

1. Derive the symmetric linear Bayesian-Nash Equilibrium of this game.

2. In this equilibrium, compare the equilibrium bidding price of bidder θ_i with that of the above question. Explain your conclusions.

Exercise 21.21 (Common Value Auctions) Given the following joint density function of the two bidders' signals θ_1 and θ_2 for the value of an object:

$$\varphi(\theta_1, \theta_2) = 1, \forall(\theta_1, \theta_2) \in [0, 1]^2$$

and that the common value of the object is $u = \theta_1 + \theta_2$. For first-price and second-price auctions, find out their Bayesian-Nash equilibrium strategies and calculate equilibrium expected revenues, respectively.

Exercise 21.22 (Common Value Auctions) Given the following joint density function of the two bidders' signals θ_1 and θ_2 for the value of an item

$$\varphi(\theta_1, \theta_2) = \frac{4(1 + \theta_1\theta_2)}{5}, \forall(\theta_1, \theta_2) \in [0, 1]^2,$$

and that the common value of the item is $u = \theta_1 + \theta_2$.

1. What is the Bayesian-Nash equilibrium strategy of bidders when using the second-price auction? And what is the expected revenue of the auctioneer?

2. What is the Bayesian-Nash equilibrium strategy of bidders if we utilize the first-price auction? And what is the expected revenue of the auctioneer? Compared with the second-price auction, which format will the auctioneer prefer?

Exercise 21.23 (Three Formats of Multiple Auctions) Consider the multiple objects auction model with independent private values, in which there are K identical objects to be sold to n bidders. Three sealed-bid auction formats are considered: (1) discriminatory auction; (2) uniform-price auction; and (3) Vickrey auction. In each of these auctions, a bidder is asked to submit K bids b_i^k, satisfying $b_i^1 \geqq b_i^2 \geqq \ldots \geqq b_i^K$, to indicate how much she is willing to pay for each additional unit. We refer to $\mathbf{b}_i = (b_i^1, b_i^2, \ldots, b_i^K)$ as a **bid vector**. Therefore, a total of $n \times K$ bids $\{b_i^k : i = 1, 2, \ldots, n; k = 1, 2, \ldots, K\}$ are collected, and K units are awarded to the K highest of these bids.

1. Define the discriminatory auction, the uniform-price auction, and the Vickrey auction, respectively.
2. Now, consider a situation in which there are seven units ($K = 7$) to be sold to three bidders, and the submitted bid vectors are

$$\mathbf{b}_1 = (64, 46, 43, 40, 32, 15, 5)$$
$$\mathbf{b}_2 = (60, 55, 47, 35, 27, 13, 8)$$
$$\mathbf{b}_3 = (50, 45, 38, 24, 14, 9, 6)$$

   (a) What are the seven highest bids? How many units does each bidder win?
   (b) What is the K-vector of competing bids $\boldsymbol{c}_{-i}$ of bidder i for $i = 1, 2, 3$?
   (c) What is the payment of each bidder under a discriminatory auction?
   (d) What is the market-clearing price and the payment of each bidder under a uniform-price auction?
   (e) What is the payment of each bidder under a Vickrey auction?
   (f) Rank the discriminatory, uniform-price, and Vickrey auctions in terms of the seller's expected revenue.

Exercise 21.24 (Uniform-Price Auctions) Consider a uniform-price auction with 3 items and 2 bidders as follows. The marginal value vector $\mathbf{V}_i = (V_i^1, V_i^2, V_i^3)$ of each bidder i is independently and identically distributed on the set $\{v \in [0,1]^3 | v^1 \geqq v^2 \geqq v^3\}$. The marginal distribution is given as follows:

$$\Phi^1(v^1) = (v^1)^2,$$
$$\Phi^2(v^2) = (2 - v^2)v^2,$$

and Φ^3 is not given. Prove that the bidding strategy $\beta(v^1, v^2, v^3) = (v^1, (v^2)^2, 0)$ constitutes a symmetric Bayesian-Nash equilibrium for this uniform-price auction.

Exercise 21.25 (Sequential Second-Price Auctions) There are 2 homogenous items to be auctioned. There are two bidders, whose types are independently distributed. The type of bidder i is $v_i \in [0, 1]$, $i = 1, 2$, which follows the uniform distribution $\Phi(v)$. Suppose that bidder i's utility for the first unit item is $u_i(v_i^1) = v_i^1$, while for the second unit item is $u_i(v_i^2) = \frac{1}{2}v_i^2$. Assuming that we adopt the sequential second-price auction, answer the following questions:

1. Find the Bayesian-Nash equilibrium strategy for bidders.
2. Derive the seller's expected revenue.
3. Is the Revenue-Equivalence Theorem still valid? Why?

Exercise 21.26 (Discriminatory Auctions) Consider an economic environment that consists of 2 items and 2 bidders. The assumptions are the same as those of the previous question. We now consider the discriminatory auction. In a simultaneous discriminatory auction, bidder i who has won k_i units needs to pay her bid, i.e., the total payment price is $b_i^1 + \cdots + b_i^{k_i}$. Answer the following questions:

1. For the increasing equilibrium, i.e., the higher type v is, the higher the bid is, what is the Bayesian-Nash equilibrium strategy?
2. Derive the seller's expected revenue.

Exercise 21.27 (Uniform-Price Auctions) The economic environment is the same as that in the above two questions. Now consider the case of using a uniform price multi-unit auction. For this uniform price auction with two items and two bidders, answer the following questions:

1. If the bidding price is equal to the highest losing bid, what is the Bayesian-Nash equilibrium strategy? What is the expected revenue of the seller?
2. If $u_i(v_i^2) = v_i^2$, i.e., the value of the item does not decrease, what are the corresponding equilibrium and revenue?

Exercise 21.28 (Sequential Second-Price Auctions, Katzman (1999)) Consider an economic environment in which 2 units of homogenous goods are sold to two bidders in two sequential second-price auctions. Each bidder draws independently θ^1 and θ^2 from the uniform distribution $\Phi(\theta) = \theta,\ \theta \in [0, 1]$. The bidder's value for the first item is $V^1 = \max\{\theta^1, \theta^2\}$, and her value for the second item is $V^2 = \min\{\theta^1, \theta^2\}$.

1. Prove that the following strategy forms a symmetric Bayesian-Nash equilibrium:
   (a) In the first round of auction, bidding $\beta^1(\theta^1, \theta^2) = \frac{1}{2}V^1$;
   (b) Bid truthfully in the second round of auction. In other words, if she wins the auction in the first round, then $\beta^2(\theta^1, \theta^2) = V^2$; otherwise, $\beta^2(\theta^1, \theta^2) = V^1$.

2. Prove that the sequence of the equilibrium bidding price is a submartingale, i.e., $E(P^2|P^1 = p^1) \geqq p^1$, and it must hold in strict inequality with a positive probability.

Exercise 21.29 Consider the setting of the previous question. Answer the following questions:

1. Prove that the following strategy is another symmetric Bayesian-Nash equilibrium:
   (a) In the first round of the auction, bidding $\beta^1(\theta^1, \theta^2) = V^2$;
   (b) In the second round of the auction, bidding the true value.
2. Comment on the sequence that consists of symmetric Bayesian-Nash equilibrium prices (P^1, P^2) generated as above.

Exercise 21.30 Consider the following optimal auction model. There are 2 bidders, and each bidder has two types. For the value of the item, two bidders have the following distribution of prior beliefs.

	θ^1	θ^2
θ^1	$\frac{1}{3}$	$\frac{1}{6}$
θ^2	$\frac{1}{6}$	$\frac{1}{3}$

Table 1 Type Space I

Assume that $\theta^1 = 1, \theta^2 = 2$. The seller has two units of this item, and each buyer only needs one unit at most. Therefore, this is a multi-unit auction model. Answer the following questions:

1. Given the distribution of prior beliefs, derive the expected social surplus.
2. Use the Bayesian-Nash equilibrium solution concept to give a revenue-maximizing mechanism. Under this mechanism, what is the maximum revenue of the seller?
3. Use the ex-post equilibrium solution concept to give a revenue-maximizing mechanism. Under this mechanism, what is the maximum revenue?

Exercise 21.31 Consider a multi-unit auction model with two bidders and the following type-distribution characteristics, where t_i^j represents the participant type and θ^k represents the reward type of the corresponding participant.

	t_2^1	t_2^2	t_2^3	
t_1^1	$\frac{1}{30}$	$\frac{1}{40}$	$\frac{1}{40}$	θ^1
t_1^2	$\frac{1}{10}$	$\frac{1}{30}$	$\frac{1}{30}$	θ^1
t_1^3	$\frac{1}{5}$	$\frac{1}{20}$	0	θ^1
t_1^4	$\frac{1}{20}$	$\frac{1}{5}$	0	θ^2
t_1^5	$\frac{1}{15}$	$\frac{1}{20}$	$\frac{1}{20}$	θ^2
t_1^6	$\frac{1}{20}$	$\frac{1}{60}$	$\frac{1}{60}$	θ^2
	θ^1	θ^2	θ^2	

Table 2 Type Space II

1. Calculate the distribution of reward types. What is the difference compared with the above question? Can you solve for the expected social surplus? If you can, how much is it?

2. Characterize the revenue-maximizing mechanism in Bayesian-Nash equilibrium. Under this mechanism, what is the maximum revenue?

Exercise 21.32 What is the difference between the distribution of reward types corresponding to type space III and Exercise 21.30? In this case, is it possible to characterize a similar revenue-maximizing mechanism that corresponds to the Table in Exercise 21.31? Why?

	t_2^1	t_2^2	
t_1^1	$\frac{1}{30}$	$\frac{1}{20}$	θ^1
t_1^2	$\frac{1}{10}$	$\frac{1}{15}$	θ^1
t_1^3	$\frac{1}{5}$	$\frac{1}{20}$	θ^1
t_1^4	$\frac{1}{20}$	$\frac{1}{5}$	θ^2
t_1^5	$\frac{1}{15}$	$\frac{1}{10}$	θ^2
t_1^6	$\frac{1}{20}$	$\frac{1}{30}$	θ^2
	θ^1	θ^2	

Table 3 Type Space III

Exercise 21.33 (Auctions for Complementary Items) Consider the case of auctioning 4 items $K = \{a_1, a_2, b_1, b_2\}$ to 5 buyers. Buyer 1 is only interested in receiving a_1 and b_1 simultaneously; buyers 2 and 3 are only interested in receiving a_2 and b_2 simultaneously; buyer 4 is only interested in receiving b_1 and b_2 simultaneously; and buyer 5 is only interested in receiving both a_1 and a_2 simultaneously. Specifically, the values are as follows:

$$x^1(a_1b_1) = 10;$$

$$x^2(a_2b_2) = 20;$$

$$x^3(a_2b_2) = 25;$$

$$x^4(b_1b_2) = 10;$$

$$x^5(a_1a_2) = 10.$$

The values of all other combinations are 0s. Use the VCG mechanism to find the efficient allocation and the corresponding payment.

Exercise 21.34 (Combinatorial Auctions, Mochon and Saez (2015)) Three items, A, B, and C, are to be sold to four bidders 1, 2, 3, and 4. The bids are as follows:

S	$b_1(S)$	$b_2(S)$	$b_3(S)$	$b_4(S)$
A	200	150	200	100
B	100	250	100	100
C	100	100	300	100
AB	200	200	200	200
AC	250	300	275	200
ABC	500	400	400	500
BC	150	150	150	200

Given the above information, answer the following questions:

1. What is the revenue-maximizing sale combination for the seller?

2. Find the allocation outcome of the VCG mechanism. Is it a core allocation? If not, give a blocking coalition.

Exercise 21.35 The following table shows the bids for a combinatorial auction with two items and two bidders.

S	$b_1(S)$	$b_2(S)$
A	30	10
B	20	15
AB	40	20

According to the above information, answer the following questions:

1. List all possible allocation outcomes. What is the allocation of the highest value?

2. If we adopt a uniform price rule, how much will the winning bidder pay? How much revenue will the seller receive?

3. If we adopt the VCG mechanism, how much will the winning bidder pay? How much revenue will the seller receive?

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